

axiomTM



The 30 Year Horizon

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New Foreword

On October 1, 2001 Axiom was withdrawn from the market and ended life as a commercial product. On September 3, 2002 Axiom was released under the Modified BSD license, including this document. On August 27, 2003 Axiom was released as free and open source software available for download from the Free Software Foundation's website, Savannah.

Work on Axiom has had the generous support of the Center for Algorithms and Interactive Scientific Computation (CAISS) at City College of New York. Special thanks go to Dr. Gilbert Baumslag for his support of the long term goal.

The online version of this documentation is roughly 1000 pages. In order to make printed versions we've broken it up into three volumes. The first volume is tutorial in nature. The second volume is for programmers. The third volume is reference material. We've also added a fourth volume for developers. All of these changes represent an experiment in print-on-demand delivery of documentation. Time will tell whether the experiment succeeded.

Axiom has been in existence for over thirty years. It is estimated to contain about three hundred man-years of research and has, as of September 3, 2003, 143 people listed in the credits. All of these people have contributed directly or indirectly to making Axiom available. Axiom is being passed to the next generation. I'm looking forward to future milestones.

With that in mind I've introduced the theme of the "30 year horizon". We must invent the tools that support the Computational Mathematician working 30 years from now. How will research be done when every bit of mathematical knowledge is online and instantly available? What happens when we scale Axiom by a factor of 100, giving us 1.1 million domains? How can we integrate theory with code? How will we integrate theorems and proofs of the mathematics with space-time complexity proofs and running code? What visualization tools are needed? How do we support the conceptual structures and semantics of mathematics in effective ways? How do we support results from the sciences? How do we teach the next generation to be effective Computational Mathematicians?

The "30 year horizon" is much nearer than it appears.

Tim Daly
CAISS, City College of New York
November 10, 2003 ((iHy))

Chapter 1

The Axiom Bibliography

This bibliography serves two purposes. It generates output for the ACM bibliography (beebe.bib) and Axiom (axiom.bib).

This bibliography covers areas of computational mathematics. Papers which mention Axiom have a “keyword=” entry of “axiomref”. Papers we have on site have a “paper=” entry. Papers which are referenced in the algebra have the “algebra” tag.

The authors are listed in the index. The topic keywords are listed in the index. Algorithms are mentioned in the index.

The **TO** index entry tries to say that the first named algorithm or author has been updated or improved by the second named algorithm or author.

Introduction of special terms (e.g. Toeplitz matrix) may include a paragraph for those unfamiliar with the terms.

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  acknowledgement = ack-nhfb,
}

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— axiom.bib —

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@book{Book12,
  author = "Axiom Authors",
  title = "Volume 12: Axiom Crystal",
  link = "\url{http://axiom-developer.org/axiom-website/bookvol12.pdf}",
  publisher = "Axiom Project",
  year = "2016",
  keywords = "axiomref",
  beebe = "Daly:2005:AVAi"
}

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— Daly:2005:AVAi —

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@Book{Daly:2005:AVAi,
  author =      "Tim Daly and Martin Dunstan",
  title =       "{Axiom} Volume 12: {Axiom} Crystal",
  pages =       "iv + 9",
  year =        "2005",
  LCCN =        "????",
  bibdate =     "Tue Mar 30 08:43:19 2010",
  bibsource =   "http://www.math.utah.edu/pub/tex/bib/axiom.bib",
  link =        "\url{http://www.axiom-developer.org/axiom-website/bookvol12.pdf}",
  acknowledgement = ack-nhfb,
}

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— axiom.bib —

```

@book{Book13,
  author = "Axiom Authors",
  title = "Volume 13: Proving Axiom Correct",
  link = "\url{http://axiom-developer.org/axiom-website/bookvol13.pdf}",
  publisher = "Axiom Project",
  year = "2016"
}

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— axiom.bib —

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@book{Book14,
  author = "Axiom Authors",
  title = "Volume 14: Algorithms",
  link = "\url{http://axiom-developer.org/axiom-website/bookvol14.pdf}",
  publisher = "Axiom Project",
  year = "2016"
}

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— axiom.bib —

```

@book{Bookbib,
  author = "Axiom Authors",
  title = "Volume Bibliography: Axiom Literature Citations",

```

```

    link = "\url{http://axiom-developer.org/axiom-website/bookvolbib.pdf}",
    publisher = "Axiom Project",
    year = "2016"
}

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— axiom.bib —

```

@book{Bookbug,
  author = "Axiom Authors",
  title = "Volume BugList: Axiom Bugs",
  link = "\url{http://axiom-developer.org/axiom-website/bookvolbug.pdf}",
  publisher = "Axiom Project",
  year = "2016"
}

```

1.2 Algebra Documentation References

— axiom.bib —

```

@misc{Cant16,
  author = "Cantrill, Bryan",
  title = "Oral Tradition in Software Engineering",
  link = "\url{https://www.youtube.com/watch?v=4PaWFYm0kEw}",
  year = "2016"
}

```

— axiom.bib —

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@article{Gons71,
  author = "Gonshor, H.",
  title = "Contributions to Genetic Algebras",
  journal = "Proc. Edinburgh Mathematical Society (Series 2)",
  volume = "17",
  number = "4",
  month = "December",

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year = "1971",
issn = "1464-3839",
pages = "289--298",
doi = "10.1017/S0013091500009548",
link = "\url{http://journals.cambridge.org/article_S0013091500009548}",
algebra = "\newline\refto{domain ALGSC AlgebraGivenByStructuralConstants}",
abstract =
  "Etherington introduced certain algebraic methods into the study of
  population genetics. It was noted that algebras arising in genetic
  systems tend to have certain abstract properties and that these can be
  used to give elegant proofs of some classical stability theorems in
  population genetics."
}

```

— axiom.bib —

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@article{Reed97,
  author = "Reed, Mary Lynn",
  title = "Algebraic Structure of Genetic Inheritance",
  journal = "Bulletin of the American Mathematical Society",
  year = "1997",
  volume = "34",
  number = "2",
  month = "April",
  pages = "107--130",
  algebra = "\newline\refto{domain ALGSC AlgebraGivenByStructuralConstants}",
  link = "\url{http://www.ams.org/bull/1997-34-02/S0273-0979-97-00712-X/S0273-0979-97-00712-X.pdf}",
  abstract =
    "In this paper we will explore the nonassociative algebraic structure
    that naturally occurs as genetic information gets passed down through
    the generations. While modern understanding of genetic inheritance
    initiated with the theories of Charles Darwin, it was the Augustinian
    monk Gregor Mendel who began to uncover the mathematical nature of the
    subject. In fact, the symbolism Mendel used to describe his first
    results (e.g. see his 1866 paper {\sl Experiments in
    Plant-Hybridization}) is quite algebraically suggestive. Seventy four
    years later, I.M.H. Etherington introduced the formal language of
    abstract algebra to the study of genetics in his series of seminal
    papers. In this paper we will discuss the concepts of genetics that
    suggest the underlying algebraic structure of inheritance, and we will
    give a brief overview of the algebras which arise in genetics and some
    of their basic properties and relationships. With the popularity of
    biologically motivated mathematics continuing to rise, we offer this
    survey article as another example of the breadth of mathematics that
    has biological significance. The most comprehensive reference for the
    mathematical research done in this area (through 1980) is

```

```

    Worz-Busekros.",
    paper = "Reed97.pdf"
}

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— axiom.bib —

```

@article{Sims71,
  author = "Sims, Charles",
  title = "Determining the Conjugacy Classes of a Permutation Group",
  journal = "Computers in Algebra and Number Theory, SIAM-AMS Proc.",
  volume = "4",
  publisher = "American Math. Soc.",
  year = "1991",
  pages = "191--195",
  algebra = "\newline\refto{domain PERMGRP PermutationGroup}"
}

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— axiom.bib —

```

@article{Worz80,
  author = {W\orz-Busekros, A.},
  title = "Algebra in Genetics",
  publisher = "Springer-Verlag",
  journal = "Lecture Notes in Biomathematics",
  volume = "36",
  year = "1980",
  algebra = "\newline\refto{domain ALGSC AlgebraGivenByStructuralConstants}"
}

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1.3 Linear Algebra

— axiom.bib —

```

@Unpublished{Kalt01,
  author = "Kaltfen, E.",

```



```

title = "Algorithms for sparse and black box matrices
        over finite fields (Invited talk)",
note = "unknown",
year = "2001",
link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/01/Ka01_Fq6.pdf}",
keywords = "survey",
abstract = "
    Sparse and structured matrices over finite fields occur in many
    settings. Sparse linear systems arise in sieve-based integer factoring
    and discrete logarithm algorithms. Structured matrices arise in
    polynomial factoring algorithms; one example is the famous Q-matrix
    from Berlekamp's method. Sparse diophantine linear problems, like
    computing the Smith canonical form of an integer matrix or computing
    an integer solution to a sparse linear system, are reduced via p-adic
    lifting to sparse matrix analysis over a finite field.

    In the past 10 years there has been substantial activity on the
    improvement of a solution proposed by Wiedemann in 1986. The main new
    ingredients are faster pre-conditioners, projections by an entire
    block of random vectors, Lanczos recurrences, and a connection to
    Kalman realizations of control theory. My talk surveys these
    developments and describe some major unresolved problems.",
paper = "Kalt01.pdf"
}

```

— axiom.bib —

```

@Article{Chen02,
  author = "Chen, L. and Eberly, W. and Kaltofen, E.
           and Saunders, B. D. and Turner, W. J. and Villard, G.",
  title = "Efficient Matrix Preconditioners for Black Box Linear Algebra",
  journal = "Linear Algebra and Applications",
  year = "2002",
  volume = "343--344",
  pages = "119--146",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/02/CEKSTV02.pdf}",
  abstract = "
    The main idea of the ‘‘black box’’ approach in exact linear algebra is
    to reduce matrix problems to the computation of minimum polynomials.
    In most cases preconditioning is necessary to obtain the desired
    result. Here good preconditioners will be used to ensure geometrical
    / algebraic properties on matrices, rather than numerical ones, so we
    do not address a condition number. We offer a review of problems for
    which (algebraic) preconditioning is used, provide a bestiary of
    preconditioning problems, and discuss several preconditioner types to
    solve these problems. We present new conditioners, including

```

conditioners to preserve low displacement rank for Toeplitz-like matrices. We also provide new analyses of preconditioner performance and results on the relations among preconditioning problems and with linear algebra problems. Thus, improvements are offered for the efficiency and applicability of preconditioners. The focus is on linear algebra problems over finite fields, but most results are valid for entries from arbitrary fields.",
 paper = "Chen02.pdf"
 }

— axiom.bib —

```
@InCollection{Kalt11d,
  author = "Kaltofen, Erich and Storjohann, Arne",
  title = "The Complexity of Computational Problems in Exact Linear Algebra",
  booktitle = "Encyclopedia of Applied and Computational Mathematics",
  publisher = "Springer",
  year = "2011",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/11/KS11.pdf}",
  abstract = "
    Computational problems in exact linear algebra including computing an
    exact solution of a system of linear equations with exact scalars,
    which can be exact rational numbers, integers modulo a prime number,
    or algebraic extensions of those represented by their residues modulo
    a minimum polynomial. Classical linear algebra problems are computing
    for a matrix its rank, determinant, characteristic and minimal
    polynomial, and rational canonical form (= Frobenius normal form). For
    matrices with integer and polynomial entries one computes the Hermite
    and Smith normal forms. If a rational matrix is symmetric, one
    determines if the matrix is definite.",
  paper = "Kalt11d.pdf"
}
```

— axiom.bib —

```
@Article{Come12,
  author = "Comer, Matthew T. and Kaltofen, Erich L.",
  title = "On the {Berlekamp}/{Massey} Algorithm and Counting Singular {Hankel}
    Matrices over a Finite Field",
  year = "2012",
  month = "April",
  journal = "Journal of Symbolic Computation",
```

```

volume = "47",
number = "4",
pages = "480--491",
link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/10/CoKa10.pdf}",
abstract = "
  We derive an explicit count for the number of singular  $n \times n$ 
  Hankel (Toeplitz) matrices whose entries range over a finite field
  with  $q$  elements by observing the execution of the Berlekamp / Massey
  algorithm on its elements. Our method yields explicit counts also when
  some entries above or on the anti-diagonal (diagonal) are fixed. For
  example, the number of singular  $n \times n$  Toeplitz matrices with 0's
  on the diagonal is  $q^{2n-3} + q^{n-1} - q^{n-2}$ .

  We also derive the count for all  $n \times n$  Hankel matrices of rank
   $r$  with generic rank profile, I.e., whose first  $r$  leading principal
  submatrices are non-singular and the rest are singular, namely
   $q^r(q-1)^{r-1}$  in the case  $r < n$  and  $q^{r-1}(q-1)^{r-1}$  in the case
   $r=n$ . This result generalizes to block-Hankel matrices as well.",
paper = "Come12.pdf"
}

```

— axiom.bib —

```

@Article{Kalt13a,
  author = "Kaltofen, Erich and Yuhasz, George",
  title = "A Fraction Free Matrix {Berlekamp}/{Massey} Algorithm",
  journal = "Linear Algebra and Applications",
  year = "2013",
  volume = "439",
  number = "9",
  month = "November",
  pages = "2515--2526",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/08/KaYu08.pdf}",
  abstract = "
    We describe a fraction free version of the Matrix Berlekamp / Massey
    algorithm. The algorithm computes a minimal matrix generator of
    linearly generated square matrix sequences over an integral
    domain. The algorithm performs all operations in the integral domain,
    so all divisions performed are exact. For scalar sequences, the matrix
    algorithm specializes to a more efficient algorithm than the algorithm
    currently in the literature. The proof of integrality of the matrix
    algorithm gives a new proof of integrality for the scalar
    specialization.",
  paper = "Kalt13a.pdf"
}

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— axiom.bib —

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@Article{Kalt13,
  author = "Kaltofen, Erich and Yuhasz, George",
  title = "On The Matrix {Berlekamp}--{Massey} Algorithm",
  year = "2013",
  volume = "9",
  number = "4",
  month = "September",
  journal = "ACM Trans. Algorithms",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/06/KaYu06.pdf}",
  abstract = "
    We analyze the Matrix Berlekamp / Massey algorithm, which generalizes
    the Berlekamp / Massey algorithm [Massey 1969] for computing linear
    generators of scalar sequences. The Matrix Berlekamp / Massey
    algorithm computes a minimal matrix generator of a linearly generated
    matrix sequence and has been first introduced by Rissanen [1972a],
    Dickinson et al. [1974], and Coppersmith [1994]. Our version of the
    algorithm makes no restrictions on the rank and dimensions of the
    matrix sequence. We also give new proofs of correctness and complexity
    for the algorithm, which is based on self-contained loop invariants
    and includes an explicit termination criterion for a given
    determinantal degree bound of the minimal matrix generator",
  paper = "Kalt13.pdf"
}
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— axiom.bib —

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@InProceedings{Kalt02a,
  author = "Kaltofen, Erich",
  title = "An output-sensitive variant of the baby steps/\allowbreak
    giant steps determinant algorithm",
  booktitle = "Proc. 2002 Internat. Symp. Symbolic Algebraic Comput.",
  pages = "138--144",
  year = "2002",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/02/Ka02.pdf}",
  paper = "Kalt02a.pdf"
}
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— axiom.bib —

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@InProceedings{Kalt01a,
  author = "Kaltofen, E. and Villard, G.",
  title = "On the complexity of computing determinants",
  booktitle = "Proc. Fifth Asian Symposium on Computer Mathematics
    (ASCM 2001)",
  pages = "13--27",
  isbn = "981-02-4763-X",
  year = "2001",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/01/KaVi01.pdf}",
  abstract = "
    The computation of the determinant of an  $n \times n$  matrix  $A$  of
    numbers or polynomials is a challenge for both numerical and symbolic
    methods. Numerical methods, such as Clarkson's algorithm [10,7] for
    the sign of the determinant must deal with conditionedness that
    determines the number of mantissa bits necessary for obtaining a
    correct sign. Symbolic algorithms that are based on Chinese
    remaindering [6,17,Chapter 5.5] must deal with the fact that the
    length of the determinant in the worse case grows linearly in the
    dimension of the matrix. Hence the number of modular operations is  $n^2$ 
    times the number of arithmetic operations in a given algorithm.
    Hensel lifting combined with rational number recovery [14,1] has cubic
    bit complexity in  $n$ , but the algorithm can only determine a factor
    of the determinant, namely the largest invariant factor. If the matrix
    is similar to a multiple of the identity matrix, the running time is
    again that of Chinese remaindering.",
  paper = "Kalt01a.pdf"
}
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— axiom.bib —

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@Article{Kalt04a,
  author = "Kaltofen, Erich and Villard, Gilles",
  title = "On the Complexity of Computing Determinants",
  journal = "Computational Complexity",
  volume = "13",
  number = "3-4",
  year = "2004",
  pages = "91--130",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/04/KaVi04_2697263.pdf}",
  abstract = "
    We present new baby steps / giant steps algorithms of asymptotically
    fast running time for dense matrix problems. Our algorithms compute
    the determinant, characteristic polynomial, Frobenius normal form and
    Smith normal form of a dense  $n \times n$  matrix  $A$  with integer
```

entries in $(x^{3.2} \log ||A||)^{1+o(1)}$ and $(x^{2.697263} \log ||A||)^{1+o(1)}$ bit operations; here $||A||$ denotes the largest entry in absolute value and the exponent adjustment by ‘‘ $+o(1)$ ’’ captures additional factors $(\log n)^{C_2}(\log \log ||A||)^{C_3}$ for positive real constants C_1, C_2, C_3 . The bit complexity $(n^{3.2} \log ||A||)^{1+o(1)}$ results from using the classical cubic matrix multiplication algorithm. Our algorithms are randomized, and we can certify that the output is the determinant of A in a Las Vegas fashion. The second category of problems deals with the setting where the matrix A has elements from an abstract commutative ring, that is, when no divisions in the domain of entries are possible. We present algorithms that deterministically compute the determinant, characteristic polynomial and adjoint of A with $(n^{3.2+o(1)})$ and $(n^{2.697263})$ ring additions, subtractions, and multiplications.”,

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paper = "Kalt04a.pdf"
}
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— axiom.bib —

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@InProceedings{Kalt97b,
  author = "Eberly, W. and Kaltofen, E.",
  title = "On Randomized {Lanczos} Algorithms",
  booktitle = "Proc. 1997 Internat. Symp. Symbolic Algebraic Comput.",
  year = "1997",
  pages = "176--183",
  link = "url{http://www.math.ncsu.edu/~kaltfen/bibliography/97/EbKa97.pdf}",
  abstract = "
    Las Vegas algorithms that are based on Lanczo's method for solving
    symmetric linear systems are presented and analyzed. These are
    compared to a similar randomized Lanczos algorithm that has been used
    for integer factorization, and to the (provably reliable) algorithm of
    Wiedemann. The analysis suggests that our Lanczos algorithms are
    preferable to several versions of Wiedemann's method for computations
    over large fields, especially for certain symmetric matrix
    computations.",
  paper = "Kalt97b.pdf"
}
```

The Sylvester matrix is used to compute the **resultant** of two polynomials. The Sylvester matrix is formed from the coefficients of the two polynomials. Given a polynomial with degree m and another of degree n form an $(m+n) \times (m+n)$ matrix by filling the matrix from the upper left corner with the coefficients of the first polynomial then shifting down one

row and one column to the right and filling in the coefficients starting there until they hit the right column. Starting at the next row, do the same process for the second polynomial. The determinant of this matrix is the **resultant** of the two polynomials.

For example, given $a_3x^3 + a_2x^2 + a_1x + a_0$ and $b_2x^2 + b_1x + b_0$ the Sylvester matrix is a $(3 + 2) \times (3 + 2)$ matrix:

$$\begin{bmatrix} a_3 & a_2 & a_1 & a_0 & 0 \\ 0 & a_3 & a_2 & a_1 & a_0 \\ b_2 & b_1 & b_0 & 0 & 0 \\ 0 & b_2 & b_1 & b_0 & 0 \\ 0 & 0 & b_2 & b_1 & b_0 \end{bmatrix}$$

The resultant of these two polynomials (assuming a leading coefficient of 1), is the product of the differences $p_i - q_i$ between the roots of the polynomials. If there are roots in common then the product will contain a 0 and the whole equation reduces to 0. This can be used to determine if two polynomials have common roots.

For example, given a polynomial in x with distinct roots a_1 and a_2 it can be factored as $t1 := (x - a_1)(x - a_2)$.

Given a second polynomial in x with distinct roots b_1 , b_2 , and b_3 it can be factored as $t2 := (x - b_1)(x - b_2)(x - b_3)$.

The Axiom call of *resultant*($t1, t2, x$) is

$$(b_1 - a_2)(b_1 - a_1)(b_2 - a_2)(b_2 - a_1)(b_3 - a_2)(b_3 - a_1)$$

In symbolic form the resultant can show the multiplicity of roots when shown in factored form.

— axiom.bib —

```
@InProceedings{Kalt94c,
  author = "Kaltofen, E.",
  title = "Asymptotically fast solution of {Toeplitz}-like singular
    linear systems",
  booktitle = "Proc. 1994 Internat. Symp. Symbolic Algebraic Comput.",
  pages = "297--304",
  year = "1994",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/94/Ka94_issac.pdf}",
  abstract = "
    The Toeplitz-likeness of a matrix (Kailath et al. 1979) is the
    generalization of the notion that a matrix is Toeplitz. Block matrices
    with Toeplitz blocks, such as the Sylvester matrix corresponding to
    the resultant of two univariate polynomials, are Toeplitz-like, as are
    products and inverses of Toeplitz-like matrices. The displacement rank
    of a matrix is a measure for the degree of being Toeplitz-like. For
    example, an  $r \times s$  block matrix with Toeplitz blocks has
    displacement rank  $r+s$  whereas a generic  $N \times N$  matrix has
    displacement rank  $N$ . A matrix of displacement rank  $\alpha$  can be
```

implicitly represented by a sum of α matrices, each of which is the product of a lower triangular and an upper triangular Toeplitz matrices. Such a σ LU representation can usually be obtained efficiently.",
 paper = "Kalt94c.pdf"
 }

— axiom.bib —

```
@Article{Kalt99,
  author = "Kaltofen, E. and Lobo, A.",
  title = "Distributed matrix-free solution of large sparse linear systems over
    finite fields",
  journal = "Algorithmica",
  year = "1999",
  pages = "331--348",
  month = "July--Aug.",
  volume = "24",
  number = "3--4",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/99/KaLo99.pdf}",
  abstract = "
    We describe a coarse-grain parallel approach for the homogeneous
    solution of linear systems. Our solutions are symbolic, i.e., exact
    rather than numerical approximations. We have performed an outer loop
    parallelization that works well in conjunction with a black box
    abstraction for the coefficient matrix. Our implementation can be run
    on a network cluster of UNIX workstations as well as on an SP-2
    multiprocessor. Task distribution and management are effected through
    MPI and other packages. Fault tolerance, checkpointing, and recovery
    are incorporated. Detailed timings are presented for experiments with
    systems that arise in RSA challenge integer factoring efforts. For
    example, we can solve a  $252,222 \times 252,222$  system with about 11.04
    million nonzero entries over the Galois field with two elements using
    four processors of an SP-2 multiprocessor, in about 26.5 hours CPU time.",
  paper = "Kalt99.pdf"
}
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— axiom.bib —

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@InProceedings{Kalt96a,
  author = "Kaltofen, E. and Lobo, A.",
  title = "Distributed matrix-free solution of large sparse linear systems"
```



```

        over finite fields",
booktitle = "Proc. High Performance Computing '96",
year = "1996",
editor = "A. M. Tentner",
pages = "244--247",
organization = "Society for Computer Simulation",
publisher = "Simulation Councils, Inc.",
address = "San Diego, CA",
link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/96/KaLo96_hpc.pdf}",
abstract = "
    We describe a coarse-grain parallel software system for the
    homogeneous solution of linear systems. Our solutions are symbolic,
    i.e., exact rather than numerical approximations. Our implementation
    can be run on a network cluster of SPARC-20 computers and on an SP-2
    multiprocessor. Detailed timings are presented for experiments with
    systems that arise in RSA challenge integer factoring efforts. For
    example, we can solve a  $252,222 \times 252,222$  system with about 11.04
    million non-zero entries over the Galois field with 2 elements using 4
    processors of an SP-2 multiprocessor, in about 26.5 hours CPU time.",
paper = "Kalt96a.pdf"
}

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— axiom.bib —

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@InProceedings{Kalt94a,
author = "Kaltofen, E. and Lobo, A.",
title = "Factoring high-degree polynomials by the black box
        Berlekamp algorithm",
booktitle = "Proc. 1994 Internat. Symp. Symbolic Algebraic Comput.",
pages = "90--98",
year = "1994",
link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/94/KaLo94.ps.gz}",
abstract = "
    Modern techniques for solving structured linear systems over finite
    fields, which use the coefficient matrix as a black box and require an
    efficient algorithm for multiplying this matrix by a vector, are
    applicable to the classical algorithm for factoring a univariate
    polynomial over a finite field by Berlekamp (1967 and 1970). We report
    on a computer implementation of this idea that is based on the
    parallel block Wiedemann linear system solver (Coppersmith 1994 and
    Kaltofen 1993 and 1995). The program uses randomization and we also
    study the expected run time behavior of our method.",
paper = "Kalt94a.ps"
}

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— axiom.bib —

```
@Article{Kalt95,
  author = "Kaltofen, E.",
  title = "Analysis of {Coppersmith}'s block {Wiedemann} algorithm for the
    parallel solution of sparse linear systems",
  journal = "Math. Comput.",
  year = "1995",
  volume = "64",
  number = "210",
  pages = "777--806",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/95/Ka95_mathcomp.pdf}",
  abstract = "
    By using projections by a block of vectors in place of a single vector
    it is possible to parallelize the outer loop of iterative methods for
    solving sparse linear systems. We analyze such a scheme proposed by
    Coppersmith for Wiedemann's coordinate recurrence algorithm, which is
    based in part on the Krylov subspace approach. We prove that by use of
    certain randomizations on the input system the parallel speed up is
    roughly by the number of vectors in the blocks when using as many
    processors. Our analysis is valid for fields of entries that have
    sufficiently large cardinality. Our analysis also deals with an
    arising subproblem of solving a singular block Toeplitz system by use
    of the theory of Toeplitz-like matrices.",
  paper = "Kalt95.pdf"
}
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— axiom.bib —

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@Article{Kalt90a,
  author = "Kaltofen, E. and Krishnamoorthy, M.S. and Saunders, B.D.",
  title = "Parallel algorithms for matrix normal forms",
  journal = "Linear Algebra and Applications",
  year = "1990",
  volume = "136",
  pages = "189--208",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/90/KKS90.pdf}",
  abstract = "
    Here we offer a new randomized parallel algorithm that determines the
    Smith normal form of a matrix with entries being univariate
    polynomials with coefficients in an arbitrary field. The algorithm has
    two important advantages over our previous one: the multipliers
    related the Smith form to the input matrix are computed, and the
```

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algorithm is probabilistic of Las Vegas type, i.e., always finds the
correct answer. The Smith form algorithm is also a good sequential
algorithm. Our algorithm reduces the problem of Smith form
computations to two Hermite form computations. Thus the Smith form
problem has complexity asymptotically that of the Hermite form
problem. We also construct fast parallel algorithms for Jordan normal
form and testing similarity of matrices. Both the similarity and
non-similarity problems are in the complexity class RNC for the usual
coefficient fields, i.e., they can be probabilistically decided in
poly-logarithmic time using polynomially many processors.",
paper = "Kalt90a.pdf"
}

```

— axiom.bib —

```

@Article{Kalt87,
  author = "Kaltofen, E. and Krishnamoorthy, M.S. and Saunders, B.D.",
  title = "Fast parallel computation of Hermite and Smith forms of
    polynomial matrices",
  journal = "SIAM J. Alg. Discrete Math.",
  year = "1987",
  volume = "8",
  pages = "683--690",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/87/KKS87.pdf}",
  abstract = "
    Boolean circuits of polynomial size and poly-logarithmic depth are
    given for computing the Hermite and Smith normal forms of polynomial
    matrices over finite fields and the field of rational numbers. The
    circuits for the Smith normal form computation are probabilistic ones
    and also determine very efficient sequential algorithms. Furthermore,
    we give a polynomial-time deterministic sequential algorithm for the
    Smith normal form over the rationals. The Smith normal form algorithms
    are applied to the Rational canonical form of matrices over finite
    fields and the field of rational numbers.",
  paper = "Kalt87.pdf"
}

```

— axiom.bib —

```

@InProceedings{Kalt92,
  author = "Kaltofen, E. and Pan, V.",
  title = "Processor-efficient parallel solution of linear systems {II}:

```

```

        the positive characteristic and singular cases",
booktitle = "Proc. 33rd Annual Symp. Foundations of Comp. Sci.",
year = "1992",
pages = "714--723",
publisher = "IEEE Computer Society Press",
address = "Los Alamitos, California",
link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/92/KaPa92.pdf}",
abstract = "
    We show that over any field, the solution set to a system of  $n$ 
    linear equations in  $n$  unknowns can be computed in parallel with
    randomization simultaneously in poly-logarithmic time in  $n$  and with
    only as many processors as are utilized to multiply two  $n \times n$ 
    matrices. A time unit represents an arithmetic operation in the
    field. For singular systems our parallel timings are asymptotically as
    fast as those for non-singular systems, due to our avoidance of binary
    search in the matrix rank problem, except when the field has small
    positive characteristic; in that case, binary search is avoided to a
    somewhat higher processor count measure.",
paper = "Kalt92.pdf"
}

```

— axiom.bib —

```

@InProceedings{Kalt91c,
  author = "Kaltofen, E. and Pan, V.",
  title = "Processor efficient parallel solution of linear systems over
          an abstract field",
  booktitle = "Proc. SPAA '91 3rd Ann. ACM Symp. Parallel Algor. Architecture",
  pages = "180--191",
  publisher = "ACM Press",
  year = "1991",
  address = "New York, N.Y.",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/91/KaPa91.pdf}",
  abstract = "
    Parallel randomized algorithms are presented that solve
     $n$ -dimensional systems of linear equations and compute inverses of
     $n \times n$  non-singular matrices over a field in  $O((\log n)^2)$  time,
    where each time unit represents an arithmetic operation in the field
    generated by the matrix entries. The algorithms utilize with a  $O(\log n)$ 
    factor as many processors as are needed to multiply two  $n \times n$ 
    matrices. The algorithms avoid zero divisions with controllably
    high probability provided the  $O(n)$  random elements used are selected
    uniformly from a sufficiently large set. For fields of small positive
    characteristics, the processor count measures of our solutions are
    somewhat higher.",
  paper = "Kalt91c.pdf"
}

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}

 — axiom.bib —

```
@InProceedings{Kalt91,
  author = "Kaltofen, E. and Saunders, B.D.",
  editor = "H. F. Mattson and T. Mora and T. R. N. Rao",
  title = "On {Wiedemann's} method of solving sparse linear systems",
  booktitle = "Proc. AAECC-9",
  series = "Lect. Notes Comput. Sci.",
  volume = "539",
  pages = "29--38",
  publisher = "Springer-Verlag",
  year = "1991",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/91/KaSa91.pdf}",
  abstract = "
    Douglas Wiedemann's (1986) landmark approach to solving sparse linear
    systems over finite fields provides the symbolic counterpart to
    non-combinatorial numerical methods for solving sparse linear systems,
    such as the Lanczos or conjugate gradient method (see Golub and van
    Loan (1983)). The problem is to solve a sparse linear system, when the
    individual entries lie in a generic field, and the only operations
    possible are field arithmetic; the solution is to be exact. Such is
    the situation, for instance, if one works in a finite field. Wiedemann
    bases his approach on Krylov subspaces, but projects further to a
    sequence of individual field elements. By making a link to the
    Berlekamp / Massey problem from coding theory -- the coordinate
    recurrences -- and by using randomization an algorithm is obtained
    with the following property. On input of an  $n \times n$  coefficient
    matrix  $A$  given by a so-called black box, which is a program that can
    multiply the matrix by a vector (see Figure 1), and of a vector  $b$ ,
    the algorithm finds, with high probability in case the system is
    solvable, a random solution vector  $x$  with  $Ax=b$ . It is assumed that
    the field has sufficiently many elements, say no less than  $50n^2 \log(x)$ ,
    otherwise one goes to a finite algebraic extension. The
    complexity of the method is in the general singular case  $O(n \log(n))$ 
    calls to the black box for  $A$  and an additional  $O(n^2 \log(n)^2)$ 
    field arithmetic operations.",
    paper = "Kalt91.pdf"
```

}

 — axiom.bib —

```

@article{Wied86,
  author = "Wiedemann, Douglas H.",
  title = "Solving sparse linear equations over finite fields",
  journal = "IEEE Transactions on Information Theory",
  year = "1986",
  volume = "32",
  number = "1",
  pages = "54-62",
  link = "\url{http://www.csd.uwo.ca/~moreno/CS424/Ressources/WIEDEMANN-IEE-1986.pdf}",
  abstract = "
    A 'coordinate recurrence' method for solving sparse systems of
    linear equations over finite fields is described. The algorithms
    discussed all require  $O(n_1(\omega+n_1)\log^k n_1)$  field operations,
    where  $n_1$  is the maximum dimension of the coefficient matrix,
     $\omega$  is approximately the number of field operations required to
    apply the matrix to a test vector, and the value of  $k$  depends on the
    algorithm. A probabilistic algorithm is shown to exist for finding the
    determinant of a square matrix. Also, probabilistic algorithms are
    shown to exist for finding the minimum polynomial and rank with some
    arbitrarily small possibility of error.",
  paper = "Wied86.pdf"
}

```

1.4 Algebraic Algorithms

— axiom.bib —

```

@InCollection{Diaz97,
  author = "Diaz, A. and Kaltofen, E. and Pan, V.",
  title = "Algebraic Algorithms",
  booktitle = "The Computer Science and Engineering Handbook",
  publisher = "CRC Press",
  year = "1997",
  editor = "A. B. Tucker",
  pages = "226--248",
  address = "Boca Raton, Florida",
  chapter = "10",
  keywords = "survey",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/97/DKP97.ps.gz}",
  ref = "00965",
  abstract = "
    The title's subject is the algorithmic approach to algebra: arithmetic
    with numbers, polynomials, matrices, differential polynomials, such as
     $y^{\prime\prime} + (1/2 + x^{4/4})y$ , truncated series,

```

```

and algebraic sets, i.e.,
quantified expressions such as  $\exists x \in \mathbb{R}: x^4 + p \cdot x + q = 0$ ,
which describes a subset of the two-dimensional space with
coordinates  $p$  and  $q$  for which the given quartic equation has a
real root. Algorithms that manipulate such objects are the backbone
of modern symbolic mathematics software such as the Maple and
Mathematica systems, to name but two among many useful systems. This
chapter restricts itself to algorithms in four areas: linear matrix
algebra, root finding of univariate polynomials, solution of systems
of nonlinear algebraic equations, and polynomial factorization.",
paper = "Diaz97.ps"
}



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— axiom.bib —

@InCollection{Diaz99,
  author = "Diaz, A. and Emiris, I. and Kaltofen, E. and Pan, V.",
  title = "Algebraic Algorithms",
  booktitle = "Algorithms & Theory of Computation Handbook",
  publisher = "CRC Press",
  year = "1999",
  editor = "M. J. Atallah",
  address = "Boca Raton, Florida",
  pages = "16.1--16.27",
  isbn = "0-8493-2649-4",
  keywords = "survey",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/99/DEKP99.ps.gz}",
  abstract = "
    The title's subject is the algorithmic approach to algebra: arithmetic
    with numbers, polynomials, matrices, differential polynomials, such as
     $y^n + (1/2 + x^{4/4})y$ , truncated series, and algebraic sets, i.e.,
    quantified expressions such as  $\exists x \in \mathbb{R}: x^4 + p \cdot x + q = 0$ ,
    which describes a subset of the two-dimensional space with
    coordinates  $p$  and  $q$  for which the given quartic equation has a
    real root. Algorithms that manipulate such objects are the backbone
    of modern symbolic mathematics software such as the Maple and
    Mathematica systems, to name but two among many useful systems. This
    chapter restricts itself to algorithms in four areas: linear algebra,
    root finding for univariate polynomials, solution of systems of
    nonlinear algebraic equations, and polynomial factorization (see
    section 5 on some pointers to the vast further material on algebraic
    algorithms and section 2.2 and [Pan 1993] on further applications to
    the graph and combinatorial computations).",
  paper = "Diaz99.ps"
}

```

— axiom.bib —

```
@InCollection{Kalt87a,
  author = "Kaltofen, E.",
  editor = "J. F. Traub",
  title = "Computer algebra algorithms",
  booktitle = "Annual Review in Computer Science",
  pages = "91--118",
  publisher = "Annual Reviews Inc.",
  year = "1987",
  volume = "2",
  address = "Palo Alto, California",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/87/Ka87_annrev.pdf}",
  abstract = "
    The origins of the discipline of computer algebra can be found in
    Issac Newton's {\sl Universal Arithmetic} (1728) [130], where methods
    for methods for manipulating universal mathematical expressions (i.e.
    formulas containing symbolic indeterminates) and algorithms for
    solving equations built with these expressions are systematically
    discussed. One can interpret the misson of computer algebra as the
    construction of computer systems that enable scientific or engineering
    users, for instance, to carry out mathematical manipulation
    automatically. Indeed, systems with this goal already exist, among
    them {MACSYMA}, {MAPLE}, {muMATH}, {REDUCE}, {SAC/2}, {SCRATCHPAD/II},
    and {SMP}. These systems carry out scientific computing tasks, whose
    results are distinguished from numerical computing in two principle
    aspects.",
  paper = "Kalt87a.pdf",
  keywords = "survey,axiomref"
}
```

— axiom.bib —

```
@book{Somm01,
  author = "Sommer, Gerald",
  title = "Geometric Computing with Clifford Algebras",
  year = "2001",
  publisher = "Springer",
  isbn = "3-540-41198-4"
}
```

— axiom.bib —

```
@misc{Wang90a,
  author = "Wang, Dongming",
  title = "Some Notes on Algebraic Methods for Geometry Theorem Proving",
  link = "\url{http://www-polsys.lip6.fr/~wang/papers/GTPnote.ps.gz}",
  year = "1990",
  abstract =
    "A new geometry theorem prover which provides the first complete
    implementation of Wu's method and includes several Groebner bases
    based methods is reported. This prover has been used to prove a number
    of non-trivial geometry theorems including several {\sl large} ones
    with less space and time cost than using the existing provers. The
    author presents a new technique by introducing the notion of {\sl
    normal ascending set}. This technique yields in some sense {\sl
    simpler} non-degenerate conditions for Wu's method and allows one to
    prove geometry theorems using characteristic sets but Groeber bases
    type reduction. Parallel variants of Wu's method are discussed; an
    implementation of the parallelized version of his algorithm utilizing
    workstation networks has also been included in our prover. Timing
    statistics for a set of typical examples is given.",
  paper = "Wang90a.pdf"
}
```

— axiom.bib —

```
@inproceedings{Wang92,
  author = "Wang, Dongming",
  title = "A Method for Factorizing Multivariate Polynomials over Successive
    Algebraic Extension Fields",
  booktitle = "Mathematics and Mathematics-Mechanization (2001)",
  pages = "138-172",
  institution = "Johannes Kepler University",
  link = "\url{http://www-polsys.lip6.fr/~wang/papers/Factor.ps.gz}",
  year = "1992",
  abstract =
    "We present a method for factorizing multivariate polynomials over
    algebraic fields obtained from successive extensions of the rational
    number field. The basic idea underlying this method is the reduction
    of polynomial factorization over algebraic extension fields to the
    factorization over the rational number field via linear transformation
    and the computation of characteristic sets with respect to a proper
    variable ordering. The factors over the algebraic extension fields are
    finally determined via GCD (greatest common divisor) computations. We
    have implemented this method in the Maple system. Preliminary
```

```

    experiments show that it is rather efficient. We give timing
    statistics in Maple 4.3 on 40 test examples which were partly taken
    from the literature and partly randomly generated. For all those
    examples to which Maple built-in algorithm is applicable, our
    algorithm is always faster.",
    paper = "Wang92.pdf"
}

```

1.5 Sparse Linear Systems

— axiom.bib —

```

@InProceedings{Kalt96b,
  author = "Kaltofen, E.",
  title = "Blocked iterative sparse linear system solvers for finite fields",
  booktitle = "Proc. Symp. Parallel Comput. Solving Large Scale Irregular
    Applic. (Stratagem '96)",
  editor = "C. Roucairol",
  publisher = "INRIA",
  address = "Sophia Antipolis, France",
  pages = "91--95",
  year = "1996",
  keywords = "survey",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/96/Ka96_stratagem.ps.gz}",
  abstract = "
    The problem of solving a large sparse or structured system of linear
    equations in the symbolic context, for instance when the coefficients
    lie in a finite field, has arisen in several applications. A famous
    example are the linear systems of  $\mathbb{F}_2$ , the field with 2
    elements, that arise in sieve based integer factoring algorithms. For
    example, for the factorization of the RSA-130 challenge number several
    column dependencies of a  $3504823 \times 3516502$  matrix with an
    average of  $39.4$  non-zero entries per column needed to be computed
    [10]. A second example is the Berlekamp polynomial factorization
    algorithm [6]. In that example, the matrix is not explicitly
    constructed, but instead a fast algorithm for performing the matrix
    times vector product is used. Further examples for such ‘black box
    matrices’ arise in the power series solution of algebraic or
    differential equations by undetermined coefficients. The arising
    linear systems for the coefficients usually have a distinct structure
    that allows a fast coefficient matrix times vector product.",
  paper = "Kalt96b.ps"
}

```

1.6 Matrix Determinants

— axiom.bib —

```
@Article{Kalt04,
  author = "Kaltofen, E. and Villard, G.",
  title = "Computing the sign or the value of the determinant of an integer
          matrix, a complexity survey",
  journal = "J. Computational Applied Math.",
  volume = "162",
  number = "1",
  month = "January",
  pages = "133--146",
  year = "2004",
  keywords = "survey",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/02/KaVi02.pdf}",
  abstract = "
    Computation of the sign of the determinant of a matrix and the
    determiniant itself is a challenge for both numerical and exact
    methods. We survey the complexity of existing methods to solve these
    problems when the input is an  $n \times n$  matrix  $A$  with integer
    entries. We study the bit-complexities of the algorithms
    asymptotically in  $n$  and the norm  $\|A\|$ . Existing approaches rely on
    numerical approximate computations, on exact computations, or on both
    types of arithmetic in combination.",
  paper = "Kalt04.pdf"
}
```

1.7 Open Problems

— axiom.bib —

```
@Article{Kalt00,
  author = "Kaltofen, E.",
  title = "Challenges of Symbolic Computation My Favorite Open Problems",
  journal = "Journal of Symbolic Computation",
  volume = "29",
  number = "6",
  pages = "891--919",
```

```

year = "2000",
keywords = "survey",
link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/2K/Ka2K.pdf}",
abstract = "
  The success of the symbolic mathematical computation discipline is
  striking. The theoretical advances have been continuous and significant:
  Gr{\o}bner bases, the Risch integration algorithm, integer lattice
  basis reduction, hypergeometric summation algorithms, etc. From the
  beginning in the early 60s, it has been the tradition of our discipline
  to create software that makes our ideas readily available to a
  scientists, engineers, and education: {SAC-1}, {Reduce}, {Macsyma}, etc.
  The commercial viability of our system products is proven by Maple and
  Mathematica.

  Today's user communities of symbolic computation systems are diverse:
  educators, engineers, stock market analysts, etc. The mathematics and
  computer science in the design and implementation of our algorithms are
  sophisticated. The research challenges in symbolic computation at the
  close of the 20th century are formidable.

  I state my favorite eight open problems in symbolic computations. They
  range from problems in symbolic /numeric computing, symbolic algorithm
  synthesis, to system component construction. I have worked on seven of
  my problems and borrowed one from George Collins. I present background
  to each of my problems and a clear-cut test that evaluates whether a
  proposed attack has solved one of my problems. An additional ninth
  open problem by Rob Corless and David Jeffrey on complex function
  semantics is given in an appendix.",
paper = "Kalt00.pdf"
}

```

1.8 Parallel Evaluation

— axiom.bib —

```

@InCollection{Kalt93a,
  author = "Kaltofen, E.",
  editor = "J. Reif",
  title = "Dynamic parallel evaluation of computation {DAG}s",
  booktitle = "Synthesis of Parallel Algorithms",
  pages = "723--758",
  publisher = "Morgan Kaufmann Publ.",
  year = "1993",
  address = "San Mateo, California",

```

```

keywords = "survey",
link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/93/Ka93_synthesis.ps.gz}",
abstract = "
  One generic parallel evaluation scheme for algebraic objects, that of
  evaluating algebraic computation trees or formulas, is presented by
  Miller in a preceding chapter of this book. However, there are basic
  algebraic functions for which the tree model of computation seems not
  sufficient to allow an efficient -- even sequential -- decision-free
  algebraic computation. The formula model essentially restricts the use
  of an intermediate result to a single place, because the parse tree
  nodes have fan-out one. If an intermediate result participates in the
  computations of several further nodes, in the tree model it must be
  recomputed anew for each of these nodes. It is a small formal change
  to allow node values to propagate to more than one node deeper level
  of the computation. Thus we obtain the {\sl algebraic circuit model},
  which is equivalent to the {\sl straight-line program model}.",
paper = "Kalt93a.ps"
}

```

1.9 Hybrid Symbolic/Numeric

— axiom.bib —

```

@misc{Corl93,
  author = "Corless, R. M. and Gonnet, G. H. and Hare, D. E. G. and
           Jeffrey, D. J. and Knuth, D. E.",
  title = "On the Lambert W Function",
  year = "1993",
  link = "\url{http://cs.uwaterloo.ca/research/tr/1993/03/W.pdf}",
  abstract =
    "The Lambert W function is defined to be the multivalued inverse of
    the function $w \mapsto we^w$. It has many applications in pure
    and applied mathematics, some of which are briefly described here. We
    present a new discussion of the complex branches of $W$, an asymptotic
    expansion valid for all branches, an efficient numerical procedure for
    evaluating the function to arbitrary precision, and a method for the
    symbolic integration of expressions containing $W$.",
  paper = "Corl93.pdf"
}

```

— axiom.bib —

```
@InProceedings{Hutt10,
  author = "Hutton, Sharon E. and Kaltofen, Erich L. and Zhi, Lihong",
  title = "Computing the radius of positive semidefiniteness of a
    multivariate real polynomial via a dual of {Seidenberg}'s method",
  year = "2010",
  booktitle = "Internat. Symp. Symbolic Algebraic Comput. ISSAC'10",
  pages = "227--234",
  month = "July",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/10/HKZ10.pdf}",
  abstract = "
    We give a stability criterion for real polynomial inequalities with
    floating point or inexact scalars by estimating from below or
    computing the radius of semdefiniteness. That radius is the maximum
    deformation of the polynomial coefficient vector measured in a weighted
    Euclidean vector norm within which the inequality remains true. A
    large radius means that the inequalities may be considered numerically
    valid.

    The radius of positive (or negative) semidefiniteness is the distance
    to the nearest polynomial with a real root, which has been thoroughly
    studied before. We solve this problem by parameterized Lagrangian
    multipliers and Karush-Kuhn-Tucker conditions. Our algorithms can
    compute the radius for several simultaneous inequalities including
    possibly additional linear coefficient constraints. Our distance
    measure is the weighted Euclidean coefficient norm, but we also
    discuss several formulas for the weighted infinity and 1-norms.

    The computation of the nearest polynomial with a real root can be
    interpreted as a dual of Seidenberg's method that decides if a real
    hypersurface contains a real point. Sums-of-squares rational lower
    bound certificates for the radius of semidefiniteness provide a new
    approach to solving Seidenberg's problem, especially when the
    coefficients are numeric. They also offer a surprising alternative
    sum-of-squares proof for those polynomials that themselves cannot be
    represented by a polynomial sum-of-squares but that have a positive
    distance to the nearest indefinite polynomial.",
  paper = "Hutt10.pdf"
}
```

— axiom.bib —

```
@InProceedings{Kalt06,
  author = "Kaltofen, Erich and Zhi, Lihong",
```

```

title = "Hybrid Symbolic-Numeric Computation",
year = "2006",
booktitle = "Internat. Symp. Symbolic Algebraic Comput. ISSAC'06",
pages = "7",
link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/06/KaZhi06.pdf}",
abstract = "
  Several standard problems in symbolic computation, such as greatest
  common divisors and factorization of polynomials, sparse
  interpolation, or computing solutions to overdetermined systems of
  polynomial equations have non-trivial solutions only if the input
  coefficients satisfy certain algebraic constraints. Errors in the
  coefficients due to floating point round-off or through physical
  measurement thus render the exact symbolic algorithms unusable. By
  symbolic-numeric methods one computes minimal deformations of the
  coefficients that yield non-trivial results. We will present hybrid
  algorithms and benchmark computations based on Gauss-Newton
  optimization, singular value decomposition (SVD) and
  structure-preserving total least squares (STLS) fitting for several of
  the above problems.

  A significant body of results to solve those ‘‘approximate computer
  algebra’’ problems has been discovered in the past 10 years. In the
  Computer Algebra Handbook the section on ‘‘Hybrid Methods’’ concludes
  as follows [2]: ‘‘The challenge of hybrid symbolic-numeric algorithms
  is to explore the effects of imprecision, discontinuity, and
  algorithmic complexity by applying mathematical optimization,
  perturbation theory, and inexact arithmetic and other tools in order
  to solve mathematical problems that today are not solvable by
  numerical or symbolic methods alone.’’ The focus of our tutorial is
  on how to formulate several approximate symbolic computation problems
  as numerical problems in linear algebra and optimization and on
  software that realizes their solutions."
paper = "Kalt06.pdf"
}

```

— axiom.bib —

```

@misc{Hjor10,
  author = "Hjorth-Jensen, Morten",
  title = "Computational Physics",
  year = "2010",
  link = "\url{http://depts.washington.edu/ph506/Hjorth-JensenLectures2010.pdf}",
  paper = "Hjor10.pdf"
}

```

— axiom.bib —

```
@InProceedings{Kalt09,
  author = "Kaltofen, Erich and Yang, Zhengfeng and Zhi, Lihong",
  title = "A Proof of the {Monotone Column Permanent (MCP) Conjecture} for
    Dimension 4 via Sums-Of-Squares of Rational Functions",
  year = "2009",
  booktitle = "Proc. 2009 Internat. Workshop on Symbolic-Numeric Comput.",
  pages = "65--69",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/09/KYZ09.pdf}",
  abstract = "
    For a proof of the monotone column permanent (MCP) conjecture for
    dimension 4 it is sufficient to show that 4 polynomials, which come
    from the permanents of real matrices, are nonnegative for all real
    values of the variables, where the degrees and the number of the
    variables of these polynomials are all 8. Here we apply a hybrid
    symbolic-numerical algorithm for certifying that these polynomials can
    be written as an exact fraction of two polynomial sums-of-squares
    (SOS) with rational coefficients.",
  paper = "Kalt09.pdf"
}
```

— axiom.bib —

```
@Article{Kalt12,
  author = "Kaltofen, Erich L. and Li, Bin and Yang, Zhengfeng and
    Zhi, Lihong",
  title = "Exact Certification in Global Polynomial Optimization
    Via Sums-Of-Squares of Rational Functions
    with Rational Coefficients",
  year = "2012",
  month = "January",
  journal = "Journal of Symbolic Computation",
  volume = "47",
  number = "1",
  pages = "1--15",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/09/KLYZ09.pdf}",
  abstract = "
    We present a hybrid symbolic-numeric algorithm for certifying a
    polynomial or rational function with rational coefficients to be
    non-negative for all real values of the variables by computing a
    representation for it as a fraction of two polynomial sum-of-squares
    (SOS) with rational coefficients. Our new approach turns the earlier
```


methods by Peyrl and Parrilo and SCN'07 and ours at ISSAC'08 both based on polynomial SOS, which do not always exist, into a universal algorithm for all inputs via Artin's theorem.

Furthermore, we scrutinize the all-important process of converting the numerical SOS numerators and denominators produced by block semidefinite programming into an exact rational identity. We improve on our own Newton iteration-based high precision refinement algorithm by compressing the initial Gram matrices and by deploying rational vector recovery aside from orthogonal projection. We successfully demonstrate our algorithm on 1. various exceptional SOS problems with necessary polynomial denominators from the literature and on 2. very large (thousands of variables) SOS lower bound certificates for Rump's model problem (up to $n=18$, factor degree $=17$).",
 paper = "Kalt12.pdf"
 }
 }
 —

— axiom.bib —

```
@InProceedings{Kalt08b,
  author = "Kaltofen, Erich and Li, Bin and Yang, Zhengfeng and Zhi, Lihong",
  title = "Exact Certification of Global Optimality of Approximate
    Factorizations Via Rationalizing Sums-Of-Squares
    with Floating Point Scalars",
  year = "2008",
  booktitle = "Internat. Symp. Symbolic Algebraic Comput. ISSAC'08",
  pages = "155--163",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/08/KLYZ08.pdf}",
  abstract = "
```

We generalize the technique by Peyrl and Parrilo [Proc. SNC 2007] to computing lower bound certificates for several well-known factorization problems in hybrid symbolic-numeric computation. The idea is to transform a numerical sum-of-squares (SOS) representation of a positive polynomial into an exact rational identity. Our algorithms successfully certify accurate rational lower bounds near the irrational global optima for benchmark approximate polynomial greatest common divisors and multivariate polynomial irreducibility radii from the literature, and factor coefficient bounds in the setting of a model problem by Rump (up to $n=14$, factor degree $=13$).

The numeric SOSes produced by the current fixed precision semi-definite programming (SDP) packages (SeDuMi, SOSTOOLS, YALMIP) are usually too coarse to allow successful projection to exact SOSes via Maple 11's exact linear algebra. Therefore, before projection we refine the SOSes by rank-preserving Newton iteration. For smaller problems the starting SOSes for Newton can be guessed without SDP

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    ("SDP-free SOS"), but for larger inputs we additionally appeal to
    sparsity techniques in our SDP formulation.",
    paper = "Kalt08b.pdf"
}

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@InProceedings{Kalt06b,
  author = "Kaltofen, Erich and Yang, Zhengfeng and Zhi, Lihong",
  title = "Approximate greatest common divisors of several polynomials
          with linearly constrained coefficients and singular polynomials",
  year = "2006",
  booktitle = "Internat. Symp. Symbolic Algebraic Comput. ISSAC'06",
  pages = "169--176",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/06/KYZ06.pdf}",
  abstract = "
    We consider the problem of computing minimal real or complex
    deformations to the coefficients in a list of relatively prime real or
    complex multivariate polynomials such that the deformed polynomials
    have a greatest common divisor (GCD) of a least a given degree $k$. In
    addition, we restrict the deformed coefficients by a given set of
    linear constraints, thus introducing the {\sl linearly constrained
    approximate GCD} problem. We present an algorithm based on a version
    of the structured total least norm (STLN) method and demonstrate, on a
    diverse set of benchmark polynomials, that the algorithm in practice
    computes globally minimal approximations. As an application of the
    linearly constrained approximate GCD problem, we present an STLN-based
    method that computes for a real or complex polynomial the nearest real
    or complex polynomial the nearest real or complex polynomial that has
    a root of multiplicity at least $k$. We demonstrate that the
    algorithm in practice computes, on the benchmark polynomials given in
    the literature, the known globally optimal nearest singular
    polynomials. Our algorithms can handle, via randomized
    preconditioning, the difficult case when the nearest solution to a
    list of real input polynomials actually has non-real complex
    coefficients.",
  paper = "Kalt06b.pdf"
}

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```

@InCollection{Kalt05,

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author = "Kaltofen, Erich and Yang, Zhengfeng and Zhi, Lihong",
title = "Structured Low Rank Approximation of a {Sylvester} Matrix",
booktitle = "Symbolic-Numeric Computation",
publisher = "Springer",
pages = "69--83",
year = "2005",
link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/05/KYZ05.pdf}",
abstract = "
    The task of determining the approximate greatest common divisor (GCD)
    of univariate polynomials with inexact coefficients can be formulated
    as computing for a given Sylvester matrix a new Sylvester matrix of
    lower rank whose entries are near the corresponding entries of that
    input matrix. We solve the approximate GCD problem by a new method
    based on structured total least norm (STLN) algorithms, in our case
    for matrices with Sylvester structure. We present iterative algorithms
    that compute an approximate GCD and that can certify an approximate
     $\epsilon$ -GCD when a tolerance  $\epsilon$  is given on input. Each
    single iteration is carried out with a number of floating point
    operations that is of cubic order in the input degrees. We also
    demonstrate the practical performance of our algorithms on a diverse
    set of univariate pairs of polynomials.",
paper = "Kalt05.pdf"
}

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@InProceedings{Kalt03a,
  author = "Kaltofen, Erich and May, John",
  title = "On Approximate Irreducibility of Polynomials in Several Variables",
  year = "2003",
  booktitle = "Internat. Symp. Symbolic Algebraic Comput. ISSAC'03",
  pages = "161--168",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/03/KM03.pdf}",
  abstract = "
    We study the problem of bounding all factorizable polynomials away
    from a polynomial that is absolutely irreducible. Such separation
    bounds are useful for testing whether a numerical polynomial is
    absolutely irreducible, given a certain tolerance on its coefficients
    Using an absolute irreducibility criterion due to Ruppert, we are able
    to find useful separation bounds, in several norms, for bivariate
    polynomials. We also use Ruppert's criterion to derive new, more
    effective Noether forms for polynomials of arbitrarily many
    variables. These forms lead to small separation bounds for polynomials
    of arbitrarily many variables.",
  paper = "Kalt03a.pdf"
}

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@InProceedings{Gao04a,
  author = "Gao, Shuhong and Kaltofen, Erich and May, John P. and
           Yang, Zhengfeng and Zhi, Lihong",
  title = "Approximate factorization of multivariate polynomials via
           differential equations",
  year = "2004",
  booktitle = "Internat. Symp. Symbolic Algebraic Comput. ISSAC'04",
  pages = "167--174",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/04/GKMYZ04.pdf}",
  abstract = "
    The input to our algorithm is a multivariate polynomial, whose complex
    rational coefficient are considered imprecise with an unknown error
    that causes  $f$  to be irreducible over the complex numbers  $\mathbb{C}$ .
    We seek to perturb the coefficients by a small quantity such that the
    resulting polynomial factors over  $\mathbb{C}$ . Ideally, one would like to
    minimize the perturbation in some selected distance measure, but no
    efficient algorithm for that is known. We give a numerical
    multivariate greatest common divisor algorithm and use it on a
    numerical variant of algorithms by W. M. Ruppert and S. Gao. Our
    numerical factorizer makes repeated use of singular value
    decompositions. We demonstrate on a significant body of experimental
    data that our algorithm is practical and can find factorizable
    polynomials within a distance that is about the same in relative
    magnitude as the input error, even when the relative error in the
    input is substantial ( $10^{-3}$ ).",
  paper = "Gao04a.pdf"
}
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— axiom.bib —

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@Article{Kalt08,
  author = "Kaltofen, Erich and May, John and Yang, Zhengfeng and Zhi, Lihong",
  title = "Approximate Factorization of Multivariate Polynomials Using
           Singular Value Decomposition",
  year = "2008",
  journal = "Journal of Symbolic Computation",
  volume = "43",
  number = "5",
  pages = "359--376",
```

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    link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/07/KMYZ07.pdf}",
    paper = "Kalt08.pdf"
}

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— axiom.bib —

```

@InProceedings{Hitz99,
  author = "Hitz, M.A. and Kaltofen, E. and Lakshman, Y.N.",
  title = "Efficient Algorithms for Computing the Nearest Polynomial
    With A Real Root and Related Problems",
  booktitle = "Proc. 1999 Internat. Symp. Symbolic Algebraic Comput.",
  pages = "205--212",
  year = "1999",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/99/HKL99.pdf}",
  paper = "Hitz99.pdf"
}

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— axiom.bib —

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@InProceedings{Hitz98,
  author = "Hitz, M. A. and Kaltofen, E.",
  title = "Efficient Algorithms for Computing the Nearest Polynomial
    with Constrained Roots",
  booktitle = "Proc. 1998 Internat. Symp. Symbolic Algebraic Comput.",
  year = "1998",
  pages = "236--243",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/98/HiKa98.pdf}",
  paper = "Hitz98.pdf"
}

```

1.10 Software Systems

— axiom.bib —

```

@InProceedings{Diaz91,

```

```

author = "Diaz, A. and Kaltofen,E. and Schmitz, K. and Valente, T.",
title = "DSC A System for Distributed Symbolic Computation",
booktitle = "Proc. 1991 Internat. Symp. Symbolic Algebraic Comput.",
pages = "323--332",
year = "1991",
link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/91/DKSV91.pdf}",
paper = "Diaz91.pdf"
}

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— axiom.bib —

```

@InProceedings{Chan94,
  author = "Chan, K.C. and Diaz, A. and Kaltofen, E.",
  editor = "R. J. Lopez",
  title = "A distributed approach to problem solving in Maple",
  booktitle = "Maple V: Mathematics and its Application",
  pages = "13--21",
  publisher = {Birkh\ "auser},
  year = "1994",
  series = "Proceedings of the Maple Summer Workshop and Symposium (MSWS'94)",
  address = "Boston",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/94/CDK94.ps.gz}",
  paper = "Chan94.ps"
}

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— axiom.bib —

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@InProceedings{Duma02,
  author = "Dumas, J.-G. and Gautier, T. and Giesbrecht, M. and Giorgi, P.
           and Hovinen, B. and Kaltofen, E. and Saunders, B.D. and
           Turner, W.J. and Villard, G.",
  title = "{LinBox}: A Generic Library for Exact Linear Algebra",
  booktitle = "Proc. First Internat. Congress Math. Software ICMS 2002,
              Beijing, China",
  pages = "40--50",
  year = "2002",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/02/Detal02.pdf}",
  paper = "Duma02.pdf"
}

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— axiom.bib —

```
@InProceedings{Kalt05a,
  author = "Kaltofen, Erich and Morozov, Dmitriy and Yuhasz, George",
  title = "Generic Matrix Multiplication and Memory Management in {LinBox}",
  year = "2005",
  booktitle = "Internat. Symp. Symbolic Algebraic Comput. ISSAC'05",
  pages = "216--223",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/05/KMY05.pdf}",
  paper = "Kalt05a.pdf"
}
```

— axiom.bib —

```
@InProceedings{Diaz98,
  author = "Diaz, A. and Kaltofen, E.",
  title = "{FoxBox}, a System for Manipulating Symbolic Objects in Black Box Representation",
  booktitle = "Proc. 1998 Internat. Symp. Symbolic Algebraic Comput.",
  publisher = "ACM Press",
  year = "1998",
  pages = "30--37",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/98/DiKa98.pdf}",
  abstract =
    "The FOXBOX system puts in practice the black box representation of symbolic objects and provides algorithms for performing the symbolic calculus with such representations. Black box objects are stored as functions. For instance: a black box polynomial is a procedure that takes values for the variables as input and evaluates the polynomial at that given point. FOXBOX can compute the greatest common divisor and factorize polynomials in black box representation, producing as output new black boxes. It also can compute the standard sparse distributed representation of a black box polynomial, for example, one which was computed for an irreducible factor. We establish that the black box representation of objects can push the size of symbolic expressions far beyond what standard data structures could handle before."
```

Furthermore, FOXBOX demonstrates the generic program design methodology. The FOXBOX system is written in C++. C++ template arguments provide for abstract domain types. Currently, FOXBOX can be compiled with SACLIB 1.1, Gnu-MP 1.0, and NTL 2.0 as its underlying field and polynomial arithmetic. Multiple arithmetic plugins can be used in the same computation. FOXBOX provides an MPI compliant distribution mechanism that allows for parallel and distributed

```

    execution of FOXBOX programs. Finally, FOXBOX plugs into a
    server/client-style Maple application interface.",
    paper = "Diaz98.pdf"
}

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— axiom.bib —

```

@InProceedings{Diaz93,
  author = "Diaz, A. and Kaltofen, E. and Lobo, A. and Valente, T.",
  editor = "A. Miola",
  title = "Process scheduling in {DSC} and the large sparse linear
    systems challenge",
  booktitle = "Proc. DISCO '93",
  series = "Lect. Notes Comput. Sci.",
  pages = "66--80",
  year = "1993",
  volume = "722",
  publisher = "Springer-Verlag",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/93/DHKL93.pdf}",
  paper = "Diaz93.pdf"
}

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— axiom.bib —

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@Article{Diaz95a,
  author = "Diaz, A. and Hitz, M. and Kaltofen, E. and Lobo, A. and
    Valente, T.",
  title = "Process scheduling in {DSC} and the large sparse linear
    systems challenge",
  journal = "Journal of Symbolic Computing",
  year = "1995",
  volume = "19",
  number = "1--3",
  pages = "269--282",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/95/DHKL95.pdf}",
  paper = "Diaz95a.pdf"
}

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— axiom.bib —


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@Article{Free88,
  author = "Freeman, T.S. and Imirzian, G. and Kaltofen, E. and
           Yagati, Lakshman",
  title = "DAGWOOD: A system for manipulating polynomials given by
           straight-line programs",
  journal = "ACM Trans. Math. Software",
  year = "1988",
  volume = "14",
  number = "3",
  pages = "218--240",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/88/FIKY88.pdf}",
  paper = "Free88.pdf"
}

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1.11 The Seven Dwarfs

— axiom.bib —

```

@InCollection{Kalt10a,
  author = "Kaltofen, Erich L.",
  title = "The ‘‘{Seven} {Dwarfs}’’ of Symbolic Computation",
  booktitle = "Numeric and Symbolic Scientific Computing
              Progress and Prospects",
  publisher = "Springer",
  pages = "95--104",
  year = "2010",
  keywords = "survey",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/10/Ka10_7dwarfs.pdf}",
  paper = "Kalt10a.pdf"
}

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1.12 Solving Systems of Equations

— axiom.bib —

```

@inproceedings{Bro86,
  author = "Bronstein, Manuel",
  title = "Gsolve: a faster algorithm for solving systems of algebraic

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        equations",
booktitle = "Proc of 5th ACM SYMSAC",
year = "1986",
pages = "247-249",
isbn = "0-89791-199-7",
abstract = "
    We apply the elimination property of Gr{\o}bner bases with respect to
    pure lexicographic ordering to solve systems of algebraic equations.
    We suggest reasons for this approach to be faster than the resultant
    technique, and give examples and timings that show that it is indeed
    faster and more correct, than MACSYMA's solve."
}

```

— axiom.bib —

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@inproceedings{Corl95,
  author = "Corless, Robert M. and Gianni, Patrizia, M. and Trager, Barry M.
    and Watt, Stephen M.",
  title = "The Singular Value Decomposition for Polynomial Systems",
  booktitle = "ISSAC 95",
  year = "1995",
  pages = "195-207",
  publisher = "ACM",
  abstract =
    "This paper introduces singular value decomposition (SVD) algorithms
    for some standard polynomial computations, in the case where the
    coefficients are inexact or imperfectly known. We first give an
    algorithm for computing univariate GCD's which gives {\sl exact}
    results for interesting {\sl nearby} problems, and give efficient
    algorithms for computing precisely how nearby. We generalize this to
    multivariate GCD computations. Next, we adapt Lazard's $$-resultant
    algorithm for the solution of overdetermined systems of polynomial
    equations to the inexact-coefficient case. We also briefly discuss an
    application of the modified Lazard's method to the location of
    singular points on approximately known projections of algebraic curves.",
  paper = "Corl95.pdf",
  keywords = "axiomref",
}

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— axiom.bib —

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@article{Lixx10,

```

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author = "Li, Xiaoliang and Mou, Chenqi and Wang, Dongming",
title = "Decomposing polynomial sets into simple sets over finite fields:
        The zero-dimensional case",
comment = "Provides clear polynomial algorithms",
journal = "Computers and Mathematics with Applications",
volume = "60",
pages = "2983-2997",
year = "2010",
abstract =
    "This paper presents algorithms for decomposing any zero-dimensional
    polynomial set into simple sets over an arbitrary finite field, with
    an associated ideal or zero decomposition. As a key ingredient of
    these algorithms, we generalize the squarefree decomposition approach
    for univariate polynomials over a finite field to that over the field
    product determined by a simple set. As a subprocedure of the
    generalized squarefree decomposition approach, a method is proposed to
    extract the  $p$ th root of any element in the field
    product. Experiments with a preliminary implementation show the
    effectiveness of our algorithms.",
paper = "Lixx10.pdf"
}

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1.13 Numerical Algorithms

— axiom.bib —

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@book{Acto70,
  author = "Acton, F.S.",
  title = "Numerical Methods that (Usually) Work",
  year = "1970",
  publisher = "Harper and Row",
  address = "New York, USA"
}

```

— axiom.bib —

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@book{Acto96,
  author = "Acton, F.S.",
  title = "Real Computing Made Real: Preventing Errors in Scientific
          and Engineering Calculations",
}

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year = "1996",
publisher = "Princeton University Press",
address = "Princeton, N.J. USA",
isbn = "0-691-03663-2"
}

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— axiom.bib —

```

@techreport{Ahre15,
  author = "Ahrens, Peter and Nguyen, Hong Diep and Demmel, James",
  title = "Efficient Reproducible Floating Point Summation and BLAS",
  institution = "University of California, Berkeley",
  year = "2015",
  month = "December",
  type = "technical report",
  number = "229",
  link = "\url{http://www.eecs.berkeley.edu/Pubs/TechRpts/2015/EECS-2015-229.pdf}",
  abstract =
    "We define reproducibility to mean getting bitwise identical results
    from multiple runs of the same program, perhaps with different
    hardware resources or other changes that should ideally not change the
    answer. Many users depend on reproducibility for debugging or
    correctness. However, dynamic scheduling of parallel computing
    resources, combined with nonassociativity of floating point addition,
    makes attaining reproducibility a challenge even for simple operations
    like summing a vector of numbers, or more complicated operations like
    Basic Linear Algebra Subprograms (BLAS). We describe an algorithm that
    computes a reproducible sum of floating point numbers independent of
    the order of summation. The algorithm depends only on a subset of the
    IEEE Floating Point Standard 754-2008. It is communication-optimal, in
    the sense that it does just one pass over the data in the sequential
    case, or one reduction operation in the parallel case, requiring an
    ‘‘accumulator’’ represented by just 6 floating point words (more can
    be used if higher precision is desired). The arithmetic code with a
    6-word accumulator is  $7n$  floating point additions to sum  $n$  words,
    and (in IEEE double precision) the final error bound can be up to
     $10^8$  times smaller than the error bound for conventional
    summation. We describe the basic summation algorithm, the software
    infrastructure used to build reproducible BLAS (ReproBLAS), and
    performance results. For example, when computing the dot product of
    4096 double precision floating point numbers, we get a  $4\times$  slowdown
    compared to Intel Math Kernel Library (MKL) running on an Intel Core
    i7-2600 CPU operating at 3.4 GHz and 256 KB L2 Cache.",
  paper = "Ahre15.pdf"
}

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— axiom.bib —

```
@article{Alef00,
  author = "Alefeld, G. and Mayer, G.",
  title = "Interval analysis: Theory and applications",
  journal = "J. Comput. Appl. Math.",
  volume = "121",
  pages = "421-464",
  year = "2000"
}
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— axiom.bib —

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@article{Anda94,
  author = "Anda, A.A. and Park, H.",
  title = "Fast plane rotations with dynamic scaling",
  journal = "SIAM J. matrix Anal. Appl.",
  volume = "15",
  year = "1994",
  pages = "162-174"
}
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— axiom.bib —

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@phdthesis{Anda95,
  author = "Anda, Andrew Allen",
  title = "Self-Scaling Fast Plane Rotation Algorithms",
  school = "University of Minnesota",
  year = "1995",
  month = "February",
  abstract =
    "A suite of {\sl self-scaling} fast circular plane rotation algorithms
    is developed which obviates the monitoring and periodic rescaling
    necessitated by the standard set of fast plane rotation
    algorithms. Self-Scaling fast rotations dynamically preserve the
    normed scaling of the diagonal factor matrix whereas standard fast
    rotations exhibit divergent scaling. Variations on standard fast
    rotation matrices are developed and algorithms which implement them
    are offered. Self-Scaling fast rotations are shown to be essentially
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as accurate as slow rotations and at least as efficient as standard fast rotations. Computational experimental results utilizing the Cray-2 illustrate the effectively stable scaling exhibited by self-scaling fast rotations. Jacobi-class algorithms with one-sided alterations are developed for the algebraic eigenvalue decomposition using self-scaling fast plane rotations and one-sided modifications. The new algorithms are shown to be both accurate and efficient on both vector and parallel architectures. The utility is described of applying fast plane rotations towards the rank-one update and downdate of least squares factorizations. The equivalence is illuminated of LINPACK, hyperbolic rotation, and fast negatively weighted plane rotation downdating. Algorithms are presented which apply self-scaling fast plane rotations to the QR factorization for stiff least squares problems. Both fast and standard Givens rotation-based algorithms are shown to produce accurate results regardless of row sorting even with extremely heavily row weighted matrices. Such matrices emanate, e.g. from equality constrained least squares problems solved via the weighting method. The necessity of column sorting is emphasized. Numerical tests expose the Householder QR factorization algorithm to be sensitive to row sorting and it yields less accurate results for greater weights. Additionally, the modified Gram-Schmidt algorithm is shown to be sensitive to row sorting to a notably significant but lesser degree. Self-Scaling fast plane rotation algorithms, having competitive computational complexities, must therefore be the method of choice for the QR factorization of stiff matrices. Timing results on the Cray 2 [XY]M/P, and C90, of rotations both in and out of a matrix factorization context are presented. Architectural features that can best exploit the advantageous features of the new fast rotations are subsequently discussed.",

paper = "Anda95.pdf"

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— axiom.bib —

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@misc{LAPA99,
  author = "Anderson E. et al.",
  title = "LAPACK User's Guide Third Addition",
  year = "1999",
  month = "August",
  link = "\url{http://www.netlib.org/lapack/lug/}"
}
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— axiom.bib —

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@book{Ande99,
  author = "Anderson, E. and Bai, Z. and Bischof, S. and Blackford, S. and
    Demmel, J. and Dongarra, J. J. and DuCroz, J. and Greenbaum, A.
    and Hammarling, S. and McKenney, A. Sorensen, D. C.",
  title = "LAPACK Users' Guide",
  publisher = "SIAM",
  year = "1999",
  isbn = "0-89871-447-8",
  link = "\url{http://www.netlib.org/lapack/lug/}"
}
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— axiom.bib —

```
@misc{Baud12,
  author = "Baudin, Michael and Smith, Robert L.",
  title = "A Robust Complex Division in Scilab",
  link = "\url{http://arxiv.org/pdf/1210.4539.pdf}",
  year = "2012",
  month = "October",
  abstract =
    "The most widely used algorithm for floating point complex division,
    known as Smith's method, may fail more often than expected. This
    document presents two improved complex division algorithms. We present
    a proof of robustness of the first improved algorithm. Numerical
    simulations show that this algorithm performs well in practice and is
    significantly more robust than other known implementations. By
    combining additional scaling methods with this first algorithm, we
    were able to create a second algorithm, which rarely fails.",
  paper = "Baud12.pdf"
}
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— axiom.bib —

```
@article{Bind02,
  author = "Bindel, D. and Demmel, J. and Kahan, W. and Marques, O.",
  title = "On computing Givens rotations reliably and efficiently",
  journal = "ACM Trans. Math. Software",
  volume = "28",
}
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    pages = "206-238",
    year = "2002"
}

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— axiom.bib —

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@article{Blac97,
  author = "Blackford, L. S. and Cleary, A. and Demmel, J. and Dhillon, I.
    and Dongarra, J. J. and Hammarling, S. and Petitet, A. and
    Ren, H. and Stanley, K. and Whaley, R. C.",
  title = "Practical experience in the numerical dangers of heterogeneous
    computing",
  journal = "ACM Trans. Math. Software",
  volume = "23",
  pages = "133-147",
  year = "1997"
}

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— axiom.bib —

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@techreport{MMEF96,
  author = "Boisvert, Ronald F. and Pozo, Roldan and Remington, Karin A.",
  title = "The Matrix Market Exchange Formats: Initial Design",
  year = "1996",
  month = "December",
  institution = "National Institute of Standards and Technology",
  type = "Technical Report",
  link = "\url{http://math.nist.gov/MatrixMarket/reports/MMformat.ps}",
  abstract =
    "We propose elementary ASCII exchange formats for matrices. Specific
    instances of the format are defined for dense and sparse matrices with
    real, complex, integer and pattern entries, with special cases for
    symmetric, skew-symmetric and Hermitian matrices. Sparse matrices are
    represented in a coordinate storage format. The overall file structure
    is designed to allow future definition of other specialized matrix
    formats, as well as for objects other than matrices.",
  paper = "MMEF96.pdf"
}

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— axiom.bib —

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@article{Bran97,  
  author = "Brankin, R. W. and Gladwell, I.",  
  title = "rksuite\_90: Fortran 90 software for ordinary differential  
          equation initial-value problems",  
  journal = "ACM Trans. Math. Software",  
  volume = "23",  
  pages = "402-415",  
  year = "1997"  
}
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— axiom.bib —

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@techreport{Bran92,  
  author = "Brankin, R. W. and Gladwell, I. and Shampine, L. F.",  
  title = "RKSUITE: A suite of runge-kutta codes for the initial value  
          problem for ODEs",  
  year = "1992",  
  institution = "Southern Methodist University, Dept of Math.",  
  number = "Softreport 92-S1",  
  type = "Technical Report"  
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@article{Bran93,  
  author = "Brankin, R. W. and Gladwell, I. and Shampine, L. F.",  
  title = "RKSUITE: A suite of explicit runge-kutta codes",  
  journal = "Contributions of Numerical Mathematics",  
  pages = "41-53",  
  publisher = "World Scientific",  
  year = "1993"  
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@book{Brit92,
  author = "Britton, J. L.",
  title = "Collected Works of A. M. Turing: Pure Mathematics",
  publisher = "North-Holland",
  year = "1992",
  isbn = "0-444-88059-3"
}
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— axiom.bib —

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@misc{Bron99,
  author = "Bronstein, Manuel",
  title = "Fast Deterministic Computation of Determinants of Dense Matrices",
  link = "\url{http://www-sop.inria.fr/cafe/Manuel.Bronstein/publications/mb_papers.html}",
  abstract = "
    In this paper we consider deterministic computation of the exact
    determinant of a dense matrix  $M$  of integers. We present a new
    algorithm with worst case complexity
     $[O(n^4(\log n + \log \|M\|) + n^3 \log^2 \|M\|)]$ ,
    where  $n$  is the dimension of the matrix
    and  $\|M\|$  is a bound on the entries in  $M$ , but with
    average expected complexity
     $[O(n^4 + m^3(\log n + \log \|M\|)^2)]$ ,
    assuming some plausible properties about the distribution of  $M$ .
    We will also describe a practical version of the algorithm and include
    timing data to compare this algorithm with existing ones. Our result
    does not depend on ‘fast’ integer or matrix techniques.",
  paper = "Bron99.pdf"
}
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— axiom.bib —

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@misc{Broo11,
  author = "Brookes, Mike",
  title = "The Matrix Reference Manual",
  year = "2011",
  link = "\url{http://www.ee.ic.ac.uk/hp/staff/dmb/matrix/intro.html}"
}
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— axiom.bib —

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@misc{Chel06,
  author = "Chellappa, Srinivas and Franchetti, Franz and Puschel, Markus",
  title = "How To Write Fast Numerical Code: A Small Introduction",
  year = "2006",
  institution = "Carnegie Mellon University",
  link = "\url{https://users.ece.cmu.edu/~franzf/papers/gttse07.pdf}",
  algebra = "\newline\refto{package BLAS1 BlasLevelOne}",
  abstract =
    "The complexity of modern computing platforms has made it extremely
    difficult to write numerical code that achieves the best possible
    performance. Straightforward implementations based on algorithms that
    minimize the operations count often fall short in performance by at
    least one order of magnitude. This tutorial introduces the reader to a
    set of general techniques to improve the performance of numerical
    code, focusing on optimizations for the computer's memory hierarchy.
    Further, program generators are discussed as a way to reduce the
    implementation and optimization effort. Two running examples are
    used to demonstrate these techniques: matrix-matrix multiplication
    and the discrete Fourier transform.",
  paper = "Chel06.pdf"
}
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— axiom.bib —

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@book{Chai96,
  author = "Chaitin-Chatelin, F. and Fraysse, V.",
  title = "Lectures on Finite Precision Computations",
  publisher = "SIAM",
  year = "1996",
  isbn = "0-89871-358-7"
}
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— axiom.bib —

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@article{Chan83,
  author = "Chan, T. F. and Golub, G. H. and LeVeque, R. J.",
  title = "Algorithms for computing the sample variance: Analysis and
    recommendations",
}
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journal = "The American Statistician",  
volume = "37",  
pages = "242-247",  
year = "1983"  
}
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— axiom.bib —

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@article{Cool03,  
author = "Cools, R. and Haegemans, A.",  
title = "Algorithm 824: CUBPACK: A package for automatic cubature;  
framework description",  
journal = "ACM Trans. Math. Software",  
volume = "29",  
pages = "287-296",  
year = "2003"  
}
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— axiom.bib —

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@techreport{Cox00,  
author = "Cox, M. G. and Dainton, M. P. and Harris, P. M.",  
title = "Testing spreadsheets and other packages used in metrology:  
Testing functions for the calculation of standard deviation",  
year = "2000",  
institution = "National Physical Lab, Teddington, Middlesex UK",  
type = "Technical Report",  
number = "NPL Report CMSC07/00"  
}
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— axiom.bib —

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@article{Davi11,  
author = "Davis, Timothy A. and Hu, Yifan",  
title = "The University of Florida Sparse Matrix Collection",  
journal = "ACM Trans. on Math. Software",  
volume = "38",
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number = "1",
year = "2011",
month = "November",
link = "\url{http://yifanhu.net/PUB/matrices.pdf}",
abstract =
    "We describe the University of Florida Sparse Matrix Collection, a large
    and actively growing set of sparse matrices that arise in real
    applications. The Collection is widely used by the numerical linear
    algebra community for the development and performance evaluation of
    sparse matrix algorithms. It allows for robust and repeatable
    experiments: robust because performance results with artificially
    generated matrices can be misleading, and repeatable because matrices
    are curated and made publicly available in many formats. Its matrices
    cover a wide spectrum of domains, including those arising from
    problems with underlying 2D or 3D geometry (as structural engineering,
    computational fluid dynamics, model reduction, electromagnetics,
    semiconductor devices, thermodynamics, materials, acoustics, computer
    graphics/vision, robotics/kinematics, and other discretizations) and
    those that typically do not have such geometry (optimization, circuit
    simulation, economic and financial modeling, theoretical and quantum
    chemistry, chemical process simulation, mathematics and statistics,
    power networks, and other networks and graphs). We provide software
    for accessing and managing the Collection, from MATLAB, Mathematica,
    Fortran, and C, as well as an online search capability. Graph
    visualization of the matrices is provided, and a new multilevel
    coarsening scheme is proposed to facilitate this task.",
paper = "Davi11.pdf"
}

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— axiom.bib —

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@techreport{Davi16,
  author = "Davis, Timothy and Rajamanickam, Sivasankaran and
    Sid-Lakhdar, Wissam M.",
  title = "A survey of direct methods for sparse linear systems",
  year = "2016",
  month = "April",
  institution = "Texas A and M",
  type = "Technical Report",
  link = "\url{http://faculty.cse.tamu.edu/davis/publications_files/survey_tech_report.pdf}",
  abstract =
    "Wilkinson defined a sparse matrix as one with enough zeros that it
    pays to take advantage of them. This informal yet practical definition
    captures the essence of the goal of direct methods for solving sparse
    matrix problems. They exploit the sparsity of a matrix to solve
    problems economically: much faster and using far less memory than if

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all the entries of a matrix were stored and took part in explicit computations. These methods form the backbone of a wide range of problems in computational science. A glimpse of the breadth of applications relying on sparse solvers can be seen in the origins of matrices in published matrix benchmark collections. The goal of this survey article is to impart a working knowledge of the underlying theory and practice of sparse direct methods for solving linear systems and least-squares problems, and to provide an overview of the algorithms, data structures, and software available to solve these problems, so that the reader can both understand the methods and know how best to use them.",
 paper = "Davi16.pdf"
}

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@techreport{Demm88,
  author = "Demmel, James and Kahan, W.",
  title = "Computing Small Singular Values of Bidiagonal Matrices with
    Guaranteed High Relative Accuracy",
  year = "1988",
  institution = "New York University",
  type = "Technical Report",
  number = "326",
  abstract =
    "Computing the singular values of a bidiagonal matrix is the final
    phase of the standard algorithm for the singular value decomposition
    of a general matrix. We present a new algorithm which computes all the
    singular values of a bidiagonal matrix to high relative accuracy
    independent of their magnitudes. In contrast, the standard algorithm
    for bidiagonal matrices may compute small singular values with no
    relative accuracy at all. Numerical experiments show that the new
    algorithm is comparable in speed to the standard algorithm, and
    frequently faster. We also show how to accurately compute tiny
    eigenvalues of some classes of symmetric tridiagonal matrices using
    the same technique.",
  paper = "Demm88.pdf"
}
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@techreport{Demm05,
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author = "Demmel, James and Hida, Yozo and Kahan, W. and Li, Xiaoye S.
        and Mukherjee, Soni and Riedy, E. Jason",
title = "Error Bounds from Extra Precise Iterative Refinement",
year = "2005",
institution = "Univerity of California, Berkeley",
type = "Technical Report",
number = "165",
link = "\url{http://www.netlib.org/lapack/lawnspdf/lawn165.pdf}",
abstract =
    "We present the design and testing of an algorithm for iterative
    refinement of the solution of linear equations, where the residual is
    computed with extra precision. This algorithm was originally proposed
    in the 1960s as a means to compute very accurate solutions to all but
    the most ill-conditioned linear systems of equations. However two
    obstacles have until now prevented its adoption in standard subroutine
    libraries like LAPACK: (1) There was no standard way to access the
    higher precision arithmetic needed to compute residuals, and (2) it
    was unclear how to compute a reliable error bound for the computed
    solution. The completion of the new BLAS Technical Forum Standard has
    recently removed the first obstacle. To overcome the second obstacle,
    we show how a single application of iterative refinement can be used
    to compute an error bound in any norm at small cost, and use this to
    compute both an error bound in the usual infinity norm, and a
    componentwise relative error bound.

    We report extensive test results on over 6.2 million matrices of
    dimension 5, 10, 100, and 1000. As long as a normwise
    (resp. componentwise) condition number computed by the algorithm is
    less than  $1/\max(10, \sqrt{n})\epsilon_w$ , the computed normwise
    (resp. componentwise) error bound is at most
     $2\max(10, \sqrt{n})\epsilon_w$ , and indeed bounds the true error. Here,
     $n$  is the matrix dimension and  $\epsilon_w$  is a single precision
    roundoff error. For worse conditioned problems, we get similarly small
    correct error bounds in over 89.4% of cases.",
paper = "Demm05.pdf"
}

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@techreport{Demm08,
  author = "Demmel, James and Hoemmen, Mark and Hida, Yozo
          and Riedy, E. Jason",
  title = "Non-Negative Diagonals and High Performance on Low-Profile
          Matrices from Householder QR",
  year = "2008",
  institution = "Univerity of California, Berkeley",
}

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type = "Technical Report",
number = "203",
link = "\url{http://www.netlib.org/lapack/lawnspdf/lawn203.pdf}",
abstract =
  "The Householder reflections used in LAPACK's QR factorization leave
  positive and negative real entries along R's diagonal. This is
  sufficient for most applications of QR factorizations, but a few
  require that R have a non-negative diagonal. This note provides a new
  Householder generation routine to produce a non-negative
  diagonal. Additionally, we find that scanning for trailing zeros in
  the generated reflections leads to large performance improvements when
  applying reflections with many trailing zeros. Factoring low-profile
  matrices, those with non-zero entries mostly near the diagonal (e.g
  band matrices), now requires far fewer operations. For example, QR
  factorization of matrices with profile width  $b$  that are stored
  densely in an  $n \times n$  matrix improves from  $O(n^3)$ 
  to  $O(n^2 + nb^2)$ .",
paper = "Demm08.pdf"
}

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— axiom.bib —

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@article{Demm90,
  author = "Demmel, James and Kahan, W.",
  title = "Accurate Singular Values of Bidiagonal Matrices",
  journal = "SIAM J. Sci. Stat. Comput.",
  volume = "11",
  number = "5",
  pages = "873-912",
  year = "1990",
  link = "\url{http://www.netlib.org/lapack/lawnspdf/lawn03.pdf}",
  abstract =
    "Computing the singular values of a bidiagonal matrix is the final
    phase of the standard algorithm for the singular value decomposition
    of a general matrix. We present a new algorithm which computes all the
    singular values of a bidiagonal matrix to high relative accuracy
    independent of their magnitudes. In contrast, the standard algorithm
    for bidiagonal matrices may compute small singular values with no
    relative accuracy at all. Numerical experiments show that the new
    algorithm is comparable in speed to the standard algorithm, and
    frequently faster.",
  paper = "Demm90.pdf"
}

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@article{Demm92,
  author = "Demmel, James and Veselic, Kresimir",
  title = "Jacobi's Method is More Accurate than QR",
  journal = "SIAM J. Matrix Anal. and Appl",
  volume = "13",
  number = "4",
  year = "1992",
  pages = "1204-1245",
  link = "\url{http://www.netlib.org/lapack/lawnspdf/lawn15.pdf}",
  abstract =
    "We show that Jacobi's method (with a proper stopping criterion)
    computes small eigenvalues of symmetric positive definite matrices
    with a uniformly better relative accuracy bound than QR, divide and
    conquer, traditional bisection, or any algorithm which first involves
    tridiagonalizing the matrix. In fact, modulo an assumption based on
    extensive numerical tests, we show that Jacobi's method is optimally
    accurate in the following sense: if the matrix is such that small
    relative errors in its entries cause small relative errors in its
    eigenvalues, Jacobi will compute them with nearly this accuracy. In
    other words, as long as the initial matrix has small relative errors
    in each component, even using infinite precision will not improve on
    Jacobi (modulo factors of dimensionality). We also show the
    eigenvectors are computed more accurately by Jacobi than previously
    thought possible. We prove similar results for using one-sided Jacobi
    for the singular value decomposition of a general matrix.",
  paper = "Demm92.pdf"
}
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@article{Rijk98,
  author = "de Rijk, P.P.",
  title = "A One-sided Jacobi Algorithm for Computing the Singular Value
    Decomposition on a vector computer",
  journal = "SIAM J. Sci. Stat. Comput.",
  volume = "10",
  number = "2",
  month = "March",
  year = "1989",
  pages = "359-371",
  abstract =
    "An old algorithm for computing the singular value decomposition,
    which was first mentioned by Hestenes, has gained renewed interest
```

because of its properties of parallelism and vectorizability. Some computational modifications are given and a comparison with the well-known Golub-Reinsch algorithm is made. Comparative experiments on a CYBER 205 are reported."

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@phdthesis{Dhil97,
  author = "Dhillon, Inderjit Singh",
  title = "A New  $O(n^2)$  Algorithm for the Symmetric Tridiagonal
          Eigenvalue/Eigenvector Problem",
  school = "University of California, Berkeley",
  year = "1997",
  link = "\url{http://www.eecs.berkeley.edu/Pubs/TechRpts/1997/CSD-97-971.pdf}",
  abstract =
```

```
  "Computing the eigenvalues and orthogonal eigenvectors of an  $n \times n$ 
  symmetric tridiagonal matrix is an important task that arises while
  solving any symmetric eigenproblem. All practical software requires
   $O(n^3)$  time to compute all the eigenvectors and ensure their
  orthogonality when eigenvalues are close. In the first part of this
  thesis we review earlier work and show how some existing
  implementations of inverse iteration can fail in surprising ways.
```

The main contribution of this thesis is a new $O(n^2)$, easily parallelizable algorithm for solving the tridiagonal eigenproblem. Three main advances lead to our new algorithm. A tridiagonal matrix is traditionally represented by its diagonal and off-diagonal elements. Our most important advance is in recognizing that its bidiagonal factors are ‘better’ for computational purposes. The use of bidiagonals enables us to invoke a relative criterion to judge when eigenvalues are ‘close’. The second advance comes with using multiple bidiagonal factorizations in order to compute different eigenvectors independently of each other. Thirdly, we use carefully chosen dqds-like transformations as inner loops to compute eigenpairs that are highly accurate and ‘faithful’ to the various bidiagonal representations. Orthogonality of the eigenvectors is a consequence of this accuracy. Only $O(n)$ work per eigenpair is needed by our new algorithm.

Conventional wisdom is that there is usually a trade-off between speed and accuracy in numerical procedures, i.e., higher accuracy can be achieved only at the expense of greater computing time. An interesting aspect of our work is that increased accuracy in the eigenvalues and eigenvectors obviates the need for explicit orthogonalization and leads to greater speed.

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We present timing and accuracy results comparing a computer
implementation of our new algorithm with four existing EISPACK and
LAPACK software routines. Our test-bed contains a variety of
tridiagonal matrices, some coming from quantum chemistry
applications. The numerical results demonstrate the superiority of
our new algorithm. For example, on a matrix of order 966 that occurs in
the modeling of a biphenyl molecule our method is about 10 times
faster than LAPACK's inverse iteration on a serial IBM RS/6000
processor and nearly 100 times faster on a 128 processor IBM SP2
parallel machine.",
paper = "Dhil97.pdf"
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@article{Dhil04a,
  author = "Dhillon, Inderjit S. and Parlett, Beresford N.",
  title = "Multiple representations to compute orthogonal eigenvectors
    of symmetric tridiagonal matrices",
  journal = "Linear Algebra and its Applications",
  volume = "387",
  number = "1",
  pages = "1-28",
  year = "2004",
  month = "August",
  abstract =
    "In this paper we present an  $O(nk)$  procedure, Algorithm  $SMR^3$ , for
    computing  $k$  eigenvectors of an  $n \times n$  symmetric tridiagonal
    matrix  $T$ . A salient feature of the algorithm is that a number of
    different  $LDL^t$  products ( $L$  unit lower triangular,  $D$  diagonal)
    are computed. In exact arithmetic each  $LDL^t$  is a factorization of a
    translate of  $T$ . We call the various  $LDL^t$  productions
    {\sl representations} (of  $T$ ) and, roughly speaking, there is a
    representation for each cluster of close eigenvalues. The unfolding of
    the algorithm, for each matrix, is well described by a
    {\sl representation tree}. We present the tree and use it to show that if
    each representation satisfies three prescribed conditions then the
    computed eigenvectors are orthogonal to working accuracy and have
    small residual norms with respect to the original matrix  $T$ .",
  paper = "Dhil04a.pdf"
}

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— axiom.bib —

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@article{Dhil04,
  author = "Dhillon, Inderjit S. and Parlett, Beresford N.",
  title = "Orthogonal Eigenvectors and Relative Gaps",
  journal = "SIAM Journal on Matrix Analysis and Applications",
  volume = "25",
  year = "2004",
  abstract =
    "Let  $LDL^t$  be the triangular factorization of a real symmetric
     $n \times n$  tridiagonal matrix so that  $L$  is a unit lower bidiagonal
    matrix,  $D$  is diagonal. Let  $(\lambda, \nu)$  be an eigenpair,
     $\lambda \neq 0$ , with the property that both  $\lambda$  and  $\nu$  are
    determined to high relative accuracy by the parameters in  $L$  and  $D$ .
    Suppose also that the relative gap between  $\lambda$  and its nearest
    neighbor  $\mu$  in the spectrum exceeds  $1/n$ ;  $n|\lambda - \mu| > |\lambda|$ .

    This paper presents a new  $O(n)$  algorithm and a proof that, in the
    presence of round-off errors, the algorithm computes an approximate
    eigenvector  $\hat{\nu}$  that is accurate to working precision
     $|\sin \angle(\nu, \hat{\nu})| = O(n\epsilon)$ , where  $\epsilon$  is the
    round-off unit. It follows that  $\hat{\nu}$  is numerically orthogonal to
    all the other eigenvectors. This result forms part of a program to
    compute numerically orthogonal eigenvectors without resorting to the
    Gram-Schmidt process.

    The contents of this paper provide a high-level description and
    theoretical justification for LAPACK (version 3.0) subroutine DLAR1V.",
  paper = "Dhil04.pdf"
}
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@article{Dods83,
  author = "Dodson, D. S.",
  title = "Corrigendum: Remark on 'Algorithm 539: Basic Linear Algebra
    Subroutines for FORTRAN usage",
  journal = "ACM Trans. Math. Software",
  volume = "9",
  pages = "140",
  year = "1983"
}
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— axiom.bib —

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@article{Dods82,  
  author = "Dodson, D. S. and Grimes, R. G.",  
  title = "Remark on algorithm 539: Basic Linear Algebra Subprograms for  
          Fortran usage",  
  journal = "ACM Trans. Math. Software",  
  volume = "8",  
  pages = "403-404",  
  year = "1982"  
}
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— axiom.bib —

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@article{Dong88,  
  author = "Dongarra, J. J. and DuCroz, J. and Hammarling, S. and  
          Hanson, R. J.",  
  title = "An extended set of FORTRAN Basic Linear Algebra Subprograms",  
  journal = "ACM Trans. Math. Software",  
  volume = "14",  
  pages = "1-32",  
  year = "1988"  
}
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— axiom.bib —

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@article{Dong88a,  
  author = "Dongarra, J. J. and DuCroz, J. and Hammarling, S. and  
          Hanson, R. J.",  
  title = "Corrigenda: 'An extended set of FORTRAN Basic Linear Algebra  
          Subprograms",  
  journal = "ACM Trans. Math. Software",  
  volume = "14",  
  pages = "399",  
  year = "1988"  
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@article{Dong90,
  author = "Dongarra, J. and DuCroz, J. and Duff, I. S. and Hammarling, S.",
  title = "A set of Level 3 Basic Linear Algebra Subprograms",
  journal = "ACM Trans. Math. Software",
  volume = "16",
  pages = "1-28",
  year = "1990"
}
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— axiom.bib —

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@article{Drma97,
  author = "Drmac, Zlatko",
  title = "Implementation of Jacobi Rotations for Accurate Singular Value
          Computation in Floating Point Arithmetic",
  journal = "SIAM Journal on Scientific Computing",
  volume = "18",
  number = "4",
  year = "1997",
  month = "July",
  pages = "1200-1222",
  abstract =
    "In this paper we consider how to compute the singular value
    decomposition (SVD)  $A = U\Sigma T^{\tau}$  of
     $A=[a_1, a_2] \in R^{m \times 2}$ 
    accurately in floating point arithmetic. It is shown
    how to compute the Jacobi rotation  $V$  (the right singular vector
    matrix) and how to compute  $AV=U\Sigma$  even if the floating point
    representation of  $V$  is the identity matrix. In the case
    underflow can produce the identity matrix as
    the floating point value of  $V$  even for  $a_1, a_2$  that are far from
    being mutually orthogonal. This can cause loss of accuracy and failure
    of convergence of the floating point implementation of the Jacobi
    method for computing the SVD. The modified Jacobi method recommended
    in this paper can be implemented as a reliable and highly accurate
    procedure for computing the SVD of general real matrices whenever the
    exact singular values do not exceed the underflow or overflow limits."
}
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— axiom.bib —

```
@article{Drma08a,
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author = "Drmac, Zlatko and Veselic, Kresimir",
title = "New fast and accurate Jacobi SVD algorithm I",
journal = "SIAM J. Matrix Anal. Appl.",
volume = "35",
number = "2",
year = "2008",
pages = "1322-1342",
comment = "LAPACK Working note 169",
link = "\url{http://www.netlib.org/lapack/lawnspdf/lawn169.pdf}",
abstract =
  "This paper is the result of contrived efforts to break the barrier
  between numerical accuracy and run time efficiency in computing the
  fundamental decomposition of numerical linear algebra - the singular
  value decomposition (SVD) of a general dense matrix. It is an
  unfortunate fact that the numerically most accurate one-sided Jacobi
  SVD algorithm is several times slower than generally less accurate
  bidiagonalization based methods such as the QR or the divide and
  conquer algorithm. Despite its sound numerical qualities, the Jacobi
  SVD is not included in the state of the art matrix computation
  libraries and it is even considered obsolete by some leading
  researchers. Our quest for a highly accurate and efficient SVD
  algorithm has led us to a new, superior variant of the Jacobi
  algorithm. The new algorithm has inherited all good high accuracy
  properties, and it outperforms not only the best implementations of
  the one-sided Jacobi algorithm but also the QR algorithm. Moreover,
  it seems that the potential of the new approach is yet to be fully
  exploited.",
paper = "Drma08a.pdf"
}

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@article{Drma08b,
  author = "Drmac, Zlatko and Veselic, Kresimir",
  title = "New fast and accurate Jacobi SVD algorithm II",
  journal = "SIAM J. Matrix Anal. Appl.",
  volume = "35",
  number = "2",
  year = "2008",
  pages = "1343-1362",
  comment = "LAPACK Working note 170",
  link = "\url{http://www.netlib.org/lapack/lawnspdf/lawn170.pdf}",
  abstract =
    "This paper presents new implementation of one-sided Jacobi SVD for
    triangular matrices and its use as the core routine in a new
    preconditioned Jacobi SVD algorithm, recently proposed by the

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authors. New pivot strategy exploits the triangular form and uses the fact that the input triangular matrix is the result of rank revealing QR factorization. If used in the preconditioned Jacobi SVD algorithm, described in the first part of this report, it delivers superior performance leading to the currently fastest method for computing SVD decomposition with high relative accuracy. Furthermore, the efficiency of the new algorithm is comparable to the less accurate bidiagonalization based methods. The paper also discusses underflow issues in floating point implementation, and shows how to use perturbation theory to fix the imperfectness of machine arithmetic on some systems.",
 paper = "Drma08b.pdf"
}

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@article{Drma08c,
  author = "Drmac, Zlatko and Bujanovic, Zvonimir",
  title = "On the failure of rank-revealing QR factorization software -
    a case study.",
  journal = "ACM Trans. math. Softw.",
  volume = "35",
  number = "2",
  year = "2008",
  pages = "1-28",
  comment = "LAPACK Working note 176",
  link = "\url{http://www.netlib.org/lapack/lawnspdf/lawn176.pdf}",
  abstract =
    "This note reports an unexpected and rather erratic behavior of the
    LAPACK software implementation of the QR factorization with
    Businger-Golub column pivoting. It is shown that, due to finite
    precision arithmetic, software implementation of the factorization can
    catastrophically fail to produce triangular factor with the structure
    characteristic to the Businger-Golub pivot strategy. The failure of
    current {\sl state of the art} software, and a proposed alternative
    implementations are analyzed in detail.",
  paper = "Drma08c.pdf"
}
```

— axiom.bib —

```
@article{Dubr83,
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```

author = "Dubrulle, A. A.",
title = "A class of numerical methods for the computation of Pythagorean
        sums",
journal = "IBM J. Res. Develop.",
volume = "27",
number = "6",
pages = "582-589",
year = "1983"
}

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— axiom.bib —

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@book{Eina05,
author = "Einarsson, B.",
title = "Accuracy and Reliability in Scientific Computing",
publisher = "SIAM",
year = "2005",
isbn = "0-89871-584-9",
link = "\url{http://www.nsc.liu.se/wg25/book/}"
}

```

— axiom.bib —

```

@article{Elmr00,
author = "Elmroth, E. and Gustavson, F. G.",
title = "Applying recursion to serial and parallel QR factorization leads
        to better performance",
journal = "IBM Journal of Research and Development",
volume = "44",
number = "4",
month = "July",
year = "2000",
pages = "605--624",
doi = "10.1.1.33.1820",
link = "\url{http://citeseerx.ist.psu.edu/viewdoc/download?doi=10.1.1.33.1820}",
abstract =
    "We present new recursive serial and parallel algorithms for QR
    factorization of an  $m \times n$  matrix. They improve performance. The
    recursion leads to an automatic variable blocking, and it also
    replaces a Level 2 part in a standard algorithm with Level 3
    operations. However, there are significant additional costs for
    creating and performing the updates, which prohibit the efficient use

```

of the recursion for large n . We present a quantitative analysis of these extra costs. This analysis leads us to introduce a hybrid recursive algorithm that outperforms the LAPACK algorithm DGEQRF by about 20% for large square matrices up to almost a factor of 3 for tall thin matrices. Uniprocessor performance results are presented for two IBM RS/6000 SP nodes -- a 120-Mhz IBM POWER2 node and one processor of a four-way 332-Mhz IBM PowerPC 604e SMP node. The hybrid recursive algorithm reaches more than 90% of the theoretical peak performance of the POWER2 node. Compared to standard block algorithms, the recursive approach also shows a significant advantage in the automatic tuning obtained from its automatic variable blocking. A successful parallel implementation on a four-way 332-MHz IBM PPC604e SMP node based on dynamic load balancing is presented. For two, three, and four processors it shows speedups of up to 1.97, 299, and 3.97."

— axiom.bib —

```
@misc{Fate13,
  author = "Fateman, Richard J.",
  title = "Interval Arithmetic, Extended Numbers and Computer Algebra Systems",
  year = "2013",
  link = "url{http://www.cs.berkeley.edu/~fateman/papers/interval.pdf}",
  abstract =
    "Many ambitious computer algebra systems were initially designed in a
    flush of enthusiasm, with the goal of automating any symbolic
    mathematical manipulation ‘‘correctly’’. Historically, this approach
    results in programs that implicitly used certain identities to
    simplify expressions. These identities, which very likely seemed
    universally true to the programmers in the heat of writing the CAS
    (and often were true in well-known abstract algebraic domains) later
    needed re-examination when such systems were extended for dealing with
    kinds of objects unanticipated in the original design. These new
    objects are generally introduced to the CAS by extending
    ‘‘generically’’ the arithmetic of other operations. For example,
    approximate floats do not have the mathematical properties of exact
    integers or rationals. Complex numbers may strain a system designed
    for real-valued variables. In the situation examined here, we consider
    two categories of ‘‘extended’’ numbers:  $\infty$  and  $\text{\textbackslashsl undefined}$ ,
    and real intervals. We comment on issues raised by these two
    troublesome notions, how their introduction into a computer algebra
    system may require a (sometimes painful) reconsideration and redesign
    of parts of the program, and how they are related. An alternative
    (followed most notably by the Axiom system is to essentially envision
    a ‘‘meta’’ CAS defined in terms of categories and inheritance with
    only the most fundamental built-in concepts; from these one can build
```

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many variants of specific CAS features. This approach is appealing but
can fail to accommodate extensions that violate some mathematical
tenets in the cause of practicality.",
paper = "Fate13.pdf",
keywords = "axiomref"
}

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— axiom.bib —

```

@techreport{Fern92,
  author = "Fernando, K. Vince and Parlett, Beresford N.",
  title = "Accurate Singular Values and Differential qd Algorithms",
  year = "1992",
  institution = "University of California, Berkeley",
  type = "Technical Report",
  number = "PAM-554",
  link = "\url{http://www.dtic.mil/dtic/tr/fulltext/u2/a256582.pdf}",
  abstract =
    "We have discovered a new implementaiton of the qd algorithm that has
    a far wider domain of stability than Rutishauser's version. Our
    algorithm was developed from an examination of the LR-Cholesky
    transformation and can be adapted to parallel computation in stark
    contrast to traditional qd. Our algorithm also yields useful a
    posteriori upper and lower bounds on the smallest singular value of a
    bidiagonal matrix.

    The zero-shift bidiagonal QR of Demmel and Kahan computes the smallest
    singlar values to maximal relative accuracy and the others to maximal
    absolute accuracy with little or no degradation in efficiency when
    compared with the LINPACK code. Our algorithm obtains maximal relative
    accuracy for all the singlar values and runs at least four times
    faster than the LINPACK code.",
  paper = "Fern92.pdf"
}

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— axiom.bib —

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@article{Fors70,
  author = "Forsythe, G. E.",
  title = "Pitfalls in computations, or why a math book isn't enough",
  journal = "Amer. Math. Monthly",
  volume = "9",

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```
pages = "931-995",  
year = "1970"  
}
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— axiom.bib —

```
@incollection{Fors69,  
  author = "Forsythe, G. E.",  
  title = "What is a satisfactory quadratic equation solver",  
  booktitle = "Constructive Aspects of the Fundamental Theorem of Algebra",  
  pages = "53-61",  
  publisher = "Wiley",  
  year = "1969"  
}
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— axiom.bib —

```
@article{Foxx71,  
  author = "Fox, L.",  
  title = "How to get meaningless answers in scientific computations (and  
    what to do about it)",  
  journal = "IMA Bulletin",  
  volume = "7",  
  pages = "296-302",  
  year = "1971"  
}
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— axiom.bib —

```
@article{Gent74,  
  author = "Gentlman, W. Morven and Marovich, Scott B.",  
  title = "More on algorithms that reveal properties of floating point  
    arithmetic units.",  
  journal = "Comm. of the ACM",  
  year = "1974",  
  month = "April",  
  volume = "17",
```

```

number = "5",
pages = "276-277",
abstract =
  "In the interests of producing portable mathematical software, it is
  highly desirable for a program to be able directly to obtain
  fundamental properties of the environment in which it is to run. The
  installer would then not be obliged to change appropriate magic
  constants in the source code, and the user would not have to provide
  information he may very well not understand. Until the standard
  definitions of programming languages are changed to require builtin
  functions that provide this information, we will have to resort to
  writing routines that discover it."
}

```

— axiom.bib —

```

@techreport{Give54,
  author = "Givens, W.",
  title = "Numerical computation of the characteristic values of a real
  symmetric matrix",
  year = "1954",
  institution = "Oak Ridge National Laboratory",
  type = "Technical Report",
  number = "ORNL-1574"
}

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— axiom.bib —

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@article{Golu65,
  author = "Golub, G.H.",
  title = "Numerical methods for solving linear least squares problems",
  journal = "Numer. Math.",
  volume = "7",
  pages = "206-216",
  year = "1965"
}

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— axiom.bib —

```
@book{Golu89,  
  author = "Golub, Gene H. and Van Loan, Charles F.",  
  title = "Matrix Computations",  
  publisher = "Johns Hopkins University Press",  
  year = "1989",  
  isbn = "0-8018-3772-3"  
}
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— axiom.bib —

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@book{Golu96,  
  author = "Golub, Gene H. and Van Loan, Charles F.",  
  title = "Matrix Computations",  
  publisher = "Johns Hopkins University Press",  
  isbn = "978-0-8018-5414-9",  
  year = "1996"  
}
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— axiom.bib —

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@article{Hamm85,  
  author = "Hammarling S.",  
  title = "The Singular Value Decomposition in Multivariate Statistics",  
  journal = "ACM Signum Newsletter",  
  volume = "20",  
  number = "3",  
  pages = "2--25",  
  year = "1985"  
}
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— axiom.bib —

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@book{Hamm05,  
  author = "Hammarling, Sven",  
  title = "An Introduction to the Quality of Computed Solutions",  
  booktitle = "Accuracy and Reliability in Scientific Computing",  
  year = "2005",
```

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    publisher = "SIAM",
    pages = "43-76",
    link = "\url{http://eprints.ma.man.ac.uk/101/}",
    paper = "Hamm05.pdf"
}

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— axiom.bib —

```

@mastersthesis{Harg02,
  author = "Hargreaves, G.",
  title = "Interval analysis in MATLAB",
  school = "University of Manchester, Dept. of Mathematics",
  year = "2002"
}

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— axiom.bib —

```

@book{High05,
  author = "Higham, D. J. and Higham, N. J.",
  title = "MATLAB Guide",
  publisher = "SIAM",
  year = "2002",
  isbn = "0-89871-521-0"
}

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— axiom.bib —

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@article{High88,
  author = "Higham, Nicholas J.",
  title = "FORTRAN codes for estimating the one-norm of a real or complex
          matrix, with applications to condition estimation",
  journal = "ACM Trans. Math. Soft",
  volume = "14",
  number = "4",
  pages = "381-396",
  year = "1988"
}

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— axiom.bib —

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@misc{High98,
  author = "Higham, Nicholas J.",
  title = "Can you 'count' on your computer?",
  link = "\url{http://www.maths.man.ac.uk/~higham/talks/}",
  year = "1998"
}
```

— axiom.bib —

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@book{High02,
  author = "Higham, Nicholas J.",
  title = "Accuracy and stability of numerical algorithms",
  publisher = "SIAM",
  isbn = "0-89871-521-0",
  year = "2002"
}
```

— axiom.bib —

```
@article{High86,
  author = "Higham, Nicholas J.",
  title = "Efficient Algorithms for Computing the Condition Number of a
          Tridiagonal Matrix",
  journal = "SIAM J. Sci. Stat. Comput.",
  volume = "7",
  number = "1",
  year = "1986",
  month = "January",
  abstract =
    "Let  $A$  be a tridiagonal matrix of order  $n$ . We show that it is
    possible to compute  $\|A^{-1}\|_{\infty}$  and hence
     $\kappa_{\infty}(A)$  in  $O(n)$  operations. Several algorithms
    which perform this task are given and their numerical properties are
    investigated.

    If  $A$  is also positive definite then  $\|A^{-1}\|_{\infty}$  can be
    computed as the norm of the solution to a positive definite
```



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tridiagonal linear system whose coefficient matrix is closely related
to $A$. We show how this computation can be carried out in parallel
with the solution of a linear system $Ax=b$. In particular we describe
some simple modifications to the LINPACK routine SPTSL which enable
this routine to compute  $\|A\|_1$ , efficiently, in addition to
solving $Ax=b$.",
paper = "High86.pdf"
}

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— axiom.bib —

```

@misc{IEEE85,
  author = "IEEE",
  title = "ANSI/IEEE Standard for Binary Floating Point Arithmetic:
          Std 754-1985",
  publisher = "IEEE Press",
  year = "1985"
}

```

— axiom.bib —

```

@misc{IEEE87,
  author = "IEEE",
  title = "ANSI/IEEE Standard for Radix Independent Floating Point Arithmetic:
          Std 854-1987",
  publisher = "IEEE Press",
  year = "1987"
}

```

— axiom.bib —

```

@book{Isaa94,
  author = "Isaacson, E. and Keller, H. B.",
  title = "Analysis of Numerical Methods",
  publisher = "Dover",
  year = "1994",
  isbn = "0-486-68029-0"
}

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}

— axiom.bib —

```
@article{Kags89,
  author = "Kagstrom, Bo and Westin, L.",
  title = "Generalized Schur Methods with Condition Estimators for Solving
    the Generalized Sylvester Equation",
  journal = "IEEE Transactions on Automatic Control",
  volume = "34",
  number = "7",
  year = "1989",
  month = "July",
  pages = "745-751",
  abstract =
    "Stable algorithms are presented for solving the generalized Sylvester
    equation. They are based on orthogonal equivalence transformations of
    the original problem. Perturbation theory and rounding error analysis
    are included. Condition estimators ( $\{\rm Dif\}^{-1}$ -estimators) are
    developed which when substituted into derived error bounds give
    accuracy estimates of a computed solution. Results from numerical
    experiments on well-conditioned and ill-conditioned problems are
    reported."
}
```

— axiom.bib —

```
@book{Kags93a,
  author = "Kagstrom, B.",
  title = "A Direct Method for Reordering Eigenvalues in the
    Generalized Real Schur Form of a Regular Matrix Pair (A, B)",
  year = "1993",
  pages = "195-218",
  booktitle = "Linear Algebra for Large Scale and Real-Time Applications",
  volume = "232",
  journal = "NATO ASI Series",
  publisher = "NATO ASI Series",
  isbn = "978-90-481-4246-0",
  abstract =
    "A direct orthogonal equivalence transformation method for reordering
    the eigenvalues along the diagonal in the generalized real Schur form
    of a regular matrix pair  $(A,B)$  is presented. Each swap of two
```

adjacent eigenvalues (real, or complex conjugate pairs) involves solving a generalized Sylvester equation and the construction of two orthogonal transformation matrices from certain eigenspaces associated with the corresponding diagonal blocks. An error analysis of the direct reordering method is presented. Results from numerical experiments on well-conditioned as well as ill-conditioned problems illustrate the stability and the accuracy of the method. Finally, a direct reordering algorithm with controlled backward error is described."

}

— axiom.bib —

```
@article{Kags93,
  author = "Kagstrom, Bo and Poromaa, Peter",
  title = "LAPACK-Style Algorithms and Software for Solving the Generalized
          Sylvester Equation and Estimating the Separation between
          Regular Matrix Pairs",
  year = "1993",
  month = "December",
  link = "\url{http://www.netlib.org/lapack/lawnspdf/lawn75.pdf}",
  comment = "LAPACK Working Note 75",
  abstract =
    "Level 3 algorithms for solving the generalized Sylvester equation
    $(AR-LB,DR-LE)=(C,F)$ and the transposed analogue
    $(A^T U + D^T V, -UB^T - VE^T)=(C,F)$ are presented. These blocked algorithms
    permit reuse of data in complex memory hierarchies of current advanced
    computer architectures. The separation of two regular matrix pairs
    $(A,D)$ and $(B,E)$,  $\text{Dif}[(A,D),(B,E)]$ , is defined in terms of the
    generalized Sylvester operator  $(AR-LB,DR-LE)$ . Robust, efficient and
    reliable Dif-estimators are presented. The basic problem is to find a
    lower bound on  $\text{Dif}^{-1}$ , which can be done by solving generalized
    Sylvester equations in triangular form. Frobenius norm-based and one
    norm based Dif estimators are described and evaluated. These estimates
    lead to computable error bounds for the generalized Sylvester
    equation. The one-norm-based estimator makes the condition estimation
    uniform with LAPACK. Fortran 77 software that implements our
    algorithms for solving generalized Sylvester equations, and for
    computing error bounds and Dif-estimators are presented. Computational
    experiments that illustrate the accuracy, efficiency and reliability
    of our software are also described.",
  paper = "Kags93.pdf"
}
```

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```
@misc{Kags94,
  author = "Kagstrom, Bo and Poromaa, Peter",
  title = "Computing Eigenspaces with Specified Eigenvalues of a Regular
          Matrix Pair (A, B) and Condition Estimation: Theory,
          Algorithms and Software",
  year = "1994",
  month = "April",
  link = "\url{http://www.netlib.org/lapack/lawns/lawn87.ps}",
  abstract =
    "Theory, algorithms and LAPACK style software for computing a pair of
    deflating subspaces with specified eigenvalues of a regular matrix
    pair  $(A,B)$  and error bounds for computed quantities (eigenvalues and
    eigenspaces) are presented. The {\sl reordering} of specified
    eigenvalues is performed with a direct orthogonal transformation
    method with guaranteed numerical stability. Each swap of two adjacent
    diagonal blocks in the real generalized Schur form, where at least one
    of them corresponds to a complex conjugate pair of eigenvalues,
    involves solving a generalized Sylvester equation and the construction
    of two orthogonal transformation matrices from certain eigenspaces
    associated with the diagonal blocks. The swapping of two  $\times 1$ 
    blocks is performed using orthogonal (unitary) Givens rotations. The
    {\sl error bounds} are based on estimates of condition numbers for
    eigenvalues and eigenspaces. The software computes reciprocal values
    of a condition number for an individual eigenvalue (or a cluster of
    eigenvalues), a condition number for an eigenvector (or eigenspace),
    and spectral projectors onto a selected cluster. By computing
    reciprocal values we avoid overflow. Changes in eigenvectors and
    eigenspaces are measured by their change in angle. The condition
    numbers yield both {\sl asymptotic} and {\sl global} error bounds. The
    asymptotic bounds are only accurate for small perturbations  $(E,F)$  of
     $(A,B)$ , while the global bounds work for all  $\|(E,F)\|$  up to a
    certain bound, whose size is determined by the conditioning of the
    problem. It is also shown how these upper bounds can be
    estimated. Fortran 77 {\sl software} that implements our algorithms
    for reordering eigenvalues, computing (left and right) deflating
    subspaces with specified eigenvalues and condition number estimation
    are presented. Computational experiments that illustrate the accuracy,
    efficiency and reliability of our software are also described.",
  paper = "Kags94.pdf"
}
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— axiom.bib —

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@article{Kags94b,
  author = "Kagstrom, Bo",
  title = "A Perturbation Analysis of the Generalized Sylvester Equation
          $(AR-LB,DR-LE)=(C,F)$",
  journal = "SIAM J. Matrix Anal. and Appl.",
  volume = "15",
  number = "4",
  pages = "1045-1060",
  year = "1994",
  abstract =
    "Perturbation and error bounds for the generalized Sylvester equation
    $(AR-LB,DR-LE)=(C,F)$ are presented. An explicit expression for the
    normwise relative backward error associated with an approximate
    solution of the generalized Sylvester equation is derived and
    conditions when it can be much greater than the relative residual are
    given. This analysis is applicable to any method that solves the
    generalized Sylvester equation. A condition number that reflects the
    structure of the problem and a normwise forward error bound based on
    $\{\rm Dif\}^{-1}[(A,D),B,E]$ and the residual are derived. The
    structure-preserving condition number can be arbitrarily smaller than
    a $\{\rm Dif\}^{-1}$-based condition number. The normwise error bound
    can be evaluated robustly and at moderate cost by using a reliable
    $\{\rm Dif\}^{-1}$ estimator. A componentwise LAPACK-style forward error
    bound that can be stronger than the normwise error bound is
    presented. A componentwise approximate error bound that can be
    evaluated to a much lower cost is also proposed. Finally, some
    computational experiments that validate and evaluate the perturbation
    and error bounds are presented."
}

```

— axiom.bib —

```

@techreport{Kaha66,
  author = "Kahan, W.",
  title = "Accurate Eigenvalues of a Symmetric Tri-Diagonal Matrix",
  year = "1966",
  month = "July",
  institution = "Stanford University",
  type = "Technical Report",
  number = "CS41",
  abstract =
    "Having established tight bounds for the quotient of two different
    lub-norms of the same tri-diagonal matrix J, the author observes that
    these bounds could be of use in an error-analysis provided a suitable
    algorithm were found. Such an algorithm is exhibited, and its errors
    are thoroughly accounted for, including the effects of scaling,

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over/underflow and roundoff. A typical result is that, on a computer using rounded floating point binary arithmetic, the biggest eigenvalue of J can be computed easily to within 2.5 units in its last place, and the smaller eigenvalues will suffer absolute errors which are no larger. These results are somewhat stronger than had been known before.",
 paper = "Kaha66.pdf"
 }

— axiom.bib —

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@misc{Kel100,
  author = "Kelsey, Tom",
  title = "Exact Numerical Computation via Symbolic Computation",
  link = "\url{http://tom.host.cs.st-andrews.ac.uk/pub/ccapaper.pdf}",
  abstract = "
    We provide a method for converting any symbolic algebraic expression
    that can be converted into a floating point number into an exact
    numeric representation. We use this method to demonstrate a suite of
    procedures for the representation of, and arithmetic over, exact real
    numbers in the Maple computer algebra system. Exact reals are
    represented by potentially infinite lists of binary digits, and
    interpreted as sums of negative powers of the golden ratio.",
  paper = "Kel100.pdf"
}
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— axiom.bib —

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@article{Knus98,
  author = {Kn\"usel, L.},
  title = "On the accuracy of statistical distributions in Microsoft
    Excel 97",
  journal = "Comput. Statist. Data Anal.",
  volume = "26",
  pages = "375-377",
  year = "1998"
}
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— axiom.bib —

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@misc{Kohl14,
  author = {K\"ohler, Martin and Saak, Jens},
  title = "On BLAS Level-3 Implementations of Common Solvers for
          (Quasi-)Triangular Generalized Lyapunov Equations",
  year = "2014",
  link = "\url{http://slicot.org/objects/software/reports/SLWN2014_1.pdf}",
  note = "SLICOT Working Note 2014-1",
  abstract =
    "The solutions of Lyapunov and generalized Lyapunov equations are a
    key player in many applications in systems and control theory. Their
    stable numerical computation, when the full solution is sought, is
    considered solved since the seminal work of Bartels and Stewart. A
    number of variants of their algorithm have been proposed, but none of
    them goes beyond BLAS level-2 style implementation. On modern
    computers, however, the formulation of BLAS level-3 type
    implementations is crucial to enable optimal usage of cache
    hierarchies and modern block scheduling methods based on directed
    acyclic graphs describing the interdependence of single block
    computations. Our contribution closes this gap by a transformation of
    the aforementioned level-2 variants to level-3 versions and a
    comparison on a standard multicore machine.",
  paper = "Kohl14.pdf"
}

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— axiom.bib —

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@misc{Krei05,
  author = "Kreinovich, V.",
  title = "Interval cmputations",
  year = "2005",
  link = "\url{http://www.cs.utep.edu/interval-comp/}"
}

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— axiom.bib —

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@article{Kuki72a,
  author = "Kuki, Hirono",
  title = "Complex Gamma Function with Error Control",
  year = "1972",
  publisher = "ACM",
  journal = "Communications of the ACM",
  volume = "15",

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number = "4",
pages = "262-267",
abstract =
  "An algorithm to compute the gamma function and the loggamma function
  of a complex variable is presented. The standard algorithm is modified
  in several respects to insure the continuity of the function value and
  to reduce accumulation of round-off errors. In addition to computation
  of function values, this algorithm includes an object-time estimation
  of round-off errors. Experimental data with regard to the effectiveness
  of this error control are presented. A Fortran program for the algorithm
  appears in the algorithms section of this issue."
}

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— axiom.bib —

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@article{Kuki72b,
  author = "Kuki, Hirono",
  title = "Algorithm 421: Complex Gamma Function with Error Control",
  year = "1972",
  publisher = "ACM",
  journal = "Communications of the ACM",
  volume = "15",
  number = "4",
  pages = "271-272",
  abstract =
    "This Fortran program computes either the gamma function or the
    loggamma function of a complex variable in double precision.
    In addition, it provides an error estimate of the computed answer.
    The calling sequences are:
    \verb|CALL CDLGAM (X, W, E, 0)|
    for the loggamma, and
    \verb|CALL CDLGAM (X, W, E, 1)|
    for the gamma, where Z is the double precision argument, W is the
    answer of the same type, and E is a single precision real variable.
    Before the call, the value of E is an estimate of the error in Z,
    and after the call, it is an estimate of the error in W."
}

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— axiom.bib —

```

@book{Laws74,
  author = "Lawson, C. L. and Hanson, R. J.",

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    title = "Solving Least Squares Problems",
    publisher = "Prentice-Hall",
    year = "1974"
}

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— axiom.bib —

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@book{Laws95,
  author = "Lawson, C. L. and Hanson, R. J.",
  title = "Solving Least Squares Problems",
  publisher = "SIAM",
  isbn = "0-89871-356-0",
  year = "1995"
}

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— axiom.bib —

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@article{Livn04,
  author = "Livne, Oren E. and Golub, Gene H.",
  title = "Scaling by Binormalization",
  journal = "Numerical Algorithms",
  volume = "35",
  number = "1",
  pages = "97-120",
  year = "2004",
  month = "January",
  abstract =
    "We present an iterative algorithm (BIN) for scaling all the rows and
    columns of a real symmetric matrix to unit 2-norm. We study the
    theoretical convergence properties and its relation to optimal
    conditioning. Numerical experiments show that BIN requires 2-4
    matrix-vector multiplications to obtain an adequate scaling, and in
    many cases significantly reduces the condition number, more than other
    scaling algorithms. We present generalizations to complex,
    non-symmetric and rectangular matrices.",
  paper = "Livn04.pdf"
}

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— axiom.bib —

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@article{Malc72,
  author = "Malcolm, Michael A.",
  title = "Algorithms to reveal properties of floating-point arithmetic",
  journal = "Comms of the ACM",
  volume = "15",
  year = "1972",
  pages = "949-951",
  link = "\url{http://www.dtic.mil/dtic/tr/fulltext/u2/727104.pdf}",
  abstract =
    "Two algorithms are presented in the form of Fortran subroutines. Each
    subroutine computes the radix and number of digits of the floating
    point numbers and whether rounding or chopping is done by the machine
    on which it is run. The methods are shown to work on any ‘reasonable’
    floating-point computer.",
  paper = "Malc72.pdf"
}

```

— axiom.bib —

```

@article{Marq06,
  author = "Marques, Osni A. and Reidy, Jason and Vomel, Christof",
  title = "Benefits of IEEE-754 Features in Modern Symmetric Tridiagonal
    Eigensolvers",
  journal = "SIAM Journal on Scientific Computing",
  volume = "28",
  number = "5",
  year = "2006",
  link = "\url{http://www.netlib.org/lapack/lawnspdf/lawn172.pdf}",
  abstract =
    "Bisection is one of the most common methods used to compute the
    eigenvalues of symmetric tridiagonal matrices. Bisection relies on the
    {\sl Sturm count}: for a given shift  $\sigma$ , the number of negative
    pivots in the factorization  $T - \sigma I = LDL^T$  equals the number of
    eigenvalues of  $T$  that are smaller than  $\sigma$ . In IEEE-754
    arithmetic, the value  $\infty$  permits the computation to continue
    past a zero pivot, producing a correct Sturm count when  $T$  is
    unreduced. Demmel and Li showed in the 90s that using  $\infty$  rather
    than testing for zero pivots within the loop could improve performance
    significantly on certain architectures.

```

When eigenvalues are to be computed to high relative accuracy, it is often preferable to work with LDL^T factorizations instead of the original tridiagonal T , see for example the MRRR algorithm. In these cases, the Sturm count has to be computed from LDL^T . The differential stationary and progressive qds algorithms are the method of choice.

While it seems trivial to replace T by LDL^T , in reality these algorithms are more complicated: in IEEE-754 arithmetic, a zero pivot produces an overflow, followed by an invalid exception (NaN), that renders the Sturm count incorrect.

We present alternative, safe formulations that are guaranteed to produce the correct result.

Benchmarking these algorithms on a variety of platforms shows that the original formulation without tests is always faster provided no exception occurs. The transforms see speed-ups of up to 2.6x over the careful formulation.

Tests on industrial matrices show that encountering exceptions in practice is rare. This leads to the following design: First, compute the Sturm count by the fast but unsafe algorithm. Then, if an exception occurred, recompute the count by a safe, slower alternative.

The new Sturm count algorithms improve the speed of bisection by up to 2x on our test matrices. Furthermore, unlike the traditional tiny-pivot substitutions, proper use of IEEE-754 features provides a careful formulation that imposed no input range restrictions.",
 paper = "Marq06.pdf"

}

— axiom.bib —

```
@article{Mart68,
  author = "Martin, R. S. and Wilkinson, J. H.",
  title = "Similarity reduction of a general matrix to Hessenberg form",
  journal = "Numer. Math.",
  volume = "12",
  pages = "349-368",
  year = "1968"
}
```

— axiom.bib —

```
@misc{Math05,
  author = "MathWorks",
  title = "MATLAB",
```

```

publisher = "The Mathworks, Inc.",
link = "\url{http://www.mathworks.com}"
}

```

— axiom.bib —

```

@article{Mccu02,
  author = "McCullough, B. D. and Wilson, B.",
  title = "On the accuracy of statistical procedures in Microsoft Excel
          2000 and Excel XP",
  journal = "Comput. Statist. Data Anal.",
  volume = "40",
  pages = "713-721",
  year = "2002"
}

```

— axiom.bib —

```

@article{Mccu99,
  author = "McCullough, B. D. and Wilson, B.",
  title = "On the accuracy of statistical procedures in Microsoft Excel 97",
  journal = "Comput. Statist. Data Anal.",
  volume = "31",
  pages = "27-37",
  year = "1999"
}

```

— axiom.bib —

```

@book{Metc96,
  author = "Metcalf, M. and Reid, J. K.",
  title = "Fortran 90/95 Explained",
  publisher = "Oxford University Press",
  year = "1996"
}

```

— axiom.bib —

```
@book{Metc04,
  author = "Metcalfe, M. and Reid, J. K. and Cohen, M.",
  title = "Fortran 95/2003 Explained",
  publisher = "Oxford University Press",
  year = "2004",
  isbn = "0-19-852693-8"
}
```

— axiom.bib —

```
@article{Mole83,
  author = "Moler, C. and Morrison, D.",
  title = "Replacing square roots by Pythagorena sums",
  journal = "IBM J. Res. Develop.",
  volume = "27",
  number = "6",
  pages = "577-581",
  year = "1983"
}
```

— axiom.bib —

```
@article{Mole71,
  author = "Moler, C.B. and Stewart, G.W.",
  title = "An Algorithm for Generalized Matrix Eigenvalue Problems",
  journal = "SIAM J. Numer. Anal",
  volume = "10",
  year = "1973",
  pages = "241--256",
  abstract =
    "A new method, called the QZ algorithm, is presented for the solution
    of the matrix eigenvalue problem  $Ax=\lambda Bx$  with general square
    matrices  $A$  and  $B$ . Particluar attention is paid to the degeracies
    which result when  $B$  is singular. No inversions of  $B$  or its
    submatrices are used. The algorithm is a generalization of the QR
    algorithm, and reduces to it when  $B=I$ . A Fortran program and some
    illustrative examples are included."
}
```

— axiom.bib —

```
@book{Moon93,  
  author = "Moonen, Marc S. and Golub, Gene H. and De Moor, Bart L.R.",  
  title = "Linear Algebra and Large Scale and Real-Time Applications",  
  year = "1993",  
  publisher = "NATO ASI Series",  
  isbn = "978-904814246-0"  
}
```

— axiom.bib —

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@book{Moor79,  
  author = "Moore, R. E.",  
  title = "methods and Applications of Interval Analysis",  
  publisher = "SIAM",  
  year = "1979"  
}
```

— axiom.bib —

```
@misc{NAGa05,  
  author = "Numerical Algorithms Group",  
  title = "The NAG Library",  
  link = "\url{http://www.nag.co.uk/numeric}",  
  year = "2005"  
}
```

— axiom.bib —

```
@misc{NAGb05,  
  author = "Numerical Algorithms Group",  
  title = "The NAG Fortran Library Manual",  
  link = "\url{http://www.nag.co.uk/numeric/fl/manual/html/FLlibrarymanual.asp}",  
}
```

```

    year = "2005"
}

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— axiom.bib —

```

@book{Over01,
  author = "Overton, M. L.",
  title = "Numerical Computing with IEEE Floating Point Arithmetic",
  publisher = "SIAM",
  year = "2001",
  isbn = "0-89871-482-6"
}

```

,
— axiom.bib —

```

@book{Pies83,
  author = {Piessens, R. and de Doncker-Kapenga, E. and "Uberhuber, C. W.
            and Kahaner, D. K.},
  title = "QUADPACK - A Subroutine Package for Automatic Integration",
  publisher = "Springer-Verlag",
  year = "1983"
}

```

— axiom.bib —

```

@misc{Pete12,
  author = "Petersen, Kaare Brandt and Pedersen, Michael Syskind",
  title = "The Matrix Cookbook",
  link = "\url{http://www2.imm.dtu.dk/pubdb/views/edoc_download.php/3274/pdf/imm3274.pdf}",
  year = "2012",
  month = "November"
}

```

— axiom.bib —

```
@article{Prie04,  
  author = "Priest, D. M.",  
  title = "Efficient scaling for complex division",  
  journal = "ACM Trans. Math. Software",  
  volume = "30",  
  pages = "389-401",  
  year = "2004"  
}
```

— axiom.bib —

```
@InProceedings{Rump99,  
  author = "Rump, S. M.",  
  title = "INTLAB - INTerval LABoratory",  
  booktitle = "Developments in Reliable Computing",  
  pages = "77-104",  
  publisher = "Kluwer Academic",  
  year = "1999"  
}
```

— axiom.bib —

```
@InProceedings{Sham92,  
  author = "Shampine, L. F. and Gladwell, I.",  
  title = "The next generation of runge-kutta codes",  
  booktitle = "Computational Ordinary Differential Equations",  
  pages = "145-164",  
  publisher = "Oxford University Press",  
  year = "1992"  
}
```

— axiom.bib —

```
@article{Smit62,  
  author = "Smith, R. L.",  
  title = "Algorithm 116: Complex division",  
  journal = "Communs. Ass. comput. Mach.",
```



```
    volume = "5",  
    pages = "435",  
    year = "1962"  
}
```

— axiom.bib —

```
@book{Stew98,  
  author = "Stewart, G. W.",  
  title = "Matrix Algorithms: Basic Decompositions, volume I",  
  publisher = "SIAM",  
  year = "1998",  
  isbn = "0-89871-414-1"  
}
```

— axiom.bib —

```
@article{Stew85,  
  author = "Stewart, G. W.",  
  title = "A note on complex division",  
  journal = "ACM Trans. Math. Software",  
  volume = "11",  
  pages = "238-241",  
  year = "1985"  
}
```

— axiom.bib —

```
@book{Stew90,  
  author = "Stewart, G. W. and Sun, J.",  
  title = "Matrix Perturbation Theory",  
  publisher = "Academic Press",  
  year = "1990"  
}
```

— axiom.bib —

```
@article{Stou07,
  author = "Stoutemyer, David R.",
  title = "Useful Computations Need Useful Numbers",
  year = "2007",
  publisher = "ACM",
  journal = "Communications in Computer Algebra",
  volume = "41",
  number = "3",
  abstract =
    "Most of us have taken the exact rational and approximate numbers in
    our computer algebra systems for granted for a long time, not thinking
    to ask if they could be significantly better. With exact rational
    arithmetic and adjustable-precision floating-point arithmetic to
    precision limited only by the total computer memory or our patience,
    what more could we want for such numbers? It turns out that there is
    much that can be done that permits us to obtain exact results more
    often, more intelligible results, approximate results guaranteed to
    have requested error bounds, and recovery of exact results from
    approximate ones."
}
```

— axiom.bib —

```
@article{Sutt13,
  author = "Sutton, Brian D.",
  title = "Computing the Complete CS Decomposition",
  journal = "Numerical Algorithms",
  volume = "50",
  pages = "33-65",
  year = "2013",
  month = "February",
  link = "\url{http://arxiv.org/pdf/0707.1838v3.pdf}",
  abstract =
    "An algorithm is developed to compute the complete CS decomposition
    (CSD) of a partitioned unitary matrix. Although the existence of the
    CSD has been recognized since 1977, prior algorithms compute only a
    reduced version (the 2-by-1 CSD) that is equivalent to two
    simultaneous singular value decompositions. The algorithm presented
    here computes the complete 2-by-2 CSD, which requires the simultaneous
    diagonalization of all four blocks of a unitary matrix partitioned
    into a 2-by-2 block structure. The algorithm appears to be the only
    fully specified algorithm available. The computation occurs in two
    phases. In the first phase, the unitary matrix is reduced to
```

```

    bidiagonal block form, as described by Sutton and Edelman. In the
    second phase, the blocks are simultaneously diagonalized using
    techniques from bidiagonal SVD algorithms of Golub, Kahan, and
    Demmel. The algorithm has a number of desirable numerical features.",
    paper = "Sutt13.pdf"
}

```

— axiom.bib —

```

@article{Turi48,
  author = "Turing, A. M.",
  title = "Rounding-off errors in matrix processes",
  journal = "Q. J. Mech. Appl. Math.",
  volume = "1",
  pages = "287-308",
  year = "1948"
}

```

— axiom.bib —

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@article{Vign93,
  author = "Vignes, J.",
  title = "A stochastic arithmetic for reliable scientific computation",
  journal = "Math. and Comp. in Sim.",
  volume = "25",
  pages = "233-261",
  year = "1993"
}

```

— axiom.bib —

```

@article{Ward81,
  author = "Ward, Robert C.",
  title = "Balancing the generalized eigenvalue problem",
  journal = "SIAM J. Sci. and Stat. Comput.",
  volume = "2",
  number = "2",

```

```

year = "1981",
pages = "141-152",
abstract =
  "An algorithm is presented for balancing the  $A$  and  $B$  matrices
  prior to computing the eigensystem of the generalized eigenvalue
  problem  $Ax = \lambda Bx$ . The three-step algorithm is specifically
  designed to precede the  $QR$ -type algorithms, but improved performance
  is expected from most eigensystem solvers. Permutations and two-sided
  diagonal transformations are applied to  $A$  and  $B$  to produce
  matrices with certain desirable properties. Test cases are presented
  to illustrate the improved accuracy of the computed eigenvalues."
}

```

— axiom.bib —

```

@book{Wilk63,
  author = "Wilkinson, J. H.",
  title = "Rounding Errors in Algebraic Processes",
  publisher = "HMSO",
  series = "Notes on Applied Science, No. 32",
  year = "1963"
}

```

— axiom.bib —

```

@book{Wilk65,
  author = "Wilkinson, J. H.",
  title = "The Algebraic Eigenvalue Problem",
  publisher = "Oxford University Press",
  year = "1965"
}

```

— axiom.bib —

```

@InProceedings{Wilk84,
  author = "Wilkinson, J. H.",
  title = "The perfidious polynomial",

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```
booktitle = "Studies in Numerical Analysis",  
volume = "24",  
chapter = "1",  
pages = "1-28",  
year = "1984"  
}
```

— axiom.bib —

```
@article{Wilk86,  
author = "Wilkinson, J. H.",  
title = "Error analysis revisited",  
journal = "IMA Bulletin",  
volume = "22",  
pages = "192-200",  
year = "1986"  
}
```

— axiom.bib —

```
@article{Wilk61,  
author = "Wilkinson, J. H.",  
title = "Error analysis of direct methods of matrix inversion",  
journal = "J. ACM",  
volume = "8",  
pages = "281-330",  
year = "1961"  
}
```

— axiom.bib —

```
@article{Wilk85,  
author = "Wilkinson, J. H.",  
title = "The state of the art in error analysis",  
journal = "NAG Newsletter",  
volume = "2/85",  
pages = "5-28",
```

```

    year = "1985"
}

```

— axiom.bib —

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@article{Wilk60,
  author = "Wilkinson, J. H.",
  title = "Error analysis of floating-point computation",
  journal = "Numer. Math.",
  volume = "2",
  pages = "319-340",
  year = "1960"
}

```

— axiom.bib —

```

@book{Wilk71,
  author = "Wilkinson, J. H.",
  title = "Handbook for Automatic Computation, V2, Linear Algebra",
  publisher = "Springer-Verlag",
  year = "1971"
}

```

— ignore —

```

{Yang14,
  author = "Yang, Xiang and Mittal, Rajat",
  title = "Acceleration of the Jacobi iterative method by factors exceeding 100
          using scheduled relation",
  link = "\url{http://engineering.jhu.edu/fsag/wp-content/uploads/sites/23/2013/10/JCP_revised_WebPost.pdf}",
  paper = "Yang14.pdf"
}

```

1.14 Special Functions

— axiom.bib —

```
@misc{Corl05,
  author = "Corless, Robert M. and Jeffrey, David J. and Watt, Stephen M.
           and Bradford, Russell and Davenport, James H.",
  title = "Reasoning about the elementary functions of complex analysis",
  link = "\url{http://www.csd.uwo.ca/~watt/pub/reprints/2002-amai-reasoning.pdf}",
  abstract = "
    There are many problems with the simplification of elementary
    functions, particularly over the complex plane. Systems tend to make
    ‘‘howlers’’ or not to simplify enough. In this paper we outline the
    ‘‘unwinding number’’ approach to such problems, and show how it can be
    used to prevent errors and to systematise such simplification, even
    though we have not yet reduced the simplification process to a
    complete algorithm. The unsolved problems are probably more amenable
    to the techniques of artificial intelligence and theorem proving than
    the original problem of complex-variable analysis.",
  paper = "Corl05.pdf"
}
```

— ignore —

```
{Ng68,
  author = "Ng, Edward W. and Geller, Murray",
  title = "A Table of Integrals of the Error functions",
  link = "\url{http://nvlpubs.nist.gov/nistpubs/jres/73B/jresv73Bn1p1_A1b.pdf}",
  abstract = "
    This is a compendium of indefinite and definite integrals of products
    of the Error functions with elementary and transcendental functions.",
  paper = "Ng68.pdf"
}
```

1.15 Exponential Integral $E_1(x)$

— ignore —

```
{Gell69,
```

```

author = "Geller, Murray and Ng, Edward W.",
title = "A Table of Integrals of the Exponential Integral",
link = "\url{http://nvlpubs.nist.gov/nistpubs/jres/73B/jresv73Bn3p191_A1b.pdf}",
abstract = "
  This is a compendium of indefinite and definite integrals of products
  of the Exponential Integral with elementary or transcendental functions.",
paper = "Gell169.pdf"
}

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— axiom.bib —

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@techreport{Seg198,
  author = "Segletes, S.B.",
  title = "A compact analytical fit to the exponential integral  $E_1(x)$ ",
  year = "1998",
  institution = "U.S. Army Ballistic Research Laboratory,
    Aberdeen Proving Ground, MD",
  type = "Technical Report",
  number = "ARL-TR-1758",
  algebra = "\newline\refto{package DFSFUN DoubleFloatSpecialFunctions}",
  abstract = "
    A four-parameter fit is developed for the class of integrals known as
    the exponential integral (real branch). Unlike other fits that are
    piecewise in nature, the current fit to the exponential integral is
    valid over the complete domain of the function (compact) and is
    everywhere accurate to within  $\pm 0.0052\%$  when evaluating the first
    exponential integral,  $E_1$ . To achieve this result, a methodology
    that makes use of analytically known limiting behaviors at either
    extreme of the domain is employed. Because the fit accurately captures
    limiting behaviors of the  $E_1$  function, more accuracy is retained
    when the fit is used as part of the scheme to evaluate higher-order
    exponential integrals,  $E_n$ , as compared with the use of brute-force
    fits to  $E_1$ , which fail to accurately model limiting
    behaviors. Furthermore, because the fit is compact, no special
    accommodations are required (as in the case of spliced piecewise fits)
    to smooth the value, slope, and higher derivatives in the transition
    region between two piecewise domains. The general methodology employed
    to develop this fit is outlined, since it may be used for other
    problems as well.",
  paper = "Seg198.pdf"
}

```


— axiom.bib —

```
@techreport{Se09,
  author = "Segletes, S.B.",
  title = "Improved fits for  $E_1(x)$  those presented
    in ARL-TR-1758",
  type = "Technical Report",
  number = "ARL-TR-1758",
  institution = "U.S. Army Ballistic Research Laboratory,
    Aberdeen Proving Ground, MD",
  year = "1998",
  month = "September",
  abstract = "
    This is a writeup detailing the more accurate fits to  $E_1(x)$ ,
    relative to those presented in ARL-TR-1758. My actual fits are to
     $[F1 = [x \exp(x) E_1(x)]]$  which spans a functional range from 0 to 1.
    The best accuracy I have been yet able to achieve, defined by limiting
    the value of  $|(F1_{fit} - F1)/F1|$  over the domain, is
    approximately  $3.1E-07$  with a 12-parameter fit, which unfortunately
    isn't quite to 32-bit floating-point accuracy. Nonetheless, the fit
    is not a piecewise fit, but rather a single continuous function over
    the domain of nonnegative  $x$ , which avoids some of the problems
    associated with piecewise domain splicing.",
  paper = "Se09.pdf"
}
```

1.16 Polynomial GCD

— axiom.bib —

```
@InProceedings{Kalt99a,
  author = "Kaltofen, E. and Monagan, M.",
  title = "On the Genericity of the Modular Polynomial {GCD} Algorithm",
  booktitle = "Proc. 1999 Internat. Symp. Symbolic Algebraic Comput.",
  year = "1999",
  pages = "59--66",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/99/KaMo99.pdf}",
  paper = "Kalt99a.pdf"
}
```

— axiom.bib —

```
@book{Knut71,
  author = "Knuth, Donald",
  title = "The Art of Computer Programming Vol. 2 (Seminumerical Algorithms)",
  year = "1971",
  publisher = "Addison-Wesley"
}
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— axiom.bib —

```
@article{Ma90,
  author = "Ma, Keju and {von zur Gathen}, Joachim",
  title =
    "Analysis of Euclidean Algorithms for Polynomials over Finite Fields",
  journal = "J. Symbolic Computation",
  year = "1990",
  volume = "9",
  pages = "429-455",
  link = "\url{http://www.researchgate.net/publication/220161718_Analysis_of_Euclidean_Algorithms_for_Polyn",
  abstract = "
    This paper analyzes the Euclidean algorithm and some variants of it
    for computing the greatest common divisor of two univariate polynomials
    over a finite field. The minimum, maximum, and average number of
    arithmetic operations both on polynomials and in the ground field
    are derived.",
  paper = "Ma90.pdf"
}
```

— ignore —

```
\bibitem[Naylor 00a]{N00} Naylor, Bill
  title = "Polynomial GCD Using Straight Line Program Representation",
  PhD. Thesis, University of Bath, 2000
  link = "\url{http://www.sci.csd.uwo.ca/~bill/thesis.ps}",
  abstract = "
    This thesis is concerned with calculating polynomial greatest common
    divisors using straight line program representation.
```

In the Introduction chapter, we introduce the problem and describe some of the traditional representations for polynomials, we then talk about some of the general subjects central to the thesis, terminating with a synopsis of the category theory which is central to the Axiom computer algebra system used during this research.

The second chapter is devoted to describing category theory. We follow with a chapter detailing the important sections of computer code written in order to investigate the straight line program subject. The following chapter on evaluation strategies and algorithms which are dependant on these follows, the major algorithm which is dependant on evaluation and which is central to our thesis being that of equality checking. This is indeed central to many mathematical problems. Interpolation, that is the determination of coefficients of a polynomial is the subject of the next chapter. This is very important for many straight line program algorithms, as their non-canonical structure implies that it is relatively difficult to determine coefficients, these being the basic objects that many algorithms work on. We talk about three separate interpolation techniques and compare their advantages and disadvantages. The final two chapters describe some of the results we have obtained from this research and finally conclusions we have drawn as to the viability of the straight line program approach and possible extensions.

Finally we terminate with a number of appendices discussing side subjects encountered during the thesis.",
 paper = "N00.pdf"

— ignore —

```
\bibitem[Shoup 93]{ST-PGCD-Sh93} Shoup, Victor
  title = "Factoring Polynomials over Finite Fields: Asymptotic Complexity vs Reality*",
  Proc. IMACS Symposium, Lille, France, (1993)
  link = "\url{http://www.shoup.net/papers/lille.pdf}",
  abstract = "
    This paper compares the algorithms by Berlekamp, Cantor and
    Zassenhaus, and Gathen and Shoup to conclude that (a) if large
    polynomials are factored the FFT should be used for polynomial
    multiplication and division, (b) Gathen and Shoup should be used if
    the number of irreducible factors of  $f$  is small. (c) if nothing is
    known about the degrees of the factors then Berlekamp's algorithm
    should be used.",
  paper = "ST-PGCD-Sh93.pdf"
```

— axiom.bib —

```
@inproceedings{Hoei04,
```

```

author = "van Hoeij, Mark and Monagan, Michael",
title = "Algorithms for Polynomial GCD Computation over Algebraic
        Function Fields",
booktitle = "Proc. ISSAC 04",
year = "2004",
isbn = "1-58113-827-X",
link = "\url{http://www.cecm.sfu.ca/personal/mmonagan/papers/AFGCD.pdf}",
abstract =
    "Let  $L$  be an algebraic function field in  $k$  parameters
      $t_1, \dots, t_k$ . Let  $f_1, f_2$  be non-zero polynomials in
      $L[x]$ . We give two algorithms for computing their gcd. The first, a
     modular GCD algorithm, is an extension of the modular GCD algorithm
     for Brown for  $\mathbb{Z}[x_1, \dots, x_n]$  and Encarnacion for  $\mathbb{Q}(\alpha[x])$ 
     to function fields. The second, a fraction-free
     algorithm, is a modification of the Moreno Maza and Rioboo algorithm
     for computing gcds over triangular sets. The modification reduces
     coefficient growth in  $L$  to be linear. We give an empirical
     comparison of the two algorithms using implementations in Maple.",
paper = "Hoei04.pdf"
}

```

— ignore —

```

\bibitem[Wang 78]{Wang78} Wang, Paul S.
    title = "An Improved Multivariate Polynomial Factoring Algorithm",
    Mathematics of Computation, Vol 32, No 144 Oct 1978, pp1215-1231
    link = "\url{http://www.ams.org/journals/mcom/1978-32-144/S0025-5718-1978-0568284-3/S0025-5718-1978-0568284-3.pdf}",
    abstract = "
        A new algorithm for factoring multivariate polynomials over the
        integers based on an algorithm by Wang and Rothschild is described.
        The new algorithm has improved strategies for dealing with the known
        problems of the original algorithm, namely, the leading coefficient
        problem, the bad-zero problem and the occurrence of extraneous factors.
        It has an algorithm for correctly predetermining leading coefficients
        of the factors. A new and efficient p-adic algorithm named EEZ is
        described. Basically it is a linearly convergent variable-by-variable
        parallel construction. The improved algorithm is generally faster and
        requires less store than the original algorithm. Machine examples with
        comparative timing are included.",
    paper = "Wang78.pdf"

```

— ignore —

```
\bibitem[Wiki 4]{Wiki4}.
  title = "Polynomial greatest common divisor",
  link = "\url{http://en.wikipedia.org/wiki/Polynomial_greatest_common_divisor}",
```

1.17 Category Theory

— ignore —

```
\bibitem[Baez 09]{Baez09} Baez, John C.; Stay, Mike
  title = "Physics, Topology, Logic and Computation: A Rosetta Stone",
  link = "\url{http://arxiv.org/pdf/0903.0340v3.pdf}",
  abstract = "
    In physics, Feynman diagrams are used to reason about quantum
    processes. In the 1980s, it became clear that underlying these
    diagrams is a powerful analogy between quantum physics and
    topology. Namely, a linear operator behaves very much like a
    ‘‘cobordism’’: a manifold representing spacetime, going between two
    manifolds representing space. But this was just the beginning: similar
    diagrams can be used to reason about logic, where they represent
    proofs, and computation, where they represent programs. With the rise
    of interest in quantum cryptography and quantum computation, it became
    clear that there is an extensive network of analogies between physics,
    topology, logic and computation. In this expository paper, we make
    some of these analogies precise using the concept of ‘‘closed
    symmetric monoidal category’’. We assume no prior knowledge of
    category theory, proof theory or computer science.",
  paper = "Baez09.pdf"
```

— axiom.bib —

```
@misc{Meij91,
  author = "Meijer, Erik and Fokkinga, Maarten and Paterson, Ross",
  title = "Functional Programming with Bananas, Lenses, Envelopes and
    Barbed Wire",
  link = "\url{http://eprints.eemcs.utwente.nl/7281/01/db-utwente-40501F46.pdf}",
  abstract = "
    We develop a calculus for lazy functional programming based on
    recursion operators associated with data type definitions. For these
    operators we derive various algebraic laws that are useful in deriving
    and manipulating programs. We shall show that all example functions in
```

```

    Bird and Wadler's "Introduction to Functional Programming" can be
    expressed using these operators.",
    paper = "Meij91.pdf"
}

```

— axiom.bib —

```

@misc{Yous04,
  author = "Youssef, Saul",
  title = "Prospects for Category Theory in Aldor",
  year = "2004",
  link = "\url{http://axiom-wiki.newsynthesis.org/public/refs/Youssef-ProspectsForCategoryTheoryInAldor.pdf}",
  abstract =
    "Ways of incorporating category theory constructions and results into
    the Aldor language are discussed. The main features of Aldor which
    make this possible are identified, examples of categorical
    constructions are provided and a suggestion is made for a foundation
    for rigorous results.",
  paper = "Yous04.pdf"
}

```

1.18 Proving Axiom Correct

— axiom.bib —

```

@misc{COQnat,
  author = "COQ Proof Assistant",
  title = "{Library} {Coq}.{Init}.{Nat}",
  link = "\url{https://coq.inria.fr/library/Coq.Init.Nat.html}",
  abstract = "Peano natural numbers, definitions of operations",
  year = "2017"
}

```

— axiom.bib —

```

@book{Acze13,

```

```

author = "Aczel, Peter et al.",
title = "Homotopy Type Theory: Univalent Foundations of Mathematics",
publisher = "Institute for Advanced Study",
year = "2013",
link = "\url{https://hott.github.io/book/nightly/hott-letter-1075-g3c53219.pdf}",
paper = "Acze13.pdf"
}

```

— ignore —

```

\bibitem[Adams 99]{Adam99} Adams, A.A.; Gottlieben, H.; Linton, S.A.;
Martin, U.
  title = "Automated theorem proving in support of computer algebra:
    symbolic definite integration as a case study",
  abstract = "
    We assess the current state of research in the application of computer
    aided formal reasoning to computer algebra, and argue that embedded
    verification support allows users to enjoy its benefits without
    wrestling with technicalities. We illustrate this claim by considering
    symbolic definite integration, and present a verifiable symbolic
    definite integral table look up: a system which matches a query
    comprising a definite integral with parameters and side conditions,
    against an entry in a verifiable table and uses a call to a library of
    lemmas about the reals in the theorem prover PVS to aid in the
    transformation of the table entry into an answer. We present the full
    model of such a system as well as a description of our prototype
    implementation showing the efficacy of such a system: for example, the
    prototype is able to obtain correct answers in cases where computer
    algebra systems [CAS] do not. We extend upon Fateman's web-based table
    by including parametric limits of integration and queries with side
    conditions.",
  paper = "Adam99.pdf"

```

— axiom.bib —

```

@InProceedings{Adam01,
  author = "Adams, Andrew A. and Dunstan, Martin and Gottlieben, Hanne and
    Kelsey, Tom and Martin, Ursula and Owre, Sam",
  title = "Computer algebra meets automated theorem proving: Integrating
    Maple and PVS",
  booktitle = "Theorem proving in higher order logics",
  series = "TPHOLs 2001",

```

```

year = "2001",
location = "Edinburgh, Scotland",
pages = "27-42",
abstract =
  "We describe an interface between version 6 of the Maple computer
  algebra system with the PVS automated theorem prover. The interface is
  designed to allow Maple users access to the robust and checkable proof
  environment of PVS. We also extend this environment by the provision
  of a library of proof strategies for use in real analysis. We
  demonstrate examples using the interface and the real analysis
  library. These examples provide proofs which are both illustrative and
  applicable to genuine symbolic computation problems.",
paper = "Adam01.pdf",
keywords = "axiomref"
}

```

— axiom.bib —

```

@misc{Avig14,
  author = "Avigad, Jeremy",
  title = "LEAN proof of GCD",
  link =
    "\url{http://github.com/leanprover/lean2/blob/master/library/data/nat/gcd.lean}",
  year = "2014"
}

```

— axiom.bib —

```

@misc{Avig16,
  author = "Avigad, Jeremy",
  title = "LEAN github repository",
  link = "\url{http://github.com/leanprover}",
  year = "2016"
}

```

— axiom.bib —

```

@misc{Avig17,

```



```

author = "Avigad, Jeremy and de Moura, Leonardo and Kong, Soonho",
title = "Theorem Proving in Lean",
year = "2017",
link = "\url{https://leanprover.github.io/tutorial/tutorial.pdf}",
paper = "Avig17.pdf"
}

```

— axiom.bib —

```

@misc{Ball98,
author = "Ballarin, Clemens and Paulson, Lawrence C.",
title = "Reasoning about Coding Theory: The Benefits We Get from
        Computer Algebra",
year = "1998",
link = "\url{http://www21.in.tum.de/~ballarin/publications/aisc98.pdf}",
abstract =
    "The use of computer algebra is usually considered beneficial for
    mechanised reasoning in mathematical domains. We present a case study,
    in the application domain of coding theory, that supports this claim:
    the mechanised proof depends on non-trivial algorithms from computer
    algebra and increase the reasoning power of the theorem prover. The
    unsoundness of computer algebra systems is a major problem in
    interfacing them to theorem provers. Our approach to obtaining a sound
    overall system is not blanket distrust but based on the distinction
    between algorithms we call sound and {\sl ad hoc} respectively. This
    distinction is blurred in most computer algebra systems Our
    experimental interface therefore uses a computer algebra library. It
    is based on theorem templates, which provide formal specifications for
    the algorithms.",
paper = "Ball98.pdf"
}

```

— axiom.bib —

```

@article{Ball99,
author = "Ballarin, Clemens and Paulson, Lawrence C.",
title = "A Pragmatic Approach to Extending Provers by Computer Algebra --
        with Applications to Coding Theory",
journal = "Fundam. Inform.",
volume = "39",
number = "1-2",
pages = "1-20",

```

```

year = "1999",
link = "\url{http://www.cl.cam.ac.uk/~lp15/papers/Isabelle/coding.pdf}",
abstract = "
  The use of computer algebra is usually considered beneficial for
  mechanised reasoning in mathematical domains. We present a case study,
  in the application domain of coding theory, that supports this claim:
  the mechanised proofs depend on non-trivial algorithms from computer
  algebra and increase the reasoning power of the theorem prover.

  The unsoundness of computer algebra systems is a major problem in
  interfacing them to theorem provers. Our approach to obtaining a sound
  overall system is not blanket distrust but based on the distinction
  between algorithms we call sound and {\sl ad hoc} respectively. This
  distinction is blurred in most computer algebra systems. Our
  experimental interface therefore uses a computer algebra library. It
  is based on formal specifications for the algorithms, and links the
  computer algebra library Sumit to the prover Isabelle.

  We give details of the interface, the use of the computer algebra
  system on the tactic-level of Isabelle and its integration into proof
  procedures.",
paper = "Ball99.pdf",
keywords = "axiomref"
}

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— axiom.bib —

```

@article{Baue16,
  author = "Bauer, Andrej",
  title = "Five Stages of Accepting Constructive Mathematics",
  year = "2016",
  journal = "Bull. of the American Mathematical Society",
  link = "\url{http://dx.doi.org/10.1090/bull/1556}",
  abstract =
    "On the odd day, a mathematician might wonder what constructive
    mathematics is all about. They may have heard arguments in favor of
    constructivism but are not at all convinced by them, and in any case
    they may care little about philosophy. A typical introductory text
    about constructivism spends a great deal of time explaining the
    principles and contains only trivial mathematics, while advanced
    constructive texts are impenetrable, like all unfamiliar mathematics.
    How then can a mathematician find out what constructive mathematics
    feels like? What new and relevant ideas does constructive mathematics
    have to offer, if any? I shall attempt to answer these questions.",
  paper = "Baue16.pdf"
}

```

— axiom.bib —

```
@misc{Beesxx,
  author = "Beeson, Michael",
  title = "Automatic Generation of Epsilon-Delta Proofs of Continuity",
  link = "\url{http://www.michaelbeeson.com/research/papers/aisc.pdf}",
  abstract =
    "As part of a project on automatic generation of proofs involving both
    logic and computation, we have automated the production of some proofs
    involving epsilon-delta arguments. These proofs involve two or three
    quantifiers on the logical side, and on the computational side, they
    involve algebra, trigonometry, and some calculus. At the border of
    logic and computation, they involve several types of arguments
    involving inequalities, including transitivity chaining and several
    types of bounding arguments, in which bounds are sought that do not
    depend on certain variables. Control mechanisms have been developed
    for intermixing logical deduction steps with computational steps and
    with inequality reasoning. Problems discussed here as examples involve
    the continuity and uniform continuity of various specific functions.",
  paper = "Beesxx.pdf"
}
```

— axiom.bib —

```
@book{Bert04,
  author = {Bertot, Yves Cast\'eran, Pierre},
  title = "Interactive Theorem Proving and Program Development",
  publisher = "Springer",
  year = "2004",
  isbn = "3-540-20854-2",
  abstract = "
    Coq is an interactive proof assistant for the development of
    mathematical theories and formally certified software. It is based on
    a theory called the calculus of inductive constructions, a variant of
    type theory.

    This book provides a pragmatic introduction to the development of
    proofs and certified programs using Coq. With its large collection of
    examples and exercices it is an invaluable tool for researchers,
    students, and engineers interested in formal methods and the
    development of zero-fault software."
```

}

— axiom.bib —

```
@misc{Bold07,
  author = "Boldo, Sylvie and Filliatre, Jean-Christophe",
  title = "Formal Verification of Floating-Point programs",
  link = "\url{http://www-lipn.univ-paris13.fr/CerPAN/files/ARITH.pdf}",
  paper = "Bold07.pdf"
}
```

— axiom.bib —

```
@misc{Bold07a,
  author = "Boldo, Sylvie and Filliatre, Jean-Christophe",
  title = "Formal Verification of Floating-Point programs",
  link = "\url{http://www.lri.fr/~filliatr/ftp/publis/caduceus-floats.pdf}",
  abstract =
    "This paper introduces a methodology to perform formal verification of
    floating-point C programs. It extends an existing tool for
    verification of C programs, Caduceus, with new annotations for
    specific floating-point arithmetic. The Caduceus first-order logic
    model for C programs is extended accordingly. Then verification
    conditions are obtained in the usual way and can be discharged
    interactively with the Coqa proof assistant, using an existing Coq
    formalization of floating-point arithmetic. This methodology is
    already implemented and has been successfully applied to several short
    floating-point programs, which are presented in this paper.",
  paper = "Bold07a.pdf"
}
```

— axiom.bib —

```
@article{Bold11,
  author = "Boldo, Sylvie and Marche, Claude",
  title = "Formal verification of numerical programs: from C annotated
    programs to mechanical proofs",
}
```

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year = "2011",
publisher = "Springer",
journal = "Mathematics in Computer Science",
volume = "5",
pages = "377-393",
link = "\url{https://hal.archives-ouvertes.fr/hal-00777605/document}",
abstract =
  "Numerical programs may require a high level of guarantee. This can be
  achieved by applying formal methods, such as machine-checked proofs.
  But these tools handle mathematical theorems while we are interested
  in C code, in which numerical computations are performed using
  floating-point arithmetic, whereas proof tools typically handle exact
  real arithmetic. To achieve this high level of confidence on C programs,
  we use a chain of tools: Frama-C, its Jessie plugin, Why and provers
  among Coq, Gappa, Alt-Ergo, CVC3 and Z3. This approach requires the C
  program to be annotated; each function must be precisely specified, and
  we prove the correctness of the program by proving both that it meets its
  specifications and that no runtime error may occur. The purpose of this
  paper is to illustrate, on various examples, the features of this
  approach.",
paper = "Bold11.pdf"
}

```

— axiom.bib —

```

@misc{Bold16,
  author = "Boldo, Sylvie",
  title = "Formal verification of numerical analysis programs",
  year = "2016",
  link = "\url{https://www.youtube.com/watch?v=7MDwpwD6Ts4}"
}

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— axiom.bib —

```

@misc{Boul00,
  author = "Boulme, S. and Hardin, T. and Rioboo, R.",
  title = "Polymorphic Data Types, Objects, Modules and Functors, :
  is it too much?",
  link = "\url{ftp://ftp.lip6.fr/lip6/reports/2000/lip6.2000.014.ps.gz}",
  abstract = "
    Abstraction is a powerful tool for developers and it is offered by
    numerous features such as polymorphism, classes, modules, and

```

```

functors, $\\ldots$ A working programmer may be confused by this
abundance. We develop a computer algebra library which is being
certified. Reporting this experience made with a language (Ocaml)
offering all these features, we argue that the are all needed
together. We compare several ways of using classes to represent
algebraic concepts, trying to follow as close as possible mathematical
specification. Then we show how to combine classes and modules to
produce code having very strong typing properties. Currently, this
library is made of one hundred units of functional code and behaves
faster than analogous ones such as Axiom.",
paper = "Boul00.pdf",
keywords = "axiomref"
}

```

— axiom.bib —

```

@InProceedings{Boul99,
  author = "Boulme, S. and Hardin, T. and Hirschkoff, D. and Rioboo, Renaud",
  title = "On the way to certify Computer Algebra Systems",
  booktitle = "Systems for integrated computation and deduction",
  series = "Calculemus 99",
  year = "1999",
  publisher = "Elsevier",
  location = "Trento, Italy",
  pages = "11-12",
  abstract =
    "The FOC project aims at supporting, within a coherent software system,
    the entire process of mathematical computation, starting with proved
    theories, ending with certified implementations of algorithms. In this
    paper, we explain our design requirements for the implementation,
    using polynomials as a running example. Indeed, proving correctness of
    implementations depends heavily on the way this design allows
    mathematical properties to be truly handled at the programming level.

    The FOC project, started at the fall of 1997, is aimed to build a
    programming environment for the development of certified symbolic
    computation. The working languages are Coq and Ocaml. In this paper,
    we present first the motivations of the project. We then explain why
    and how our concern for proving properties of programs has led us to
    certain implementation choices in Ocaml. This way, the sources express
    exactly the mathematical dependencies between different structures.

    This may ease the achievement of proofs.",
  paper = "Boul99.pdf",
  keywords = "axiomref"
}

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— axiom.bib —

```
@book{Brad07,  
  author = "Bradley, Aaron R. and Manna, Zohar",  
  title = "The Calculus of Computation",  
  year = "2007",  
  publisher = "Springer",  
  isbn = "978-3-540-74112-1",  
}
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— axiom.bib —

```
@article{Bres93,  
  author = "Bressoud, David",  
  title = "Review of The problems of mathematics",  
  journal = "Math. Intell.",  
  volume = "15",  
  number = "4",  
  year = "1993",  
  pages = "71-73"  
}
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— axiom.bib —

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@article{Buch97,  
  author = "Buchberger, Bruno",  
  title = "Mathematica: doing mathematics by computer?",  
  journal = "Advances in the design of symbolic computation systems",  
  year = "1997",  
  publisher = "Springer-Verlag",  
  pages = "2-20",  
  isbn = "978-3-211-82844-1",  
  paper = "Buch97.pdf"  
}
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— axiom.bib —

```
@article{Bulo04,
  author = {Medina-Bulo, I. and Palomo-Lozano, F. and Alonso-Jim\`enez, J.A.
            and Ruiz-Reina, J.L.},
  title = "Verified Computer Algebra in ACL2",
  journal = "ASIC 2004, LNAI 3249",
  year = "2004",
  pages = "171-184",
  abstract = "In this paper, we present the formal verification of a
              Common Lisp implementation of Buchberger's algorithm for computing
              Groebner bases of polynomial ideals. This work is carried out in the
              ACL2 system and shows how verified Computer Algebra can be achieved
              in an executable logic.",
  paper = "Bulo04.pdf"
}
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— axiom.bib —

```
@article{Bulo10,
  author = "Medina-Bulo, I. and Palomo-Lozano, F. and Alonso-Jim\`enez, J.A.
            and Ruiz-Reina, J.L.",
  title = "A verified Common Lisp implementation of Buchberger's algorithm
            in ACL2",
  journal = "Journal of Symbolic Computation",
  year = "2010",
  pages = "96-123",
  abstract = "In this article, we present the formal verification of a
              Common Lisp implementation of Buchberger's algorithm or computing
              Groebner bases of polynomial ideals. This work is carried out in ACL2,
              a system which provides an integrated environment where programming
              (in a pure functional subset of Common Lisp) and formal verification
              of programs, with the assistance of a theorem prover, are possible. Our
              imlementation is written in a real programming language and it is
              directly executable within the ACL2 system or any compliant Common Lisp
              system. We provide here snippets o real verified code, discuss the
              formalization details in depth, and present quantitative data about
              the proof effort.",
  paper = "Bulo10.pdf"
}
```

— axiom.bib —

```
@article{Chli10,
```



```

author = "Chlipala, Adam",
title = "An Introduction to Programming and Proving with Dependent Types
        in Coq",
journal = "Journal of Formalized Reasoning",
volume = "3",
number = "2",
pages = "1-93",
year = "2010",
paper = "Chli10.pdf"
}

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— axiom.bib —

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@misc{Chli14,
  author = "Chlipala, Adam and Braibant, Thomas and Cuellar, Santiago and
            Delaware, Benjamin and Gross, Jason and Malecha, Gregory and
            Clement,-Claudel, Pit and Wang, Peng",
  title = "Bedrock: A Software Development Ecosystem Inside a Proof Assistant",
  year = "2014",
  link = "\url{https://www.youtube.com/watch?v=BSyrrp-iYBMo}",
  abstract =
    "The benefits of formal correctness proofs for software are clear
    intuitively, but the high human costs of proof construction have
    generally been viewed as prohibitive. To support that integration, we
    need to rethink the familiar programming toolchains. The new world
    needn't be all about doing prodigious extra work to achieve the virtue
    of correct programs; formal methods also suggest new programming
    approaches that better support abstraction and modularity than do
    coarser-grained specification styles like normal static types. This
    talk overviews Bedrock, a framework for certified programming inside
    of the Coq proof assistant. Bedrock programs are implemented,
    specified, verified, and compiled inside of Coq. A single program may
    be divided into modules with formal interfaces, written in different
    programming languages and verified with different proof styles. The
    common foundation is an assembly language with an operational
    semantics (serving as the trusted code base) and a semantic module
    system (orchestrating linking of code and proofs across source
    languages). A few different programming styles have been connects to
    the shared foundation, including a C-like language with an ‘‘array of
    bytes’’ memory model, higher-level more C++-like languages with ‘‘array
    of abstract data types’’ memory models, a domain-specific language for
    XML processing, standard Coq functional programs, and even declarative
    specifications that are refined automatically into assembly code with
    correctness proofs. The talk will present Bedrock's shared foundation
    and sketch the pieces that go into refining declarative specifications
    into closed assembly programs, covering joint work with Thomas

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Braibant, Santiago Cuellar, Benjamin Delaware, Jason Gross, Gregory Malecha, Clement Pit-Claudel, and Peng Wang."

}

— axiom.bib —

```
@techreport{Coqu86,
  author = {Coquand, Thierry and Huet, G\'erard},
  title = "The Calculus of Constructions",
  year = "1986",
  institution = "INRIA Centre de Rocquencourt",
  number = "530",
  link = "\url{https://hal.inria.fr/inria-00076024/document}",
  abstract =
    "The Calculus of Constructions is a higher-order formalism for
    constructive proofs in natural deduction style. Every proof is a
     $\lambda$ -expression, typed with propositions of the underlying
    logic. By removing types we get a pure  $\lambda$ -expression,
    expressing its associated algorithm. Computing this
     $\lambda$ -expression corresponds roughly to cut-elimination. It is our
    thesis that (as already advocated by Martin-Lof) the Curry-Howard
    correspondence between propositions and types is a powerful paradigm
    for Computer Science. In the case of Constructions, we obtain the
    notion of a very high-level functional programming language, with
    complex polymorphism well-suited for modules specification. The notion
    of type encompasses the usual notion of data type, but allows as well
    arbitrarily complex algorithmic specifications. We develop the basic
    theory of a Calculus of Constructions, and prove a strong
    normalization theorem showing that all computations terminate.
    Finally, we suggest various extensions to stronger calculi.",
  paper = "Coqu86.pdf"
}
```

— axiom.bib —

```
@misc{Coqu16,
  author = {Coquand, Thierry and Huet, G\'erard and Paulin, Christine},
  title = "The COQ Proof Assistant",
  year = "2016",
  link = "\url{https://coq.inria.fr}"
}
```

— axiom.bib —

```
@misc{COQR16,
  author = {Coquand, Thierry and Huet, G\'erard and Paulin, Christine},
  title = "The COQ Proof Assistant Reference Manual",
  year = "2016",
  link="\url{https://coq.inria.fr/distrib/current/files/Reference-Manual.pdf}",
  paper = "COQR16.pdf"
}
```

— axiom.bib —

```
@misc{Coqui6a,
  author = {Coquand, Thierry and Huet, G\'erard and Paulin, Christine},
  title = "COQ Proof Assistant Library Coq.ZArith.Znumtheory",
  year = "2016",
  link = "\url{https://coq.inria.fr/library/Coq.ZArith.Znumtheory.html}"
}
```

— axiom.bib —

```
@book{Chli15,
  author = "Chlipala, Adam",
  title = "Certified Programming with Dependent Types",
  year = "2015",
  link = "\url{http://adam.chlipala.net/cpdt/cpdt.pdf}",
  publisher = "MIT Press",
  isbn = "9780262026659",
  paper = "Chli15.pdf"
}
```

— axiom.bib —

```

@misc{Cons98,
  author = "Constable, Robert L. and Jackson, Paul B.",
  title = "Towards Integrated Systems for Symbolic Algebra and Formal
          Constructive Mathematics",
  link = "\url{http://www.nuprl.org/documents/Constable/towardsintegrated.pdf}",
  year = "1998",
  abstract =
    "The purpose of this paper is to report on our efforts to give e a
    formal account of some of the algebra used in Computer Algebra Systems
    (CAS). In particular, we look at the concepts used in the so called
    3rd generation algebra systems, such as Axiom[4] and Weyl[9]. It is
    our claim that the Nuprl proof development system is especially well
    suited to support this kind of mathematics.",
  paper = "Cons98.pdf"
}

```

— ignore —

```

\bibitem[Daly 10]{Daly10} Daly, Timothy
  title = "Intel Instruction Semantics Generator",
  link = "\url{http://daly.axiom-developer.org/TimothyDaly_files/publications/sei/intel/intel.pdf}",
  abstract = "
    Given an Intel x86 binary, extract the semantics of the instruction
    stream as Conditional Concurrent Assignments (CCAs). These CCAs
    represent the semantics of each individual instruction. They can be
    composed to represent higher level semantics.",
  paper = "Daly10.pdf"

```

— axiom.bib —

```

@InProceedings{Dani06,
  author = "Danielsson, Nils Anders and Hughes, John and Jansson, Patrik and
          Gibbons, Jeremy",
  title = "Fast and Loose Reasoning is Morally Correct",
  booktitle = "Proc. of ACM POPL '06",
  series = "POPL '06",
  year = "2006",
  location = "Charleston, South Carolina",
  abstract =
    "Functional programmers often reason about programs as if they were
    written in a total language, expecting the results to carry over to
    non-toal (partial) languages. We justify such reasoning.

```

Two languages are defined, one total and one partial, with identical syntax. The semantics of the partial language includes partial and infinite values, and all types are lifted, including the function spaces. A partial equivalence relation (PER) is then defined, the domain of which is the total subset of the partial language. For types not containing function spaces the PER relates equal values, and functions are related if they map related values to related values.

It is proved that if two closed terms have the same semantics in the total language, then they have related semantics in the partial language. It is also shown that the PER gives rise to a bicartesian closed category which can be used to reason about values in the domain of the relation.",

```
paper = "Dani06.pdf",
keywords = "axiomref"
}
```

— ignore —

```
\bibitem[Davenport 12]{Davenp12} Davenport, James H.; Bradford, Russell;
England, Matthew; Wilson, David
```

```
title = "Program Verification in the presence of complex numbers,
functions with branch cuts etc.",
```

```
link = "\url{http://arxiv.org/pdf/1212.5417.pdf}",
```

```
abstract = "
```

In considering the reliability of numerical programs, it is normal to ‘‘limit our study to the semantics dealing with numerical precision’’. On the other hand, there is a great deal of work on the reliability of programs that essentially ignores the numerics. The thesis of this paper is that there is a class of problems that fall between these two, which could be described as ‘‘does the low-level arithmetic implement the high-level mathematics’’. Many of these problems arise because mathematics, particularly the mathematics of the complex numbers, is more difficult than expected: for example the complex function log is not continuous, writing down a program to compute an inverse function is more complicated than just solving an equation, and many algebraic simplification rules are not universally valid.

The good news is that these problems are {\sl theoretically} capable of being solved, and are {\sl practically} close to being solved, but not yet solved, in several real-world examples. However, there is still a long way to go before implementations match the theoretical possibilities.",

```
paper = "Davenp12.pdf"
```

— axiom.bib —

```
@article{Demi79,
  author = "DeMilo, Richard A. and Lipton, Richard J. and Perlis, Alan J.",
  title = "Social Processes and Proofs of Theorems and Programs",
  journal = "Communications of the ACM",
  volume = "22",
  number = "5",
  year = "1979",
  pages = "271-280",
  abstract =
    "It is argued that formal verifications of programs, no matter how
    obtained, will not play the same key role in the development of
    computer science and software engineering as proofs do in mathematics.
    Furthermore the absence of continuity, the inevitability of change, and
    the complexity of specification of significantly many real programs
    make the formal verification process difficult to justify and manage.
    It is felt that ease of formal verification should not dominate program
    language design.",
  paper = "Demi79.pdf"
}
```

— ignore —

```
\bibitem[Dolzmann 97]{Dolz97} Dolzmann, Andreas; Sturm, Thomas
  title = "Guarded Expressions in Practice",
  link = "\url{http://redlog.dolzmann.de/papers/pdf/MIP-9702.pdf}",
  abstract = "
    Computer algebra systems typically drop some degenerate cases when
    evaluating expressions, e.g.  $x/x$  becomes 1 dropping the case
     $x=0$ . We claim that it is feasible in practice to compute also the
    degenerate cases yielding \sl guarded expressions. We work over real
    closed fields but our ideas about handling guarded expressions can be
    easily transferred to other situations. Using formulas as guards
    provides a powerful tool for heuristically reducing the combinatorial
    explosion of cases: equivalent, redundant, tautological, and
    contradictory cases can be detected by simplification and quantifier
    elimination. Our approach allows to simplify the expressions on the
    basis of simplification knowledge on the logical side. The method
    described in this paper is implemented in the REDUCE package GUARDIAN,
    which is freely available on the WWW.",
  paper = "Dolz97.pdf"
```

— ignore —

```
\bibitem[Dos Reis 11]{DR11} Dos Reis, Gabriel; Matthews, David; Li, Yue
  title = "Retargeting OpenAxiom to Poly/ML: Towards an Integrated Proof Assistants and Computer
  Calculamus (2011) Springer
  link = "\url{http://paradise.caltech.edu/~yli/paper/oa-polym1.pdf}",
  abstract = "
    This paper presents an ongoing effort to integrate the Axiom family of
    computer algebra systems with Poly/ML-based proof assistants in the
    same framework. A long term goal is to make a large set of efficient
    implementations of algebraic algorithms available to popular proof
    assistants, and also to bring the power of mechanized formal
    verification to a family of strongly typed computer algebra systems at
    a modest cost. Our approach is based on retargeting the code generator
    of the OpenAxiom compiler to the Poly/ML abstract machine.",
  paper = "DR11.pdf",
  keywords = "axiomref"
```

— ignore —

```
\bibitem[Dunstan 00a]{Dun00a} Dunstan, Martin N.
  title = "Adding Larch/Aldor Specifications to Aldor",
  abstract = "
    We describe a proposal to add Larch-style annotations to the Aldor
    programming language, based on our PhD research. The annotations
    are intended to be machine-checkable and may be used for a variety
    of purposes ranging from compiler optimizations to verification
    condition (VC) generation. In this report we highlight the options
    available and describe the changes which would need to be made to
    the compiler to make use of this technology.",
  paper = "Dunxx.pdf"
```

— axiom.bib —

```
@InProceedings{Duns98,
  author = "Dunstan, Martin and Kelsey, Tom and Linton, Steve and
  Martin, Ursula",
  title = "Lightweight Formal Methods For Computer Algebra Systems",
  publisher = "ACM Press",
```

```

booktitle = "Proc. ISSAC 1998",
year = "1998",
location = "Rostock, Germany",
pages = "80-87",
link = "\url{http://www.cs.st-andrews.ac.uk/~tom/pub/issac98.pdf}",
abstract =
  "Demonstrates the use of formal methods tools to provide a semantics
  for the type hierarchy of the Axiom computer algebra system, and a
  methodology for Aldor program analysis and verification. There are
  examples of abstract specifications of Axiom primitives.",
paper = "Duns98.pdf",
keywords = "axiomref"
}

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— axiom.bib —

```

@phdthesis{Duns99a,
  author = "Dunstan, Martin N.",
  title = "Larch/Aldor - A Larch BSL for AXIOM and Aldor",
  school = "University of St. Andrews",
  year = "1999",
  abstract = "
    In this thesis we investigate the use of lightweight formal methods
    and verification conditions (VCs) to help improve the reliability of
    components constructed within a computer algebra system. We follow the
    Larch approach to formal methods and have designed a new behavioural
    interface specification language (BSL) for use with Aldor: the
    compiled extension language of Axiom and a fully-featured programming
    language in its own right. We describe our idea of lightweight formal
    methods, present a design for a lightweight verification condition
    generator and review our implementation of a prototype verification
    condition generator for Larch/Aldor.",
  paper = "Duns99a.pdf",
  keywords = "axiomref"
}

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— axiom.bib —

```

@InProceedings{Duns99,
  author = "Dunstan, Martin and Kelsey, Tom and Martin, Ursula and
    Linton, Steve A.",
  title = "Formal Methods for Extensions to CAS",

```



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booktitle = "Proc. of FME'99",
series = "FME'99",
location = "Toulouse, France",
year = "1999",
pages = "1758-1777",
link = "\url{http://tom.host.cs.st-andrews.ac.uk/pub/fm99.ps}",
abstract =
    "We demonstrate the use of formal methods tools to provide a semantics
    for the type hierarchy of the AXIOM computer algebra system, and a
    methodology for Aldor program analysis and verification. We give a
    case study of abstract specifications of AXIOM primitives, and provide
    an interface between these abstractions and Aldor code.",
paper = "Duns99.pdf",
keywords = "axiomref"
}

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— axiom.bib —

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@misc{Fate02a,
author = "Fateman, Richard J.",
title = "Symbolic Execution Merges Construction, Debugging and Proving",
link = "\url{http://www.cs.berkeley.edu/~fateman/papers/symex.pdf}",
year = "2002",
abstract =
    "There is naturally an interest in any technology which promises to
    assist us in producing correct programs. Some efforts attempt to
    insure correct programs by making their construction simpler. Some
    efforts are oriented toward increasing the effectiveness of testing to
    make the programs appear to perform as required. Other efforts are
    directed to prove the correctness of the resulting program. Symbolic
    execution, in which symbols instead of numbers are used in what
    appears to be a numerical program, is an old but to-date still not
    widely-used technique. It has been available in various forms for
    decades from the computer algebra community. Symbolic execution has
    the potential to assist in all these phases: construction, debugging,
    and proof. We describe how this might work specifically with regard to
    our own recent experience in the construction of correct linear
    algebra programs for structured matrices and LU factorization. We show
    how developing these programs with a computer algebra system, and then
    converting incrementally to use more efficient forms. Frequent symbolic
    execution of the algorithms, equivalent to testing over infinite test
    sets, aids in debugging, while strengthening beliefs that the correctness
    of results is an algebraic truth rather than an accident.",
paper = "Fate02a.pdf"
}

```

— axiom.bib —

```
@inproceedings{Fate03a,
  author = "Fateman, Richard J.",
  title = "High-level proofs of mathematical programs using automatic
    differentiation, simplification, and some common sense",
  booktitle = "Proc. ISSAC 2003",
  pages = "88-94",
  year = "2003",
  isbn = "1-58113-641-2",
  abstract =
    "One problem in applying elementary methods to prove correctness of
    interesting scientific programs is the large discrepancy in level of
    discourse between low-level proof methods and the logic of scientific
    calculation, especially that used in a complex numerical program. The
    justification of an algorithm typically relies on algebra or analysis,
    but the correctness of the program requires that the arithmetic
    expressions are written correctly and that iterations converge to
    correct values in spite of truncation of infinite processes or series
    and the commission of numerical roundoff errors. We hope to help
    bridge this gap by showing how we can, in some cases, state a
    high-level requirement and by using a computer algebra system (CAS)
    demonstrate that a program satisfies that requirement. A CAS can
    contribute program manipulation, partial evaluation, simplification or
    other algorithmic methods. A novelty here is that we add to the usual
    list of techniques automatic differentiation, a method already widely
    used in optimization contexts where algorithms are differentiated. We
    sketch a proof of a numerical program to compute sine, and display a
    related approach to a version of a Bessel function algorithm for J0(x)
    based on a recurrence.",
  paper = "Fate03a.pdf"
}
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— axiom.bib —

```
@article{Frad08,
  author = "Frade, Maria Joao",
  title = "Calculus of Inductive Construction. Software Formal Verification",
  year = "2008",
  link = "\url{http://www4.di.uminho.pt/~jno/mfes/0809/SFV-CIC.pdf}",
  journal = "MFES",
  paper = "Frad08.pdf"
}
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— axiom.bib —

```
@misc{Geuv92,
  author = "Geuvers, Herman",
  title = "The Calculus of Constructions and Higher Order Logic",
  year = "1992",
  link = "\url{http://www.cs.ru.nl/~herman/PUBS/CC\_CHiso.pdf}",
  abstract =
    "The Calculus of Constructions (CC) is a typed lambda calculus for
    higher order intuitionistic logic: proofs of the higher order logic
    are interpreted as lambda terms and formulas as types. It is also the
    union of Girard's system  $\mathcal{F}_\omega$ , a higher order typed lambda
    calculus, and a first order dependent typed lambda calculus in the
    style of de Bruijn's Automath or Martin-Lof's intuitionistic theory
    of types. Using the impredicative coding of data types in  $\mathcal{F}_\omega$ ,
    the Calculus of Constructions thus becomes a higher order language for
    the typing of functional programs. We shall introduce and try to
    explain CC by exploiting especially the first point of view, by
    introducing a typed lambda calculus that faithfully represent higher
    order predicate logic (so for this system the Curry-Howard
    'formulas-as-types isomorphism' is really an isomorphism.) Then we
    discuss some propositions that are provable in CC but not in the
    higher order logic, showing that the formulas-as-types embedding of
    higher order predicate logic into CC is not an isomorphism. It is our
    intention that this chapter can be read without any specialist
    knowledge of higher order logic or higher order typed lambda calculi.",
  paper = "Geuv92.pdf"
}
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— axiom.bib —

```
@article{Geuv02,
  author = "Geuvers, Herman and Pollack, Randy and Wiedijk, Freek and
    Zwanenburg, Jan",
  title = "A Constructive Algebraic Hierarchy in Coq",
  year = "2002",
  journal = "Journal of Symbolic Computation",
  abstract =
    "We describe a framework of algebraic structures in the proof assistant
    Coq. We have developed this framework as part of the FTA project in
    Nijmegen, in which a constructive proof of the Fundamental Theorem of
```

Algebra has been formalized in Coq.

The algebraic hierarchy that is described here is both abstract and structured. Structures like groups and rings are port of it in an abstract way, defining e.g. a ring as a tuple consisting of a group, a binary operation and a constant that together satisfy the properties of a ring. In this way, a ring automatically inherits the group properties of the additive subgroup. The algebraic hierarchy is formalized in Coq by applying a combination of labeled record types and coercions. In the labeled record types of Coq, one can use `{\sl dependent types}`: the type of one label may depend on another label. This allows to give a type to a dependent-typed tuple like `\langle A, f, a \rangle`, where `A` is a set, `f` an operation on `A` and `a` an element of `A`. Coercions are functions that are used implicitly (they are inferred by the type checker) and allow, for example, to use the structure `\mathcal{A} := \langle A, f, a \rangle` as a synonym or the carrier set `A`, as is often done in mathematical practice. Apart from the inheritance and reuse of properties, the algebraic hierarchy has proven very useful for reusing notations.",
 paper = "Geuv02.pdf",
 keywords = "axiomref"
}

— axiom.bib —

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@misc{Gim16,
  author = "Gimenez, Eduardo and Casteran, Pierre",
  title = "A Tutorial on [Co-]Inductive Types in Coq",
  year = "2016",
  link = "\url{https://coq.inria.fr/distrib/current/files/RecTutorial.pdf}",
  abstract =
    "This document is an introduction to the definition and use of
    inductive and co-inductive types in the {\sl Coq} proof environment.
    It explains how types like natural numbers and infinite streams are
    defined in {\sl Coq}, and the kind of proof techniques that can be
    used to reason about them (case analysis, induction, inversion of
    predicates, co-induction, etc.) Each technique is illustrated
    through an executable and self-contained {\sl Coq} script.",
  paper = "Gim16.pdf"
}
```

— axiom.bib —

```

@article{Gott05,
  author = "Gottliebsen, Hanne and Kelsey, Tom and Martin, Ursula",
  title = "Hidden verification for computational mathematics",
  journal = "Journal of Symbolic Computation",
  volume = "39",
  number = "5",
  pages = "539-567",
  year = "2005",
  link = "\url{http://www.sciencedirect.com/science/article/pii/S0747717105000295}",
  abstract =
    "We present hidden verification as a means to make the power of
    computational logic available to users of computer algebra systems
    while shielding them from its complexity. We have implemented in PVS a
    library of facts about elementary and transcendental function, and
    automatic procedures to attempt proofs of continuity, convergence and
    differentiability for functions in this class. These are called
    directly from Maple by a simple pipe-lined interface. Hence we are
    able to support the analysis of differential equations in Maple by
    direct calls to PVS for: result refinement and verification, discharge
    of verification conditions, harnesses to ensure more reliable
    differential equation solvers, and verifiable look-up tables.",
  paper = "Gott05.pdf",
  keywords = "axiomref"
}

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@misc{Hard13,
  author = "Hardin, David S. and McClurg, Jedidiah R. and Davis, Jennifer A.",
  title = "Creating Formally Verified Components for Layered Assurance with an LLVM to ACL2 Tran",
  link = "\url{http://www.jrmccclurg.com/papers/law_2013_paper.pdf}",
  abstract = "
    This paper describes an effort to create a library of formally
    verified software component models from code that have been compiled
    using the Low-Level Virtual Machine (LLVM) intermediate form. The idea
    is to build a translator from LLVM to the applicative subset of Common
    Lisp accepted by the ACL2 theorem prover. They perform verification of
    the component model using ACL2's automated reasoning capabilities.",
  paper = "Hard13.pdf"
}

```

— axiom.bib —

```

@misc{Hard14,
  author = "Hardin, David S. and Davis, Jennifer A. and Greve, David A. and
           McClurg, Jedidiah R.",
  title = "Development of a Translator from LLVM to ACL2",
  link = "\url{http://arxiv.org/pdf/1406.1566}",
  abstract = "
    In our current work a library of formally verified software components
    is to be created, and assembled, using the Low-Level Virtual Machine
    (LLVM) intermediate form, into subsystems whose top-level assurance
    relies on the assurance of the individual components. We have thus
    undertaken a project to build a translator from LLVM to the
    applicative subset of Common Lisp accepted by the ACL2 theorem
    prover. Our translator produces executable ACL2 formal models,
    allowing us to both prove theorems about the translated models as well
    as validate those models by testing. The resulting models can be
    translated and certified without user intervention, even for code with
    loops, thanks to the use of the def::ung macro which allows us to
    defer the question of termination. Initial measurements of concrete
    execution for translated LLVM functions indicate that performance is
    nearly 2.4 million LLVM instructions per second on a typical laptop
    computer. In this paper we overview the translation process and
    illustrate the translator's capabilities by way of a concrete example,
    including both a functional correctness theorem as well as a
    validation test for that example.",
  paper = "Hard14.pdf"
}

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— axiom.bib —

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@inproceedings{Huet87,
  author = {Huet, G\'erard},
  title = "Induction Principles Formalized in the Calculus of Constructions",
  booktitle = "TAPSOFT 87",
  publisher = "Springer-Verlag",
  series = "LNCS 249",
  year = "1987",
  pages = "276-286",
  abstract =
    "The Calculus of Constructions is a higher-order formalism for writing
    constructive proofs in a natural deduction style, inspired from work
    by de Bruijn, Girard, Martin-Lof, and Scott. The calculus and its
    syntactic theory were presented in Coquand's thesis, and an
    implementation by the author was used to mechanically verify a
    substantial number of proofs demonstrating the power of expression of
    the formalism. The Calculus of Constructions is proposed as a
    foundation for the design of programming environments where programs

```

are developed consistently with formal specifications. The current paper shows how to define inductive concepts in the calculus.

A very general induction schema is obtained by postulating all elements of the type of interest to belong to the standard interpretation associated with a predicate map. This is similar to the treatment of D. Park, but the power of expression of the formalism permits a very direct treatment, in a language that is formalized enough to be actually implemented on a computer. Special instances of the induction schema specialize to Noetherian induction and Structural induction over any algebraic type. Computational Induction is treated in an axiomatization of Domain Theory in Constructions. It is argued that the resulting principle is more powerful than LCF's, since the restriction on admissibility is expressible in the object language.",
 paper = "Huet87.pdf"
 }

— axiom.bib —

```
@misc{Heut16,
  author = {Huet, G\'erard and Kahn, Gilles and Paulin-Mohring, Christine},
  title = "The COQ Proof Assistant. A Tutorial",
  year = "2016",
  link = "\url{https://coq.inria.fr/distrib/current/files/Tutorial.pdf}",
  paper = "Heut16.pdf"
}
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— axiom.bib —

```
@phdthesis{Jack95,
  author = "Jackson, Paul Bernard",
  title = "Enhancing the NUPRL Proof Development System and Applying it to
    Computational Abstract Algebra",
  school = "Cornell University",
  year = "1995",
  month = "1",
  abstract = "
    This thesis describes substantial enhancements that were made to the
    software tools in the Nuprl system that are used to interactively
    guide the production of formal proofs. Over 20,000 lines of code were
    written for these tools. Also, a corpus of formal mathematics was
    created that consists of roughly 500 definitions and 1300
```

theorems. Much of this material is of a foundational nature and supports all current work in Nuprl. This thesis concentrates on describing the half of this corpus that is concerned with abstract algebra and that covers topics central to the mathematics of the computations carried out by computer algebra systems.

The new proof tools include those that solve linear arithmetic problems, those that apply the properties of order relations, those that carry out inductive proof to support recursive definitions, and those that do sophisticated rewriting. The rewrite tools allow rewriting with relations of differing strengths and take care of selecting and applying appropriate congruence lemmas automatically. The rewrite relations can be order relations as well as equivalence relations. If they are order relations, appropriate monotonicity lemmas are selected.

These proof tools were heavily used throughout the work on computational algebra. Many examples are given that illustrate their operation and demonstrate their effectiveness.

The foundation for algebra introduced classes of monoids, groups, ring and modules, and included theories of order relations and permutations. Work on finite sets and multisets illustrates how a quotienting operation hides details of datatypes when reasoning about functional programs. Theories of summation operators were developed that drew indices from integer ranges, lists and multisets, and that summed over all the classes mentioned above. Elementary factorization theory was developed that characterized when cancellation monoids are factorial. An abstract data type for the operations of multivariate polynomial arithmetic was defined and the correctness of an implementation of these operations was verified. The implementation is similar to those found in current computer algebra systems.

This work was all done in Nuprl's constructive type theory. The thesis discusses the appropriateness of this foundation, and the extent to which the work relied on it.",

```
paper = "Jack95.pdf",
keyword = "axiomref"
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}
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```
@misc{Kauf98,
  author = "Kaufmann, Matt and Moore, J Strother",
  title = "A Precise Description of the ACL2 Logic",
  year = "1998",
```



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link = "\url{www.cs.utexas.edu/users/moore/publications/km97a.pdf}",
abstract = "The ACL2 logic is a first-order, essentially quantifier-free
logic of total recursive functions providing mathematical induction
and several extension principles, including symbol package definition
and recursive function definition. In this document we describe the
logic more precisely.",
paper = "km97a.pdf"
}

```

— axiom.bib —

```

@misc{Kenn13,
  author = "Kennedy, Andrew and Benton, Nick and Jensen, Jonas B. and
Dagand, Pierre-Evariste",
  title = "Coq: The world's best macro assembler?",
  year = "2013",
  link = "\url{http://research.microsoft.com/en-us/um/people/nick/coqasm.pdf}",
  abstract =
    "We describe a Coq formalization of a subset of the x86
architecture. One emphasis of the model is brevity: using dependent
types, type classes and notation we give the x86 semantics a makeover
that counters its reputation for baroque-ness. We model bits, bytes,
and memory concretely using functions that can be computed inside Coq
itself; concrete representations are mapped across to mathematical
objects in the SSREFLECT library (naturals, and integers modulo 2n)
to prove theorems. Finally, we use notation to support conventional
assembly code syntax inside Coq, including lexically-scoped
labels. Ordinary Coq definitions serve as a powerful ‘macro’ feature
for everything from simple conditionals and loops to stack-allocated
local variables and procedures with parameters. Assembly code can be
assembled within Coq, producing a sequence of hex bytes. The assembler
enjoys a correctness theorem relating machine code in memory to a
separation-logic formula suitable for program verification.",
  paper = "Kenn13.pdf"
}

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— axiom.bib —

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@phdthesis{Kreb15,
  author = "Krebbers, Robbert Jan",
  title = "The C standard formalized in Coq",
  school = "Radboud University",
}

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year = "2015",
file = "Kreb15.pdf",
link = "\url{http://robbertkrebbers.nl/thesis.html}"
}

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— axiom.bib —

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@article{Kupe14,
  author = "Kuper, Jan and Wester, Rinse",
  title = "N Queens on an FPGA: Mathematics, Programming, or Both?",
  year = "2014",
  link = "\url{http://doc.utwente.nl/94663/1/NQueensOnFPGA.pdf}",
  publisher = "Open Channel Publishing Ltd",
  journal = "Communicating Process Architectures 2014",
  abstract =
    "This paper presents a design methodology for deriving an FPGA
    implementation directly from a mathematical specification, thus
    avoiding the switch in semantic perspective as is present in widely
    applied methods which include an imperative implementation as an
    intermediate step.

    The first step in the method presented in this paper is to transform a
    mathematical specification into a Haskell program. The next step is to
    make repetition structures explicit by higher order functions, and
    after that rewrite the specification in the form of a Mealy
    Machine. Finally, adaptations have to be made in order to comply to
    the fixed nature of hardware. The result is then given to
    C$\lambda$aSH, a compiler which generates synthesizable VHDL from the
    resulting Haskell code. An advantage of the approach presented here is
    that in all phases of the process the design can be directly simulated
    by executing the defining code in a standard Haskell environment.

    To illustrate the design process, the N queens problem is chosen as a
    running example."
}

```

— axiom.bib —

```

@article{Kell12,
  author = "Keller, Chantal and Lasson, Marc",
  title = "The Refined Calculus of Inductive Construction: Parametricity and
    Abstraction",

```

```

journal = "arXiv",
year = "2012",
link = "\url{http://arxiv.org/pdf/1211.6341v1.pdf}",
abstract =
  "We present a refinement of the Calculus of Inductive Constructions in
  which one can easily define a notion of relational parametricity. It
  provides a new way to automate proofs in an interactive theorem prover
  like Coq.",
paper = "Kell12.pdf"
}

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— axiom.bib —

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@book{Lamp02,
  author = "Lamport, Leslie",
  title = "Specifying Systems",
  year = "2002",
  link = "\url{http://research.microsoft.com/en-us/um/people/lamport/tla/book-02-08-08.pdf}",
  publisher = "Addison-Wesley",
  isbn = "0-321-14306-X",
  paper = "Lamp02.pdf"
}

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— axiom.bib —

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@misc{Lamp13,
  author = "Lamport, Leslie",
  title = "Errata to Specifying Systems",
  year = "2013",
  link = "\url{http://research.microsoft.com/en-us/um/people/lamport/tla/errata-1.pdf}",
  publisher = "Microsoft",
  abstract = "
    These are all the errors and omissions to the first printing (July
    2002) of the book {\sl Specifying Systems} reported as of 29 October
    2013. Positions in the book are indicated by page and line number,
    where the top line of a page is number 1 and the bottom line is number
    $-1$. A running head and a page number are not considered to be lines,
    but all other lines are. Please report any additional errors to the
    author, whose email address is posted on {\tt http://lamport.org}. The
    first person to report an error will be acknowledged in any revised
    edition.",
  paper = "Lamp13.pdf"
}

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}

— axiom.bib —

```
@misc{Lamp14,
  author = "Lamport, Leslie",
  title = "How to Write a  $21^{\text{st}}$  Century Proof",
  year = "2014",
  link = "\url{http://research.microsoft.com/en-us/um/people/lamport/pubs/paper.pdf}",
  publisher = "Microsoft",
  abstract = "
    A method of writing proofs is described that makes it harder to prove
    things that are not true. The method, based on hierarchical
    structuring, is simple and practical. The author's twenty years of
    experience writing such proofs is discussed.",
  paper = "Lamp14.pdf"
}
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— axiom.bib —

```
@misc{Lamp14a,
  author = "Lamport, Leslie",
  title = "Talk: How to Write a  $21^{\text{st}}$  Century Proof",
  year = "2014",
  link = "\url{http://hits.mediasite.com/mediasite/Play/29d825439b3c49f088d35555426fbdf81d}",
  comment = "2nd Heidelberg Laureate Forum Lecture Tuesday Sep 23, 2014"
}
```

— axiom.bib —

```
@book{Mahb16,
  author = "Mahboubi, Assia and Tassi, Enrico and Bertot, Yves and
    Gonthier, Georges",
  title = "Mathematical Components",
  year = "2016",
  publisher = "math-comp.github.io/mcb",
  link = "\url{https://math-comp.github.io/mcb/book.pdf}",
}
```

```

abstract =
  "{\sl Mathematical Components} is the name of a library of formalized
  mathematic for the COQ system. It covers a verietiy of topics, from the
  theory of basic data structures (e.g. numbers, lists, finite sets) to
  advanced results in various flavors of algebra. This library
  constitutes the infrastructure for the machine-checked proofs of the
  Four Color Theorem and the Odd Order Theorem.

  The reason of existence of this books is to break down the barriers to
  entry. While there are several books around covering the usage of the
  COQ system and the theory it is based on, the Mathematical Components
  library is build in an unconventional way. As a consequence, this book
  provides a non-standard presentation of COQ, putting upfront the
  formalization choices and the proof style that are the pillars of the
  library.

  This book targets two classes of public. On one hand, newcomers, even
  the more mathematically inclined ones, find a soft introduction to the
  programming language of COQ, Gallina, and the Ssreflect proof
  language. On the other hand accustomed COQ users find a substantial
  account of the formalization style that made the Mathematical
  Components library possible.

  By no means does this book pretend to be a complete description of COQ
  or Ssreflect: both tools already come with a comprehensive user
  manual. In the course of the book, the reader is nevertheless invited
  to experiment with a large library of formalized concepts and she is
  given as soon as possible sufficient tools to prove non-trivial
  mathematical results by reusing parts of the library. By the end of
  the first part, the reader has learnt how to prove formally the
  infinitude of prime numbers, or the correctness of the Euclidean's
  division algorithm, in a few lines of proof text.",
paper = "Mahb16.pdf"
}

```

— axiom.bib —

```

@article{Mart96,
  author = {Martin-L\"of, Per},
  title = "On the Meaning of the Logical Constants and the Justifications
    of the Logical Laws",
  year = "1996",
  journal = "Nordic Journal of Philosophical Logic",
  volume = "1",
  number = "1",
  pages = "11-60",

```

```

abstract =
  "The following three lectures were given in the form of a short course
  at the meeting Teoria della Dimostrazione e Filosofia della Logica,
  organized in Siena, 6-9 April 1983, by the Scuola di Specializzazione
  in Logica Matematica of the Universit  degli Studi di Siena. I am
  very grateful to Giovanni Sambin and Aldo Ursini of that school, not
  only for recording the lectures on tape, but, above all, for
  transcribing the tapes produced by the recorder: no machine could have
  done that work. This written version of the lectures is based on their
  transcription. The changes that I have been forced to make have mostly
  been of a stylistic nature, except at one point. In the second
  lecture, as I actually gave it, the order of conceptual priority
  between the notions of proof and immediate inference was wrong. Since
  I discovered my mistake later the same month as the meeting was held,
  I thought it better to let the written text diverge from the oral
  presentation rather than possibly confusing others by letting the
  mistake remain. The oral origin of these lectures is the source of the
  many redundancies of the written text. It is also my sole excuse for
  the lack of detailed references.",
paper = "Mart96.pdf"
}

```

— axiom.bib —

```

@misc{Mart97,
  author = "Martin, Ursula and Shand, D",
  title = "Investigating some Embedded Verification Techniques for
  Computer Algebra Systems",
  link = "\url{http://www.risc.jku.at/conferences/Theorema/papers/shand.ps.gz}",
  abstract = "
    This paper reports some preliminary ideas on a collaborative project
    between St. Andrews University in the UK and NAG Ltd. The project aims
    to use embedded verification techniques to improve the reliability and
    mathematical soundness of computer algebra systems. We give some
    history of attempts to integrate computer algebra systems and
    automated theorem provers and discuss possible advantages and
    disadvantages of these approaches. We also discuss some possible case
    studies.",
  paper = "Mart97.ps"
}

```

— axiom.bib —

```

@book{Maso86,
  author = "Mason, Ian A.",
  title = "The Semantics of Destructive Lisp",
  publisher = "Center for the Study of Language and Information",
  year = "1986",
  isbn = "0-937073-06-7",
  abstract = "
    Our basic premise is that the ability to construct and modify programs
    will not improve without a new and comprehensive look at the entire
    programming process. Past theoretical research, say, in the logic of
    programs, has tended to focus on methods for reasoning about
    individual programs; little has been done, it seems to us, to develop
    a sound understanding of the process of programming -- the process by
    which programs evolve in concept and in practice. At present, we lack
    the means to describe the techniques of program construction and
    improvement in ways that properly link verification, documentation and
    adaptability."
}

```

— axiom.bib —

```

@misc{Mohr14,
  author = "Mohring-Paulin, Christine",
  title = "Introduction to the Calculus of Inductive Constructions",
  year = "2014",
  link = "\url{https://hal.inria.fr/hal-01094195/file/CIC.pdf}",
  paper = "Mohr14.pdf"
}

```

— axiom.bib —

```

@misc{Mour16,
  author = "de Moura, Leonardo and Avigad, Jeremy and Kong, Soonho and
    Roux, Cody",
  title = "Elaboration in Dependent Type Theory",
  year = "2016",
  link = "\url{http://leodemoura.github.io/files/elaboration.pdf}",
  abstract =
    "We describe the elaboration algorithm that is used in {\sl Lean}, a
    new interactive theorem prover based on dependent type theory. To be
    practical, interactive theorem provers must provide mechanisms to
    resolve ambiguities and infer implicit type information, thereby

```

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supporting convenient input of expressions and proofs. Lean's
elaborator supports higher-order unification, ad-hoc overloading,
insertion of coercions, type class inference, the use of tactics, and
the computational reduction of terms. The interactions between these
components are subtle and complex, and Lean's elaborator has been
carefully designed to balance efficiency and usability.",
paper = "Mour16.pdf"
}

```

— ignore —

```

\bibitem[Newcombe 13]{Newc13} Newcombe, Chris; Rath, Tim; Zhang, Fan;
Munteanu, Bogdan; Brooker, Marc; Deardeuff, Michael
title = "Use of Formal Methods at Amazon Web Services",
link = "\url{http://research.microsoft.com/en-us/um/people/lamport/tla/formal-methods-amazon.pdf}",
abstract = "

```

In order to find subtle bugs in a system design, it is necessary to have a precise description of that design. There are at least two major benefits to writing a precise design; the author is forced to think more clearly, which helps eliminate ‘‘plausible hand-waving’’, and tools can be applied to check for errors in the design, even while it is being written. In contrast, conventional design documents consist of prose, static diagrams, and perhaps pseudo-code in an ad hoc untestable language. Such descriptions are far from precise; they are often ambiguous, or omit critical aspects such as partial failure or the granularity of concurrency (i.e. which constructs are assumed to be atomic). At the other end of the spectrum, the final executable code is unambiguous, but contains an overwhelming amount of detail. We needed to be able to capture the essence of a design in a few hundred lines of precise description. As our designs are unavoidably complex, we need a highly-expressive language, far above the level of code, but with precise semantics. That expressivity must cover real-world concurrency and fault-tolerance. And, as we wish to build services quickly, we wanted a language that is simple to learn and apply, avoiding esoteric concepts. We also very much wanted an existing ecosystem of tools. We found what we were looking for in TLA+, a formal specification language."

— axiom.bib —

```

@inproceedings{Pare93,
author = "Parent, Catherine",

```



```

title = "Developing Certified Programs in the System Coq: The Program
        Tactic",
booktitle = "Proc. Int. Workshop on Types for Proofs and Programs",
publisher = "Springer-Verlag",
isbn = "3-540-58085-9",
pages = "291-312",
year = "1993",
abstract =
  "The system {\sl Coq} is an environment for proof development based on
  the Calculus of Constructions extended by inductive definitions. The
  specification of a program can be represented by a logical formula and
  the program itself can be extracted from the constructive proof of the
  specification. In this paper, we look at the possibility of inverting
  the specification and a program, builds the logical condition to be
  verified in order to obtain a correctness proof of the program. We
  build a proof of the specification from the program from which the
  program can be extracted. Since some information cannot automatically
  be inferred, we show how to annotate the program by specifying some of
  its parts in order to guide the search for the proof.",
paper = "Pare93.ps"
}

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— axiom.bib —

```

@techreport{Pare94,
  author = "Parent, Catherine",
  title = "Synthesizing proofs from programs in the Calculus of Inductive
          Constructions",
  year = "1994",
  institution = {Ecole Normale Sup\'erieure de Lyon},
  abstract =
    "In type theory, a proof can be represented as a typed  $\lambda$ -term.
    There exist methods to mark logical parts in proofs and extract their
    algorithmic contents. The result is a correct program with respect to
    a specification. This paper focuses on the inverse problem: how to
    generate a proof from its specification. The framework is the Calculus
    of Inductive Constructions. A notion of coherence is introduced between
    a specification and a program containing types but no logical proofs.
    This notion is based on the definition of an extraction function called
    the weak extraction. Such a program can give a method to reconstruct a
    set of logical properties needed to have a proof of the initial
    specification. This can be seen either as a method of proving programs
    or as a method of synthetically describing proofs.",
  paper = "Pare94.pdf"
}

```

— axiom.bib —

```
@misc{Pare96,
  author = "Parent-Vigouroux, Catherine",
  title = "Natural proofs versus programs optimization in the
          Calculus of Inductive Constructions",
  year = "1996",
  abstract =
    "This paper presents how to automatically prove that an 'optimized'
    program is correct with respect to a set of given properties that is a
    specification. Proofs of specifications contain logical and
    computational parts. Programs can be seen as computational parts of
    proofs. They can thus be extracted from proofs and be certified to be
    correct. The inverse problem can be solved: it is possible to
    reconstruct proof obligations from a program and its specification.
    The framework is a type theory where a proof can be represented as a
    typed  $\lambda$ -term and, particularly, the Calculus of Inductive
    Constructions. This paper shows how programs can be simplified in
    order to be written in a much closer way to the ML one's. Indeed,
    proofs structures are often much more heavy than program structures.
    The problem is consequently to consider natural programs (in a ML sense)
    and see how to retrieve natural structures of proofs from them.",
  paper = "Pare96.pdf"
}
```

— axiom.bib —

```
@article{Pare97,
  author = "Parent-Vigouroux, Catherine",
  title = "Verifying programs in the Calculus of Inductive Constructions",
  year = "1997",
  journal = "Formal Aspects of Computing",
  volume = "9",
  number = "5",
  pages = "484-517",
  abstract =
    "This paper deals with a particular approach to the verification of
    functional programs. A specification of a program can be represented
    by a logical formula. In a constructive framework, developing a program
    then corresponds to proving this formula. Given a specification and a
    program, we focus on reconstructing a proof of the specification whose
    algorithmic contents corresponds to the given program. The best we can
    hope is to generate proof obligations on atomic parts of the program"
```

corresponding to logical properties to be verified. First, this paper studies a weak extraction of a program from a proof that keeps track of intermediate specifications. From such a program, we prove the determinism of retrieving proof obligations. Then, heuristic methods are proposed for retrieving the proof from a natural program containing only partial annotations. Finally, the implementation of this method as a tactic of the `{\sl Coq}` proof assistant is presented.",
 paper = "Pare97.pdf"
}

— axiom.bib —

@techreport{Pfen89,
 author = "Pfenning, Frank and Paulin-Mohring, Christine",
 title = "Inductively Defined Types in the Calculus of Constructions",
 institution = "Carnegie-Mellon University",
 year = "1989",
 number = "CMU-CS-89-209",
 link = "\url{http://repository.cmu.edu/cgi/viewcontent.cgi?article=2907&context=compsci}",
 abstract =
 "We define the notion of an `{\sl inductively defined type}` in the Calculus of Constructions and show how inductively defined types can be represented by closed types. We show that all primitive recursive functional over these inductively defined types are also representable. This generalizes work by Bohm and Berarducci on synthesis of functions on term algebras in the second-order polymorphic λ -calculus (F_2). We give several applications of this generalization, including a representation of F_2 -programs in F_3 , along with a definition of functions `{\bf reify}`, `{\bf reflect}`, and `{\bf eval}` for F_2 in F_3 . We also show how to define induction over inductively defined types and sketch some results that show that the extension of the Calculus of Construction by induction principles does not alter the set of functions in its computational fragment, F_ω . This is because a proof by induction can be `{\bf realized}` by primitive recursion, which is already definable in F_ω .",
 paper = "Pfen89.pdf"
}

— axiom.bib —

@book{Pier00,

```

author = "Pierce, Benjamin C.",
title = "Type Systems for Programming Languages",
year = "2000",
publisher = "MIT Press",
link = "\url{http://ropas.snu.ac.kr/~kwang/S20/pierce\_book.pdf}",
paper = "Pier00.pdf"
}

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— axiom.bib —

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@misc{Poll98,
  author = "Poll, Erik and Thompson, Simon",
  title = "Adding the axioms to Axiom. Toward a system of automated
          reasoning in Aldor",
  year = "1998",
  link = "\url{http://citeseerx.ist.psu.edu/viewdoc/download?doi=10.1.1.7.1457}",
  abstract = "
    This paper examines the proposal of using the type system of Axiom to
    represent a logic, and thus to use the constructions of Axiom to
    handle the logic and represent proofs and propositions, in the same
    way as is done in theorem provers based on type theory such as Nuprl
    or Coq.

    The paper shows an interesting way to decorate Axiom with pre- and
    post-conditions.

    The Curry-Howard correspondence used is
    \begin{tabular}{lcl}
    PROGRAMMING          & & & LOGIC\\
    Type                 & & & Formula\\
    Program              & & & Proof\\
    Product/record type  & & (...) & Conjunction\\
    Sum/union type       & & \vee   & Disjunction\\
    Function type        & & ->    & Implication\\
    Dependent function type & (x:A) -> B(x) & & Universal quantifier\\
    Dependent product type & (x:A,B(x))   & & Existential quantifier\\
    Empty type           & Exit         & & Contradictory proposition\\
    One element type     & Triv         & & True proposition\\
    \end{tabular}",
  paper = "Poll98.pdf",
  keywords = "axiomref"
}

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— axiom.bib —

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@misc{Poll199,
  author = "Poll, Erik",
  title = "The Type System of Axiom",
  year = "1999",
  link = "\url{http://www.cs.ru.nl/E.Poll/talks/axiom.pdf}",
  abstract =
    "This is a slide deck from a talk on the correspondence between
    Axiom/Aldor types and Logic.",
  paper = "Poll199.pdf",
  keywords = "axiomref"
}
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— axiom.bib —

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@misc{Poll199a,
  author = "Poll, Erik and Thompson, Simon",
  title = "The Type System of Aldor",
  link = "\url{http://www.cs.kent.ac.uk/pubs/1999/874/content.ps}",
  abstract =
    "This paper gives a formal description of -- at least a part of --
    the type system of Aldor, the extension language of the Axiom.
    In the process of doing this a critique of the design of the system
    emerges.",
  paper = "Poll199a.pdf",
  keywords = "axiomref"
}
```

— axiom.bib —

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@misc{Poll100,
  author = "Poll, Erik and Thompson, Simon",
  title = "Integrating Computer Algebra and Reasoning through the Type
    System of Aldor",
  year = "2000",
  abstract =
    "A number of combinations of reasoning and computer algebra systems
    have been proposed; in this paper we describe another, namely a way to
    incorporate a logic in the computer algebra system Axiom. We examine
```

```

the type system of Aldor -- the Axiom Library Compiler -- and show
that with some modifications we can use the dependent types of the
system to model a logic, under the Curry-Howard isomorphism. We give
a number of example applications of the logic we construct and explain
a prototype implementation of a modified type-checking system written
in Haskell.",
paper = "Poll100.pdf",
keywords = "axiomref"
}

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— axiom.bib —

```

@InProceedings{Poll100a,
  author = "Poll, Erik and Thompson, Simon",
  title = "Integrating Computer Algebra and Reasoning through the Type
    System of Aldor",
  booktitle = "Frontiers of Combining Systems",
  series = "Lecture Notes in Artificial Intelligence",
  year = "2000",
  isbn = "3-540-67281-8",
  location = "Nancy, France",
  pages = "136-150",
  keywords = "axiomref"
}

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— axiom.bib —

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@article{Prev02,
  author = "Prevosto, Virgile and Doligez, Damien",
  title = "Algorithms and proofs inheritance in the FOC language",
  journal = "J. Autom. Reasoning",
  volume = "29",
  number = "3-4",
  year = "2002",
  pages = "337-363",
  abstract =
    "In this paper, we present the FOC language, dedicated to the
    development of certified computer algebra libraries (that is sets of
    programs). These libraries are based on a hierarchy of implementations
    of mathematical structures. After presenting the core set of features
    of our language, we describe the static analyses, which reject
    inconsistent programs. We then show how we translate FOC definitions

```

```

    into OCAML and COQ, our target languages for the computational part
    and the proof checking, respectively.",
    paper = "Prev02.pdf",
    keywords = "axiomref"
}

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— axiom.bib —

```

@misc{Robe15,
  author = "Roberts, Siobhan",
  title = "In Mathematics, Mistakes Aren't What They Used To Be",
  year = 2015,
  link = "\url{http://nautil.us/issue/24/error/In-mathematics-mistakes-arent-what-they-used-to-be}"
}

```

— axiom.bib —

```

@misc{Lamp16,
  author = "Lamport, Leslie",
  title = "TLA+ Proof System",
  year = "2016",
  link = "\url{https://tla.msr-inria.inria.fr/tlaps/content/Documentation/Tutorial/The_example.html}"
  abstract = "Demonstration of Euclid Algorithm Proof in TLA+"
}

```

— axiom.bib —

```

@article{Mahb06,
  author = "Mahboubi, Assia",
  title = "Proving Formally the Implementation of an Efficient gcd
    Algorithm for Polynomials",
  journal = "Lecture Notes in Computer Science",
  volume = "4130",
  year = "2006",
  pages = "438-452",
  abstract = "
    We describe here a formal proof in the Coq system of the structure

```

```

theorem for subresultants which allows to prove formally the
correctness of our implementation of the subresultants algorithm.
Up to our knowledge it is the first mechanized proof of this result.",
paper = "Mahb06.pdf"
}

```

— axiom.bib —

```

@mastersthesis{Mort10,
  author = {M\ortbert, Anders},
  title = "Constructive Algebra in Functional Programming and Type Theory",
  school = "University of Gothenburg, Department of Computer Science",
  year = "2010",
  month = "5",
  file = "Mort10.pdf",
  abstract =
    "This thesis considers abstract algebra from a constructive point
    of view. The central concept of study is coherent rings -- algebraic
    structures in which it is possible to solve homogeneous systems of
    linear equations. Three different algebraic theories are considered;
    Bezout domains, Prufer domains and polynomial rings. The first two
    of these are non-Noetherian analogues of classical notions. The
    polynomial rings are presented from a constructive point of view with a
    treatment of Groebner bases. The goal of the thesis is to study the
    proofs that these theories are coherent and explore how the proofs can
    be implemented in functional programming and type theory."
}

```

— axiom.bib —

```

@article{Nguy16,
  author = "Nguyen, Phuc C. and Tobin-Hochstadt, Sam and van Horn, David",
  title = "Higher-order symbolic execution for contract verification and
    refutation",
  journal = "arXiv",
  link = "\url{http://arxiv.org/pdf/1507.04817v2.pdf}",
  year = "2016",
  month = "February",
  abstract =
    "We present a new approach to automated reasoning about higher-order
    programs by endowing symbolic execution with a notion of higher-order,
    symbolic values."
}

```


To validate our approach, we use it to develop and evaluate a system for verifying and refuting behavioral software contracts of components in a functional language, which we call {\sl soft contract verification}. In doing so, we discover a mutually beneficial relation between behavioral contracts and higher-order symbolic execution. Contracts aid symbolic execution by providing a rich language of specifications that can serve as the basis of symbolic higher-order values; the theory of blame enables modular verification and leads to the theorem that {\sl verified compnents can't be blamed}; and the run-time monitoring of contracts enables {\sl soft} verification whereby verified and unverified components can safely interact and verification is not an all-or-nothing proposition. Conversely, symbolic execution aids contracts by providing compile-time verification which increases assurance and enables optimizations; automated test-case generation for contracts with counter-examples; and engendering a virtuous cycle between verification and the gradual spread of contracts.

Our system uses higher-order symbolic execution, leveraging contracts as a source of symbolic values including unknown behavioral values, and employs an updatable heap of contract invariants to reason about flow-sensitive facts. Whenever a contract is refuted, it reports a concrete {\sl counterexample} reproducing the error, which may involve solving for an unknown function. The approach is able to analyze first-class contracts, recursive data structures, unknown functions, and control-flow-sensitive refinement of values, which are all idiomatic in dynamic languages. It makes effective use of an off-the-shelf solver to decide problems without heavy encodings. Our counterexample search is sound and relatively complete with respect to a first-order solver for base type values. Therefore, it can form the basis of automated verification and bug-finding tools for higher-order programs. The approach is competitive with a wide range of existing tools -- including type systems, flow analyzers, and model checkers -- on their own benchmarks. We have built a tool which analyzes programs written in Racket, and report on its effectiveness in verifying and refuting contracts."

```
paper = "Nguy16.pdf"
}
```

— ignore —

```
@misc{OCon15,
  author = {O'Connor, Liam},
  title = "Write Your Compiler by Proving It Correct",
  year = "2015",
```

```

link = "\url{http://liamoc.net/posts/2015-08-23-verified-compiler.html}",
abstract =
  "Recently my research has been centered around the development of a
  self-certifying compiler for a functional language with linear types
  called Cogent (see O'Connor et al. [2016]). The compiler works by
  emitting, along with generated low-level code, a proof in Isabelle/HOL
  (see Nipkow et al. [2002]) that the generated code is a refinement of
  the original program, expressed via a simple functional semantics in HOL.

  As dependent types unify for us the language of code and proof, my
  current endeavour has been to explore how such a compiler would look
  if it were implemented and verified in a dependently typed programming
  language instead. In this post, I implement and verify a toy compiler
  for a language of arithmetic expressions and variables to an idealised
  assembly language for a virtual stack machine, and explain some of the
  useful features that dependent types give us for writing verified
  compilers."}

```

— axiom.bib —

```

@misc{Pier15,
  author = {Pierce, Benjamin C. and Casinghino, Chris and Gaboardi, Marco and
    Greenberg, Michael and Hritcu, Catalin and Sjoberg, Vilhelm and
    Yorgey, Brent},
  title = "Software Foundations",
  year = "2015",
  file = "Pier15.tgz",
  abstract =
    "This electronic book is a course on Software Foundations, the
    mathematical underpinnings of reliable software. Topics include basic
    concepts of logic, computer-assisted theorem proving, the Coq proof
    assistant, functional programming, operational semantics, Hoare logic,
    and static type systems. The exposition is intended for a broad range
    of readers, from advanced undergraduates to PhD students and
    researchers. No specific background in logic or programming languages
    is assumed, though a degree of mathematical maturity will be helpful.

    The principal novelty of the course is that it is one hundred per cent
    formalized and machine-checked: the entire text is literally a script
    for Coq. It is intended to be read alongside an interactive session
    with Coq. All the details in the text are fully formalized in Coq, and
    the exercises are designed to be worked using Coq.

    The files are organized into a sequence of core chapters, covering
    about one semester's worth of material and organized into a coherent
    linear narrative, plus a number of appendices covering additional

```

```

    topics. All the core chapters are suitable for both upper-level
    undergraduate and graduate students."
}

```

— axiom.bib —

```

@article{Rudn01,
  author = "Rudnicki, Piotr and Schwarzweller, Christoph and
    Trybulec, Andrzej",
  title = "Commutative algebra in the Mizar system",
  journal = "J. Symb. Comput.",
  volume = "32",
  number = "1-2",
  pages = "143-169",
  year = "2001",
  link = "\url{https://inf.ug.edu.pl/~schwarz/papers/jsc01.pdf}",
  abstract =
    "We report on the development of algebra in the Mizar system. This
    includes the construction of formal multivariate power series and
    polynomials as well as the definition of ideals up to a proof of the
    Hilbert basis theorem. We present how the algebraic structures are
    handled and how we inherited the past developments from the Mizar
    Mathematical Library (MML). The MML evolves and past contributions are
    revised and generalized. Our work on formal power series caused a
    number of such revisions. It seems that revising past developments
    with an intent to generalize them is a necessity when building a
    database of formalized mathematics. This poses a question: how much
    generalization is best?",
  paper = "Rudn01.pdf",
  keywords = "axiomref"
}

```

— axiom.bib —

```

@phdthesis{Schw97,
  author = "Schwarzweller, Christoph",
  title = "MIZAR verification of generic algebraic algorithms",
  school = "University of Tübingen",
  year = "1997",
  abstract =
    "Although generic programming founds more and more attention
    nowadays generic programming languages as well as generic libraries

```

exist there are hardly approaches for the verification of generic algorithms or generic libraries. This thesis deals with generic algorithms in the field of computer algebra. We propose the Mizar system as a theorem prover capable of verifying generic algorithms on an appropriate abstract level. The main advantage of the MIZAR theorem prover is its special input language that enables textbook style presentation of proofs. For generic versions of Brown/Henrici addition and of Euclidean algorithm we give complete correctness proofs written in the MIZAR language.

Moreover, we do not only prove algorithms correct in the usual sense. In addition we show how to check, using the MIZAR system, that a generic algebraic algorithm is correctly instantiated with a particular domain. Answering this question that especially arises if one wants to implement generic programming languages, in the field of computer algebra requires nontrivial mathematical knowledge.

To build a verification system using the MIZAR theorem prover, we also implemented a generator which almost automatically computes for a given algorithm a set of theorems that imply the correctness of this algorithm."

```
paper = "Schw97.pdf",
keywords = "axiomref"
}
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— axiom.bib —

```
@misc{Seli13,
  author = "Selinger, Peter",
  title = "Lecture Notes on the Lambda Calculus",
  year = "2013",
  link = "\url{https://www.irif.fr/~mellies/mpri/mpri-ens/biblio/Selinger-Lambda-Calculus-Notes.pdf}",
  abstract =
    "This is a set of lecture notes that developed out of courses on the
    lambda calculus that I taught at the University of Ottawa in 2001 and
    at Dalhousie University in 2007 and 2013. Topics covered in these
    notes include the untyped lambda calculus, the Church-Rosser theorem,
    combinatory algebras, the simply-typed lambda calculus, the
    Curry-Howard isomorphism, weak and strong normalization, polymorphism,
    type inference, denotational semantics, complete partial orders, and
    the language PCF.",
  paper = "Seli13.pdf"
}
```

— axiom.bib —

```
@book{Simm00,
  author = "Simmons, Harold",
  title = "Derivation and Computation: Taking the Curry-Howard correspondence
    seriously",
  year = "2000",
  publisher = "Cambridge University Press",
  isbn = "0-521-77173-0"
}
```

— axiom.bib —

```
@misc{Stac17,
  author = "csttheory.stackexchange.com",
  title = "Why does Coq have Prop?",
  link = "\url{http://csttheory.stackexchange.com/questions/21836/why-does-coq-have-prop/21878\#21878}",
  year = "2017",
}
```

— axiom.bib —

```
@article{Ther01,
  author = "Th\'ery, Laurent",
  title = "A Machine-Checked Implementation of Buchberger's Algorithm",
  journal = "Journal of Automated Reasoning",
  volume = "26",
  year = "2001",
  pages = "107-137",
  abstract = "We present an implementation of Buchberger's algorithm that
    has been proved correct within the proof assistant Coq. The
    implementation contains the basic algorithm plus two standard
    optimizations.",
  paper = "Ther01.pdf"
}
```

— axiom.bib —

```
@misc{Troe97,
  author = "Troelstra, A.S.",
  title = "From constructivism to computer science",
  volume = "211",
  year = "1999",
  pages = "232-252",
  abstract =
    "My field is mathematical logic, with a special interest in
    constructivism, and I would not dare to call myself a computer
    scientist. But some computer scientists regard my work as a
    contribution to their field; and in this text I shall try to explain
    how this is possible, by taking a look at the history of ideas.

    I want to describe how two interrelated ideas, connected with the
    constructivistic trend in the foundations of mathematics, developed
    within mathematical logic and ultimately diffused into computer
    science.

    It will be seen that this development has not been a quite
    straightforward one. In the history of ideas it often looks as if a
    certain idea has to be discovered several times, by different people,
    before it really enters in the 'consciousness' of science",
  paper = "Troe99.pdf"
}
```

— axiom.bib —

```
@misc{Wadl14,
  author = "Wadler, Philip",
  title = "Propositions as Types",
  year = "2014",
  link = "\url{http://homepages.inf.ed.ac.uk/wadler/papers/propositions-as-types/propositions-as-types.pdf}",
  paper = "Wadl14.pdf"
}
```

— axiom.bib —

```
@misc{Warn17,
  author = "Warner, Evan",
  title = "Splash Talk: The Foundational Crisis of Mathematics",
  year = "2017",
  link = "\url{web.stanford.edu/~ebwarner/SplashTalk.pdf}",
}
```

```

abstract =
  "This class will cover some of the mathematics, history, and
  philosophy of the co-called {\sl foundational crisis in mathematics}.
  Broadly speaking, mathematics in the late nineteenth and early
  twentieth centuries was marked by an increased awareness of
  ‘‘foundational issues,’’ prompted by a number of problems in the
  practice of mathematics that had accumulated over the years. We will
  discuss a few examples of some of these problems, and then discuss the
  three major schools of thought that emerged to deal with them and
  provide a coherent philosophical and methodological underpinning for
  mathematics.",
paper = "Warn17.pdf"
}

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— axiom.bib —

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@misc{Wiki17,
  author = "Wikipedia",
  title = "Calculus of constructions",
  year = "2017",
  link = "\url{https://en.wikipedia.org/wiki/Calculus\_of\_constructions}"
}

```

— axiom.bib —

```

@misc{WikiED,
  author = "Wikipedia",
  title = "Euclidean Domain",
  year = "2017",
  link = "\url{https://en.wikipedia.org/wiki/Euclidean\_domain}"
}

```

— axiom.bib —

```

@book{Wins93,
  author = "Winskel, Glynn",
  title = "The Formal Semantics of Programming Languages",

```

```

    isbn = "978-0262731034",
    year = "1993",
    publisher = "MIT"
}

```

— axiom.bib —

```

@book{Yasu71,
  author = "Yasuhara, Ann",
  title = "Recursive Function Theory and Logic",
  year = "1971",
  publisher = "Academic Press",
  isbn = "0-12-768950-8"
}

```

1.19 Interval Arithmetic

— axiom.bib —

```

@misc{Boeh86,
  author = "Boehm, Hans-J. and Cartwright, Robert and Riggles, Mark and
    O'Donnell, Michael J.",
  title = "Exact Real Arithmetic: A Case Study in Higher Order Programming",
  year = "1986",
  link = "\url{http://dev.acm.org/pubs/citations/proceedings/lfp/319838/p162-boehm}",
  abstract =
    "Two methods for implementing {\sl exact} real arithmetic are explored.
    One method is based on formulating real numbers as functions that map
    rational tolerances to rational approximations. This approach, which
    was developed by constructive mathematicians as a concrete
    formalization of the real numbers, has lead to a surpris- ingly
    successful implementation. The second method formulates real numbers
    as potentially infinite sequences of digits, evaluated on demand.
    This approach has frequently been advocated by proponents of lazy
    functional languages in the computer science community. Ironically,
    it leads to much less satisfactory implementations. We discuss the
    theoretical problems involved in both methods, give algorithms for the
    basic arithmetic operations, and give an empirical comparison of the
    two techniques. We conclude with some general observations about the
    lazy evaluation paradigm and its implementation."

```



```

    paper = "Boeh86.pdf"
}

```

— ignore —

```

\bibitem[Briggs 04]{Bri04} Briggs, Keith
  title = "Exact real arithmetic",
  link = "\url{http://keithbriggs.info/documents/xr-kent-talk-pp.pdf}",
  paper = "Bri04.pdf"

```

— ignore —

```

\bibitem[Fateman 94]{Fat94} Fateman, Richard J.; Yan, Tak W.
  title = "Computation with the Extended Rational Numbers and an Application to Interval Arithmetic",
  link = "\url{http://www.cs.berkeley.edu/~fateman/papers/extrat.pdf}",
  abstract = "
    Programming languages such as Common Lisp, and virtually every
    computer algebra system (CAS), support exact arbitrary-precision
    integer arithmetic as well as exact rational number computation.
    Several CAS include interval arithmetic directly, but not in the
    extended form indicated here. We explain why changes to the usual
    rational number system to include infinity and ‘not-a-number’ may be
    useful, especially to support robust interval computation. We describe
    techniques for implementing these changes.",
  paper = "Fat94.pdf"

```

— axiom.bib —

```

@misc{Gust16,
  author = "Gustafson, John",
  title = "A Radical Approach to Computation with Real Numbers",
  link = "\url{http://www.johngustafson.net/presentations/Multicore2016-JLG.pdf}",
  ppt = "Gust16.pptx",
  abstract =
    "If we are willing to give up compatibility with IEEE 754 floats and
    design a number format with goals appropriate to 2016, we can achieve
    several goals simultaneously: Extremely high energy efficiency and

```

information-per-bit, no penalty for decimal operations instead of binary, rigorous bounds on answers without the overly pessimistic bounds produced by interval methods, and unprecedented high speed up to some precision. This approach extends the ideas of unum arithmetic introduced two years ago by breaking completely from the IEEE float-type format, resulting in fixed bit size values, fixed execution time, no exception values or 'gradual underflow' issues, no wasted bit patterns, and no redundant representations (like 'negative zero'). As an example of the power of this format, a difficult 12-dimensional nonlinear robotic kinematics problem that has defied solvers to date is quickly solvable with absolute bounds. Also unlike interval methods, it becomes possible to operate on arbitrary disconnected subsets of the real number line with the same speed as operating on a simple bound.",
 paper = "Gust16.pdf",
}

— axiom.bib —

```
@book{Gust16a,
  author = "Gustafson, John",
  title = "The End of Error: Unum Computing",
  publisher = "Chapman and Hall / CRC Computational Series",
  year = "2016",
  isbn = "978-1482239867"
}
```

— axiom.bib —

```
@article{Gust16b,
  author = "Gustafson, John",
  title = "Unums 2.0 An Interview with John L. Gustafson",
  publisher = "ACM",
  journal = "Ubiquity",
  year = "2016",
  paper = "Gust16b.pdf"
}
```

— axiom.bib —

```

@incollection{Lamb06,
  author = "Lambov, Branimir",
  title = "Interval Arithmetic Using SSE-2",
  booktitle = "Lecture Notes in Computer Science",
  publisher = "Springer-Verlag",
  year = "2006",
  isbn = "978-3-540-85520-0",
  pages = "102-113"
}

```

1.20 Numerics

— ignore —

```

\bibitem[Atkinson 09]{Atk09} Atkinson, Kendall; Han, Welmin; Stewear, David
  title = "Numerical Solution of Ordinary Differential Equations",
  link = "\url{http://homepage.math.uiowa.edu/~atkinson/papers/NAODE_Book.pdf}",
  abstract = "
    This book is an expanded version of supplementary notes that we used
    for a course on ordinary differential equations for upper-division
    undergraduate students and beginning graduate students in mathematics,
    engineering, and sciences. The book introduces the numerical analysis
    of differential equations, describing the mathematical background for
    understanding numerical methods and giving information on what to
    expect when using them. As a reason for studying numerical methods as
    a part of a more general course on differential equations, many of the
    basic ideas of the numerical analysis of differential equations are
    tied closely to theoretical behavior associated with the problem being
    solved. For example, the criteria for the stability of a numerical
    method is closely connected to the stability of the differential
    equation problem being solved.",
  paper = "Atk09.pdf"

```

— ignore —

```

\bibitem[Crank 96]{Cran96} Crank, J.; Nicolson, P.
  title = "A practical method for numerical evaluations of solutions of partial differential equations",
  booktitle = "Advances in Computational Mathematics Vol 6 pp207-226 (1996)",
  link = "\url{http://www.acms.arizona.edu/FemtoTheory/MK_personal/opti547/literature/CNMethod-o}",
  paper = "Cran96.pdf",

```

— ignore —

```
\bibitem[Lef\`evre 06]{Lef06} Lef\`evre, Vincent; Stehl\`e, Damien;
Zimmermann, Paul
  title = "Worst Cases for the Exponential Function in the IEEE-754r decimal64 Format",
  in Lecture Notes in Computer Science, Springer ISBN 978-3-540-85520-0
  (2006) pp114-125
  abstract = "
    We searched for the worst cases for correct rounding of the
    exponential function in the IEEE 754r decimal64 format, and computed
    all the bad cases whose distance from a breakpoint (for all rounding
    modes) is less than  $10^{-15}$  ulp, and we give the worst ones. In
    particular, the worst case for
     $\|x\| \geq 3 \times 10^{-11}$  is
    \[
    \exp(9.407822313572878 \times 10^{-2}) =
    1.0986456820663385000000000000000000278 \ldots
    \]
    This work can be extended to other elementary functions in the decimal64
    format and allows the design of reasonably fast routines that will
    evaluate these functions with correct rounding, at least in some
    situations."
```

— axiom.bib —

```
@book{Hamm62,
  author = "Hamming, R W.",
  title = "Numerical Methods for Scientists and Engineers",
  publisher = "Dover",
  year = "1973",
  isbn = "0-486-65241-6"
}
```

1.21 Advanced Documentation

— ignore —

```
\bibitem [Bostock 14]{Bos14} Bostock, Mike
  title = "Visualizing Algorithms",
  link = "\url{http://bost.ocks.org/mike/algorithms}",
  abstract = "
    This website hosts various ways of visualizing algorithms. The hope is
    that these kind of techniques can be applied to Axiom."
```

— axiom.bib —

```
@misc{Kama15,
  author = "Kamareddine, Fairouz and Wells, Joe and Zengler, Christoph and
    Barendregt, Henk",
  title = "Computerising Mathematical Text",
  year = "2015",
  abstract =
    "Mathematical texts can be computerised in many ways that capture
    differing amounts of the mathematical meaning. At one end, there is
    document imaging, which captures the arrangement of black marks on
    paper, while at the other end there are proof assistants (e.g. Mizar,
    Isabelle, Coq, etc.), which capture the full mathematical meaning and
    have proofs expressed in a formal foundation of mathematics. In
    between, there are computer typesetting systems (e.g. Latex and
    Presentation MathML) and semantically oriented systems (e.g. Content
    MathML, OpenMath, OMDoc, etc.). In this paper we advocate a style of
    computerisation of mathematical texts which is flexible enough to
    connect the different approaches to computerisation, which allows
    various degrees of formalisation, and which is compatible with
    different logical frameworks (e.g. set theory, category theory, type
    theory, etc.) and proof systems. The basic idea is to allow a
    man-machine collaboration which weaves human input with machine
    computation at every step in the way. We propose that the huge step from
    informal mathematics to fully formalised mathematics be divided into
    smaller steps, each of which is a fully developed method in which
    human input is minimal."
}
```

— axiom.bib —

```
@misc{Leeuux,
  author = {van Leeuwen, Andr'e M.A.},
  title = "Representation of mathematical object in interactive books",
  abstract = "
```

We present a model for the representation of mathematical objects in structured electronic documents, in a way that allows for interaction with applications such as computer algebra systems and proof checkers. Using a representation that reflects only the intrinsic information of an object, and storing application-dependent information in so-called {\sl application descriptions}, it is shown how the translation from the internal to an external representation and {\sl vice versa} can be achieved. Hereby a formalisation of the concept of {\sl context} is introduced. The proposed scheme allows for a high degree of application integration, e.g., parallel evaluation of subexpressions (by different computer algebra systems), or a proof checker using a computer algebra system to verify an equation involving a symbolic computation.",

```
paper = "Leeuxx.pdf"
}
```

— ignore —

```
\bibitem[Soiffer 91]{Soif91} Soiffer, Neil Morrell
title = "The Design of a User Interface for Computer Algebra Systems",
link = "\url{http://www.eecs.berkeley.edu/Pubs/TechRpts/1991/CSD-91-626.pdf}",
abstract = "
  This thesis discusses the design and implementation of natural user
  interfaces for Computer Algebra Systems. Such an interface must not
  only display expressions generated by the Computer Algebra System in
  standard mathematical notation, but must also allow easy manipulation
  and entry of expressions in that notation. The user interface should
  also assist in understanding of large expressions that are generated
  by Computer Algebra Systems and should be able to accommodate new
  notational forms.",
paper = "Soif91.pdf"
```

— ignore —

```
\bibitem[Victor 11]{Vict11} Victor, Bret
title = "Up and Down the Ladder of Abstraction",
link = "\url{http://worrydream.com/LadderOfAbstraction}",
abstract = "
  This interactive essay presents the ladder of abstraction, a technique for
  thinking explicitly about these levels, so a designer can move among
  them consciously and confidently. "
```

— ignore —

```
\bibitem[Victor 12]{Vict12} Victor, Bret
  title = "Inventing on Principle",
  link = "\url{http://www.youtube.com/watch?v=PUv66718DII}",
  abstract = "
    This video raises the level of discussion about human-computer
    interaction from a technical question to a question of effectively
    capturing ideas. In particular, this applies well to Axiom's focus on
    literate programming."
```

1.22 Differential Equations

— axiom.bib —

```
@InProceedings[Kalt84,
  author = "Kaltofen, E.",
  title = "A Note on the {Risch} Differential Equation",
  booktitle = "Proc. EUROSAM '84",
  pages = "359--366",
  year = "1984",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/84/Ka84_risch.ps.gz}",
  paper = "Kalt84.ps"
]
```

— ignore —

```
\bibitem[Abramov 95]{Abra95} Abramov, Sergei A.; Bronstein, Manuel;
Petkovsek, Marko
  title = "On Polynomial Solutions of Linear Operator Equations",
  link = "\url{http://www-sop.inria.fr/cafe/Manuel.Bronstein/publications/mb_papers.html}",
  paper = "Abra95.pdf"
```

— axiom.bib —

```

@misc{Abra01,
  author = "Abramov, Sergei and Bronstein, Manuel",
  title = "On Solutions of Linear Functional Systems",
  year = "2001",
  link = "\url{http://www-sop.inria.fr/cafe/Manuel.Bronstein/publications/mb_papers.html}",
  algebra =
    "\newline\refto{category OREPCAT UnivariateSkewPolynomialCategory}
    \newline\refto{category LODOCAT LinearOrdinaryDifferentialOperatorCategory}
    \newline\refto{domain AUTOMOR Automorphism}
    \newline\refto{domain ORESUP SparseUnivariateSkewPolynomial}
    \newline\refto{domain OREUP UnivariateSkewPolynomial}
    \newline\refto{domain LODO LinearOrdinaryDifferentialOperator}
    \newline\refto{domain LODO1 LinearOrdinaryDifferentialOperator1}
    \newline\refto{domain LODO2 LinearOrdinaryDifferentialOperator2}
    \newline\refto{package APPLYORE ApplyUnivariateSkewPolynomial}
    \newline\refto{package OREPCTO UnivariateSkewPolynomialCategoryOps}
    \newline\refto{package LODOF LinearOrdinaryDifferentialOperatorFactorizer}
    \newline\refto{package LODOOPS LinearOrdinaryDifferentialOperatorsOps}",
  abstract = "
    We describe a new direct algorithm for transforming a linear system of
    recurrences into an equivalent one with nonsingular leading or
    trailing matrix. Our algorithm, which is an improvement to the EG
    elimination method, uses only elementary linear algebra operations
    (ranks, kernels, and determinants) to produce an equation satisfied by
    the degree of the solutions with finite support. As a consequence, we
    can bound and compute the polynomial and rational solutions of very
    general linear functional systems such as systems of differential or
    ($q$)-difference equations.",
  paper = "Abra01.pdf"
}

```

— axiom.bib —

```

@inproceedings{Bron96b,
  author = "Bronstein, Manuel",
  title = "On the Factorization of Linear Ordinary Differential Operators",
  booktitle = "Mathematics and Computers in Simulation",
  volume = "42",
  pages = "387-389",
  year = "1996",
  abstract =
    "After reviewing the arithmetic of linear ordinary differential
    operators, we describe the current status of the factorisation
    algorithm, specially with respect to factoring over non-algebraically
    closed constant fields. We also describe recent results from Singer
    and Ulmer that reduce determining the differential Galois group of an

```



```

        operator to factoring.",
    paper = "Bro96b.pdf"
}

```

— axiom.bib —

```

@article{Bron96a,
    author = "Bronstein, Manuel and Petkovsek, Marko",
    title = "An introduction to pseudo-linear algebra",
    journal = "Theoretical Computer Science",
    volume = "157",
    pages = "3-33",
    year = "1966",
    link = "\url{http://www-sop.inria.fr/cafe/Manuel.Bronstein/publications/mb_papers.html}",
    algebra = "\newline\ref{category LORER LeftOreRing}",
    abstract =
        "Pseudo-linear algebra is the study of common properties of linear
        differential and difference operators. We introduce in this paper its
        basic objects (pseudo-derivations, skew polynomials, and pseudo-linear
        operators) and describe several recent algorithms on them, which, when
        applied in the differential and difference cases, yield algorithms for
        uncoupling and solving systems of linear differential and difference
        equations in closed form.",
    paper = "Bron96a.pdf"
}

```

— axiom.bib —

```

@inproceedings{Bron01,
    author = "Bronstein, Manuel",
    title = "Computer Algebra Algorithms for Linear Ordinary Differential
        and Difference equations",
    link = "\url{http://www-sop.inria.fr/cafe/Manuel.Bronstein/publications/ecm3.pdf}",
    booktitle = "European Congress of Mathematics",
    series = "Progress in Mathematics",
    volume = "202",
    year = "2001",
    pages = "105-119",
    abstract = "
        Galois theory has now produced algorithms for solving linear ordinary
        differential and difference equations in closed form. In addition,
        recent algorithmic advances have made those algorithms effective and
    "
}

```

```

implementable in computer algebra systems. After introducing the
relevant parts of the theory, we describe the latest algorithms for
solving such equations.",
paper = "Bron01.pdf"
}

```

— ignore —

```

\bibitem[Bronstein 94]{Bro94} Bronstein, Manuel
title = "An improved algorithm for factoring linear ordinary differential operators",
link = "\url{http://www-sop.inria.fr/cafe/Manuel.Bronstein/publications/mb_papers.html}",
abstract = "
We describe an efficient algorithm for computing the associated
equations appearing in the Beke-Schlesinger factorisation method for
linear ordinary differential operators. This algorithm, which is based
on elementary operations with sets of integers, can be easily
implemented for operators of any order, produces several possible
associated equations, of which only the simplest can be selected for
solving, and often avoids the degenerate case, where the order of the
associated equation is less than in the generic case. We conclude with
some fast heuristics that can produce some factorizations while using
only linear computations."

```

— ignore —

```

\bibitem[Bronstein 90]{Bro90} Bronstein, Manuel
title = "On Solutions of Linear Ordinary Differential Equations in their Coefficient Field",
link = "\url{http://www-sop.inria.fr/cafe/Manuel.Bronstein/publications/mb_papers.html}",
abstract = "
We describe a rational algorithm for finding the denominator of any
solution of a linear ordinary differential equation in its coefficient
field. As a consequence, there is now a rational algorithm for finding
all such solutions when the coefficients can be built up from the
rational functions by finitely many algebraic and primitive
adjunctions. This also eliminates one of the computational bottlenecks
in algorithms that either factor or search for Liouvillian solutions
of such equations with Liouvillian coefficients.",
paper = "Bro90.pdf"

```

— ignore —

```
\bibitem[Bronstein 96]{Bro96} Bronstein, Manuel
  title =
    "$\sum^{IT}$ -- A strongly-typed embeddable computer algebra library",
  link = "\url{http://www-sop.inria.fr/cafe/Manuel.Bronstein/publications/mb_papers.html}",
  abstract = "
    We describe the new computer algebra library $\sum^{IT}$ and its
    underlying design. The development of $\sum^{IT}$ is motivated by the
    need to provide highly efficient implementations of key algorithms for
    linear ordinary differential and ($q$)-difference equations to
    scientific programmers and to computer algebra users, regardless of
    the programming language or interactive system they use. As such,
    $\sum^{IT}$ is not a computer algebra system per se, but a library (or
    substrate) which is designed to be ‘‘plugged’’ with minimal efforts
    into different types of client applications.",
  paper = "Bro96.pdf"
```

— ignore —

```
\bibitem[Bronstein 99a]{Bro99a} Bronstein, Manuel
  title = "Solving linear ordinary differential equations over $\mathbb{C}(x, e^{\int f(x)dx})$",
  link = "\url{http://www-sop.inria.fr/cafe/Manuel.Bronstein/publications/mb_papers.html}",
  abstract = "
    We describe a new algorithm for computing the solutions in
    $\mathbb{C}(x, e^{\int f(x)dx})$ of linear ordinary differential equations
    with coefficients in $\mathbb{F}$. Compared to the general algorithm, our
    algorithm avoids the computation of exponential solutions of equations
    with coefficients in $\mathbb{C}(x)$, as well as the solving of linear
    differential systems over $\mathbb{C}(x)$. Our method is effective and has been
    implemented.",
  paper = "Bro99a.pdf"
```

— ignore —

```
\bibitem[Bronstein 00]{Bro00} Bronstein, Manuel
  title = "On Solutions of Linear Ordinary Differential Equations in their Coefficient Field",
  link = "\url{http://www-sop.inria.fr/cafe/Manuel.Bronstein/publications/mb_papers.html}",
  abstract = "
    We extend the notion of monomial extensions of differential fields,
```

i.e. simple transcendental extensions in which the polynomials are closed under differentiation, to difference fields. The structure of such extensions provides an algebraic framework for solving generalized linear difference equations with coefficients in such fields. We then describe algorithms for finding the denominator of any solution of those equations in an important subclass of monomial extensions that includes transcendental indefinite sums and products. This reduces the general problem of finding the solutions of such equations in their coefficient fields to bounding their degrees. In the base case, this yields in particular a new algorithm for computing the rational solutions of q -difference equations with polynomial coefficients.",
 paper = "Bro00.pdf"
 }
 —————
 — axiom.bib —
 @inproceedings{Bron02,
 author = "Bronstein, Manuel and Lafaille, S\'ebastien",
 title = "Solutions of linear ordinary differential equations in terms
 of special functions",
 booktitle = "Proc. ISSAC '02",
 publisher = "ACM Press",
 pages = "23-28",
 year = "2002",
 isbn = "1-58113-484-3",
 link = "\url{http://www-sop.inria.fr/cafe/Manuel.Bronstein/publications/issac2002.pdf}",
 url2 = "http://xena.hunter.cuny.edu/ksda/papers/bronstein2.pdf",
 paper2 = "Bron02x.pdf",
 abstract =
 "We describe a new algorithm for computing special function solutions
 of the form $y(x) = m(x)F(\eta(x))$ of second order linear ordinary
 differential equations, where $m(x)$ is an arbitrary Liouvillian
 function, $\eta(x)$ is an arbitrary rational function, and F
 satisfies a given second order linear ordinary differential
 equations. Our algorithm, which is base on finding an appropriate
 point transformation between the equation defining F and the one to
 solve, is able to find all rational transformations for a large class
 of functions F , in particular (but not only) the $_0F_1$ and $_1F_1$
 special functions of mathematical physics, such as Airy, Bessel,
 Kummer and Whittaker functions. It is also able to identify the values
 of the parameters entering those special functions, and can be
 generalized to equations of higher order.",
 paper = "Bron02.pdf"
 }

— ignore —

```
\bibitem[Bronstein 03]{Bro03} Bronstein, Manuel; Trager, Barry M.
  title = "A Reduction for Regular Differential Systems",
  link = "\url{http://www-sop.inria.fr/cafe/Manuel.Bronstein/publications/mega2003.pdf}",
  abstract = "
    We propose a definition of regularity of a linear differential system
    with coefficients in a monomial extension of a differential field, as
    well as a global and truly rational (i.e. factorisation-free)
    iteration that transforms a system with regular finite singularities
    into an equivalent one with simple finite poles. We then apply our
    iteration to systems satisfied by bases of algebraic function fields,
    obtaining algorithms for computing the number of irreducible
    components and the genus of algebraic curves.",
  paper = "Bro03.pdf"
```

— ignore —

```
\bibitem[Bronstein 03a]{Bro03a} Bronstein, Manuel; Sol\'e, Patrick
  title = "Linear recurrences with polynomial coefficients",
  link = "\url{http://www-sop.inria.fr/cafe/Manuel.Bronstein/publications/mb_papers.html}",
  abstract = "
    We relate sequences generated by recurrences with polynomial
    coefficients to interleaving and multiplexing of sequences generated
    by recurrences with constant coefficients. In the special case of
    finite fields, we show that such sequences are periodic and provide
    linear complexity estimates for all three constructions.",
  paper = "Bro03a.pdf"
```

— ignore —

```
\bibitem[Bronstein 05]{Bro05} Bronstein, Manuel; Li, Ziming; Wu, Min
  title = "Picard-Vessiot Extensions for Linear Functional Systems",
  link = "\url{http://www-sop.inria.fr/cafe/Manuel.Bronstein/publications/issac2005.pdf}",
  abstract = "
    Picard-Vessiot extensions for ordinary differential and difference
    equations are well known and are at the core of the associated Galois
    theories. In this paper, we construct fundamental matrices and
    Picard-Vessiot extensions for systems of linear partial functional
```

equations having finite linear dimension. We then use those extensions to show that all the solutions of a factor of such a system can be completed to solutions of the original system.",
 paper = "Bro05.pdf"

— axiom.bib —

```
\bibitem[Davenport 86]{Dav86} Davenport, J.H.
@article{Dave86,
  author = "Davenport, James H.",
  title = "The Risch Differential Equation Problem",
  year = "1986",
  journal = "SIAM J. COMPUT.",
  volume = "15",
  number = "4",
  comment = "Technical Report 83-4, Dept. Comp. Sci, Univ. Delaware",
  abstract = "
    We propose a new algorithm, similar to Hermite's method for the
    integration of rational functions, for the resolution of Risch
    differential equations in closed form, or proving that they have no
    resolution. By requiring more of the presentation of our differential
    fields (in particular that the exponentials be weakly normalized), we
    can avoid the introduction of arbitrary constants which have to be
    solved for later.

    We also define a class of fields known as exponentially reduced, and
    show that solutions of Risch differential equations which arise from
    integrating in these fields satisfy the 'natural' degree constraints
    in their main variables, and we conjecture (after Risch and Norman)
    that this is true in all variables.",
  paper = "Dave86.pdf"
}
```

— ignore —

```
\bibitem[Singer 9]{Sing91.pdf} Singer, Michael F.
  title = "Liouvillian Solutions of Linear Differential Equations with Liouvillian Coefficients",
  J. Symbolic Computation V11 No 3 pp251-273
  year = "1991",
  link = "\url{http://www.sciencedirect.com/science/article/pii/S074771710880048X}",
  abstract = "
    Let  $L(y)=b$  be a linear differential equation with coefficients in a
```

differential field K . We discuss the problem of deciding if such an equation has a non-zero solution in K and give a decision procedure in case K is an elementary extension of the field of rational functions or is an algebraic extension of a transcendental liouvillian extension of the field of rational functions. We show how one can use this result to give a procedure to find a basis for the space of solutions, liouvillian over K , of $L(y)=0$ where K is such a field and $L(y)$ has coefficients in K .",
 paper = "Sing91.pdf"

— ignore —

\bibitem[Von Mohrenschildt 94]{Mohr94} {von Mohrenschildt}, Martin
 title = "Symbolic Solutions of Discontinuous Differential Equations",
 link = "\url{http://e-collection.library.ethz.ch/eserv/eth:39463/eth-39463-01.pdf}",
 paper = "Mohr94.pdf"

— ignore —

\bibitem[Von Mohrenschildt 98]{Mohr98} von Mohrenschildt, Martin
 title = "A Normal Form for Function Rings of Piecewise Functions",
 J. Symbolic Computation (1998) Vol 26 pp607-619
 link = "\url{http://www.cas.mcmaster.ca/~mohrens/JSC.pdf}",
 abstract = "
 Computer algebra systems often have to deal with piecewise continuous functions. These are, for example, the absolute value function, signum, piecewise defined functions but also functions that are the supremum or infimum of two functions. We present a new algebraic approach to these types of problems. This paper presents a normal form for a function ring containing piecewise polynomial functions of an expression. The main result is that this normal form can be used to decide extensional equality of two piecewise functions. Also we define supremum and infimum for piecewise functions; in fact, we show that the function ring forms a lattice. Additionally, a method to solve equalities and inequalities in this function ring is presented. Finally, we give a ‘‘user interface’’ to the algebraic representation of the piecewise functions.",
 paper = "Mohr98.pdf"

— ignore —

```
\bibitem[Weber 06]{Webe06} Weber, Andreas
  title = "Quantifier Elimination on Real Closed Fields and Differential Equations",
  link = "\url{http://cg.cs.uni-bonn.de/personal-pages/weber/publications/pdf/WeberA/Weber2006a.pdf}",
  keywords = "survey",
  abstract = "
    This paper surveys some recent applications of quantifier elimination
    on real closed fields in the context of differential
    equations. Although polynomial vector fields give rise to solutions
    involving the exponential and other transcendental functions in
    general, many questions can be settled within the real closed field
    without referring to the real exponential field. The technique of
    quantifier elimination on real closed fields is not only of
    theoretical interest, but due to recent advances on the algorithmic
    side including algorithms for the simplification of quantifier-free
    formulae the method has gained practical applications, e.g. in the
    context of computing threshold conditions in epidemic modeling.",
  paper = "Webe06.pdf"
```

— ignore —

```
\bibitem[Ulmer 03]{Ulm03} Ulmer, Felix
  title = "Liouvillian solutions of third order differential equations",
  J. Symbolic COmputations 36 pp 855-889
  year = "2003",
  link = "\url{http://www.sciencedirect.com/science/article/pii/S0747717103000658}",
  abstract = "
    The Kovacic algorithm and its improvements give explicit formulae for
    the Liouvillian solutions of second order linear differential
    equations. Algorithms for third order differential equations also
    exist, but the tools they use are more sophisticated and the
    computations more involved. In this paper we refine parts of the
    algorithm to find Liouvillian solutions of third order equations. We
    show that, except for four finite groups and a reduction to the second
    order case, it is possible to give a formula in the imprimitive
    case. We also give necessary conditions and several simplifications
    for the computation of the minimal polynomial for the remaining finite
    set of finite groups (or any known finite group) by extracting
    ramification information from the character table. Several examples
    have been constructed, illustrating the possibilities and limitations.",
  paper = "Ulm03.pdf"
```


1.23 Expression Simplification

— ignore —

```
\bibitem[Car04]{Car04} Carette, Jacques
  title = "Understanding Expression Simplification",
  link = "\url{http://www.cas.mcmaster.ca/~carette/publications/simplification.pdf}",
  abstract = "
    We give the first formal definition of the concept of {\sl
    simplification} for general expressions in the context of Computer
    Algebra Systems. The main mathematical tool is an adaptation of the
    theory of Minimum Description Length, which is closely related to
    various theories of complexity, such as Kolmogorov Complexity and
    Algorithmic Information Theory. In particular, we show how this theory
    can justify the use of various ‘magic constants’ for deciding
    between some equivalent representations of an expression, as found in
    implementations of simplification routines.",
  paper = "Car04.pdf"
```

— axiom.bib —

```
@inproceedings{Cavi76,
  author = "Caviness, Bob F. and Fateman, Richard J.",
  title = "Simplification of Radical Expressions",
  booktitle = "Proc. 1976 SYMSAC",
  pages = "329-338",
  year = "1976",
  abstract =
    "In this paper we discuss the problem of simplifying unnested radical
    expressions. We describe an algorithm implemented in MACSYMA that
    simplifies radical expressions and then follow this description with
    a formal treatment of the problem. Theoretical computing times for some
    of the algorithms are briefly discussed as is related work of other
    authors",
  paper = "Cavi76.pdf",
  keywords = "axiomref"
}
```

— axiom.bib —

```
@article{Land93,
```

```

author = "Landau, Susan",
title = "How to Tangle with a Nested Radical",
institution = "University of Massachusetts",
journal = "The Mathematical Intelligencer",
year = "1993",
paper = "Land93.pdf"
}

```

1.24 Integration

— axiom.bib —

```

@TechReport{Kalt84b,
author = "Kaltofen, E.",
title = "The Algebraic Theory of Integration",
institution = "RPI",
address = "Dept. Comput. Sci., Troy, New York",
year = "1984",
link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/84/Ka84_integration.pdf}",
paper = "Kalt84b.pdf"
}

```

— ignore —

```

\bibitem[Adamchik xx]{Adamxx} Adamchik, Victor
title = "Definite Integration",
link = "\url{http://www.cs.cmu.edu/~adamchik/articles/integr/mj.pdf}",
paper = "Adamxx.pdf"

```

— ignore —

```

\bibitem[Adamchik 97]{Adam97} Adamchik, Victor
title = "A Class of Logarithmic Integrals",
link = "\url{http://www.cs.cmu.edu/~adamchik/articles/issac/issac97.pdf}",
abstract = "

```

A class of definite integrals involving cyclotomic polynomials and nested logarithms is considered. The results are given in terms of derivatives of the Hurwitz Zeta function. Some special cases for which such derivatives can be expressed in closed form are also considered.",
 paper = "Adam97.pdf"

— ignore —

```
\bibitem[Avgoustis 77]{Avgo77} Avgoustis, Ioannis Dimitrios
  title =
    "Definite Integration using the Generalized Hypergeometric Functions",
  link = "\url{http://dspace.mit.edu/handle/1721.1/16269}",
  abstract = "
    A design for the definite integration of approximately fifty Special
    Functions is described. The Generalized Hypergeometric Functions are
    utilized as a basis for the representation of the members of the above
    set of Special Functions. Only a relatively small number of formulas
    that generally involve Generalized Hypergeometric Functions are
    utilized for the integration stage. A last and crucial stage is
    required for the integration process: the reduction of the Generalized
    Hypergeometric Function to Elementary and/or Special Functions.

    The result of an early implementation which involves Laplace
    transforms are given and some actual examples with their corresponding
    timing are provided.",
  paper = "Avgo77.pdf"
```

— ignore —

```
\bibitem[Baddoura 89]{Bad89} Baddoura, Jamil
  title = "A Dilogarithmic Extension of Liouville's Theorem on Integration in
    Finite Terms",
  link = "\url{http://www.dtic.mil/dtic/tr/fulltext/u2/a206681.pdf}",
  abstract = "
    The result obtained generalizes Liouville's Theorem by allowing, in
    addition to the elementary functions, dilogarithms to appear in the
    integral of an elementary function. The basic conclusion is that an
    associated function to the dilogarithm, if dilogarithms appear in the
    integral, appears linearly, with logarithms appearing in a non-linear
    way.",
  paper = "Bad89.pdf"
```

— ignore —

```
\bibitem[Baddoura 94]{Bad94} Baddoura, Mohamed Jamil
  title = "Integration in Finite Terms with Elementary Functions and
    Dilogarithms",
  link = "\url{http://dspace.mit.edu/bitstream/handle/1721.1/26864/30757785.pdf}",
  abstract = "
    In this thesis, we report on a new theorem that generalizes
    Liouville's theorem on integration in finite terms. The new theorem
    allows dilogarithms to occur in the integral in addition to elementary
    functions. The proof is base on two identities for the dilogarithm,
    that characterize all the possible algebraic relations among
    dilogarithms of functions that are built up from the rational
    functions by taking transcendental exponentials, dilogarithms, and
    logarithms.",
  paper = "Bad94.pdf"
```

— ignore —

```
\bibitem[Baddoura 10]{Bad10} Baddoura, Jamil
  title = "A Note on Symbolic Integration with Polylogarithms",
  year = "2011",
  J. Math Vol 8 pp229-241 (2011)
  abstract = "
    We generalize partially Liouville's theorem on integration in finite
    terms to allow polylogarithms of any order to occur in the integral in
    addition to elementary functions. The result is a partial
    generalization of a theorem proved by the author for the
    dilogarithm. It is also a partial proof of a conjecture postulated by
    the author in 1994. The basic conclusion is that an associated
    function to the nth polylogarithm appears linearly with logarithms
    appearing possibly in a polynomial way with non-constant coefficients.",
  paper = "Bad10.pdf"
```

— ignore —

```
\bibitem[Bajpai 70]{Bajp70} Bajpai, S.D.
  title = "A contour integral involving legendre polynomial and Meijer's G-function",
  link = "\url{http://link.springer.com/article/10.1007/BF03049565}",
```

```

abstract = "
  In this paper a contour integral involving Legendre polynomial and
  Meijer's G-function is evaluated. the integral is of general character
  and it is a generalization of results recently given by Meijer,
  MacRobert and others. An integral involving regular radial Coulomb
  wave function is also obtained as a particular case.",
paper = "Bajp70.pdf"

```

— ignore —

```

\bibitem[Bronstein 89]{Bro89a} Bronstein, M.
  title = "An Algorithm for the Integration of Elementary Functions",
  Lecture Notes in Computer Science Vol 378 pp491-497
  year = "1989",
  abstract = "
    Trager (1984) recently gave a new algorithm for the indefinite
    integration of algebraic functions. His approach was ‘‘rational’’ in
    the sense that the only algebraic extension computed in the smallest
    one necessary to express the answer. We outline a generalization of
    this approach that allows us to integrate mixed elementary
    functions. Using only rational techniques, we are able to normalize
    the integrand, and to check a necessary condition for elementary
    integrability.",
  paper = "Bro89a.pdf"

```

— axiom.bib —

```

@article{Bron90a,
  author = "Bronstein, Manuel",
  title = "Integration of Elementary Functions",
  journal = "J. Symbolic Computation",
  volume = "9",
  pages = "117-173",
  year = "1990",
  abstract =
    "We extend a recent algorithm of Trager to a decision procedure for the
    indefinite integration of elementary functions. We can express the
    integral as an elementary function or prove that it is not
    elementary. We show that if the problem of integration in finite terms
    is solvable on a given elementary function field  $k$ , then it is
    solvable in any algebraic extension of  $k(\theta)$ , where  $\theta$  is
    a logarithm or exponential of an element of  $k$ . Our proof considers

```

an element of such an extension field to be an algebraic function of one variable over k .

In his algorithm for the integration of algebraic functions, Trager describes a Hermite-type reduction to reduce the problem to an integrand with only simple finite poles on the associated Riemann surface. We generalize that technique to curves over liouvillian ground fields, and use it to simplify our integrands. Once the multiple finite poles have been removed, we use the Puiseux expansions of the integrand at infinity and a generalization of the residues to compute the integral. We also generalize a result of Rothstein that gives us a necessary condition for elementary integrability, and provide examples of its use.",
 paper = "Bro90a.pdf"
 }

— axiom.bib —

```
@article{Bron90c,
  author = "Bronstein, Manuel",
  title = "On the integration of elementary functions",
  journal = "Journal of Symbolic Computation",
  volume = "9",
  number = "2",
  pages = "117-173",
  year = "1990",
  month = "February"
}
```

— axiom.bib —

```
@inproceedings{Bron93,
  author = "Bronstein, Manuel and Salvy, Bruno",
  title = "Full partial fraction decomposition of rational functions",
  booktitle = "Proc. ISSAC 1993",
  year = "1993",
  pages = "157-160",
  isbn = "0-89791-604-2",
  link = "\url{http://www.acm.org/pubs/citations/proceedings/issac/164081/}",
  algebra = "\newline\ref{domain FPARFRAC FullPartialFractionExpansion}",
  abstract =
    "We describe a rational algorithm that computes the full partial
```

```

fraction expansion of a rational function over the algebraic closure
of its field of definition. The algorithm uses only gcd operations
over the initial field but the resulting decomposition is expressed
with linear denominators. We give examples from its Axiom and Maple
implementations.",
paper = "Bron93.pdf",
keywords = "axiomref",
beebe = "Bronstein:1993:FPF"
}

```

— Bronstein:1993:FPF —

```

@InProceedings{Bronstein:1993:FPF,
  author =      "Manuel Bronstein and Bruno Salvy",
  title =      "Full Partial Fraction Decomposition of Rational
                Functions",
  crossref =    "Bronstein:1993:IPI",
  pages =      "157--160",
  year =       "1993",
  bibdate =    "Thu Mar 12 08:40:26 MST 1998",
  bibsource =   "http://www.acm.org/pubs/toc/;
                http://www.math.utah.edu/pub/tex/bib/axiom.bib",
  link =       "\url{http://www.acm.org:80/pubs/citations/proceedings/issac/164081/p157-brons",
  abstract =    "We describe a rational algorithm that computes the
                full partial fraction expansion of a rational function
                over the algebraic closure of its field of definition.
                The algorithm uses only gcd operations over the initial
                field but the resulting decomposition is expressed with
                linear denominators. We give examples from its Axiom
                and Maple implementations.",
  acknowledgement = ack-nhfb,
  affiliation = "Wissenschaftliches Rechnen, Eidgenossische Tech.
                Hochschule, Zurich, Switzerland",
  classification = "B0290D (Functional analysis); B0290H (Linear
                  algebra); B0290M (Numerical integration and
                  differentiation); C4120 (Functional analysis); C4140
                  (Linear algebra); C4160 (Numerical integration and
                  differentiation); C7310 (Mathematics computing)",
  keywords =    "ACM; Algebraic closure; algebraic computation; Axiom;
                Decomposition; Full partial fraction decomposition; Gcd
                operations; Maple; Polynomial; Rational functions;
                SIGSAM; symbolic computation; Symbolic integration,
                ISSAC; theory; verification",
  subject =     "{\bf I.1.0} Computing Methodologies, SYMBOLIC AND
                ALGEBRAIC MANIPULATION, General. {\bf G.1.0}
                Mathematics of Computing, NUMERICAL ANALYSIS, General.

```

```

        {\bf I.1.2} Computing Methodologies, SYMBOLIC AND
        ALGEBRAIC MANIPULATION, Algorithms.",
thesaurus = "Function evaluation; Integration; Matrix
             decomposition; Polynomial matrices; Symbol
             manipulation",
}

```

— ignore —

```

\bibitem[Bronstein 90]{Bro90b} Bronstein, Manuel
  title = "A Unification of Liouvillian Extensions",
  abstract = "
    We generalize Liouville's theory of elementary functions to a larger
    class of differential extensions. Elementary, Liouvillian and
    trigonometric extensions are all special cases of our extensions. In
    the transcendental case, we show how the rational techniques of
    integration theory can be applied to our extensions, and we give a
    unified presentation which does not require separate cases for
    different monomials.",
  paper = "Bro90b.pdf"

```

— axiom.bib —

```

@book[Bron97,
  author = "Bronstein, Manuel",
  title = "Symbolic Integration I--Transcendental Functions",
  publisher = "Springer, Heidelberg",
  year = "1997",
  isbn = "3-540-21493-3",
  link = "\url{http://evil-wire.org/arrXiv/Mathematics/Bronstein,_Symbolic_Integration_I,1997.pdf}",
  paper = "Bron97.pdf"
}

```

— ignore —

```

\bibitem[Bronstein 05a]{Bro05a} Bronstein, Manuel
  title = "The Poor Man's Integrator, a parallel integration heuristic",

```



```

link = "\url{http://www-sop.inria.fr/cafe/Manuel.Bronstein/pmint/pmint.txt}",
url2 = "http://www-sop.inria.fr/cafe/Manuel.Bronstein/pmint/examples",
paper = "Bro05a.txt"

```

— axiom.bib —

```

@article{Bron06,
  author = "Bronstein, M.",
  title = "Parallel integration",
  journal = "Programming and Computer Software",
  year = "2006",
  issn = "0361-7688",
  volume = "32",
  number = "1",
  doi = "10.1134/S0361768806010075",
  link = "\url{http://dx.doi.org/10.1134/S0361768806010075}",
  publisher = "Nauka/Interperiodica",
  pages = "59-60",
  abstract = "
    Parallel integration is an alternative method for symbolic
    integration. While also based on Liouville's theorem, it handles all
    the generators of the differential field containing the integrand 'in
    parallel', i.e. all at once rather than considering only the topmost
    one in a recursive fasion. Although it still contains heuristic
    aspects, its ease of implementation, speed, high rate of success, and
    ability to integrate functions that cannot be handled by the Risch
    algorithm make it an attractive alternative.",
  paper = "Bron06.pdf"
}

```

— axiom.bib —

```

@article{Bron07,
  author = "Bronstein, Manuel",
  title = "Structure theorems for parallel integration",
  journal = "Journal of Symbolic Computation",
  volume = "42",
  number = "7",
  pages = "757-769",
  year = "2007",
  month = "July",
  abstract = "

```

We introduce structure theorems that refine Liouville's Theorem on integration in closed form for general derivations on multivariate rational function fields. By predicting the arguments of the new logarithms that appear in integrals, as well as the denominator of the rational part, those theorems provide theoretical backing for the Risch-Norman integration method. They also generalize its applicability to non-monomial extensions, for example the Lambert W function.",
 paper = "Bron07.pdf"
 }

— ignore —

\bibitem[Charlwood 07]{Charl07} Charlwood, Kevin
 title = "Integration on Computer Algebra Systems",
 The Electronic J of Math. and Tech. Vol 2, No 3, ISSN 1933-2823
 link = "\url{http://12000.org/my_notes/ten_hard_integrals/paper.pdf}",
 abstract = "
 In this article, we consider ten indefinite integrals and the ability
 of three computer algebra systems (CAS) to evaluate them in
 closed-form, appealing only to the class of real, elementary
 functions. Although these systems have been widely available for many
 years and have undergone major enhancements in new versions, it is
 interesting to note that there are still indefinite integrals that
 escape the capacity of these systems to provide antiderivatives. When
 this occurs, we consider what a user may do to find a solution with
 the aid of a CAS.",
 paper = "Charl07.pdf"

— ignore —

\bibitem[Charlwood 08]{Charl08} Charlwood, Kevin
 title = "Symbolic Integration Problems",
 link = "\url{http://www.apmaths.uwo.ca/~arich/IndependentTestResults/CharlwoodIntegrationProblems.pdf}",
 abstract = "
 A list of the 50 example integration problems from Kevin Charlwood's 2008
 article 'Integration on Computer Algebra Systems'. Each integral along
 with its optimal antiderivative (that is, the best antiderivative found
 so far) is shown.",
 paper = "Charl08.pdf"

— ignore —

```
\bibitem[Cherry 84]{Che84} Cherry, G.W.
  title = "Integration in Finite Terms with Special Functions: The Error Function",
  J. Symbolic Computation (1985) Vol 1 pp283-302
  abstract = "
    A decision procedure for integrating a class of transcendental
    elementary functions in terms of elementary functions and error
    functions is described. The procedure consists of three mutually
    exclusive cases. In the first two cases a generalised procedure for
    completing squares is used to limit the error functions which can
    appear in the integral of a finite number. This reduces the problem
    to the solution of a differential equation and we use a result of
    Risch (1969) to solve it. The third case can be reduced to the
    determination of what we have termed  $\sum$ -decompositions. The result
    presented here is the key procedure to a more general algorithm which
    is described fully in Cherry (1983).",
  paper = "Che84.pdf"
```

— ignore —

```
\bibitem[Cherry 86]{Che86} Cherry, G.W.
  title = "Integration in Finite Terms with Special Functions: The Logarithmic Integral",
  SIAM J. Comput. Vol 15 pp1-21 February 1986
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— ignore —

```
\bibitem[Cherry 89]{Che89} Cherry, G.W.
  title = "An Analysis of the Rational Exponential Integral",
  SIAM J. Computing Vol 18 pp 893-905 (1989)
  abstract = "
    In this paper an algorithm is presented for integrating expressions of
    the form  $\int g e^{f dx}$ , where  $f$  and  $g$  are rational functions of
     $x$ , in terms of a class of special functions called the special
    incomplete  $\Gamma$  functions. This class of special functions
    includes the exponential integral, the error functions, the sine and
    cosine integrals, and the Fresnel integrals. The algorithm presented
    here is an improvement over those published previously for integrating
    with special functions in the following ways: (i) This algorithm
    combines all the above special functions into one algorithm, whereas
```

previously they were treated separately, (ii) Previous algorithms require that the underlying field of constants be algebraically closed. This algorithm, however, works over any field of characteristic zero in which the basic field operations can be carried out. (iii) This algorithm does not rely on Risch's solution of the differential equation $y' + fy = g$. Instead, a more direct method of undetermined coefficients is used.",

paper = "Che89.pdf"

— ignore —

```
\bibitem[Churchill 06]{Chur06}Churchill, R.C.
  title = "Liouville's Theorem on Integration Terms of Elementary Functions",
  link = "\url{http://www.sci.ccny.cuny.edu/~ksda/PostedPapers/liouv06.pdf}",
  abstract = "
    This talk should be regarded as an elementary introduction to
    differential algebra. It culminates in a purely algebraic proof, due
    to M. Rosenlicht, of an 1835 theorem of Liouville on the existence of
    ‘‘elementary’’ integrals of ‘‘elementary’’ functions. The precise
    meaning of elementary will be specified. As an application of that
    theorem we prove that the indefinite integral  $\int e^{x^2} dx$ 
    cannot be expressed in terms of elementary functions.
  \begin{itemize}
    \item Preliminaries on Meromorphic Functions
    \item Basic (Ordinary) Differential Algebra
    \item Differential Ring Extensions with No New Constants
    \item Extending Derivations
    \item Integration in Finite Terms
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  paper = "Chur06.pdf"
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  author = "Davenport, James H.",  
  title = "On the Integration of Algebraic Functions",  
  publisher = "Springer-Verlag",  
  series = "Lecture Notes in Computer Science 102",  
  isbn = "0-387-10290-6",  
  year = "1981"  
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  title = "Algorithms for the Integration of Algebraic Functions",
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  abstract = "
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    long been recognized as difficult, and has sometimes been thought
    insoluble. Risch stated a theorem characterising the integrands with
    elementary integrals, and we can use the language of algebraic
    geometry and the techniques of Davenport to yield an algorithm that will
    always produce the integral if it exists. We explain the difficulty in
    the way of extending this algorithm, and outline some ways of solving
    it. Using work of Manin we are able to solve the problem in all cases
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  title = "The Parallel Risch Algorithm (I)"
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  title = "On the Parallel Risch Algorithm (III): Use of Tangents",
  SIGSAM V16 no. 3 pp3-6 August 1982
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  title = "The Difficulties of Definite Integration",
  link = "\url{http://www.researchgate.net/publication/247837653_The_Diculties_of_Definite_Integration/file,
  abstract = "
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    and, before we can understand what we mean by indefinite integration,
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  paper = "Dav03.pdf"
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\bibitem[Fateman 02]{Fat02} Fateman, Richard
  title = "Symbolic Integration",
  link = "\url{http://inst.eecs.berkeley.edu/~cs282/sp02/lects/14.pdf}",
  paper = "Fat02.pdf"
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@inproceedings{Gedd89,
  author = "Geddes, K. O. and Stefanus, L. Y.",
  title = "On the Risch-norman Integration Method and Its Implementation
    in MAPLE",
  booktitle = "Proc. of the ACM-SIGSAM 1989 Int. Symp. on Symbolic and
    Algebraic Computation",
  series = "ISSAC '89",
  year = "1989",
  isbn = "0-89791-325-6",
  location = "Portland, Oregon, USA",
  pages = "212--217",
  numpages = "6",
  link = "\url{http://doi.acm.org/10.1145/74540.74567}",
  doi = "10.1145/74540.74567",
  acmid = "74567",
  publisher = "ACM",
  address = "New York, NY, USA",
  abstract = "
    Unlike the Recursive Risch Algorithm for the integration of
    transcendental elementary functions, the Risch-Norman Method processes
    the tower of field extensions directly in one step. In addition to
    logarithmic and exponential field extensions, this method can handle
    extentions in terms of tangents. Consequently, it allows trigonometric
    functions to be treated without converting them to complex exponential
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    title = "The Risch Integration Algorithm",
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        integration schemes is presented. The organization of the system
        differs from that of earlier programs in its emphasis on the
        algorithmic approach to integration, its modularity and its ease of
        revision. The new Norman-Rish algorithm and its implementation at the
        University of Cambridge are employed, supplemented by a powerful
        collection of simplification and transformation rules. The facility
        for user defined integrals and functions is also included. The program
        is both fast and powerful, and can be easily modified to incorporate
        anticipated developments in symbolic integration.",
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@misc{Herm1872,
  author = "Hermite, E.",
  title = "Sur l'intégration des fractions rationnelles",
  journal = "Nouvelles Annales de Mathématiques",
  volume = "11",
  pages = "145-148",
  year = "1872"
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  pp441-457
  ref = "00018",
  abstract = "
    Algorithms for symbolic partial fraction decomposition and indefinite
    integration of rational functions are described. Two types of
    partial fraction decomposition are investigated, square-free and
    complete square-free. A method is derived, based on the solution of
    a linear system, which produces the square-free decomposition of any
    rational function, say A/B. The computing time is shown to be
     $O(n^4(\ln n)^2)$  where  $\deg(A) < \deg(B) = n$  and  $n$ 
    is a number which is closely related to the size of the coefficients
    which occur in A and B. The complete square-free partial fraction
    decomposition can then be directly obtained and it is shown that the
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  paper = "Horo71.pdf"
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\bibitem[Jeffrey 97]{Jeff97} Jeffrey, D.J.; Rich, A.D.
  title = "Recursive integration of piecewise-continuous functions",
  link = "\url{http://www.cybertester.com/data/recint.pdf}",
  abstract = "
    An algorithm is given for the integration of a class of
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piecewise-continuous functions. The integration is with respect to a real variable, because the functions considered do not in general allow integration in the complex plane to be defined. The class of integrands includes commonly occurring waveforms, such as square waves, triangular waves, and the floor function; it also includes the signum function. The algorithm can be implemented recursively, and it has the property of ensuring that integrals are continuous on domains of maximum extent.",
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 When a computer algebra system has an assumption facility, it is possible to distinguish between integration problems with respect to a real variable, and those with respect to a complex variable. Here, a class of integration problems is defined in which the integrand consists of compositions of continuous functions and signum functions, and integration is with respect to a real variable. Algorithms are given for evaluating such integrals.",
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  journal = "Programming and Computer Software",
  volume = "35",
  number = "2",
  pages = "63-78",
  year = "2009",
  issn = "0361-7688",
  abstract = "
    In this paper by means of computer experiment we study advantages and
    disadvantages of the heuristical method of 'parallel integrator'. For
    this purpose we describe and use implementation of the method in
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    one in Maple.",
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  title = "Algebra",
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  title = "A Generalization of Liouville's Theorem on Integration in Finite Terms",
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  abstract = "
    A generalization of Liouville's theorem on integration in finite
    terms, by enlarging the class of fields to an extension called
    Ei-Gamma extension is established. This extension includes the
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  title = "Symbolic definite (and indefinite) integration: methods and open issues",
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  link = "\url{http://www.sigsam.org/bulletin/articles/175/issue175.pdf}",
  abstract = "
    The computation of definite integrals presents one with a variety of
    choices. There are various methods such as Newton-Leibniz or Slater's
    convolution method. There are questions such as whether to split or
    merge sums, how to search for singularities on the path of
    integration, when to issue conditional results, how to assess
    (possibly conditional) convergence, and more. These various
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    with examples.",
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  author = "Liouville, Joseph",
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  volume = "14",
  pages = "124-128",
  year = "1833"
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  author = "Liouville, Joseph",
  title = "Second m\`{e}moire sur la d\`{e}termination des int\`{e}grales
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  title = "A note on integration of rational functions",
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  abstract = "
    Let  $P$  and  $Q$  be polynomials in one variable with complex coefficients
    and let  $n$  be a natural number. Suppose that  $Q$  is not constant and
    has only simple roots. Then there is a rational function  $\varphi$ 
    with  $\varphi' = P/Q^{n+1}$  if and only if the Wronskian of the
    functions  $Q', (Q^2)', \dots, (Q^n)', P$  is
    divisible by  $Q$ .",
  paper = "Mari91.pdf"
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  title = "An introduction to the Risch Integration Algorithm",
  ACM Proc. 1976 annual conference pp425-428
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abstract = "
  Risch's decision procedure for determining the integrability in closed
  form of the elementary functions of the calculus is presented via
  examples. The exponential and logarithmic cases of the algorithm had
  been implemented for the MACSYMA system several years ago. The
  implementation of the algebraic case of the algorithm is the subject
  of current research.",
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  title = "Symbolic Integration: The Stormy Decade",
  CACM Aug 1971 Vol 14 No 8 pp548-560
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  ref = "00017",
  abstract = "
    Three approaches to symbolic integration in the 1960's are
    described. The first, from artificial intelligence, led to Slagle's
    SAINT and to a large degree to Moses' SIN. The second, from algebraic
    manipulation, led to Monove's implementation and to Horowitz' and
    Tobey's reexamination of the Hermite algorithm for integrating
    rational functions. The third, from mathematics, led to Richardson's
    proof of the unsolvability of the problem for a class of functions and
    for Risch's decision procedure for the elementary functions.
    Generalizations of Risch's algorithm to a class of special
    functions and programs for solving differential equations and for
    finding the definite integral are also described.",
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    indefinite integrals with a competence exceeding that of typical
    undergraduates. This practical advance was backed up by algorithmic
    interpretations of a number of classical results on integration, and by
    some significant mathematical extensions to these same results. At
    that time it would have been possible to claim that all the major

```

barriers in the way of a complete system for automated analysis had been breached. In this paper we survey the work that has grown out of the above-mentioned early results, showing where the development has been smooth and where it has spurred work in seemingly unrelated fields.",
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 title = "Definite Integration in Differential Fields",
 link = "\url{http://www.risc.jku.at/publications/download/risc_4583/PhD_CGR.pdf}",
 abstract = "

The general goal of this thesis is to investigate and develop computer algebra tools for the simplification resp. evaluation of definite integrals. One way of finding the value of a definite integral is via the evaluation of an antiderivative of the integrand. In the nineteenth century Joseph Liouville was among the first who analyzed the structure of elementary antiderivatives of elementary functions systematically. In the early twentieth century the algebraic structure of differential fields was introduced for modeling the differential properties of functions. Using this framework Robert H. Risch published a complete algorithm for transcendental elementary integrands in 1969. Since then this result has been extended to certain other classes of integrands as well by Michael F. Singer, Manuel Bronstein, and several others. On the other hand, if no antiderivative of suitable form is available, then linear relations that are satisfied by the parameter integral of interest may be found based on the principle of parametric integration (often called differentiating under the integral sign or creative telescoping).

The main result of this thesis extends the results mentioned above to a complete algorithm for parametric elementary integration for a certain class of integrands covering a majority of the special functions appearing in practice such as orthogonal polynomials, polylogarithms, Bessel functions, etc. A general framework is provided

to model those functions in terms of suitable differential fields. If the integrand is Liouvillian, then the present algorithm considerably improves the efficiency of the corresponding algorithm given by Singer et al. in 1985. Additionally, a generalization of Czichowskis algorithm for computing the logarithmic part of the integral is presented. Moreover, also partial generalizations to include other types of integrands are treated.

As subproblems of the integration algorithm one also has to find solutions of linear ordinary differential equations of a certain type. Some contributions are also made to solve those problems in our setting, where the results directly dealing with systems of differential equations have been joint work with Moulay A. Barkatou.

For the case of Liouvillian integrands we implemented the algorithm in form of our Mathematica package Integrator. Parts of the implementation also deal with more general functions. Our procedures can be applied to a significant amount of the entries in integral tables, both indefinite and definite integrals. In addition, our procedures have been successfully applied to interesting examples of integrals that do not appear in these tables or for which current standard computer algebra systems like Mathematica or Maple do not succeed. We also give examples of how parameter integrals coming from the work of other researchers can be solved with the software, e.g., an integral arising in analyzing the entropy of certain processes.",
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  title = "Generalization of Risch's Algorithm to Special Functions",
  link = "\url{http://arxiv.org/pdf/1305.1481.pdf}",
  abstract = "
    Symbolic integration deals with the evaluation of integrals in closed
    form. We present an overview of Risch's algorithm including recent
    developments. The algorithms discussed are suited for both indefinite
    and definite integration. They can also be used to compute linear
    relations among integrals and to find identities for special functions
    given by parameter integrals. The aim of this presentation is twofold:
    to introduce the reader to some basic idea of differential algebra in
    the context of integration and to raise awareness in the physics
    community of computer algebra algorithms for indefinite and definite
    integration.",
  paper = "Raab13.pdf"
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  title = "Integration in finite terms for Liouvillian functions",
  link = "\url{http://www.mmrc.iss.ac.cn/~dart4/posters/Raab.pdf}",
  abstract = "
    Computing integrals is a common task in many areas of science,
    antiderivatives are one way to accomplish this. The problem of
    integration in finite terms can be states as follows. Given a
    differential field  $(F,D)$  and  $f \in F$ , compute  $g$  in some
    elementary extension of  $(F,D)$  such that  $Dg = f$  if such a  $g$ 
    exists.

    This problem has been solved for various classes of fields  $F$ . For
    rational functions  $(C(x), \frac{d}{dx})$  such a  $g$  always exists and
    algorithms to compute it are known already for a long time. In 1969
    Risch published an algorithm that solves this problem when  $(F,D)$  is
    a transcendental elementary extension of  $(C(x), \frac{d}{dx})$ . Later
    this has been extended towards integrands being Liouvillian functions
    by Singer et. al. via the use of regular log-explicit extensions of
     $(C(x), \frac{d}{dx})$ . Our algorithm extends this to handling
    transcendental Liouvillian extensions  $(F,D)$  of  $(C,0)$  directly
    without the need to embed them into log-explicit extensions. For
    example, this means that
    
$$\int (z-x)x^{z-1}e^{-x}dx = x^ze^{-x}$$

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  title = "A Knowledge Repository for Indefinite Integration Based on Transformation Rules",
  link = "\url{http://www.apmaths.uwo.ca/~arich/A%2520Rule-based%2520Knowledge%2520Repository.pdf}",
  abstract = "
    Taking the specific problem domain of indefinite integration, we
    describe the on-going development of a repository of mathematical
    knowledge based on transformation rules. It is important that the
    repository be not confused with a look-up table. The database of
    transformation rules is at present encoded in Mathematica, but this is
    only one convenient form of the repository, and it could be readily
    translated into other formats. The principles upon which the set of
    rules is compiled is described. One important principle is
    minimality. The benefits of the approach are illustrated with
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examples, and with the results of comparisons with other approaches.",
 paper = "Rich09.pdf"

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@techreport{Risc68,
  author = "Risch, Robert",
  title = "On the integration of elementary functions which are built up
    using algebraic operations",
  type = "Research Report",
  number = "SP-2801/002/00",
  institution = "System Development Corporation, Santa Monica, CA, USA",
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  author = "Risch, Robert",
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  title = "The problem of integration in finite terms",
  journal = "Transactions of the American Mathematical Society",
  volume = "139",
  year = "1969",
  pages = "167-189",
  abstract = "This paper deals with the problem of telling whether a
    given elementary function, in the sense of analysis, has an elementary
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    indefinite integral.",
    paper = "Ris69b.pdf"
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@article{Risc70,
  author = "Risch, Robert",
  title = "The Solution of the Problem of Integration in Finite Terms",
  journal = "Bull. AMS",
  year = "1970",
  issn = "0002-9904",
  volume = "76",
  number = "3",
  pages = "605-609",
  abstract = "
    The problem of integration in finite terms asks for an algorithm for
    deciding whether an elementary function has an elementary indefinite
    integral and for finding the integral if it does. 'Elementary' is
    used here to denote those functions build up from the rational
    functions using only exponentiation, logarithms, trigonometric,
    inverse trigonometric and algebraic operations. This vaguely worded
    question has several precise, but inequivalent formulations. The
    writer has devised an algorithm which solves the classical problem of
    Liouville. A complete account is planned for a future publication. The
    present note is intended to indicate some of the ideas and techniques
    involved.",
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  pages = "743-759",
  year = "1979"
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  Pacific Journal of Mathematics Vol 24 No 1 (1968)
  link = "\url{http://msp.org/pjm/1968/24-1/pjm-v24-n1-p16-p.pdf}",
  ref = "00047",
  abstract = "
    Defining a function with one variable to be elementary if it has an
    explicit representation in terms of a finite number of algebraic
    operations, logarithms, and exponentials. Liouville's theorem in its
    simplest case says that if an algebraic function has an elementary
    integral then the latter is itself an algebraic function plus a sum of
    constant multiples of logarithms of algebraic functions. Ostrowski has
    generalized Liouville's results to wider classes of meromorphic
    functions on regions of the complex plane and J.F. Ritt has given the
    classical account of the entire subject in his Integraion in Finite
    Terms, Columbia University Press, 1948. In spite of the essentially
    algebraic nature of the problem, all proofs so far have been analytic.
    This paper gives a self contained purely algebraic exposition of the
    problem, making a few new points in addition to the resulting
    simplicity and generalization.",
  paper = "Ro68.pdf"
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@article{Rose72,
  author = "Rosenlicht, Maxwell",
  title = "Integration in finite terms",
  journal = "American Mathematical Monthly",
  year = "1972",
  volume = "79",
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    pages = "963-972",
    paper = "Rose72.pdf"
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@phdthesis{Roth76,
  author = "Rothstein, Michael",
  title = "Aspects of symbolic integration and simplification of
           exponential and primitive functions",
  school = "University of Wisconsin-Madison",
  year = "1976",
  link = "\url{http://www.cs.kent.edu/~rothstei/dis.pdf}",
  abstract =
    "In this thesis we cover some aspects of the theory necessary to obtain
    a canonical form for functions obtained by integration and
    exponentiation from the set of rational functions.

    These aspects include a new algorithm for symbolic integration of
    functions involving logarithms and exponentials which avoids
    factorization of polynomials in those cases where algebraic extension
    of the constant field is not required, avoids partial fraction
    decompositions, and only solves linear systems with a small number of
    unknowns.

    We have also found a theorem which states, roughly speaking, that if
    integrals which can be represented as logarithms are represented as
    such, the only algebraic dependence that a new exponential or
    logarithm can satisfy is given by the law of exponents or the law of
    logarithms.",
  paper = "Ro76.pdf"
}

```

— axiom.bib —

```

@article{Ro76a,
  author = "Rothstein, Michael and Caviness, Bob F.",
  title = "A structure theorem for exponential and primitive functions:
           a preliminary report",
  journal = "ACM Sigsum Bulletin",
  volume = "10",
  number = "4",

```

```

year = "1976",
abstract =
  "In this paper a generalization of the Risch Structure Theorem is reported.
  The generalization applies to fields  $F(t_1, \dots, t_n)$  where  $F$ 
  is a differential field (in our applications  $F$  will be a finitely
  generated extension of  $\mathbb{Q}$ , the field of rational numbers) and each  $t_i$ 
  is either algebraic over  $F_{i-1} = F(t_1, \dots, t_{i-1})$ , is an
  exponential of an element in  $F_{i-1}$ , or is an integral of an element
  in  $F_{i-1}$ . If  $t_i$  is an integral and can be expressed using
  logarithms, it must be so expressed for the generalized structure
  theorem to apply.",
paper = "Ro76a.pdf"
}

```

— ignore —

```

\bibitem[Rothstein 76b]{Ro76b} Rothstein, Michael; Caviness, B.F.
  title = "A structure theorem for exponential and primitive functions",
  SIAM J. Computing Vol 8 No 3 (1979)
  ref = "00104",
  abstract = "
    In this paper a new theorem is proved that generalizes a result of
    Risch. The new theorem gives all the possible algebraic relationships
    among functions that can be built up from the rational functions by
    algebraic operations, by taking exponentials, and by integration. The
    functions so generated are called exponential and primitive functions.
    From the theorem an algorithm for determining algebraic dependence
    among a given set of exponential and primitive functions is derived.
    The algorithm is then applied to a problem in computer algebra.",
  paper = "Ro76b.pdf"

```

— axiom.bib —

```

@article{Roth77,
  author = "Rothstein, Michael",
  title = "A new algorithm for the integration of exponential and
    logarithmic functions",
  journal = "Proceedings of the 1977 MACSYMA Users Conference",
  year = "1977",
  pages = "263-274",
  publisher = "NASA Pub CP-2012"
}

```

— ignore —

```
\bibitem[Seidenberg 58]{Sei58} Seidenberg, Abraham
  title = "Abstract differential algebra and the analytic case",
  Proc. Amer. Math. Soc. Vol 9 pp159-164 (1958)
```

— ignore —

```
\bibitem[Seidenberg 69]{Sei69} Seidenberg, Abraham
  title = "Abstract differential algebra and the analytic case. II",
  Proc. Amer. Math. Soc. Vol 23 pp689-691 (1969)
```

— axiom.bib —

```
@article{Sing85,
  author = "Singer, Michael F. and Saunders, B. David and Caviness, Bob F.",
  title = "An extension of Liouville's theorem on integration in finite terms",
  journal = "SIAM J. of Comp.",
  volume = "14",
  pages = "965-990",
  year = "1985",
  link = "\url{http://www4.ncsu.edu/~singer/papers/singer_saunders_caviness.pdf}",
  abstract =
    "In Part 1 of this paper, we give an extension of Liouville's Theorem
    and give a number of examples which show that integration with special
    functions involves some phenomena that do not occur in integration
    with the elementary functions alone. Our main result generalizes
    Liouville's Theorem by allowing, in addition to the elementary
    functions, special functions such as the error function, Fresnel
    integrals and the logarithmic integral (but not the dilogarithm or
    exponential integral) to appear in the integral of an elementary
    function. The basic conclusion is that these functions, if they
    appear, appear linearly. We give an algorithm which decides if an
    elementary function, built up using only exponential functions and
    rational operations has an integral which can be expressed in terms of
    elementary functions and error functions."
```

```
paper = "Sing85.pdf"
}
```

— ignore —

```
\bibitem[Slagle 61]{Slag61} Slagle, J.
  title = "A heuristic program that solves symbolic integration problems in freshman calculus",
  Ph.D Diss. MIT, May 1961; also Computers and Thought, Feigenbaum and Feldman.
  % REF:00014
```

— axiom.bib —

```
@mastersthesis{Tere09,
  author = "Terelius, Bjorn",
  title = "Symbolic Integration",
  school = "Royal Institute of Technology",
  address = "Stockholm, Sweden",
  year = "2009",
  abstract =
    "Symbolic integration is the problem of expressing an indefinite integral
     $\int f$  of a given function  $f$  as a finite combination  $g$  of elementary
    functions, or more generally, to determine whether a certain class of
    functions contains an element  $g$  such that  $g' = f$ ."
```

In the first part of this thesis, we compare different algorithms for symbolic integration. Specifically, we review the integration rules taught in calculus courses and how they can be used systematically to create a reasonable, but somewhat limited, integration method. Then we present the differential algebra required to prove the transcendental cases of Risch's algorithm. Risch's algorithm decides if the integral of an elementary function is elementary and if so computes it. The presentation is mostly self-contained and, we hope, simpler than previous descriptions of the algorithm. Finally, we describe Risch-Norman's algorithm which, although it is not a decision procedure, works well in practice and is considerably simpler than the full Risch algorithm.

In the second part of this thesis, we briefly discuss an implementation of a computer algebra system and some of the experiences it has given us. We also demonstrate an implementation of the rule-based approach and how it can be used, not only to compute integrals, but also to generate readable derivations of the results."


```

    paper = "Tere09.pdf"
}

```

— axiom.bib —

```

@article{Trag76,
  author = "Trager, Barry",
  title = "Algebraic factoring and rational function integration",
  journal = "Proceedings of SYMSAC'76",
  year = "1976",
  pages = "219-226",
  abstract = "
    This paper presents a new, simple, and efficient algorithm for
    factoring polynomials in several variables over an algebraic number
    field. The algorithm is then used iteratively to construct the
    splitting field of a polynomial over the integers. Finally the
    factorization and splitting field algorithms are applied to the
    problem of determining the transcendental part of the integral of a
    rational function. In particular, a constructive procedure is given
    for finding a least degree extension field in which the integral can
    be expressed.",
  paper = "Trag76.pdf"
}

```

— ignore —

```

\bibitem[Trager 76a]{Tr76a} Trager, Barry Marshall
  title = "Algorithms for Manipulating Algebraic Functions",
  MIT Master's Thesis.
  link = "\url{http://www.dm.unipi.it/pages/gianni/public_html/Alg-Comp/fattorizzazione-EA.pdf}",
  ref = "00050",
  abstract = "
    Given a base field  $K$ , of characteristic zero, with effective
    procedures for performing arithmetic and factoring polynomials, this
    thesis presents algorithms for extending those capabilities to
    elements of a finite algebraic symbolic manipulation system. An
    algebraic factorization algorithm along with a constructive version of
    the primitive element theorem is used to construct splitting fields of
    polynomials. These fields provide a context in which we can operate
    symbolically with all the roots of a set of polynomials. One
    application for this capability is rational function integrations.
    Previously presented symbolic algorithms concentrated on finding the

```

rational part and were only able to compute the complete integral in special cases. This thesis presents an algorithm for finding an algebraic extension field of least degree in which the integral can be expressed, and then constructs the integral in that field. The problem of algebraic function integration is also examined, and a highly efficient procedure is presented for generating the algebraic part of integrals whose function fields are defined by a single radical extension of the rational functions.",
 paper = "Tr76a.pdf"

— axiom.bib —

```
@phdthesis{Trag84,
  author = "Trager, Barry",
  title = "Integration of Algebraic Functions",
  school = "MIT",
  year = "1984",
  link = "\url{http://www.dm.unipi.it/pages/gianni/public_html/Alg-Comp/thesis.pdf}",
  abstract = "
    We show how the ‘‘rational’’ approach for integrating algebraic
    functions can be extended to handle elementary functions. The
    resulting algorithm is a practical decision procedure for determining
    whether a given elementary function has an elementary antiderivative,
    and for computing it if it exists.",
  paper = "Trag76.pdf"
}
```

— ignore —

```
\bibitem[W\urfl 07]{Wurf07} W\urfl, Andreas
  title = "Basic Concepts of Differential Algebra",
  link = "\url{http://www14.in.tum.de/konferenzen/Jass07/courses/1/Wuerfl/wuerfl_paper.pdf}",
  abstract = "
    Modern computer algebra systems symbolically integrate a vast variety
    of functions. To reveal the underlying structure it is necessary to
    understand infinite integration not only as an analytical problem but
    as an algebraic one. Introducing the differential field of elementary
    functions we sketch the mathematical tools like Liouville’s Principle
    used in modern algorithms. We present Hermite’s method for integration
    of rational functions as well as the Rothstein/Trager method for
    rational and for elementary functions. Further applications of the
    mentioned algorithms in the field of ODE’s conclude this paper.",
```

paper = "Wurf07.pdf"

1.25 Partial Fraction Decomposition

— ignore —

```
\bibitem[Angell]{Angell} Angell, Tom
  title = "Guidelines for Partial Fraction Decomposition",
  link = "\url{http://www.math.udel.edu/~angell/partfrac_I.pdf}",
  paper = "Angell.pdf"
```

— ignore —

```
\bibitem[Laval 08]{Lava08} Laval, Philippe B.
  title = "Partial Fractions Decomposition",
  link = "\url{http://www.math.wisc.edu/~park/Fall2011/integration/Partial%20Fraction.pdf}",
  paper = "Lava08.pdf"
```

— ignore —

```
\bibitem[Mudd 14]{Mudd14} Harvey Mudd College
  title = "Partial Fractions",
  link = "\url{http://www.math.hmc.edu/calculus/tutorials/partial_fractions/partial_fractions.pdf}",
  paper = "Mudd14.pdf"
```

— ignore —

```
\bibitem[Rajasekaran 14]{Raja14} Rajasekaran, Raja
  title = "Partial Fraction Expansion",
  link = "\url{http://www.utdallas.edu/~raja1/EE4361%20Spring%2014/Lecture%20Notes/Partial%20Fractions.pdf}",
  paper = "Raja14.pdf"
```

— ignore —

```
\bibitem[Wootton 14]{Woot14} Wootton, Aaron
  title = "Integration of Rational Functions by Partial Fractions",
  link = "\url{http://faculty.up.edu/wootton/calc2/section7.4.pdf}",
  paper = "Woot14.pdf"
```

1.26 Ore Rings

This is used as a reference for the `LeftOreRing` category, in particular, the least left common multiple (`lcmCoef`) function.

— ignore —

```
\bibitem[Abramov 97]{Abra97} Abramov, Sergei A.; van Hoeij, Mark
  title = "A method for the Integration of Solutions of Ore Equations",
  Proc ISSAC 97 pp172-175 (1997)
  abstract = "
    We introduce the notion of the adjoint Ore ring and give a definition
    of adjoint polynomial, operator and equation. We apply this for
    integrating solutions of Ore equations.",
  paper = "Abra97.pdf"
```

— axiom.bib —

```
@misc{Dele06,
  author = "Delenclos, Jonathon and Leroy, Andr\'e",
  title = "Noncommutative Symmetric functions and  $W$ -polynomials",
  link = "\url{http://arxiv.org/pdf/math/0606614.pdf}",
  algebra = "\newline\refto{category LORER LeftOreRing}",
  abstract = "
    Let  $K$ ,  $S$ ,  $D$  be a division ring an endomorphism and a
     $S$ -derivation of  $K$ , respectively. In this setting we introduce
    generalized noncommutative symmetric functions and obtain Vi\`ete
    formula and decompositions of different operators.  $W$ -polynomials
    show up naturally, their connetions with  $P$ -independency. Vandermonde
    and Wronskian matrices are briefly studied. The different linear
```

```

factorizations of  $\mathbb{W}$ -polynomials are analysed. Connections between
the existence of LLCM (least left common multiples) of monic linear
polynomials with coefficients in a ring and the left duo property are
established at the end of the paper.",
paper = "Dele06.pdf"
}

```

— ignore —

```

\bibitem[Abramov 05]{Abra05} Abramov, S.A.; Le, H.Q.; Li, Z.
‘‘Univariate Ore Polynomial Rings in Computer Algebra’’
link = "\url{http://www.mmrc.iss.ac.cn/~zmli/papers/oretools.pdf}",
abstract = "
We present some algorithms related to rings of Ore polynomials (or,
briefly, Ore rings) and describe a computer algebra library for basic
operations in an arbitrary Ore ring. The library can be used as a
basis for various algorithms in Ore rings, in particular, in
differential, shift, and  $\mathbb{Q}$ -shift rings.",
paper = "Abra05.pdf"

```

1.27 Number Theory

— axiom.bib —

```

@mastersthesis{Bohl08,
author = "Bohler, Per Reidar",
title = "Special number field sieve",
school = "Norwegian University of Science and Technology",
year = "2008",
link = "\url{http://www.diva-portal.org/smash/get/diva2:348611/FULLTEXT01.pdf}",
abstract =
"Integer factorization is a problem not yet solved for arbitrary integers.
Huge integers are therefore widely used for encrypting, e.g. in the RSA
encryption scheme. The special number field sieve holds the current
factorization record for factoring the number  $2^{1039}+1$ . The
algorithm depends on arithmetic in algebraic number fields and is
a further development from the quadratic sieve factoring algorithm.
We therefore present the quadratic sieve first. Then the special number
field is described. The key concepts are evaluated one by one.
Everything is illustrated with the corresponding parts of an example

```

factorization. The running time of the special number field sieve is then evaluated and compared against that of the quadratic sieve. The special number field sieve only applies to integers of a special form, but a generalization has been made, the general number field sieve. It is slower but all estimates suggests it is asymptotically faster than all other existing general purpose algorithms.",
 paper = "Bohl08.pdf"
 }

— axiom.bib —

```
@misc{Case16,
  author = "Case, Michael",
  title = "A Beginner's Guide to the General Number Field Sieve",
  year = "2016",
  link = "\url{http://citeseerx.ist.psu.edu/viewdoc/download?doi=10.1.1.219.2389}",
  paper = "Case16.pdf"
}
```

— axiom.bib —

```
@misc{Hill11,
  author = "Hill, Joshua E.",
  title = "The Number Field Sieve: An Extended Abstract",
  year = "2011",
  link = "\url{http://www.untruth.org/~josh/math/NFS.pdf}",
  paper = "Hill11.pdf"
}
```

— axiom.bib —

```
@InProceedings{Kalt89d,
  author = "Kaltofen, E. and Valente, T. and Yui, N.",
  title = "An improved {Las Vegas} primality test",
  booktitle = "Proc. 1989 Internat. Symp. Symbolic Algebraic Comput.",
  pages = "26--33",
  year = "1989",
}
```

```

    link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/89/KVY89.pdf}",
    paper = "Kalt89d.pdf"
}

```

— axiom.bib —

```

@InCollection{Kalt91b,
  author = "Kaltofen, E. and Yui, N.",
  editor = "D. V. Chudnovsky and G. V. Chudnovsky and H. Cohn and
    M. B. Nathanson",
  title = "Explicit construction of {Hilbert} class fields of imaginary
    quadratic fields by integer lattice reduction",
  booktitle = "Number Theory New York Seminar 1989--1990",
  pages = "150--202",
  publisher = "Springer-Verlag",
  year = "1991",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/91/KaYui91.pdf}",
  paper = "Kalt91b.pdf"
}

```

— axiom.bib —

```

@InProceedings{Kalt84a,
  author = "Kaltofen, E. and Yui, N.",
  title = "Explicit construction of the {Hilbert} class field of imaginary
    quadratic fields with class number 7 and 11",
  booktitle = "Proc. EUROSAM '84",
  pages = "310--320",
  year = "1984",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/84/KaYui84_eurosam.ps.gz}",
  paper = "Kalt84a.ps"
}

```

— axiom.bib —

```

@article{Pome94,
  author = "Pomerance, Carl",

```

```

title = "The Number Field Sieve",
journal = "Proc. Symposia in Applied Mathematics",
volume = "48",
year = "1994",
abstract =
  "The most exciting recent development in the integer factorization
  problem is the number field sieve. It has had some spectacular successes
  with integers in certain special forms, most notably the factorization in
  1990 of the 155 decimal digit number  $2^{512}+1$ . For arbitrary hard
  integers, it now appears to threaten the quadratic sieve as the algorithm
  of choice. In this paper the number field sieve, and the ideas behind it,
  are described",
paper = "Pome94.pdf"
}

```

— axiom.bib —

```

@misc{Sho08,
  author = "Shoup, Victor",
  title = "A Computational Introduction to Number Theory",
  link = "\url{http://shoup.net/ntb/ntb-v2.pdf}",
  paper = "Sho08.pdf"
}

```

1.28 Sparse Polynomial Interpolation

— axiom.bib —

```

@InProceedings{Kalt07a,
  author = "Kaltofen, Erich and Yang, Zhengfeng and Zhi, Lihong",
  title = "On probabilistic analysis of randomization in hybrid
  symbolic-numeric algorithms",
  year = "2007",
  booktitle = "Proc. 2007 Internat. Workshop on Symbolic-Numeric Comput.",
  pages = "11--17",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/07/KYZ07.pdf}",
  paper = "Kalt07a.pdf"
}

```

— axiom.bib —

```
@InProceedings{Kalt07b,
  author = "Kaltofen, Erich and Yang, Zhengfeng",
  title = "On Exact and Approximate Interpolation of Sparse
    Rational Functions",
  year = "2007",
  booktitle = "Internat. Symp. Symbolic Algebraic Comput. ISSAC'07",
  pages = "203--210",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/07/KaYa07.pdf}",
  paper = "Kalt07b.pdf"
}
```

— axiom.bib —

```
@Article{Gies03,
  author = "Giesbrecht, Mark and Kaltofen, Erich and Lee, Wen-shin",
  title = "Algorithms for Computing Sparsest Shifts of Polynomials in
    Power, {Chebychev}, and {Pochhammer} Bases",
  year = "2003",
  journal = "Journal of Symbolic Computation",
  volume = "36",
  number = "3--4",
  pages = "401--424",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/03/GKL03.pdf}",
  paper = "Gies03.pdf"
}
```

— axiom.bib —

```
@InProceedings{Gies02,
  author = "Giesbrecht, Mark and Kaltofen, Erich and Lee, Wen-shin",
  title = "Algorithms for Computing the Sparsest Shifts for Polynomials via the
    {Berlekamp}/{Massey} Algorithm",
  booktitle = "Proc. 2002 Internat. Symp. Symbolic Algebraic Comput.",
  pages = "101--108",
  year = "2002",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/02/GKL02.pdf}",
  paper = "Gies02.pdf"
}
```

}

— axiom.bib —

```
@Article{Kalt03b,
  author = "Kaltofen, Erich and Lee, Wen-shin",
  title = "Early Termination in Sparse Interpolation Algorithms",
  year = "2003",
  journal = "Journal of Symbolic Computation",
  volume = "36",
  number = "3--4",
  pages = "365--400",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/03/KL03.pdf}",
  paper = "Kalt03b.pdf"
}
```

— axiom.bib —

```
@InProceedings{Kalt00a,
  author = "Kaltofen, E. and Lee, W.-s. and Lobo, A.A.",
  title = "Early termination in {Ben-Or/Tiwari} sparse interpolation
    and a hybrid of {Zippel}'s algorithm",
  booktitle = "Proc. 2000 Internat. Symp. Symbolic Algebraic Comput.",
  pages = "192--201",
  year = "2000",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/2K/KLL2K.pdf}",
  paper = "Kalt00a.pdf"
}
```

— axiom.bib —

```
@InProceedings{Kalt10b,
  author = "Kaltofen, Erich L.",
  title = "Fifteen years after {DSC} and {WLSS2} {What} parallel
    computations {I} do today [{Invited} Lecture at {PASC0} 2010]",
  year = "2010",
  booktitle = "Proc. 2010 Internat. Workshop on Parallel Symbolic Comput.",
}
```

```

    pages = "10--17",
    month = "July",
    link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/10/Ka10_pasco.pdf}",
    paper = "Kalt10b.pdf"
}

```

— axiom.bib —

```

@InProceedings{Kalt90,
  author = "Kaltofen, E. and Lakshman, Y.N. and Wiley, J.M.",
  editor = "S. Watanabe and M. Nagata",
  title = "Modular rational sparse multivariate polynomial interpolation",
  booktitle = "Proc. 1990 Internat. Symp. Symbolic Algebraic Comput.",
  pages = "135--139",
  publisher = "ACM Press",
  year = "1990",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/90/KLW90.pdf}",
  paper = "Kalt90.pdf"
}

```

— axiom.bib —

```

@InProceedings{Kalt88a,
  author = "Kaltofen, E. and Yagati, Lakshman",
  title = "Improved sparse multivariate polynomial interpolation algorithms",
  booktitle = "Symbolic Algebraic Comput. Internat. Symp. ISSAC '88 Proc.",
  pages = "467--474",
  year = "1988",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/88/KaLa88.pdf}",
  paper = "Kalt88a.pdf"
}

```

1.29 Divisions and Algebraic Complexity

— axiom.bib —

```

@InCollection{Gren11,
  author = "Grenet, Bruno and Kaltofen, Erich L. and Koiran, Pascal
           and Portier, Natacha",
  title = "Symmetric Determinantal Representation of Formulas and Weakly
           Skew Circuits",
  booktitle = "Randomization, Relaxation, and Complexity in Polynomial
              Equation Solving",
  year = "2011",
  editor = "Leonid Gurvits and Philippe P\`{e}bay and J. Maurice Rojas
           and David Thompson",
  pages = "61--96",
  publisher = "American Mathematical Society",
  address = "Providence, Rhode Island, USA",
  isbn = "978-0-8218-5228-6",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/10/GKKP10.pdf}",
  paper = "Gren11.pdf"
}

```

— axiom.bib —

```

@InProceedings{Kalt08a,
  author = "Kaltofen, Erich and Koiran, Pascal",
  title = "Expressing a Fraction of Two Determinants as a Determinant",
  year = "2008",
  booktitle = "Internat. Symp. Symbolic Algebraic Comput. ISSAC'08",
  pages = "141--146",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/08/KaKoi08.pdf}",
  paper = "Kalt08a.pdf"
}

```

— axiom.bib —

```

@Article{Hitz95,
  author = "Kitz, M.A. and Kaltofen, E.",
  title = "Integer division in residue number systems",
  journal = "IEEE Trans. Computers",
  year = "1995",
  volume = "44",
  number = "8",
  pages = "983--989",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/95/HiKa95.pdf}",
  paper = "Hitz95.pdf"
}

```

}

— axiom.bib —

```
@InProceedings{Kalt92a,
  author = "Kaltofen, E.",
  title = "On computing determinants of matrices without divisions",
  booktitle = "Proc. 1992 Internat. Symp. Symbolic Algebraic Comput.",
  pages = "342--349",
  year = "1992",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/92/Ka92_issac.pdf}",
  paper = "Kalt92a.pdf"
}
```

— axiom.bib —

```
@Article{Cant91,
  author = "Cantor, D.G. and Kaltofen, E.",
  title = "On fast multiplication of polynomials over arbitrary algebras",
  journal = "Acta Inform.",
  year = "1991",
  volume = "28",
  number = "7",
  pages = "693--701",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/91/CaKa91.pdf}",
  paper = "Cant91.pdf"
}
```

— axiom.bib —

```
@Article{Kalt88b,
  author = "Kaltofen, E.",
  title = "Greatest common divisors of polynomials given by
          straight-line programs",
  journal = "J. ACM",
  year = "1988",
  volume = "35",
```

```

number = "1",
pages = "231--264",
link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/88/Ka88_jacm.pdf}",
paper = "Kalt88b.pdf"
}

```

1.30 Polynomial Factorization

— axiom.bib —

```

@InProceedings{Bern97a,
  author = "Bernardin, Laurent and Monagan, Michael B.",
  title = "Efficient multivariate factorization over finite fields",
  booktitle = "Applied algebra, algebraic algorithms and error-correcting
              codes",
  series = "AAECC-12",
  year = "1997",
  location = "Toulouse, France",
  publisher = "Springer",
  pages = "15-28",
  link = "\url{http://www.cecm.sfu.ca/~monaganm/papers/AAECC.pdf}",
  abstract =
    "We describe the Maple implementation of multivariate factorization
    over general finite fields. Our first implementation is available in
    Maple V Release 3. We give selected details of the algorithms and show
    several ideas that were used to improve its efficiency. Most of the
    improvements presented here are incorporated in Maple V Release 4. In
    particular, we show that we needed a general tool for implementing
    computations in  $\text{GF}(p^k)[x_1, x_2, \dots, x_v]$ . We also needed an
    efficient implementation of our algorithms  $\mathbb{Z}_p[y][x]$  in
    because any multivariate factorization may depend on several bivariate
    factorizations. The efficiency of our implementation is illustrated by
    the ability to factor bivariate polynomials with over a million
    monomials over a small prime field.",
  paper = "Bern97a.pdf",
  keywords = "axiomref"
}

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— axiom.bib —

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@InProceedings{Diaz95,
  author = "Diaz, A. and Kaltofen, E.",
  title = "On computing greatest common divisors with polynomials given by
          black boxes for their evaluation",
  booktitle = "Proc. 1995 Internat. Symp. Symbolic Algebraic Comput.",
  pages = "232--239",
  year = "1995",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/95/DiKa95.ps.gz}",
  paper = "Diaz95.ps"
}

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— axiom.bib —

```

@article{Gath01,
  author = "von zur Gathen, Joachim and Panario, Daniel",
  title = "Factoring Polynomials Over Finite Fields: A Survey",
  journal = "J. Symbolic Computation",
  year = "2001",
  volume = "31",
  pages = "3-17",
  link = "\url{http://people.csail.mit.edu/dmoshdov/courses/codes/poly-factorization.pdf}",
  keywords = "survey",
  abstract =
    "This survey reviews several algorithms for the factorization of
    univariate polynomials over finite fields. We emphasize the main ideas
    of the methods and provide an up-to-date bibliography of the problem.
    This paper gives algorithms for {\sl squarefree factorization},
    {\sl distinct-degree factorization}, and {\sl equal-degree factorization}.
    The first and second algorithms are deterministic, the third is
    probabilistic.",
  paper = "Gath01.pdf"
}

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— axiom.bib —

```

@article{Gian88,
  author = "Gianni, Patrizia. and Trager, Barry. and Zacharias, Gail",
  title = "Groebner Bases and Primary Decomposition of Polynomial Ideals",
  journal = "J. Symbolic Computation",
  volume = "6",
  pages = "149-167",
  year = "1988",

```

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link = "\url{http://www.sciencedirect.com/science/article/pii/S0747717188800403/pdf?md5=40c29b67947035884}
algebra = "\newline\refto{package IDECOMP IdealDecompositionPackage}",
paper = "Gian88.pdf"
}

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— axiom.bib —

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@article{Gian96,
  author = "Gianni, P. and Trager, B.",
  title = "Square-free algorithms in positive characteristic",
  journal =
    "J. of Applicable Algebra in Engineering, Communication and Computing",
  volume = "7",
  pages = "1-14",
  year = "1996",
}

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— axiom.bib —

```

@PhdThesis{Kalt82,
  author = "Kaltofen, E.",
  title = "On the complexity of factoring polynomials with integer
    coefficients",
  school = "RPI",
  address = "Troy, N. Y.",
  year = "1982",
  month = "December",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/82/Ka82_thesis.pdf}",
  paper = "Kalt82.pdf"
}

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— axiom.bib —

```

@Article{Gath85,
  author = "{von zur Gathen}, Joachim and Kaltofen, E.",
  title = "Factoring sparse multivariate polynomials",
  journal = "J. Comput. System Sci.",
}

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year = "1985",
volume = "31",
pages = "265--287",
link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/85/GaKa85_mathcomp.ps.gz}",
paper = "Gath85.ps"
}

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— axiom.bib —

```

@Article{Gath85b,
author = "{von zur Gathen}, Joachim and Kaltofen, E.",
title = "Polynomial-Time Factorization of Multivariate Polynomials over
        Finite Fields",
journal = "Math. Comput.",
year = "1985",
volume = "45",
pages = "251-261",
link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/85/GaKa85\_mathcomp.ps.gz}",
paper = "Gath85.ps",
abstract =
    "We present a probabilistic algorithm that finds the irreducible
    factors of a bivariate polynomial with coefficients from a finite
    field in time polynomial in the input size, i.e. in the degree of the
    polynomial and  $\log(\text{cardinality of field})$ . The algorithm generalizes
    to multivariate polynomials and has polynomial running time for
    densely encoded inputs. Also a deterministic version of the algorithm
    is discussed whose running time is polynomial in the degree of the
    input polynomial and the size of the field."
}

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— axiom.bib —

```

@InCollection{Kalt11c,
author = "Kaltofen, Erich and Lecerf, Gr{\'e}goire",
title = "Section 11.5. {Factorization} of multivariate polynomials",
booktitle = "Handbook of Finite Fields",
publisher = "Springer",
pages = "382--392",
year = "2011",
link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/11/KL11.pdf}",
paper = "Kalt11c.pdf"
}

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— axiom.bib —

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@InProceedings{Kalt05b,
  author = "Kaltofen, Erich and Koiran, Pascal",
  title = "On the complexity of factoring bivariate supersparse
    (lacunary) polynomials",
  year = "2005",
  booktitle = "Internat. Symp. Symbolic Algebraic Comput. ISSAC'05",
  pages = "208--215",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/05/KaKoi05.pdf}",
  paper = "Kalt05b.pdf"
}
```

— axiom.bib —

```
@InProceedings{Kalt06a,
  author = "Kaltofen, Erich and Koiran, Pascal",
  title = "Finding Small Degree Factors of Multivariate Supersparse
    (Lacunary) Polynomials Over Algebraic Number Fields",
  year = "2006",
  booktitle = "Internat. Symp. Symbolic Algebraic Comput. ISSAC'06",
  pages = "162--168",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/06/KaKoi06.pdf}",
  paper = "Kalt06a.pdf"
}
```

— axiom.bib —

```
@InProceedings{Kalt97a,
  author = "Kaltofen, E. and Shoup, V.",
  title = "Fast polynomial factorization over high algebraic extensions of
    finite fields",
  booktitle = "Proc. 1997 Internat. Symp. Symbolic Algebraic Comput.",
  year = "1997",
  pages = "184--188",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/97/KaSh97.pdf}",
}
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    paper = "Kalt97a.pdf"
}

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— axiom.bib —

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@Article{Kalt98,
  author = "Kaltofen, E. and Shoup, V.",
  title = "Subquadratic-time factoring of polynomials over finite fields",
  journal = "Math. Comput.",
  month = "July",
  year = "1998",
  volume = "67",
  number = "223",
  pages = "1179--1197",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/98/KaSh98.pdf}",
  paper = "Kalt98.pdf"
}

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— axiom.bib —

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@InProceedings{Kalt95a,
  author = "Kaltofen, E. and Shoup, V.",
  title = "Subquadratic-time factoring of polynomials over finite fields",
  booktitle = "Proc. 27th Annual ACM Symp. Theory Comput.",
  year = "1995",
  publisher = "ACM Press",
  address = "New York, N.Y.",
  pages = "398--406",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/95/KaSh95.ps.gz}",
  paper = "Kalt95a.ps"
}

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— axiom.bib —

```

@InProceedings{Kalt88,
  author = "Kaltofen, E. and Trager, B.",
  title = "Computing with polynomials given by black boxes for their

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    evaluations: Greatest common divisors, factorization, separation of
    numerators and denominators",
booktitle = "Proc. 29th Annual Symp. Foundations of Comp. Sci.",
pages = "296--305",
year = "1988",
organization = "IEEE",
link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/88/focs88.ps.gz}",
paper = "Kalt88.ps"
}

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— axiom.bib —

```

@InProceedings{Kalt85b,
  author = "Kaltofen, E.",
  title = "Computing with polynomials given by straight-line programs {II};
  sparse factorization",
  booktitle = "Proc. 26th Annual Symp. Foundations of Comp. Sci.",
  year = "1985",
  pages = "451--458",
  organization = "IEEE",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/85/Ka85_focs.ps.gz}",
  paper = "Kalt85b.ps"
}

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— axiom.bib —

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@InProceedings{Kalt86,
  author = "Kaltofen, E.",
  title = "Uniform closure properties of p-computable functions",
  booktitle = "Proc. 18th Annual ACM Symp. Theory Comput.",
  year = "1986",
  pages = "330--337",
  organization = "ACM",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/86/Ka86_stoc.pdf}",
  paper = "Kalt86.pdf"
}

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— axiom.bib —

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@InProceedings{Kalt87b,
  author = "Kaltofen, E.",
  title = "Single-factor Hensel lifting and its application to the
          straight-line complexity of certain polynomials",
  booktitle = "Proc. 19th Annual ACM Symp. Theory Comput.",
  year = "1987",
  pages = "443--452",
  organization = "ACM",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/87/Ka87_stoc.pdf}",
  paper = "Kalt87b.pdf"
}

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— axiom.bib —

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@InCollection{Kalt89,
  author = "Kaltofen, E.",
  editor = "S. Micali",
  title = "Factorization of polynomials given by straight-line programs",
  booktitle = "Randomness and Computation",
  pages = "375--412",
  publisher = "JAI Press Inc.",
  year = "1989",
  volume = "5",
  series = "Advances in Computing Research",
  address = "Greenwich, Connecticut",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/89/Ka89_slpfac.pdf}",
  paper = "Kalt89.pdf"
}

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— axiom.bib —

```

@Article{Gao04,
  author = "Gao, Shuhong and Kaltofen, E. and Lauder, A.",
  title = "Deterministic distinct degree factorization for polynomials
          over finite fields",
  year = "2004",
  journal = "Journal of Symbolic Computation",
  volume = "38",
  number = "6",
  pages = "1461--1470",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/01/GKL01.pdf}",
  paper = "Gao04.pdf"
}

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}

— axiom.bib —

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@Article{Kalt87c,
  author = "Kaltofen, E.",
  title = "Deterministic irreducibility testing of polynomials over
    large finite fields",
  journal = "Journal of Symbolic Computation",
  year = "1987",
  volume = "4",
  pages = "77--82",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/87/Ka87_jsc.ps.gz}",
  paper = "Kalt87c.ps"
}
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— axiom.bib —

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@Article{Kalt95b,
  author = "Kaltofen, E.",
  title = "Effective {Noether} irreducibility forms and applications",
  journal = "J. Comput. System Sci.",
  year = "1995",
  volume = "50",
  number = "2",
  pages = "274--295",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/95/Ka95_jcss.pdf}",
  paper = "Kalt95b.pdf"
}
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@Article{Kalt85a,
  author = "Kaltofen, E.",
  title = "Fast parallel absolute irreducibility testing",
  journal = "Journal of Symbolic Computation",
  year = "1985",
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    volume = "1",
    number = "1",
    pages = "57--67",
    link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/85/Ka85_jsc.pdf}",
    paper = "Kalt85a.pdf"
}

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— axiom.bib —

```

@Article{Gath85a,
  author = "{von zur Gathen}, Joachim and Kaltofen, E.",
  title = "Factoring multivariate polynomials over finite fields",
  journal = "Math. Comput.",
  year = "1985",
  volume = "45",
  pages = "251--261",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/85/GaKa85_mathcomp.ps.gz}",
  paper = "Gath85a.ps"
}

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@Article{Kalt85e,
  author = "Kaltofen, E.",
  title = "Polynomial-time reductions from multivariate to bi- and univariate
          integral polynomial factorization",
  journal = "{SIAM} J. Comput.",
  year = "1985",
  volume = "14",
  number = "2",
  pages = "469--489",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/85/Ka85_sicomp.pdf}",
  paper = "Kalt85e.pdf"
}

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@InProceedings{Kalt82a,

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author = "Kaltofen, E.",
title = "A polynomial-time reduction from bivariate to univariate
        integral polynomial factorization",
booktitle = "Proc. 23rd Annual Symp. Foundations of Comp. Sci.",
year = "1982",
pages = "57--64",
organization = "IEEE",
link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/82/Ka82_focs.pdf}",
paper = "Kalt82a.pdf"
}

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@InProceedings{Kalt03,
  author = "Kaltofen, Erich",
  title = "Polynomial Factorization: a Success Story",
  year = "2003",
  booktitle = "Symbolic Algebraic Comput. Internat. Symp. ISSAC '88 Proc.",
  pages = "3--4",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/03/Ka03.pdf}",
  keywords = "survey",
  paper = "Kalt03.pdf"
}

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@InProceedings{Kalt92b,
  author = "Kaltofen, E.",
  title = "Polynomial factorization 1987-1991",
  booktitle = "Proc. LATIN '92",
  editor = "I. Simon",
  series = "Lect. Notes Comput. Sci.",
  volume = "583",
  pages = "294--313",
  publisher = "Springer-Verlag",
  year = "1992",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/92/Ka92_latin.pdf}",
  keywords = "survey",
  paper = "Kalt92b.pdf"
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```
@InCollection{Kalt90c,
  author = "Kaltofen, E.",
  editor = "D. V. Chudnovsky and R. D. Jenks",
  title = "Polynomial Factorization 1982-1986",
  booktitle = "Computers in Mathematics",
  pages = "285--309",
  publisher = "Marcel Dekker, Inc.",
  year = "1990",
  volume = "125",
  series = "Lecture Notes in Pure and Applied Mathematics",
  address = "New York, N. Y.",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/90/Ka90_survey.ps.gz}",
  keywords = "survey",
  paper = "Kalt90c.ps"
}
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@InCollection{Kalt82b,
  author = "Kaltofen, E.",
  title = "Polynomial factorization",
  editor = "B. Buchberger and G. Collins and R. Loos",
  booktitle = "Computer Algebra",
  edition = "2",
  pages = "95--113",
  publisher = "Springer-Verlag",
  year = "1982",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/82/Ka82_survey.ps.gz}",
  keywords = "survey",
  paper = "Kalt82b.ps"
}
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— axiom.bib —

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@phdthesis{Sale04,
  author = "Salem, Fatima Khaled Abu",
  title = "Factorisation Algorithms for Univariate and Bivariate Polynomials"
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        over Finite Fields",
school = "Meron College",
year = "2004",
link = "\url{http://www.cs.aub.edu.lb/fa21/Dissertations/My\_thesis.pdf}",
abstract =
    "In this thesis we address algorithms for polynomial factorisation
    over finite fields. In the univariate case, we study a recent
    algorithm due to Niederreiter where the factorisation problem is
    reduced to solving a linear system over the finite field in question,
    and the solutions are used to produce the complete factorisation of
    the polynomials into irreducibles. We develop a new algorithm for
    solving the linear system using sparse Gaussian elimination with the
    Markowitz ordering strategy, and conjecture that the Niederreiter
    linear system is not only initially sparse, but also preserves its
    sparsity throughout the Gaussian elimination phase. We develop a new
    bulk synchronous parallel (BSP) algorithm base on the approach of
    Gottfert for extracting the factors of a polynomial using a basis of
    the Niederreiter solution set of  $\mathbb{F}_2$ . We improve upon the
    complexity and performance of the original algorithm, and produce
    binary univariate factorisations of trinomials up to degree 400000.

    We present a new approach to multivariate polynomial factorisation
    which incorporates ideas from polyhedral geometry, and generalises
    Hensel lifting. The contribution is an algorithm for factoring
    bivariate polynomials via polytopes which is able to exploit to some
    extent the sparsity of polynomials. We further show that the polytope
    method can be made sensitive to the number of nonzero terms of the
    input polynomial. We describe a sparse adaptation of the polytope
    method over finite fields of prime order which requires fewer bit
    operations and memory references for polynomials which are known to be
    the product of two sparse factors. Using this method, and to the best
    of our knowledge, we achieve a world record in binary bivariate
    factorisation of a sparse polynomial of degree 20000. We develop a BSP
    variant of the absolute irreducibility testing via polytopes given in
    [45], producing a more memory and run time efficient method that can
    provide wider ranges of applicability. We achieve absolute
    irreducibility testing of a bivariate and trivariate polynomial of
    degree 30000, and of multivariate polynomials with up to 3000
    variables.",
paper = "Sale04.pdf"
}

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— axiom.bib —

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@InProceedings{Shou91,
  author = "Shoup, Victor",

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title = "A Fast Deterministic Algorithm for Factoring Polynomials over
        Finite Fields of Small Characteristic",
booktitle = "Proc. ISSAC 1991",
series = "ISSAC 1991",
year = "1991",
pages = "14-21",
link = "\url{http://www.shoup.net/papers/quadfactor.pdf}",
abstract =
    "We present a new algorithm for factoring polynomials over finite
    fields. Our algorithm is deterministic, and its running time is
    ‘almost’ quadratic when the characteristic is a small fixed
    prime. As such, our algorithm is asymptotically faster than previously
    known deterministic algorithms for factoring polynomials over finite
    fields of small characteristic.",
paper = "Shou91.pdf"
}

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— axiom.bib —

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@inproceedings{Trev91,
  author = "Trevisan, Vilmar and Wang, Paul",
  title = "Practical factorization of univariate polynomials over
          finite fields",
  booktitle = "Proc. ISSAC 1991",
  series = "ISSAC '91",
  publisher = "ACM",
  isbn = "0-89791-437-6",
  pages = "22-31",
  year = "1991",
  link = "\url{http://lib.org/by/_djvu/_Papers/Computer/_algebra/Algebraic/_%20numbers}",
  abstract =
    "Research presented here is part of an effort to establish
    state-of-the-art factoring routines for polynomials. The foundation of
    such algorithms lies in the efficient factorization over a finite
    field  $\mathbb{GF}(p^k)$ . The Cantor-Zassenhaus algorithm together with
    innovative ideas suggested by others is compared with the Berlekamp
    algorithm. The studies led us to design a hybrid algorithm that
    combine the strengths of the different approaches. The algorithms are
    also implemented and machine timings are obtained to measure the
    performance of these algorithms.",
  paper = "Trev91.djvu"
}

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1.31 Branch Cuts

— axiom.bib —

```
@article{Beau03,
  author = "Beaumont, James and Bradford, Russell and Davenport, James H.",
  title = "Better simplification of elementary functions through power series",
  journal = "2003 International Symposium on Symbolic and Algebraic Computation",
  series = "ISSAC'03",
  year = "2003",
  month = "August",
  abstract = "
    In [5], we introduced an algorithm for deciding whether a proposed
    simplification of elementary functions was correct in the presence of
    branch cuts. This algorithm used multivalued function simplification
    followed by verification that the branches were consistent.

    In [14] an algorithm was presented for zero-testing functions defined
    by ordinary differential equations, in terms of their power series.

    The purpose of the current paper is to investigate merging the two
    techniques. In particular, we will show an explicit reduction to the
    constant problem [16].",
  paper = "Beau03.pdf"
}
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— axiom.bib —

```
@article{Brad02,
  author = "Bradford, Russell and Corless, Robert M. and Davenport, James H.
    Jeffrey, David J. and Watt, Stephen M.",
  title = "Reasoning about the Elementary Functions of Complex Analysis",
  journal = "Annals of Mathematics and Artificial Intelligence",
  year = "2002",
  issn = "1012-2443",
  volume = "36",
  number = "3",
  doi = "10.1023/A:1016007415899",
  link = "\url{http://dx.doi.org/10.1023/A%3A1016007415899}",
  publisher = "Kluwer Academic Publishers",
  keywords = "elementary functions; branch cuts; complex identities",
  pages = "303-318",
  abstract =
    "There are many problems with the simplification of elementary
```

```

functions, particularly over the complex plane, though not
exclusively. Systems tend to make ‘‘howlers’’ or not to simplify
enough. In this paper we outline the ‘‘unwinding number’’ approach to
such problems, and show how it can be used to prevent errors and to
systematise such simplification, even though we have not yet reduced
the simplification process to a complete algorithm. The unsolved
problems are probably more amenable to the techniques of artificial
intelligence and theorem proving than the original problem of complex
variable analysis.",
paper = "Brad02.pdf"
}

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— axiom.bib —

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@inproceedings{Chyz11,
  author = "Chyzak, Frederic and Davenport, James H. and
            Koutschan, Christoph and Salvy, Bruno",
  title = "On Kahan's Rules for Determining Branch Cuts",
  booktitle = "Proc. 13th Int. Symp. on Symbolic and Numeric Algorithms
              for Scientific Computing",
  year = "2011",
  isbn = "978-1-4673-0207-4",
  location = "Timisoara",
  pages = "47-51",
  doi = "10.1109/SYNASC.2011.51",
  acmid = "258794",
  publisher = "IEEE",
  abstract =
    "In computer algebra there are different ways of approaching the
    mathematical concept of functions, one of which is by defining them as
    solutions of differential equations. We compare different such
    appraoches and discuss the occurring problems. The main focus is on
    the question of determining possible branch cuts. We explore the
    extent to which the treatment of branch cuts can be rendered (more)
    algorithmic, by adapting Kahan's rules to the differential equation
    setting.",
  paper1 = "Chyz11a.pdf",
  paper = "Chyz11.pdf"
}

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— axiom.bib —

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@article{Dave10,
  author = "Davenport, James",
  title = {The Challenges of Multivalued "Functions"},
  journal = "Lecture Notes in Computer Science",
  volume = "6167",
  year = "2010",
  pages = "1-12",
  abstract = "
    Although, formally, mathematics is clear that a function is a
    single-valued object, mathematical practice is looser, particularly
    with n-th roots and various inverse functions. In this paper, we point
    out some of the looseness, and ask what the implications are, both for
    Artificial Intelligence and Symbolic Computation, of these practices.
    In doing so, we look at the steps necessary to convert existing tests
    into
    \begin{itemize}
    \item (a) rigorous statements
    \item (b) rigorously proved statements
    \end{itemize}
    In particular we ask whether there might be a constant ‘de Bruij factor’
    [18] as we make these texts more formal, and conclude that the answer
    depends greatly on the interpretation being placed on the symbols.",
  paper = "Dave10.pdf"
}

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— axiom.bib —

```

@article{Phis11,
  author = "Phisanbut, Nalina and Bradford, Russell J. and
    Davenport, James H.",
  title = "Geometry of branch cuts",
  journal = "ACM Communications in Computer Algebra",
  volume = "44",
  number = "3-4",
  pages = "132-135",
  year = "2011",
  abstract =
    "'Simplification' is a key concept in Computer Algebra. But many
    simplification rules, such as  $\sqrt{x}\sqrt{y} \rightarrow \sqrt{xy}$ 
    are not universally valid, due to the fact that many elementary
    functions are multi-valued. Hence a key question is ‘Is this
    simplification correct?’, which involves algorithmic analysis of the
    branch cuts involved. In this paper, we look at variable ordering and
    pre-conditioning as supporting technologies for this analysis.",
  paper = "Phis11.pdf"
}

```

— axiom.bib —

```
@article{Dave12,
  author = "Davenport, James H. and Bradford, Russell and England, Matthew
           and Wilson, David",
  title = "Program Verification in the presence of complex numbers, functions
           with branch cuts etc",
  journal = "14th Int. Symp. on Symbolic and Numeric Algorithms for
            Scientific Computing",
  year = "2012",
  series = "SYNASC'12",
  pages = "83-88",
  publisher = "IEEE",
  abstract = "
    In considering the reliability of numerical programs, it is normal to
    ‘‘limit our study to the semantics dealing with numerical precision’’.
    On the other hand, there is a great deal of work on the reliability of
    programs that essentially ignores the numerics. The thesis of this
    paper is that there is a class of problems that fall between the two,
    which could be described as ‘‘does the low-level arithmetic implement
    the high-level mathematics’’. Many of these problems arise because
    mathematics, particularly the mathematics of the complex numbers, is
    more difficult than expected; for example the complex function log is
    not continuous, writing down a program to compute an inverse function
    is more complicated than just solving an equation, and many algebraic
    simplification rules are not universally valid.

    The good news is that these problems are theoretically capable of
    being solved, and are practically close to being solved, but not yet
    solved, in several real-world examples. However, there is still a long
    way to go before implementations match the theoretical possibilities.",
  paper = "Dave12.pdf"
}
```

— axiom.bib —

```
@article{Eng113,
  author = "England, M. and Bradford, R. and Davenport, J. H. and
           Wilson, D.",
  title = "Understanding Branch Cuts of Expressions",
  journal = "Intelligent Computer Mathematics",
```

```

year = "2013",
series = " LNCS 7961",
publisher = "Springer, Berlin",
pages = "136-151",
isbn = "9783642393198",
abstract = "
  We assume some standard choices for the branch cuts of a group of
  functions and consider the problem of then calculating the branch cuts
  of expressions involving those functions. Typical examples include the
  addition formulae for inverse trigonometric functions. Understanding
  these cuts is essential for working with the single-valued
  counterparts, the common approach to encoding multi-valued functions
  in computer algebra systems. While the defining choices are usually
  simple (typically portions of either the real or imaginary axes) the
  cuts induced by the expression may be surprisingly complicated. We
  have made explicit and implemented techniques for calculating the cuts
  in the computer algebra programme Maple. We discuss the issues raised,
  classifying the different cuts produced. The techniques have been
  gathered in the BranchCuts package, along with tools for visualising
  the cuts. The package is included in Maple 17 as part of the
  FunctionAdvisor tool.",
paper = "Engl13.pdf"
}

```

— axiom.bib —

```

@article{Jeff04,
  author = "Jeffrey, D. J. and Norman, A. C.",
  title = "Not Seeing the Roots for the Branches: Multivalued Functions in
    Computer Algebra",
  journal = "SIGSAM Bull.",
  issue_date = "September 2004",
  volume = "38",
  number = "3",
  month = "September",
  year = "2004",
  issn = "0163-5824",
  pages = "57--66",
  numpages = "10",
  link = "\url{http://doi.acm.org/10.1145/1040034.1040036}",
  doi = "10.1145/1040034.1040036",
  acmid = "1040036",
  publisher = "ACM",
  address = "New York, NY, USA",
  abstract = "
    We discuss the multiple definitions of multivalued functions and their

```



```

suitability for computer algebra systems. We focus the discussion by
taking one specific problem and considering how it is solved using
different definitions. Our example problem is the classical one of
calculating the roots of a cubic polynomial from the Cardano formulae,
which contains fractional powers. We show that some definitions of
these functions result in formulae that are correct only in the sense
that they give candidates for solutions; these candidates must then be
tested. Formulae that are based on single-valued functions, in
contract, are efficient and direct.",
paper = "Jeff04.pdf"
}

```

— axiom.bib —

```

@inproceedings{Kaha86,
  author = "Kahan, W.",
  title = "Branch cuts for complex elementary functions",
  booktitle = "The State of the Art in Numerical Analysis",
  year = "1986",
  month = "April",
  editor = "Powell, M.J.D and Iserles, A.",
  publisher = "Oxford University Press",
  paper1 = "Kaha86a.pdf",
  paper = "Kaha86.pdf"
}

```

— axiom.bib —

```

@article{Rich96,
  author = "Rich, Albert D. and Jeffrey, David J.",
  title = "Function Evaluation on Branch Cuts",
  journal = "SIGSAM Bull.",
  issue_date = "June 1996",
  volume = "30",
  number = "2",
  month = "June",
  year = "1996",
  issn = "0163-5824",
  pages = "25--27",
  numpages = "3",
  link = "\url{http://doi.acm.org/10.1145/235699.235704}",
  doi = "10.1145/235699.235704",
}

```

```

acmid = "235704",
publisher = "ACM",
address = "New York, NY, USA",
abstract = "
  Once it is decided that a CAS will evaluate multivalued functions on
  their principal branches, questions arise concerning the branch
  definitions. The first questions concern the standardization of the
  positions of the branch cuts. These questions have largely been
  resolved between the various algebra systems and the numerical
  libraries, although not completely. In contrast to the computer
  systems, many mathematical textbooks are much further behind: for
  example, many popular textbooks still specify that the argument of a
  complex number lies between 0 and  $2\pi$ . We do not intend to discuss
  these first questions here, however. Once the positions of the branch
  cuts have been fixed, a second set of questions arises concerning the
  evaluation of functions on their branch cuts."
}

```

— axiom.bib —

```

@article{Patt96,
  author = "Patton, Charles M.",
  title = "A Representation of Branch-cut Information",
  journal = "SIGSAM Bull.",
  issue_date = "June 1996",
  volume = "30",
  number = "2",
  month = "June",
  year = "1996",
  issn = "0163-5824",
  pages = "21--24",
  numpages = "4",
  link = "\url{http://doi.acm.org/10.1145/235699.235703}",
  doi = "10.1145/235699.235703",
  acmid = "235703",
  publisher = "ACM",
  address = "New York, NY, USA",
  abstract = "
    Handling (possibly) multi-valued functions is a problem in all current
    computer algebra systems. The problem is not an issue of technology.
    Its solution, however, is tied to a uniform handling of the issues by
    the mathematics community.",
  paper = "Patt96.pdf"
}

```

— axiom.bib —

```
@article{Squi91,
  author = "Squire, Jon S.",
  title = "Rationale for the Proposed Standard for a Generic Package of
    Complex Elementary Functions",
  journal = "Ada Lett.",
  issue_date = "Fall 1991",
  volume = "XI",
  number = "7",
  month = "September",
  year = "1991",
  issn = "1094-3641",
  pages = "166--179",
  numpages = "14",
  link = "\url{http://doi.acm.org/10.1145/123533.123545}",
  doi = "10.1145/123533.123545",
  acmid = "123545",
  publisher = "ACM",
  address = "New York, NY, USA",
  abstract = "
    This document provides the background on decisions that were made
    during the development of the specification for Generic Complex
    Elementary functions. It also provides some information that was used to
    develop error bounds, range, domain and definitions of complex
    elementary functions.",
  paper = "Squi91.pdf"
}
```

— axiom.bib —

```
@article{Squi91a,
  author = "Squire, Jon S.",
  title = "Proposed Standard for a Generic Package of Complex
    Elementary Functions",
  journal = "Ada Lett.",
  issue_date = "Fall 1991",
  volume = "XI",
  number = "7",
  month = "September",
  year = "1991",
  issn = "1094-3641",
  pages = "140--165",
```

```

numpages = "26",
link = "\url{http://doi.acm.org/10.1145/123533.123544}",
doi = "10.1145/123533.123544",
acmid = "123544",
publisher = "ACM",
address = "New York, NY, USA",
abstract = "
  This document defines the specification of a generic package of
  complex elementary functions called Generic Complex Elementary
  Functions. It does not provide the body of the package."
}

```

— axiom.bib —

```

@misc{Unkn15,
  author = "Unknown",
  title = "Branches of Functions",
  link = "\url{http://scipp.ucsc.edu/~haber/ph116A/ComplexFunBranchTheory.pdf}",
  paper = "Unkn15.pdf"
}

```

1.32 Square-free Decomposition

— axiom.bib —

```

@article{Bern97,
  author = "Bernardin, Laurent",
  title = "On square-free factorization of multivariate polynomials over a
    finite field",
  journal = "Theoretical Computer Science",
  volume = "187",
  number = "1-2",
  year = "1997",
  month = "November",
  pages = "105-116",
  abstract = "
    In this paper we present a new deterministic algorithm for computing
    the square-free decomposition of multivariate polynomials with
    coefficients from a finite field.
  "
}

```

Our algorithm is based on Yun's square-free factorization algorithm for characteristic 0. The new algorithm is more efficient than existing, deterministic algorithms based on Musser's squarefree algorithm

We will show that the modular approach presented by Yun has no significant performance advantage over our algorithm. The new algorithm is also simpler to implement and it can rely on any existing GCD algorithm without having to worry about choosing 'good' evaluation points.

To demonstrate this, we present some timings using implementations in Maple (Char et al. 1991), where the new algorithm is used for Release 4 onwards, and Axiom (Jenks and Sutor, 1992) which is the only system known to the author to use and implementation of Yun's modular algorithm mentioned above.",

```
paper = "Bern97.pdf",
keywords = "axiomref"
}
```

— axiom.bib —

```
@article{Chez07,
  author = "Ch'eze, Guillaume and Lecerf, Gr'egoire",
  title = "Lifting and recombination techniques for absolute factorization",
  journal = "Journal of Complexity",
  volume = "23",
  number = "3",
  year = "2007",
  month = "June",
  pages = "380-420",
  abstract = "
    In the vein of recent algorithmic advances in polynomial factorization
    based on lifting and recombination techniques, we present new faster
    algorithms for computing the absolute factorization of a bivariate
    polynomial. The running time of our probabilistic algorithm is less
    than quadratic in the dense size of the polynomial to be factored.",
  paper = "Chez07.pdf"
}
```

— axiom.bib —

```

@article{Lece07,
  author = "Lecerf, Gr\'egoire",
  title = "Improved dense multivariate polynomial factorization algorithms",
  journal = "Journal of Symbolic Computation",
  volume = "42",
  number = "4",
  year = "2007",
  month = "April",
  pages = "477-494",
  abstract = "
    We present new deterministic and probabilistic algorithms that reduce
    the factorization of dense polynomials from several variables to one
    variable. The deterministic algorithm runs in sub-quadratic time in
    the dense size of the input polynomial, and the probabilistic
    algorithm is softly optimal when the number of variables is at least
    three. We also investigate the reduction from several to two variables
    and improve the quantitative versions of Bertini's irreducibility
    theorem.",
  paper = "Lece07.pdf"
}

```

— axiom.bib —

```

@article{Wang77,
  author = "Wang, Paul S.",
  title = "An efficient squarefree decomposition algorithm",
  journal = "ACM SIGSAM Bulletin",
  volume = "11",
  number = "2",
  year = "1977",
  month = "May",
  pages = "4-6",
  abstract = "
    The concept of polynomial squarefree decomposition is an important one
    in algebraic computation. The squarefree decomposition process has
    many uses in computer symbolic computation. A recent survey by D. Yun
    [3] describes many useful algorithms for this purpose. All of these
    methods depend on computing the greatest common divisor (gcd) of the
    polynomial to be decomposed and its first derivative (with respect to
    some variable). In the multivariate case, this gcd computation is
    non-trivial and dominates the cost for the squarefree decomposition.",
  paper = "Wang77.pdf"
}

```

— axiom.bib —

```
@article{Wang79,
  author = "Wang, Paul S. and Trager, Barry M.",
  title = "New Algorithms for Polynomial Square-Free Decomposition
    over the Integers",
  journal = "SIAM Journal on Computing",
  volume = "8",
  number = "3",
  year = "1979",
  publisher = "Society for Industrial and Applied Mathematics",
  issn = "00975397",
  abstract = "
    Previously known algorithms for polynomial square-free decomposition
    rely on greatest common divisor (gcd) computations over the same
    coefficient domain where the decomposition is to be performed. In
    particular, gcd of the given polynomial and its first derivative (with
    respect to some variable) is obtained to begin with. Application of
    modular homomorphism and  $p$ -adic construction (multivariate case) or
    the Chinese remainder algorithm (univariate case) results in new
    square-free decomposition algorithms which, generally speaking, take
    less time than a single gcd between the given polynomial and its first
    derivative. The key idea is to obtain one or several ‘‘correct’’
    homomorphic images of the desired square-free decomposition
    first. This provides information as to how many different square-free
    factors there are, their multiplicities and their homomorphic
    images. Since the multiplicities are known, only the square-free
    factors need to be constructed. Thus, these new algorithms are
    relatively insensitive to the multiplicities of the square-free factors.",
  paper = "Wang79.pdf"
}
```

— axiom.bib —

```
@inproceedings{Yun76,
  author = "Yun, D.Y.Y",
  title = "On square-free decomposition algorithms",
  booktitle = "Proceedings of SYMSAC'76",
  year = "1976",
  keywords = "survey",
  pages = "26-35"
}
```

1.33 Symbolic Summation

— axiom.bib —

```
@article{Abra71,
  author = "Abramov, S.A.",
  title = "On the summation of rational functions",
  year = "1971",
  journal = "USSR Computational Mathematics and Mathematical Physics",
  volume = "11",
  number = "4",
  pages = "324--330",
  abstract = "
    An algorithm is given for solving the following problem: let
     $F(x_1, \dots, x_n)$  be a rational function of the variables
     $x_i$  with rational (real or complex) coefficients; to see if
    there exists a rational function  $G(v, w, x_2, \dots, x_n)$  with
    coefficients from the same field, such that
    
$$\sum_{x_1=v}^w F(x_1, \dots, x_n) = G(v, w, x_2, \dots, x_n)$$

    for all integral values of  $v \leq w$ . If  $G$  exists, to obtain it.
    Realization of the algorithm in the LISP language is discussed.",
  paper = "Abra71.pdf"
}
```

— axiom.bib —

```
@article{Gosp78,
  author = "Gosper, R. William",
  title = "Decision procedure for indefinite hypergeometric summation",
  year = "1978",
  journal = "Proc. Natl. Acad. Sci. USA",
  volume = "75",
  number = "1",
  pages = "40--42",
  month = "January",
  abstract = "
    Given a summand  $a_n$ , we seek the ‘indefinite sum’  $S(n)$ 
    determined (within an additive constant) by
    
$$\sum_{n=1}^m a_n = S(m) - S(0)$$

    or, equivalently, by
    
$$a_n = S(n) - S(n-1)$$

    An algorithm is exhibited which, given  $a_n$ , finds those  $S(n)$ 
    with the property
    
$$\frac{S(n)}{S(n-1)} = \text{a rational function of } n$$


```



```

With this algorithm, we can determine, for example, the three
identities
\[\displaystyle\sum_{n=1}^m\{
\frac{\displaystyle\prod_{j=1}^{n-1}\{bj^2+cj+d\}}
{\displaystyle\prod_{j=1}^n\{bj^2+cj+e\}}=
\frac{1-\{\displaystyle\prod_{j=1}^m\{\frac{bj^2+cj+d}{bj^2+cj+e}\}\}\{e-d\}}{\}
\[\displaystyle\sum_{n=1}^m\{
\frac{\displaystyle\prod_{j=1}^{n-1}\{aj^3+bj^2+cj+d\}}
{\displaystyle\prod_{j=1}^n\{aj^3+bj^2+cj+e\}}=
\frac{1-\{\displaystyle\prod_{j=1}^m\{
\frac{aj^3+bj^2+cj+d}{aj^3+bj^2+cj+e}\}\}\{e-d\}}{\}
\[\displaystyle\sum_{n=1}^m\{
\displaystyle\frac{\displaystyle\prod_{j=1}^{n-1}\{bj^2+cj+d\}}
{\displaystyle\prod_{j=1}^{n+1}\{bj^2+cj+e\}}=
\displaystyle\frac{
\displaystyle\frac{2b}{\{e-d\}}-
\displaystyle\frac{3b+c+d-e}{\{b+c+e\}}-
\left(
\displaystyle\frac{2b}{\{e-d\}}-\frac{b(2m+3)+c+d-e}{\{b(m+1)^2+c(m+1)+e\}}
\right)
\displaystyle\prod_{j=1}^m\{\frac{bj^2+cj+d}{bj^2+cj+e}\}\}
\{b^2-c^2+d^2+e^2+2bd-2de+2eb\}}{\},
paper = "Gosp78.pdf"
}

```

— axiom.bib —

```

@article{Karr81,
  author = "Karr, Michael",
  title = "Summation in Finite Terms",
  journal = "Journal Association for Computing Machinery",
  year = "1981",
  volume = "28",
  number = "2",
  month = "April",
  issn = "0004-5411",
  pages = "305--350",
  link = "\url{http://doi.acm.org/10.1145/322248.322255}",
  publisher = "ACM",
  abstract = "
    Results which allow either the computation of symbolic solutions to
    first-order linear difference equations or the determination that
    solutions of a certain form do not exist are presented. Starting with
    a field of constants, larger fields may be constructed by the formal
    adjunction of symbols which behave like solutions to first-order
    linear equations (with a few restrictions). It is in these extension
  "
}

```

```

fields that the difference equations may be posed and in which the
solutions are requested. The principal application of these results is
in finding formulas for a broad class of finite sums or in showing the
nonexistence of such formula.",
paper = "Karr81"
}

```

— axiom.bib —

```

@book{Lafo82,
  author = "Lafon, J.C.",
  title = "Summation in Finite Terms",
  year = "1982",
  publisher = "Springer-Verlag",
  isbn = "3-211-81776-X",
  pages = "71--77",
  abstract = "
    A survey on algorithms for summation in finite terms is given. After a
    precise definition of the problem the cases of polynomial and rational
    summands are treated. The main concern of this paper is a description
    of Gosper's algorithm, which is applicable for a wide class of
    summands. Karr's theory of extension difference fields and some
    heuristic techniques are touched on briefly.",
  keywords = "axiomref,survey"
}

```

— axiom.bib —

```

@article{Abra85,
  author = "Abramov, S.A.",
  title = "Separation of variables in rational functions",
  year = "1985",
  journal = "USSR Computational Mathematics and Mathematical Physics",
  volume = "25",
  number = "5",
  pages = "99--102",
  abstract = "
    The problem of expanding a rational function of several variables into
    terms with separable variables is formulated. An algorithm for solving
    this problem is given. Programs which implement this algorithm can
    occur in sets of algebraic alphabetical transformations on a computer
    and can be used to reduce the multiplicity of sums and integrals of

```

```

    rational functions for investigating differential equations with
    rational right-hand sides etc.",
    paper = "Abra85.pdf"
}

```

— axiom.bib —

```

@Article{Karr85,
  author = "Karr, Michael",
  title = "Theory of Summation in Finite Terms",
  year = "1985",
  journal = "Journal of Symbolic Computation",
  volume = "1",
  number = "3",
  month = "September",
  pages = "303-315",
  abstract = "
    This paper discusses some of the mathematical aspects of an algorithm
    for finding formulas for finite sums. The results presented here
    concern a property of difference fields which show that the algorithm
    does not divide by zero, and an analogue to Liouville's theorem on
    elementary integrals.",
  paper = "Karr85.pdf"
}

```

— axiom.bib —

```

@book{Koep98,
  author = "Koepf, Wolfram",
  title = "Hypergeometric Summation",
  publisher = "Springer",
  year = "1998",
  isbn = "978-1-4471-6464-7",
  abstract = "
    Modern algorithmic techniques for summation, most of which were
    introduced in the 1990s, are developed here and carefully implemented
    in the computer algebra system Maple.

    The algorithms of Fasenmyer, Gosper, Zeilberger, Petkovsek and van
    Hoeij for hypergeometric summation and recurrence equations, efficient
    multivariate summation as well as q-analogues of the above algorithms
    are covered. Similar algorithms concerning differential equations are
  "
}

```

considered. An equivalent theory of hyperexponential integration due to Almkvist and Zeilberger completes the book.

The combination of these results gives orthogonal polynomials and (hypergeometric and q-hypergeometric) special functions a solid algorithmic foundation. Hence, many examples from this very active field are given.

The materials covered are suitable for an introductory course on algorithmic summation and will appeal to students and researchers alike.",

```
paper = "Koep98.pdf"
}
```

— axiom.bib —

```
@InProceedings{Schn00,
  author = "Schneider, Carsten",
  title = "An implementation of Karr's summation algorithm in Mathematica",
  year = "2000",
  booktitle = "S\'eminaire Lotharingien de Combinatoire",
  volume = "S43b",
  pages = "1-10",
  abstract = "
    Implementations of the celebrated Gosper algorithm (1978) for
    indefinite summation are available on almost any computer algebra
    platform. We report here about an implementation of an algorithm by
    Karr, the most general indefinite summation algorithm known. Karr's
    algorithm is, in a sense, the summation counterpart of Risch's
    algorithm for indefinite integration. This is the first implementation
    of this algorithm in a major computer algebra system. Our version
    contains new extensions to handle also definite summation problems. In
    addition we provide a feature to find automatically appropriate
    difference field extensions in which a closed form for the summation
    problem exists. These new aspects are illustrated by a variety of
    examples.",
  paper = "Schn00.pdf"
}
```

— axiom.bib —

```
@phdthesis{Schn01,
```

```

author = "Schneider, Carsten",
title = "Symbolic Summation in Difference Fields",
school = "RISC Research Institute for Symbolic Computation",
year = "2001",
link = "\url{http://www.risc.jku.at/publications/download/risc_3017/SymbSumTHESIS.pdf}",
abstract =
  "There are implementations of the celebrated Gosper algorithm (1978) on
  almost any computer algebra platform. Within my PhD thesis work I
  implemented Karr's Summation Algorithm (1981) based on difference
  field theory in the Mathematica system. Karr's algorithm is, in a
  sense, the summation counterpart of Risch's algorithm for indefinite
  integration. Besides Karr's algorithm which allows us to find closed
  forms for a big class of multisums, we developed new extensions to
  handle also definite summation problems. More precisely we are able to
  apply creative telescoping in a very general difference field setting
  and are capable of solving linear recurrences in its context.

  Besides this we find significant new insights in symbolic summation by
  rephrasing the summation problems in the general difference field
  setting. In particular, we designed algorithms for finding appropriate
  difference field extensions to solve problems in symbolic summation.
  For instance we deal with the problem to find all nested sum
  extensions which provide us with additional solutions for a given
  linear recurrence of any order. Furthermore we find appropriate sum
  extensions, if they exist, to simplify nested sums to simpler nested
  sum expressions. Moreover we are able to interpret creative
  telescoping as a special case of sum extensions in an indefinite
  summation problem. In particular we are able to determine sum
  extensions, in case of existence, to reduce the order of a recurrence
  for a definite summation problem.",
paper = "Schn01.pdf"
}

```

— axiom.bib —

```

@phdthesis{Scho95,
  author = "Schorn, Markus",
  title = "Contributions to Symbolic Summation",
  school = "Johannes Kepler University, RISC",
  year = "1995",
  link = "\url{http://www.risc.jku.at/publications/download/risc_2246/diplom.pdf}",
  paper = "Scho95.pdf"
}

```

— axiom.bib —

```
@inproceedings{Gerh03,
  author = "Gerhard, J. and Giesbrecht, M. and Storjohann, A. and Zima, E.V.",
  title = "Shiftless decomposition and polynomial-time rational summation",
  booktitle = "Proceedings of ISSAC'03",
  year = "2003",
  pages = "119--126",
  abstract = "
    New algorithms are presented for computing the dispersion set of two
    polynomials over  $\mathbb{Q}$  and for shiftless factorization. Together
    with a summability criterion by Abramov, these are applied to get a
    polynomial-time algorithm for indefinite rational summation, using a
    sparse representation of the output.",
  paper = "Gerh03.pdf"
}
```

— axiom.bib —

```
@article{Schn05,
  author = "Schneider, Carsten",
  title = "A new Sigma approach to multi-summation",
  year = "2005",
  journal = "Advances in Applied Mathematics",
  volume = "34",
  number = "4",
  pages = "740--767",
  abstract = "
    We present a general algorithmic framework that allows not only to
    deal with summation problems over summands being rational expressions
    in indefinite nested syms and products (Karr, 1981), but also over
     $\delta$ -finite and holonomic summand expressions that are given by a
    linear recurrence. This approach implies new computer algebra tools
    implemented in Sigma to solve multi-summation problems efficiently.
    For instance, the extended Sigma package has been applied successively
    to provide a computer-assisted proof of Stembridge's TSPP Theorem.",
  paper = "Schn05.pdf"
}
```

— axiom.bib —

```

@article{Kaue08a,
  author = "Kauers, Manuel and Schneider, Carsten",
  title = "Indefinite summation with unspecified summands",
  year = "2006",
  journal = "Discrete Mathematics",
  volume = "306",
  number = "17",
  pages = "2073--2083",
  abstract = "
    We provide a new algorithm for indefinite nested summation which is
    applicable to summands involving unspecified sequences  $x(n)$ . More
    than that, we show how to extend Karr's algorithm to a general
    summation framework by which additional types of summand expressions
    can be handled. Our treatment of unspecified sequences can be seen as
    a first illustrative application of this approach.",
  paper = "Kaue08a.pdf"
}

```

— axiom.bib —

```

@article{Kaue07,
  author = "Kauers, Manuel",
  title = "Summation algorithms for Stirling number identities",
  year = "2007",
  journal = "Journal of Symbolic Computation",
  volume = "42",
  number = "10",
  month = "October",
  pages = "948--970",
  abstract = "
    We consider a class of sequences defined by triangular recurrence
    equations. This class contains Stirling numbers and Eulerian numbers
    of both kinds, and hypergeometric multiples of those. We give a
    sufficient criterion for sums over such sequences to obey a recurrence
    equation, and present algorithms for computing such recurrence
    equations efficiently. Our algorithms can be used for verifying many
    known summation identities on Stirling numbers instantly, and also for
    discovering new identities.",
  paper = "Kaue07.pdf"
}

```

— axiom.bib —

```

@InProceedings{Schn07,
  author = "Schneider, Carsten",
  title = "Symbolic Summation Assists Combinatorics",
  year = "2007",
  booktitle = "S\'eminaire Lotharingien de Combinatoire",
  volume = "56",
  article = "B56b",
  abstract = "
    We present symbolic summation tools in the context of difference
    fields that help scientists in practical problem solving. Throughout
    this article we present multi-sum examples which are related to
    combinatorial problems.",
  paper = "Schn07.pdf"
}

```

— axiom.bib —

```

@article{Schn08,
  author = "Schneider, Carsten",
  title = "A refined difference field theory for symbolic summation",
  year = "2008",
  journal = "Journal of Symbolic Computation",
  volume = "43",
  number = "9",
  pages = "611--644",
  abstract = "
    In this article we present a refined summation theory based on Karr's
    difference field approach. The resulting algorithms find sum
    representations with optimal nested depth. For instance, the
    algorithms have been applied successively to evaluate Feynman
    integrals from Perturbative Quantum Field Theory",
  paper = "Schn08.pdf"
}

```

— axiom.bib —

```

@article{Schn09,
  author = "Schneider, Carsten",
  title = "Structural theorems for symbolic summation",
  journal = "Proc. AAECC-2010",
  year = "2010",
  volume = "21",

```



```

pages = "1--32",
abstract = "
    Starting with Karr's structural theorem for summation - the discrete
    version of Liouville's structural theorem for integration - we work
    out crucial properties of the underlying difference fields. This leads
    to new and constructive structural theorems for symbolic summation.
    E.g., these results can be applied for harmonic sums which arise
    frequently in particle physics.",
paper = "Schn09.pdf"
}

```

— axiom.bib —

```

@article{Eroc10,
  author = {Er\ocal, Bur\c{in}},
  title = "Summation in Finite Terms Using Sage",
  journal = "ACM Commun. Comput. Algebra",
  volume = "44",
  number = "3/4",
  month = "January",
  year = "2011",
  issn = "1932-2240",
  pages = "190--193",
  link = "\url{http://doi.acm.org/10.1145/1940475.1940517}",
  publisher = "ACM",
  abstract = "
    The summation analogue of the Risch integration algorithm developed by
    Karr uses towers of difference fields to model nested indefinite sums
    and products, as the Risch algorithm uses towers of differential
    fields to model the so called {\sl elementary functions}. The
    algorithmic machinery developed by Karr, and later generalized and
    extended, allows one to find solutions of first order difference
    equations over such towers of difference fields, in turn simplifying
    expressions involving sums and products.

    We present an implementation of this machinery in the open source
    computer algebra system Sage. Due to the nature of open source
    software, this allows direct experimentation with the algorithms and
    structures involved while taking advantage of the state of the art
    primitives provided by Sage. Even though these methods are used behind
    the scenes in the summation package Sigma and they were previously
    implemented, this is the first open source implementation.",
  paper = "Eroc10.pdf"
}

```

— axiom.bib —

```
@phdthesis{Eroc11,
  author = {Er\"ocal, Bur\"c{c}in},
  title = "Algebraic Extensions for Symbolic Summation",
  school = "RISC Research Institute for Symbolic Computation",
  year = "2011",
  link = "\url{http://www.risc.jku.at/publications/download/risc_4320/erocal_thesis.pdf}",
  abstract =
    "The main result of this thesis is an effective method to extend Karr's
    symbolic summation framework to algebraic extensions. These arise, for
    example, when working with expressions involving  $(-1)^n$ . An
    implementation of this method, including a modernised version of
    Karr's algorithm is presented.

    Karr's algorithm is the summation analogue of the Risch algorithm for
    indefinite integration. In the summation case, towers of specialized
    difference fields called  $\text{\$}\backslash\text{prod}\backslash\text{sum}\text{\$}$ -fields are used to model nested
    sums and products. This is similar to the way elementary functions
    involving nested logarithms and exponentials are represented in
    differential fields in the integration case.

    In contrast to the integration framework, only transcendental
    extensions are allowed in Karr's construction. Algebraic extensions of
     $\text{\$}\backslash\text{prod}\backslash\text{sum}\text{\$}$ -fields can even be rings with zero divisors. Karr's
    methods rely heavily on the ability to solve first-order linear
    difference equations and they are no longer applicable over these
    rings.

    Based on Bronstein's formulation of a method used by Singer for the
    solution of differential equations over algebraic extensions, we
    transform a first-order linear equation over an algebraic extension to
    a system of first-order equations over a purely transcendental
    extension field. However, this domain is not necessarily a
     $\text{\$}\backslash\text{prod}\backslash\text{sum}\text{\$}$ -field. Using a structure theorem by Singer and van der
    Put, we reduce this system to a single first-order equation over a
     $\text{\$}\backslash\text{prod}\backslash\text{sum}\text{\$}$ -field, which can be solved by Karr's algorithm. We also
    describe how to construct towers of difference ring extensions on an
    algebraic extension, where the same reduction methods can be used.

    A common bottleneck for symbolic summation algorithms is the
    computation of nullspaces of matrices over rational function
    fields. We present a fast algorithm for matrices over  $\text{\$}\backslash\text{mathbb{Q}}\text{\$}(x)\text{\$}$ 
    which uses fast arithmetic at the hardware level with calls to BLAS
    subroutines after modular reduction. This part is joint work with Arne
    Storjohann.",
  paper = "Eroc11.pdf"
```

}

 — axiom.bib —

```
@article{Poly11,
  author = "Polyadov, S.P.",
  title = "Indefinite summation of rational functions with factorization
    of denominators",
  year = "2011",
  month = "November",
  journal = "Programming and Computer Software",
  volume = "37",
  number = "6",
  pages = "322--325",
  abstract = "
    A computer algebra algorithm for indefinite summation of rational
    functions based on complete factorization of denominators is
    proposed. For a given  $f$ , the algorithm finds two rational functions
     $g$ ,  $r$  such that  $f=g(x+1)-g(x)+r$  and the degree of the denominator
    of  $r$  is minimal. A modification of the algorithm is also proposed
    that additionally minimizes the degree of the denominator of
     $g$ . Computational complexity of the algorithms without regard to
    denominator factorization is shown to be  $O(m^2)$ , where  $m$  is the
    degree of the denominator of  $f$ .",
  paper = "Poly11.pdf"
}
```

 — axiom.bib —

```
@article{Schn13,
  author = "Schneider, Carsten",
  title =
    "Fast Algorithms for Refined Parameterized Telescoping in Difference Fields",
  journal = "CoRR",
  year = "2013",
  volume = "abs/1307.7887",
  keywords = "survey",
  abstract = "
    Parameterized telescoping (including telescoping and creative
    telescoping) and refined versions of it play a central role in the
    research area of symbolic summation. In 1981 Karr introduced
     $\sum$ -fields, a general class of difference fields, that enables
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one to consider this problem for indefinite nested sums and products
covering as special cases, e.g., the (q-)hypergeometric case and their
mixed versions. This survey article presents the available algorithms
in the framework of  $\prod/\sum$ -extensions and elaborates new results
concerning efficiency.",
paper = "Schn13.pdf"
}

```

— axiom.bib —

```

@article{Zima13,
  author = "Zima, Eugene V.",
  title = "Accelerating Indefinite Summation: Simple Classes of Summands",
  journal = "Mathematics in Computer Science",
  year = "2013",
  month = "December",
  volume = "7",
  number = "4",
  pages = "455--472",
  abstract = "
    We present the history of indefinite summation starting with classics
    (Newton, Montmort, Taylor, Stirling, Euler, Boole, Jordan) followed by
    modern classics (Abramov, Gosper, Karr) to the current implementation
    in computer algebra system Maple. Along with historical presentation
    we describe several ‘‘acceleration techniques’’ of algorithms for
    indefinite summation which offer not only theoretical but also
    practical improvements in running time. Implementations of these
    algorithms in Maple are compared to standard Maple summation tools",
  paper = "Zima13.pdf"
}

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— axiom.bib —

```

@misc{Schn14,
  author = "Schneider, Carsten",
  title = "A Difference Ring Theory for Symbolic Summation",
  year = "2014",
  abstract = "
    A summation framework is developed that enhances Karr's difference
    field approach. It covers not only indefinite nested sums and products
    in terms of transcendental extensions, but it can treat, e.g., nested
    products defined over roots of unity. The theory of the so-called

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```

$R\prod\sum*$_extensions is supplemented by algorithms that support the
construction of such difference rings automatically and that assist in
the task to tackle symbolic summation problems. Algorithms are
presented that solve parameterized telescoping equations, and more
generally parameterized first-order difference equations, in the given
difference ring. As a consequence, one obtains algorithms for the
summation paradigms of telescoping and Zeilberger's creative
telescoping. With this difference ring theory one obtains a rigorous
summation machinery that has been applied to numerous challenging
problems coming, e.g., from combinatorics and particle physics.",
paper = "Schn14.pdf"
}

```

— axiom.bib —

```

@phdthesis{Vazq14,
  author = "Vazquez-Trejo, Javier",
  title = "Symbolic Summation in Difference Fields",
  year = "2014",
  school = "Carnegie-Mellon University",
  abstract = "
    We seek to understand a general method for finding a closed form for a
    given sum that acts as its antidifference in the same way that an
    integral has an antiderivative. Once an antidifference is found, then
    given the limits of the sum, it suffices to evaluate the
    antidifference at the given limits. Several algorithms (by Karr and
    Schneider) exist to find antidifferences, but the papers describing
    these algorithms leave out several of the key proofs needed to
    implement the algorithms. We attempt to fill in these gaps and find
    that many of the steps to solve difference equations rely on being
    able to solve two problems: the equivalence problem and the homogenous
    group membership problem. Solving these two problems is essential to
    finding the polynomial degree bounds and denominator bounds for
    solutions of difference equations. We study Karr and Schneider's
    treatment of these problems and elaborate on the unproven parts of
    their work. Section 1 provides background material; section 2 provides
    motivation and previous work; Section 3 provides an outline of Karr's
    Algorithm; section 4 examines the Equivalence Problem, and section 5
    examines the Homogeneous Group Membership Problem. Section 6 presents
    some proofs for the denominator and polynomial bounds used in solving
    difference equations, and Section 7 gives some directions for future
    work.",
  paper = "Vazq14.pdf"
}

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— axiom.bib —

```
@book{Petk97,
  author = "Petkov\v{s}ek, Marko and Wilf, Herbert S. and
           Zeilberger, Doran",
  title = "A=B",
  publisher = "A.K. Peters, Ltd",
  year = "1997",
  paper = "Petk97.pdf"
}
```

— axiom.bib —

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@misc{Temml4,
  author = "Temme, N.M.",
  title = "Bernoulli Polynomials Old and New",
  abstract =
    "We consider two problems on generalized Bernoulli polynomials
     $B_n^u(z)$ . One is connected with defining functions instead of
    polynomials by making the degree  $n$  of the polynomial a complex
    variable. In the second problem we are concerned with the asymptotic
    behaviour of  $B_n^u(z)$  when the degree  $n$  tends to infinity.",
  paper = "Temml4.pdf"
}
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1.34 Differential Forms

— axiom.bib —

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@book{Cart06,
  author = {Cartan, Henri},
  title = {Differential Forms},
  year = "2006",
  location = {Mineola, N.Y},
  edition = {Auflage: Tra},
  isbn = {9780486450100},
  pagetotal = {166},
}
```

```

    publisher = {Dover Pubn Inc},
    date = {2006-05-26}
}

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— axiom.bib —

```

@book{Flan03,
  author = "Flanders, Harley",
  title = "Differential Forms with Applications to the Physical Sciences",
  year = "2003",
  location = "Mineola, N.Y",
  isbn = "9780486661698",
  pagetotal = "240",
  publisher = "Dover Pubn Inc",
  date = "2003-03-28",
  algebra = "\newline\refto{domain DERHAM DeRhamComplex}"
}

```

— axiom.bib —

```

@book{Whit12,
  author = {Whitney, Hassler},
  title =
    {Geometric Integration Theory: Princeton Mathematical Series, No. 21},
  year = "2012",
  isbn = {9781258346386},
  shorttitle = {Geometric Integration Theory},
  pagetotal = {402},
  publisher = {Literary Licensing, {LLC}},
  date = {2012-05-01}
}

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— axiom.bib —

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@book{Fede13,
  author = {Federer, Herbert},
  title = {Geometric Measure Theory},

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year = "2013",
location = {Berlin ; New York},
edition = {Reprint of the 1st ed. Berlin, Heidelberg, New York 1969},
isbn = {9783540606567},
pagetotal = {700},
publisher = {Springer},
date = {2013-10-04},
abstract =
  "This book is a major treatise in mathematics and is essential in the
  working library of the modern analyst. (Bulletin of the London
  Mathematical Society)"
}

```

— axiom.bib —

```

@book{Abra93,
  author = "Abraham, Ralph and Marsden, Jerrold E. and Ratiu, Tudor",
  title = "Manifolds, Tensor Analysis, and Applications",
  year = "1993",
  location = "New York",
  edition = "2nd Corrected ed. 1988. Corr. 2nd printing 1993",
  isbn = "9780387967905",
  pagetotal = "656",
  publisher = "Springer",
  date = "1993-08-26",
  abstract = "
    The purpose of this book is to provide core material in nonlinear
    analysis for mathematicians, physicists, engineers, and mathematical
    biologists. The main goal is to provide a working knowledge of
    manifolds, dynamical systems, tensors, and differential forms. Some
    applications to Hamiltonian mechanics, fluid mechanics,
    electromagnetism, plasma dynamics and control theory are given using
    both invariant and index notation. The prerequisites required are
    solid undergraduate courses in linear algebra and advanced calculus."
}

```

— axiom.bib —

```

@book{Lamb97a,
  author = {Lambe, L. A. and Radford, D. E.},
  title = {Introduction to the Quantum Yang-Baxter Equation and
    Quantum Groups: An Algebraic Approach},

```



```

year = "1997",
location = {Dordrecht ; Boston},
edition = {Auflage: 1997},
isbn = {9780792347217},
shorttitle = {Introduction to the Quantum Yang-Baxter Equation and
              Quantum Groups},
abstract = {
  Chapter 1 The algebraic prerequisites for the book are covered here
  and in the appendix. This chapter should be used as reference material
  and should be consulted as needed. A systematic treatment of algebras,
  coalgebras, bialgebras, Hopf algebras, and representations of these
  objects to the extent needed for the book is given. The material here
  not specifically cited can be found for the most part in [Sweedler,
  1969] in one form or another, with a few exceptions. A great deal of
  emphasis is placed on the coalgebra which is the dual of  $n \times n$ 
  matrices over a field. This is the most basic example of a coalgebra
  for our purposes and is at the heart of most algebraic constructions
  described in this book. We have found pointed bialgebras useful in
  connection with solving the quantum Yang-Baxter equation. For this
  reason we develop their theory in some detail. The class of examples
  described in Chapter 6 in connection with the quantum double consists
  of pointed Hopf algebras. We note the quantized enveloping algebras
  described Hopf algebras. Thus for many reasons pointed bialgebras are
  elsewhere are pointed of fundamental interest in the study of the
  quantum Yang-Baxter equation and objects quantum groups.},
pagetotal = {300},
publisher = {Springer},
date = {1997-10-31}
}

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— axiom.bib —

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@misc{Pag16,
  author = "Pagani, Kurt",
  title = "SurfaceComplex",
  year = "2016",
  link = "\url{https://groups.google.com/forum/\#!topic/fricas-devel/FRDGVFsoAKw}",
  abstract =
    "This manual describes the FriCAS domains {\bf CellMap} and
    {\bf SurfaceComplex}. These domains provide methods to compute various
    differential geometric properties of so-called  $\mathbb{R}^n$ -surfaces in
     $\mathbb{R}^n$ , a notion which is used by Walter Rudin in his famous
    {\sl Principles of Mathematical Analysis}."
  paper = "Pag16.pdf"
}

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— axiom.bib —

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@misc{Pagal6a,
  author = "Pagani, Kurt",
  title = "DifferentialGeometry1",
  year = "2016",
  link = "\url{https://groups.google.com/forum/\#!topic/fricas-devel/FRDGVFsoAKw}",
  abstract =
    "This manual describes the FriCAS package {\bf DifferentialGeometry1}.
    This package combines differential forms and cell mappings to provide
    methods which compute {\bf pull backs}, {\bf integrals} as well as some
    other quantities of differential forms living on a surface complex.",
  paper = "Pagal6a.pdf"
}
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— axiom.bib —

```
@misc{Pagal6b,
  author = "Pagani, Kurt",
  title = "DifferentialForms",
  year = "2016",
  link = "\url{https://groups.google.com/forum/\#!topic/fricas-devel/FRDGVFsoAKw}",
  abstract = "Reference manual for the package {\bf {\tt DifferentialForms}}",
  paper = "Pagal6b.pdf"
}
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— axiom.bib —

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@misc{Wheel2,
  author = "Wheeler, James T.",
  title = "Differential Forms",
  year = "2012",
  month = "September",
  link = "\url{http://www.physics.usu.edu/Wheeler/ClassicalMechanics/CMDifferentialForms.pdf}",
  paper = "Wheel2.pdf"
}
```

1.35 Cylindrical Algebraic Decomposition

— ignore —

```
\bibitem[Arnon 88]{Arno88} Arno, D.S.; Mignotte, M.
  title = "On Mechanical Quantifier Elimination for Elementary Algebra and Geometry",
  J. Symbolic Computation 5, 237-259 (1988)
  link = "\url{http://http://www.sciencedirect.com/science/article/pii/S0747717188800142/pdf?md5}
  abstract = "
    We give solutions to two problems of elementary algebra and geometry:
    (1) find conditions on real numbers  $p, q, r$  so that the
    polynomial function  $f(x)=x^4+px^2+qx+r$  is nonnegative for all real
     $x$  and (2) find conditions on real numbers  $a, b, c$  so that
    the ellipse  $\frac{(x-e)^2}{q^2}+\frac{y^2}{b^2}-1=0$  lies inside the
    unit circle  $y^2+x^2-1=0$ . Our solutions are obtained by following the
    basic outline of the method of quantifier elimination by cylindrical
    algebraic decomposition (Collins, 1975), but we have developed, and
    have been considerably aided by, modified versions of certain of its
    steps. We have found three equally simple but not obviously equivalent
    solutions for the first problem, illustrating the difficulty of
    obtaining unique ‘‘simplest’’ solutions to quantifier elimination
    problems of elementary algebra and geometry.",
  paper = "Arno88.pdf"
```

— axiom.bib —

```
@article{Beau07,
  author = "Beaumont, James C. and Bradford, Russell J. and
    Davenport, James H. and Phisanbut, Nalina",
  title = "Testing elementary function identities using CAD",
  journal = "Applicable Algebra in Engineering, Communication and Computing",
  year = "2007",
  volume = "18",
  number = "6",
  issn = "0938-1279",
  publisher = "Springer-Verlag",
  pages = "513-543",
  abstract = "
    One of the problems with manipulating function identities in computer
    algebra systems is that they often involve functions which are
    multivalued, whilst most users tend to work with single-valued
    functions. The problem is that many well-known identities may no
    longer be true everywhere in the complex plane when working with their
    single-valued counterparts. Conversely, we cannot ignore them, since
```

in particular contexts they may be valid. We investigate the practicality of a method to verify such identities by means of an experiment; this is based on a set of test examples which one might realistically meet in practice. Essentially, the method works as follows. We decompose the complex plane via means of cylindrical algebraic decomposition into regions with respect to the branch cuts of the functions. We then test the identity numerically at a sample point in the region. The latter step is facilitated by the notion of the {\sl adherence} of a branch cut, which was previously introduced by the authors. In addition to presenting the results of the experiment, we explain how adherence relates to the proposal of {\sl signed zeros} by W. Kahan, and develop this idea further in order to allow us to cover previously untreatable cases. Finally, we discuss other ways to improve upon our general methodology as well as topics for future research.",

```
paper = "Beau07.pdf"
}
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— axiom.bib —

```
@misc{Brad14,
  author = "Bradford, Russell and Chen, Changbo and Davenport, James H. and
    England, Matthew and Maza, Marc Moreno and Wilson, David",
  title = "Truth Table Invariant Cylindrical Algebraic Decomposition by
    Regular Chains",
  link = "\url{https://arxiv.org/pdf/1401.6310.pdf}",
  year = "2014",
  abstract =
    "A new algorithm to compute cylindrical algebraic decompositions
    (CADs) is presented, building on two recent advances. Firstly, the
    output is truth table invariant (a TTICAD) meaning given formulae have
    constant truth value on each cell of the decomposition. Secondly, the
    computation uses regular chains theory to first build a cylindrical
    decomposition of complex space (CCD) incrementally by polynomial.
    Significant modification of the regular chains technology was used to
    achieve the more sophisticated invariance criteria. Experimental
    results on an implementation in the {\tt RegularChains} Library for Maple
    verify that combining these advances gives an algorithm superior to
    its individual components and competitive with the state of the art.",
  paper = "Brad14.pdf"
}
```

— axiom.bib —

```
@misc{Brad15,
  author = "Bradford, Russell and Davenport, James H. and England, Matthew and
           McCallum, Scott",
  title = "Truth Table Invariant Cylindrical Algebraic Decomposition",
  link = "\url{https://arxiv.org/pdf/1401.0645.pdf}",
  year = "2015",
  abstract =
    "When using cylindrical algebraic decomposition (CAD) to solve a
    problem with respect to a set of polynomials, it is likely not the
    signs of those polynomials that are of paramount importance but rather
    the truth values of certain quantifier free formulae involving
    them. This observation motivates our article and definition of a Truth
    Table Invariant CAD (TTICAD). In ISSAC 2013 the current authors
    presented an algorithm that can efficiently and directly construct a
    TTICAD for a list of formulae in which each has an equational
    constraint. This was achieved by generalising McCallum's theory of
    reduced projection operators. In this paper we present an extended
    version of our theory which can be applied to an arbitrary list of
    formulae, achieving savings if at least one has an equational
    constraint. We also explain how the theory of reduced projection
    operators can allow for further improvements to the lifting phase of
    CAD algorithms, even in the context of a single equational constraint.
    The algorithm is implemented fully in Maple and we present both
    promising results from experimentation and a complexity analysis
    showing the benefits of our contributions.",
  paper = "Brad15.pdf"
}
```

— axiom.bib —

```
@phdthesis{Brow99,
  author = "Brown, Christopher W.",
  title = "Solution Formula Construction for Truth Invariant CADs",
  school = "University of Delaware",
  year = "1999",
  website = "http://www.usna.edu/CS/qepcadweb/B/impl/Implementation.html",
  link = "\url{http://www.usna.edu/Users/cs/wcbrown/research/thesis.ps.gz}",
  abstract =
    "The CAD-based quantifier elimination algorithm takes a formula from
    the elementary theory of real closed fields as input, and constructs a
    CAD of the space of the formula's unquantified variables. This
    decomposition is truth invariant with respect to the input formula,
    meaning that the formula is either identically true or identically
```

false in each cell of the decomposition. The method determines the truth of the input formula for each cell of the CAD, and then uses the CAD to construct a solution formula -- a quantifier free formula that is equivalent to the input formula. This final phase of the algorithm, the solution formula construction phase, is the focus of this thesis.

An optimal solution formula construction algorithm would be {\sl complete} -- i.e. applicable to any truth-invariant CAD, would be {\sl efficient}, and would produce {\sl simple} solution formulas. Prior to this thesis, no method was available with even two of these three properties. Several algorithms are presented, all addressing problems related to solution formula construction. In combination, these provide an efficient and complete method for constructing solution formulas that are simple in a variety of ways.

Algorithms presented in this thesis have been implemented using the SACLIB library, and integrated into QEPCAD, a SACLIB-based implementation of quantifier elimination by CAD. Example computations based on these implementations are discussed.",

```
paper = "Brow99.pdf"
}
```

— axiom.bib —

```
@misc{Brow01,
  author="Brown, Christopher W.",
  title="The McCallum projection, lifting, and order-invariance",
  year="2001",
  link = "\url{http://www.usna.edu/Users/cs/wcbrown/research/MOTS2001.1.ps.gz}",
  abstract =
    "The McCallum Projection for Cylindrical Algebraic Decomposition (CAD)
    produces a smaller projection factor set than previous projections,
    however it does not always produce a sign-invariant CAD for the set of
    input polynomials. Problems may arise when a ($k+1$)-level projection
    factor vanishes identically over a k-level cell. According to
    McCallum's paper, when this happens (and $k+1$ is not the highest
    level in the CAD) we do not know whether the projection is valid,
    i.e. whether or not a sign-invariant CAD for the set of input
    polynomials will be produced when lifting is performed in the usual
    way. When the $k$-level cell in question has dimension 0, McCallum
    suggests a modification of the lifting method that will ensure the
    validity of his projection, although to my knowledge this has never
    been implemented.
```

In this paper we give easily computable criteria that often allow us to conclude that McCallum's projection is valid even though a

projection factor vanishes identically over a cell. We also improve on McCallum's modified lifting method.

We've incorporated the ideas contained in the paper into QEPCAD, the most complete implementation of CAD. When McCallum's projection is invalid because of a projection factor not being order-invariant over a region on which it vanishes identically, at least a warning message ought to be issued. Currently, QEPCAD may print warning messages that are not needed, and may fail to print warning messages when they are needed. Our implementation in QEPCAD ensures that warning messages are printed when needed, and reduces the number of times warning messages are printed when not needed. Neither McCallum's modified lifting method nor our improvement of it have been implemented in QEPCAD. The design of the system would make implementing such a feature quite difficult.",

```
paper = "Brow01.pdf"
}
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— axiom.bib —

```
@misc{Brow02,
  author = "Brown, Christopher W.",
  title = "QEPCAD B -- A program for computing with semi-algebraic sets
          using CADs",
  year = "2002",
  abstract =
    "This report introduces QEPCAD B, a program for computing with real
    algebraic sets using cylindrical algebraic decomposition (CAD). QEPCAD
    B both extends and improves upon the QEPCAD system for quantifier
    elimination by partial cylindrical algebraic decomposition written by
    Hoon Hong in the early 1990s. This paper briefly discusses some of the
    improvements in the implementation of CAD and quantifier elimination
    vis CAD, and provides somewhat more detail on extensions to the system
    that go beyond quantifier elimination. The author is responsible for
    most of the extended features of QEPCAD B, but improvements to the
    basic CAD implementation and to the SACLIB library on which QEPCAD is
    based are the results of many people's work.",
  paper = "Brow02.pdf"
}
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— axiom.bib —

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@misc{Chen10,
  author = "Chen, Changbo and Davenport, James H. and May, John P. and
           Maza, Marc Moreno and Xia, Bican and Xiao, Rong",
  title = "Triangular Decomposition of Semi-algebraic Systems",
  year = "2010",
  link = "\url{https://arxiv.org/pdf/1002.4784.pdf}",
  abstract =
    "Regular chains and triangular decompositions are fundamental and
    well-developed tools for describing the complex solutions of
    polynomial systems. This paper proposes adaptations of these tools
    focusing on solutions of the real analogue: semi-algebraic systems.

    We show that any such system can be decomposed into finitely many
    regular semi-algebraic systems. We propose two specifications of such
    a decomposition and present corresponding algorithms. Under some
    assumptions, one type of decomposition can be computed in singly
    exponential time w.r.t. the number of variables. We implement our
    algorithms and the experimental results illustrate their
    effectiveness.",
  paper = "Chen10.pdf"
}
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— axiom.bib —

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@misc{Chen12,
  author = "Chen, Changbo and Maza, Marc Moreno",
  title = "An Incremental Algorithm for Computing Cylindrical Algebraic
           Decompositions",
  link = "\url{https://arxiv.org/pdf/1210.5543.pdf}",
  year = "2012",
  abstract =
    "In this paper, we propose an incremental algorithm for computing
    cylindrical algebraic decompositions. The algorithm consists of two
    parts: computing a complex cylindrical tree and refining this complex
    tree into a cylindrical tree in real space. The incrementality comes
    from the first part of the algorithm, where a complex cylindrical tree
    is constructed by refining a previous complex cylindrical tree with a
    polynomial constraint. We have implemented our algorithm in Maple. The
    experimentation shows that the proposed algorithm outperforms existing
    ones for many examples taken from the literature",
  paper = "Chen12.pdf"
}
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— axiom.bib —

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@InProceedings{Emir04,
  author = "Emiris, Ioannis Z. and Tsigaridas, Elias P.",
  title = "Comparing real algebraic numbers of small degree",
  booktitle = "12th annual European symposium",
  series = "ESA 2004",
  year = "2004",
  isbn = "3-540-23025-4",
  location = "Bergen, Norway",
  pages = "652-663",
  link = "\url{http://www-polsys.lip6.fr/~elias/files//et-esa-04.pdf}",
  algebra = "\newline\ref{domain RECLOSE RealClosure}",
  abstract =
    "We study polynomials of degree up to 4 over the rationals or a
    computable real subfield. Our motivation comes from the need to
    evaluate predicates in nonlinear computational geometry efficiently
    and exactly. We show a new method to compare real algebraic numbers by
    precomputing generalized Sturm sequences, thus avoiding iterative
    methods; the method, moreover handles all degenerate cases. Our first
    contribution is the determination of rational isolating points, as
    functions of the coefficients, between any pair of real roots. Our
    second contribution is to exploit invariants and Bezoutian
    subexpressions in writing the sequences, in order to reduce bit
    complexity. The degree of the tested quantities in the input
    coefficients is optimal for degree up to 3, and for degree 4 in
    certain cases. Our methods readily apply to real solving of pairs of
    quadratic equations, and to sign determination of polynomials over
    algebraic numbers of degree up to 4. Our third contribution is an
    implementation in a new module of library SYNAPS v2.1. It improves
    significantly upon the efficiency of certain publicly available
    implementations: Rioboos approach on AXIOM, the package of
    Guibas-Karavelas-Russel, and CORE v1.6, MAPLE v9, and SYNAPS
    v2.0. Some existing limited tests had shown that it is faster than
    commercial library LEDA v4.5 for quadratic algebraic numbers.",
  paper = "Emir04.pdf",
  keywords = "axiomref"
}
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— axiom.bib —

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@misc{Engl14,
  author = "England, Matthew and Wilson, David and Bradford, Russell and
    Davenport, James H.",
  title = "Using the Regular Chains Library to build cylindrical algebraic
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        decompositions by projecting and lifting",
link = "\url{https://arxiv.org/pdf/1405.6090.pdf}",
year = "2014",
abstract =
    "Cylindrical algebraic decomposition (CAD) is an important tool, both
    for quantifier elimination over the reals and a range of other
    applications. Traditionally, a CAD is built through a process of
    projection and lifting to move the problem within Euclidean spaces of
    changing dimension. Recently, an alternative approach which first
    decomposes complex space using triangular decomposition before
    refining to real space has been introduced and implemented within the
    Regular-Chains Library of Maple. We here describe a freely available
    package ProjectionCAD which utilises the routines within the
    RegularChains Library to build CADs by projection and lifting. We
    detail how the projection and lifting algorithms were modified to
    allow this, discuss the motivation and survey the functionality of the
    package.",
paper = "Engl14.pdf"
}

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— axiom.bib —

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@misc{Engl14a,
  author = "England, Matthew and Bradford, Russell and Davenport, James H. and
    Wilson, David",
  title = "Choosing a variable ordering for truth-table invariant cylindrical
    algebraic decomposition by incremental triangular decomposition",
  link = "\url{https://arxiv.org/pdf/1405.6094.pdf}",
  year = "2014",
  abstract =
    "Cylindrical algebraic decomposition (CAD) is a key tool for solving
    problems in real algebraic geometry and beyond. In recent years a new
    approach has been developed, where regular chains technology is used
    to first build a decomposition in complex space. We consider the
    latest variant of this which builds the complex decomposition
    incrementally by polynomial and produces CADs on whose cells a
    sequence of formulae are truth-invariant. Like all CAD algorithms the
    user must provide a variable ordering which can have a profound impact
    on the tractability of a problem. We evaluate existing heuristics to
    help with the choice for this algorithm, suggest improvements and then
    derive a new heuristic more closely aligned with the mechanics of the
    new algorithm.",
  paper = "Engl14a.pdf"
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@misc{Engl14b,
  author = "England, Matthew and Bradford, Russell and Chen, Changbo and
           Davenport, James H. and Maza, Marc Moreno",
  title = "Problem formulation for truth-table invariant cylindrical
           algebraic decomposition by incremental triangular decomposition",
  link = "\url{https://arxiv.org/pdf/1404.6371.pdf}",
  year = "2014",
  abstract =
    "Cylindrical algebraic decompositions (CADs) are a key tool for
    solving problems in real algebraic geometry and beyond. We recently
    presented a new CAD algorithm combining two advances: truth-table
    invariance, making the CAD invariant with respect to the truth of
    logical formulae rather than the signs of polynomials; and CAD
    construction by regular chains technology, where first a complex
    decomposition is constructed by refining a tree incrementally by
    constraint. We here consider how best to formulate problems for input
    to this algorithm. We focus on a choice (not relevant for other CAD
    algorithms) about the order in which constraints are presented. We
    develop new heuristics to help make this choice and thus allow the
    best use of the algorithm in practice. We also consider other choices
    of problem formulation for CAD, as discussed in C13M 2013, revisiting
    these in the context of the new algorithm.",
  paper = "Engl14b.pdf"
}
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— axiom.bib —

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@article{Engl15,
  author = "England, M. and Bradford, R. and Davenport, J. H.",
  title = "Improving the use of equational constraints in cylindrical
           algebraic decomposition",
  journal = "ISSAC 15",
  year = "2015",
  series = "LNCS 7961",
  publisher = "ACM",
  link = "\url{http://opus.bath.ac.uk/42451/}",
  abstract = "
    When building a cylindrical algebraic decomposition (CAD) savings can
    be made in the presence of an equational constraint (EC): an equation
    logically implied by a formula.
  "
```

```

The present paper is concerned with how to use multiple ECs,
propagating those in the input throughout the projection set. We
improve on the approach of McCallum in ISSAC 2001 by using the reduced
projection theory to make savings in the lifting phase (both to the
polynomials we lift with and the cells lifted over). We demonstrate
the benefits with worked examples and a complexity analysis.",
paper = "Engl15.pdf"
}

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— axiom.bib —

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@book{LaVa06,
  author = "LaValle, Steven M.",
  title = "Planning Algorithms",
  year = "2006",
  publisher = "Cambridge University Press"
}

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— axiom.bib —

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@article{Mahb07,
  author = "Mahboubi, Assia",
  title = "Implementing the cylindrical algebraic decomposition within the Coq system",
  journal = "Math. Struct. in Comp. Science",
  volume = "17",
  year = "2007",
  pages = "99-127",
  abstract = "The Coq system is a Curry-Howard based proof assistant.
  Therefore, it contains a full functional, strongly typed programming
  language, which can be used to enhance the system with powerful
  automation tools through the implementation of reflexive tactics.
  We present the implementation of a cylindrical algebraic decomposition
  algorithm within the Coq system, whose certification leads to a proof
  producing decision procedure for the first-order theory of real numbers.",
  paper = "Mahb07.pdf"
}

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— axiom.bib —

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@misc{Riobxx,
  author = "Rioboo, Renaud",
  title = "Cylindrical Algebraic Decomposition class notes",
  link = "\url{http://people.bath.ac.uk/masjhd/TRITA.pdf}",
  year = "2016",
  abstract =
    "This report describes techniques for resolving systems of polynomial
    equations and inequalities. The general technique is {\sl cylindrical
    algebraic decomposition}, which decomposes space into a number of
    regions, on each of which the equations and inequalities have the same
    sign. Most of the report is spent describing the algebraic and
    algorithmic pre-requisites (resultants, algebraic numbers, Sturm
    sequences, etc.), and then describing the method, first in two
    dimensions and then in an arbitrary number of dimensions",
  paper = "Riobxx.pdf"
}

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— axiom.bib —

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@InProceedings{Riob92,
  author = "Rioboo, Renaud",
  title = "Real algebraic closure of an ordered field,
    Implementation in Axiom",
  booktitle = "Proc. ISSAC 92",
  series = "ISSAC 92",
  year = "1992",
  isbn = "978-3-540-75186-1",
  location = "Berkeley, California",
  pages = "206-215",
  algebra = "\newline\refto{domain RECLOSE RealClosure}",
  abstract =
    "Real algebraic numbers appear in many Computer Algebra problems. For
    instance the determination of a cylindrical algebraic decomposition
    for an euclidean space requires computing with real algebraic numbers.
    This paper describes an implementation for computations with the real
    roots of a polynomial. This process is designed to be recursively
    used, so the resulting domain of computation is the set of all real
    algebraic numbers. An implementation for the real algebraic closure
    has been done in Axiom (previously called Scratchpad).",
  paper = "Riob92.pdf",
  keywords = "axiomref",
  beebe = "Rioboo:1992:RAC"
}

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— Rioboo:1992:RAC —

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@InProceedings{Rioboo:1992:RAC,
  author =      "Renaud Rioboo",
  title =      "Real algebraic closure of an ordered field,
                implementation in {Axiom}",
  crossref =    "Wang:1992:ISS",
  pages =      "206--215",
  year =       "1992",
  bibdate =    "Tue Sep 17 06:35:39 MDT 1996",
  bibsource =   "http://www.math.utah.edu/pub/tex/bib/axiom.bib",
  abstract =    "Real algebraic numbers appear in many computer algebra
                problems. For instance the determination of a
                cylindrical algebraic decomposition for an Euclidian
                space requires computing with real algebraic numbers.
                This paper describes an implementation for computations
                with the real roots of a polynomial. This process is
                designed to be recursively used, so the resulting
                domain of computation is the set of all real algebraic
                numbers. An implementation for the real algebraic
                closure has been done in Axiom (previously called
                Scratchpad).",
  acknowledgement = ack-nhfb,
  affiliation = "LITP, Univ. Pierre et Marie Curie, Paris, France",
  classification = "C4130 (Interpolation and function approximation);
                  C6130 (Data handling techniques); C7310 (Mathematics)",
  keywords =    "Axiom; Computer algebra; Cylindrical algebraic
                decomposition; Euclidian space; Ordered field;
                Polynomial; Real algebraic closure",
  language =    "English",
  thesaurus =    "Polynomials; Symbol manipulation",
}
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— axiom.bib —

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@misc{Wils14,
  author = "Wilson, David and Bradford, Russell and Davenport, James H. and
            England, Matthew",
  title = "Cylindrical Algebraic Sub-Decompositions",
  link = "\url{https://arxiv.org/pdf/1401.0647.pdf}",
  year = "2014",
  abstract = "Cylindrical algebraic decompositions (CADs) are a key tool in real
            algebraic geometry, used primarily for eliminating quantifiers over
            the reals and studying semi-algebraic sets. In this paper we
```

```

introduce cylindrical algebraic sub-decompositions (sub-CADs), which
are subsets of CADs containing all the information needed to specify a
solution for a given problem. We define two new types of sub-CAD:
variety sub-CADs which are those cells in a CAD lying on a designated
variety; and layered sub-CADs which have only those cells of
dimension higher than a specified value. We present algorithms to
produce these and describe how the two approaches may be combined with
each other and the recent theory of truth-table invariant CAD. We
give a complexity analysis showing that these techniques can offer
substantial theoretical savings, which is supported by experimentation
using an implementation in Maple.",
paper = "Wils14.pdf"
}

```

— axiom.bib —

```

@article{Zhao11,
  author = "Zhao, Ting and Wang, Dongming and Hong, Hoon",
  title = "Solution formulats for cubic equations without or with constraints",
  journal = "J. Symbolic Computation",
  volume = "46",
  pages = "904-918",
  year = "2011",
  abstract =
    "We present a convention (for square/cubic roots) which provides
    correct interpretations of the Lagrange formula for all cubic
    polynomial equations with real coefficients. Using this convention, we
    also present a real solution formula for the general cubic equation
    with real coefficients under equality and inequality constraints.",
  paper = "Zhao11.pdf"
}

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1.36 Comparison of Computer Algebra System

— axiom.bib —

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@article{Bern96,
  author = "Benardin, Laurent",
  title = "A review of symbolic solvers",
  journal = "SIGSAM Bull.",

```

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volume = "30",
number = "1",
pages = "9-20",
year = "1996",
abstract =
  "Solving equations and systems of equations symbolically is a key
  feature of every Computer Algebra System. This review examines the
  capabilities of the six best known general purpose systems to date in
  the area of general algebraic and transcendental equation
  solving. Areas explicitly not covered by this review are differential
  equations and numeric or polynomial system solving as special purpose
  systems exist for these kinds of problems. The aim is to provide a
  benchmark for comparing Computer Algebra Systems in a specific
  domain. We do not intend to give a rating of overall capabilities as
  for example in [9]. 1 The Contestants We compare six major Computer
  Algebra Systems. Axiom 2.0 [7], Derive 3.06 [1], Macsyma 420 [8],
  Maple V R4 [3], Mathematica 2.2 [10], MuPAD 1.2.9 [5] and Reduce 3.6
  [6]. When available, we tried to use the latest shipping version of
  each system. 2 The Problem Set The following table presents the set of
  80 problems that we used to evaluate the different solvers...",
paper = "Bern96.pdf",
keywords = "axiomref"
}

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— axiom.bib —

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@misc{West95,
  author = "Wester, Michael J.",
  title = "A Review of CAS Mathematical Capabilities",
  year = "1995",
  link = "\url{http://math.unm.edu/~wester/cas/Paper.ps}",
  abstract =
    "Computer algebra systems (CASs) have become an important
    computational tool in the last decade. General purpose CASs, which are
    designed to solve a wide variety of problems, have gained special
    prominence. In this paper, the capabilities of seven major general
    purpose CASs (Axiom, Derive, Macsyma, Maple, Mathematica, MuPAD, and
    Reduce) are reviewed on 131 short problems covering a broad range of
    (primarily) symbolic mathematics.

    A demo was developed for each CAS, run and the results
    evaluated. Problems were graded in terms of whether it was easy or
    difficult or possible to produce an answer and if an answer was
    produced, whether it was correct. It is the author's hope that this
    review will encourage the development of a comprehensive CAS test
    suite."

```



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    paper = "West95.pdf",
    keywords = "axiomref"
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— axiom.bib —

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@misc{West99a,
  author = "Wester, Michael J.",
  title = "A Critique of the Mathematical Abilities of CA Systems",
  year = "1999",
  link = "\url{http://math.unm.edu/~wester/cas/book/Wester.pdf}",
  url2 = "http://math.unm.edu/~wester/cas_review.html",
  abstract =
    "Computer algebra systems (CASs) have become an essential computational
    tool in the last decade. General purpose CASs, which are designed to
    solve a wide variety of problems, have gained special prominence. In
    this chapter, the capabilities of seven major general purpose CASs
    (Axiom, Derive, Macsyma, Maple, Mathematica, MuPAD and Reduce) are
    reviewed on 542 short problems covering a broad range of (primarily)
    symbolic mathematics.",
  paper = "West99a.pdf",
  keywords = "axiomref"
}

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1.37 Finite Fields

— axiom.bib —

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@article{Ashx89,
  author = "Ash, D.W. and Black, I.F. and Vanstone, S.A.",
  title = "Low Complexity Normal Bases",
  journal = "Discrete Applied Mathematics",
  volume = "25",
  pages = "191-210",
  year = "1989"
}

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— axiom.bib —

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@article{Beth91,  
  author = "Beth, T. and Geiselmann, W. and Meyer, F.",  
  title = "Finding (Good) Normal Bases in Finite Fields",  
  journal = "Proc. ISSAC '91",  
  year = "1991"  
}
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— axiom.bib —

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@article{Gath90,  
  author = "Gathen, J. von zur and Giesbrecht, M.",  
  title = "Constructing normal bases in finite fields",  
  journal = "J. Symb. Comp.",  
  volume = "10",  
  pages = "547-570",  
  year = "1990"  
}
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— axiom.bib —

```
@article{Geis89,  
  author = "Geiselmann, W. and Gollmann, D.",  
  title = "Symmetry and Duality in Normal Basis Multiplication",  
  journal = "Proc. AAECC-6, LNCS",  
  volume = "357",  
  year = "1989"  
}
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— axiom.bib —

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@article{Hube90,  
  author = "Huber, K.",  
  title = "Some Comments on Zech's Logarithm",  
  journal = "IEEE Trans. Information Theory",
```

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    volume = "IT-36",
    pages = "946-950",
    year = "1990"
}

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— axiom.bib —

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@article{Itoh88,
  author = "Itoh, T. and Tsujii, S.",
  title = "A fast algorithm for computing multiplicative inverses in
           $GF(2^m)$ using normal bases",
  journal = "Inf. and Comp.",
  volume = "78",
  pages = "171-177",
  year = "1988",
  algebra = "\newline\ref{package INBFF InnerNormalBasisFieldFunctions}",
  abstract =
    "This paper proposes a fast algorithm for computing multiplicative
    inverses in $GF(2^m)$ using normal bases. Normal bases have the
    following useful property: In the case that an element $x$ in
    $GF(2^m)$ is represented by normal bases, $2^k$ power operation of an
    element $x$ in $GF(2^m)$ can be carried out by $k$ times cyclic shift
    of its vector representation. C.C. Wang et al. proposed an algorithm
    for computing multiplicative inverses using normal bases, which
    requires $(m-2)$ multiplications in $GF(2^m)$ and $(m-1)$ cyclic
    shifts. The fast algorithm proposed in this paper also uses normal
    bases, and computes multiplicative inverses iterating multiplications
    in $GF(2^m)$. It requires at most $2[\log_2(m-1)]$ multiplications in
    $GF(2^m)$ and $(m-1)$ cyclic shifts, which are much less than those
    required in Wang's method. The same idea of the proposed fast
    algorithm is applicable to the general power operation in $GF(2^m)$
    and the computation of multiplicative inverses in $GF(q^m)$
    $(q=2^n)$.",
  paper = "Itoh88.pdf"
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— axiom.bib —

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@book{Jaco85,
  author = "Jacobson, N.",
  title = "Basic Algebra I, 2nd ed.",
  publisher = "W.H. Freeman and Co.",

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    year = "1985"  
}
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— axiom.bib —

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@article{Lens82,  
  author = "Lenstra, A.K.",  
  title = "Factorization of Polynomials, Comp. Methods in Number Theory  
          (part 1)",  
  journal = "Math. Centre Tracts",  
  volume = "154",  
  year = "1982"  
}
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— axiom.bib —

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@article{Lens91,  
  author = "Lenstra Jr., H.W.",  
  title = "Finding Isomorphisms between Finite Fields",  
  journal = "Math. of Comp.",  
  volume = "56",  
  number = "193",  
  pages = "329-347",  
  year = "1991"  
}
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— axiom.bib —

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@article{Lens87,  
  author = "Lenstra, H. W. and Schoof, R. J.",  
  title = "Primitive Normal Bases for Finite Fields",  
  journal = "Mathematics of Computation",  
  volume = "48",  
  number = "177",  
  year = "1987",  
  pages = "217-231",  
  link = "\url{http://www.math.leidenuniv.nl/~hwl/PUBLICATIONS/}",
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algebra = "\newline\refto{package FFPOLY FiniteFieldPolynomialPackage}",
abstract =
  "It is proved that any finite extension of a finite field has a normal
  basis consisting of primitive roots",
paper = "Lens87.pdf"
}

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— axiom.bib —

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@book{Lidl83,
  author = "Lidl, Rudolf and Niederreiter, Harald",
  title = "Finite Field, Encyclopedia of Mathematics and Its Applications",
  volume = "20",
  publisher = "Cambridge Univ. Press",
  year = "1983",
  isbn = "0-521-30240-4",
  algebra =
    "\newline\refto{category FAXF FiniteAlgebraicExtensionField}
    \newline\refto{domain FF FiniteField}
    \newline\refto{domain FFCG FiniteFieldCyclicGroup}
    \newline\refto{domain FFCGX FiniteFieldCyclicGroupExtension}
    \newline\refto{domain FFCGP FiniteFieldCyclicGroupExtensionByPolynomial}
    \newline\refto{domain FFX FiniteFieldExtension}
    \newline\refto{domain FFP FiniteFieldExtensionByPolynomial}
    \newline\refto{domain FFNB FiniteFieldNormalBasis}
    \newline\refto{domain FFNBX FiniteFieldNormalBasisExtension}
    \newline\refto{domain FFNBP FiniteFieldNormalBasisExtensionByPolynomial}
    \newline\refto{domain IFF InnerFiniteField}
    \newline\refto{domain IPF InnerPrimeField}
    \newline\refto{domain PF PrimeField}
    \newline\refto{package INBFF InnerNormalBasisFieldFunctions}
    \newline\refto{package FFPOLY2 FiniteFieldPolynomialPackage2}
    \newline\refto{package FFPOLY FiniteFieldPolynomialPackage}
    \newline\refto{package FFHOM FiniteFieldHomomorphisms}
    \newline\refto{package FFF FiniteFieldFunctions}"
}

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— axiom.bib —

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@book{Lips81,
  author = "Lipson, John D.",
  title = "Elements of Algebra and Algebraic Computing",

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publisher = "Addison-Wesley Educational Publishers",
year = "1981",
isbn = "978-0201041156",
algebra = "\newline\refto{category FFIELDC FiniteFieldCategory}"
}

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— axiom.bib —

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@misc{Lune87,
  author = {L\uneburg, H},
  title = "On the Rational Normal Form of Endomorphisms",
  comment = "BI-Wissenschaftsverlag",
  year = "1987"
}

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— axiom.bib —

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@techreport{Grab92,
  author = "Grabmeier, Johannes and Scheerhorn, Alfred",
  title = "Finite fields in Axiom",
  type = "technical report",
  number = "AXIOM Technical Report TR7/92 (ATR/5)(NP2522)",
  institution = "Numerical Algorithms Group, Inc.",
  address = "Downer's Grove, IL, USA and Oxford, UK",
  year = "1992",
  link = "\url{http://www.nag.co.uk/doc/TechRep/axiomtr.html}",
  algebra =
    "\newline\refto{category CHARNZ CharacteristicNonZero}
    \newline\refto{category FPC FieldOfPrimeCharacteristic}
    \newline\refto{category XF ExtensionField}
    \newline\refto{category FFIELDC FiniteFieldCategory}
    \newline\refto{category FAXF FiniteAlgebraicExtensionField}
    \newline\refto{domain SAE SimpleAlgebraicExtension}
    \newline\refto{domain IPF InnerPrimeField}
    \newline\refto{domain PF PrimeField}
    \newline\refto{domain FFP FiniteFieldExtensionByPolynomial}
    \newline\refto{domain FF CGP FiniteFieldCyclicGroupExtensionByPolynomial}
    \newline\refto{domain FFNB FiniteFieldNormalBasisExtensionByPolynomial}
    \newline\refto{domain FFX FiniteFieldExtension}
    \newline\refto{domain FF CGX FiniteFieldCyclicGroupExtension}
    \newline\refto{domain FFNBX FiniteFieldNormalBasisExtension}
    \newline\refto{domain IFF InnerFiniteField}

```

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\newline\refto{domain FF FiniteField}
\newline\refto{domain FFCG FiniteFieldCyclicGroup}
\newline\refto{domain FFNB FiniteFieldNormalBasis}
\newline\refto{package DLP DiscreteLogarithmPackage}
\newline\refto{package FFF FiniteFieldFunctions}
\newline\refto{package INBFF InnerNormalBasisFieldFunctions}
\newline\refto{package FFPOLY FiniteFieldPolynomialPackage}
\newline\refto{package FFPOLY2 FiniteFieldPolynomialPackage2}
\newline\refto{package FFHOM FiniteFieldHomomorphisms}
\newline\refto
{package FFFACTSE FiniteFieldFactorizationWithSizeParseBySideEffect}",
abstract =
  "Finite fields play an important role for many applications (e.g. coding
  theory, cryptograpy). There are different ways to construct a finite
  field for a given prime power. The paper describes the different
  constructions implemented in AXIOM. These are {\sl polynomial basis
  representation}, {\sl cyclic group representation}, and {\sl normal
  basis representation}. Furthermore, the concept of the implementation,
  the used algorithms and the various datatype coercions between these
  representations are discussed.",
paper = "Grab92.pdf",
keywords = "axiomref",
beebe = "Grabmeier:1992:FFA"
}

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— Grabmeier:1992:FFA —

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@TechReport{Grabmeier:1992:FFA,
  author =      "J. Grabmeier and A. Scheerhorn",
  title =       "Finite Fields in {AXIOM}",
  type =        "AXIOM Technical Report",
  number =      "TR7/92 (ATR/5) (NP2522)",
  institution = inst-NAG,
  address =     inst-NAG:adr,
  pages =       "??",
  month =       dec,
  year =        "1992",
  bibdate =     "Fri Dec 29 16:31:49 1995",
  bibsource =   "/usr/local/src/bib/bibliography/Theory/Comp.Alg.bib;
  http://www.math.utah.edu/pub/tex/bib/axiom.bib",
  link =        "\url{http://www.nag.co.uk/doc/TechRep/axiomtr.html}",
  acknowledgement = ack-nhfb,
}

```

— axiom.bib —

```
@article{Mull88,
  author = "Mullin, R.C. and Onyszchuk, I.M. and Vanstone, S.A.",
  title = {Optimal Normal Bases in  $GF(p^n)$ },
  journal = "Discrete Applied Mathematics",
  volume = "22",
  pages = "149-161",
  year = "1988"
}
```

— axiom.bib —

```
@misc{Nick88,
  author = "Nickel, W.",
  title = {Endliche K\"orper in dem gruppentheoretischen Programmsystem GAP},
  comment = "Diplomarbeit, RWTH Aachen",
  year = "1988"
}
```

— axiom.bib —

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@article{Odly85,
  author = "Odlyzko, A.M.",
  title = "Discrete logarithms in finite fields and their cryptographic
    significance",
  journal = "Proc. Eurocrypt '84, LNCS",
  volume = "209",
  publisher = "Springer-Verlag",
  pages = "224-314",
  year = "1985"
}
```

— axiom.bib —

```
@article{Pinc89,
```



```

author = "Pincin, A",
title = "Bases for finite fields and a canonical decomposition for a
        normal basis generator",
journal = "Communications in Algebra",
volume = "17",
number = "6",
pages = "1337-1352",
year = "1989"
}

```

— axiom.bib —

```

@article{Pohl78,
author = "Pohlig, S.C. and Hellman, M.",
title = {An improved algorithm for computing logarithms over  $\text{GF}(p)$ 
        and its cryptographic significance},
journal = "IEEE Trans Information Theory",
volume = "IT-24",
pages = "106-110",
year = "1978"
}

```

— axiom.bib —

```

@article{Rybo89,
author = "Rybowicz, M",
title = "Search of primitive polynomials over finite fields",
journal = "J. Pure Appl.",
volume = "65",
pages = "139-151",
year = "1989"
}

```

— axiom.bib —

```

@misc{Saloi6a,
author = "Salomone, Matthew",

```

```

title = "Group Theory II",
year = "2016",
institution = "Bridgewater State University",
link = "\url{http://axiom-developer.org/axiom-website/GroupTheoryII/Salomone.html}"
}

```

— axiom.bib —

```

@misc{Sche92,
  author = "Scheerhorn, A.",
  title = "Trace- and Norm-Compatible Extensions of Finite Fields",
  journal = "Appl. Alg. in Eng., Comm. and Comp.",
  year = "1992"
}

```

— axiom.bib —

```

@misc{Sche93,
  author = "Scheerhorn, Alfred",
  title = "Presentation of the algebraic closure of finite fields and
          trace-compatible polynomial sequences",
  comment = "Darstellungen des algebraischen Abschlusses endlicher Korper
            und spur-kompatible Polynomfolgen",
  year = "1993",
  abstract =
    "For numerical experiments concerning various problems in a finite
    field  $\mathbb{F}_q$  it is useful to have an explicit data
    presentation  $\mathbb{F}_{q^m}$  of for large  $m$ , and a method for the
    construction of towers
    
$$[\mathbb{F}_q \subset \mathbb{F}_{q^{d_1}} \subset \cdots \subset \mathbb{F}_{q^{d_k}} = \mathbb{F}_{q^m}]$$

    In order to avoid the identification problem it is advantageous to
    have all fields in the tower presented by properly chosen normal bases,
    whereby the embedding
    
$$\mathbb{F}_{q^{d_i}} \subset \mathbb{F}_{q^{d_{i+1}}}$$

    is given by the trace function.

    The following notion is introduced: A sequence of polynomials
     $\{f_n \mid n \geq 1\}$  with  $\deg(f_n) = n$  called trace-compatible over
     $\mathbb{F}_q$  if (1)  $f_n$  is a normal polynomial over  $\mathbb{F}_q$ ,
    (2) if  $\alpha_n \in \mathbb{F}_{q^n}$  is a root of  $f_n$ , then for any
    proper divisor  $d$  of  $n$  the trace of  $\alpha_n$  over  $\mathbb{F}_{q^d}$ 

```

is a root of f_d .

The main goal of the dissertation is to give algorithms for construction of sequences of trace-compatible polynomials and to present explicit numerical data. An analogous notion of norm-compatible sequences is also introduced and studied.

The dissertation consists of four chapters and a supplement, as follows: (1) Basic notions (1-31). (2) Presentation of the algebraic closure of a finite field (32-59). (3) Sequences of polynomials and sequences of elements (60-115). (4) Implementations (118-139). (5) Supplement (142-171).

In chapters (1)(3) various known results and algorithms are collected, and new results are added and compared with those previously used.

The numerical results in the supplement contain sequences of trace-compatible polynomials of degree n , where $n \leq 100$, and $q=2,3,5,7,11,13$. For implementation, the computer-algebra system AXIOM has been used. The details contained in this dissertation are not readily describable in a short review.",

```
keywords = "axiomref"
}
```

— axiom.bib —

```
@article{Shou92,
  author = "Shoup, V.",
  title = "Searching for Primitive Roots in Finite Fields",
  journal = "Math. of Comp.",
  volume = "58",
  number = "197",
  pages = "369-380",
  year = "1992"
}
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— axiom.bib —

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@article{Stin90,
  author = "Stinson, D.R.",
  title = "Some observations on parallel Algorithms for fast exponentiation"
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        in $GF(2^n)$",
journal = "Siam J. Comp.",
volume = "19",
number = "4",
pages = "711-717",
year = "1990",
algebra = "\newline\refto{package INBFF InnerNormalBasisFieldFunctions}",
abstract =
    "A normal basis representation in $GF(2^n)$ allows squaring to be
    accomplished by a cyclic shift. Algorithms for multiplication in
    $GF(2^n)$ using a normal basis have been studied by several
    researchers. In this paper, algorithms for performing exponentiation
    in $GF(2^n)$ using a normal basis, and how they can be speeded up by
    using parallelization, are investigated.",
paper = "Stin90.pdf"
}

```

— axiom.bib —

```

@article{Vars81,
  author = "Varshamov, Gamkrelidze",
  title = "Method of construction of primitive polynomials over finite fields",
  journal = "Soobsheh. Akad. Nauk Gruzin.",
  volume = "99",
  pages = "61-64",
  year = "1981"
}

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— axiom.bib —

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@article{Wass89,
  author = "Wassermann, A.",
  title = "Konstruktion von Normalbasen",
  journal = "Bayreuther Math. Schriften",
  volume = "31",
  pages = "1-9",
  year = "1989"
}

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1.38 To Be Classified

— axiom.bib —

```
@phdthesis{Fate72,
  author = "Fateman, Richard J.",
  title = "Essays in Algebraic Simplification",
  link = "\url{http://www.cs.berkeley.edu/~fateman/papers/tr-95.pdf}",
  school = "MIT",
  comment = "MAC TR-95 technical report",
  year = "1972",
  abstract =
    "This thesis consists of essays on several aspects of the problem
    of algebraic simplification by computer. We first discuss a pattern
    matching system intended to recognize non-obvious occurrences of
    patterns within Algebraic expression. A user of such a system can
    ‘‘teach’’ the computer new simplification rules. Then we report on
    new applications of canonical simplification of rational functions.
    These applications include techniques for picking out coefficients,
    and for substituting for sums, products, quotients, etc. Our final
    essay is on a new, practical, canonical simplification algorithms
    for radical expressions (i.g. algebraic expressions including roots
    of polynomials). The effectiveness of the procedure is assured
    through proofs of appropriate properties of the simplified forms.
    Two appendices describe MACSYM, a computer system for algebraic
    manipulations, which served as the basis for this work.",
  paper = "Fate72.pdf"
}
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— axiom.bib —

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@misc{Fate08b,
  author = "Fateman, Richard J.",
  title = "Applications and Methods for Recognition of (Anti)-Symmetric
    Functions",
  link = "\url{http://www.cs.berkeley.edu/~fateman/papers/symmetry.pdf}",
  year = "2008",
  abstract =
    "One of the important advantages held by computer algebra systems (CAS)
    over purely-numerical computational frameworks is that the CAS can
    provide a higher-level ‘‘symbolic’’ viewpoint for problem
    solving. Sometimes this can convert apparently impossible problems to
    trivial ones. Sometimes the symbolic perspective can provide
    information about questions which cannot be directly answered, or
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questions which might be hard to pose. For example, we might be able to analyze the asymptotic behavior of a solution to a differential equation even though we cannot solve the equation. One route to implicitly solving problems is the use of symmetry arguments. In this paper we suggest how, through symmetry, one can solve a large class of definite integration problems, including some that we found could not be solved by computer algebra systems. One case of symmetry provides for recognition of periodicity, and this solves additional problems, since removal of periodic components can be important in integration and in asymptotic expansions.",

```
paper = "Fate08b.pdf"
}
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— axiom.bib —

```
@misc{Fate03b,
  author = "Fateman, Richard J.",
  title = "Manipulation of Matrices Symbolically",
  link = "\url{http://www.cs.berkeley.edu/~fateman/papers/symmat2.pdf}",
  year = "2003",
  abstract =
    "Traditionally, matrix algebra in computer algebra systems is
    ‘‘implemented’’ in three ways:
    \begin{itemize}
    \item numeric explicit computation in a special arithmetic domain:
    exact rational or integer, high-precision software floating-point,
    interval, or conventional hardware floating-point.
    \item symbolic explicit computation with polynomial or other
    expression entries,
    \item (implicit) matrix computation with symbols defined over a
    (non-commuting) ring.
    \end{itemize}
    Manipulations which involve matrices of indefinite size ( $n \times m$ ) or
    perhaps have components which are block submatrices of indefinite size
    have little or no support in general-purpose computer algebra systems,
    in spite of their importance in theorems, proofs, and generation of
    programs. We describe some efforts to design and implement tools for
    this mode of thinking about matrices in computer systems.",
  paper = "Fate03b.pdf"
}
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— axiom.bib —

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@misc{Fate09a,
  author = "Fateman, Richard J.",
  title = "Simplifying RootSum Expressions",
  link = "\url{http://www.cs.berkeley.edu/~fateman/papers/rootsum.pdf}",
  year = "2009",
  abstract =
    "It's useful to sum an expression with a parameter varying over all
    the roots of a given polynomial. Here's a defense of that statement
    and a method to do the task.",
  paper = "Fate09a.pdf"
}

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— axiom.bib —

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@misc{Fate09,
  author = "Fateman, Richard J.",
  title = "Rational Integration, Simplified",
  link = "\url{http://www.cs.berkeley.edu/~fateman/papers/root-integ.pdf}",
  year = "2009",
  abstract =
    "After all this computer algebra stuff, and several PhD theses in
    the last few decades, what more could we say about symbolic
    rational function integration?

    How about a closed formula for the result, subject to a few algebraic
    side-conditions, which works even with parameters in the denominator?",
  paper = "Fate09.pdf"
}

```

— axiom.bib —

```

@misc{Fate99b,
  author = "Fateman, Richard J.",
  title = "Generation and Optimization of Numerical Programs by
    Symbolic Mathematical Methods",
  link = "\url{http://www.cs.berkeley.edu/~fateman/papers/RIMS.pdf}",
  year = "1999",
  abstract =
    "Symbolic mathematical methods and systems
    \begin{itemize}
    \item support scientific and engineering ‘‘problem solving environments’’
    (PSEs),

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\item The specific manipulation of mathematical models as a precursor
to the coding of algorithms
\item Expert system selection of modules from numerical libraries and
other facilities
\item The production of custom numerical software such as derivatives
or non-standard arithmetic code-generation packages,
\item The complete solution of certain classes of mathematical problems
that simply cannot be handled solely by conventional floating-point
computation.
\end{itemize}

Viewing computational objects and algorithms from a symbolic
perspective and then specializing them to numerical or graphical views
provides substantial additional flexibility over a more conventional view.

We also consider interactive symbolic computing as a tool to provide
an organizing principle or glue among otherwise dissimilar components.",
paper = "Fate99b.pdf",
keywords = "axiomref"
}

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— axiom.bib —

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@misc{Fate07,
  author = "Fateman, Richard J.",
  title = "Rational Function Computing with Poles and Residues",
  link = "\url{http://www.cs.berkeley.edu/~fateman/papers/qd.pdf}",
  year = "2007",
  abstract =
    "In a numerical calculation sometimes we need higher-than
    double-precision floating-point arithmetic to allow us to be confident
    of a result. One alternative is to rewrite the program to use a
    software package implementing arbitrary-precision extended
    floating-point arithmetic such as ARPREC or MPFR, and try to choose a
    suitable precision.

```

Such an arithmetic scheme, in spite of helpful tools, may be inconvenient to write. There are also facilities in computer algebra systems (CAS) for such software-implemented ‘‘bigfloats.’’ These facilities are convenient if one is already using the CAS. In any of these situations the bigfloats may be rather slow, a cost of its generality.

There are possibilities intermediate between the largest hardware floating-point format and the general arbitrary-precision software which combine a considerable (but not arbitrary) amount of extra

precision with a (relatively speaking) modest factor loss in speed. Sometimes merely doubling the number of bits in a double-floating-point fraction is enough, in which case arithmetic on double-double (DD) operands would suffice. Another possibility is to go for yet another doubling to quad-double (QD) arithmetic: instead of using the machine double-floats to give about 16 decimal digits of precision, QD supplies about 64 digits. DD and QD as used here provide the same exponent range as ordinary double.

Here we describe how we incorporated QD arithmetic implemented in a library into a Common Lisp system, providing a smooth interface while adding only modest overhead to the run-time costs (compared to accessing the library from C or C++). One advantage is that we keep the program text almost untouched while switching from double to quad-double. Another is that the programs can be written, debugged, and run in an interactive environment. Most of the lessons from QD can be used for other versions of arithmetic which can be embedded in Lisp, including MPFR, for indefinite (arbitrary) precision, should QD provide inadequate precision or range.",
 paper = "Fate07.pdf",
 keywords = "axiomref"
}

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@misc{Fate13a,
  author = "Fateman, Richard J.",
  title = "Rational Function Computing with Poles and Residues",
  link = "\url{http://www.cs.berkeley.edu/~fateman/papers/openmathcrit.pdf}",
  year = "2013",
  abstract =
    "Computer algebra systems (CAS) usually support computation with exact
    or approximate rational functions as ratios of polynomials in
    'expanded form' with explicit coefficients. We examine the
    consequences of introducing a partial-fraction type of form in which
    some of the usual rational operations can be implemented in
    substantially faster times. In this form an expression in one
    variable, say  $x$ , is expressed as a polynomial in  $x$  plus a sum of
    terms each of which has a denominator  $x-c$  perhaps to an integer
    power, where  $c$  is in general a complex constant. We show that some
    common operations including rational function addition,
    multiplication, and matrix determinant calculation can be performed
    many times faster than in the conventional representation. Polynomial
    GCD operations, the costliest part of rational additions, are entirely
    eliminated. Application of Cauchy's integral theorem allow for trivial
    integration of an expression around a closed contour. In some cases
```

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    the approximate evaluation of transcendental functions can be
    accelerated, especially in parallel, by evaluation of a formula in
    pole+residue form.",
    paper = "Fate13a.pdf"
}

```

— axiom.bib —

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@misc{Fate01a,
  author = "Fateman, Richard J.",
  title = "A Critique of OpenMath and Thoughts on Encoding Mathematics",
  year = "2001",
  link = "\url{http://www.cs.berkeley.edu/~fateman/papers/openmathcrit.pdf}",
  algebra =
    "\newline\ref{category OM OpenMath}
    \newline\ref{domain COMPLEX Complex}
    \newline\ref{domain DFLOAT DoubleFloat}
    \newline\ref{domain FLOAT Float}
    \newline\ref{domain FRAC Fraction}
    \newline\ref{domain INT Integer}
    \newline\ref{domain LIST List}
    \newline\ref{domain SINT SingleInteger}
    \newline\ref{domain STRING String}
    \newline\ref{domain SYMBOL Symbol}
    \newline\ref{package OMEXPR ExpressionToOpenMath}
    \newline\ref{package OMSERVER OpenMathServerPackage}",
  abstract =
    "The OpenMath project, as portrayed in the Special Issue of the SIGSAM
    Bulletin (volume 34 no. 2), seems to have a number of problems to
    face. One of them is the (apparently implicit) assumption that
    OpenMath designers, through dint of mathematical thought and the
    advice of the members of the Open Math Society, have solved, in their
    domain, one of the most pressing problems of software engineering
    today, namely software re-use. After six years there is insufficient
    evidence on which to base any claims of success and it appears that
    most substantive practical issues of mathematical representation and
    communication have yet to be addressed. We also raise questions about
    related computational mathematical goals and mathematical encodings.",
  paper = "Fate01a.pdf"
}

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— axiom.bib —

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@misc{Fate08a,
  author = "Fateman, Richard J.",
  title = "Verbs, Nouns, and Computer Algebra, or What's Grammar Got
          to do with Math?",
  link = "\url{http://www.cs.berkeley.edu/~fateman/papers/nounverbmac.pdf}",
  year = "2008",
  abstract =
    "Computer algebra systems (CAS) are supposed to do mathematics.
    Unfortunately, much of human mathematics presentation and
    manipulation depends on humans to understand what is going on more
    generally. This context is often not made available to the computer
    system, and so it will sometimes fail to make the right guess at the
    meaning of constructions, unless they are made unambiguous. Among many
    issues, one that has been raised and resolved, is the need for
    attributing properties to operators that distinguish between 'the name
    of the operation' and 'the operation itself'.

    As a simple example, referential transparency requires that if  $x = y$ ,
    then any true statement  $P(x)$  about  $x$  must be true about  $y$ , namely
     $P(y)$  is also true.

    Consider now the well-known equation  $\sin(0) = 0$ , and the statement
     $P(r) :=$  'r = 0 is a rule to simplify the  $\sin()$  function'. The
    statement  $P(\sin(0))$  is clearly true. Therefore  $P(0)$  should be true
    as well. It is not, since  $0 = 0$  is not a rule to simplify  $\sin()$ .

    The point here is that  $\sin(0)$  is not equal to 0 if  $\sin$  is treated as
    the name of an operation. Perhaps it is a character string, or a data
    structure of some sort: a noun phrase. The corresponding expression
    including the verb form of  $\sin$  is one in which  $\sin(0)$  can be
    simplified to something equivalent, but in some sense preferable. Here
    it is reasonable to replace it by 0.",
  paper = "Fate08a.pdf"
}

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— axiom.bib —

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@misc{Fate97c,
  author = "Fateman, Richard J.",
  title = "Excerpts from a proposal to the National Science Foundation
          on Programming Environments and Tools for Advanced Scientific
          Computation",
  year = "1997",
  link = "\url{http://www.cs.berkeley.edu/~fateman/papers/newpse.pdf}",
  abstract =
    "We propose to concentrate on investigations of technology in support

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of advanced scientific computation, based on a higher-level view of the relationship of mathematical modeling to computation. This is based on the following considerations:

```

\begin{enumerate}
\item Symbolic mathematical manipulation and computer algebra systems:
These are tools and techniques that support problem solving, code
development, comprehension, debugging and partial evaluation
(the principle behind many optimization techniques).
\item Data modeling: in particular, the use of IEEE floating-point
number representations and operations.
\item Closed form or exact solutions: For some sub-problems of
scientific interest, numerical solutions are not the only paradigm.
The availability of exact solutions, or symbolic-approximation solutions
has advanced along with computing technology too.
\item The use of scripting languages: the linkage of high-performance
library routines to graphical interfaces for data analysis and model setup
can be enhanced by symbolic support and high-level scripting languages.
\item Extending the use of remote computation. We are now using standard
network connections for the acquisition of complex data (in this case,
symbolic solution of definite integrals).
\end{enumerate}",
paper = "Fate97c.pdf",
keywords = "axiomref"
}

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— axiom.bib —

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@misc{Fate98,
  author = "Fateman, Richard J.",
  title = "Open Problems in Human-Computer Interaction with Specific
    Application to Computer-Aided Mathematics",
  link = "\url{http://www.cs.berkeley.edu/~fateman/papers/open4.ps}",
  year = "1998",
  abstract =
    "It is our intention, in this short paper, to indicate some of the
    problems of human-computer interactions at the boundaries of current
    research is algebraic manipulation as we see the field now, some 35
    years from the development of its first software systems (the
    now-ancient Macsyma was announced in 1971, and it had precursors.)

    Because computers, algorithms, and software systems have advanced so
    substantially, there is a tendency for computer scientists to think
    that all of mathematics can be done by computer, or at least that we
    are on track for the continued ‘arithmetization of mathematics.’
    Working mathematicians tend to disagree with the characterization that
    we are at all close now, nor that we are headed toward any goal of

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being able to do or understand mathematics.

Most of the problems which we indicate below are actually quite general problems in computer-human communication. For specificity (and the possibility that one could quantify a solution!) we will count our discussion of in terms of applications in the area of algebraic manipulation. If we cannot handle allegedly well-formed mathematical issues, how can we expect to handle ill-formed ambiguous ‘‘natural’’ communication?

Some of the problems are representative of strategic mathematical manipulation procedures which lack totally algorithmic approaches at this time, and so are more of an interactive problem-solving protocol than a fixed recipe.",
 paper = "Fate98.pdf",
 keywords = "axiomref"
 }

— axiom.bib —

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@misc{Fate00a,
  author = "Fateman, Richard J.",
  title = "Improving Exact Integral From Symbolic Algebra Systems",
  year = "2000",
  link = "\url{http://www.cs.berkeley.edu/~fateman/papers/nform2col.pdf}",
  abstract =
    "Programs in symbolic algebraic manipulation systems can compute
    certain classes of symbolic indefinite integrals in closed form. Although
    these answers are ordinarily formally correct algebraic anti-derivatives,
    their form is often unsuitable for further numerical or even analytical
    processing. In particular, we address cases in which such ‘‘exact answers’’
    when numerically evaluated may give less-accurate answers than numerical
    approximations from first principles! The symbolic formulas may also
    behave inappropriately near singularities. We discuss techniques, based
    in part on the calculus of divided differences, for improving the form of
    results of symbolic mathematics systems. In particular, computer algebra
    systems must take explicit account of the possibility that they are
    producing not mathematics but templates of programs consisting of sequences
    of arithmetic operations. In brief, mathematical correctness is
    not enough. Forms produced by rational integration programs are used
    for examples.",
  paper = "Fate00a.pdf"
}
```

— axiom.bib —

```
@misc{Fate03,
  author = "Fateman, Richard J.",
  title = "Continuity and Limits of Programs",
  link = "\url{http://www.cs.berkeley.edu/~fateman/papers/limit.pdf}",
  year = "2003",
  abstract =
    "In demonstrations of proofs [3] of correctness for programs that are
    designed to compute mathematical functions we attempted to show that
    programs have the right properties, namely the same properties as the
    mathematical functions they are alleged to compute. In the cited paper
    we were forced to hand-wave (our excuse was the need for brevity) in
    at least two places, trying to side-step sticky issues. Here we try to
    address these issues by clarifying two concepts: (a) computational
    continuity and (b) equality in a domain where floating-point
    computations can be done to variable (presumably high) precision. We
    introduce notations p-representable and p-negligible where p denotes
    precision, and show how this helps in our applications.",
  paper = "Fate03.pdf"
}
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— axiom.bib —

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@misc{Fate15,
  author = "Fateman, Richard J.",
  title = "Interval Arithmetic, Extended Numbers and Computer Algebra
    Systems",
  link = "\url{http://www.cs.berkeley.edu/~fateman/papers/interval.pdf}",
  year = "2015",
  abstract =
    "Many ambitious computer algebra systems were initially designed in a
    flush of enthusiasm, with the goal of automating any symbolic
    mathematical manipulation correctly. Historically, this approach
    resulted in programs that implicitly used certain identities to
    simplify expressions. These identities, which very likely seemed
    universally true to the programmers in the heat of writing the CAS,
    (and often were true in well-known abstract algebraic domains) later
    needed re-examination when such systems were extended for dealing with
    kinds of objects unanticipated in the original design. These new
    objects are generally introduced to the CAS by extending generically
    the arithmetic or other operations. For example, approximate floats
    do not have the mathematical properties of exact integers or
    rationals. Complex numbers may strain a system designed for
    real-valued variables. In the situation examined here, we consider two
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categories of extended numbers: or undefined, and real intervals. We comment on issues raised by these two troublesome notions, how their introduction into a computer algebra system may require a (sometimes painful) reconsideration and redesign of parts of the program, and how they are related. An alternative (followed most notably by the Axiom system is to essentially envision a meta CAS defined in terms of categories and inheritance with only the most fundamental built-in concepts; from these one can build many variants of specific CAS features. This approach is appealing but can fails to accommodate extensions that violate some mathematical tenets in the cause of practicality.",
 paper = "Fate15.pdf"
 }

— axiom.bib —

```
@misc{Fate14,
  author = "Fateman, Richard J.",
  title = "Fun with Filon Quadrature - a Computer Algebra Perspective",
  year = "2014",
  link = "\url{http://www.cs.berkeley.edu/~fateman/papers/fun-filon.pdf}",
  abstract =
    "Filon quadrature, useful for oscillatory integrals, is easily
    implemented in a computer algebra system (CAS) using exact or
    approximate arithmetic. Although quadrature is almost always used in
    the context of approximation, we see that various properties are
    exhibited by running an exact algorithm on exact {\sl symbolic}
    inputs. This experimental approach to the mathematics allows us to
    prove the implementation of the algorithm has the expected
    mathematical correctness properties, and is therefore more likely to
    be, itself, a correct implementation.",
  paper = "Fate14.pdf"
}
```

— axiom.bib —

```
@misc{Fate97a,
  author = "Fateman, Richard J.",
  title = "The Fast Fourier Transform",
  course = "Berkeley CS 292, Fall 1997, Handout 2",
  year = "1997",
  link = "\url{http://www.cs.berkeley.edu/~fateman/papers/fftnotes.pdf}",
```

```

    paper = "Fate97a.pdf"
}

```

— axiom.bib —

```

@misc{Fate97b,
  author = "Fateman, Richard J.",
  title = "More Versatile Scientific Documents Over-Extended Abstract,
    working paper",
  year = "1997",
  link = "\url{http://www.cs.berkeley.edu/~fateman/papers/mvsd.ps}",
  abstract =
    "The electronic representation of scientific documents: journals,
    technical reports, program documentation, laboratory notebooks,
    etc. present challenges in several distinct communities. We see five
    distinct groups concerned with electronic versions of scientific
    documents:
    \begin{itemize}
    \item Publishers of journals, texts, reference works, and their authors.
    \item Software publishers for OCR/document analysis, document formatting
    \item Software publishers whose products access ‘‘contents semantics’’
    from documents, including library keyword search programs, natural language
    search programs, data-base systems, visual presentation systems,
    mathematical computation systems, etc.
    \item Institutions maintain access to electronic libraries, which must
    be broadly construed to include data and programs of all sorts.
    \item Individuals and their programs acting as their agents who need to
    use these libraries, to identify, locate, and retrieve relevant
    documents
    \end{itemize}

    We need to have a convergence in design and standards for encoding new
    or pre-existing (typically paper-based) documents in order to most
    efficiently meet the needs of all these groups. Various effort, some
    loosely coordinated, but just as often competing, are trying to set
    standards and build tools.

    We approach this by first dividing the task into visual, syntactic,
    and semantic components. These specifications can focus attention on
    the addressing requirements of the different groups. Additionally,
    these components rely on a structure for documents that incorporates
    existing library models and extends them to new modes of operation.
    Berkeley’s MVD is one plausible structure which will allow prototype
    development and explorations.",
  paper = "Fate97b.pdf"
}

```

— axiom.bib —

```
@article{Heck97,
  author = "Heckmann, Reinhold and Wilhelm, Reinhard",
  title = "A Functional Description of TeX's Formula Layout",
  journal = "J. Functional Programming",
  volume = "7",
  number = "5",
  pages = "451-485",
  year = "1997",
  link = "\url{http://http.cs.berkeley.edu/~fateman/temp/neuform.pdf}",
  comment = "ftp://ftp.cs.uni-sb.de/formulae/formulae.tar.gz",
  abstract =
    "While the quality of the results of TeX's mathematica formula layout
    algorithm is convincing, its original description is hard to
    understand since it is presented as an imperative program with complex
    control flow and destructive manipulations of the data structures
    representing formulae. In this paper, we present a re-implementation
    of TeX's formula layout algorithm in the functional language SML,
    thereby providing a more readable description of the algorithm,
    extracted from the monolithical TeX system.",
  paper = "Heck97.pdf"
}
```

— axiom.bib —

```
@misc{Kohl08,
  author = "Kohlhase, Michael",
  title = "Using Latex as a Semantic Markup Format",
  year = "2008",
  link = "\url{https://kwarc.info/kohlhase/papers/mcs08-stex.pdf}",
  abstract =
    "One of the great problems of Mathematical Knowledge Management (MKM)
    systems is to obtain access to a sufficiently large corpus of
    mathematical knowledge to allow the management / search / navigation
    techniques developed by the community to display their strength. Such
    systems usually expect the mathematical knowledge they operate on in
    the form of semantically enhanced documents, but mathematicians and
    publishers in Mathematics have heavily invested into the Tex/Latex
    format and workflow."
```

We analyze the current practice of semi-semantic markup in Latex documents and extend it by a markup infrastructure that allows to embed semantic annotations into latex documents without changing their visual appearance. This collection of tex macro packages is called stex (semantic tex) as it allows to markup latex documents semantically without leaving the time-tried tex/latex workflow, essentially turning latex into an MKM format. At the heart of stex is a definition mechanism for semantic macros for mathematical objects and a non-standard scoping construct for them, which is oriented at the semantic dependency relation rather than the document structure.

We evaluate the stex macro collection on a large case study: the course materials of a two-semester course in Computer Science was annotated semantically and converted to the OMDOC MKM format by Bruce Miller's LatexML system.",

```
paper = "Kohl08.pdf"
}
```

— axiom.bib —

```
@misc{Vaki98,
  author = "Vakil, Ravi",
  title = "A Beginner's Guide to Jet Bundles from the Point of View of
    Algebraic Geometry",
  link = "\url{http://math.stanford.edu/~vakil/files/jets.pdf}",
  year = "1998",
  paper = "Vaki98.pdf"
}
```

— axiom.bib —

```
@article{Wang74,
  author = "Wang, Paul S.",
  title = "The Undecidability of the Existence of Zeros of Real Elementary
    Functions",
  journal = "J. ACM",
  volume = "21",
  number = "4",
  pages = "586-589",
  year = "1974",
  abstract =
    "From Richardson's undecidability results, it is shown that the predicate
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    ‘‘there exists a real number  $r$  such that  $G(r)=0$ ’’ is recursively
    undecidable for  $G(x)$  in a class of functions which involves polynomials
    and the sine function. The deduction follows that the convergence of a
    class of improper integrals is recursively undecidable.",
    paper = "Wang74.djvu"
}

```

— axiom.bib —

```

@article{Lang02,
  author = "Langley, Simon and Richardson, Daniel",
  title = "What can we do with a Solution?",
  journal = "Electronic Notes in Theoretical Computer Science",
  volume = "66",
  number = "1",
  year = "2002",
  link = "\url{http://www.elsevier.nl/locate/entcs/volume66.html}",
  abstract =
    "If  $S=0$  is a system of  $n$  equations and unknowns over  $\mathbb{C}$ 
    and  $S(\alpha)=0$  to what extent can we compute with the point  $\alpha$ ?
    In particular, can we decide whether or not a polynomial expressions
    in the components of  $\alpha$  with integral coefficients is zero?
    This question is considered for both algebraic and elementary systems
    of equations.",
  paper = "Lang02.pdf"
}

```

— axiom.bib —

```

@article{John71,
  author = "Johnson, S.C.",
  title = "On the Problem of Recognizing Zero",
  journal = "J. ACM",
  volume = "18",
  number = "4",
  year = "1971",
  pages = "559-565",
  paper = "John71.djvu"
}

```

— axiom.bib —

```
@article{Rich07,
  author = "Richardson, Daniel",
  title = "How to Recognize Zero",
  journal = "J. Symbolic Computation",
  volume = "24",
  number = "6",
  year = "2007",
  pages = "627-645",
  abstract =
    "An elementary point is a point in complex  $\mathbb{C}^n$  space, which is an
    isolated, non-singular solution of  $n$  equations in  $n$  variables,
    each equation being either of the form  $p=0$ , where  $p$  is a polynomial
    in  $\mathbb{Q}[x_1, \dots, x_n]$ , or of the form  $x_j = e^{x_i} = 0$ . An
    elementary number is the polynomial image of an elementary point. In
    this article a semi-algorithm is given to decide whether or not a
    given elementary number is zero. It is proved that this semi-algorithm
    is an algorithm, i.e. that it always terminates, unless it is given
    a problem containing a counter example to Schanuel's conjecture.",
  paper = "Rich07.pdf"
}
```

— axiom.bib —

```
@inproceedings{Hurx00,
  author = "Hur, Namhyun and Davenport, James H.",
  title = "An exact real algebraic arithmetic with equality determination",
  booktitle = "Proc. ISSAC 2000",
  series = "ISSAC '00",
  pages = "169-174",
  year = "2000",
  abstract =
    "We describe a new arithmetic model for real algebraic numbers with
    an exact equality determination. The model represents a real algebraic
    number as a pair of an arbitrary precision numerical value and a
    symbolic expression. For the numerical part we currently (another
    representation could be used) use the dyadic exact real number and
    for the symbolic part we use a square-free polynomial for the real
    algebraic number. In this model we show that we can decide exactly
    the equality of real algebraic numbers.",
  paper = "Hurx00.djvu"
}
```

— axiom.bib —

```
@article{Chow99,
  author = "Chow, Timothy Y.",
  title = "What is a closed-form number?",
  journal = "The American Mathematical Monthly",
  volume = "106",
  number = "5",
  pages = "440-448",
  year = "1999",
  paper = "Chowxx.pdf"
}
```

— axiom.bib —

```
@book{Yapx00,
  author = "Yap, Chee-Keng",
  title = "Fundamental Problems in Algorithmic Algebra",
  publisher = "John Wiley and Sons",
  year = "2000",
  link = "\url{http://www.csie.nuk.edu.tw/~cychen/gcd/Fundamental%20Problems%20in%20Algorithmic%20Algebra.pdf}",
  abstract =
    "The author shows an interesting way to write arithmetic operations.
    Given
    \[z(x,y)=\frac{axy+bx+cy+d}{a^{\prime}xy+b^{\prime}x+c^{\prime}y+d^{\prime}}\]
    we call the numerical constants  $a,b,\ldots,c^{\prime},d^{\prime}$ 
    {\sl state variables} and use the compact notation
    \[z(x,y)=\frac{a,b,c,d}{a^{\prime},b^{\prime},c^{\prime},d^{\prime}}\}
    \{x \text{ \choose y}\}
    The arithmetic operations can be recovered by suitable choices for
    the state variables:
    \[x+y=\frac{(0,1,1,0)}{(0,0,0,1)}\{x \text{ \choose y}\}
    \[x-y=\frac{(0,1,-1,0)}{(0,0,0,1)}\{x \text{ \choose y}\}
    \[xy=\frac{(1,0,0,0)}{(0,0,0,1)}\{x \text{ \choose y}\}
    \[x/y=\frac{(0,1,0,0)}{(0,0,1,0)}\{x \text{ \choose y}\}
    \[\frac{ax+b}{cx+d}=\frac{(0,a,0,b)}{(0,c,0,d)}\{x \text{ \choose y}\}
    and if
    \[x=q+\frac{p}{x^{\prime}}\]
    then
    \[\begin{array}{ccc}
    z(x,y)&=&\displaystyle\frac{a(q+p/x^{\prime})y+b(q+p/x^{\prime})+cy+d}{a^{\prime}(q+p/x^{\prime})y+
    \end{array}
```

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      b^{\prime}(q+p/x^{\prime})+
      c^{\prime}y+d^{\prime})\\
&=&\displaystyle\frac{(aq+c,bq+d,ap,bp)}
      {(a^{\prime}q+c^{\prime},b^{\prime}q+d^{\prime},a^{\prime}p,
      b^{\prime}p)}{x^{\prime} \choose {y}}
\end{array}]",
paper = "Yapx00.pdf"
}

```

— axiom.bib —

```

@article{Burn00,
  author = "Burnikel, C. and Fleischer, R. and Mehlhom, K. and Schirra, S.",
  title = "A Strong and Easily Computable Separation Bound for Arithmetic
    Expressions Involving Radicals",
  journal = "Algorithmica",
  volume = "27",
  pages = "87-99",
  year = "2000",
  abstract =
    "We consider arithmetic expressions over operators $+$, $-$, $$, $/$,
    and $\sqrt{k}$, with integer operands. For an expression $E$ having
    value $\eta$, a separation bound $sep(E)$ is a positive real number with
    the property that $\eta \ne 0$ implies $|\eta| \ge sep(E)$. We
    propose a new separation bound that is easy to compute and stronger
    than previous bounds.",
  paper = "Burn00.pdf"
}

```

— axiom.bib —

```

@inproceedings{Corl97,
  author = "Corless, Robert M. and Jeffrey, David J. and Knuth, Donald E.",
  title = "A Sequence of Series for The Lambert W Function",
  booktitle = "Proc. ISSAC 1997",
  series = "ISSAC '97",
  pages = "197-204",
  year = "1997",
  abstract =
    "We give a uniform treatment of several series expansions for the
    Lambert $W$ function, leading to an infinite family of new series.
    We also discuss standardization, complex branches, a family of

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arbitrary-order iterative methods for computation of  $W_i$ , and
give a theorem showing how to correctly solve another simple and
frequently occurring nonlinear equation in terms of  $W$  and the
unwinding number",
paper = "Corl97.pdf"
}

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— axiom.bib —

```

@article{Corl96,
  author = "Corless, Robert M. and Jeffrey, David J.",
  title = "Editor's Corner: The Unwinding Number",
  journal = "SIGSAM Bulletin",
  volume = "30",
  number = "1",
  issue = "115",
  pages = "28-35",
  year = "1996",
  abstract =
    "From the Oxford English Dictionary we find that {\sl to unwind}
    can mean 'to become free from a convoluted state'. Further down
    we find the quotation 'The solution of all knots, and unwinding
    of all intricacies', from H. Brooke (The Fool of Quality, 1809).
    While we do not promise that the unwinding number, defined below,
    will solve {\sl all} intricacies, we do show that it may help for
    quite a few problems.",
  paper = "Corl96.pdf"
}

```

— axiom.bib —

```

@article{Corl98,
  author = "Corless, Robert M. and Jeffrey, David J.",
  title = "Graphing Elementary Riemann Surfaces",
  journal = "SIGSAM Bulletin",
  volume = "32",
  number = "1",
  pages = "11-17",
  year = "1998",
  abstract =
    "This paper discusses one of the prettiest pieces of elementary
    mathematics or computer algebra, that we have ever had the pleasure

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to learn. The tricks that we discuss here are certainly ‘‘well-known’’
 (that is, in the literature), but we didn’t know them until recently,
 and none of our immediate colleagues knew them either. Therefore we
 believe that it is useful to publicize them further. We hope that
 you find these ideas as pleasant and useful as we do.”,
 paper = "Cor198.pdf"
}

— axiom.bib —

```
@article{Wang93,
  author = "Wang, Dongming",
  title = "An Elimination Method for Polynomial Systems",
  journal = "J. Symbolic Computation",
  volume = "16",
  number = "2",
  pages = "83-114",
  year = "1993",
  abstract =
    "We present an elimination method for polynomial systems, in the form
    of three main algorithms. For any given system  $[\mathbb{P}], \mathbb{Q}$ 
    of two sets of multivariate polynomials, one of the algorithms computes a
    sequence of triangular forms  $\mathbb{T}_1, \dots, \mathbb{T}_e$  and
    polynomial sets  $\mathbb{U}_1, \dots, \mathbb{U}_e$  such that
     $\text{Zero}(\mathbb{P}/\mathbb{Q})$ 
 $= \bigcup_{i=1}^e \{\text{Zero}(\mathbb{T}_i/\mathbb{U}_i)\}$ ,
    where  $\text{Zero}(\mathbb{P}/\mathbb{Q})$  denotes the set of common zeros of
    the polynomials in  $\mathbb{P}$  which are not zeros of any polynomial in
     $\mathbb{Q}$ , and similarly for  $\text{Zero}(\mathbb{T}_i/\mathbb{U}_i)$ .
    The two other algorithms compute the same zero decomposition but with nicer
    properties such as  $\text{Zero}(\mathbb{T}_i/\mathbb{U}_i) \neq \emptyset$  for each  $i$ .
    One of them, for which the computed triangular systems
     $[\mathbb{T}_i, \mathbb{U}_i]$  possess the projection property, provides
    a quantifier elimination procedure for algebraically closed fields.
    For the other, the computed triangular forms  $\mathbb{T}_i$  are
    irreducible. The relationship between our method and some existing
    elimination methods is explained. Experimental data for a set of test
    examples by a draft implementation of the method are provided, and show
    that the efficiency of our method is comparable with that of some
    well-known methods. A few encouraging examples are given in detail for
    illustration."
```

paper = "Wang93.pdf"
}

— axiom.bib —

```
@article{Wang94,
  author = "Wang, Dongming",
  title = "Differentiation and Integration of Indefinite Summations with
    Respect to Indexed Variables - Some Rules and Applications",
  journal = "J. Symbolic Computation",
  volume = "18",
  number = "3",
  pages = "249-263",
  year = "1994",
  abstract =
    "In this paper we present some rules for the differentiation and
    integration of expressions involving indefinite summations with
    respect to indexed variables which have not yet been taken into
    account of current computer algebra systems. These rules, together
    with several others, have been implemented in MACSYMA and MAPLE as a
    toolkit for manipulating indefinite summations. We discuss some
    implementation issues and report our experiments with a set of typical
    examples. The present work is motivated by our investigation in the
    computer-aided analysis and derivation of artificial neural systems.
    The application of our rules to this subject is briefly explained.",
  paper = "Wang94.pdf"
}
```

— axiom.bib —

```
@article{Wang95a,
  author = "Wang, Dongming",
  title = "A Method for Proving Theorems in Differential Geometry and
    Mechanics",
  journal = "J. Universal Computer Science",
  volume = "1",
  number = "9",
  pages = "658-673",
  year = "1995",
  link = "\url{http://www.jucs.org/jucs\1\9/a\_method\_for\_proving}",
  abstract =
    "A zero decomposition algorithm is presented and used to devise a
    method for proving theorems automatically in differential geometry and
    mechanics. The method has been implemented and its practical
    efficiency is demonstrated by several non-trivial examples including
    Bertrand's theorem, Schell's theorem and Kepler-Newton's laws.",
  paper = "Wang95a.pdf"
}
```

— axiom.bib —

```
@misc{Wang98,
  author = "Wang, Dongming",
  title = "Decomposing Polynomial Systems into Simple Systems",
  volume = "25",
  number = "3",
  pages = "295-314",
  year = "1998",
  abstract =
    "A simple system is a pair of multivariate polynomial sets (one set
    for equations and the other for inequations) ordered in triangular
    form, in which every polynomial is squarefree and has non-vanishing
    leading coefficient with respect to its leading variable. This paper
    presents a method that decomposes any pair of polynomial sets into
    finitely many simple systems with an associated zero decomposition.
    The method employs top-down elimination with splitting and the
    formation of subresultant regular subchains as basic operation.",
  paper = "Wang98.pdf"
}
```

— axiom.bib —

```
@article{Wang99,
  author = "Wang, Dongming",
  title = "Polynomial Systems from Certain Differential Equations",
  journal = "J. Symbolic Computation",
  volume = "28",
  number = "1-2",
  pages = "303-315",
  year = "1999",
  abstract =
    "In this paper, combined elimination techniques are applied to
    establish relations among center conditions for certain cubic
    differential systems initially investigated by Kukles in 1944. The
    obtained relations clarify recent rediscoveries of some known
    conditions of Cherkas. The computational difficulties of establishing
    the complete center conditions for Kukles system, a problem that is
    still open, are illustrated by interactive elimination. Some generated
    polynomial systems that need be solved are made available
    electronically for other developers to test elimination algorithms and
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    their implementations.",
    paper = "Wang99.pdf"
}

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— axiom.bib —

```

@article{Wang00,
  author = "Wang, Dongming",
  title = "Computing Triangular Systems and Regular Systems",
  journal = "J. Symbolic Computation",
  volume = "30",
  number = "2",
  pages = "221-236",
  year = "2000",
  abstract =
    "A previous algorithm of computing simple systems is modified and
    extended to compute triangular systems and regular systems from any
    given polynomial system. The resulting algorithms, based on the
    computation of subresultant regular subchains, have a simple structure
    and are efficient in practice. Preliminary experiments indicate that
    they perform at least as well as some of the known algorithms. Several
    properties about regular systems are also proved.",
  paper = "Wang00.pdf"
}

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— axiom.bib —

```

@article{Wang04,
  author = "Wang, Dongming",
  title = "A simple method for implicitizing rational curves and surfaces",
  journal = "J. Symbolic Computation",
  volume = "38",
  number = "1",
  pages = "899-914",
  year = "2004",
  abstract =
    "This paper presents a simple method for converting rational
    parametric equations of curves and surfaces into implicit
    equations. The method proceeds via writing out the implicit polynomial
    $$$ of estimated degree with indeterminate coefficients $u_i$,
    substituting the rational expressions for the given parametric curve
    or surface into $$$ to yield a rational expression $g/h$ in the

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parameter $$ (or $$ and $t$), equating the coefficients of $g$ in
terms of $$ (and $t$) to 0 to generate a sparse, partially triangular
system of linear equations in $u_i$ with constant coefficients, and
finally solving the linear system for $u_i$. If a nontrivial solution is
found, then an implicit polynomial is obtained; otherwise, one repeats
the same process, increasing the degree of $F$. Our experiments show
that this simple method is efficient. It performs particularly well in
the presence of base points and may detect the dependency of
parameters incidentally.",
paper = "Wang04.pdf"
}

```

— axiom.bib —

```

@inproceedings{Brad92,
  author = "Bradford, Russell",
  title = "Algebraic Simplification of Multiple-Valued Functions",
  booktitle = "Proc. DISCO 92",
  series = "Lecture Notes in Computer Science 721",
  year = "1992",
  abstract =
    "Many current algebra systems have a lax attitude to the
    simplification of expressions involving functions like log and
    $\sqrt{\phantom{x}}$, leading the the ability to ‘‘prove’’ equalities like $-1=1$
    in such systems. In fact, only a little elementary arithmetic is
    needed to devise what the correct simplification should be. We detail
    some of these simplification rules, and outline a method for their
    incorporation into an algebra system.",
  paper = "Brad92.djvu"
}

```

— axiom.bib —

```

@misc{Kais09,
  author = "Kaisler, Stephen H. and Madey, Gregory",
  title = "Complex Adaptive Systems: Emergence and Self-Organization",
  year = "2009",
  institution = "University of Notre Dame",
  comment = "source for Sweeney.eps",
  link = "\url{http://www3.nd.edu/~gmadey/Activities/CAS-Briefing.pdf}"
}

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— axiom.bib —

```
@article{Kalm01,
  author = "Kalman, Dan",
  title = "A Generalized Logarithm for Exponential-Linear Equations",
  journal = "The College Mathematics Journal",
  volume = "32",
  number = "1",
  year = "2001",
  abstract =
    "How do you solve the equation
    \[1.6^x = 5054.4 - 122.35*x\]
    We will refer to equations of this type, with an exponential
    expression on one side and a linear one on the other, as
    {\sl exponential-linear} equations. Numerical approaches such as Newtons
    method or bisection quickly lead to accurate approximate solutions of
    exponential-linear equations. But in terms of the elementary functions
    of calculus and college algebra, there is no analytic solution.

    One approach to remedying this situation is to introduce a special
    function designed to solve exponential-linear equations. Quadratic
    equations, by way of analogy, are solvable in terms of the special
    function  $\sqrt{x}$ , which in turn is simply the inverse of a very
    special and simple quadratic function. Similarly, exponential
    equations are solvable in terms of the natural logarithm {\sl log},
    and that too is the inverse of a very special function. So it is
    reasonable to ask whether there is a special function in terms of
    which exponential-linear equations might be solved. Furthermore, an
    obvious strategy for finding such a function is to invert some simple
    function connected with exponential-linear equations.

    This line of thinking proves to be immediately successful, and leads
    to a function I call {\sl glog} (pronounced {\sl gee-log}) which is a
    kind of generalized logarithm. As intended, glog can be used to solve
    exponential-linear equations. But that is by no means all it is good
    for. For example, with glog you can write a closed-form expression for
    the iterated exponential ( $x^{x^{x^{\dots}}}$ ), and solve  $x + y = x^y$  for
     $y$ . The glog function is also closely related to another special
    function, called the Lambert  $W$  function in [3] and [6], whose study
    dates to work of Lambert in 1758 and of Euler in 1777. Interesting
    questions about glog arise at every turn, from symbolic integration,
    to inequalities and estimation, to numerical computation. Elaborating
    these points is the goal of this paper.",
  paper = "Kalm01.pdf"
}
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```
@phdthesis{Dewa91,
  author = "Dewar, Michael C.",
  title = "Interfacing algebraic and numeric computation",
  year = "1991",
  school = "University of Bath, UK, England"
}
```

— axiom.bib —

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@misc{Fate92,
  author = "Fateman, Richard J. and Einwohner, Theodore H.",
  title = "A Proposal for Automated Integral Tables (Work in Progress)",
  year = "1992",
  link = "\url{http://people.eecs.berkeley.edu/~fateman/papers/intable.pdf}",
  abstract =
    "One of the long-term general goals of algebraic manipulation systems
    has been the automation of difficult or tedious, yet common, symbolic
    mathematical operations. Prominent amount these has been symbolic
    integration. Although some effective algorithms have been devised for
    integration expecially for those problems solvable in terms of
    elementary functions and a few additional special functions, the vast
    majority of entires in large tables of indefinite and definite
    integrals remain out of reach of current machine algorithms. We
    propose techniques for introducing the information in such tables to
    computers, extending such tables, and measuring the success of such
    automation.

    Similar tabular data concerning simplifications, summation identities,
    and similar formulas could also be treated by some of the same
    techniques.",
  paper = "Fate92.pdf"
}
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— axiom.bib —

```
@misc{Atha03,
  author = "Athale, Rahul Ramesh and Diaz, Glauco Alfredo Lopez",
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title = "Explaining Schwarz's LODEF algorithm with examples",
link = "\url{http://www.risc.jku.at/publications/download/risc/_1539/03-11.ps.gz}",
year = "2003",
abstract =
  "To every linear homogeneous ordinary differential equation
  \[L(y)=y^{\{n\}}+a_1y^{\{n-1\}}+\cdots+a_{n-1}y^{\{\prime\}}+a_ny=0\]
  one can associate the linear operator:
  \[L(D)[y]=(D^n+a_1D^{n-1}+\cdots+a_{n-1}D+a_n)[y]\]
  Here,  $D^i$  is another notation for the  $i$ -th derivative of  $y$ ,
  and the coefficients  $a_i$  belong to a differential field  $K$ .
  Such an operator is called a linear homogeneous differential operator
  either over  $K$ , or with coefficients in  $K$ . Linear homogeneous
  differential operators over  $K$  form a ring under the usual addition
  of operators and composition as multiplication.

  {\bf Problem}: Decompose  $L$  as a product of operators of lower degree
  in  $D$ . We allow expressions algebraic over  $K$ .",
paper = "Atha03.pdf"
}

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— axiom.bib —

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@techreport{Chou89,
  author = "Chou, Shang-Ching and Gao, Xiao-Shan",
  title = "A Collection of 120 Computer Solved Geometry Problems in
  Mechanical Formula Derivation",
  institution = "University of Texas, Austin",
  link = "\url{http://www.cs.utexas.edu/ftp/techreports/tr89-22.pdf}",
  type = "technical report",
  number = "tr-89-22",
  year = "1989",
  abstract =
    "This is a collection of 120 geometric problems mechanically solved by
    a program based on the methods introduced by us. Researchers can use
    this collection to experiment with their methods/programs similar to
    ours. It consists of two parts: the exact specification of the input
    to our program and a collection of 120 examples. A typical example
    consists of an informal description of the geometric problem, the
    input to the program which is the exact specification of the problem,
    the result of the problem, and a diagram.",
  paper = "Chou89.pdf"
}

```

— axiom.bib —

```
@book{Chou94,
  author = "Chou, Shang-Ching and Gao, Xiao-Shan and Zhang, Jing-Zhong",
  title = "Machine Proofs in Geometry: Automated Production of Readable
    Proofs for Geometry Theorems",
  publisher = "World Scientific",
  link = "\url{http://citeseerx.ist.psu.edu/viewdoc/download?doi=10.1.1.374.11778}",
  year = "1994",
  paper = "Chou94.pdf"
}
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— axiom.bib —

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@misc{Stur00,
  author = "Sturmfels, Bernd",
  title = "Solving Systems of Polynomial Equations",
  link = "\url{https://math.berkeley.edu/~bernd/cbms.pdf}",
  year = "2000",
  abstract =
    "One of the most classical problems of mathematics is to solve systems
    of polynomial equations in several unknowns. Today, polynomial
    models are ubiquitous and widely applied across the sciences. They
    arise in robotics, coding theory, optimization, mathematical
    biology, computer vision, game theory, statistics, machine learning,
    control theory, and numerous other areas. The set of solutions to a
    system of polynomial equations is an algebraic variety, the basic
    object of algebraic geometry. The algorithmic study of algebraic
    varieties is the central theme of computational algebraic
    geometry. Exciting recent developments in symbolic algebra and
    numerical software for geometric calculations have revolutionized
    the field, making formerly inaccessible problems tractable, and
    providing fertile ground for experimentation and conjecture."
```

The first half of this book furnishes an introduction and represents a snapshot of the state of the art regarding systems of polynomial equations. Afficionados of the well-known text books by Cox, Little, and OShea will find familiar themes in the first five chapters: polynomials in one variable, Groebner bases of zero-dimensional ideals, Newton polytopes and Bernsteins Theorem, multidimensional resultants, and primary decomposition.

The second half of this book explores polynomial equations from a variety of novel and perhaps unexpected angles. Interdisciplinary connections are introduced, highlights of current research are

discussed, and the authors hopes for future algorithms are outlined. The topics in these chapters include computation of Nash equilibria in game theory, semidefinite programming and the real Nullstellensatz, the algebraic geometry of statistical models, the piecewise-linear geometry of valuations and amoebas, and the Ehrenpreis-Palamodov theorem on linear partial differential equations with constant coefficients.

Throughout the text, there are many hands-on examples and exercises, including short but complete sessions in the software systems maple, matlab, Macaulay 2, Singular, PHC, and SOSTools . These examples will be particularly useful for readers with zero background in algebraic geometry or commutative algebra. Within minutes, anyone can learn how to type in polynomial equations and actually see some meaningful results on the computer screen.",

```
paper = "Stur00.pdf"
}
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— axiom.bib —

```
@misc{NTCI16,
  author = "NTCIR",
  title = "Axiom (computer algebra system)",
  link = "\url{http://ntcir11-wmc.nii.ac.jp/index.php/Axiom\_(computer_algebra_system)}",
  year = "2016",
  keywords = "axiomref"
}
```

— axiom.bib —

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@misc{Vers16,
  author = "Verstraete, Jacques",
  title = "Combinatorial Calculus of Formal Power Series",
  comment = "264A Lecture B",
  link = "\url{http://www.math.ucsd.edu/~jverstra/264A-LECTUREB.pdf}",
  paper = "Vers16.pdf"
}
```

— axiom.bib —

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@article{Koep92,
  author = "Koepf, Wolfram",
  title = "Power Series in Computer Algebra",
  journal = "J. Symbolic Computation",
  volume = "13",
  pages = "581-603",
  year = "1992",
  abstract =
    "Formal power series (FPS) of the form
    
$$\sum_{k=0}^{\infty} a_k(x-x_0)^k$$

    are important in calculus and
    complex analysis. In some Computer Algebra Systems (CASs) it is
    possible to define an FPS by direct or recursive definition of its
    coefficients. Since some operations cannot be directly supported
    within the FPS domain, some systems generally convert FPS to finite
    truncated power series (TPS) for operations such as addition,
    multiplication, division, inversion and formal substitution. This
    results in a substantial loss of information. Since a goal of
    Computer Algebra is - in contrast to numerical programming - to work
    with formal objects and preserve such symbolic information, CAS should
    be able to use FPS when possible.

    There is a one-to-one correspondence between FPS with positive radius
    of convergence and corresponding analytic functions. It should be
    possible to automate conversion between these forms. Among CASs
    only MACSYMA provides a procedure {\tt powerseries} to calculate FPS from
    analytic expressions in certain special cases, but this is rather
    limited.

    Here we give an algorithmic approach for computing an FPS for a
    function from a very rich family of functions including all of the
    most prominent ones that can be found in mathematical dictionaries
    except those where the general coefficient depends on the Bernoulli,
    Euler, or Eulerian numbers. The algorithm has been implemented by the
    author and A. Rennoch in the CAS MATHEMATICA, and by D. Gruntz in
    MAPLE.

    Moreover, the same algorithm can sometimes be reversed to calculate a
    function that corresponds to a given FPS, in those cases when a
    certain type of ordinary differential equation can be solved.",
  paper = "Koep92.pdf"
}

```

— axiom.bib —

```

@article{Asla96,
  author = "Aslaksen, Helmer",

```

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title = "Multiple-valued complex functions and computer algebra",
journal = "SIGSAM Bulletin",
volume = "30",
number = "2",
year = "1996",
pages = "12-20",
link = "\url{http://www.math.nus.edu.sg/aslaksen/papers/cacas.pdf}",
abstract =
  "I recently taught a course on complex analysis. That forced me to
  think more carefully about branches. Being interested in computer
  algebra, it was only natural that I wanted to see how such programs
  dealt with these problems. I was also inspired by a paper by
  Stoutemyer.

  While programs like Derive, Maple, Mathematica and Reduce are very
  powerful, they also have their fair share of problems. In particular,
  branches are somewhat of an Achilles' heel for them. As is well-known,
  the complex logarithm function is properly defined as a
  multiple-valued function. And since the general power and exponential
  functions are defined in terms of the logarithm function, they are
  also multiple-valued. But for actual computations, we need to make
  them single valued, which we do by choosing a branch. In Section 2, we
  will consider some transformation rules for branches of
  multiple-valued complex functions in painstaking detail.

  The purpose of this short article is not to do a comprehensive
  comparative study of different computer algebra systems. My goal is
  simply to make the readers aware of some of the problems, and to
  encourage the readers to sit down and experiment with their favourite
  programs.",
paper = "Asla96.pdf"
}

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— axiom.bib —

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@article{Tonixx,
  author = "Tonisson, Eno",
  title = "Branch Completeness in School Mathematics and in Computer Algebra
  Systems",
  journal = "The Electronic Journal of Mathematics and Technology",
  volume = "1",
  number = "1",
  year = "2000",
  issn = "1933-2823",
  link = "\url{https://php.radford.edu/~ejmt/deliveryBoy.php?paper=eJMT_v1n3p5}",
  abstract =

```

```

    "In many cases when solving school algebra problems (e.g. simplifying
    an expression, solving an equation), the solution is separable into
    branches in some manner. The paper describes some approaches to
    branches that are used in school textbooks and computer algebra
    systems and compares them with mathematically branch-complete
    solutions. It tries to identify possible reasons behind different
    approaches and also indicate some ideas how such differences could be
    explained to the students.",
    paper = "Tonixx.pdf"
}

```

— axiom.bib —

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@article{Stou91,
  author = "Stoutemyer, David R.",
  title = "Crimes and misdemeanors in the computer algebra trade",
  journal = "Notices of the American Mathematical Society",
  volume = "38",
  number = "7",
  pages = "778-785",
  year = "1991"
}

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— axiom.bib —

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@misc{Sang10a,
  author = "Sangwin, Chris",
  title = "Intriguing Integrals: Part I and II",
  year = "2010",
  link = "\url{https://plus.maths.org/issue54/features/sangwin/2pdf/index.html/op.pdf}",
  paper = "Sang10a.pdf",
}

```

— axiom.bib —

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@misc{Sang10b,
  author = "Sangwin, Chris",

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```

title = "Intriguing Integrals: Part I and II",
year = "2010",
link = "\url{https://plus.maths.org/issue54/features/sangwin2/2pdf/index.html/op.pdf}",
paper = "Sang10b.pdf"
}

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— axiom.bib —

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@misc{Evanxx,
author = "Evans, Brian",
title = "History of CA Systems",
link = "\url{http://felix.unife.it/Root/d-Mathematics/d-The-mathematician/d-History-of-mathema",
paper = "Evanxx.txt"
}

```

— axiom.bib —

```

@misc{Apel94,
author = "Apel, Joachim and Klaus, Uwe",
title = "Representing Polynomials in Computer Algebra Systems",
year = "1994",
abstract =
  "There are discussed implementational aspects of the special-purpose
  computer algebra system FELIX designed for computations in
  constructive algebra. In particular, data types developed for the
  representation of and computation with commutative and non-commuative
  polynomials are described. Furthermore, comparison of time and memory
  requirements of different polynomial representations are reported.",
paper = "Apel94.pdf"
}

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— axiom.bib —

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@book{Juds15,
author = "Judson, Thomas W.",
title = "Abstract Algebra: Theory and Applications",
year = "2015",

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    link = "\url{http://abstract.ups.edu/aata/colophon-1.html}",
    publisher = "Website"
}

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— axiom.bib —

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@InProceedings{Kalt83,
  author = "Kaltofen, E.",
  title = "On the complexity of finding short vectors in integer lattices",
  booktitle = "Proc. EUROCAL '83",
  series = "Lect. Notes Comput. Sci.",
  year = "1983",
  volume = "162",
  pages = "236--244",
  publisher = "Springer-Verlag",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/83/Ka83_eurocal.pdf}",
  paper = "Kalt83.pdf"
}

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— axiom.bib —

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@InProceedings{Kalt85,
  author = "Kaltofen, E.",
  title = "Effective {Hilbert} Irreducibility",
  booktitle = "Proc. EUROSAM '84",
  pages = "275--284",
  year = "1985",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/85/Ka85_infcontr.ps.gz}",
  paper = "Kalt85.ps"
}

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— axiom.bib —

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@TechReport{Kalt85c,
  author = "Kaltofen, E.",
  title = "Sparse Hensel lifting",
  institution = "RPI",

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address = "Dept. Comput. Sci., Troy, N. Y.",
year = "1985",
number = "85-12",
link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/85/Ka85_techrep.pdf}",
paper = "Kalt85c.pdf"
}

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@InProceedings{Kalt85d,
author = "Kaltofen, E.",
title = "Sparse Hensel lifting",
booktitle = "EUROCAL 85 European Conf. Comput. Algebra Proc. Vol. 2",
pages = "4--17",
year = "1985",
link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/85/Ka85_eurocal.pdf}",
paper = "Kalt85d.pdf"
}

```

— axiom.bib —

```

@Article{Mill88,
author = "Miller, G.L. and Ramachandran, V. and Kaltofen, E.",
title = "Efficient parallel evaluation of straight-line code and
arithmetic circuits",
journal = "SIAM J. Comput.",
year = "1988",
volume = "17",
number = "4",
pages = "687--695",
link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/88/MRK88.pdf}",
paper = "Mill88.pdf"
}

```

— axiom.bib —

```

@Article{Greg88,

```

```

author = "Gregory, B.; Kaltofen, E.",
title = "Analysis of the binary complexity of asymptotically fast
        algorithms for linear system solving",
journal = "SIGSAM Bulletin",
year = "1988",
month = "April",
volume = "22",
number = "2",
pages = "41--49",
link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/88/GrKa88.pdf}",
paper = "Grey88.pdf"
}

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— axiom.bib —

```

@Article{Kalt89a,
author = "Kaltofen, E.; Rolletschek, H.",
title = "Computing greatest common divisors and factorizations in
        quadratic number fields",
journal = "Math. Comput.",
year = "1989",
volume = "53",
number = "188",
pages = "697--720",
link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/89/KaRo89.pdf}",
paper = "Kalt89a.pdf"
}

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— axiom.bib —

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@Unpublished{Kalt89b,
author = "Kaltofen, E.",
title = "Processor efficient parallel computation of polynomial greatest
        common divisors",
note = "unknown",
year = "1989",
month = "July",
link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/89/Ka89_gcd.ps.gz}",
paper = "Kalt89b.ps"
}

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— axiom.bib —

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@TechReport{Kalt89c,
  author = "Kaltofen, E.",
  title = "Parallel Algebraic Algorithm Design",
  institution = "RPI",
  address = "Dept. Comput. Sci., Troy, New York",
  year = "1989",
  month = "July",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/89/Ka89_parallel.ps.gz}",
  paper = "Kalt89c.ps"
}
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— axiom.bib —

```
@InProceedings{Cann89,
  author = "Canny, J. and Kaltofen, E. and Yagati, Lakshman",
  title = "Solving systems of non-linear polynomial equations faster",
  booktitle = "Proc. 1989 Internat. Symp. Symbolic Algebraic Comput.",
  pages = "121--128",
  year = "1989",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/89/CKL89.pdf}",
  paper = "Cann89.pdf"
}
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— axiom.bib —

```
@Article{Kalt90b,
  author = "Kaltofen, E.",
  title = "Computing the irreducible real factors and components of an
          algebraic curve",
  journal = "Applic. Algebra Engin. Commun. Comput.",
  year = "1990",
  volume = "1",
  number = "2",
  pages = "135--148",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/90/Ka90_aaecc.pdf}",
  paper = "Kalt90b.pdf"
}
```

— axiom.bib —

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@Article{Kalt90d,
  author = "Kaltofen, E.; Trager, B.",
  title = "Computing with polynomials given by black boxes for their
    evaluations: Greatest common divisors, factorization, separation of
    numerators and denominators",
  journal = "J. Symbolic Comput.",
  year = "1990",
  volume = "9",
  number = "3",
  pages = "301--320",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/90/KaTr90.pdf}",
  paper = "Kalt90d.pdf"
}
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— axiom.bib —

```
@InProceedings{Kalt91a,
  author = "Kaltofen, E. and Singer, M.F.",
  editor = "D. V. Shirkov and V. A. Rostovtsev and V. P. Gerdt",
  title = "Size efficient parallel algebraic circuits for partial derivatives",
  booktitle =
    "IV International Conference on Computer Algebra in Physical Research",
  pages = "133--145",
  publisher = "World Scientific Publ. Co.",
  year = "1991",
  address = "Singapore",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/91/KaSi91.pdf}",
  paper = "Kalt91a.pdf"
}
```

— axiom.bib —

```
@InProceedings{Kalt93,
  author = "Kaltofen, E.",
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```

title = "Computational Differentiation and Algebraic Complexity Theory",
booktitle = "Workshop Report on First Theory Institute on Computational
            Differentiation",
editor = "C. H. Bischof and A. Griewank and P. M. Khademi",
publisher = "Argonne National Laboratory",
address = "Argonne, Illinois",
series = "Tech. Rep.",
volume = "ANL/MCS-TM-183",
month = "December",
year = "1993",
pages = "28--30",
link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/93/Ka93_diff.pdf}",
paper = "Kalt93.pdf"
}

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— axiom.bib —

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@Article{Kalt93b,
  author = "Kaltofen, E.",
  title = "Direct proof of a theorem by Kalkbrener, Sweedler, and Taylor",
  journal = "SIGSAM Bulletin",
  year = "1993",
  volume = "27",
  number = "4",
  pages = "2",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/93/Ka93_sambull.ps.gz}",
  paper = "Kalt93b.ps"
}

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— axiom.bib —

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@InProceedings{Kalt94,
  author = "Kaltofen, E. and Pan, V.",
  title = "Parallel solution of Toeplitz and Toeplitz-like linear
            systems over fields of small positive characteristic",
  booktitle = "Proc. First Internat. Symp. Parallel Symbolic Comput.",
  pages = "225--233",
  year = "1994",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/94/KaPa94.pdf}",
  paper = "Kalt94.pdf"
}

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— axiom.bib —

```
@article{Raab13,
  author = "Raab, C.G.",
  title = "Generalization of Risch's Algorithm to Special Functions",
  journal = "CoRR",
  volume = "abs/1305.1481",
  year = "2013",
  link = "\url{http://arxiv.org/pdf/1305.1481v1.pdf}",
  abstract =
    "Symbolic integration deals with the evaluation of integrals in closed
    form. We present an overview of Risch's algorithm including recent
    developments. The algorithms discussed are suited for both indefinite
    and definite integration. They can also be used to compute linear
    relations among integrals and to find identities for special functions
    given by parameter integrals. The aim of this presentation is twofold:
    to introduce the reader to some basic ideas of differential algebra in
    the context of integration and to raise awareness in the physics
    community of computer algebra algorithms for indefinite and definite
    integration. ",
  paper = "Raab13.pdf"
}
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— axiom.bib —

```
@InProceedings{Sama95,
  author = "Samadani, M. and Kaltofen, E.",
  title = "Prediction based task scheduling in distributed computing",
  booktitle = "Languages, Compilers and Run-Time Systems for Scalable
    Computers",
  editor = "B. K. Szymanski and B. Sinharoy",
  publisher = "Kluwer Academic Publ.",
  address = "Boston",
  pages = "317--320",
  year = "1996",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/95/SaKa95_poster.ps.gz}",
  paper = "Sama95.ps"
}
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— axiom.bib —

```
@InProceedings{Erli96,
  author = "Erlingsson, U. and Kaltofen, E. and Musser, D.",
  title = "Generic {Gram}-{Schmidt} Orthogonalization by Exact Division",
  booktitle = "Proc. 1996 Internat. Symp. Symbolic Algebraic Comput.",
  year = "1996",
  pages = "275--282",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/96/EKM96.pdf}",
  paper = "Erli96.pdf"
}
```

— axiom.bib —

```
@InProceedings{Kalt96,
  author = "Kaltofen, E. and Lobo, A.",
  title = "On rank properties of {Toeplitz} matrices over finite fields",
  booktitle = "Proc. 1996 Internat. Symp. Symbolic Algebraic Comput.",
  year = "1996",
  pages = "241--249",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/96/KaLo96_issac.pdf}",
  paper = "Kalt96.pdf"
}
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— axiom.bib —

```
@Article{Kalt97,
  author = "Kaltofen, E.",
  title = "Teaching Computational Abstract Algebra",
  journal = "Journal of Symbolic Computation",
  volume = "23",
  number = "5-6",
  pages = "503--515",
  year = "1997",
  note = "Special issue on education, L. Lambe, editor.",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/97/Ka97_jsc.pdf}",
  abstract = "
    We report on the contents and pedagogy of a course in abstract algebra
    that was taught with the aid of educational software developed within
    the Mathematica system. We describe the topics covered and the
```

```

    didactical use of the corresponding Mathematica packages, as well as
    draw conclusions for future such courses from the students' comments
    and our own experience.",
    paper = "Kalt97.pdf",
    keywords = "axiomref"
}

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@Unpublished{Hitz97,
  author = "Hitz, M. A. and Kaltofen, E.",
  title = "The {Kharitonov} theorem and its applications in symbolic
    mathematical computation",
  note = "unknown",
  year = "1997",
  month = "May",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/97/HiKa97_kharit.pdf}",
  paper = "Hitz97.pdf"
}

```

— axiom.bib —

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@InProceedings{Bern99,
  author = "Bernardin, L. and Char, B. and Kaltofen, E.",
  title = "Symbolic Computation in {Java}: an Appraisalment",
  booktitle = "Proc. 1999 Internat. Symp. Symbolic Algebraic Comput.",
  year = "1999",
  pages = "237--244",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/99/BCK99.pdf}",
  paper = "Bern99.pdf"
}

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— axiom.bib —

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@InProceedings{Kalt02,
  author = "Kaltofen, Erich and McLean, Michael and Norris, Larry",
  title = "'{Using} {Maple} to Grade {Maple}' Assessment Software from

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        {North Carolina State University}",
booktitle = "Proceedings 2002 Maple Workshop",
year = "2002",
publisher = "Waterloo Maple Inc.",
address = "Waterloo, Canada",
link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/02/KMN02.pdf}",
paper = "Kalt02.pdf"
}

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— axiom.bib —

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@Book{Grab03,
  editor = "Grabmeier, J. and Kaltofen, E. and Weispfenning, V.",
  title = "Computer Algebra Handbook",
  publisher = "Springer-Verlag",
  year = "2003",
  note = "637 + xx~pages + CD-ROM. Includes E. Kaltofen and V. Weispfenning
    \S1.4 Computer algebra -- impact on research, pages 4--6;
    E. Kaltofen
    \S2.2.3 Absolute factorization of polynomials, page 26;
    E. Kaltofen and B. D. Saunders
    \S2.3.1 Linear systems, pages 36--38;
    R. M. Corless, E. Kaltofen and S. M. Watt
    \S2.12.3 Hybrid methods, pages 112--125;
    E. Kaltofen
    \S4.2.17 FoxBox and other blackbox systems, pages 383--385.",
  isbn = "3-540-65466-6",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/01/symnum.pdf}",
  paper = "Grab03.pdf",
  keywords = "axiomref",
  beebe = "Grabmeier:2003:CAH"
}

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— Grabmeier:2003:CAH —

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@Book{Grabmeier:2003:CAH,
  editor = "Johannes Grabmeier and Erich Kaltofen and Volker
    Weispfenning",
  title = "Computer algebra handbook: foundations, applications,
    systems",
  publisher = "pub-SV",
  address = "pub-SV:adr",
}

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pages =      "xx + 637",
year =       "2003",
ISBN =       "3-540-65466-6",
ISBN-13 =    "978-3-540-65466-7",
LCCN =       "QA155.7.E4 C64954 2003",
MRclass =    "68W30, 00B15, 68-06",
bibdate =    "Tue Nov 22 06:00:25 MST 2005",
bibsource =  "http://www.math.utah.edu/pub/tex/bib/axiom.bib;
              z3950.loc.gov:7090/Voyager",
note =       "Includes CD-ROM.",
link =       "\url{http://www.springer.com/sgw/cda/frontpage/0,11855,1-102-22-1477871-0,00.html}",
acknowledgement = ack-nhfb,
keywords =   "Aldor; AXIOM; Derive; exact arithmetic; Macsyma;
              Magma; Maple Mathematica; MuPAD; REDUCE; TI-92",
subject =    "Algebra; Data processing",
}

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— axiom.bib —

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@InProceedings{Kalt07,
  author = "Kaltofen, Erich and Li, Bin and Sivaramakrishnan, Kartik and
            Yang, Zhengfeng and Zhi, Lihong",
  title = "Lower bounds for approximate factorizations via semidefinite
            programming (extended abstract)",
  year = "2007",
  booktitle = "SNC'07 Proc. 2007 Internat. Workshop on Symbolic-Numeric Comput.",
  pages = "203--204",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/07/KLSYZ07.pdf}",
  paper = "Kalt07.pdf"
}

```

— axiom.bib —

```

@Article{Borw07,
  author = "Borwein, Peter and Kaltofen, Erich and Mossinghoff, Michael J.",
  title = "Irreducible Polynomials and {Barker} Sequences",
  journal = "{ACM} Communications in Computer Algebra",
  volume = "162",
  number = "4",
  year = "2007",
  pages = "118--121",
}

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```

    month = "December",
    link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/07/BKM07.pdf}",
    paper = "Borw07.pdf"
}

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— axiom.bib —

```

@Article{Kalt10,
  author = "Kaltofen, Erich and Lavin, Mark",
  title = "Efficiently Certifying Non-Integer Powers",
  journal = "Computational Complexity",
  year = "2010",
  volume = "19",
  number = "3",
  month = "September",
  pages = "355--366",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/09/KaLa09.pdf}",
  paper = "Kalt10.pdf"
}

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— axiom.bib —

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@InProceedings{Kalt11,
  author = "Kaltofen, Erich L. and Nehring, Michael",
  title = "Supersparse black box rational function interpolation",
  booktitle = "Internat. Symp. Symbolic Algebraic Comput. ISSAC'11",
  month = "June",
  year = "2011",
  pages = "177--185",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/11/KaNe11.pdf}",
  paper = "Kalt11.pdf"
}

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— axiom.bib —

```

@InProceedings{Gren11a,
  author = "Grenet, Bruno and Kaltofen, Erich L. and Koiran, Pascal

```

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        and Portier, Natacha",
title = "Symmetric Determinantal Representation of Weakly Skew Circuits",
booktitle = "Proc. 28th Internat. Symp. on Theoretical Aspects of Computer
        Science, STACS 2011",
pages = "543--554",
year = "2011",
link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/11/GKKP11.pdf}",
paper = "Gren11a.pdf"
}

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— axiom.bib —

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@InProceedings{Kalt11a,
author = "Kaltofen, Erich L. and Nehring, Michael and Saunders, David B.",
title = "Quadratic-Time Certificates in Linear Algebra",
booktitle = "Internat. Symp. Symbolic Algebraic Comput. ISSAC'11",
month = "June",
year = "2011",
pages = "171--176",
link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/11/KNS11.pdf}",
paper = "Kalt11a.pdf"
}

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— axiom.bib —

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@InProceedings{Kalt11b,
author = "Kaltofen, Erich L. and Lee, Wen-shin and Yang, Zhengfeng",
title = "Fast estimates of {Hankel} matrix condition numbers
        and numeric sparse interpolation",
booktitle = "Proc. 2011 Internat. Workshop on Symbolic-Numeric Comput.",
month = "June",
year = "2011",
pages = "130--136",
link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/11/KLY11.pdf}",
paper = "Kalt11b.pdf"
}

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— axiom.bib —

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@InProceedings{Guo12,
  author = "Guo, Feng and Kaltofen, Erich L. and Zhi, Lihong",
  title = "Certificates of Impossibility of {Hilbert}--{Artin} Representations
    of a Given Degree for Definite Polynomials and Functions",
  booktitle = "Internat. Symp. Symbolic Algebraic Comput. ISSAC'12",
  month = "July",
  year = "2012",
  pages = "195--202",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/12/GKZ12.pdf}",
  paper = "Guo12.pdf"
}

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— axiom.bib —

```

@InProceedings{Come12a,
  author = "Comer, Matthew T. and Kaltofen, Erich L. and Pernet, Cl{\'e}ment",
  title = "Sparse Polynomial Interpolation and {Berlekamp}/\allowbreak
    {Massey} Algorithms That Correct Outlier Errors in Input Values",
  booktitle = "Internat. Symp. Symbolic Algebraic Comput. ISSAC'12",
  month = "July",
  year = "2012",
  pages = "138--145",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/12/CKP12.pdf}",
  paper = "Come12a.pdf"
}

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— axiom.bib —

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@InProceedings{Boye13,
  author = "Boyer, Brice and Comer, Matthew T. and Kaltofen, Erich L.",
  title = "Sparse Polynomial Interpolation by Variable Shift in
    the Presence of Noise and Outliers in the Evaluations",
  booktitle =
    "Proc. Tenth Asian Symposium on Computer Mathematics (ASCM 2012)",
  year = "2013",
  month = "October",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/13/BCK13.pdf}",
  paper = "Boye13.pdf"
}

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@InProceedings{Kalt13b,
  author = "Kaltofen, Erich and Yang, Zhengfeng",
  title = "Sparse multivariate function recovery from values with noise and
    outlier errors",
  year = "2013",
  booktitle = "Internat. Symp. Symbolic Algebraic Comput. ISSAC'13",
  pages = "219--226",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/13/KaYa13.pdf}",
  paper = "Kalt13b.pdf"
}
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— axiom.bib —

```
@InProceedings{Kalt13c,
  author = "Kaltofen, Erich L.",
  title = "Symbolic Computation and Complexity Theory Transcript of My Talk",
  booktitle =
    "Proc. Tenth Asian Symposium on Computer Mathematics (ASCM 2012)",
  year = "2013",
  month = "October",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/13/Ka13.pdf}",
  paper = "Kalt13c.pdf"
}
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```
@InProceedings{Kalt14,
  author = "Kaltofen, Erich L. and Yang, Zhengfeng",
  title = "Sparse Multivariate Function Recovery With a High Error Rate
    in Evaluations",
  year = "2014",
  booktitle = "Internat. Symp. Symbolic Algebraic Comput. ISSAC'14",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/14/KaYa14.pdf}",
  paper = "Kalt14.pdf"
}
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— axiom.bib —

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@InProceedings{Kalt14a,
  author = "Kaltofen, Erich L. and Pernet, Clément",
  title = "Sparse Polynomial Interpolation Codes and Their Decoding
    Beyond Half the Minimal Distance",
  year = "2014",
  booktitle = "Internat. Symp. Symbolic Algebraic Comput. ISSAC'14",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/14/KaPe14.pdf}",
  paper = "Kalt14a.pdf"
}
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— axiom.bib —

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@InProceedings{Duma14,
  author = "Dumas, Jean-Guillaume and Kaltofen, Erich L.",
  title = "Essentially Optimal Interactive Certificates In Linear Algebra",
  year = "2014",
  booktitle = "Internat. Symp. Symbolic Algebraic Comput. ISSAC'14",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/14/DuKa14.pdf}",
  paper = "Duma14.pdf"
}
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— axiom.bib —

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@InProceedings{Boye14,
  author = "Boyer, Brice B. and Kaltofen, Erich L.",
  title = "Numerical Linear System Solving With Parametric Entries By
    Error Correction",
  year = "2014",
  booktitle = "SNC'14 Proc. 2014 Int. Workshop on Symbolic-Numeric Comput.",
  link = "\url{http://www.math.ncsu.edu/~kaltofen/bibliography/14/BoKa14.pdf}",
  paper = "Boye14.pdf"
}
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1.39 Axiom Citations in the Literature

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— axiom.bib —

```
@book{Abla96,
  author = "Ablamowicz, Rafal and Lounesto, Pertti and Para, Josep M.",
  title = "Clifford algebras with numeric and symbolic computations",
  publisher = "Birkhauser",
  year = "1996",
  abstract =
    "The 20 articles of this volume will be reviewed individually. They
    have been grouped into the following four sections: I) Verifying and
    falsifying conjectures (1); II) Differential geometry quantum
    mechanics, spinors and conformal group (9); III) Generalized Clifford
    algebras and number systems, projective geometry and crystallography
    (7); IV) Numerical methods in Clifford algebras (3).

    Selected electronic materials submitted by our contributors can be
    found at a Web site:
    http://www.birkhauser.com/books/ISBN/0-8176-3907-1

    These materials contain original packages, worksheets, notebooks,
    computer programs, etc., that were used in deriving results presented
    in this book.",
  keyword = "axiomref"
}
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— axiom.bib —

```
@misc{ACA00,
  author = "Kokol-voljc, Vlasta and Kutzler, Bernhard",
  title = "Computer Algebra Meets Education",
  conference = "Sessions of ACA2000",
  year = "2000",
  abstract =
    "Education has become one of the fastest growing application areas for
    computers in general and computer algebra in particular. Computer
    algebra tools such as TI-92/89, Derive, Mathematica, Maple, Axiom,
    Reduce, Macsyma, or Mupad make powerful teaching tools in mathematics,
    physics, chemistry, biology, economy.

    The goal of this session is to exchange ideas and experiences, to hear
```

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        about classroom experiments, and to discuss all issues related to the
        use of computer algebra tools in the classroom (such as assessment,
        change of curricula, new support material, ...)",
keywords = "axiomref"
}

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```

@misc{ACA15,
  author = "Martinez-Moro, Edgar Kotsireas, Ilias",
  title = "21st Conference on Applications of Computer Algebra",
  conference = "Sessions of ACA2015",
  location = "Kalamata, Greece",
  year = "2015",
  link = "\url{http://www.singacom.uva.es/ACA2015/latex/ACAproc.pdf}",
  paper = "ACA15.pdf",
  keywords = "axiomref"
}

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@misc{ACM89,
  author = "ACM",
  title = "Proceedings of the ACM-SIGSAM 1989 International
          Symposium on Symbolic and Algebraic Computation, ISSAC '89",
  year = "1989",
  isbn = "0-89791-325-6",
  link = "\url{http://doi.acm.org/10.1145/74540.74567}",
  doi = "10.1145/74540.74567",
  acmid = "74567",
  publisher = "ACM Press",
  address = "New York, NY, USA",
  keywords = "axiomref"
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@misc{ACM94,

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author = "ACM",
title = "Proceedings of the ACM-SIGSAM 1989 International
        Symposium on Symbolic and Algebraic Computation, ISSAC '94",
year = "1994",
isbn = "0-89791-638-7",
publisher = "ACM Press",
address = "New York, NY, USA",
keywords = "axiomref"
}

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@InProceedings{Adam99a,
  author = "Adams, A.A. and Gottlieb, H. and Linton, S.A. and Martin, U.",
  title = "VSDITLU: A verifiable symbolic definite integral table look-up",
  booktitle = "Automated Deduction",
  series = "CADE 16",
  year = "1999",
  location = "Trento, Italy",
  pages = "112-126",
  link = "\url{http://www.a-cubed.info/Publications/CADE99.pdf}",
  abstract =
    "We present a verifiable symbolic definite integral table lookup: a
     system which matches a query, comprising a definite integral with
     parameters and side conditions, against an entry in a verifiable table
     and uses a call to a library of facts about the reals in the theorem
     prover PVS to aid in the transformation of the table entry into an
     answer. Our system is able to obtain correct answers in cases where
     standard techniques implemented in computer algebra systems fail. We
     present the full model of such a system as well as a description of
     our prototype implementation showing the efficacy of such a system:
     for example, the prototype is able to obtain correct answers in cases
     where computer algebra systems do not. We extend upon Fatemans
     web-based table by including parametric limits of integration and
     queries with side conditions.",
  paper = "Adam99a.pdf",
  keywords = "axiomref"
}

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— axiom.bib —

```

@article{Akba08,

```



```

author = "Akbar Hussain, D.M. and Haq, Shaiq A. and Khan, Zafar Ullah
        and Ahmed, Zaki",
title = "Simple object oriented designed computer algebra system",
journal = "J. Comput. Methods Sci. Eng.",
volume = "8",
number = "3",
pages = "195-211",
year = "2008",
abstract =
    "Computer Algebra System (CAS) is a software program that facilitates
    symbolic mathematics. The core functionality of a typical CAS is
    manipulation of mathematical expressions in symbolic forms. The
    expressions manipulated by CAS normally include polynomials in
    multiple variables, standart trigonometric and exponential functions,
    various special functions for example gamma, zeta, Bessel, etc. and
    also arbitrary functions like derivatives, integrals, sums, and
    products of expressions. Our implementation of a CAS tool provides an
    object orienged design framework. The system is portable to other
    platforms and highly scalable. The other key features include a very
    simple and interactive user GUI support for a formula editor, making
    it a self contained system, additionally the formula editor provides a
    real-time syntax checking for expressions.",
keywords = "axiomref"
}

```

— axiom.bib —

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@article{Aitk99,
  author = "Aitken, William E. and Constable, Robert L. and
          Underwood, Judith L.",
  title = "Metalogical frameworks. II: Developing a reflected decision
          procedure",
  journal = "J. Autom. Reasoning",
  volume = "22",
  number = "2",
  pages = "171-221",
  year = "1999",
  link = "\url{http://www.nuprl.org/documents/Constable/MetalogicalFrameworksII.pdf}",
  abstract =
    "Proving theorems is a creative act demanding new combinations of ideas
    and on occasion new methods of argument. For this reason, theorem
    proving systems need to be extensible. The provers should also remain
    correct under extension, so there must be a secure mechanism for doing
    this. The tactic-style provers pioneered by Edinburgh LCF provide a
    very effective way to achieve secure extensions, but in such systems,
    all new methods must be reduced to tactics. This is a drawback because

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there are other useful proof generating tools such as decision procedures; these include, for example, algorithms which reduce a deduction problem, such as arithmetic provability, to a computation on graphs.

The Nuprl system pioneered the combination of fixed decision procedures with tactics, but the issue of securely adding new ones was not solved. In this paper, we show how to safely include user-defined decision procedures in theorem provers. The idea is to prove properties of the procedure inside the provers logic and then invoke a reflection rule to connect the procedure to the system. We also show that using a rich underlying logic permits an abstract account of the approach so that the results carry over to different implementations and other logics.",
 paper = "Aitk99.pdf",
 keywords = "axiomref"

}

— axiom.bib —

@misc{America,
 author = "america.pink",
 title = "Axiom (computer algebra system)",
 year = "2016",
 link = "\url{http://america.pink/axiom-computer-algebra-system_526647.html}",
 keywords = "axiomref"

}

— axiom.bib —

@inproceedings{Andr90,
 author = "Andrews, George and Baxter, R.J.",
 title = "SCRATCHPAD explorations for elliptic theta functions",
 booktitle = "Computers in Mathematics",
 series = "Lecture Notes in Pure and Appl. Math 125",
 pages = "17-33",
 year = "1990",
 keywords = "axiomref"

}

— axiom.bib —

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@article{Andr09,
  author = "Andrews, George",
  title = "The Meaning of Ramanujan Now and for the Future",
  journal = "Ramanujan Journal",
  volume = "20",
  pages = "257-273",
  year = "2009",
  link = "\url{http://www.personal.psu.edu/gea1/pdf/274.pdf}",
  abstract =
    "December 22, 2010 marks the 123th anniversary of Ramanujan's
    birth. In this paper we pay homage to this towering figure whose
    mathematical discoveries so affected mathematics throughout the
    twentieth century and into the twenty-first.",
  paper = "Andr09.pdf",
  keywords = "axiomref"
}
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— axiom.bib —

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@article{Anto01,
  author = "Antoine, Xavier",
  title = "Microlocal diagonalization of strictly hyperbolic pseudodifferential
    systems and application to the design of radiation conditions in
    electromagnetism",
  journal = "SIAM J. Appl. Math",
  volume = "61",
  number = "6",
  pages = "1877-1905",
  year = "2001",
  abstract =
    "In [Commun. Pure Appl. Math. 28, 457-478 (1975; Zbl 0332.35058)],
    M. E. Taylor described a constructive diagonalization method to
    investigate the reflection of singularities of the solution to
    first-order hyperbolic systems. According to further works initiated
    by Engquist and Majda, this approach proved to be well adapted to the
    construction of artificial boundary conditions. However, in the case
    of systems governed by pseudodifferential operators with variable
    coefficients, Taylors method involves very elaborate calculations of
    the symbols of the operators. Hence, a direct approach may be
    difficult when dealing with high-order conditions.
```

This motivates the first part of this paper, where a recursive explicit formulation of Taylors method is stated for microlocally

```
@article{Arna95a,  
author = "Arnault, Francois",  
title = "Rabin-Miller primality test: Composite numbers which pass it",  
journal = "Mathematics of Computation",  
volume = "64",  
number = "209",  
pages = "355-361",  
year = "1995",  
link = "\url{https://www.jointmathematicsm Meetings.org/mcom/1995-64-209/S0025-5718-1995-1260124-2/S0025-5718-1995-1260124-2/S0025-5718-1995-1260124-2.pdf}",  
abstract =  
    "The Rabin-Miller primality test is a probabilistic test which can be  
found in several algebraic computing systems (such as Pari, Maple,  
ScratchPad) because it is very easy to implement and, with a  
reasonable amount of computing, indicates whether a number is  
composite or ‘‘probably prime’’ with a very low probability of error. In  
this paper, we compute composite numbers which are strong pseudoprimes  
to several chosen bases. Because these bases are those used by the  
ScratchPad implementation of the test, we obtain, by a method which  
differs from a recent one by G. Jaeschke [ibid. 61, 915-926 (1993; Zbl  
0802.11001)], composite numbers which are found to be ‘‘probably prime’’  
by this test.",  
paper = "Arna95a.pdf",  
keywords = "axiomref"  
}
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@article{Atki95,
  author = "Atkinson, M. D. and Linton, S. A. and Walker, L. A.",
  title = "Priority queues and multisets",
  journal = "J. Comb",
  volume = "2",
  pages = "385-402",
  year = "1995",
  link = "\url{http://www.combinatorics.org/ojs/index.php/eljc/article/download/v2i1r24.pdf}",
  abstract =
    "A priority queue, a container data structure equipped with the
    operations insert and delete-minimum, can re-order its input in
    various ways, depending both on the input and on the sequence of
    operations used. If a given input  $\sigma$  can produce a particular
    output  $\tau$  then  $(\sigma, \tau)$  is said to be an allowable pair. It
    is shown that allowable pairs on a fixed multiset are in one-to-one
    correspondence with certain  $k$ -way trees and, consequently, the
    allowable pairs can be enumerated. Algorithms are presented for
    determining the number of allowable pairs with a fixed input
    component, or with a fixed output component. Finally, generating
    functions are used to study the maximum number of output components
    with a fixed input component, and a symmetry result is derived.",
  paper = "Atki95.pdf",
  keywords = "axiomref"
}
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— axiom.bib —

```
@inproceedings{Augo91,
  author = "Augot, D. and Charpin, P. and Sendrier, N.",
  title = "The minimum distance of some binary codes via the
    Newton's identities",
  booktitle = "Int. Symp. on Coding Theory and Applications",
  year = "1991",
  pages = "65-73",
  isbn = "0-387-54303-1",
  abstract =
    "In this paper, we give a natural way of deciding whether a given
    cyclic code contains a word of given weight. The method is based on
    the manipulation of the locators and of the locator polynomial of a
    codeword  $x$ ."

  Because of the dimensions of the problem, we need to use a symbolic
  computation software, like Maple or Scratchpad II. The method can be
```

ineffective when the length is too large.

The paper contains two parts: In the first part we will present the main definitions and properties we need.

In the second part, we will explain how to use these properties, and, as illustration, we will prove the following facts:

```
\begin{itemize}
\item The dual of the BCH code of length 63 and designed distance 9
has true minimum distance 14 (which was already known).
\item The BCH code of length 1023 and designed distance of 253 has
minimum distance 253.
\item The cyclic codes of length  $2^{11}-1$ ,  $2^{13}-1$ ,  $2^{17}-1$ , with
generator polynomial  $m_1(x)$  and  $m_7(x)$  have minimum distance 4.
\end{itemize}",
paper = "Augo91.pdf",
keywords = "axiomref",
beebe = "Augot:1991:MDS"
}
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@InProceedings{Augot:1991:MDS,
  author = "D. Augot and P. Charpin and N. Sendrier",
  title = "The minimum distance of some binary codes via the
{Newton}'s identities",
  crossref = "Cohen:1991:EIS",
  pages = "65--73",
  month = "",
  year = "1991",
  bibdate = "Tue Sep 17 06:41:20 MDT 1996",
  bibsource = "http://www.math.utah.edu/pub/tex/bib/axiom.bib",
  abstract = "The authors propose a natural way of deciding whether
a given cyclic code contains a word of given weight.
The method is based on the manipulation of the locators
and of the locator polynomial of a codeword x. Because
of the dimensions of the problem, one needs to use
symbolic computation software, like Maple or Scratchpad
II. The method can be ineffective when the length is
too large. The paper is in two parts: In the first
part, they present the main definitions and properties
needed. In the second part, they explain how to use
these properties, and, as illustration, prove the three
following facts: the dual of the BCH code of length 63
and designed distance 9 has true minimum distance 14
(which was already known). The BCH code of length 1023
```

```

        and designed distance 253 has minimum distance 253. The
        cyclic codes of length  $2^{11}$ ,  $2^{13}$ ,  $2^{17}$ ,
        with generator polynomial  $m_1(x)$  and  $m_7(x)$ 
        have minimum distance 4.",
    acknowledgement = ack-nhfb,
    affiliation = "Paris 6 Univ., France",
    classification = "B6120B (Codes)",
    keywords = "BCH code; Binary codes; Codeword; Cyclic codes;
        Generator polynomial; Locator polynomial; Minimum
        distance; Newton identities; Symbolic computation",
    language = "English",
    thesaurus = "Codes",
}

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— ignore —

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\bibitem[Adams 94]{AL94}
    author = "Adams, William W. and Loustau, Philippe",
    title = "An Introduction to Gröbner Bases",
    year = "1994",
    American Mathematical Society (1994)
    isbn = "0-8218-3804-0",
    keywords = "axiomref"

```

— axiom.bib —

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@InProceedings{Andr84,
    author = "Andrews, George E.",
    title = "Ramanujan and SCRATCHPAD",
    booktitle = "Proc. of 1984 MACSYM Users' Conference, July 1984",
    location = "General Electric, Schenectady, NY",
    year = "1984",
    pages = "383-408",
    keywords = "axiomref"
}

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— Andrews:1984:RS —

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@InProceedings{Andrews:1984:RS,
  author = "George E. Andrews",
  title = "{Ramanujan} and {SCRATCHPAD}",
  crossref = "Golden:1984:PMU",
  pages = "383--??",
  year = "1984",
  bibsource = "/usr/local/src/bib/bibliography/Theory/Comp.Alg.1.bib;
              http://www.math.utah.edu/pub/tex/bib/axiom.bib",
}

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— axiom.bib —

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@InProceedings{Andr88,
  author = "Andrews, G. E.",
  title = "Application of Scratchpad to problems in special functions and
          combinatorics",
  booktitle = "Trends in Computer Algebra",
  publisher = "Springer",
  series = "Lecture Notes in Comp. Sci. 296",
  year = "1988",
  isbn = "3-540-18928-9",
  pages = "159-166",
  abstract =
    "Within the last few years, there have been numerous applications of
    computer algebra to special functions. G. Gasper (Northwestern
    University) has studied classical hypergeometric functions, and
    W. Gosper (Symbolics, Inc.) has developed a large variety of
    spectacular transformation and summation techniques for MACSYMA. The
    purpose of this note is to explore some of the interface between
    computer algebra and special functions. In section 2 we examine an
    application of MACSYMA which inadequately relied, in my opinion, on
    what was readily available in the literature on hypergeometric
    series. In Section 3 we consider classical observations on sums of
    powers of binomial coefficients. In Section 4 we consider a problem of
    D.M. Jackson wherein SCRATCHPAD and classical hypergeometric series
    interact nicely. We close with a problem inspired by work in
    statistical mechanics which leads us to question about algorithm that
    would be useful in computer algebra applications.

```

In this brief survey, we have illustrated some of the uses of computer algebra. It might be objected that our work could well be carried out in almost any computer language; so why bother with SCRATCHPAD? The answer, of course, lies in the naturalness and simplicity of computer algebra approaches to these problems. Expressions like (2.2), (3.1) and (4.1) can be coded in SCRATCHPAD in one line exactly as they are written. They can then be studied with minimal thought about the


```

computer and maximal concentration on what is happening. Often
mathematical research consists of sifting low grade ore, and when such
sifting requires ingenious programming skills it is likely not to be
carried out.",
keywords = "axiomref",
beebe = "Andrews:1988:ASP"
}

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— Andrews:1988:ASP —

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@InProceedings{Andrews:1988:ASP,
  author =      "G. E. Andrews",
  title =      "Application of {Scratchpad} to problems in special
                functions and combinatorics",
  crossref =    "Janssen:1988:TCA",
  pages =      "158--??",
  year =       "1988",
  bibdate =    "Fri Dec 29 18:28:25 1995",
  bibsource =   "/usr/local/src/bib/bibliography/Theory/Comp.Alg.1.bib;
                http://www.math.utah.edu/pub/tex/bib/axiom.bib",
}

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— axiom.bib —

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@book{Anon91,
  author = "Anonymous",
  title = "Challenges of a Changing World (2 Volumes)",
  publisher = "American Society for Engineering Education",
  year = "1991",
  keywords = "axiomref"
}

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— axiom.bib —

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@book{Anon95,
  author = "Anonymous",
  title = "Zeitschrift fur Angewandte Mathematik und Physik, 75 (suppl. 2)",
  publisher = "Unknown",
}

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year = "1995",
issn = "0044-2267",
keywords = "axiomref"
}

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— axiom.bib —

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@Article{Arna95,
  author = "Arnault, Francois",
  title = "Constructing Carmichael numbers which are strong pseudoprimes to
          several bases",
  year = "1995",
  journal = "Journal of Symbolic Computation",
  volume = "20",
  number = "2",
  pages = "151-161",
  link = "\url{http://ac.els-cdn.com/S0747717185710425/1-s2.0-S0747717185710425-main.pdf}",
  algebra = "\newline\refto{package PRIMES IntegerPrimesPackage}",
  abstract =
    "We describe here a method of constructing Carmichael numbers which
    are strong pseudoprimes to some set of prime bases. We apply it to
    find composite numbers which are found to be prime by the Rabin-Miller
    test of packages as Axiom or Maple. We also use a variation of this
    method to construct strong Lucas pseudoprimes with respect to several
    pairs of parameters.",
  paper = "Arna95.pdf",
  keywords = "axiomref",
  beebe = "Arnault:1995:CCN"
}

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— Arnault:1995:CCN —

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@Article{Arnault:1995:CCN,
  author = "Fran{\c{c}}ois Arnault",
  title = "Constructing {Carmichael} Numbers Which are Strong
          Pseudoprimes to Several Bases",
  journal = "j-J-SYMBOLIC-COMP",
  volume = "20",
  number = "2",
  pages = "151--162 (or 151--161??)",
  month = "aug",
  year = "1995",

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CODEN =      "JSYCEH",
ISSN =       "0747-7171 (print), 1095-855X (electronic)",
ISSN-L =     "0747-7171",
MRclass =    "11Y11 (11A51)",
MRnumber =   "96k:11153",
MRreviewer = "Andrew Granville",
bibdate =    "Sat May 10 15:54:09 MDT 1997",
bibsource =  "http://www.math.utah.edu/pub/tex/bib/axiom.bib;
              http://www.math.utah.edu/pub/tex/bib/jsymcomp.bib",
acknowledgement = ack-nhfb,
classcodes = "C7310 (Mathematics computing); C1160 (Combinatorial
              mathematics)",
corpsource = "Fac. des Sci., Limoges Univ., France",
fjournal =   "Journal of Symbolic Computation",
link =       "\url{http://www.sciencedirect.com/science/journal/07477171}",
keywords =   "Axiom; Carmichael numbers; composite numbers; Lucas;
              Maple; number theory; pseudoprimes; Rabin-Miller test;
              symbol manipulation",
treatment =  "T Theoretical or Mathematical",
}

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@article{Augo97,
  author = "Augot, Daniel and Camion, Paul",
  title = "On the computation of minimal polynomials, cyclic vectors,
           and Frobenius forms",
  journal = "Linear Algebra Appl.",
  volume = "260",
  pages = "61-94",
  year = "1997",
  abstract =
    "Algorithms related to the computation of the minimal polynomial of an
     $x \times n$  matrix over a field  $K$  are introduced. The complexity of
    the first algorithm, where the complete factorization of the
    characteristic polynomial is needed, is  $O(\sqrt{n} \cdot n^3)$ . An
    iterative algorithm for finding the minimal polynomial has complexity
     $O(n^3 + n^2m^2)$ , where  $m$  is a parameter of the shift Hessenberg
    matrix used. The method does not require the knowledge of the
    characteristic polynomial. The average value of  $m$  is  $O(\log n)$ .

```

Next methods are discussed for finding a cyclic vector for a matrix. The authors first consider the case when its characteristic polynomial is squarefree. Using the shift Hessenberg form leads to an algorithm at cost $O(n^3 + n^2m^2)$. A more sophisticated recurrent procedure gives the result in $O(n^3)$ steps. In particular, a normal basis for

an extended finite field of size q^n will be obtained with complexity $O(n^3 + n^2 \log q)$.

Finally, the Frobenius form is obtained with asymptotic average complexity $O(n^3 \log n)$.

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paper = "Augo97.pdf",
keywords = "axiomref"
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@misc{AxiomPressRelease,
  author = "Numerical Algorithms Group",
  title = "Axiom Press Release",
  link = "\url{http://www.dorn.org/uni/sls/kap05/e08\_01.htm}",
  year = "1996",
  month = "July",
  day = "24",
  paper = "AxiomPressRelease.tgz",
  keywords = "axiomref"
}
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— axiom.bib —

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@article{Bacl14,
  author = "Baclawski, Krystian",
  title = "SPAD language type checker",
  journal = "unknown",
  year = "2014",
  link = "\url{http://github.com/cahirwpz/phd}",
  abstract = "
    The project aims to deliver a new type checker for SPAD language.
    Several improvements over current type checker are planned.
    \begin{itemize}
    \item introduce better type inference
    \item introduce modern language constructs
    \item produce understandable diagnostic messages
    \item eliminate well known bugs in the type system
  "
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\item find new type errors
\end{itemize}",
keywords = "axiomref"
}

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— axiom.bib —

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@misc{Bake16,
  author = "Baker, Martin",
  title = "Axiom Maths Program",
  year = "2016",
  link = "\url{http://www.euclideanspace.com/prog/scratchpad/axiom/index.htm}",
  keywords = "axiomref"
}

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— axiom.bib —

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@misc{Bake16a,
  author = "Baker, Martin",
  title = "Axioms in Axiom",
  year = "2016",
  link = "\url{http://www.euclideanspace.com/prog/scratchpad/axiomsinAxiom}",
  keywords = "axiomref"
}

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@article{Ball14,
  author = "Ballarin, Clemens",
  title = "Locales: a module system for mathematical theories",
  journal = "J. Autom. Reasoning",
  volume = "52",
  number = "2",
  pages = "123-153",
  year = "2014",
  link = "\url{http://www21.in.tum.de/~ballarin/publications/jar2013.pdf}",
  abstract =

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"Locales are a module system for managing theory hierarchies in a
theorem prover through theory interpretation. They are available for
the theorem prover Isabelle. In this paper, their semantics is defined
in terms of local theories and morphisms. Locales aim at providing
flexible means of extension and reuse. Theory modules (which are
called locales) may be extended by definitions and
theorems. Interpretation of Isabelle's global theories and proof
contexts is possible via morphisms. Even the locale hierarchy may be
changed if declared relations between locales do not adequately
reflect logical relations, which are implied by the locales'
specifications. By discussing their design and relating it to more
commonly known structuring mechanisms of programming languages and
provers, locales are made accessible to a wider audience beyond the
users of Isabelle. The discussed mechanisms include ML-style functors,
type classes and mixins (the latter are found in modern
object-oriented languages).",
paper = "Ball14.pdf",
keywords = "axiomref"
}

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— axiom.bib —

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@article{Barn02,
  author = "Barnett, Michael P.",
  title = "Computer algebra in the life sciences",
  journal = "SIGSAM Bulletin",
  volume = "36",
  number = "4",
  pages = "5-31",
  year = "2002",
  link = "\url{https://notendur.hi.is/vae11/\%C3\%9Eekking/Systems\%20Biology/Biological\%20Algebra.PDF}",
  abstract =
    "This note (1) provides references to recent work that applies computer
    algebra (CA) to the life sciences, (2) cites literature that explains
    the biological background of each application, (3) states the
    mathematical methods that are used, (4) mentions the benefits of CA,
    and (5) suggests some topics for future work.",
  paper = "Barn02.pdf",
  keywords = "axiomref"
}

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— axiom.bib —

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@misc{Beeb14,
  author = "Beebe, Nelson H. F.",
  title = "A Bibliography of Publications about the AXIOM
          (formerly, Scratchpad) Symbolic Algebra Language",
  year = "2014",
  link = "\url{http://www.netlib.org/bibnet/journals/axiom.ps.gz}",
  paper = "Beeb14.pdf"
}

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— axiom.bib —

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@book{Benk98,
  author = "Benker, Hans",
  title = "Engineering mathematics with computer algebra systems",
  publisher = "Unknown",
  year = "1998",
  comment = "german",
  keywords = "axiomref",
  beebe = "Benker:1998:ICS"
}

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— Benker:1998:ICS —

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@Book{Benker:1998:ICS,
  author = "Hans Benker",
  title = "{Ingenieurmathematik mit Computeralgebra-Systemen.
          AXIOM, DERIVE, MACSYMA, MAPLE, MATHCAD, MATHEMATICA,
          MATLAB und MuPAD in der Anwendung}. (German)
          {Engineering mathematics with computer algebra systems.
          The applications: AXIOM, DERIVE, MACSYMA, MAPLE,
          MATHCAD, MATHEMATICA, MATLAB UND MuPAD.}",
  publisher = pub-VIEWEG,
  address = pub-VIEWEG:adr,
  pages = "xiii + 439",
  year = "1998",
  MRclass = "68W30 (Symbolic computation and algebraic computation)
            68-01 (Textbooks (computer science)) 00A06 (Mathematics
            for non-mathematicians) 68-04 (Machine computation,
            programs (computer science)) 65D18 (Computer graphics
            and computational geometry)",
  bibdate = "Tue Mar 30 18:49:35 2010",
  bibsource = "http://www.math.utah.edu/pub/tex/bib/axiom.bib",
}

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ZMnumber =      "0909.68109",
acknowledgement = ack-nhfb,
keywords =      "calculus; differential equations; Fourier transform;
                  Laplace transform; linear algebra; optimization;
                  probability theory; statistics; textbook",
language =      "German",
reviewer =      "Helmut K{\o}cher (Dresden)",
}

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@misc{Bern14,
  author = "Bernard, Joey",
  title = "Open Axiom",
  link = "\url{http://www.linuxjournal.com/content/open-axiom}",
  year = "2014",
  keywords = "axiomref"
}

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— axiom.bib —

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@book{Bett99,
  author = "Betten, Anton and Kohnert, Axel and Laue, Reinhard and
            Wassermann, Alfred",
  title = "Algebraic Combinatorics and Applications",
  year = "1999",
  publisher = "Springer",
  isbn = "978-3-540-41110-9",
  keywords = "axiomref"
}

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— axiom.bib —

```

@inproceedings{Blai70,
  author = "Blair, Fred W. and Griesmer, James H. and Jenks, Richard D.",
  title = "An interactive facility for symbolic mathematics",
  booktitle = "Proc. International Computing Symposium, Bonn, Germany",
}

```



```

year = "1970",
pages = "394-419",
abstract =
  "The SCRATCHPAD/1 system is designed to provide an interactive symbolic
  coputational facility for the mathematician user. The system features
  a user language designed to capture the style and succinctness of
  mathematical notation, together with a facility for conveniently
  introducing new notations into the language. A comprehensive system
  library incorporates symbolic capabilities provided by such systems as
  SIN, MATHLAB, and REDUCE.",
keywords = "axiomref"
}

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\bibitem[Blair 70a]{BJ70}
  author = "Blair, Fred W. and Jenks, Richard D.",
  title = "LPL: LISP programming language",
  year = "1970",
  IBM Research Report, RC3062 Sept ,
  keywords = "axiomref"

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@InProceedings{Bosm94,
  author = "Bosma, Wieb and Cannon, John and Matthews, Graham",
  title = "Programming with algebraic structures: Design of the Magma
  language",
  booktitle = "Proc. ISSAC 94",
  series = "ISSAC 94",
  year = "1994",
  publisher = "ACM Press",
  location = "Baltimore, MD",
  pages = "52-57",
  link = "\url{http://www.math.ru.nl/~bosma/pubs/ISSAC94.pdf}",
  abstract =
    "MAGMA is a new software system for computational algebra, number
    theory and geometry whose design is centred on the concept of
    algebraic structure (magma). The use of algebraic structure as a
    design paradigm provides a natural strong typing mechanism. Further,
    structures and their morphisms appear in the language as first class
    objects. Standard mathematical notions are used for the basic data

```

```

types. The result is a powerful, clean language which deals with
objects in a mathematically rigorous manner. The conceptual and
implementation ideas behind MAGMA will be examined in this paper. This
conceptual base differs significantly from those underlying other
computer algebra systems.",
paper = "Bosm94.pdf",
keywords = "axiomref"
}

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— axiom.bib —

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@article{Bosm97,
  author = "Bosma, Wieb and Cannon, John and Playoust, Catherine",
  title = "The Magma Algebra System I: The User Language",
  journal = "J. Symbolic Computation",
  volume = "24",
  pages = "235-265",
  year = "1997",
  link = "\url{http://lib.org.by/_djvu/_Papers/Computer_algebra/CAS%20systems/}",
  abstract =
    "In the first of two papers on MAGMA, a new system for computational
    algebra, we present the MAGMA language, outline the design principles
    and theoretical background, and indicate its scope and use. Particular
    attention is given to the constructors for structures, maps, and sets.",
  paper = "Bosm97.djvu",
  keywords = "axiomref"
}

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— axiom.bib —

```

@misc{Boyl88,
  author = "Boyle, Ann and Caviness, B.F. and Hearn, Anthony C.",
  title = "Future Directions for Research in Symbolic Computation",
  publisher = "Soc. for Industrial and Applied Mathematics",
  year = "1988",
  link = "\url{http://www.eecis.udel.edu/~caviness/wsreport.pdf}",
  paper = "Boyl88.pdf",
  keywords = "axiomref"
}

```

— axiom.bib —

```
@InProceedings{Brad09,
  author = "Bradford, Russell and Davenport, James H. and
           Sangwin, Christopher J.",
  title = "A comparison of equality in computer algebra and correctness
           in mathematical pedagogy",
  year = "2009",
  booktitle = "Intelligent Computer Mathematics, 16th symposium",
  series = "Calculus 2009",
  pages = "75-89",
  isbn = "978-3-642-02613-3",
  link = "\url{http://opus.bath.ac.uk/15188/1/RJBHDCJSv2.pdf}",
  abstract =
    "How do we recognize when an answer is ‘‘right’’? This is a question
    that has bedevilled the use of computer systems in mathematics (as
    opposed to arithmetic) ever since their introduction. A computer
    system can certainly say that some answers are definitely wrong, in
    the sense that they are provably not an answer to the question
    posed. However, an answer can be mathematically right without being
    pedagogically right. Here we explore the differences and show that,
    despite the apparent distinction, it is possible to make many of the
    differences amenable to formal treatment, by asking ‘‘under which
    congruence is the pupil’s answer equal to the teacher’s?’’.",
  paper = "Brad09.pdf",
  keywords = "axiomref"
}
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@misc{Brem08,
  author = "Bremner, Murray R. and Murakami, Lucia I. and Shestakov, Ivan P.",
  title = "Nonassociative Algebras",
  year = "2008",
  algebra = "\newline\refto{category NARNG NonAssociativeRng}",
  link = "\url{http://citeseerx.ist.psu.edu/viewdoc/download?doi=10.1.1.146.2665}",
  abstract =
    "One of the earliest surveys on nonassociative algebras is the article
    by Shirshov which introduced the phrase ‘‘rings that are nearly
    associative’’. The first book in the English language devoted to a
    systematic study of nonassociative algebras is Schafer (Scha66). A
    comprehensive exposition of the work of the Russian School is
    Zhevlakov, Slinko, Shestakov and Shirshov. A collection of open
    research problems in algebra, including many problems on
    nonassociative algebra, is the {\sl Dniester Notebook}; the survey
```

article by Kuzmin and Shetakov is from the same period. Three books on Jordan algebras which contain substantial material on general nonassociative algebras are Braun and Koecher, Jacobson, and McCrimmon. Recent research appears in the Proceedings of the International Conferences on Nonassociative Algebras and its Applications. The present section provides very limited information on Lie algebras, since they are the subject of Section 16.4. The last part (section 9) of the present section presents three applications of computational linear algebra to the study of polynomial identities for nonassociative algebras: pseudorandom vectors in a nonassociative algebra, the expansion matrix for a nonassociative operation, and the representation theory of the symmetric group."

}

— axiom.bib —

```
@InProceedings{Breu98,
  author = "Breuer, Thomas and Linton, Steve",
  title = "The GAP 4 type system organising algebraic algorithms",
  booktitle = "Proc. ISSAC 98",
  series = "ISSAC 98",
  year = "1998",
  publisher = "ACM Press",
  location = "Rostock, Germany",
  pages = "13-15",
  link = "\url{http://www.gap-system.org/Doc/Talks/paper.ps}",
  abstract =
    "Version 4 of the GAP (Groups, Algorithms, Programming) system for
    computational discrete mathematics has a number of novel features. In
    this paper, we describe the type system, and the way in which it is
    used for method selection. This system is central to the organization
    of the library which is the main part of the GAP system. Unlike
    simpler object-oriented systems, GAP allows method selection based on
    the types of all arguments and on certain aspects of the relationship
    between the arguments. In addition, the type of an object can change,
    in a controlled way, during its life. This reflects information about
    the object which has been computed and stored. Individual methods can
    be written and installed independently. Furthermore, most checking of
    the arguments is done in a uniform way by the method selection system,
    making individual methods simpler and less prone to error. The methods
    are combined automatically to produce a powerful and usable system for
    interactive use or programming.",
  paper = "Breu98.pdf",
  keywords = "axiomref"
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@inproceedings{Broa95,
  author = "Broadbery, P. A. and Gómez-Díaz, T. and Watt, S. M.",
  title = "On the Implementation of Dynamic Evaluation",
  booktitle = "Unknown",
  publisher = "Unknown",
  year = "1995",
  pages = "77-84",
  isbn = "0-89791-699-9",
  link = "\url{http://pdf.aminer.org/000/449/014/on_the_implementation_of_dynamic_evaluation.pdf}",
  abstract = "
    Dynamic evaluation is a technique for producing multiple results
    according to a decision tree which evolves with program execution.
    Sometimes it is desired to produce results for all possible branches
    in the decision tree, while on other occasions, it may be sufficient
    to compute a single result which satisfies certain properties. This
    technique finds use in computer algebra where computing the correct
    result depends on recognizing and properly handling special cases of
    parameters. In previous work, programs using dynamic evaluation have
    explored all branches of decision trees by repeating the computations
    prior to decision points.

    This paper presents two new implementations of dynamic evaluation
    which avoid recomputing intermediate results. The first approach uses
    Scheme ‘‘continuations’’ to record state for resuming program
    execution. The second implementation uses the Unix ‘‘fork’’ operation
    to form new processes to explore alternative branches in parallel.",
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  keywords = "axiomref",
  beebe = "Broadbery:1995:IDE"
}
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— Broadbery:1995:IDE —

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@InProceedings{Broadbery:1995:IDE,
  author = "P. A. Broadbery and T. Gómez-Díaz and S. M. Watt",
  title = "On the Implementation of Dynamic Evaluation",
  crossref = "Levelt:1995:IPi",
  pages = "77--84",
  year = "1995",
  bibdate = "Thu Mar 12 08:42:30 MST 1998",
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bibsource = "http://www.acm.org/pubs/toc/;
             http://www.math.utah.edu/pub/tex/bib/axiom.bib",
link =      "\url{http://www.acm.org:80/pubs/citations/proceedings/issac/220346/p77-broadbery/}",
abstract =  "Dynamic evaluation is a technique for producing
             multiple results according to a decision tree which
             evolves with program execution. Sometimes we need to
             produce results for all possible branches in the
             decision tree, while on other occasions it may be
             sufficient to compute a single result which satisfies
             certain properties. This technique finds use in
             computer algebra where computing the correct result
             depends on recognising and properly handling special
             cases of parameters. In previous work, programs using
             dynamic evaluation have explored all branches of
             decision trees by repeating the computations prior to
             decision points. The paper presents two new
             implementations of dynamic evaluation which avoid
             recomputing intermediate results. The first approach
             uses Scheme 'continuations' to record the state for
             resuming program execution. The second implementation
             uses the Unix 'fork' operation to form new processes to
             explore alternative branches in parallel. These
             implementations are based on modifications to Lisp- and
             C-based run-time systems for the Axiom Version 2
             extension language (previously known as $ A^{\mbox{
             {Hash}}} $ ). This allows the same high-level source
             code to be compared using the 're-evaluation', the
             'continuation', and the 'fork' implementations.",
acknowledgement = ack-nhfb,
affiliation = "Numerical Algorithms Group Ltd., Oxford, UK",
classification = "C1140E (Game theory); C1160 (Combinatorial
                 mathematics); C6130 (Data handling techniques); C6150G
                 (Diagnostic, testing, debugging and evaluating
                 systems); C6150J (Operating systems)",
keywords =    "algebraic computation, Dynamic evaluation; algorithms;
               Axiom Version 2 extension language; C-based run-time
               systems; Computer algebra; Decision points; Decision
               tree; High-level source code; ISSAC; languages;
               Lisp-based run-time systems; Multiple results; Program
               execution; Re-evaluation; Scheme continuations; State
               recording; symbolic computation; Unix fork operation",
subject =     "{\bf I.1.2} Computing Methodologies, SYMBOLIC AND
               ALGEBRAIC MANIPULATION, Algorithms, Algebraic
               algorithms. {\bf D.3.2} Software, PROGRAMMING
               LANGUAGES, Language Classifications, SCHEME. {\bf
               F.2.1} Theory of Computation, ANALYSIS OF ALGORITHMS
               AND PROBLEM COMPLEXITY, Numerical Algorithms and
               Problems, Computations on polynomials. {\bf D.3.2}
               Software, PROGRAMMING LANGUAGES, Language
               Classifications, C.",

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thesaurus = "Decision theory; Symbol manipulation; System
             monitoring; Trees [mathematics]; Unix",
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— axiom.bib —

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@misc{Bronxx,
  author = "Bronstein, Manuel",
  title = "Symbolic Integration in Computer Algebra",
  link = "\url{http://www.iaea.org/inis/collection/NCLCollectionStore/_Public/26/042/26042580.pdf}",
  year = "1990",
  abstract =
    "One major goal of symbolic integrators is to determine under what
    circumstances the integral of the elementary functions of calculus can
    themselves be expressed as elementary functions. While using tables
    and the ad hoc tricks taught in calculus courses can have some limited
    success, a decision procedure is necessary in all but the most trivial
    cases. The first complete algorithm for solving this problem was
    presented by Risch in 1969, but its complexity, specially when
    algebraic functions are present in the integrand, has prevented it
    from being fully implemented. Over the past 20 years, the Risch
    integration algorithm has been completed, extended, and improved to
    such a point that recent computer algebra systems can integrate
    elementary functions without using any of the heuristics traditionally
    taught in calculus courses and used by older systems. In this talk,
    we give an overview and description of the algorithms used in the
    Scratchpad symbolic integrator, and illustrate them with integrals
    drawn from the physical sciences.",
  paper = "Bron90.pdf",
  keywords = "axiomref"
}

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— axiom.bib —

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\article{Brun04,
  author = "Brunelli, J.C.",
  title = "PSEUDO: applications of streams and lazy evaluation to integrable
           models",
  journal = "Comput. Phys. Commun.",
  volume = "163",
  number = "1",
  pages = "22-40",
}

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year = "2004",
abstract =
  "Procedures to manipulate pseudo-differential operators in MAPLE are
  implemented in the program PSEUDO to perform calculations with
  integrable models. We use lazy evaluation and streams to represent and
  operate with pseudo-differential operators. No order of truncation is
  needed since terms are produced on demand. We give a series of
  concrete examples.",
keywords = "axiomref"
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— axiom.bib —

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@article{Boeh89,
  author = "Boehm, Hans-J.",
  title = "Type Inference in the Presence of Type Abstraction",
  journal = "ACM SIGPLAN Notices",
  volume = "24",
  number = "7",
  month = "July",
  year = "1989",
  pages = "192-206",
  link = "\url{http://www.acm.org/pubs/citations/proceedings/pldi/73141/p192-boehm}",
  abstract = "
    A number of recent programming language designs incorporate a type
    checking system based on the Girard-Reynolds polymorphic
     $\lambda$ -calculus. This allows the construction of general purpose,
    reusable software without sacrificing compile-time type checking. A
    major factor constraining the implementation of these languages is the
    difficulty of automatically inferring the lengthy type information
    that is otherwise required if full use is made of these
    languages. There is no known algorithm to solve any natural and fully
    general formulation of the ‘type inference’ problem. One very
    reasonable formulation of the problem is known to be undecidable.

    Here we define a restricted version of the type inference problem and
    present an efficient algorithm for its solution. We argue that the
    restriction is sufficiently weak to be unobtrusive in practice.",
  paper = "Boeh89.pdf",
  keywords = "axiomref",
  beebe = "Boehm:1989:TIP"
}

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— Boehm:1989:TIP —

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@Article{Boehm:1989:TIP,
  author =      "Hans-J. Boehm",
  title =       "Type inference in the presence of type abstraction",
  journal =     j-SIGPLAN,
  volume =      "24",
  number =      "7",
  pages =       "192--206",
  month =       jul,
  year =        "1989",
  CODEN =       "SINODQ",
  ISSN =        "0362-1340 (print), 1523-2867 (print), 1558-1160
                 (electronic)",
  ISSN-L =      "0362-1340",
  bibdate =     "Thu May 13 12:31:07 MDT 1999",
  bibsource =   "http://www.acm.org/pubs/contents/proceedings/pldi/73141/index.html;
                 http://www.math.utah.edu/pub/tex/bib/axiom.bib",
  link =        "\url{http://www.acm.org:80/pubs/citations/proceedings/pldi/73141/p192-boehm/}",
  abstract =    "A number of recent programming language designs
                 incorporate a type checking system based on the
                 Girard--Reynolds polymorphic  $\lambda$ -calculus. This
                 allows the construction of general purpose, reusable
                 software without sacrificing compile-time type
                 checking. A major factor constraining the
                 implementation of these languages is the difficulty of
                 automatically inferring the lengthy type information
                 that is otherwise required if full use is made of these
                 languages. There is no known algorithm to solve any
                 natural and fully general formulation of this 'type
                 inference' problem. One very reasonable formulation of
                 the problem is known to be undecidable. Here we define
                 a restricted version of the type inference problem and
                 present an efficient algorithm for its solution. We
                 argue that the restriction is sufficiently weak to be
                 unobtrusive in practice.",
  acknowledgement = ack-nhfb,
  affiliationaddress = "Houston, TX, USA",
  annote =       "Published as part of the Proceedings of PLDI'89.",
  classification = "723",
  conference =   "Proceedings of the SIGPLAN '89 Conference on
                 Programming Language Design and Implementation",
  fjournal =     "ACM SIGPLAN Notices",
  link =         "\url{http://portal.acm.org/browse_dl.cfm?idx=J706}",
  journalabr =   "SIGPLAN Not",
  keywords =     "Abstract Data Types (ADT); algorithms; Computer
                 Programming Languages--Design; Data Processing; Data
                 Structures; design; languages; Scratchpad; theory",
  meetingaddress = "Portland, OR, USA",

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meetingdate = "Jun 21--23 1989",
meetingdate2 = "06/21--23/89",
sponsor =      "ACM, Special Interest Group on Programming Languages,
                New York; SS NY, USA",
subject =      "{\bf F.4.1} Theory of Computation, MATHEMATICAL LOGIC
                AND FORMAL LANGUAGES, Mathematical Logic. {\bf F.3.3}
                Theory of Computation, LOGICS AND MEANINGS OF PROGRAMS,
                Studies of Program Constructs, Type structure.",
}

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— axiom.bib —

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@inproceedings{BHGM04,
  author = "Boulton, Richard and Hardy, Ruth and Gottliebsen, Hanne
            and Martin, Ursula",
  title = "Design verification for control engineering",
  year = "2004",
  month = "April",
  booktitle = "Proc 4th Int. Conf. on Integrated Formal Methods",
  abstract = "
    We introduce control engineering as a new domain of application for
    formal methods. We discuss design verification, drawing attention to
    the role played by diagrammatic evaluation criteria involving numeric
    plots of a design, such as Nichols and Bode plots. We show that
    symbolic computation and computational logic can be used to discharge
    these criteria and provide symbolic, automated, and very general
    alternatives to these standard numeric tests. We illustrate our work
    with reference to a standard reference model drawn from military
    avionics.",
  keywords = "axiomref"
}

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@misc{Boul91,
  author = "Boulanger, Jean-Louis",
  title = "Etude de la compilation de scratchpad 2",
  year = "1991",
  month = "September",
  publisher = "Rapport de DEA Universite dl lille 1",
  keywords = "axiomref"
}

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— axiom.bib —

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@misc{Bou93a,
  author = "Boulanger, Jean-Louis",
  title = "Axiom, language fonctionnel \‘a d\’eveloppement objet",
  year = "1993",
  month = "October",
  paper = "Bou93a.pdf",
  keywords = "axiomref"
}
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— axiom.bib —

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@misc{Boul93b,
  author = "Boulanger, Jean-Louis",
  title = "AXIOM, A Functional Language with Object Oriented Development",
  year = "1993",
  abstract =
    "We present in this paper, a study about the computer algebra system
    Axiom, which gives us many very interesting Software engineering
    concepts. This language is a functional language with an Object
    Oriented Development. This feature is very important for modeling the
    mathematical world (Hierarchy) and provides a running with
    mathematical sense. (All objects are functions). We present many
    problems of running and development in Axiom. We can note that Aiom is
    the only system of this category.",
  paper = "Boul93b.pdf",
  keywords = "axiomref"
}
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— axiom.bib —

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@misc{Boul95,
  author = "Boulanger, J.L.",
  title = "Object Oriented Method for Axiom",
  year = "1995",
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month = "February",
pages = "33-41",
publisher = "ACM SIGPLAN Notices, 30(2) CODEN SINODQ ISSN 0362-1340",
abstract = "
    Axiom is a very powerful computer algebra system which combines two
    language paradigms (functional and OOP). Mathematical world is complex
    and mathematicians use abstraction to design it. This paper presents
    some aspects of the object oriented development in Axiom. The Axiom
    programming is based on several new tools for object oriented
    development, it uses two levels of class and some operations such that
    {\sl coerce}, {\sl retract}, or {\sl convert} which permit the type
    evolution. These notions introduce the concept of multi-view.",
paper = "Boul95.pdf",
keywords = "axiomref",
beebe = "Boulanger:1995:OOM"
}

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— Boulanger:1995:OOM —

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@Article{Boulanger:1995:OOM,
  author = "J.-L. Boulanger",
  title = "Object oriented method for {Axiom}",
  journal = j-SIGPLAN,
  volume = "30",
  number = "2",
  pages = "33--41",
  month = feb,
  year = "1995",
  CODEN = "SINODQ",
  ISSN = "0362-1340 (print), 1523-2867 (print), 1558-1160
    (electronic)",
  ISSN-L = "0362-1340",
  bibdate = "Tue Sep 17 06:32:41 MDT 1996",
  bibsource = "http://www.math.utah.edu/pub/tex/bib/axiom.bib",
  abstract = "Axiom is a very powerful computer algebra system which
    combines two language paradigms (functional and OOP).
    The mathematical world is complex and mathematicians
    use abstraction to design it. The paper presents some
    aspects of object oriented development in Axiom. Axiom
    programming is based on several new tools for object
    oriented development, it uses two levels of class and
    some operations such as coerce, retract or convert
    which permit the type evolution. These notions
    introduce the concept of multi-view.",
  acknowledgement = ack-nhfb,
  affiliation = "Lab. d'Inf. Fondamentale de Lille, Lille I Univ.,

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Villeneuve d'Ascq, France",
classification = "C4240 (Programming and algorithm theory); C6110J
                  (Object-oriented programming); C6120 (File
                  organisation); C6130 (Data handling techniques); C6140D
                  (High level languages); C7310 (Mathematics computing)",
fjournal =      "ACM SIGPLAN Notices",
link = "\url{http://portal.acm.org/browse_dl.cfm?idx=J706}",
keywords =      "Abstraction; Axiom; Axiom programming; Class; Coerce;
                  Computer algebra system; Convert; Functional language;
                  Multiview concept; Object oriented development; Object
                  oriented method; Object-oriented language; Retract;
                  Tools; Type evolution",
language =      "English",
pubcountry =    "USA",
thesaurus =     "Abstract data types; Functional languages; Functional
                  programming; Mathematics computing; Object-oriented
                  languages; Object-oriented programming; Symbol
                  manipulation; Type theory",
}

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— ignore —

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\bibitem[Bronstein 87]{Bro87}
  author = "Bronstein, Manuel",
  title = "Integration of Algebraic and Mixed Functions",
  year = "1987",
in [Wit87], p18
  keywords = "axiomref"

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— axiom.bib —

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@inproceedings{Bron89,
  author = "Bronstein, Manuel",
  title = "Simplification of real elementary functions",
  booktitle = "Proc. ISSAC 1989",
  series = "ISSAC 1989",
  year = "1989",
  pages = "207-211",
  isbn = "0-89791-325-6",
  doi = "https://dx.doi.org/10.1145/74540.74566",
  abstract = "
    We describe an algorithm, based on Risch's real structure theorem, that
  "

```

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determines explicitly all the algebraic relations among a given set of
real elementary functions. We also provide examples from its
implementation that illustrate the advantages over the use of complex
logarithms and exponentials.",
paper = "Bron88.djvu",
keywords = "axiomref",
beebe = "Bronstein:1989:SRE"
}

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— Bronstein:1989:SRE —

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@InProceedings{Bronstein:1989:SRE,
  author = "M. Bronstein",
  title = "Simplification of real elementary functions",
  crossref = "ACM:1989:PAI",
  pages = "207--211",
  month = "",
  year = "1989",
  bibdate = "Tue Sep 17 06:46:18 MDT 1996",
  bibsource = "http://www.math.utah.edu/pub/tex/bib/axiom.bib",
  abstract = "The author describes an algorithm, based on Risch's
real structure theorem, that determines explicitly all
the algebraic relations among a given set of real
elementary functions. He provides examples from its
implementation in the Scratchpad computer algebra
system that illustrate the advantages over the use of
complex logarithms and exponentials.",
  acknowledgement = ack-nhfb,
  affiliation = "IBM Res. Div., T. J. Watson Res. Center, Yorktown
Heights, NY, USA",
  classification = "C1110 (Algebra); C7310 (Mathematics)",
  keywords = "Computer algebra system; Real elementary functions;
Real structure theorem; Scratchpad",
  language = "English",
  thesaurus = "Functions; Mathematics computing; Symbol
manipulation",
}

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— axiom.bib —

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@inproceedings{Bron91a,
  author = "Bronstein, M.",

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title = "The Risch Differential Equation on an Algebraic Curve",
booktitle = "Proc. 1991 Int. Symp. on Symbolic and Algebraic Computation",
series = "ISSAC'91",
year = "1991",
pages = "241-246",
isbn = "0-89791-437-6",
publisher = "ACM, NY",
abstract =
    "We present a new rational algorithm for solving Risch differential
    equations over algebraic curves. This algorithm can also be used to
    solve  $n^{\text{th}}$ -order linear ordinary differential equations with
    coefficients in an algebraic extension of the rational functions. In
    the general ('mixed function') case, this algorithm finds the
    denominator of any solution of the equation.",
paper = "Bro91a.pdf",
keywords = "axiomref",
beebe = "Bronstein:1991:RDE"
}

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— Bronstein:1991:RDE —

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@InProceedings{Bronstein:1991:RDE,
  author = "M. Bronstein",
  title = "The {Risch} differential equation on an algebraic
    curve",
  crossref = "Watt:1991:PIS",
  pages = "241--246",
  month = "",
  year = "1991",
  bibdate = "Tue Sep 17 06:44:07 MDT 1996",
  bibsource = "http://www.math.utah.edu/pub/tex/bib/axiom.bib",
  abstract = "The author presents a new rational algorithm for
    solving Risch differential equations over algebraic
    curves. This algorithm can also be used to solve  $n/\text{sup}$ 
     $\text{th}/$ -order linear ordinary differential equations with
    coefficients in an algebraic extension of the rational
    functions. In the general ('mixed function') case, this
    algorithm finds the denominator of any solution of the
    equation. The algorithm has been implemented in the
    Maple and Scratchpad computer algebra systems.",
  acknowledgement = ack-nhfb,
  affiliation = "Inf. ETH-Zentrum, Zurich, Switzerland",
  classification = "C4170 (Differential equations); C7310
    (Mathematics)",
  keywords = "Algebraic curve; Computer algebra systems; Maple;
     $N/\text{sup}$   $\text{th}/$ -order linear ordinary differential equations;

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        Rational algorithm; Rational functions; Risch
        differential equation; Scratchpad",
    language =      "English",
    thesaurus =     "Differential equations; Symbol manipulation",
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\bibitem[Bronstein 91c]{Bro91c}
    author = "Bronstein, Manuel",
    title = "Computer Algebra and Indefinite Integrals",
    year = "1991",
in Computer Aided Proofs in Analysis, K.R. Meyers et al. (eds)
Springer-Verlag, NY (1991)
    abstract = "
        We give an overview, from an analytical point of view, of decision
        procedures for determining whether an elementary function has an
        elementary function has an elementary antiderivative. We give examples
        of algebraic functions which are integrable and non-integrable in
        closed form, and mention the current implementation of various computer
        algebra systems.",
    paper = "Bro91c.pdf",
    keywords = "axiomref"
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— axiom.bib —

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@misc{Bron92,
    author = "Bronstein, Manuel",
    title = "Linear Ordinary Differential Equations: Breaking Through the
        Order 2 Barrier",
    booktitle = "Proc. ISSAC 1992",
    series = "ISSAC 92",
    publisher = "ACM",
    pages = "42-48",
    year = "1992",
    link = "\url{http://www-sop.inria.fr/cafe/Manuel.Bronstein/publications/issac92.ps.gz}",
    algebra =
        "\newline\refto{package ODECONST ConstantLODE}
        \newline\refto{package LODEEF ElementaryFunctionLODESolver}
        \newline\refto{package ODEPAL PureAlgebraicLODE}
        \newline\refto{package ODERAT RationalLODE}

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\newline\ref{package ODERED ReduceLODE}",
abstract = "
  A major subproblem for algorithms that either factor ordinary linear
  differential equations or compute their closed form solutions is to
  find their solutions  $y$  which satisfy  $y'/y \in \overline{K}(x)$ 
  where  $K$  is the constant field for the coefficients of the equation.
  While a decision procedure for this subproblem was known in the
  19th century, it requires factoring polynomials over
   $\overline{K}$  and has not been implemented in full generality. We
  present here an efficient algorithm for this subproblem, which has
  been implemented in the AXIOM computer algebra system for equations of
  arbitrary order over arbitrary fields of characteristic 0. This
  algorithm never needs to compute with the individual complex
  singularities of the equation, and algebraic numbers are added only
  when they appear in the potential solutions. Implementation of the
  complete Singer algorithm for  $n=2,3$  based on this building block is
  in progress.",
paper = "Bron92.pdf",
keywords = "axiomref"
}

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— axiom.bib —

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@article{Bron92a,
  author = "Bronstein, Manuel",
  title = "Formulas for Series Computation",
  journal = "Applied Algebra in Engineering, Communication and Computing",
  volume = "2",
  pages = "195-206",
  year = "1992",
  abstract =
    "We describe an algorithm that computes polynomials whose roots are
    the coefficients of any specific order of the Laurent series of a
    rational function. The algorithm uses only rational operations in the
    coefficient field of the function. This allows us to compute with the
    principal parts of the Laurent series, in particular with the residues
    of the function, without factoring or splitting its denominator. As
    applications, we get a generalisation of the residue formulas used in
    symbolic integration algorithms to  $n^{\text{th}}$ -order formulas. We also
    get an algorithm that computes explicitly the principal parts at all
    the poles simultaneously, while computing in the field generated by
    the coefficients of the series instead of the one generated by the
    poles of the function. This yields an improved version of the
    necessary conditions for the various cases of Kovacic's algorithm,
    that expresses those conditions in the smallest possible extension
    field."
}

```

}

— ignore —

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\bibitem[Bronstein 93]{Bro93}
  author = "Bronstein, Manuel (ed)",
  year = "1993",
  month = "July"
  isbn = "0-89791-604-2",
ISSAC'93: proceedings of the 1993 International Symposium on Symbolic
and Algebraic Computation, Kiev, Ukraine,
ACM Press New York, NY 10036, USA, ISBN
LCCN QA76.95 I59 1993 ACM order number 505930
  keywords = "axiomref"
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\bibitem[Brunelli 08]{Brun08}
  author = "Brunelli, J.C.",
  title = "Streams and Lazy Evaluation Applied to Integrable Models",
  year = "2008",
  link = "\url{http://arxiv.org/PS_cache/nlin/pdf/0408/0408058v1.pdf}",
  abstract = "
    Computer algebra procedures to manipulate pseudo-differential
    operators are implemented to perform calculations with integrable
    models. We use lazy evaluation and streams to represent and operate
    with pseudo-differential operators. No order of truncation is needed
    since terms are produced on demand. We give a series of concrete
    examples using the computer algebra language MAPLE.",
  paper = "Brun08.pdf",
  keywords = "axiomref"
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\bibitem[Bronstein 93]{BS93}
  author = "Bronstein, Manuel and Salvy, Bruno",
  title = "Full Partial Fraction Decomposition of Rational Functions",
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year = "1993",
pages = "157-160",
isbn = "0-89791-604-2",
In Bronstein [Bro93]   LCCN QA76.95 I59 1993
keywords = "axiomref"

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— axiom.bib —

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@misc{Bro92a,
  author = "Bronstein, Manuel",
  title = "Integration and Differential Equations in Computer Algebra",
  year = "1992",
  link = "\url{http://citeseerx.ist.psu.edu/viewdoc/summary?doi=10.1.1.42.576}",
  abstract = "
    We describe in this paper how the problems of computing indefinite
    integrals and solving linear ordinary differential equations in closed
    form are now solved by computer algebra systems. After a brief review
    of the mathematical history of those problems, we outline the two
    major algorithms for them (respectively the Risch and Singer
    algorithms) and the recent improvements on those algorithms which has
    allowed them to be implemented.",
  paper = "Bro92a.pdf",
  keywords = "axiomref"
}

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— axiom.bib —

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@article{Bene94,
  author = "Beneke, T. and Schwippert, W.",
  title = "Double-track into the future: MathCAD will gain new users with
    Standard and Plus versions",
  journal = "Elektronik",
  volume = "43",
  number = "15",
  pages = "107-110",
  year = "1994",
  month = "July",
  abstract =
    "MathCAD software is a type of 'intelligent scratchpad with a pocket
    calculator function'. Hitherto it has been suitable only to a limited
    extent for engineering mathematics. The new Version 5.0 is now offered
    in two implementations: as an inexpensive basic package and in a

```

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considerably more costly Plus version. The authors question whether
MathCAD can catch up with the classical Maple and Mathematica products.",
keywords = "axiomref",
beebe = "Beneke:1994:DFM",
}

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— Beneke:1994:DFM —

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@Article{Beneke:1994:DFM,
  author =      "T. Beneke and W. Schwippert",
  title =      "Double-track into the future: {MathCAD} will gain new
                users with {Standard} and {Plus} versions",
  journal =     j-ELECTRONIK,
  volume =     "43",
  number =     "15",
  pages =      "107--110",
  month =      jul,
  year =       "1994",
  CODEN =      "EKRKAR",
  ISSN =       "0013-5658",
  bibdate =    "Tue Sep 17 06:35:39 MDT 1996",
  bibsource =  "http://www.math.utah.edu/pub/tex/bib/axiom.bib",
  abstract =    "MathCAD software is a type of 'intelligent scratchpad
                with a pocket calculator function'. Hitherto it has
                been suitable only to a limited extent for engineering
                mathematics. The new Version 5.0 is now offered in two
                implementations: as an inexpensive basic package and in
                a considerably more costly Plus version. The authors
                question whether MathCAD can catch up with the
                classical Maple and Mathematica products.",
  acknowledgement = ack-nhfb,
  classification = "C7310 (Mathematics computing)",
  fjournal =    "Elektronik",
  keywords =    "Engineering mathematics; Intelligent scratchpad;
                MathCAD software",
  language =    "German",
  pubcountry =  "Germany",
  thesaurus =   "CAD; Mathematics computing; Software packages",
}

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— axiom.bib —

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@inproceedings{Bron97a,

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author = "Bronstein, Manuel and Weil, Jacques-Arthur",
title = "On Symmetric Powers of Differential Operators",
booktitle = "Proc. of ISSAC 1997",
series = "ISSAC'97",
year = "1997",
pages = "156-163",
link = "\url{http://www-sop.inria.fr/cafe/Manuel.Bronstein/publications/mb_papers.html}",
publisher = "ACM, NY",
abstract = "
    We present alternative algorithms for computing symmetric powers of
    linear ordinary differential operators. Our algorithms are applicable
    to operators with coefficients in arbitrary integral domains and
    become faster than the traditional methods for symmetric powers of
    sufficiently large order, or over sufficiently complicated coefficient
    domains. The basic ideas are also applicable to other computations
    involving cyclic vector techniques, such as exterior powers of
    differential or difference operators.",
paper = "Bro97a.pdf",
keywords = "axiomref"
}

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\bibitem[Borwein 00]{Bor00}
author = "Borwein, Jonathan",
title = "Multimedia tools for communicating mathematics",
year = "2000",
pages = "58",
isbn = "3-540-42450-4",
publisher = "Springer-Verlag",
keywords = "axiomref"
}

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— axiom.bib —

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@article{Brow94,
author = "Brown, Ronald and Tonks, Andrew",
title = "Calculations with simplicial and cubical groups in AXIOM",
journal = "Journal of Symbolic Computation",
volume = "17",
number = "2",
pages = "159-179",

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year = "1994",
month = "February",
misc = "CODEN JSYCEH ISSN 0747-7171",
abstract =
  "Work on calculations with simplicial and cubical groups in AXIOM was
  carried out using loan equipment and software from IBM UK and guidance
  from L. A. Lambe. We report on the results of this work, and present
  the AXIOM code written by the second author during this period. This
  includes an implementation of the monoids which model cubes and
  simplices, together with a new AXIOM category of near-rings with which
  to carry out non-abelian calculations. Examples of the use of this
  code in interactive AXIOM sessions are also given.",
paper = "Brow94.pdf",
keywords = "axiomref"
}

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— Brown:1994:CSC —

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@Article{Brown:1994:CSC,
  author =      "R. Brown and A. Tonks",
  title =      "Calculations with simplicial and cubical groups in
                {AXIOM}",
  journal =    j-J-SYMBOLIC-COMP,
  volume =     "17",
  number =     "2",
  pages =      "159--179",
  month =      feb,
  year =       "1994",
  CODEN =      "JSYCEH",
  ISSN =       "0747-7171 (print), 1095-855X (electronic)",
  ISSN-L =     "0747-7171",
  bibdate =    "Tue Sep 17 06:35:39 MDT 1996",
  bibsource =  "http://www.math.utah.edu/pub/tex/bib/axiom.bib",
  abstract =   "Work on calculations with simplicial and cubical
                groups in AXIOM was carried out using loan equipment
                and software from IBM UK and guidance from L. A. Lambe.
                We report on the results of this work, and present the
                AXIOM code written by the second author during this
                period. This includes an implementation of the monoids
                which model cubes and simplices, together with a new
                AXIOM category of near-rings with which to carry out
                non-abelian calculations. Examples of the use of this
                code in interactive AXIOM sessions are also given.",
  acknowledgement = ack-nhfb,
  affiliation = "Sch. of Math., Univ. of Wales, Bangor, UK",
  classification = "C6130 (Data handling techniques); C7310

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        (Mathematics)",
fjournal = "Journal of Symbolic Computation",
link = "\url{http://www.sciencedirect.com/science/journal/07477171}",
keywords = "AXIOM; Cubical groups; Monoids; Nonabelian
            calculations; Simplicial groups; Software",
language = "English",
pubcountry = "UK",
thesaurus = "Mathematics computing; Symbol manipulation",
}

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@misc{Brow95,
  author = "Brown, Ronald and Dreckmann, Winfried",
  title = "Domains of data and domains of terms in AXIOM",
  year = "1995",
  link = "\url{http://axiom-wiki.newsynthesis.org/public/refs/brown-free-c-g.pdf}",
  abstract = "
    The main new concept we wish to illustrate in this paper is a
    distinction between ‘‘domains of data’’ and ‘‘domains of terms’’, and
    its use in the programming of certain mathematical structures.
    Although this distinction is implicit in much of the programming work
    that has gone into the construction of Axiom categories and domains,
    we believe that a formalisation of this is new, that standards and
    conventions are necessary and will be useful in various other
    contexts. We shall show how this concept may be used for the coding of
    free categories and groupoids on directed graphs.",
  paper = "Brow95.pdf",
  keywords = "axiomref"
}

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— axiom.bib —

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@misc{Buch11,
  author = "Buchberger, Bruno",
  title = "Groebner Bases: A Short Introduction for System Theorists",
  year = "2011",
  abstract = "
    In this paper, we give a brief overview on Groebner bases theory,
    addressed to novices without prior knowledge in the field. After
    explaining the general strategy for solving problems via the Groebner
    approach, we develop the concept of Groebner bases by studying

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    uniqueness of polynomial division ('reduction'). For explicitly
    constructing Groebner bases, the crucial notion of S-polynomials is
    introduced, leading to the complete algorithmic solution of the
    construction problem. The algorithm is applied to examples from
    polynomial equation solving and algebraic relations. After a short
    discussion of complexity issues, we conclude the paper with some
    historical remarks and references."
}

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@book{Buch85,
  author = "Buchberger, Bruno and Caviness, Bob F. (eds)",
  title = "Lecture Notes in Computer Science, Vol 204",
  isbn = "0-387-15983-5 (vol 1), 0-387-15984-3 (vol 2)",
  year = "1985",
  month = "April",
  publisher = "Springer-Verlag, Berlin, Germany",
  keywords = "axiomref"
}

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— axiom.bib —

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@misc{Buh05,
  author = "Buhl, Soren L.",
  title = "Some Reflections on Integrating a Computer Algebra System in R",
  year = "2005",
  keywords = "axiomref"
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@InProceedings{Burg91,
  author = "Burge, W.H.",
  title = "Scratchpad and the Rogers-Ramanujan identities",
  booktitle = "Proc. ISSAC 1991",
  year = "1991",

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pages = "189-190",
abstract =
    "This note sketches the part played by Scratchpad in obtaining new
    proofs of Euler's theorem and the Rogers-Ramanujan Identities.",
paper = "Burg91.pdf",
keywords = "axiomref",
beebe = "Burge:1991:SRI"
}

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— Burge:1991:SRI —

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@InProceedings{Burge:1991:SRI,
  author = "W. H. Burge",
  title = "{Scratchpad} and the {Rogers--Ramanujan} identities",
  crossref = "Watt:1991:PIS",
  pages = "189--190",
  month = "",
  year = "1991",
  bibdate = "Tue Sep 17 06:44:07 MDT 1996",
  bibsource = "http://www.math.utah.edu/pub/tex/bib/axiom.bib",
  abstract = "This note sketches the part played by Scratchpad in
    obtaining new proofs of Euler's theorem and the
    Rogers--Ramanujan Identities.",
  acknowledgement = ack-nhfb,
  affiliation = "IBM Thomas J. Watson Res. Center, Yorktown Heights,
    NY, USA",
  classification = "C1160 (Combinatorial mathematics); C7310
    (Mathematics)",
  keywords = "Euler theorem; Infinite series; Restricted partition
    pairs; Rogers--Ramanujan identities; Scratchpad",
  language = "English",
  thesaurus = "Mathematics computing; Number theory; Symbol
    manipulation",
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@techreport{Burg87,
  author = "Burge, William and Watt, Stephen",
  title = "Infinite structures in SCRATCHPAD II",
  year = "1987",
  institution = "IBM Research",
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type = "Technical Report",
number = "RC 12794 (\#57573)",
link = "\url{http://www.csd.uwo.ca/~watt/pub/reprints/1987-eurocal-infinite.pdf}",
abstract =
  "An {\sl infinite structure} is a data structure which cannot be fully
  constructed in any fixed amount of space. Several varieties of
  infinite structures are currently supported in Scratchpad II: infinite
  sequences, radix expansions, power series and continued fractions. Two
  basic methods are employed to represent infinite structures:
  self-referential data structures and lazy evaluation. These may be
  employed separately or in conjunction.

  This paper presents recently developed facilities in Scratchpad II for
  manipulating infinite structures. General techniques for manipulating
  infinite structures are covered, as well as the higher level
  manipulations on the various types of mathematical objects represented
  by infinite structures.",
paper = "Burge87.pdf",
keywords = "axiomref"
}

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— Burge:1987:ISS —

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@TechReport{Burge:1987:ISS,
  author = "W. Burge and S. Watt",
  title = "Infinite Structures in {SCRATCHPAD II}",
  number = "RC 12794 (\#57573)",
  institution = "IBM Thomas J. Watson Research Center",
  address = "Bos 218, Yorktown Heights, NY 10598, USA",
  pages = "??",
  year = "1987",
  bibsource = "/usr/local/src/bib/bibliography/Theory/Comp.Alg.bib;
  http://www.math.utah.edu/pub/tex/bib/axiom.bib",
}

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\bibitem[Burge 87a]{BWM87}
  author = "Burge, William H. and Watt, Stephen M. and Morrison, Scott C.",
  title = "Streams and Power Series",
  year = "1987",
  pages = "9-12",

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in [Wit87], pp9-12
  keywords = "axiomref"
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— axiom.bib —

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@inproceedings{Burg87b,
  author = "Burge, William H. and Watt, Stephen M.",
  title = "Infinite structures in Scratchpad II",
  booktitle = "EUROCAL '87. European Conference on Computer Algebra
    Proceedings",
  year = "1987",
  pages = "138-148",
  isbn = "3-540-51517-8",
  abstract =
    "An infinite structure is a data structure which cannot be fully
    constructed in any fixed amount of space. Several varieties of
    infinite structures are currently supported in Scratchpad II: infinite
    sequences, radix expansions, power series and continued fractions. Two
    basic methods are employed to represent infinite structures:
    self-referential data structures and lazy evaluation. These may be
    employed either separately or in conjunction. This paper presents
    recently developed facilities in Scratchpad II for manipulating
    infinite structures. General techniques for manipulating infinite
    structures are covered, as well as the higher level manipulations on
    the various types of mathematical objects represented by infinite
    structures.",
  keywords = "axiomref",
  beebe = "Burge:1989:ISS"
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— Burge:1989:ISS —

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@InProceedings{Burge:1989:ISS,
  author = "W. H. Burge and S. M. Watt",
  title = "Infinite structures in {Scratchpad} {II}",
  crossref = "Davenport:1989:EEC",
  pages = "138--148",
  month = "",
  year = "1989",
  bibdate = "Tue Sep 17 06:46:18 MDT 1996",
  bibsource = "http://www.math.utah.edu/pub/tex/bib/axiom.bib",
  abstract = "An infinite structure is a data structure which cannot
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be fully constructed in any fixed amount of space.
 Several varieties of infinite structures are currently
 supported in Scratchpad II: infinite sequences, radix
 expansions, power series and continued fractions. Two
 basic methods are employed to represent infinite
 structures: self-referential data structures and lazy
 evaluation. These may be employed either separately or
 in conjunction. This paper presents recently developed
 facilities in Scratchpad II for manipulating infinite
 structures. General techniques for manipulating
 infinite structures are covered, as well as the higher
 level manipulations on the various types of
 mathematical objects represented by infinite
 structures.",
 acknowledgement = ack-nhfb,
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 NY, USA",
 classification = "C6120 (File organisation); C7310 (Mathematics)",
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 Infinite sequences; Infinite structure; Lazy
 evaluation; Mathematical objects; Power series; Radix
 expansions; Scratchpad II; Self-referential data
 structures",
 language = "English",
 thesaurus = "Algebra; Data structures; Mathematics computing;
 Series [mathematics]; Software packages; Symbol
 manipulation",
 }

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— ignore —

\bibitem[Calmet 94]{Cal94} Calmet, J. (ed)
 Rhine Workshop on Computer Algebra, Proceedings.
 Universit{"a"}t Karlsruhe, Karlsruhe, Germany 1994
 keywords = "axiomref"

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@article{Calm97,

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author = "Calmet, J. and Campbell, J.A.",
title = "A perspective on symbolic mathematical computing and
        artificial intelligence",
journal = "Ann. Math. Artif. Intell.",
volume = "19",
number = "3-4",
pages = "261-277",
year = "1997",
link = "\url{http://citeseerx.ist.psu.edu/viewdoc/download?doi=10.1.1.52.5425}",
abstract =
    "The nature and history of the research area common to artificial
    intelligence and symbolic mathematical computation are examined, with
    particular reference to the topics having the greatest current amount
    of activity or potential for further development: mathematical
    knowledge-based computing environments, autonomous agents and
    multi-agent systems, transformation of problem descriptions in logics
    into algebraic forms, exploitation of machine learning, qualitative
    reasoning, and constraint-based programming. Knowledge representation,
    for mathematical knowledge, is identified as a central focus for much
    of this work. Several promising topics for further research are stated.",
paper = "Calm97.pdf",
keywords = "axiomref"
}

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— axiom.bib —

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@techreport{Cami92,
author = "Camion, Paul and Courteau, Bernard and Montpetit, Andre",
title = "A combinatorial problem in Hamming Graphs and its solution
        in Scratchpad",
comment = {Un probl\`eme combinatoire dans les graphies de Hamming et sa
        solution en Scratchpad},
year = "1992",
month = "January",
link = "\url{https://hal.inria.fr/inria-00074974/document}",
type = "Research report",
number = "1586",
institution = "Institut National de Recherche en Informatique et en
        Automatique, Le Chesnay, France",
abstract =
    "We present a combinatorial problem which arises in the determination
    of the complete weight coset enumerators of error-correcting codes
    [1]. In solving this problem by exponential power series with
    coefficients in a ring of multivariate polynomials, we fall on a
    system of differential equations with coefficients in a field of
    rational functions. Thanks to the abstraction capabilities of

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Scratchpad this differential equation may be solved simply and
naturally, which seems not to be the case for the other computer
algebra systems now available.",
paper = "Cami92.pdf",
keywords = "axiomref",
beebe = "Camion:1992:PCG"
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— Camion:1992:PCG —

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@TechReport{Camion:1992:PCG,
  author = "Paul Camion and Bernard Courteau and Andre Montpetit",
  title = "Un probl{\e}me combinatoire dans les graphes de
           {Hamming} et sa solution en {Scratchpad}. ({English}:
           {A} combinatorial problem in {Hamming} graphs and its
           solution in {Scratchpad})",
  type = "Rapports de recherche",
  number = "1586",
  institution = "Institut National de Recherche en Informatique et en
                Automatique",
  address = "Le Chesnay, France",
  pages = "12",
  month = jan,
  year = "1992",
  bibdate = "Sat Dec 30 08:42:16 1995",
  bibsource = "http://www.math.utah.edu/pub/tex/bib/axiom.bib",
  abstract = "We present a combinatorial problem which arises in the
              determination of the complete weight coset enumerators
              of error-correcting codes [1]. In solving this problem
              by exponential power series with coefficients in a ring
              of multivariate polynomials, we fall on a system of
              differential equations with coefficients in a field of
              rational functions. Thanks to the abstraction
              capabilities of Scratchpad this differential equation
              may be solved simply and naturally, which seems not to
              be the case for the other computer algebra systems now
              available.",
  acknowledgement = ack-nhfb,
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\bibitem[Capriotti 00]{CCR00}

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author = "Capriotti, Olga and Cohen, Arjeh M. and Riem, Manfred",
title = "Java Phrasebooks for Computer Algebra and Automated Deduction",
link = "\url{http://www.sigsam.org/bulletin/articles/132/paper8.pdf}",
paper = "CCR00.pdf",
keywords = "axiomref"

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— axiom.bib —

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@misc{CC99,
  author = "Capriotti, O. and Carlisle, D.",
  title = "OpenMath and MathML: Semantic Mark Up for Mathematics",
  year = "1999",
  link = "\url{http://www.acm.org/crossroads/xrds6-2/openmath.html}",
  keywords = "axiomref"
}

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@misc{Capr99,
  author = "Capriotti, Olga and Cohen, Arjeh M. and Cuypers, Hans and
    Sterk, Hans",
  title = "OpenMath Technology for Interactive Mathematical Documents",
  year = "2002",
  pages = "51-66",
  publisher = "Springer-Verlag, Berlin, Germany",
  link = "\url{http://www.win.tue.nl/~hansc/lisbon.pdf}",
  misc = "in Multimedia Tools for Communicating Mathematics",
  paper = "Capr99.pdf",
  keywords = "axiomref"
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@article{Care06,
  author = "Carette, Jacques",
  title = "Gaussian elimination: a case study in efficient genericity
    with MetaOCaml",

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journal = "Sci. Comput. Program.",
volume = "62",
number = "1",
pages = "3-24",
year = "2006",
link = "\url{http://www.cas.mcmaster.ca/~carette/publications/ge.pdf}",
abstract =
  "The Gaussian Elimination algorithm is in fact an algorithm family
  common implementations contain at least six (mostly independent)
  ‘‘design choices’’. A generic implementation can easily be parametrized
  by all these design choices, but this usually leads to slow and
  bloated code. Using MetaOCaml's staging facilities, we show how we can
  produce a natural and type-safe implementation of Gaussian Elimination
  which exposes its design choices at code-generation time, so that
  these choices can effectively be specialized away, and where the
  resulting code is quite efficient.",
paper = "Care06.pdf",
keywords = "axiomref"
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@article{Care11,
  author = "Carette, Jacques and Kiselyov, Oleg",
  title = "Multi-stage programming with functors and monads: eliminating
    abstraction overhead from generic code",
  journal = "Sci. Comput. Program",
  volume = "76",
  number = "5",
  pages = "349-375",
  year = "2011",
  link = "\url{http://www.cas.mcmaster.ca/~carette/metamonads/metamonads.pdf}",
  abstract =
    "We use multi-stage programming, monads and OCaml's advanced module
    system to demonstrate how to eliminate all abstraction overhead from
    generic programs, while avoiding any inspection of the resulting code.
    We demonstrate this clearly with Gaussian Elimination as a
    representative family of symbolic and numeric algorithms. We
    parameterize our code to a great extent -- over domain, input and
    permutation matrix representations, determinant and rank tracking,
    pivoting policies, result types, etc. -- at no run-time cost. Because
    the resulting code is generated just right and not changed afterward,
    MetaOCaml guarantees that the generated code is well-typed. We further
    demonstrate that various abstraction parameters (aspects) can be made
    orthogonal and compositional, even in the presence of name-generation
    for temporaries, and ‘‘interleaving’’ of aspects. We also show how to

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    encode some domain-specific knowledge so that ‘clearly wrong’
    compositions can be rejected at or before generation time, rather than
    during the compilation or running of the generated code.",
    paper = "Care11.pdf",
    keywords = "axiomref"
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@misc{Carp04,
  author = "Carpent, Quentin and Conil, Christophe",
  title = "Utilisation de logiciels libres pour la r\'ealisation de TP MT26",
  year = "2004",
  link = "\url{http://axiom-wiki.newsynthesis.org/public/refs/ac20.pdf}",
  comment = "french",
  abstract = "radicalSolve(x**3+x**2-7=0,x)",
  paper = "Carp04.pdf",
  keywords = "axiomref"
}

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@misc{Caru10,
  author = "Caruso, Fabrizio",
  title = "Factorization of Non-Commutative Polynomials",
  link = "\url{https://arxiv.org/pdf/1002.3108.pdf}",
  year = "2010",
  abstract =
    "We describe an algorithm for the factorization of non-commutative
    polynomials over a field. The first sketch of this algorithm appeared
    in an unpublished manuscript (literally hand written notes) by James
    H. Davenport more than 20 years ago. This version of the algorithm
    contains some improvements with respect to the original sketch. An
    improved version of the algorithm has been fully implemented in the
    Axiom computer algebra system.",
  paper = "Caru10.pdf",
  keywords = "axiomref"
}

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— axiom.bib —

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@misc{Cavi03,
  author = "Caviness, Bob and Trager, Barry and Gianni, Patrizia",
  title = "Dedicated to the Memory of Richard Dimick Jenks",
  year = "2003",
  link = "\url{https://www.eecis.udel.edu/~caviness/jenks/issac-ded.pdf}",
  abstract =
```

```
  "On December 30, 2003, Dick Jenks died at the age of 66, after an
  extended and courageous battle with multiple system atrophy.
```

He received his PhD in mathematics from the University of Illinois at Urbana-Champaign in 1966. The title of his dissertation was Quadratic Differential Systems for Mathematical Models and was written under the supervision of Donald Gilles. After completing his PhD, he was a post-doctoral fellow at Brookhaven National Laboratory on Long Island. In 1968 he joined IBM Research where he worked until his retirement in 2002.

At IBM he was one of the principal architects of the Scratchpad system, one of the earliest computer algebra systems(1971). Dick always believed that natural user interfaces were essential and developed a user-friendly rule-based system for Scratchpad. Although this rule-based approach was easy to use, as algorithms for computer algebra became more complicated, he began to understand that an abstract data type approach would give sophisticated algorithm development considerably more leverage. In 1977 he began the Axiom development (originally called Scratchpad II) with the design of MODLISP, a merger of Lisp with types (modes). In 1980, with the help of many others, he completed an initial prototype design based on categories and domains that were intended to be natural for mathematically sophisticated users.

During this period many researchers in computer algebra visited IBM Research in Yorktown Heights and contributed to the development of the Axiom system. All this activity made the computer algebra group at IBM one of the leading centers for research in this area and Dick was always there to organize the visits and provide a stimulating and pleasant working environment for everyone. He had a good perspective on the most important research directions and worked to attract world-renowned experts to visit and interact with his group. He was an ideal manager for whom to work, one who always put the project and the needs of the group members first. It was a joy to work in such a vibrant and stimulating environment.

After many years of development, a decision was made to rename Scratch- pad II to Axiom and to release it as a product. Dick and Robert Sutor were the primary authors of the book {\sl Axiom: The Scientific Computation System}. In the foreword of the book, written

by David and Gregory Chudnovsky, it is stated that ‘‘The Scratchpad system took its time to blossom into the beautiful Axiom product. There is no rival to this powerful environment in its scope and, most importantly, in its structure and organization.’’ Axiom was recently made available as free software. See <http://savannah.nongnu.org/projects/axiom>

Dick was active in service to the computer algebra community as well. Here are some highlights. He served as Chair of ACM SIGSAM (1979–81) and Conference Co-chair (with J. A. van Hulzen) of EUROSAM 84, a precursor of the ISSAC meetings. Dick also had a long period of service on the editorial board of the *Journal of Symbolic Computation*. At ISSAC 95 in Montreal, Dick was elected to the initial ISSAC Steering Committee and was elected as the second Chair of the Committee in 1997. He, along with David Chudnovsky, organized the highly successful meetings on Computers and Mathematics that were held at Stanford in 1986 and MIT in 1989. As a legacy of those meetings, the Jenks Prize for outstanding contributions to software engineering in computer algebra has been established.

Dick had many interests outside of his professional pursuits including reading, travel, physical fitness, and especially music. Dick was an accomplished pianist, organist, and vocalist. At one point he was the organist and choir master of the Church of the Holy Communion in Mahopac, NY. In the 1980s and 1990s, he sang in choral groups under the direction of Dr. Dennis Keene that performed at Lincoln Center in New York city.

Personally, Dick was warm, generous, and outgoing with many friends. He will be missed for his technical accomplishments, his artist talents, and most of all for his positive, gentle, charming spirit.”,

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paper = "Cavi03.pdf",
keywords = "axiomref",
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— Caviness:2004:MRD —

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@Article{Caviness:2004:MRD,
  author = "Bob Caviness and Barbara Gatje and James H. Griesmer
    and Tony Hearn and Manual Bronstein and Erich
    Kaltofen",
  title = "In Memoriam: {Richard Dimick Jenks}: {Axiom} Developer
    and Computer Algebra Pioneer",
  journal = j-SIGSAM,
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volume =      "38",
number =      "1",
pages =       "30--30",
month =       mar,
year =        "2004",
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ISSN =        "0163-5824 (print), 1557-9492 (electronic)",
ISSN-L =      "0163-5824",
bibdate =     "Sat Apr 17 11:49:58 MDT 2004",
bibsource =   "http://portal.acm.org/;
               http://www.math.utah.edu/pub/tex/bib/axiom.bib",
link =        "\url{http://savannah.nongnu.org/projects/axiom/}",
acknowledgement = ack-nhfb,
fjournal =    "SIGSAM Bulletin",
issue =       "147",
}

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@misc{Cert16,
  author = "Certik, Ondrej",
  title = "SymPy vs. Axiom",
  link = "\url{https://github.com/sympy/sympy/wiki/SymPy-vs.-Axiom}",
  keywords = "axiomref"
}

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@inproceedings{Chan04,
  author = "Chan, L. and Cheb-Terrab, E.S.",
  title = "Non-Liouvillian solutions for second order linear ODEs",
  booktitle = "Proc. ISSAC 04",
  pages = "80-86",
  year = "2004",
  isbn = "1-58113-827-X",
  link = "\url{http://www.cecm.sfu.ca/CAG/papers/edgardoIS04.pdf}",
  abstract =
    "There exist sound literature and algorithms for computing Liouvillian
    solutions for the important problem of linear ODEs with rational
    coefficients. Taking as sample the 363 second order equations of that
    type found in Kamke's book, for instance, 51\% of them admit Liouvillian
    solutions and so are solvable using Kovacic's algorithm. On the other

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hand, special function solutions not admitting Liouvillian form appear frequently in mathematical physics, but there are not so general algorithms for computing them. In this paper we present an algorithm for computing special function solutions which can be expressed using the ${}_2F_1$, ${}_1F_1$ or ${}_0F_1$ hypergeometric functions. The algorithm is easy to implement in the framework of a computer algebra system and systematically solves 91\% of the 363 Kamke's linear ODE examples mentioned.",

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paper = "Chan04.pdf",
keywords = "axiomref"
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@misc{Chic04,
  author = "Chicha, Yannis and Lloyd, Michael Oancea, Cosmin",
  title = "Parametric Polymorphism for Computer Algebra Software Components",
  booktitle = "6th Int. Symp. on Symbolic and Numeric Algorithms for
    Scientific Computing",
  series = "SYNASC 04",
  location = "Iisoara, Romania",
  year = "2004",
  link = "\url{http://www.csd.uwo.ca/~watt/pub/reprints/2004-synasc-ppca.pdf}",
  abstract =
    "This paper presents our experiments in providing mechanisms for
    parametric polymorphism for computer algebra software components.
    Specific interfaces between Aldor and C++ and between Aldor and Maple
    are described. We then present a general solution, Generic IDL (GIDL),
    an extension to CORBA IDL supporting generic types. We describe our
    language bindings for C++, Java 1.5 and Aldor as well as aspects of
    our implementation, consisting of a GIDL to IDL compiler and tools for
    generating interface code for the various language bindings.",
  paper = "Chic04.pdf",
  keywords = "axiomref"
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@misc{Choux,
  author = "Chou, Shang-Ching and Gao, Xiao-Shan and Liu, Zhuo-Jun and
    Wang, Ding-Kang and Wang, Dongming",
  title = "Geometric Theorem Provers and Algebraic Equations Solvers",
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comment = "Chapter 20",
link = "\url{http://www.mmrc.iss.ac.cn/~xgao/papers/soft1.pdf}",
abstract =
  "The methods of mechanizing mathematics are realized by means of
  computer software for solving scientific and engineering problems via
  symbolic and hybrid computation. This chapter provides a collection of
  geometric theorem provers and algebraic equations solvers that are
  pieces of mathematical software based mostly on Wu's method and were
  developed mainly by members of the extended Wu group. The early
  theorem provers, though efficient, were written in basic programming
  languages and on primitive computers. Now there exist more powerful
  and mature geometric theorem provers of which some have already been
  published as commercial software. On the other hand, building effective
  equations solvers is still at the experimental stage and
  remains for further research and development.",
paper = "Chouxx.pdf"
}

```

— axiom.bib —

```

@misc{Chu85,
  author = "Chudnovsky, D.V and Chudnovsky, G.V.",
  title = "Elliptic Curve Calculations in Scratchpad II",
  year = "1985",
  institution = "Mathematics Dept., IBM Research",
  type = "Scratchpad II Newsletter 1 (1)",
  keywords = "axiomref"
}

```

— ignore —

```

\bibitem[Chudnovsky 87]{Chu87}
  author = "Chudnovsky, D.V and Chudnovsky, G.V.",
  title = "New Analytic Methods of Polynomial Root Finding",
  year = "1987",
  pages = "2",
  keywords = "axiomref"
in [Wit87]

```

— ignore —

```
\bibitem[Chudnovsky 89]{Chu89}
  author = "Chudnovsky, D.V. and Chudnovsky, G.V.",
  title = "The computation of classical constants",
  year = "1989",
  month = "November",
  pages = "8178-8182",
  Proc. Natl. Acad. Sci. USA Vol 86 ,
  keywords = "axiomref"
```

— axiom.bib —

```
@proceedings{CJ86,
  editor = "Chudnovsky, David and Jenks, Richard",
  title = "Computers in Mathematics",
  year = "1986",
  month = "July",
  isbn = "0-8247-8341-7",
  note = "International Conference on Computers and Mathematics",
  publisher = "Marcel Dekker, Inc",
  keywords = "axiomref"
}
```

— axiom.bib —

```
@InProceedings{Coli95,
  author = "Colin, Antoine",
  title = "Formal computation of Galois groups with relative resolvants",
  booktitle = "Proc. AAECC-11",
  series = "AAECC-11",
  year = "1995",
  publisher = "Springer-Verlag",
  location = "Paris, France",
  pages = "169-182",
  abstract =
    "Let  $k$  be a field and  $f \in k[x]$  be a polynomial of degree  $n$ . The
    permutation action of  $G$  on the roots  $\{\alpha_i\}_{i=1}^n$  of  $f$ 
    can be determined by an algorithm suggested by R. Stauduhar
    [Math. Comput. 27, 981-996 (1973; Zbl 0282.12004)] that approximates  $G$ "
```

via successive steps in a chain of subgroups
 $S_n = H_0 > H_1 > \dots > H_k = G$. In each step $H_{i-1} > H_i$
 it checks as a test for $G \leq H_i$ whether a relative invariant k_i
 $\in k[x_1, \dots, x_n]$ yields a value under the specialization
 $\varphi : g(x_1, \dots, x_n) \mapsto g(\alpha_1, \dots, \alpha_n)$. In
 implementations this evaluation has been done using p -adic
 [H. Darmon and D. Ford, Commun. Algebra 17, No. 12, 2941-2943 (1989;
 Zbl 0693.12010)] or numerical (R. Stauduhar [ibid.]; Y. Eichenlaub and
 M. Olivier [preprint]) approximation of the roots.

The paper under review presents a new approach which avoids all
 approximations: If $G \leq H_i$ and H_i is maximal in H_{i-1} the
 invariant h_i is a primitive element of the invariant field
 $k_i = k(x_1, \dots, x_n)^{H_i}$ as an extension of
 $k_{i-1} = K(x_1, \dots, x_n)^{H_{i-1}}$.
 The author develops an algorithm to express the specialized values
 $\varphi(g)$ of elements $g \in k_i$ in terms of k_{i-1} and the
 specialization $\varphi(h_i)$.

This algorithm then is applied to the relative resolvent polynomial
 $s_i = \prod_a (y - a(x_1, \dots, x_n))$
 where a runs through the images of h_i under H_{i-1} .
 It has y -coefficients which are in k_{i-1} .
 The algorithm then permits to express the coefficients of the
 specialization $r_i(y) = \varphi(s_i) \in k[y]$ recursively in the
 (already known) specializations $\varphi(h_j)$ for $j \leq i-1$,
 using the coefficients of f (as S_n -invariants in the roots)
 as a seed. A root of $r_i(y)$ in the base field then proves that G
 is contained in (a conjugate of) H_i , and this value of the root can
 be used as specialized $\varphi(h_{i+1})$ in the next step of the
 algorithm. Special care is given to the case when denominators of
 elements in $k(x_1, \dots, x_n)$ evaluate to zero after specialization.

The paper closes with a short discussion of applicability. An
 implementation using AXIOM and GAP is in process but has not yet been
 completed.",

keywords = "axiomref"

}

— axiom.bib —

```
@article{Coli97,
  author = "Colin, Antoine",
  title = "Solving a system of algebraic equations with symmetries",
  journal = "J. Pure Appl. Algebra",
  volume = "117-118",
```



```

pages = "195-215",
year = "1997",
abstract =
  "Let  $(F)$  be a system of  $p$  polynomial equations
 $F_i(\mathbf{X}) \in k[\mathbf{X}]$ , where  $k$  is a commutative field and
 $\mathbf{X} := (X_1, \dots, X_n)$  are indeterminates. Let  $G$  be a subgroup
of  $GL_n(k)$ . A polynomial  $P \in k[\mathbf{X}]$  (resp. rational function
 $P \in k(\mathbf{X})$ ) is an invariant of  $G$  if and only if for all
 $A \in G$  we have  $A \cdot P = P$ . We denote  $k[\mathbf{X}]^G$  by (resp.
 $k(\mathbf{X})^G$ ) the algebra of polynomial (resp. rational function)
invariants of  $G$ . If  $L$  is another subgroup of  $GL_n(k)$  such that
 $G \subset L$ ,  $P$  is called a primary invariant of  $G$  relative to  $L$  if
and only if  $\text{Stab}_L(P) = G$  (where  $\text{Stab}_L(P)$  is the stabilizer of
 $P$  in  $L$ ).
```

The paper describes the algebra of the invariants of a finite group and how to express these invariants in terms of a small number of them, from both the Cohen-Macaulay algebra and the field theory points of view. A method is proposed to solve (F) by expressing it in terms of primary invariants $\pi_1, \dots, \pi_n$ (e.g. the elementary symmetric polynomials) and one ‘‘primitive’’ secondary invariant.

The main thrust of the paper is contained in the following theorem. Let (F) be a set of invariants of G . Let L be a subgroup of $GL_n(k)$ such that $G \subset L$ and $k(\mathbf{X})^L$ is a purely transcendental extension of k_i , let $\pi_1, \dots, \pi_n$ be polynomials such that $k(\mathbf{X})^L = k(\pi_1, \dots, \pi_n)$, and let $\theta \in k[\mathbf{X}]^G$ be a primitive polynomial invariant of G relative to L .

When possible, it is convenient to choose θ to be one of the polynomials in (F) . An algorithm is given that allows each polynomial F_i to be expressed as $F_i(\mathbf{X}) = H_i(\pi_1, \dots, \pi_n, \theta)$, an algebraic fraction in $\pi_1, \dots, \pi_n$ and a polynomial in θ . Now let L be the minimal polynomial of θ over $k[\mathbf{X}]^L$; we have

$$[L(\mathbf{X}), T] = \prod_{\theta' \in L \cdot \theta} (T - \theta') \in k[\mathbf{X}]^L[T]$$

(where L is called a generic Lagrange resolvent).

As $k(\pi_1, \dots, \pi_n) = k(\mathbf{X})^L$, we can write

$L(\mathbf{X}, T) = H_0(\pi_1, \dots, \pi_n, T)$ where H_0 is some rational function. The question

$H_0(\pi_1, \dots, \pi_n, \theta) = 0$ is always satisfied because

θ is a root of L . Then, we solve the system of ($p+1$)

algebraic equations $H_i(\pi_1, \dots, \pi_n, \theta) = 0$,

$0 \leq i \leq p$ for $\pi_1, \dots, \pi_n, \theta$ as indeterminates.

Theorem 1: Let $D \in k[\pi_1, \dots, \pi_n]$ be the LCM of the denominators of all the fractions H_i , $0 \leq i \leq p$ and let $H_i' = DH_i$. For every solution

$x := (x_1, \dots, x_n)$ of the system $(F): F_i(\mathbf{X}) = 0$,
 $1 \leq i \leq p$, there exists a solution $(\pi_1, \dots, \pi_n, \theta)$
of the system
 $(H'): H_i(\pi_1, \dots, \pi_n, \theta) = 0$, $0 \leq i \leq p$ such
that x is a solution of the system
 $(P_\pi): P_i(\mathbf{X}) = \pi_i$, $1 \leq i \leq n$, and of the equation
 $\theta(\mathbf{X}) = 0$. Conversely, for any solution
 $(\pi_1, \dots, \pi_n, \theta)$ of the system (H') such that
 $D(\pi_1, \dots, \pi_n) \neq 0$, if x is a solution of the system
 (P_π) relative to $(\pi_1, \dots, \pi_n)$, then there exists
some $A \in L$ such that $\theta(A \cdot x) = \theta$, and then for all
 $B \in G$, $BA \cdot x$, is a solution of the system (F) .

A slightly more general version of this theorem is also given. The
paper then presents an algorithm that applies the theory and has been
implemented in AXIOM. It is followed by several examples.",
keywords = "axiomref"

}

— axiom.bib —

```

@article{Coll82,
  author = "Collins, G.E. and Mignotte, M. and Winkler, F.",
  title = "Arithmetic in Basic Algebraic Domains",
  publisher = "Springer-Verlag",
  journal = "Computing, Supplement 4",
  pages = "189-220",
  year = "1982",
  abstract =
    "This chapter is devoted to the arithmetic operations, essentially
    addition, multiplication, exponentiation, division, gcd calculations
    and evaluation, on the basic algebraic domains. The algorithms for
    these basic domains are those most frequently used in any computer
    algebra system. Therefore the best known algorithms, from a
    computational point of view, are presented. The basic domains
    considered here are the rational integers, the rational numbers,
    integers modulo  $m$ , Gaussian integers, polynomials, rational
    functions, power series, finite fields and  $p$ -adic numbers. Bounds on
    the maximum, minimum and average computing time ( $t^+$ ,  $t^-$ ,  $t^*$ ) for
    the various algorithms are given."
}

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— axiom.bib —

```
@misc{Cohe03,
  author = "Cohen, Arjeh and Cuypers, H. and Barreiro, Hans and
           Reinaldo, Ernesto and Sterk, Hans",
  title = "Interactive Mathematical Documents on the Web",
  year = "2003",
  pages = "289-306",
  editor = "Joswig, M. and Takayma, N.",
  publisher = "Springer-Verlag, Berlin, Germany",
  misc = "in Algebra, Geometry and Software Systems",
  keywords = "axiomref"
}
```

— ignore —

```
\bibitem[Cohen 91]{CC91} Cohen, G.; Charpin, P.; (ed)
EUROCODE '90 International Symposium on
Coding Theory and Applications Proceedings. Springer-Verlag, Berlin, Germany
/ Heidelberg, Germany / London, UK / etc., 1991 ISBN 0-387-54303-1
(New York), 3-540-54303-1 (Berlin), LCCN QA268.E95 1990
keywords = "axiomref"
```

— axiom.bib —

```
@book{Cohe03a,
  author = "Cohen, Joel S.",
  title = "Computer algebra and symbolic computation. Mathematical Methods",
  year = "2003",
  publisher = "A. K. Peters",
  isbn = "1-56881-159-4",
  keywords = "axiomref"
}
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— axiom.bib —

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@book{Cohe03b,
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```

author = "Cohen, Joel S.",
title = "Computer algebra and symbolic computation. Elementary Algorithms",
year = "2003",
publisher = "A. K. Peters",
isbn = "1-56881-159-4",
paper = "Cohe03b.pdf",
keywords = "axiomref"
}

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@misc{Conrxxa,
author = "Conrad, Marc and French, Tim and Maple, Carsten and Pott, Sandra",
title = "Approaching Inheritance from a Natural Mathematical Perspective
        and from a Java Driven Viewpoint: a Comparative Review",
link = "\url{http://axiom-wiki.newsynthesis.org/public/refs/McTfCmSp-axiom.pdf}",
abstract = "
    It is well-known that few object-oriented programming languages allow
    objects to change their nature at run-time. There have been a number
    of reasons presented for this, but it appears that there is a real
    need for matters to change. In this paper we discuss the need for
    object-oriented programming languages to reflect the dynamic nature of
    problems, particularly those arising in a mathematical context. It is
    from this context that we present a framework that realistically
    represents the dynamic and evolving characteristic of problems and
    algorithms.",
paper = "Conrxxa.pdf",
keywords = "axiomref"
}

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@misc{Conrxxb,
author = "Conrad, Marc and French, Tim and Maple, Carsten and Pott, Sandra",
title = "Mathematical Use Cases lead naturally to non-standard Inheritance
        Relationships: How to make them accessible in a mainstream language?",
abstract = "
    Conceptually there is a strong correspondence between Mathematical
    Reasoning and Object-Oriented techniques. We investigate how the ideas
    of Method Renaming, Dynamic Inheritance and Interclassing can be used
    to strengthen this relationship. A discussion is initiated concerning
    the feasibility of each of these features.",

```

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    paper = "Conrxxb.pdf",
    keywords = "axiomref"
}

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@article{Cor100,
  author = "Corless, Robert M. and Jeffrey, David J. and Watt, Stephen M. and
    Davenport, James H.",
  title = "'According to Abramowitz and Stegun' or
    arccoth needn't be Uncouth",
  journal = "SIGSAM Bulletin - Special Issue on OpenMath",
  volume = "34",
  number = "2",
  pages = "58-65",
  year = "2000",
  algebra =
    "\newline\ref{category OM OpenMath}
    \newline\ref{domain COMPLEX Complex}
    \newline\ref{domain DFLOAT DoubleFloat}
    \newline\ref{domain FLOAT Float}
    \newline\ref{domain FRAC Fraction}
    \newline\ref{domain INT Integer}
    \newline\ref{domain LIST List}
    \newline\ref{domain SINT SingleInteger}
    \newline\ref{domain STRING String}
    \newline\ref{domain SYMBOL Symbol}
    \newline\ref{package OMEXPR ExpressionToOpenMath}
    \newline\ref{package OMSERVER OpenMathServerPackage}",
  abstract =
    "This paper addresses the definitions in OpenMath of the elementary
    functions. The original OpenMath definitions, like most other sources,
    simply cite [2] as the definition. We show that this is not adequate,
    and propose precise definitions, and explore the relationships between
    these definitions. In particular, we introduce the concept of a couth
    pair of definitions, e.g. of arcsin and arcsinh, and show that the
    pair arccot and {\sl arccoth} can be couth.",
  paper = "Cor100.pdf"
}

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@book{Coxx07,
  author = "Cox, David and Little, John and O'Shea, Donald",
  title = "Ideals, varieties and algorithms. An introduction to computational
    algebraic geometry and commutative algebra",
  publisher = "Springer",
  isbn = "978-0-387-35650-1",
  year = "2007",
  link = "\url{http://www.dm.unipi.it/~caboara/Misc/Cox,\%20Little,\%20O'Shea\%20-\%20Ideals,\%20varieties\%20and\%20algorithms}",
  algebra = "\newline\refto{package GB GroebnerPackage}
    \newline\refto{package PSEUDLIN PseudoLinearNormalForm}
    \newline\refto{package PGROEB PolyGroebner}
    \newline\refto{domain DMP DistributedMultivariatePolynomial}
    \newline\refto{domain GDMP GeneralDistributedMultivariatePolynomial}
    \newline\refto{domain HDMP HomogeneousDistributedMultivariatePolynomial}",
  abstract =
    "Around 1980 two new directions in science and technique came
    together. One was Buchbergers algorithms in order to handle Groebner
    bases in an effective way for solving polynomial equations. The second
    one was the development of the personal computers. This was the
    starting point of a computational perspective in commutative algebra
    and algebraic geometry. In 1991 the three authors invented the first
    edition of their book as an introduction for undergraduates to some
    interesting ideas in commutative algebra and algebraic geometry with a
    strong perspective to practical and computational aspects. A second
    revised edition appeared in 1996. That means from the very beginning
    the book provides a bridge for the new, computational aspects in the
    field of commutative algebra and algebraic geometry.
```

To be more precise, the book gives an introduction to Buchbergers algorithm with applications to syzygies, Hilbert polynomials, primary decompositions. There is an introduction to classical algebraic geometry with applications to the ideal membership problem, solving polynomial equations, and elimination theory. Some more spectacular applications are about robotics, automatic geometric theorem proving, and invariants of finite groups. It seems to the reviewer to carry coals to Newcastle for estimating the importance and usefulness of the book. It should be of some interest to ask how many undergraduates have been introduced to algorithmic aspects of commutative algebra and algebraic geometry following the line of the book. The reviewer will be sure that this will continue in the future too.

What are the changes to the previous editions? There is a significant shorter proof of the Extension Theorem, see 3.6 in Chapter 3, suggested by A.H.M. Levelt. A major update has been done in Appendix C 'Computer Algebra Systems'. This concerns in the main the section about MAPLE. Some minor updated information concern the use of AXIOM, CoCoA, Macaulay2, Magma, Mathematica, and SINGULAR. This reflects about the recent developments in Computer Algebra Systems. It encourages an interested reader to more practical exercises. The authors have made changes on over 200 pages to enhance clarity and

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correctness. Many individuals have reported typographical errors and
gave the authors feedback on the earlier editions. The book is
well-written. The reviewer guesses that it will become more and more
difficult to earn 1 dollar (sponsored by the authors) for every new
typographical error as it was the case also with the first and second
edition. The reviewer is sure that it will be a excellent guide to
introduce further undergraduates in the algorithmic aspect of
commutative algebra and algebraic geometry.",
paper = "Coxx07.pdf",
keywords = "axiomref"
}

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— axiom.bib —

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@article{Crou95,
  author = "Crouch, Peter E. and Lamnabhi-Lagarrigue, Francoise and
           Pinchon, Didier",
  title = "Some realizations of algorithms for nonlinear input-output systems",
  journal = "Int. J. Control",
  volume = "62",
  number = "4",
  pages = "941-960",
  year = "1995",
  abstract =
    "The first two authors previously developed an algorithm for
    constructing a parametrization of the observation space of a nonlinear
    control system directly from the differential equation representation
    of the input-output behaviour. This paper extends the previous
    algorithm by including settings where a set of implicit input-output
    differential equations is given as well as more general state-space
    representations in which the controls enter nonlinearly. Various
    state-space realizations, including bilinear, polynomial and nilpotent
    approximating realizations are discussed. The final section of the
    paper sketches the implementation of the algorithm using the symbolic
    manipulation package AXIOM to find the realizations mentioned above in
    feasible cases.",
  keywords = "axiomref"
}

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```

@misc{Cuyp10,

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author = "Cuypers, Hans and Hendriks, Maxim and Knopper, Jan Willem",
title = "Interactive Geometry inside MathDox",
year = "2010",
link = "\url{http://www.win.tue.nl/~hansc/MathDox_and_InterGeo_paper.pdf}",
paper = "Cuyp10",
keywords = "axiomref"
}

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— axiom.bib —

```

@inproceedings{Dalm97,
  author = {Dalmas, St\'ephane and Gaetano, Marc and Watt, Stephen},
  title = "An OpenMath 1.0 Implementation",
  booktitle = "Proc. 1997 Int. Symp. on Symbolic and Algebraic Computation",
  series = "ISSAC'97",
  year = "1997",
  isbn = "0-89791-875-4",
  location = "Kihei, Maui, Hawaii, USA",
  pages = "241-248",
  numpages = "8",
  link = "\url{http://doi.acm.org/10.1145/258726.258794}",
  doi = "10.1145/258726.258794",
  acmid = "258794",
  publisher = "ACM, New York, NY USA",
  keywords = "axiomref"
}

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— axiom.bib —

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@inproceedings{Dalm92,
  author = "Dalmas, Stephane",
  title = "A polymorphic functional language applied to symbolic computation",
  year = "1992",
  booktitle = "Proc. ISSAC 1992",
  series = "ISSAC 1992",
  pages = "369-375",
  isbn = "0-89791-489-9 (soft cover) 0-89791-490-2 (hard cover)",
  abstract =

```



```

"The programming language in which to describe mathematical objects
and algorithms is a fundamental issue in the design of a symbolic
computation system. XFun is a strongly typed functional programming
language. Although it was not designed as a specialized language, its
sophisticated type system can be successfully applied to describe
mathematical objects and structures. After illustrating its main
features, the author sketches how it could be applied to symbolic
computation. A comparison with Scratchpad II is attempted. XFun seems
to exhibit more flexibility simplicity and uniformity.",
keywords = "axiomref",
beebe = "Dalmas:1992:PFL"
}

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— Dalmas:1992:PFL —

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@InProceedings{Dalmas:1992:PFL,
  author = "S. Dalmas",
  title = "A polymorphic functional language applied to symbolic
computation",
  crossref = "Wang:1992:ISS",
  pages = "369--375",
  month = "",
  year = "1992",
  bibdate = "Tue Sep 17 06:35:39 MDT 1996",
  bibsource = "http://www.math.utah.edu/pub/tex/bib/axiom.bib",
  abstract = "The programming language in which to describe
mathematical objects and algorithms is a fundamental
issue in the design of a symbolic computation system.
XFun is a strongly typed functional programming
language. Although it was not designed as a specialized
language, its sophisticated type system can be
successfully applied to describe mathematical objects
and structures. After illustrating its main features,
the author sketches how it could be applied to symbolic
computation. A comparison with Scratchpad II is
attempted. XFun seems to exhibit more flexibility
simplicity and uniformity.",
  acknowledgement = ack-nhfb,
  affiliation = "Inst. Nat. de Recherche d'Inf. et d'Autom., Valbonne,
France",
  classification = "C6140D (High level languages); C7310 (Mathematics)",
  keywords = "Mathematical objects; Polymorphic functional language;
Scratchpad II; Symbolic computation; XFun",
  language = "English",
  thesaurus = "Functional programming; High level languages; Symbol
manipulation",

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}

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@misc{Daly08,
  author = "Daly, Timothy",
  title = "Axiom Computer Algebra System Information Sources",
  video = "https://www.youtube.com/watch?v=CV8y3Urpady",
  year = "2008",
  keywords = "axiomref"
}
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@misc{Daly88,
  author = "Daly, Timothy",
  title = "Axiom in an Educational Setting, Axiom course slide deck",
  year = "1988",
  month = "January",
  keywords = "axiomref"
}
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— axiom.bib —

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@article{Daly02,
  author = "Daly, Timothy",
  title = "Axiom as open source",
  journal = "SIGSAM Bulletin",
  volume = "36",
  number = "1",
  pages = "28-28",
  month = "March",
  year = "2002",
  keywords = "axiomref",
  beebe = "Daly:2002:AOS"
}
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— Daly:2002:AOS —

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@Article{Daly:2002:AOS,
  author =      "T. Daly",
  title =       "{Axiom} as Open Source",
  journal =     "j-SIGSAM",
  volume =      "36",
  number =      "1",
  pages =       "28--??",
  month =       mar,
  year =        "2002",
  CODEN =       "SIGSBZ",
  ISSN =        "0163-5824 (print), 1557-9492 (electronic)",
  ISSN-L =      "0163-5824",
  bibdate =     "Mon Apr 29 07:16:09 MDT 2002",
  bibsource =   "http://portal.acm.org/;
                http://www.math.utah.edu/pub/tex/bib/axiom.bib",
  acknowledgement = ack-nhfb,
  fjournal =    "SIGSAM Bulletin",
  issue =       "139",
}
```

— ignore —

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\bibitem[Daly 03]{Dal03} Daly, Timothy
  title = "The Axiom Wiki Website",
  link = "\url{http://axiom.axiom-developer.org}",
  keywords = "axiomref",
```

— axiom.bib —

```
@misc{Daly05,
  author = "Daly, Timothy",
  title = "LispNYC Presentation at Trinity",
  year = "2005",
  month = "May",
  day = "10",
  abstract =
    "Timothy Daly presents Axiom"
```

Timothy Daly, published author, academic researcher, open source programmer and lead developer of Axiom will be presenting about his role as the driving force behind Axiom. With over 70 developers and 200 researchers worldwide it can best be described as:

Axiom is a general purpose Computer Algebra system. It is useful for research and development of mathematical algorithms providing a very high level way to express abstract mathematical concepts. The Axiom Library defines over 1,000 strongly-typed mathematical domains and categories.

Axiom consists of an interpreter and compiler, a browser, a graphical interface, and a new online wiki that allows users to create web pages that inline computations.

```
Axiom is built upon Common Lisp.",
keywords = "axiomref"
}
```

— axiom.bib —

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@misc{Dalyxx,
  author = "Daly, Timothy",
  title = "Tim Daly on Lisp in Industry",
  link = "\url{https://news.ycombinator.com/item?id=1580904}",
  keywords = "axiomref"
}
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— ignore —

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@book{Daly06,
  author = "Daly, Timothy",
  title = "Axiom Volume 1: Tutorial",
  publisher = "Lulu, Inc.",
  address = "860 Aviation Parkway, Suite 300, Morrisville, NC 27560 USA",
  year = "2006",
  isbn = "1-4116-6597-X",
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}
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— Daly:2006:AVA —

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@Book{Daly:2006:AVA,
  author = "Timothy Daly",
  title = "{Axiom} Volume 1: {Axiom} Tutorial",
  publisher = "Lulu, Inc.",
  address = "860 Aviation Parkway, Suite 300, Morrisville, NC
            27560, USA",
  pages = "iv + 285",
  year = "2006",
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  bibdate = "Thu Mar 23 05:24:19 2006",
  bibsource = "http://www.math.utah.edu/pub/tex/bib/axiom.bib",
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  acknowledgement = ack-nhfb,
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\bibitem[Daly 09]{Dal09} Daly, Timothy
  title = "The Axiom Literate Documentation",
  link = "\url{http://axiom-developer.org/axiom-website/documentation.html}",
  keywords = "axiomref"
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— ignore —

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\bibitem[Daly 13]{Dal13} Daly, Timothy
  title = "Literate Programming in the Large'",
  April 8-9, 2013 Portland Oregon
  url1 = "http://conf.writethedocs.org",
  url2 = "http://daly.axiom-developer.org",
  link = "\url{http://www.youtube.com/watch?v=AvOPQDVTP4A}",
  keywords = "axiomref"
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— axiom.bib —

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@misc{Daly13a,
  author = "Daly, Timothy and Barnes, Nick",
  title = "Ten reasons you must publish your code",
  year = "2013",
  link = "\url{http://climatecode.org/blog/2013/07/ten-reasons-you-must-publish-your-code/}",
  keywords = "axiomref"
}
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— axiom.bib —

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@book{Dave05,
  author = "Davenport, James H.",
  title = "Integration -- What do we want from the theory?",
  booktitle = "Computer Algebra",
  publisher = "Springer",
  series = "Lecture Notes in Computer Science 162",
  pages = "2-11",
  year = "2005",
  abstract =
    "The theory of integration has moved a long way in the last fourteen
    years, thought not far enough to satisfy the demands placed on it by
    its customers. This paper outlines what problems have yet to be solved,
    and tries to explain why they are not trivial."
}
```

— ignore —

```
\bibitem[Davenport 79a]{Dav79a} Davenport, J.H.
  title = "What can SCRATCHPAD/370 do?",
  VM/370 SPAD.SCRIPTS August 24, 1979 SPAD.SCRIP
  keywords = "axiomref"
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— axiom.bib —

```
@techreport{Dave80,
  author = "Davenport, James H. and Jenks, Richard D.",
  title = "MODLISP: A Preliminary Design",
  institution = "IBM Research",
  type = "Research Report",
  year = "1980",
  number = "RC 8073",
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— axiom.bib —

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@techreport{Dave80a,
  author = "Davenport, James H. and Jenks, Richard D.",
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  institution = "IBM Research",
  type = "Research Report",
  year = "1980",
  number = "RC 8537 (\#37198)",
  comment = "http://www.computerhistory.org/collections/catalog/102719109",
  keywords = "axiomref"
}
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— axiom.bib —

```
@article{Dave81a,
  author = "Davenport, James H. and Jenks, Richard D.",
  title = "MODLISP",
  year = "1981",
  journal = "ACM SIGSAM Bulletin",
  volume = "15",
  issue = "1",
  pages = "11-20",
  publisher = "ACM",
  abstract =
    "This paper discusses the design and implementation of MODLISP, a
    LISP-like language enhanced with the idea of MODEs. This extension
    permits, but does not require, the user to declare the types of
    various variables, and to compile functions with the arguments
    declared to be of a particular type. It is possible to declare several
    functions of the same name, with arguments of different type
    (e.g. PLUS could be declared for Integer arguments, or Rational, or
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    Real, or even Polynomial arguments) and the system will apply the
    correct function for the types of the arguments.",
keywords = "axiomref"
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— axiom.bib —

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@manual{Dave84,
  author = "Davenport, James H. and Gianni, Patrizia and Jenks, Richard D. and
           Miller, Victor and Morrison, Scott C. and Rothstein, Michael and
           Sundaresan, Christine and Sutor, Robert S. and Trager, Barry M.",
  title = "Scratchpad",
  organization = "Mathematical Sciences Department",
  address = "IBM Thomas Watson Research Center, Yorktown Heights, NY",
  year = "1984",
  keywords = "axiomref",
  beebe = "Davenport:1984:S"
}

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— Davenport:1984:S —

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@Manual{Davenport:1984:S,
  author = "J. Davenport and P. Gianni and R. Jenks and V. Miller
           and S. Morrison and M. Rothstein and C. Sundaresan and
           R. Sutor and B. Trager",
  title = "{Scratchpad}",
  organization = "Mathematical Sciences Department",
  address = "IBM Thomas Watson Research Center",
  pages = "??",
  year = "1984",
  bibsource = "/usr/local/src/bib/bibliography/Theory/Comp.Alg.1.bib;
              http://www.math.utah.edu/pub/tex/bib/axiom.bib",
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— axiom.bib —

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@misc{Dave84a,
  author = "Davenport, James H.",

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title = "A New Algebra System",
link = "\url{http://axiom-wiki.newsynthesis.org/public/refs/Davenport-1984-a\_new\_algebra\_sy
abstract = "Seminal internal paper discussing Axiom design decisions.",
paper = "Dave84a.pdf",
keywords = "axiomref"
}

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— axiom.bib —

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@misc{Dave84b,
author = "Davenport, James H. and Gianni, Patrizia and Jenks, Ricard D. and
Miller, Victor and Morrison, Scott and Rothstein, Michael and
Sundaresan, Christine J. and Sutor, Robert S. and Trager, Barry",
title = "SCRATCHPAD System Programming Language Manual",
year = "1984",
keywords = "axiomref"
}

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— axiom.bib —

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@article{Dave85,
author = "Davenport, James H.",
title = "The LISP/VM Foundation of Scratchpad II",
journal = "The Scratchpad II Newsletter",
volume = "1",
number = "1",
year = "1985",
month = "September",
institution = "IBM Research",
keywords = "axiomref"
}

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— axiom.bib —

```

@book{Dave88,
author = "Davenport, James H. and Siret, Y. and Tournier, E.",
title = "Computer Algebra: Systems and Algorithms for Algebraic

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        Computation",
    publisher = "Academic Press",
    year = "1988",
    isbn = "0-12-204230-1",
    link = "\url{http://staff.bath.ac.uk/masjhd/masternew.pdf}",
    paper = "Dave88.pdf",
    keywords = "axiomref",
    beebe = "Davenport:1988:CA"
}

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— Davenport:1988:CA —

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@Book{Davenport:1988:CA,
  author =      "J. H. Davenport and Y. Siret and E. Tournier",
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    Computation",
  publisher =   pub-AP,
  address =     pub-AP:adr,
  pages =       "xix + 267",
  year =        "1988",
  ISBN =        "0-12-204230-1",
  ISBN-13 =     "978-0-12-204230-0",
  LCCN =         "QA155.7.E4 D38 1988",
  bibdate =     "Fri Dec 29 18:14:51 1995",
  bibsource =   "/usr/local/src/bib/bibliography/Theory/Comp.Alg.bib;
    http://www.math.utah.edu/pub/tex/bib/axiom.bib",
  notes =       "{\footnotesize Dies ist die englische Ausgabe des
    urspr{\u}nglich bei Masson 1987 erschienen Buches {\em
    Calcul Formel}. Es ist die erste Monographie {\u}ber
    Computeralgebra. Es wird in etwa die Theorie behandelt,
    die heute in den gr{o}{ss}eren Systemen wie MACSYMA,
    MAPLE, REDUCE oder SCRATCHPAD II realisiert ist. Das
    erste Kapitel ist der Diskussion verschiedener \CA\
    Systeme mit Beispielen gewidmet. Die wichtige Frage der
    Repr{a}sentation der mathematischen Objekte auf einem
    Computer ist das Thema des zweiten Kapitels. Der
    Algorithmus von Buchberger, zylindrische Dekomposition,
    Berechnung von gr{o}{ss}ten gemeinsamen Teilern,
    p-adische Methoden und Faktorisierung,
    Differentialgleichungen und Stammfunktionen sind die
    wichtigsten behandelten Gegenst{a}nde des Buches, das
    mit einer ausf{u}hrlichen Bibliographie und einer
    Beschreibung von REDUCE im Anhang endet. \hfill J.
    Grabmeier}",
}

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— axiom.bib —

```
@techreport{Dave89,
  author = "Davenport, James H.",
  title = "Looking at a set of equations",
  institution = "University of Bath, School of Mathematical Sciences",
  year = "1989",
  type = "technical report",
  link = "\url{http://citeseerx.ist.psu.edu/viewdoc/download?doi=10.1.1.261.767}",
  abstract =
    "This working paper describes our experiences with using the
    Groebner-basis method [Buchberger, 1985] to solve some related systems
    of polynomial equations. While we have not yet been able to solve the
    system that was our primary motivation, we feel that these experiences
    may prove useful to others investigating Buchberger's algorithm in
    this context, especially when, as is the case for the system under
    investigation, the equations are highly structured. We conclude with
    some examples of the polynomials that we factored in the course of
    this investigation.",
  paper = "Dave89.pdf"
}
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— axiom.bib —

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@misc{Dave93,
  author = "Davenport, James H.",
  title = "The PoSSo Project",
  paper = "Dave93.pdf",
  keywords = "axiomref"
}
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— axiom.bib —

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@InProceedings{Dave00,
  author = "Davenport, James H.",
  title = "Abstract data types in computer algebra",
  booktitle = "Mathematical foundations of computer science",
  series = "MFCS 2000",
  year = "2000",
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location = "Bratislava, Slovakia",
pages = "21-35",
abstract =
  "The theory of abstract data types was developed in the late 1970s and
  the 1980s by several people, including the ‘‘ADJ’’ group, whose work
  influenced the design of Axiom. One practical manifestation of this
  theory was the OBJ-3 system. An area of computing that cries out for
  this approach is computer algebra, where the objects of discourse are
  mathematical, generally satisfying various algebraic rules. There have
  been various theoretical studies of this in the literature. The aim of
  this paper is to report on the practical applications of this theory
  within computer algebra, and also to outline some of the theoretical
  issues raised by this practical application. We also give a
  substantial bibliography.",
keywords = "axiomref"
}

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— axiom.bib —

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@article{Dave02,
  author = "Davenport, James H.",
  title = "Equality in computer algebra and beyond",
  journal = "J. Symbolic Computing",
  volume = "34",
  number = "4",
  pages = "259-270",
  year = "2002",
  link = "\url{http://www.calculumus.net/meetings/siena01/Papers/Davenport.pdf}",
  abstract =
    "Equality is such a fundamental concept in mathematics that, in
    fact, we seldom explore it in detail, and tend to regard it as
    trivial. When it is shown to be non-trivial, we are often
    surprised. As is often the case, the computerization of
    mathematical computation in computer algebra systems on the one
    hand, and mathematical reasoning in theorem provers on the other
    hand, forces us to explore the issue of equality in greater
    detail. In practice, there are also several ambiguities in the
    definition of equality. For example, we refer to  $\mathbb{Q}(x)$ 
    as ‘‘rational functions’’, even though  $\frac{x^2-1}{x-1}$  and
     $x+1$  are not equal as functions from  $\mathbb{R}$  to
     $\mathbb{R}$ , since the former is not defined at  $x=1$ , even
    though they are equal as elements of  $\mathbb{Q}(x)$ . The aim of
    this paper is to point out some of the problems, both with
    mathematical equality and with data structure equality, and to
    explain how necessary it is to keep a clear distinction between the two.",
  paper = "Dave02.pdf",

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    keywords = "axiomref"
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— axiom.bib —

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@misc{Dave07,
  author = "Davenport, James H. and Fitch, John",
  title = "Computer Algebra and the three 'E's: Efficiency, Elegance, and
    Expressiveness",
  link = "\url{http://staff.bath.ac.uk/masjhd/Drafts/PLMMS2007}",
  abstract =
    "What author of a programming language would not claim that the 3 'E's
    were the goals? Nevertheless, we claim that computer algebra does lead
    to particular emphases, and constraints, in these areas.

    We restrict 'efficiency' to mean machine efficiency, since the other
    'E's cover programmer efficiency. For the sake of clarity, we describe
    as 'expressiveness', what can be expressed in the language, and
    'elegance' as how it can be expressed.",
  paper = "Dave07.pdf",
  keywords = "axiomref"
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— ignore —

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\bibitem[Davenport 14]{Dav14} Davenport, James H.
  title = "Computer Algebra textbook",
  link = "\url{http://staff.bath.ac.uk/masjhd/JHD-CA.pdf}",
  paper = "Dav14.pdf",
  keywords = "axiomref"

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— ignore —

```

\bibitem[Davenport 89]{Dav89} Davenport, J.H. (ed)
EUROCAL '87 European Conference on Computer Algebra Proceedings
Springer-Verlag, Berlin, Germany / Heidelberg, Germany / London,
UK / etc., 1989 ISBN 3-540-51517-8 LCCN QA155.7.E4E86 1987

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keywords = "axiomref"

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@InProceedings{Dave90,
  author = "Davenport, James H. and Trager, Barry M.",
  title = "Scratchpad's view of algebra I: Basic commutative algebra",
  booktitle = "Design and Implementation of Symbolic Computation Systems",
  year = "1990",
  pages = "40-54",
  series = "DISCO '90",
  location = "Capri, Italy",
  publisher = "Springer-Verlag",
  isbn = "0-387-52531-9",
  link = "\url{http://opus.bath.ac.uk/32336/1/Davenport\_DISCO\_1990.pdf}",
  comment = "AXIOM Technical Report, ATR/1, NAG Ltd., Oxford, 1992",
  keywords = "axiomref",
  abstract =
    "While computer algebra systems have dealt with polynomials and
    rational functions with integer coefficients for many years, dealing
    with more general constructs from commutative algebra is a more recent
    problem. In this paper we explain how one system solves this problem,
    what types and operators it is necessary to introduce and, in short,
    how one can construct a computational theory of commutative
    algebra. Of necessity, such a theory is rather different from the
    conventional, non-constructive, theory. It is also somewhat different
    from the theories of Seidenberg [1974] and his school, who are not
    particularly concerned with practical questions of efficiency.",
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— Davenport:1990:SVA —

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@InProceedings{Davenport:1990:SVA,
  author = "J. H. Davenport and B. M. Trager",
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  crossref = "Miola:1990:DIS",
  pages = "40--54",
  month = "",
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year = "1990",
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bibsource = "/usr/local/src/bib/bibliography/Theory/Comp.Alg.bib;
             http://www.math.utah.edu/pub/tex/bib/axiom.bib",
note = "auch in: {AXIOM} Technical Report, ATR/1, NAG Ltd.,
        Oxford, 1992.",
abstract = "The paper describes the constructive theory of
            commutative algebra which underlies that part of
            Scratchpad which deals with commutative algebra. The
            authors begin by explaining the background that led the
            Scratchpad group to construct such a general theory.
            They contrast the general theory in Scratchpad with
            Reduce-3's theory of domains, which is in many ways
            more limited, but is the closest approach to an
            implemented general theory to be found outside
            Scratchpad. This leads them to describe the general
            Scratchpad view of data types and categories, and the
            possibilities it offers. They then digress a little to
            ask what criteria should be adopted in choosing what
            types to define. Having discussed the philosophical
            issues, they then discuss commutative algebra proper,
            breaking this up into the sections 'up to Ring',
            'Integral Domain', 'Gcd Domain' and 'Euclidean
            Domain'.",
acknowledgement = ack-nhfb,
affiliation = "Sch. of Math. Sci., Bath Univ., UK",
classification = "C1110 (Algebra); C7310 (Mathematics)",
keywords = "Categories; Commutative algebra; Constructive theory;
            Data types; Euclidean Domain; Gcd Domain; Greatest
            common divisors; Integral Domain; Philosophical issues;
            Ring; Scratchpad",
language = "English",
thesaurus = "Algebra; Software packages; Symbol manipulation",
}

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@inproceedings{Dave91,
  author = "Davenport, J. H. and Gianni, P. and Trager, B. M.",
  title = "Scratchpad's View of Algebra II:
           A Categorical View of Factorization",
  booktitle = "Proc. 1991 Int. Symp. on Symbolic and Algebraic Computation",
  series = "ISSAC '91",
  year = "1991",
  isbn = "0-89791-437-6",
  location = "Bonn, West Germany",
}

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pages = "32--38",
numpages = "7",
link = "\url{http://doi.acm.org/10.1145/120694.120699}",
doi = "10.1145/120694.120699",
acmid = "120699",
publisher = "ACM",
address = "New York, NY, USA",
abstract = "
  This paper explains how Scratchpad solves the problem of presenting a
  categorical view of factorization in unique factorization domains,
  i.e. a view which can be propagated by functors such as
  SparseUnivariatePolynomial or Fraction. This is not easy, as the
  constructive version of the classical concept of
  UniqueFactorizationDomain cannot be so propagated. The solution
  adopted is based largely on Seidenberg's conditions (F) and (P), but
  there are several additional points that have to be borne in mind to
  produce reasonably efficient algorithms in the required generality.

  The consequence of the algorithms and interfaces presented in this
  paper is that Scratchpad can factorize in any extension of the
  integers or finite fields by any combination of polynomial, fraction
  and algebraic extensions: a capability far more general than any other
  computer algebra system possesses. The solution is not perfect: for
  example we cannot use these general constructions to factorize
  polynomials in  $\overline{\mathbb{Z}[\sqrt{-5}]}[x]$  since the domain
 $\mathbb{Z}[\sqrt{-5}]$  is not a unique factorization domain, even though
 $\overline{\mathbb{Z}[\sqrt{-5}]}$  is, since it is a field. Of course, we can
  factor polynomials in  $\overline{\mathbb{Z}[\sqrt{-5}]}[x]$ ",
paper = "Dave91.pdf",
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@InProceedings{Davenport:1991:SVA,
  author = "J. H. Davenport and P. Gianni and B. M. Trager",
  title = "{Scratchpad}'s view of algebra. {II}. {A} categorical
  view of factorization",
  crossref = "Watt:1991:PIS",
  pages = "32--38",
  month = "",
  year = "1991",
  bibdate = "Tue Sep 17 06:44:07 MDT 1996",
  bibsource = "/usr/local/src/bib/bibliography/Theory/Comp.Alg.bib;
  http://www.math.utah.edu/pub/tex/bib/axiom.bib",

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note = "auch in: {AXIOM} Technical Report, ATR/2, NAG Ltd.,
        Oxford, 1992.",
abstract = "For pt.I see Proc. DISCO 1990 (p.40-54). The paper
            explains how Scratchpad solves the problem of
            presenting a categorical view of factorization in
            unique factorization domains, i.e. a view which can be
            propagated by functors such as
            SparseUnivariatePolynomial or Fraction. This is not
            easy, as the constructive version of the classical
            concept of UniqueFactorizationdomain cannot be so
            propagated. The solution adopted is based largely on
            the Seidenberg conditions (F) and (P), but there are
            several additional points that have to be borne in mind
            to produce reasonably efficient algorithms in the
            required generality. The consequence of the algorithms
            and interfaces presented is that Scratchpad can
            factorize in any extension of the integers or finite
            fields by any combination of polynomial, fraction and
            algebraic extensions: a capability far more general
            than any other computer algebra system possesses.",
acknowledgement = ack-nhfb,
affiliation = "Sch. of Math., Bath Univ., Claverton Down, UK",
classification = "C4130 (Interpolation and function approximation);
                  C7310 (Mathematics)",
keywords = "Algebraic extensions; Categorical view; Computer
            algebra system; Factorization; Finite fields; Fraction;
            Integers; Polynomial; Scratchpad; Seidenberg
            conditions",
language = "English",
thesaurus = "Mathematics computing; Polynomials; Symbol
            manipulation",
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@techreport{Dave92c,
  author = "Davenport, James H. and Trager, Barry M.",
  title = "Scratchpad's view of algebra I: Basic commutative algebra",
  number = "TR3/92 (ATR/1) (NP2490)",
  institution = "Numerical Algorithm Group (NAG) Ltd.",
  year = "1992",
  abstract = "While computer algebra systems have dealt with polynomials and
            rational functions with integer coefficients for many years, dealing
            with more general constructs from commutative algebra is a more recent
            problem. In this paper we explain how one system solves this problem,

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what types and operators it is necessary to introduce and, in short,
 how one can construct a computational theory of commutative
 algebra. Of necessity, such a theory is rather different from the
 conventional, non-constructive, theory. It is also somewhat different
 from the theories of Seidenberg [1974] and his school, who are not
 particularly concerned with practical questions of efficiency.",
 paper = "Dave90.pdf",
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— Davenport:1992:SVAA —

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  author = "J. H. Davenport and B. M. Trager",
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  number = "TR3/92 (ATR/1) (NP2490)",
  institution = inst-NAG,
  address = inst-NAG:adr,
  pages = "??",
  month = dec,
  year = "1992",
  bibdate = "Fri Dec 29 16:31:49 1995",
  bibsource = "http://www.math.utah.edu/pub/tex/bib/axiom.bib",
  link = "\url{http://www.nag.co.uk/doc/TechRep/axiomtr.html}",
  acknowledgement = ack-nhfb,
}
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— axiom.bib —

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@techreport{Dave92d,
  author = "Davenport, James H. and Gianni, Patrizia and Trager, Barry M.",
  title = "Scratchpad's view of algebra II:
           A categorical view of factorization",
  type = "Technical Report",
  number = "TR4/92 (ATR/2) (NP2491)",
  institution = "Numerical Algorithms Group, Inc.",
  address = "Downer's Grove, IL, USA and Oxford, UK",
  year = "1992",
  link = "\url{http://www.nag.co.uk/doc/TechRep/axiomtr.html}",
  abstract = "
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This paper explains how Scratchpad solves the problem of presenting a categorical view of factorization in unique factorization domains, i.e. a view which can be propagated by functors such as `SparseUnivariatePolynomial` or `Fraction`. This is not easy, as the constructive version of the classical concept of `UniqueFactorizationDomain` cannot be so propagated. The solution adopted is based largely on Seidenberg's conditions (F) and (P), but there are several additional points that have to be borne in mind to produce reasonably efficient algorithms in the required generality.

The consequence of the algorithms and interfaces presented in this paper is that Scratchpad can factorize in any extension of the integers or finite fields by any combination of polynomial, fraction and algebraic extensions: a capability far more general than any other computer algebra system possesses. The solution is not perfect: for example we cannot use these general constructions to factorize polynomials in $\overline{\mathbb{Z}[\sqrt{-5}]}[x]$ since the domain $\mathbb{Z}[\sqrt{-5}]$ is not a unique factorization domain, even though $\overline{\mathbb{Z}[\sqrt{-5}]}$ is, since it is a field. Of course, we can factor polynomials in $\overline{\mathbb{Z}[\sqrt{-5}]}[x]$,

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keywords = "axiomref",
beebe = "Davenport:1992:SVAb"
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— Davenport:1992:SVAb —

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@TechReport{Davenport:1992:SVAb,
  author = "J. H. Davenport and P. Gianni and B. M. Trager",
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           View of Factorization",
  number = "TR4/92 (ATR/2) (NP2491)",
  institution = inst-NAG,
  address = inst-NAG:adr,
  pages = "??",
  month = dec,
  year = "1992",
  bibdate = "Fri Dec 29 16:31:49 1995",
  bibsource = "http://www.math.utah.edu/pub/tex/bib/axiom.bib",
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  acknowledgement = ack-nhfb,
}
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— axiom.bib —

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@techreport{Dave92a,
  author = "Davenport, James H.",
  title = "The AXIOM system",
  type = "technical report",
  number = "TR5/92 (ATR/3) (NP2492)",
  institution = "Numerical Algorithms Group, Inc.",
  year = "1992",
  abstract =
    "AXIOM is a computer algebra system superficially like many others,
    but fundamentally different in its internal construction, and
    therefore in the possibilities it offers to its users. In these
    lecture notes, we will
    \begin{itemize}
    \item outline the high-level design of the AXIOM kernel and the AXIOM type
    system,
    \item explain some of the algebraic facilities implemented in AXIOM,
    which may be more general than the reader is used to,
    \item show how the type system and the information system interact,
    \item give some references to the literature on particular aspects of
    AXIOM and,
    \item suggest the way forward.
    \end{itemize}",
  paper = "Dave92a.pdf",
  keywords = "axiomref",
  beebe = "Davenport:1992:AS"
}
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— Davenport:1992:AS —

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@TechReport{Davenport:1992:AS,
  author = "J. H. Davenport",
  title = "The {AXIOM} System",
  type = "AXIOM Technical Report",
  number = "TR5/92 (ATR/3) (NP2492)",
  institution = inst-NAG,
  address = inst-NAG:adr,
  pages = "??",
  month = dec,
  year = "1992",
  bibdate = "Fri Dec 29 16:31:49 1995",
  bibsource = "/usr/local/src/bib/bibliography/Theory/Comp.Alg.bib;
  http://www.math.utah.edu/pub/tex/bib/axiom.bib",
  link = "\url{http://www.nag.co.uk/doc/TechRep/axiomtr.html}",
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    acknowledgement = ack-nhfb,
}

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— axiom.bib —

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@techreport{Dave92b,
  author = "Davenport, James H.",
  title = "How does one program in the AXIOM system?",
  institution = "Numerical Algorithms Group, Inc.",
  year = "1992",
  type = "technical report",
  number = "TR6/92 (ATR/4) (NP2493)",
  link = "\url{http://www.nag.co.uk/doc/TechRep/axiomtr.html}",
  abstract =
    "Axiom is a computer algebra system superficially like many others, but
    fundamentally different in its internal construction, and therefore in
    the possibilities it offers to its users and programmers. In these
    lecture notes, we will explain, by example, the methodology that the
    author uses for programming substantial bits of mathematics in Axiom.",
  paper = "Dave92b.pdf",
  keywords = "axiomref",
  beebe = "Davenport:1992:HDO"
}

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— Davenport:1992:HDO —

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@TechReport{Davenport:1992:HDO,
  author = "J. H. Davenport",
  title = "How Does One Program in the {AXIOM} System?",
  type = "AXIOM Technical Report",
  number = "TR6/92 (ATR/4) (NP2493)",
  institution = inst-NAG,
  address = inst-NAG:adr,
  pages = "??",
  month = dec,
  year = "1992",
  bibdate = "Fri Dec 29 16:31:49 1995",
  bibsource = "/usr/local/src/bib/bibliography/Theory/Comp.Alg.bib;
    http://www.math.utah.edu/pub/tex/bib/axiom.bib",
  link = "\url{http://www.nag.co.uk/doc/TechRep/axiomtr.html}",
  acknowledgement = ack-nhfb,
}

```

— ignore —

```
\bibitem[Davenport 92c]{DT92} Davenport, J. H.; Trager, B. M.
  title = "Scratchpad's view of algebra I: Basic commutative algebra",
  DISCO 90 Capri, Italy April 1990 ISBN 0-387-52531-9 pp40-54
  Technical Report TR3/92 (ATR/1)(NP2490), Numerical
  Algorithms Group, Inc., Downer's Grove, IL, USA and Oxford, UK,
  December 1992.
  link = "\url{http://www.nag.co.uk/doc/TechRep/axiomtr.html}",
  keywords = "axiomref"
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— axiom.bib —

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@inproceedings{Dave92,
  author = "Davenport, James H.",
  title = "Primality Testing Revisited",
  link = "\url{http://staff.bath.ac.uk/masjhd/ISSACs/ISSAC1992.pdf}",
  booktitle = "Proc. ISSAC 1992",
  series = "ISSAC 92",
  publisher = "ACM",
  pages = "123-129",
  year = "1992",
  report = "Technical Report TR2/93 Numerical Algorithms Group, Inc",
  algebra = "\newline\refto{package PRIMES IntegerPrimesPackage}",
  abstract =
    "Rabin's algorithm is commonly used in computer algebra systems and
    elsewhere for primality testing. This paper presents an experience
    with this in the Axiom computer algebra system. As a result of this
    experience, we suggest certain strengthenings of the algorithm.",
  paper = "Dave92.pdf",
  keywords = "axiomref",
  beebe = "Davenport:1993:PTR"
}
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— Davenport:1993:PTR —

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@TechReport{Davenport:1993:PTR,
  author = "J. H. Davenport",
  title = "Primality Testing Revisited",
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number =      "TR2/93 (ATR/6) (NP2556)",
institution =  inst-NAG,
address =      inst-NAG:adr,
pages =        "??",
month =        aug,
year =         "1993",
bibdate =      "Fri Dec 29 16:31:49 1995",
bibsource =    "http://www.math.utah.edu/pub/tex/bib/axiom.bib",
link =         "\url{http://www.nag.co.uk/doc/TechRep/axiomtr.html}",
abstract =     "Rabin's algorithm is commonly used in computer algebra
systems and elsewhere for primality testing. This paper
presents an experience with this in the Axiom computer
algebra system. As a result of this experience, we
suggest certain strengthenings of the algorithm.",
acknowledgement = ack-nhfb,
}

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— axiom.bib —

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@misc{Davexx,
  author = {Davenport, James; Faure, Christ'ele},
  title = "The Unknown in Computer Algebra",
  link = "\url{http://axiom-wiki.newsynthesis.org/public/refs/TheUnknownInComputerAlgebra.pdf}",
  abstract = "
    Computer algebra systems have to deal with the confusion between
    'programming variables' and 'mathematical symbols'. We claim that
    they should also deal with 'unknowns', i.e. elements whose values
    are unknown, but whose type is known. For examples  $x^p \neq x$  if  $x$ 
    is a symbol, but  $x^p = x$  if  $x \in GF(p)$ . We show how we have
    extended Axiom to deal with this concept.",
  paper = "Davexx.pdf",
  keywords = "axiomref"
}

```

— ignore —

```

\bibitem[Davenport 00]{Dav00} Davenport, James
  '13th OpenMath Meeting'
  James H. Davenport
    title = "A New Algebra System",
    May 1984
    link = "\url{http://xml.coverpages.org/openmath13.html}",

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paper = "Dav00.pdf",
keywords = "axiomref"
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— axiom.bib —

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@article{Dave11,
  author = "Davenport, James H.",
  title = "CICM 2011: Conferences on Intelligent Computer Mathematics 2011",
  journal = "Springer Lecture Notes in Artificial Intelligence 6824",
  pages = "1-67",
  link = "\url{http://people.bath.ac.uk/masjhd/Meetings/CICM2011.pdf}",
  comment = "http://www.springerlink.com/content/978-3-642-22672-4",
  year = "2011",
  paper = "Dave11.pdf",
  keywords = "axiomref"
}
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@misc{Dave12a,
  author = "Davenport, James H.",
  title = "Computer Algebra or Computer Mathematics?",
  year = "2012",
  link = "\url{http://people.bath.ac.uk/masjhd/Slides/CalculemusSchool2002.pdf}",
  abstract =
    "Scope: ‘‘polynomial-type’’ systems: Axiom, Macsyma/Maxima, Maple,
    Mathematica, and Reduce.",
  paper = "Dave12a.pdf"
}
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— ignore —

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\bibitem[Davenport 12]{Dav12} Davenport, J.H.
  title = "Computer Algebra",
  link = "\url{http://staff.bath.ac.uk/masjhd/JHD-CA.pdf}",
  keywords = "axiomref"}
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— axiom.bib —

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@misc{Dave15,
  author = "Davenport, James H.",
  title = "SIAM AAG 15 and ICIAM 2015",
  link = "\url{http://people.bath.ac.uk/masjhd/Meetings/AAG-ICIAM15.pdf}",
  paper = "Dave15.pdf",
  keywords = "axiomref"
}
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— axiom.bib —

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@misc{Deckxx,
  author = "Decker, Wolfram",
  title = "Some Introductory Remarks on Computer Algebra",
  link = "\url{https://www.math.uni-bielefeld.de/~rehmann/ECM/cdrom/3ecm/pdfs/pant3/decker.pdf}",
  abstract =
    "Computer algebra is a relatively young but rapidly growing field. In
    this introductory note to the mini-symposium on computer algebra
    organized as part of the third European Congress of Mathematics I will
    not even attempt to adress all major streams of research and the many
    applications of computer algebra. I will concentrate on a few aspects,
    mostly from a mathematical point of view, and I will discuss a few
    typical applications in mathematics. I will present a couple of
    examples which underline the fact that computer algebra systems
    provide easy access to powerful computing tools. And, I will quote
    from and refer to a couple of survey papers, textbooks and web-pages
    which I recommend for further reading.",
  paper = "Deckxx.pdf",
  keywords = "axiomref"
}
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— axiom.bib —

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@phdthesis{Dell99,
  author = "Delliere, Stephane",
  title = {Trangularisation de syst\`emes constructibles Application \`a
    l'\`evaluation dynamique},
  school = {L'Universit\`e de Limoges},
}
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year = "1999",
link = "\url{http://www.unilim.fr/laco/theses/1999/T1999_03.pdf}",
paper = "Dell99.pdf",
keywords = "axiomref"
}

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— axiom.bib —

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@techreport{Dell00,
  author = "Delliere, Stephane and Wang, Dongming",
  title = "simple systems and dynamic constructible closure",
  institution = "Universite de Limoges",
  year = "2000",
  type = "technical report",
  number = "2000-16",
  link = "\url{http://www.unilim.fr/laco/rapports/2000/R2000\_16.pdf}",
  abstract =
    "Dynamic evaluation is a general method for computing with parameters
    [6, 9]. In 1994, T. Gomez-Diaz implemented the dynamic constructible
    closure in the scientific computation system Axiom [17]: by simulating
    dynamic evaluation, it offers the possibility to compute with
    parameters in a very large way [13]. The outputs of a calculus with
    T. Gomez-Diaz programs are represented by a finite collection of
    constructible triangular systems defined in [12, definition
    p.106]. Though there are numerous applications of these programs
    (notably polynomial system solving with parameters [11], automatic
    geometric theorem proving [14, 15], computation of Jordan forms with
    parameters [16]), nobody gives theoretical interest to this kind of
    triangular systems. The main reason of this phenomenon is that they
    are defined in [12] within the dynamic evaluation context. On the
    opposite, most notions of triangular systems (J.F. Ritt-W.T. Wu
    characteristic sets [24, 28], M. Kalkbrener regular chains [18],
    D. Lazard triangular sets [20], M. Moreno Maza regular sets [22],
    D.M. Wang simple systems [26, 27]) are defined in terms of commutative
    algebra. This problem is at the origin of the work done in [7] where
    we give a relevant algebraic model of T. Gomez-Diaz systems within
    commutative algebra terminology. This allows us to relate them to many
    concepts of triangular systems [7]. Thus, we give interest to the
    connections with D. Lazard triangular sets in [8]. In a way, this
    paper is the continuation of this previous work. This time, we study
    relationships between T. Gomez-Diaz systems and D.M. Wang simple
    systems. The paper is structured as follows. We have collected in
    section 2 some needed notations. In section 3, we give all the
    terminology related to our algebraic model of T. Gomez-Diaz
    systems. Thus, we define the notion of weak constructible triangular
    systems and introduce the properties of normalization and

```

squarefreeness. Section 4 is more detailed. First of all, we study a weaker form of normalization called $\$L\$$ -normalization. Then we give many properties of constructible triangular systems verifying this new notion. We obtain an algebraic and geometric framework which permits, in section 5, to explore the connections between T. Gomez-Diaz systems and D.M. Wang simple systems. In particular, this last section will demonstrate well the importance of our $\$L\$$ -normalization property. Indeed, we show that simple systems and squarefree $\$L\$$ -normalized constructible triangular systems are equivalent.",
 paper = "Dell00.pdf",
 keywords = "axiomref"
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@techreport{Dell00a,
  author = "Delliere, Stephane",
  title = {A first course to  $\$D_7\$$  with examples},
  institution = "Universite de Limoges",
  year = "2000",
  type = "technical report",
  number = "2000-17",
  link = "\url{http://www.unilim.fr/laco/rapports/2000/R2000_17.pdf}",
  paper = "Dell00a.pdf",
  keywords = "axiomref"
}
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@article{Dell01,
  author = "Delliere, Stephane",
  title = "On the links between triangular sets and dynamic constructable closure",
  journal = "J. Pure Appl. Algebra",
  volume = "163",
  number = "1",
  pages = "49-68",
  year = "2001",
  abstract =
    "Two kinds of triangular systems are studied: normalized triangular polynomial systems (a weaker form of Lazards triangular sets [D. Lazard, Discrete Appl. Math. 33, No. 1-3, 147-160 (1991); Zbl
```

0753.13013)] and constructible triangular systems (involved in the dynamic constructible closure programs of T. Gomez-Daz [Quelques applications de l'évaluation dynamique, Ph.D. Thesis, Université de Limoges (1994)]. This paper shows that these notions are strongly related. In particular, combining the two points of view (constructible and polynomial) on the subject of square-free conditions, it allows us to effect dramatic improvements in the dynamic constructible closure programs.",
 keywords = "axiomref"
 }

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@InProceedings{Dewa92,
  author = "Dewar, Michael C.",
  title = "Using Computer Algebra to Select Numerical Algorithms",
  booktitle = "Proc. ISSAC 1992",
  series = "ISSAC 1992",
  year = "1992",
  location = "Berkeley, CA",
  pages = "1-8",
  algebra =
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\newline\refto{domain NIPROB NumericalIntegrationProblem}
\newline\refto{domain ODEPROB NumericalODEProblem}
\newline\refto{domain OPTPROB NumericalOptimizationProblem}
\newline\refto{domain PDEPROB NumericalPDEProblem}",
abstract =
  "Many real-life problems require a combination of both symbolic and
  numerical methods for their solution. This has led to the development
  of integrated, interactive symbolic / numeric packages which use a
  computer algebra system for the former and a standard subroutine
  library for the latter. These systems may also be viewed as simplified
  front-ends to the numerical library. To use these packages, however, a
  user must be able to select which of the many available routines is
  the most appropriate for his or her problem, which contrasts with the
  'black-box' style interfaces available in computer algebra
  systems. This paper describes how a computer algebra system can be
  used to make this decision, thus providing a much-simplified and more
  orthogonal interface.",
paper = "Dewa92.pdf"
}

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\bibitem[Dewar 94]{Dew94} Dewar, M. C.
  title = "Manipulating Fortran Code in AXIOM and the AXIOM-NAG Link",
  Proceedings of the Workshop on Symbolic and Numeric Computing, ed by Apiola, H.
  and Laine, M. and Valkeila, E. pp1-12 University of Helsinki, Finland (1994)
  keywords = "axiomref",

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@misc{Dewa95,
  author = "Dewar, Mike C.",
  title = "AXIOM and A\#: Current Status and Future Plans",
  year = "1995",
  link = "\url{ftp://ftp.inf.ethz.ch/org/cathode/workshops/jan95/abstracts/dewar.ps}",
  paper = "Dewa95.pdf",
  keywords = "axiomref"
}

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@misc{Dewa,
  author = "Dewar, Mike",
  title = "OpenMath: An Overview",
  link = "\url{http://www.sigsam.org/bulletin/articles/132/paper1.pdf}",
  paper = "Dewa.pdf",
  keywords = "axiomref"
}
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@phdthesis{Diaz06,
  author = "Diaz, Glauco Alfredo Lopez",
  title = "Symbolic Methods for Factoring Linear Differential Operators",
  school = "Johannes Kepler Universitat, Linz",
  year = "2006",
  month = "February",
  abstract =
    "A survey of symbolic methods for factoring linear differential
    operators is given. Starting from basic notions ring of operators,
    differential Galois theory methods for finding rational and
    exponential solutions that can provide first order right-hand factors
    are considered. Subsequently several known algorithms for
    factorization are presented. These include Singers eigenring
    factorization algorithm, factorization via Newton polygons, van
    Hoeijs methods for local factorization, and an adapted version of
    Pade approximation.
```

In addition a procedure based on pure algebraic methods for factoring second order linear partial differential operators is developed. Splitting an operator of this kind reduces to solving a system of linear algebraic equations. Those solutions which satisfy a certain differential condition, immediately produce linear factors of the operator. The method applies also to operators of third order, thereby resulting in a more complicated system of equations. In contrast to the second order case, differential equations must also be solved, which, in particular cases, are simplified with the aid of characteristic sets.

Finally, complete decomposition into linear factors of ordinary differential operators of arbitrary order is discussed. A splitting formula is developed, provided that a linear basis of solutions is available. This theoretical representation is valuable in understanding the nature of the classical Beke algorithm and its

```

    variants like the algorithm LODEF by Schwarz and the Beke-Bronstein
    algorithm.",
    paper = "Diaz06.pdf",
    keywords = "axiomref"
}

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@article{DiBl95,
  author = "DiBlasio, Paolo and Temperini, Marco",
  title = "Subtyping Inheritance and Its Application in Languages for
    Symbolic Computation Systems",
  journal = "J. Symbolic Computation",
  volume = "19",
  pages = "39-63",
  year = "1995",
  abstract =
    "Application of object-oriented programming techniques to design and
    implementation of symbolic computation is investigated. We show the
    significance of certain correctness problems, occurring in programming
    environments based on specialization inheritance, due to use of method
    redefinition and polymorphism. We propose a solution to these
    problems, by defining a mechanism of subtyping inheritance and the
    prototype of an object-oriented programming language for a symbolic
    computation system. We devise the subtyping inheritance {\sl ESI}
    (Enhanced String Inheritance)} by lifting to programming language
    constructs a given model of subtyping, which is established by a
    monotonic (covariant) subtyping rule. Type safeness of language
    instructions is proved.

    The adoption of {\sl ESI} allows to model method and class
    specialization in a natural way. The {\sl ESI} mechanism verifies the
    type correctness of language statements by means of type checking
    rules and preserves their correctness at run-time by a suitable method
    lookup algorithm.",
  paper = "DiBl95.pdf",
  keywords = "axiomref"
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@InProceedings{DiBl97,

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author = "DiBlasio, Paolo and Temperini, Marco",
title = "On subtyping in languages for symbolic computation systems",
booktitle = "Advances in the design of symbolic computation systems",
series = "Monographs in Symbolic Computation",
year = "1997",
publisher = "Springer",
pages = "164-178",
abstract =
  "We want to define a strongly typed OOP language suitable as the
  software development tool of a symbolic computation system, which
  provides class structure to manage ADTs and supports multiple
  inheritance to model specialization hierarchies. In this paper, we
  provide the theoretical background for such a task.",
keywords = "axiomref"
}

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@InProceedings{Dicr88,
  author = "Dicrescenzo, C. and Duval, D.",
  title = "Algebraic extensions and algebraic closure in Scratchpad II",
  booktitle = "Proc. ISSAC 1988",
  series = "ISSAC 1998",
  year = "1998",
  pages = "440-446",
  isbn = "3-540-51084-2",
  abstract =
    "Many problems in computer algebra, as well as in high-school
    exercises, are such that their statement only involves integers but
    their solution involves complex numbers. For example, the complex
    numbers  $\sqrt{2}$  and  $-\sqrt{2}$  appear in the solutions of
    elementary problems in various domains.
    \begin{itemize}
    \item in {\bf integration}:
    \[ \int \frac{dx}{x^2-2} = \frac{\text{Log}(x-\sqrt{2})}{2\sqrt{2}}
    + \frac{\text{Log}(x-(-\sqrt{2}))}{2(-\sqrt{2})} \]
    \item in {\bf linear algebra}: the eigenvalues of the matrix
    \[ \left( \begin{array}{cc} 1 & 1 \\ 1 & -1 \end{array} \right) = \sqrt{2} \text{ } \text{rm} \text{ and } -\sqrt{2} \]
    \item in {\bf geometry}: the line  $y=x$  intersects the circle
     $y^2+x^2=1$  at the points
    \[ (\sqrt{2}, \sqrt{2}) \text{ } \text{rm} \text{ and } (-\sqrt{2}, -\sqrt{2}) \]
    \end{itemize}
    Of course, more ‘‘complicated’’ complex numbers appear in more

```


complicated examples.

But two facts have to be emphasized:

```
\begin{itemize}
\item in general, if a problem is stated over the integers (or over
the field  $\mathbb{Q}$  of rational numbers), the complex numbers that
appear are {\sl algebraic} complex numbers, which means that they are
roots of some polynomial with rational coefficients, like  $\sqrt{2}$ 
and  $-\sqrt{2}$  are roots of  $T^2-2$ .
\item Similar problems appear with base fields different from
 $\mathbb{Q}$ . For example finite fields, or fields of rational
functions over  $\mathbb{Q}$  or over a finite field. The general
situation is that a given problem is stated over some ‘‘small field’’
 $K$ , and its solution is expressed in an {\sl algebraic closure}
 $\overline{K}$  of  $K$ , which means that this solution involves numbers
which are roots of polynomials with coefficients in  $K$ .
\end{itemize}
```

The aim of this paper is to describe an implementation of an algebraic closure domain constructor in the language Scratchpad II, simply called Scratchpad below. In the first part we analyze the problem, and in the second part we describe a solution based on the D5 system.",

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keywords = "axiomref",
beebe = "Dicrescenzo:1989:AEA"
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— Dicrescenzo:1989:AEA —

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@InProceedings{Dicrescenzo:1989:AEA,
  author = "C. Dicrescenzo and D. Duval",
  title = "Algebraic extensions and algebraic closure in
{Scratchpad} {II}",
  crossref = "Gianni:1989:SAC",
  pages = "440--446",
  year = "1989",
  bibdate = "Tue Sep 17 06:46:18 MDT 1996",
  bibsource = "/usr/local/src/bib/bibliography/Theory/Comp.Alg.bib;
http://www.math.utah.edu/pub/tex/bib/axiom.bib",
  abstract = "Many problems in computer algebra, as well as in
high-school exercises, are such that their statement
only involves integers but their solution involves
complex numbers. For example, the complex numbers  $\sqrt{2}$ 
and  $-\sqrt{2}$  appear in the solutions of
elementary problems in various domains. The authors
describe an implementation of an algebraic closure
domain constructor in the language Scratchpad II. In
```

```

        the first part they analyze the problem, and in the
        second part they describe a solution based on the D5
        system.",
    acknowledgement = ack-nhfb,
    affiliation = "TIM3, INPG, Grenoble, France",
    classification = "C7310 (Mathematics)",
    keywords = "Algebraic closure domain constructor; D5 system;
        Language Scratchpad II",
    language = "English",
    thesaurus = "Mathematics computing; Symbol manipulation",
}

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@misc{Dicr95,
  author = "Dicrescenzo, C. and Jung, Francoise",
  title = "COMPASS package",
  year = "1995",
  link = "\url{ftp://ftp.inf.ethz.ch/org/cathode/workshops/jan95/abstracts/bronstein.ps}",
  paper = "Dicr95.pdf",
  keywords = "axiomref"
}

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— axiom.bib —

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@book{Dicr05,
  author = "Dicrescenzo, C. and Duval, D.",
  title = "Algebraic extensions and algebraic closure in Scratchpad II",
  booktitle = "Symbolic and Algebraic Computation",
  series = "Lecture Notes in Computer Science 358",
  year = "2005",
  publisher = "Springer",
  pages = "440-446",
  keywords = "axiomref"
}

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```

@InProceedings{Ding94,
  author = "Dingle, Adam and Fateman, Richard",
  title = "Branch Cuts in Computer Algebra",
  year = "1994",
  booktitle = "Proc. ISSAC 1994",
  series = "ISSAC 94",
  link = "\url{http://www.cs.berkeley.edu/~fateman/papers/ding.ps}",
  abstract =
    "Many standard functions, such as the logarithms and square root
    functions, cannot be defined continuously on the complex
    plane. Mistaken assumptions about the properties of these functions
    lead computer algebra systems into various conundrums. We discuss how
    they can manipulate such functions in a useful fashion.",
  paper = "Ding94.pdf",
  keywords = "axiomref"
}

```

— ignore —

```

\bibitem[DLMF]{DLMF}.
  title = "Digital Library of Mathematical Functions",
  link = "\url{http://dlmf.nist.gov/software/#T1}",
  keywords = "axiomref"

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@InProceedings{Dool98,
  author = "Dooley, Samuel S.",
  title = "Coordinating mathematical content and presentation markup in
    interactive mathematical documents",
  booktitle = "Proc. ISSAC 1998",
  series = "ISSAC 98",
  year = "1998",
  publisher = "ACM Press",
  location = "Rostock, Germany",
  pages = "13-15",
  abstract =
    "This paper presents a method for representing mathematical content
    and presentation markup in interactive mathematical documents that
    treats each view of the information on a separate and equal
    footing. By providing extensible, overridable, default mappings from
    content to presentation in a way that supports efficient mappings back

```

from the presentation to the underlying content, a user interface for an interactive textbook has been implemented where the user interacts with high-quality presentation markup that supports user operations defined in terms of the mathematical content. In addition, the user interface can be insulated from content-specific information, while still being enabled to transfer that information to other programs for computation. This method has been employed to embed interactive mathematical content into the IBM techexplorer Interactive Textbook for Linear Algebra. The issues involved in the implementation of the interactive textbook also shed some light on the problems faced by the MathML working group in representing both presentation and content for mathematics for interactive web documents.",

```
keywords = "axiomref"
}
```

— ignore —

```
\bibitem[Dooley 99]{Doo99} Dooley, Sam editor.
ISSAC 99: July 29-31, 1999, Simon Fraser University,
Vancouver, BC, Canada: proceedings of the 1999 International Symposium on
Symbolic and Algebraic Computation. ACM Press, New York, NY 10036, USA, 1999.
ISBN 1-58113-073-2 LCCN QA76.95.I57 1999
keywords = "axiomref"
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— axiom.bib —

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@article{Reis11,
  author = "Dos Reis, Gabriel",
  title = "Retargeting OpenAxiom to Poly/ML: towards an integrated proof
          assistants and computer algebra system framework",
  journal = "Intelligent computer mathematics (MKM 2011)",
  year = "2011",
  isbn = "978-3-642-22672-4",
  link = "\url{https://www.semanticscholar.org/paper/Retargeting-OpenAxiom-to-PolyML-Towards-an-Reis-Matthe",
  abstract =
    "This paper presents an ongoing effort to integrate the AXIOM family
    of computer algebra systems with Poly/ML-based proof assistants in the
    same framework. A long-term goal is to make a large set of efficient
    implementations of algebraic algorithms available to popular proof
    assistants, and also to bring the power of mechanized formal
    verification to a family of strongly typed computer algebra systems at
    a modest cost. Our approach is based on retargeting the code generator"
```

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    of the OpenAxiom compiler to the Poly/ML abstract machine.",
    paper = "Reis11.pdf",
    keywords = "axiomref"
}

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— axiom.bib —

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@article{Reis06,
  author = "Dos Reis, Gabriel and Stroustrup, Bjarne",
  title = "Specifying C++ Concepts",
  journal = "POPL",
  publisher = "ACM",
  year = "2006",
  link = "\url{http://www.stroustrup.com/pop106.pdf}",
  abstract =
    "C++ templates are key to the design of current successful mainstream
    libraries and systems. They are the basis of programming techniques in
    diverse areas ranging from conventional general-purpose programming to
    software for safety-critical embedded systems. Current work on improving
    templates focuses on the notion of {\sl concepts} (a type system for
    templates), which promises significantly improved error diagnostics and
    increased expressive power such as concept-based overloading and function
    template partial specialization. This paper presents C++ templates with
    an emphasis on problems related to separate compilation. We consider the
    problem of how to express concepts in a precise way that is simple enough
    to be usable by ordinary programmers. In doing so, we expose a few
    weaknesses of the current specification of the C++ standard library and
    suggest a far more precise and complete specification. We also present
    a systematic way of translating our proposed concept definitions, based
    on use-patterns rather than function signatures, into constraint sets
    that can serve as convenient basis for concept checking in a compiler.",
  paper = "Reis06.pdf",
  keywords = "axiomref"
}

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— axiom.bib —

```

@article{Reis12,
  author = "Dos Reis, Gabriel",
  title = "A System for Axiomatic Programming",
  journal = "Proc. Conf. on Intelligent Computer Mathematics",
  publisher = "Springer",

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year = "2012",
link = "\url{http://www.axiomatics.org/~gdr/liz/cicm-2012.pdf}",
abstract = "
    We present the design and implementation of a system for axiomatic
    programming, and its application to mathematical software
    construction. Key novelties include a direct support for user-defined
    axioms establishing local equality between types, and overload
    resolution based on equational theories and user-defined local
    axioms. We illustrate uses of axioms, and their organization into
    concepts, in structured generic programming as practiced in
    computational mathematical systems.",
paper = "Reis12.pdf",
keywords = "axiomref"
}

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— axiom.bib —

```

@phdthesis{Doye97,
  author = "Doye, Nicolas James",
  title = "Order Sorted Computer Algebra and Coercions",
  school = "University of Bath",
  year = "1997",
  abstract =
    "Computer algebra systems are large collections of routines for solving
    mathematical problems algorithmically, efficiently and above all,
    symbolically. The more advanced and rigorous computer algebra systems
    (for example, Axiom) use the concept of strong types based on
    order-sorted algebra and category theory to ensure that operations are
    only applied to expressions when they ‘‘make sense’’."

```

In cases where Axiom uses notions which are not covered by current mathematics we shall present new mathematics which will allow us to prove that all such cases are reducible to cases covered by the current theory. On the other hand, we shall also point out all the cases where Axiom deviates undesirably from the mathematical ideal. Furthermore we shall propose solutions to these deviations.

Strongly typed systems (especially of mathematics) become unusable unless the system can change the type in a way a user expects. We wish any change expected by a user to be automated, ‘‘natural’’, and unique. ‘‘Coercions’’ are normally viewed as ‘‘natural type changing maps’’. This thesis shall rigorously define the word ‘‘coercion’’ in the context of computer algebra systems.

We shall list some assumptions so that we may prove new results so that all coercions are unique. This concept is called ‘‘coherence’’.

```

    We shall give an algorithm for automatically creating all coercions in
    type system which adheres to a set of assumptions. We shall prove that
    this is an algorithm and that it always returns a coercion when one
    exists. Finally, we present a demonstration implementation of this
    automated coercion algorithm in Axiom.",
    paper = "Doye97.pdf",
    keywords = "axiomref"
}

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— axiom.bib —

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@inproceedings{Doye99,
  author = "Doye, Nicolas James",
  title = "Automated coercion for Axiom",
  booktitle = "Proc. ISSAC 1999",
  pages = "229-235",
  year = "1999",
  isbn = "1-58113-073-2",
  link = "\url{http://www.acm.org/citation.cfm?id=309944}",
  keywords = "axiomref",
  beebe = "Doye:1999:ACA"
}

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— Doye:1999:ACA —

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@InProceedings{Doye:1999:ACA,
  author = "Nicolas J. Doye",
  title = "Automated coercion for {Axiom}",
  crossref = "Dooley:1999:IJS",
  pages = "229--235",
  year = "1999",
  bibdate = "Sat Mar 11 16:39:42 MST 2000",
  bibsource = "http://www.math.utah.edu/pub/tex/bib/axiom.bib",
  link = "\url{http://www.acm.org/pubs/contents/proceedings/issac/309831/}",
  acknowledgement = ack-nhfb,
}

```

— axiom.bib —

```

@InProceedings{Domi01,
  author = {Dom\`inguez, C\`esar; Rubio, Julio},
  title = "Modeling Inheritance as Coercion in a Symbolic Computation System",
  booktitle = "Proc. ISSAC 2001",
  series = "ISSAC 2001",
  year = "2001",
  abstract = "
    In this paper the analysis of the data structures used in a symbolic
    computation system, called Kenzo, is undertaken. We deal with the
    specification of the inheritance relationship since Kenzo is an
    object-oriented system, written in CLOS, the Common Lisp Object
    System. We focus on a particular case, namely the relationship between
    simplicial sets and chain complexes, showing how the order-sorted
    algebraic specifications formalisms can be adapted, through the
    'inheritance as coercion' metaphor, in order to model this Kenzo
    fragment.",
  paper = "Domi01.pdf",
  keywords = "axiomref"
}

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— axiom.bib —

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@InProceedings{Drag10,
  author = "Dragan, Laurentiu and Watt, Stephen",
  title = "Type Specialization in Aldor",
  booktitle = "Computer algebra in scientific computing",
  series = "CASC 2010",
  year = "2010",
  location = "Tsakhadzor, Armenia",
  pages = "73-84",
  link = "\url{http://www.csd.uwo.ca/~watt/pub/reprints/2010-casc-specdom.pdf}",
  abstract =
    "Computer algebra in scientific computation squarely faces the dilemma
    of natural mathematical expression versus efficiency. While
    higher-order programming constructs and parametric polymorphism
    provide a natural and expressive language for mathematical
    abstractions, they can come at a considerable cost. We investigate how
    deeply nested type constructions may be optimized to achieve
    performance similar to that of hand-tuned code written in lower-level
    languages.",
  paper = "Drag10.pdf",
  keywords = "axiomref"
}

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— ignore —

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\bibitem[Dunstan 97]{Dun97} Dunstan, Martin and Martin, Ursula and
  Linton, Steve
  title = "Embedded Verification Techniques for Computer Algebra Systems",
  Grant citation GR/L48256 Nov 1, 1997-Feb 28, 2001
  link = "\url{http://www.cs.st-andrews.ac.uk/research/output/detail?output=ML97.php}",
  keywords = "axiomref"
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— axiom.bib —

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@misc{Duns99b,
  author = "Dunstan, Martin",
  title = "An Introduction to Aldor and its Type System",
  year = "1999",
  link = "\url{http://www.alдор.org/docs/reports/cfc99/aldor-cfc99.pdf}",
  comment = "slides",
  paper = "Duns99b.pdf"
}
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@misc{Dupe95,
  author = "Dupee, Brian J. and Davenport, James H.",
  title = "Using Computer Algebra to Choose and Apply Numerical Routines",
  year = "1995",
  link = "\url{http://citeseerx.ist.psu.edu/viewdoc/download?doi=10.1.1.33.5645}",
  algebra =
    "\newline\ref{domain D01AJFA d01ajfAnnaType}
    \newline\ref{domain D01AKFA d01akfAnnaType}
    \newline\ref{domain D01ALFA d01alfAnnaType}
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    \newline\ref{domain D01FCFA d01fcfAnnaType}
    \newline\ref{domain D01GBFA d01gbfAnnaType}
    \newline\ref{domain D01TRNS d01TransformFunctionType}
    \newline\ref{domain D02BBFA d02bbfAnnaType}"
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\newline\refto{domain E04NAFA e04nafAnnaType}
\newline\refto{domain E04UCFA e04ucfAnnaType}
\newline\refto{domain NIPROB NumericalIntegrationProblem}
\newline\refto{domain ODEPROB NumericalODEProblem}
\newline\refto{domain OPTPROB NumericalOptimizationProblem}
\newline\refto{domain PDEPROB NumericalPDEProblem}",
abstract =
  "In applied mathematics, electronic and chemical engineering, the
  modelling process can produce a number of mathematical problems which
  require numerical solutions for which symbolic methods are either not
  possible or not obvious. With the plethora of numerical library
  routines for the solution of these problems often the numerical
  analyst has to answer the question {\sl Which routine to choose?} and
  {\sl How do I use it?}. Some analysis needs to be carried out before
  the appropriate routine can be identified, i.e. {\sl How stiff is this
  ODE?} and {\sl Is this function continuous?}. It may well be the case
  that more than one routine is applicable to the problem. So the
  question may become {\sl Which is likely to be the best?}. Such a
  choice may be critical for both accuracy and efficiency.

  An expert system is thus required to make this choice based on the
  results of its own analysis of the problem, call the routine and act
  on the outcome. This may be to put the answer in a relevant form or
  react to an apparent failure of the chosen routine and thus choose and
  call an alternative. It should also have sufficient explanation
  mechanisms to inform on the choice of routine and the reasons for that
  choice. Much of this work can be achieved using computer algebra and
  symbolic algebra packages.

  This paper describes an expert system currently in prototype in terms
  of both its object-based structure and its computational agents. Some
  of these agents are described in detail, paying particular attention
  to the practical aspects of their algorithms and the use of computer
  algebra.

  The {\bf axiom2} Symbolic Algebra System is used as a user interface
  as well as the link to the NAG Foundation Library for the numerical
  routines and the inference mechanisms for the expert system.",
paper = "Dupe95.pdf",
keywords = "axiomref"

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}

— axiom.bib —

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@inproceedings{Dupe99,
  author = "Dupee, Brian J. and Davenport, James H.",
  title = "An Automatic Symbolic-Numeric Taylor Series ODE Solver",
  booktitle = "Computer Algebra in Scientific Computing, CASC'99",
  isbn = "978-3-540-66047-7",
  pages = "37-50",
  year = "1999",
  link = "\url{http://people.eecs.berkeley.edu/~fateman/papers/casc99-34.pdf}",
  comment = "Contains FORTRAN Code of Taylor Series",
  algebra = "\newline\refto{package EXPRODE ExpressionSpaceODESolver}",
  abstract =
    "One of the basic techniques in every mathematician's toolkit is the
    Taylor series representation of functions. It is of such fundamental
    importance and it is so well understood that its use is often a first
    choice in numerical analysis. This faith has not, unfortunately, been
    transferred to the design of computer algorithms.

    Approximation by use of Taylor series methods is inherently partly a
    symbolic process and partly numeric> This aspect has often, with
    reason, been regared as a major hindrance in algorithm design. Whilst
    attempts have been made in the past to build a consistent set of
    programs for the symbolic and numeric paradigms, these have been
    necessarily multi-stage processes.

    Using current technology it has at last become possible to integrate
    these two concepts and build an automatic adaptive symbolic-numeric
    algorithm within a uniform framework which can hide the internal
    workings behind a modern interface.",
  paper = "Dupe99.pdf",
  keywords = "axiomref"
}
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```
@inproceedings{Dupe05,
  author = "Dupee, Brian J. and Davenport, James H.",
  title = "An Intelligent Interface to Numerical Routines",
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booktitle = "Design and Implementation of Symbolic Computation Systems",
series = "Lecture Notes in Computer Science 1128",
pages = "252-262",
publisher = "Springer",
year = "2005",
abstract =
  "Links from Computer Algebra Systems to Numerical Libraries have been
  increasingly made available. However, the remain, like the numerical
  routines which comprise those libraries, difficult to use by a novice
  and there is little help in choosing the appropriate routine for any
  given problem, should there be a choice.
```

Computer Algebra Systems use generic names for each problem area. For examples, 'integrate' (or 'int') is used for integration of a function, whatever method the code may use. Numeric interfaces still use different names for each method together with a variety of extra parameters, some of which may be optional. Ideally, we should extend the generic name structure to cover numerical routines. This would then, necessarily, require algorithms for making an assessment of the efficacy of different methods where such a choice exists.

This paper considers the link to the NAG Fortran Library from version 2.0 of Axiom and shows how we can build on this to extend and simplify the interface using an expert system for choosing and using the numerical routines."

```
}
```

— ignore —

```
\bibitem[Adams 01]{DGKM01} Adams, Andrew; Dunstan, Martin; Gottlieb, Hanne;
Kelsey, Tom; Martin, Ursula; Owre, Sam
  title = "Computer Algebra meets Automated Theorem Proving: Integrating Maple and PVS",
TPHOLS 2001, Edinburgh
  link = "\url{http://www.csl.sri.com/~owre/papers/tphols01/tphols01.pdf}",
  abstract = "
    We describe an interface between version 6 of the Maple computer
    algebra system with the PVS automated theorem prover. The interface is
    designed to allow Maple users access to the robust and checkable proof
    environment of PVS. We also extend this environment by the provision
    of a library of proof strategies for use in real analysis. We
    demonstrate examples using the interface and the real analysis
    library. These examples provide proofs which are both illustrative and
    applicable to genuine symbolic computation problems.",
  paper = "DGKM01.pdf",
  keywords = "axiomref"
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```
@article{Duva92,
  author = "Duval, Anne and Loday-Richaud, Michele",
  title = "Kovacic's Algorithm and Its Application to Some Families
    of Special Functions",
  journal = "Applicable Algebra in Engineering, Communication, and Computing",
  series = "AAECC 3",
  pages = "211-246",
  year = "1992",
  publisher = "Springer-Verlag",
  abstract =
    "We apply the Kovacic algorithm to some families of special functions,
    mainly the hypergeometric one and that of Heun, in order to discuss
    the existence of closed-form solutions. We begin by giving a slightly
    modified version of the kovacic algorithm and a sketch proof.",
  keywords = "axiomref"
}
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— axiom.bib —

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@inproceedings{Duva92a,
  author = "Duval, Dominique and Jung, F.",
  title = "Examples of problem solving using computer algebra",
  booktitle = "Programming environments for high-level scientific problem
    solving",
  series = "IFIP Transactions",
  editor = "Gaffney, Patrick W. and Houstis, Elias N.",
  publisher = "North-Holland",
  pages = "133-143",
  year = "1992",
  keywords = "axiomref",
  beebe = "Duval:1992:EPS"
}
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— Duval:1992:EPS —

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@Article{Duval:1992:EPS,
  author = "D. Duval and F. Jung",
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title =      "Examples of problem solving using computer algebra",
journal =    j-IFIP-TRANS-A,
volume =     "A-2",
pages =      "133--141, 143",
month =      "",
year =       "1992",
CODEN =      "ITATEC",
ISSN =       "0926-5473",
bibdate =    "Tue Sep 17 06:41:20 MDT 1996",
bibsource =  "http://www.math.utah.edu/pub/tex/bib/axiom.bib",
abstract =   "Computer algebra, in contrast with numerical analysis,
              aims at returning exact solutions to given problems.
              One consequence is that the shape of the solutions may,
              at first, look somewhat surprising. The authors present
              two examples of problem solving using computer algebra,
              with emphasis on the shape of the solutions. The first
              example is the resolution of linear differential
              equations with polynomial coefficients, and the second
              one is the resolution of polynomial equations in one
              variable. In the first example the solution may look
              useless since it makes use of divergent series, and in
              the second example the solution may look rather
              awkward. But in both examples it is shown that these
              solutions are in the right shape for a lot of
              applications, including numerical ones. It is also
              shown that some features of the computer algebra system
              Scratchpad, especially strong typing and genericity,
              are useful for the implementation of a method for a
              second problem, i.e. for the implementation of the
              'dynamic' algebraic closure of a field.",
acknowledgement = ack-nhfb,
affiliation = "Lab. de Theorie des Nombres et Algorithmique, Limoges
              Univ., France",
classification = "C7310 (Mathematics)",
fjournal =    "IFIP Transactions. A. Computer Science and
              Technology",
keywords =    "Algebraic closure; Computer algebra; Divergent series;
              Exact solutions; Genericity; Linear differential
              equations; Polynomial coefficients; Polynomial
              equations; Problem solving; Scratchpad; Strong typing",
language =    "English",
thesaurus =   "Linear differential equations; Polynomials; Symbol
              manipulation",
}

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@misc{Duva94e,
  author = "Duval, Dominique",
  title = "Symbolic or algebraic computation?",
  booktitle = "Publication du LACO",
  year = "1995",
  location = "Madrid Spain",
  comment = "NAG conference",
  keywords = "axiomref"
}

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— axiom.bib —

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@article{Duva94d,
  author = "Duval, Dominique and Senechaud, Pascale",
  title = "Sketches and parametrization",
  journal = "Theor. Comput. Sci.",
  volume = "123",
  number = "1",
  pages = "117-130",
  year = "1994",
  abstract =
    "The paper deals with problems about conception and design of
    high-level computer algebra systems. Here we use a categorical
    approach given by the notion of sketches. Sketches allow to describe
    computation mechanisms in a syntactic way, well adapted to
    implementation.

    A computer algebra system must allow the manipulation of algebraic
    structures, in particular, the construction of new structures from
    known ones. In the paper we give a definition, at the sketch level, of
    parametrization of a structure by another one.",
  keywords = "axiomref"
}

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— axiom.bib —

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@article{Duva95,
  author = "Duval, Dominique",
  title = "Dynamic evaluation and algebraic closure in Axiom",
  comment = "Evaluation dynamique et cl\`oture alg\`ebrique en Axiom",
  journal = "Journal of Pure and Applied Algebra",
  volume = "99",

```

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year = "1995",
pages = "267--295",
abstract =
  "Dynamic evaluation allows to compute with algebraic numbers without
  factorizing polynomials. It also allows to manipulate parameters in a
  flexible and user-friendly way. The aim of this paper is the
  following: Explain what is dynamic evaluation, with its basic notions
  of dynamic set and splitting. Present its application to computations
  involving algebraic numbers, which amounts to defining the dynamic
  algebraic closure of a field. Describe the Axiom program which
  implements this, and give a user guide for it (only this last point
  assumes some knowledge of Axiom) Dynamic evaluation is described here
  without any reference to sketch theory, however our presentation, less
  rigorous, may be considered as more accessible.",
paper = "Duva95.pdf",
keywords = "axiomref",
beebe = "Duval:1995:DEA"
}

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— Duval:1995:DEA —

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@Article{Duval:1995:DEA,
  author =      "Dominique Duval",
  title =      "{{\'}E}}valuation dynamique et cl{\^o}ture
                alg{\'}e}brique en {Axiom}. (French) {Dynamic evaluation
                and algebraic closure in Axiom}",
  journal =    "J. Pure Appl. Algebra",
  volume =     "99",
  number =     "3",
  pages =      "267--295",
  year =       "1995",
  DOI =        "http://dx.doi.org/10.1016/0022-4049(94)00053-L",
  ISSN =       "0022-4049",
  MRclass =    "11Y40 (Algebraic number theory computations) 68W30
                (Symbolic computation and algebraic computation) 68Q65
                (Abstract data types; algebraic specification) 18C10
                (Algebraic theories, etc.)",
  bibdate =    "Tue Mar 30 18:47:12 2010",
  bibsource =  "http://www.math.utah.edu/pub/tex/bib/axiom.bib",
  ZMnumber =   "0851.1107",
  abstract =    "Dynamic evaluation is a method of computing that
                permits the computation to be refined into different
                cases that are considered separately. A precise
                description and mathematical foundation was given in
                terms of sketch theory by {\it D. Duval} and {\it J.-C.
                Reynaud} [Math. Structures Comput. Sci. 4, 239-271

```


(1994; Zbl 0822.68063)]. In the present paper, the mechanism of dynamic evaluation is explained without reference to sketch theory in order to make it accessible to a wider audience. Next, it is shown how dynamic evaluation can be employed to compute with algebraic numbers without having to do explicit factorization of polynomials. The essential step here is to define the dynamic algebraic closure of a field. Finally, a program for the Axiom system implementing dynamic algebraic closure is presented.",

```

acknowledgement = ack-nhfb,
keywords =      "algebraic numbers; Axiom; dynamic algebraic closure of
a field; dynamic evaluation; sketches",
language =      "French",
reviewer =      "A. Bijlsma (Eindhoven)",
}

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— axiom.bib —

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@mastersthesis{ElAl01,
  author = "El-Alfy, Hazem Mohamed",
  title = "Computer Algebra and its Applications",
  school = "Alexandria University, Department of Engineering, Mathematics,
and Physics",
  year = "2001",
  link = "\url{http://www.umiacs.umd.edu/~helalfy/pub/mscthesi01.pdf}",
  file = "ElAl01.pdf",
  abstract =
    "In the recent decades, it has been more and more realized that
computers are of enormous importance for numerical
computations. However, these powerful general-purpose machines can
also be used for transforming, combining and computing symbolic
algebraic expressions. In other words, computers can not only deal
with numbers, but also with abstract symbols representing mathematical
formulas. This fact has been realized much later and is only now
gaining acceptance among mathematicians and engineers. [Franz Winkler,
1996].

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Computer Algebra is that field of computer science and mathematics, where computation is performed on symbols representing mathematical objects rather than their numeric values.

This thesis attempts to present a definition of computer algebra by

means of a survey of its main topics, together with its major application areas. The survey includes necessary algebraic basics and fundamental algorithms, essential in most computer algebra problems, together with some problems that rely heavily on these algorithms. The set of applications, presented from a range of fields of engineering and science, although very short, indicates the applied nature of computer algebra systems.

A recent research area, central in most computer algebra software packages and in geometric modeling, is the implicitization problem. Curves and surfaces are naturally represented either parametrically or implicitly. Both forms are important and have their uses, but many design systems start from parametric representations. Implicitization is the process of converting curves and surfaces from parametric form into implicit form.

We have surveyed the problem of implicitization and investigated its currently available methods. Algorithms for such methods have been devised, implemented and tested for practical examples. In addition, a new method has been devised for curves for which a direct method is not available. The new method has been called `{\sl near implicitization}` since it relies on an approximation of the input problem. Several variants of the method try to compromise between accuracy and complexity of the designed algorithms.

The problem of implicitization is an active topic where research is still taking place. Examples of further research points are included in the conclusion",
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@misc{Ency16,
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  title = "Encyclopedia of Mathematics",
  link = "\url{https://www.encyclopediaofmath.org/index.php/Computer\_algebra\_package}",
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\bibitem[Erocal 10]{ES10} Er\ "ocal, Burcin; Stein, William
  title = "The Sage Project",
  link = "\url{http://wstein.org/papers/icms/icms_2010.pdf}",
  abstract = "
    Sage is a free, open source, self-contained distribution of
    mathematical software, including a large library that provides a
    unified interface to the components of this distribution. This library
    also builds on the components of Sage to implement novel algorithms
    covering a broad range of mathematical functionality from algebraic
    combinatorics to number theory and arithmetic geometry.",
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@article{Fakl97,
  author = "Fakler, Winfried",
  title = "On second order homogeneous linear differential equations with
    Liouvillian solutions",
  journal = "Theor. Comput. Sci.",
  volume = "187",
  number = "1-2",
  pages = "27-48",
  year = "1997",
  abstract =
    "We determine all minimal polynomials for second order homogeneous
    linear differential equations with algebraic solutions decomposed into
    invariants and we show how easily one can recover the known conditions
    on differential Galois groups [J. Kovacic, J. Symb. Comput. 2, 3-43
    (1986; Zbl 0603.68035), M. F. Singer and F. Ulmer,
    J. Symb. Comput. 16, 9-36, 37-73 (1993; Zbl 0802.12004, Zbl
    0802.12005), F.Ulmer and J. A. Weil, J. Symb. Comput. 22, 179-200
    (1996; Zbl 0871.12008)] using invariant theory. Applying these
    conditions and the differential invariants of a differential equation
    we deduce an alternative method to the algorithms given in (loc. cit.)
    for computing Liouvillian solutions. For irreducible second order
    equations our method determines solutions by formulas in all but three
    cases.",
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@article{Farm03,
  author = "Farmer, William M. and von Mohrenschildt, Martin",
  title = "An overview of a formal framework for managing mathematics",
  journal = "Ann. Math. Artif. Intell.",
  volume = "38",
  number = "1-3",
  pages = "165-191",
  year = "2003",
  link = "\url{https://www.emis.de/proceedings/MKM2001/farmer.ps}",
  abstract =
    "Mathematics is a process of creating, exploring, and connecting
    mathematical models. This paper presents an overview of a formal
    framework for managing the mathematics process as well as the
    mathematical knowledge produced by the process. The central idea of
    the framework is the notion of a biform theory which is simultaneously
    an axiomatic theory and an algorithmic theory. Representing a
    collection of mathematical models, a biform theory provides a formal
    context for both deduction and computation. The framework includes
    facilities for deriving theorems via a mixture of deduction and
    computation, constructing sound deduction and computation rules, and
    developing networks of biform theories linked by interpretations. The
    framework is not tied to a specific underlying logic; indeed, it is
    intended to be used with several background logics
    simultaneously. Many of the ideas and mechanisms used in the framework
    are inspired by the IMPS Interactive Mathematical Proof System and the
    Axiom computer algebra system.",
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@inproceedings{Fate79,
  author = "Fateman, Richard J.",
  title = "MACSYMA's General Simplifier: Philosophy and Operation",
  booktitle = "Proc. Macsyma Users' Conference 1979",
  year = "1979",
  link = "\url{http://people.eecs.berkeley.edu/~fateman/papers/simplifier.txt}",
  abstract =
    "Ideally the transformations performed by MACSYMA's simplification
    program on algebraic expressions correspond to those simplifications"
```

desired by each user and each program. Since it is impossible for a program to intuit all users' requirements simultaneously, explicit control of the simplifier is necessary to override default transformations. A model of the simplification process is helpful in controlling this large and complex program.

Having examined several algebraic simplification programs, it appears that to date no program has been written which combines a conceptually simple and useful view of simplification with a program nearly as powerful as MACSYMA's. {note, 1979. not clear this would be different in 2001. RJF} Rule-directed transformation schemes struggle to approach the power of the varied control structures in more usual program schemes [Fenichel, 68]. {note, Mathematica pushes rules further. RJF}

It is our belief that a thorough grasp of the decision and data structures of the MACSYMA simplifier program itself is the most direct way of understanding its potential for algebraic expression transformation. This is an unfortunate admission to have to make, but it appears to reflect the state of the art in dealing with formalizations of complex programs. Simplification is a perplexing task. Because of this, we feel it behooves the 'guardians of the simplifier' to try to meet the concerned MACSYMA users part-way by documenting the program as it has evolved. We hope this paper continues to grow to reflect a reasonably accurate, complete, and current description.

Of course Lisp program details are available to the curious, but even for those without a working knowledge of the Lisp language (in which the simplifier is written) we expect this paper to be of some help in answering questions which arise perennially as to why MACSYMA deals with some particular class of expressions in some unanticipated fashion, or is inefficient in performing some set of transformations. Most often difficulties such as these are accounted for by implicit design decisions which are not evident from mere descriptions of what is done in the anticipated and usual cases. We also hope that improvements or revisions of the simplifier will benefit from the more centralized treatment of issues given here. We also provide additional commentary which reflects our current outlook on how simplification programs should be written, and what capabilities they should have.",

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@inproceedings{Fate90,
  author = "Fateman, Richard J.",
  title = "Advances and trends in the design and construction of algebraic
    manipulation systems",
  booktitle = "Proc. ISSAC 1990",
  publisher = "ACM",
  pages = "60-67",
  isbn = "0-89791-401-5",
  year = "1990",
  link = "\url{http://people.eecs.berkeley.edu/~fateman/papers/advances.pdf}",
  abstract =
    "We compare and contrast several techniques for the implementation of
    components of an algebraic manipulation system. On one hand is the
    mathematical-algebraic approach which characterizes (for example)
    IBM's Axiom. On the other hand is the more {\sl ad hoc} approach which
    characterizes many other popular systems (for example, Macsyma,
    Reduce, Maple, and Mathematica). While the algebraic approach has
    generally positive results, careful examination suggests that there
    are significant remaining problems, especially in the representation
    and manipulation of analytical, as opposed to algebraic,
    mathematics. We describe some of these problems and some general
    approaches for solutions.",
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@InProceedings{Fateman:1990:ATD,
  author = "R. J. Fateman",
  title = "Advances and trends in the design and construction of
    algebraic manipulation systems",
  crossref = "Watanabe:1990:IPI",
  pages = "60--67",
  year = "1990",
  DOI = "http://dx.doi.org/10.1145.96895",
  bibdate = "Thu Jul 26 09:04:25 2001",
  bibsource = "http://www.math.utah.edu/pub/tex/bib/axiom.bib",
  abstract = "Compares and contrasts several techniques for the
    implementation of components of an algebraic
    manipulation system. On one hand is the mathematical
    algebraic approach which characterizes (for example)
    IBM's Scratchpad II. On the other hand is the more ad
    hoc approach which characterizes many other popular
    systems (for example, Macsyma, Reduce, Maple, and

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Mathematica). While the algebraic approach has
generally positive results, careful examination
suggests that there are significant remaining problems,
especially in the representation and manipulation of
analytical, as opposed to algebraic mathematics. The
author describes some of these problems, and some
general approaches for solutions.",
acknowledgement = ack-nhfb,
affiliation = "California Univ., Berkeley, CA, USA",
classification = "C4240 (Programming and algorithm theory); C7310
(Mathematics)",
keywords = "Algebraic manipulation systems; Algebraic mathematics;
Macsyma; Maple; Mathematica; Mathematical algebraic;
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@misc{Fate94,
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abstract =
    "We compare and contrast several techniques for the implementation of
    components of an algebraic manipulation system. On one hand is the
    mathematical-algebraic approach which characterizes (for example)
    IBM's Axiom. On the other hand is the more {\sl ad-hoc} approach which
    characterizes many other popular systems (for example, Macsyma,
    Reduce, Maple, and Mathematica). While the algebraic approach has
    generally positive results, careful examination suggests that there
    are significant remaining problems, especially in the representation
    and manipulation of analytical, as opposed to algebraic,
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@InProceedings{Fate96,
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  booktitle = "Proc 1996 ISSAC",
  series = "ISSAC 96",
  year = "1996",
  pages = "86-94",
  link = "\url{http://http.cs.berkeley.edu/~fateman/papers/eval.ps}",
  abstract =
    "'Evaluation' of expressions and programs in a computer algebra
    system is central to every system, but inevitably fails to provide
    complete satisfaction. Here we explain the conflicting requirements,
    describe some solutions from current systems, and propose alternatives
    that might be preferable sometimes. We give examples primarily from
    Axiom, Macsyma, Maple, Mathematica, with passing mention of a few other
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@inproceedings{Fate97,
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  title = "Network Servers for Symbolic Mathematics",
  booktitle = "Proc. ISSAC 1997",
  pages = "249-256",
  year = "1997",
  isbn = "0-89791-875-4",
  link = "\url{http://http.cs.berkeley.edu/~fateman/papers/cas-serve.ps}",
  abstract =
    "We describe advantages to using network socket facilities for
    communication and distributed computing from the perspective of
    symbolic mathematics systems. For some applications, an easily
    constructed Lisp server model provides a flexible portal between
    computer algebra and other programs, and one need not use new
    languages or new systems or write new stand-alone web-specific cgi-bin
    applications. Such socket programs can use, if necessary, HTML as a
    common transport encoding, but more efficient means are possible. We
    show that, rather than distributing all information to each computer
    algebra user's system, it makes sense to consider networking for
    accessing tables of information maintained at one or a few
    sites. Finally, we mention some consequences of the economic value of
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@inproceedings{Fate99,
  author = "Fateman, Richard J.",
  title = "Symbolic mathematics system evaluators",
  booktitle = "Proc. ISSAC 1996",
  pages = "86-94",
  year = "1999",
  link = "\url{http://people.eecs.berkeley.edu/~fateman/papers/evalnew.pdf}",
  abstract =
    "'Evaluation' of expressions and programs in a computer algebra
    system is central to every system, but inevitably fails to provide
    complete satisfaction. Here we explain the conflicting requirements,
    describe some solutions from current systems, and propose alternatives
    that might be preferable sometimes. We give examples primarily from
    Axiom, Macsyma, Maple, Mathematica, with passing mention of a few other
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@misc{Fate99a,
  author = "Fateman, Richard J. and Caspi, Eylon",
  title = "Parsing TeX into Mathematics",
  year = "1999",
  link = "\url{http://lib.org.by/_djvu/_Papers/Computer\_algebra/CAS%20systems/}",
  abstract =
    "Communication, storage, transmission, and searching of complex
    material has become increasingly important. Mathematical computing in
    a distributed environment is also becoming more plausible as libraries
    and computing facilities are connected with each other and with user
    facilities. TeX is a well-known mathematical typesetting language, and
    from the display perspective it might seem that it could be used for
    communication between computer systems as well as an intermediate form
    for the results of OCR (optical character recognition) of mathematical
    expressions. There are flaws in this reasoning, since exchanging
    mathematical information requires a system to parse and semantically
    'understand' the TeX, even if it is 'ambiguous' notationally. A
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    program we developed can handle 43\% of 10,740 TeX formulas in a
    well-known table of integrals. We expect that a higher success rate can
    be achieved easily.",
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    keywords = "axiomref"
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@InProceedings{Fate00,
  author = "Fateman, Richard J.",
  title = "Problem solving environments and symbolic computing",
  booktitle = "Enabling technologies for computational science",
  publisher = "Kluwer Academic Publishers",
  year = "2000",
  pages = "91-102",
  link = "url{http://people.eecs.berkeley.edu/~fateman/papers/pse-kluwer.pdf}",
  abstract =
    "What role should be played by symbolic mathematical computation
    facilities in scientific and engineering ‘‘problem solving
    environments’’? Drawing upon standard facilities such as numerical and
    graphical libraries, symbolic computation should be useful for: The
    creation and manipulation of mathematical models; The production of
    custom optimized numerical software; The solution of delicate classes
    of mathematical problems that require handling beyond that available
    in traditional machine-supported floating-point computation. Symbolic
    representation and manipulation can potentially play a central
    organizing role in PSEs since their more general object representation
    allows a program to deal with a wider range of computational
    issues. In particular numerical, graphical, and other processing can
    be viewed as special cases of symbolic manipulation with interactive
    symbolic computing providing both an organizing backbone and the
    communication ‘‘glue’’ among otherwise dissimilar components",
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@misc{Fate91,
  author = "Fateman, Richard J.",
  title = "A Review of Macsyma",

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year = "1991",
link = "\url{http://people.eecs.berkeley.edu/~fateman/papers/mma.pdf}",
abstract =
    "The Mathematica computer system is reviewed from the perspective of
    its contributions to symbolic and algebraic computation, as well as
    its stated goals. Design and implementation issues are discussed.",
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@article{Fate01,
  author = "Fateman, Richard J.",
  title = "A Review of Macsyma",
  journal = "IEEE Trans. Knowl. Eng.",
  volume = "1",
  number = "1",
  year = "2001",
  link = "\url{http://people.eecs.berkeley.edu/~fateman/papers/mac82b.pdf}",
  abstract =
    "We review the successes and failures of the Macsyma algebraic
    manipulation system from the point of view of one of the original
    contributors. We provide a retrospective examination of some of the
    controversial ideas that worked, and some that did not. We consider
    input/output, language semantics, data types, pattern matching,
    knowledge-adjunction, mathematical semantics, the user community,
    and software engineering. We also comment on the porting of this
    system to a variety of computing systems, and possible future
    directions for algebraic manipulation system-building.",
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@misc{Fate02,
  author = "Fateman, Richard J.",
  title = "Comparing the speed of programs for sparse polynomial
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  link = "\url{http://www.cs.berkeley.edu/~fateman/papers/fastmult.pdf}",
  paper = "Fate02.pdf",
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@misc{Fate05,
  author = "Fateman, Richard J.",
  title = "An incremental approach to building a mathematical
    expert out of software",
  conference = "Axiom Computer Algebra Conference",
  location = "City College of New York, CAISS project",
  year = "2005",
  month = "April",
  day = "19",
  link = "\url{http://www.cs.berkeley.edu/~fateman/papers/axiom.pdf}",
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@misc{Fate05a,
  author = "Fateman, Richard J.",
  title = "Haddock's eyes and computer algebra systems: Some essays",
  link = "\url{http://www.cs.berkeley.edu/~fateman/papers/haddock.pdf}",
  year = "2005",
  abstract =
    "(From {\sl Through the Looking Glass} by Lewis Carroll)

    The White Knight proposes to comfort Alice by singing her a song:

    ‘‘Is it very long?’’ Alice asked, for she had heard a good deal of
    poetry that day.

    ‘‘It’s long,’’ said the Knight, ‘‘but it’s very, very beautiful.
    Everybody that hears me sing it--either it brings tears into the
    eyes, or else --

    ‘‘Or else what?’’ said Alice, for the Knight had made a sudden pause.

    ‘‘Or else it doesn’t, you know. The name of the song is called
    ‘Haddock’s Eyes’.’’
```

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‘‘Oh, that’ the name of the song, is it?’’ Alice said, trying to feel
interested.

‘‘No, you don’t understand,’’ the Knight said, looking a little vexed.

‘‘That’s what the name is called. The name really is ‘The Aged Aged Man’’’

‘‘Then I ought to have said ‘That’s what the song is called?’’ Alice
corrected herself.

‘‘No, you oughtn’t: that’s quite another thing! The song is called
‘Ways and Means’: but that’s only what it’s called, you know!’’

‘‘Well, what is the song, then?’’ said Alice, who was by this time
completely bewildered.

‘‘I was coming to that,’’ the Knight said. ‘‘The song really is
‘A-sitting on a Gate’: and the tune’s my own invention.’’”,
paper = "Fate05a.pdf"
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\bibitem[Fateman 06]{Fat06} Fateman, R. J.
  title = "Building Algebra Systems by Overloading Lisp",
  link = "\url{http://www.cs.berkeley.edu/~fateman/generic/overload-small.pdf}",
  abstract = "
    Some of the earliest computer algebra systems (CAS) looked like
    overloaded languages of the same era. FORMAC, PL/I FORMAC, Formula
    Algol, and others each took advantage of a pre-existing language base
    and expanded the notion of a numeric value to include mathematical
    expressions. Much more recently, perhaps encouraged by the growth in
    popularity of C++, we have seen a renewal of the use of overloading to
    implement a CAS.

    This paper makes three points. 1. It is easy to do overloading in
    Common Lisp, and show how to do it in detail. 2. Overloading per se
    provides an easy solution to some simple programming problems. We show
    how it can be used for a ‘‘demonstration’’ CAS. Other simple and
    plausible overloadings interact nicely with this basic system. 3. Not
    all goes so smoothly: we can view overloading as a case study and
    perhaps an object lesson since it fails to solve a number of
    fairly-well articulated and difficult design issues in CAS for which
    other approaches are preferable.",
  paper = "Fat06.pdf",

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keywords = "axiomref"

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@misc{Fate06a,
  author = "Fateman, Richard J.",
  title = "Comments on Extending Macsyma with New Data Types",
  link = "\url{http://www.cs.berkeley.edu/~fateman/papers/addformat.pdf}",
  abstract =
    "Any design for a computer algebra system (CAS) naturally includes a
    set of data layouts for symbolic or mathematical algebraic expressions
    intended for use by built-in or user-written programs. The CAS cannot
    build in all plausible data designs, but supports those of most
    interest to the programmers. In such a situation it is almost
    inevitable that some new data encoding idea will come to mind and with
    it an interest in adding additional data forms. The motivation may be
    for compact representation, or efficient (fast) manipulation, or for
    other reasons such as interchange with other programs. Most CAS
    therefore include at least one way to extend the base set of
    operations. We comment on the kinds of extensions possible, using
    Macsyma as an example CAS. The particular interest in Macsyma and its
    open-source soruceforge variant 'Maxima' is that a substantial group
    of very-losely coupled independent researchers are approaching this
    problem and may benefit from some guidance. Some the observations
    apply to other CAS, even though they are not open-source.",
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@misc{Fate06b,
  author = "Fateman, Richard J.",
  title = "Building Algebra Systems by Overloading Lisp: Automatic
    Differentiation",
  link = "\url{http://www.cs.berkeley.edu/~fateman/papers/overload-AD.pdf}",
  year = "2006",
  abstract =
    "In an earlier paper we began a discussion of the use of overloaded
    languages for support of computer algebra systems. Here we extend that
    notion to provide a more detailed approach to Automatic
    Differentiation or Algorithm Differentiation (AD)."
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This paper makes three points. 1. It is extremely easy to do express
AD by overloading in Common Lisp. 2. While the resulting program is
not the most efficient approach in terms of run-time, it is quite
small and very general. It also interacts nicely with some other kinds
of generic arithmetic. 3. A more efficient AD compile-time program
generation approach is described as well.",
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keywords = "axiomref"
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@article{Faug94,
author = "Faug\`ere, J.C. and Gianni, P. and Lazard, D. and Mora, T.",
title = "Efficient Computation of Zero-dimensional Grobner Bases by
        Change of Ordering",
journal = "J. Symbolic Computation",
issue_date = "February 1994",
volume = "11",
number = "2",
month = "February",
year = "1984",
link = "\url{http://www-polsys.lip6.fr/~jcf/Papers/FGLM.pdf}",
publisher = "Academic Press Limited",
algebra = "\newline\refto{package LEXTRIPK LexTriangularPackage}",
abstract = "
    We present an efficient algorithm for the transformation of a
    Grobner basis of a zero-dimensional ideal with respect to any given
    ordering into a Grobner basis with respect to any other
    ordering. This algorithm is polynomial in the degree of the idea. In
    particular, the lexicographical Grobner basis can be obtained by
    applying this algorithm after a total degree Grobner basis
    computation: it is usually much faster to compute the basis this way
    than with a direct application of Buchberger's algorithm.",
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keywords = "axiomref"
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\bibitem[Faure 00a]{FDN00a} Faure, Christ\`ele; Davenport, James

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title = "Parameters in Computer Algebra",
keywords = "axiomref"

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\bibitem[Faure 00b]{FDN00b} Faure, Christ\`ele; Davenport, James;
Naciri, Hanane
  title = "Multi-values Computer Algebra",
  ISSN 0249-6399 Institut National De Recherche en Informatique et en
  Automatique Sept. 2000 No. 4001
  link = "url{http://hal.inria.fr/inria-00072643/PDF/RR-4401.pdf}",
  abstract = "
    One of the main strengths of computer algebra is being able to solve a
    family of problems with one computation. In order to express not only
    one problem but a family of problems, one introduces some symbols
    which are in fact the parameters common to all the problems of the
    family.

    The user must be able to understand in which way these parameters
    affect the result when he looks at the answer. Otherwise it may lead
    to completely wrong calculations, which when used for numerical
    applications bring nonsensical answers. This is the case in most
    current Computer Algebra Systems we know because the form of the
    answer is never explicitly conditioned by the values of the
    parameters. The user is not even informed that the given answer may be
    wrong in some cases then computer algebra systems can not be entirely
    trustworthy. We have introduced multi-valued expressions called {\sl
    conditional} expressions, in which each potential value is associated
    with a condition on some parameters. This is used, in particular, to
    capture the situation in integration, where the form of the answer can
    depend on whether certain quantities are positive, negative or
    zero. We show that it is also necessary when solving modular linear
    equations or deducing congruence conditions from complex expressions.",
  paper = "FDN00b.pdf",
  keywords = "axiomref"

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\bibitem[Fitch 84]{Fit84} Fitch, J. P. (ed)
EUROSAM '84: International Symposium on Symbolic and
Algebraic Computation, Cambridge, England, July 9-11, 1984, volume 174 of
Lecture Notes in Computer Science. Springer-Verlag, Berlin, Germany /

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Heidelberg, Germany / London, UK / etc., 1984 ISBN 0-387-13350-X
 LCCN QA155.7.E4 I57 1984
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@InProceedings{Fitc93,
  author = "Fitch (ed), John P.",
  title = "Design and Implementation of Symbolic Computation Systems",
  year = "1992",
  booktitle = "Int. Symp. DISCO '92 Proceedings",
  series = "DISCO 92",
  publisher = "Springer-Verlag, Berlin",
  isbn = "0-387-57272-4",
  paper = "Fitc93.tex",
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@book{Flei94,
  author = "Fleischer, J. and Grabmeier, J. and Hehl, F.W. and
           Kuchlin, W. (eds)",
  title = "Proc. Conf. Computer Algebra in Science and Engineering",
  booktitle = "Computer Algebra in Science and Engineering",
  year = "1994",
  location = "Bielefeld, Germany",
  publisher = "World Scientific, River Edge, NJ",
  abstract =
    "Systems and tools of computer algebra (Like AXIOM, Derive, FORM,
     Mathematica, Maple, Mupad, REDUCE, Macsyma) let us manipulate
     extremely complex algebraic formulae symbolically on a
     computer. Contrary to numerics these computations are exact and there
     is no loss of accuracy. After decades of research and development,
     these tools are now becoming as indispensable in Science and
     Engineering as traditional number crunching already is.

     The ZiF'94 workshop is amongst the first devoted specifically to
     applications of computer algebra (CA) in Science and Engineering. The
     book documents the state of the art in this area and serves as an
     important reference for future work."
}
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— ignore —

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\bibitem[Fogus 11]{Fog11} Fogus, Michael
  title = "UnConj",
  link = "\url{http://clojure.com/blog/2011/11/22/unconj.html}",
  keywords = "axiomref"
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— axiom.bib —

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@techreport{Fort85,
  author = "Fortenbacher, A. and Jenks, Richard and Lucks, Michael and
           Sutor, Robert and Trager, Barry and Watt, Stephen",
  title = "An Overview of the Scratchpad II Language and System",
  institution = "IBM",
  year = "1985",
  type = "Research Report",
  publisher = "IBM Research Computer Algebra Group",
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@inproceedings{Fort90,
  author = "Fortenbacher, Albrecht",
  title = "Efficient type inference and coercion in computer algebra",
  booktitle = "Design and Implementation of Symbolic Computation Systems",
  series = "Lecture Notes in Computer Science 429",
  pages = "56-60",
  isbn = "0-387-52531-9",
  year = "1990",
  abstract =
    "Computer algebra systems of the new generation, like SCRATCHPAD, are
    characterized by a very rich type concept, which models the
    relationship between mathematical domains of computation. To use these
    systems interactively, however, the user should be freed of type
    information. A type inference mechanism determines the appropriate"
```

```

function to call. All known models which allow to define a semantics
for type inference cannot express the rich ‘‘mathematical’’ type
structure, so presently type inference is done heuristically. The
following paper defines a semantics for a subproblem thereof, namely
coercion, which is based on rewrite rules. From this definition, an
efficient coercion algorithm for SCRATCHPAD is constructed using graph
techniques.",
keywords = "axiomref",
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@InProceedings{Fortenbacher:1990:ETI,
  author = "A. Fortenbacher",
  title = "Efficient type inference and coercion in computer
algebra",
  crossref = "Miola:1990:DIS",
  pages = "56--60",
  month = "",
  year = "1990",
  bibdate = "Tue Sep 17 06:44:07 MDT 1996",
  bibsource = "http://www.math.utah.edu/pub/tex/bib/axiom.bib",
  abstract = "Computer algebra systems of the new generation, like
Scratchpad, are characterized by a very rich type
concept, which models the relationship between
mathematical domains of computation. To use these
systems interactively, however, the user should be
freed of type information. A type inference mechanism
determines the appropriate function to call. All known
models which define a semantics for type inference
cannot express the rich ‘mathematical’ type structure,
so presently type inference is done heuristically. The
following paper defines a semantics for a subproblem,
namely coercion, which is based on rewrite rules. From
this definition, an efficient coercion algorithm for
Scratchpad is constructed using graph techniques.",
  acknowledgement = ack-nhfb,
  affiliation = "Sci. Center Heidelberg, IBM Deutschland GmbH,
Germany",
  classification = "C1110 (Algebra); C4210 (Formal logic); C6120 (File
organisation); C7310 (Mathematics)",
  keywords = "Coercion algorithm; Computer algebra; Graph
techniques; Rewrite rules; Scratchpad; Type inference
mechanism",
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thesaurus =    "Algebra; Data structures; Inference mechanisms;
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@article{Fort05,
  author = "Fortuna, E. and Gianni, P. and Luminati, D. and Parenti, P.",
  title = "The adjacency graph of a real algebraic surface",
  journal = "Appl. Algebra Eng. Commun. Comput.",
  volume = "16",
  number = "5",
  pages = "271-292",
  year = "2005",
  link = "\url{http://eprints.biblio.unitn.it/788/1/UTM671.pdf}",
  abstract =
    "The paper deals with the question of recognizing the mutual positions
    of the connected components of a non-singular real projective surface
    $$$ in the real projective 3-space. We present an algorithm that
    answers this question through the computation of the adjacency graph
    of the surface; it also allows to decide whether each connected
    component is contractible or not. The algorithm, combined with a
    previous one returning as an output the topology of the surface,
    computes a set of data invariant up to ambient-homeomorphism which,
    though not sufficient to determine the pair  $(\mathbb{R}, \mathbb{P}^3, S)$ ,
    give information about the nature of the surface as an embedded object.",
  paper = "Fort05.pdf",
  keywords = "axiomref"
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@techreport{Fouc90,
  author = "Fouche, Francois",
  title = "Une implantation de l'algorithme de Kovacic en Scratchpad",
  type = "technical report",
  number = "ULP-IRMA-447-P-254",
  year = "1990",
  institution = "{Institut de Recherche Mathématique Avancée}",
  location = "Strasbourg, France",
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— Fouche:1990:ILK —

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@TechReport{Fouche:1990:ILK,
  author =      "Francois Fouche",
  title =      "Une Implantation de l'algorithme de {Kovacic} en
                {Scratchpad}",
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                Avanc{\'}{e}}e",
  address =     "Strasbourg, France",
  pages =      "31",
  year =       "1990",
  bibdate =    "Sat Dec 30 08:25:26 MST 1995",
  bibsource =   "http://www.math.utah.edu/pub/tex/bib/axiom.bib",
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\bibitem[FSF 14]{FSF14} FSF
  title = "Free Software Directory",
  link = "\url{http://directory.fsf.org/wiki/Axiom}",
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\bibitem[Frisco ]{Fris} Frisco
  title = "Objectives and Results",
  link = "\url{http://www.nag.co.uk/projects/frisco/frisco/node3.htm}",
  keywords = "axiomref"

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@article{Frit94,
  author = "Fritzson, D. and Fritzson, P. and Viklund, L. and Herber, J.",
  title = "Object-oriented mathematical modelling - applied to machine
          elements",
  journal = "Comput. Struct.",
  volume = "51",
  number = "3",
  pages = "241-253",
  year = "1994",
  abstract =
    "Machine element analysis has a goal of describing function and other
    aspects of machine elements in a theoretical form. This paper shows
    how ideas from object-oriented modelling can be applied to machine
    element analysis. The models thus obtained are both easier to
    understand, better structured, and allow a higher degree of re-use
    than conventional models. An object-oriented model description is
    natural and suitable for machine element analysis. As a realistic
    example an equational model of rolling bearings is presented. The
    structure of the model is general, and applies to many types of
    rolling bearings. The model and one solution require approximately
    200+200 equations. The model is extensible, e.g. simple submodels of
    detailed properties can be made more complex without altering the
    overall structure. The example model has been implemented in a
    language of our own design. ObjectMath (Object-oriented Mathematical
    language for scientific computing). Using ObjectMath, it is possible
    to model classes of equation objects, to support multiple and single
    inheritance of equations, to support composition of equations, and to
    solve systems of equations. Algebraic transformations can conveniently
    be done since ObjectMath models are translated into the Mathematica
    computer algebra language. When necessary, equations can be
    transformed into C++ code for efficient numerical solution. The re-use
    of equations through inheritance reduced the size of the model by a
    factor of two, compared to a direct representation of the model in the
    Mathematica computer algebra language.",
  paper = "Frit94.pdf",
  keywords = "axiomref"
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@techreport{Gall92,
  author = "Gallopoulos, Stratis and Houstis, Elias and Rice, John",
  title = "Future Research Directions in Problem Solving Environments for

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        Computational Science",
institution = "Purdue University",
year = "1992",
type = "technical report",
number = "CSD-TR-92-032",
link = "\url{http://docs.lib.purdue.edu/cgi/viewcontent.cgi?article=1953\&context=cstech}",
abstract =
    "During the early 1960s some were visualizing that computers could
    provide a powerful problem solving environment (PSE) which would
    interact with scientists on their own terms. By the mid 1960s there
    were many attempts underway to create these PSEs, but the early
    1970s almost all of these attempts had been abandoned, because the
    technological infrastructure could not yet support PSEs in
    computational science. The dream of the 1960s can be the reality of
    the 1990s; high performance computers combined with better
    understanding of computing and computational science have put PSEs
    well within our reach.",
paper = "Gall92.pdf",
keywords = "axiomref"
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@book{Ganz00,
author = "Ganzha, Victor G. and Vorozhtsov, Evgenii V. and Wester, Michael",
title = "An Assessment of the Efficiency of Computer Algebra Systems in
        the Solution of Scientific Computing Problems",
booktitle = "Computer Algebra in Scientific Computing",
year = "2000",
isbn = "978-3-540-41040-9",
publisher = "Springer",
pages = "145-166",
abstract =
    "Computer algebra systems (CASs) have become an important tool for the
    solution of scientific computing problems. With the increasing number
    of general purpose CASs, there is now a need for an assessment of the
    efficiency of these systems. We discuss some peculiarities associated
    with the analysis of CPU time efficiency in CASs, and then present
    results from three specific systems (Maple Vr5, Mathematics 4.0 and
    MuPAD 1.4) on a sample of intermediate size problems. These results
    show that Maple Vr5 is generally the speediest on our
    examples. Finally, we formulate some requirements for developing a
    comprehensive suite for analyzing the efficiency of CASs.",
keywords = "axiomref"
}

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@inproceedings{Gebu86,
  author = "Gebauer, R{\u}diger and M{\o}ller, H. Michael",
  title = "Buchberger's algorithm and staggered linear bases",
  booktitle = "Proc. 1986 Symposium on Symbolic and Algebraic Computation",
  series = "SYMSAC '86",
  year = "1986",
  pages = "218-221",
  publisher = "ACM Press",
  isbn = "0-89791-199-7",
  doi = "http://dx.doi.org/10.1145/32439.32482",
  keywords = "axiomref",
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}
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— Gebauer:1986:BAS —

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@InProceedings{Gebauer:1986:BAS,
  author = "R{\u}diger Gebauer and H. Michael M{\o}ller",
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  booktitle = "Proceedings of the 1986 Symposium on Symbolic and
    Algebraic Computation: Symsac '86, July 21--23, 1986,
    Waterloo, Ontario",
  title = "{Buchberger}'s algorithm and staggered linear bases",
  publisher = pub-ACM,
  address = pub-ACM:adr,
  pages = "218--221",
  year = "1986",
  DOI = "http://dx.doi.org/10.1145.32482",
  ISBN = "0-89791-199-7",
  ISBN-13 = "978-0-89791-199-3",
  LCCN = "QA155.7.E4 A281 1986",
  bibdate = "Thu Jul 26 09:06:12 2001",
  bibsource = "http://www.math.utah.edu/pub/tex/bib/axiom.bib",
  note = "ACM order number 505860.",
  acknowledgement = ack-nhfb,
  bookpages = "254",
}
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@article{Geba88,
  author = "Gebauer, Rudiger and Moller, H. Michael",
  title = "On an installation of Buchberger's algorithm",
  journal = "Journal of Symbolic Computation",
  volume = "6",
  number = "2-3",
  pages = "275-286",
  year = "1988",
  abstract =
    "Buchberger's algorithm calculates Groebner bases of polynomial
    ideals. Its efficiency depends strongly on practical criteria for
    detecting superfluous reductions. Buchberger recommends two
    criteria. The more important one is interpreted in this paper as a
    criterion for detecting redundant elements in a basis of a module of
    syzygies. We present a method for obtaining a reduced, nearly minimal
    basis of that module. The simple procedure for detecting (redundant
    syzygies and )superfluous reductions is incorporated now in our
    installation of Buchberger's algorithm in SCRATCHPAD II and REDUCE
    3.3. The paper concludes with statistics stressing the good
    computational properties of these installations.",
  paper = "GM88.pdf",
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}
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@Article{Gebauer:1988:IBA,
  author = "R. Gebauer and H. M. Moller",
  title = "On an installation of {Buchberger}'s algorithm",
  journal = "J-SYMBOLIC-COMP",
  volume = "6",
  number = "2-3",
  pages = "275--286",
  month = "oct # "-" # dec",
  year = "1988",
  CODEN = "JSYCEH",
  ISSN = "0747-7171 (print), 1095-855X (electronic)",
  ISSN-L = "0747-7171",
  bibdate = "Tue Sep 17 08:24:38 1996",
  bibsource = "http://www.math.utah.edu/pub/tex/bib/axiom.bib",
  abstract = "Buchberger's algorithm calculates Gr{o}bner bases of
    polynomial ideals. Its efficiency depends strongly on
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practical criteria for detecting superfluous
reductions. Buchberger recommends two criteria. The
more important one is interpreted in this paper as a
criterion for detecting redundant elements in a basis
of a module of syzygies The authors present a method
for obtaining a reduced, nearly minimal basis of that
module. The simple procedure for detecting (redundant
syzygies and) superfluous reductions is incorporated
now in the installation of Buchberger's algorithm in
SCRATCHPAD II and REDUCE 3.3. The paper concludes with
statistics stressing the good computational properties
of these installations.",
acknowledgement = ack-nhfb,
affiliation = "Springer-Verlag, New York, NY, USA",
classification = "C4130 (Interpolation and function approximation);
                  C6130 (Data handling techniques); C7310 (Mathematics)",
fjournal = "Journal of Symbolic Computation",
link = "\url{http://www.sciencedirect.com/science/journal/07477171}",
keywords = "Buchberger algorithm installation; Gr{\o}bner bases;
            Polynomial ideals; Superfluous reductions; Redundant
            elements; Module of syzygies; SCRATCHPAD II; REDUCE
            3.3; Computational properties",
language = "English",
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thesaurus = "Polynomials; Symbol manipulation",
}

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@book{Ged92,
  author = "Geddes, Keith and Czapor, O. and Stephen R. and Labahn, George",
  title = "Algorithms For Computer Algebra",
  year = "1992",
  publisher = "Kluwer Academic Publishers",
  isbn = "0-7923-9259-0",
  month = "September",
  abstract =
    "Computer Algebra (CA) is the name given to the discipline of
    algebraic, rather than numerical, computation. There are a number of
    computer programs Computer Algebra Systems (CASs) available for
    doing this. The most widely used general-purpose systems that are
    currently available commercially are Axiom, Derive, Macsyma, Maple,
    Mathematica and REDUCE. The discipline of computer algebra began in
    the early 1960s and the first version of REDUCE appeared in 1968.

```

A large class of mathematical problems can be solved by using a CAS

purely interactively, guided only by the user documentation. However, sophisticated use requires an understanding of the considerable amount of theory behind computer algebra, which in itself is an interesting area of constructive mathematics. For example, most systems provide some kind of programming language that allows the user to expand or modify the capabilities of the system.

This book is probably the most general introduction to the theory of computer algebra that is written as a textbook that develops the subject through a smooth progression of topics. It describes not only the algorithms but also the mathematics that underlies them. The book provides an excellent starting point for the reader new to the subject, and would make an excellent text for a postgraduate or advanced undergraduate course. It is probably desirable for the reader to have some background in abstract algebra, algorithms and programming at about second-year undergraduate level.

The book introduces the necessary mathematical background as it is required for the algorithms. The authors have avoided the temptation to pursue mathematics for its own sake, and it is all sharply focused on the task of performing algebraic computation. The algorithms are presented in a pseudo-language that resembles a cross between Maple and C. They provide a good basis for actual implementations although quite a lot of work would still be required in most cases. There are no code examples in any actual programming language except in the introduction.

The authors are all associated with the group that began the development of Maple. Hence, the book reflects the approach taken by Maple, but the majority of the discussion is completely independent of any actual system. The authors experience in implementing a practical CAS comes across clearly.

The book focuses on the core of computer algebra. The first chapter introduces the general concept and provides a very nice historical survey. The next three chapters discuss the fundamental topics data structures, representations and the basic arithmetic of integers, rational numbers, multivariate polynomials and rational functions on which the rest of the book is built.

A major technique in CA involves projection onto one or more homomorphic images, for which the ground ring is usually chosen to be a finite field. The image solution is lifted back to the original problem domain by means of the Chinese Remainder Theorem in the case of multiple homomorphic images, or the Hensel (-adic or ideal-adic) construction in the case of a single image. The next two chapters are devoted to these techniques in a fairly general setting. The two subsequent chapters specialise them to GCD computation and factorisation for multivariate polynomials; the first of these chapters also discusses the important but difficult topic of

subresultants.

The next two chapters describe the use of fraction-free Gaussian elimination, resultants and Grbner Bases for manipulation and exact solution of linear and nonlinear polynomial equations. The two final chapters describe ‘classical’ algorithms and the more recent Risch algorithm for symbolic indefinite integration, and provide an introduction to differential algebra.

The book does not consider more specialised problem areas such as symbolic summation, definite integration, differential equations, group theory or number theory. Nor does it consider more applied problem areas such as vectors, tensors, differential forms, special functions, geometry or statistics, even though Maple and other CASs provide facilities in all or many of these areas. It does not consider questions of CA programming language design, nor any of the important but non-algebraic facilities provided by current CASs such as their user interfaces, numerical and graphical facilities.

This is a long book (nearly 600 pages); it is generally very well presented and the three authors have merged their contributions seamlessly. I noticed very few typographical errors, and none of any consequence. I have only two complaints about the book. The typeface is too small, particularly for the relatively large line spacing used, and it is much too expensive, particularly for a book that would otherwise be an excellent student text. I recommend it highly to anyone who can afford it.",

keywords = "axiomref"

}

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\bibitem[Gianni 87]{Gia87} Gianni, Patrizia
 title = "Primary Decomposition of Ideals",
 in [Wit87], pp12-13
 keywords = "axiomref"

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\bibitem[Gianni 89a]{Gia89} Gianni, P. (Patrizia) (ed)
 Symbolic and Algebraic Computation.
 International Symposium ISSAC '88, Rome, Italy, July 4-8, 1988. Proceedings,

volume 358 of Lecture Notes in Computer Science. Springer-Verlag, Berlin, Germany / Heidelberg, Germany / London, UK / etc., 1989. ISBN 3-540-51084-2
 LCCN QA76.95.I57 1988 Conference held jointly with AAECC-6
 keywords = "axiomref"

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@inproceedings{Gian89,
  author = "Gianni, Patrizia and Mora, T.",
  title = "Algebraic solution of systems of polynomial equations
          using Groebner bases.",
  booktitle = "Applied Algebra, Algebraic Algorithms and Error-Correcting
              Codes",
  series = "AAECC-5",
  pages = "247-257",
  year = "1989",
  isbn = "3-540-51082-6",
  abstract =
    "One of the most important applications of Buchberger's algorithm for
    Groebner basis computation is the solution of systems of polynomial
    equations (having finitely many roots), i.e. the computation of zeros
    of 0-dimensional polynomial ideals. It is based on a relation between
    Groebner bases w.r.t. a lexicographical ordering and elimination
    ideals, which was discovered by Trinks.
```

Packages for isolation of real roots of systems of polynomial equations using Groebner basis computation are currently available in different computer algebra systems, including SAC-2, Reduce, Scratchpad II, Maple.

In principle, Buchberger-Trinks algorithm should allow to compute solutions of such systems in the algebraic closure of the coefficient field k (usually the rational numbers), in the sense that it is possible to represent explicitly a finite extension of k containing all solutions and to express the roots in this field.

However, this requires several factorisations of polynomials over a tower of algebraic extensions of k , which is usually very costly, so that the resulting algorithm is not very feasible and, as far as we know, no implementation is available.

The results of [GT2] on primary decomposition of ideals include a thorough study on the structure of Groebner bases for 0-dimensional ideals; in particular, the paper shows, that after a ‘generic’ linear change of coordinates, the roots of a system of polynomial equations can be expressed in a simple extension of k . Therefore, in

this case, no factorisation of polynomials over towers of algebraic extensions is needed.

However performing a change of coordinates has the undesirable effects of introducing dense polynomials and of increasing the size of coefficients.

The problem then arises of producing strategies to compute Groebner bases for (0-dimensional) ideals, which at least are able to control the influence of these side-effects: two such strategies are presented in this paper, together with the application to the present problem of an algorithm by Gianni that computes the radical of a 0-dimensional ideal after a ‘generic’ change of coordinates.

A different approach, based on her ‘splitting algorithm’, to compute solutions of systems of polynomial equations without the need of polynomial factorisations has been proposed by D. Duval; also her algorithm should be simplified by a ‘generic’ change of coordinates.

The algorithms discussed in this paper are implemented in SCRATCHPAD II.

In the first section we recall some well-known properties of Groebner bases and properties on the structure of Groebner bases of zero-dimensional ideals from [GT2]; in the second section we recall the Groebner basis algorithm for solving systems of algebraic equations.

The original results are contained in Sections 3 to 5; in Section 3 we take advantage of the obvious fact that density can be controlled by performing ‘small’ changes of coordinates: we show that such approach is possible during a Groebner basis computation, in such a way that computations done before a change of coordinates are valid also after it; in Section 4 we propose a ‘linear algebra’ approach to obtain the Groebner basis w.r.t the lexicographical ordering from the one w.r.t the total-degree ordering; in Section 5, we present a zero-dimensional radical algorithm and show how to apply it to the present problem.”,

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@InProceedings{Gianni:1989:ASS,
  author = "P. Gianni and T. Mora",
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title = "Algebraic solution of systems of polynomial equations
        using {Gr{"o}bner} bases",
crossref = "Huguet:1989:AAA",
pages = "247--257",
month = "",
year = "1989",
bibdate = "Tue Sep 17 06:46:18 MDT 1996",
bibsource = "http://www.math.utah.edu/pub/tex/bib/axiom.bib",
abstract = "One of the most important applications of Buchberger's
            algorithm for Gr{"o}bner basis computation is the
            solution of systems of polynomial equations (having
            finitely many roots), i.e. the computation of zeros of
            0-dimensional polynomial ideals. It is based on a
            relation between Gr{"o}bner bases w.r.t. a
            lexicographical ordering and elimination ideals. The
            algorithms discussed in this paper are implemented in
            SCRATCHPAD II. In the first section the authors recall
            some well-known properties of Gr{"o}bner bases and
            properties on the structure of Gr{"o}bner bases of
            zero-dimensional ideals; in the second section they
            recall the Gr{"o}bner basis algorithm for solving
            systems of algebraic equations. The original results
            are then presented. The authors first take advantage of
            the obvious fact that density can be controlled
            performing 'small' changes of coordinates: they show
            that such approach is possible during a Gr{"o}bner
            basis computation, in such a way that computations done
            before a change of coordinates are valid also after it;
            they propose a 'linear algebra' approach to obtain the
            Gr{"o}bner basis w.r.t. the lexicographical ordering
            from the one w.r.t. the total-degree ordering; and
            finally they present a zero-dimensional radical
            algorithm and show how to apply it to the present
            problem.",
acknowledgement = ack-nhfb,
affiliation = "Pisa Univ., Italy",
classification = "C1110 (Algebra); C4140 (Linear algebra); C7310
                (Mathematics)",
keywords = "Coordinate changes; Polynomial equations; Gr{"o}bner
            bases; Buchberger's algorithm; Gr{"o}bner basis
            computation; Zeros; 0-Dimensional polynomial ideals;
            Lexicographical ordering; Elimination ideals;
            SCRATCHPAD II; Algebraic equations; Linear algebra;
            Total-degree ordering; Zero-dimensional radical
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language = "English",
thesaurus = "Equations; Linear algebra; Mathematics computing;
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@inproceedings{Gilx92,
  author = "Gil, Isabelle",
  title = "Computation of the Jordan canonical form of a square matrix
          (using the Axiom programming language)",
  booktitle = "Proc ISSAC 1992",
  series = "ISSAC '92",
  year = "1992",
  publisher = "ACM",
  pages = "138-145",
  isbn = "0-89791-489-9 (soft cover), 0-89791-490-2 (hard cover)",
  abstract =
    "Presents an algorithm for computing: the Jordan form of a square
    matrix with coefficients in a field K using the computer algebra
    system Axiom. This system presents the advantage of allowing generic
    programming. That is to say, the algorithm can first be implemented
    for matrices with rational coefficients and then generalized to
    matrices with coefficients in any field. Therefore the author
    presents the general method which is essentially based on the use of
    the Frobenius form of a matrix in order to compute its Jordan form;
    and then restricts attention to matrices with rational
    coefficients. On the one hand the author streamlines the algorithm
    froben which computes the Frobenius form of a matrix, and on the other
    she examines in some detail the transformation from the Frobenius form
    to the Jordan form, and gives the so called algorithm Jordform. The
    author studies in particular, the complexity of this algorithm and
    proves that it is polynomial when the coefficients of the matrix are
    rational. Finally the author gives some experiments and a conclusion.",
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  beebe = "Gil:1992:CJC"
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— Gil:1992:CJC —

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@InProceedings{Gil:1992:CJC,
  author = "I. Gil",
  title = "Computation of the {Jordan} canonical form of a square
          matrix (using the {Axiom} programming language)",
  crossref = "Wang:1992:ISS",
  pages = "138--145",
  month = "",
  year = "1992",
}
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bibdate = "Tue Sep 17 06:35:39 MDT 1996",
bibsource = "http://www.math.utah.edu/pub/tex/bib/axiom.bib",
abstract = "Presents an algorithm for computing: the Jordan form
of a square matrix with coefficients in a field K using
the computer algebra system Axiom. This system presents
the advantage of allowing generic programming. That is
to say, the algorithm can first be implemented for
matrices with rational coefficients and then
generalized to matrices with coefficients in any field.
Therefore the author presents the general method which
is essentially based on the use of the Frobenius form
of a matrix in order to compute its Jordan form; and
then restricts attention to matrices with rational
coefficients. On the one hand the author streamlines
the algorithm froben which computes the Frobenius form
of a matrix, and on the other she examines in some
detail the transformation from the Frobenius form to
the Jordan form, and gives the so called algorithm
Jordform. The author studies in particular, the
complexity of this algorithm and proves that it is
polynomial when the coefficients of the matrix are
rational. Finally the author gives some experiments and
a conclusion.",
acknowledgement = ack-nhfb,
affiliation = "LMC, IMAG, Grenoble, France",
classification = "C4130 (Interpolation and function approximation);
C4140 (Linear algebra); C4240 (Programming and
algorithm theory); C7310 (Mathematics)",
keywords = "Axiom programming language; Complexity; Computer
algebra system; Froben; Frobenius form; Generic
programming; Jordan canonical form; Jordform;
Polynomial; Rational coefficients; Square matrix",
language = "English",
thesaurus = "Computational complexity; Matrix algebra; Polynomials;
Symbol manipulation",
}

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@phdthesis{Gome92,
  author = "Gomez-Dias, Teresa",
  title = {Quelques applications de l'\`evaluation dynamique},
  school = "L'Universite de Limoges",
  year = "1992",
  month = "March",
  paper = "Gome92.pdf"
}

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— ignore —

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@article{Gome93
\bibitem[Gomez-Diaz 93]{Gom93} G\omeg-D\iaz, Teresa
  title = "Examples of using Dynamic Constructible Closure",
  IMACS Symposium SC-1993
  abstract = "
    We present here some examples of using the ‘‘Dynamic Constructible
    Closure’’ program, which performs automatic case distinction in
    computations involving parameters over a base field  $K$ . This program
    is an application of the ‘‘Dynamic Evaluation’’ principle, which
    generalizes traditional evaluation and was first used to deal with
    algebraic numbers.",
  paper = "Gom93.pdf",
  keywords = "axiomref"
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@article{Gome96,
  author = "Gomez-Diaz, Theresa",
  title = "Examples of using dynamic constructible closure",
  journal = "Math. Comput. Simul.",
  volume = "42",
  number = "4-6",
  pages = "375-383",
  year = "1996",
  abstract =
    "We present here some examples of using the ‘‘Dynamic Constructible
    Closure’’ program, which performs automatic case distinctions in
    computations involving parameters over a base field  $K$ . This
    program is an application of the ‘‘Dynamic Evaluation’’ principle
    which generalizes traditional evaluation and was first used to deal with
    algebraic numbers.",
  keywords = "axiomref"
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@misc{Gonn05,
  author = "Gonnet, Gaston and Haigh, Thomas",
  title = "An Interview with Gaston Gonnet",
  year = "2005",
  publisher = "SIAM",
  link = "\url{http://history.siam.org/pdfs2/Gonnet_final.pdf}",
  abstract =
    "Born in Uruguay, Gonnet was first exposed to computers while working
    for IBM in Montevideo as a young man. This led him to a position at
    the university computer center, and in turn to an undergraduate degree
    in computer science in 1973. In 1974, following a military coup, he
    left for graduate studies in computer science at the University of
    Waterloo. Gonnet earned an M.Sc. and a Ph.D. in just two and a half
    years, writing a thesis on the analysis of search algorithms under the
    supervision of Alan George. After one year teaching in Rio de Janeiro
    he returned to Waterloo, as a faculty member.
```

In 1980, Gonnet began work with a group including Morven Gentleman and Keith Geddes to produce an efficient interactive computer algebra system able to work well on smaller computers: Maple. Gonnet discusses in great detail the goals and organization of the Maple project, its technical characteristics, the Maple language and kernel, the Maple library, sources of funding, the contributions of the various team members, and the evolution of the system over time. He compares the resulting system to MACSYMA, Mathematica, Reduce, Scratchpad and other systems. Gonnet also examines the licensing and distribution of Maple and the projects relations to its users. Maple was initially used for teaching purposes within the university, but soon found users in other institutions. From 1984, distribution was handled by Watcom, a company associated with the university, and 1988, Gonnet and Geddes created a new company, Waterloo Maple Software, Inc. to further commercialize Maple, which established itself as the leading commercial computer algebra system. However, during the mid-1990s the company ran into trouble and disagreements with his colleagues caused Gonnet to withdraw from managerial involvement. Since then, he feels that Maple has lost its battle with Mathematica. Gonnet also discusses Maples relation to Matlab and its creator, Cleve Moler.

From 1984 onward with Frank Tompa, Tim Bray, and other Waterloo colleagues, Gonnet worked on the production of computer software to support the creation of the second edition of the Oxford English Dictionary. This led to the creation of another startup company, Open Text, producing software for the searching and indexing of textual information within large corporations. Gonnet explains his role in the firm, including his departure and his feeling that it made a strategic blunder by not exploiting its early lead in Internet search.

Gonnet continued to work in a number of areas of computer science, including analysis of algorithms. In 1990, Gonnet moved from Waterloo to ETH in Switzerland. Among his projects since then have been Darwin,

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a bioinformatics system for the manipulation of genetic data, and
leadership of the OpenMath project to produce a standard
representation for mathematical objects. He has been involved in
several further startup companies, including Aruna, a relational
database company focused on business intelligence applications.",
keywords = "axiomref",
paper = "Gonn05.pdf"
}

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— axiom.bib —

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@inproceedings{Good91,
  author = "Goodwin, B. M. and Buonopane, R. A. and Lee, A.",
  title = "Using MathCAD in teaching material and energy balance concepts",
  booktitle = "Challenges of a Changing World",
  comment = "Proc. 1991 Ann. Conf., Amer. Soc. for Engineering Education",
  pages = "345-349",
  year = "1991",
  abstract =
    "We show how PC-based applications software, specifically MathCAD, is
    used in the teaching of material and energy balance concepts. MathCAD
    is a microcomputer software package which acts as a mathematical
    scratchpad. It has proven to be a very useful instructional tool in
    introductory chemical engineering courses. MathCAD solutions to
    typical course problems are presented.",
  keywords = "axiomref",
  beebe = "Goodwin:1991:UMT"
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— Goodwin:1991:UMT —

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@InProceedings{Goodwin:1991:UMT,
  author = "B. M. Goodwin and R. A. Buonopane and A. Lee",
  title = "Using {MathCAD} in teaching material and energy
    balance concepts",
  crossref = "Anonymous:1991:PAC",
  pages = "345--349 (vol. 1)",
  year = "1991",
  bibdate = "Tue Sep 17 06:37:45 MDT 1996",
  bibsource = "http://www.math.utah.edu/pub/tex/bib/axiom.bib",
  abstract = "The authors show how PC-based applications software,
    specifically MathCAD, is used in the teaching of

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material and energy balance concepts. MathCAD is a
microcomputer software package which acts as a
mathematical scratchpad. It has proven to be a very
useful instructional tool in introductory chemical
engineering courses. MathCAD solutions to typical
course problems are presented.",
acknowledgement = ack-nhfb,
affiliation = "Northeastern Univ., Boston, MA, USA",
classification = "C7450 (Chemical engineering); C7810C (Computer-aided
instruction)",
keywords = "Energy balance concepts; Instructional tool;
Introductory chemical engineering courses; MathCAD;
Mathematical scratchpad; PC-based applications
software",
language = "English",
thesaurus = "Chemical engineering computing; Computer aided
instruction; Microcomputer applications; Spreadsheet
programs",
}

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— ignore —

```

\bibitem[Golden 4]{GH84} Golden, V. Ellen; Hussain, M. A. (eds)
Proceedings of the 1984 MACSYMA Users' Conference:
Schenectady, New York, July 23-25, 1984, General Electric,
Schenectady, NY, USA, 1984
keywords = "axiomref"

```

— ignore —

```

\bibitem[Gonnet 96]{Gon96} Gonnet, Gaston H.
title = "Official version 1.0 of the Meta Content Dictionary",
link = "\url{http://www.inf.ethz.ch/personal/gonnet/ContDict/Meta}",
keywords = "axiomref"

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— axiom.bib —

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@inproceedings{Good93,

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author = "Goodloe, A. and Loustauanau, Philippe",
title = "An abstract data type development of graded rings",
booktitle = "Design and Implementation of Symbolic Computation Systems",
series = "Lecture Notes in Computer Science 721",
pages = "193-202",
isbn = "0-387-57272-4 (New York), 3-540-57272-4 (Berlin)",
year = "1993",
abstract =
    "Recently new computer algebra systems such as Scratchpad and Weyl have
    been developed with built in mechanisms for expressing abstract data types.
    These systems are object oriented in that they incorporate multiple
    inheritance and polymorphic types. Davenport and Trager have build much
    of the framework for basic commutative algebra in Scratchpad II
    utilizing its rich set of abstraction mechanisms. Davenport and Trager
    concentrated on developing factorization algorithms on domains which
    were abstract data types.

    We are taking a similar approach to the development of algorithms for
    computing in graded rings. The purpose of this paper is to develop the
    tools required to compute with polynomials with coefficients in a graded
    ring  $R[x]$ . We focus on graded rings  $R[x]$  which are polynomial rings graded
    by a monoid, and we allow partial orders on the monomials. The ideas
    presented here can be applied to more general graded rings  $R[x]$ , such as
    associated graded rings to filtered rings, as long as certain computational
    'requirements' are satisfied",
keywords = "axiomref",
beebe = "Goodloe:1993:ADT"
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— Goodloe:1993:ADT —

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@InProceedings{Goodloe:1993:ADT,
  author = "A. Goodloe and P. Loustauanau",
  title = "An abstract data type development of graded rings",
  crossref = "Fitch:1993:DIS",
  pages = "193--202",
  month = "",
  year = "1993",
  bibdate = "Tue Sep 17 06:37:45 MDT 1996",
  bibsource = "http://www.math.utah.edu/pub/tex/bib/axiom.bib",
  abstract = "Novel computer algebra systems such as Scratchpad and
    Weyl have been developed with built in mechanisms for
    expressing abstract data types. These systems are
    object oriented in that they incorporate multiple
    inheritance and polymorphic types. The authors are
    taking a similar approach to the development of

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        algorithms for computing in graded rings. They develop
        the tools required to compute with polynomials with
        coefficients in a graded ring R. They focus on graded
        rings R which are polynomial rings graded by a monoid,
        and allow partial orders on the monomials. The ideas
        presented can be applied to more general graded rings
        R, such as associated graded rings to filtered rings,
        as long as certain computational 'requirements' are
        satisfied.",
    acknowledgement = ack-nhfb,
    affiliation = "Dept. of Math., George Mason Univ., Fairfax, VA, USA",
    classification = "C4130 (Interpolation and function approximation);
        C6120 (File organisation); C7310 (Mathematics)",
    keywords = "Abstract data types; Computer algebra systems;
        Filtered rings; Graded rings; Monoid; Multiple
        inheritance; Object oriented; Partial orders;
        Polymorphic types; Polynomial rings; Scratchpad; Weyl",
    language = "English",
    thesaurus = "Abstract data types; Polynomials; Symbol
        manipulation",
}

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@misc{Grab98,
    author = "Grabe, Hans-Gert",
    title = "About the Polynomial System Solve Facility of Axiom, Macsyma,
        Maple Mathematica, MuPAD, and Reduce",
    link = "\url{https://www.informatik.uni-leipzig.de/~graebe/ComputerAlgebra/Publications/Wester",
    abstract =
        "We report on some experiences with the general purpose Computer
        Algebra Systems (CAS) Axiom, Macsyma, Maple, Mathematica, MuPAD, and
        Reduce solving systems of polynomial equations and the way they
        present their solutions. This snapshot (taken in the spring of 1996)
        of the current power of the different systems in a special area
        concentrates on both CPU-times and the quality of the output.",
    paper = "Grab98.pdf",
    keywords = "axiomref"
}

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— axiom.bib —

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@InProceedings{Grab02,
  author = "Grabe, Hans-Gert",
  title = "The SymbolicData Benchmark Problems Collection of Polynomial
    Systems",
  booktitle = "Workshop on Under- and Overdetermined Systems of Algebraic or
    Differential Equations",
  location = "Karlsruhe, Germany",
  pages = "57-76",
  year = "2002",
  link = "\url{http://symbolicdata.org/Papers/karlsruhe-02.pdf}",
  paper = "Grab02.pdf",
  keywords = "axiomref"
}

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— axiom.bib —

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@misc{Grab06,
  author = "Grabe, Hans-Gert",
  title = "The Groebner Factorizer and Polynomial System Solving",
  year = "2006",
  report = "Special Semester on Groebner Bases",
  location = "Linz",
  link = "\url{https://www.ricam.oeaw.ac.at/specsem/srs/groeb/download/06\_02\_Solver.pdf}",
  abstract =
    "Let  $S := k[x_1, \dots, x_n]$  be the polynomial ring in the
    variables  $x_1, \dots, x_n$  over the field  $k$  and
     $B := \{f_1, \dots, f_m\} \subset S$ 
    be a finite system of polynomials. Denote by  $I(B)$  the
    ideal generated by these polynomials. One of the major tasks of
    constructive commutative algebra is the derivation of information
    about the structure of
     $V(B) := \{a \in K^n : \text{for all } f \in B \text{ such that } f(a) = 0\}$ 
    the set of common zeroes of the system  $B$  over an
    algebraically closed extension  $K$  of  $k$ . Splitting the system into
    smaller ones, solving them separately, and patching all solutions
    together is often a good guess for a quick solution of even highly
    nontrivial problems. This can be done by several techniques, e.g.,
    characteristic sets, resultants, the Groebner factorizer or some ad
    hoc methods. Of course, such a strategy makes sense only for problems
    that really will split, i.e., for reducible varieties of
    solutions. Surprisingly often, problems coming from 'real life'
    fulfill this condition.

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Among the methods to split polynomial systems into smaller pieces probably the Groebner factorizer method attracted the most theoretical attention, see Czapor ([4, 5]), Davenport ([6]), Melenk, M

oller and Neun ([16, 17]) and Grabe ([13, 14]). General purpose Computer Algebra Systems (CAS) are well suited for such an approach, since they make available both a (more or less) well tuned implementation of the classical Groebner algorithm and an effective multivariate polynomial factorizer.

Furthermore it turned out that the Groebner factorizer is not only a good heuristic approach for splitting, but its output is also usually a collection of almost prime components. Their description allows a much deeper understanding of the structure of the set of zeroes compared to the result of a sole Groebner basis computation.

Of course, for special purposes a general CAS as a multipurpose mathematical assistant can't offer the same power as specialized software with efficiently implemented and well adapted algorithms and data types. For polynomial system solving, such specialized software has to implement two algorithmically complex tasks, solving and splitting, and until recently none of the specialized systems (as e.g., GB, Macaulay, Singular, CoCoA, etc.) did both efficiently. Meanwhile, being very efficient computing (classical) Groebner bases, development efforts are also directed, not only for performance reasons, towards a better inclusion of factorization into such specialized systems. Needless to remark that it needs some skill to force a special system to answer questions and the user will probably first try his 'home system' for an answer. Thus the polynomial systems solving facility of the different CAS should behave especially well on such polynomial systems that are hard enough not to be done by hand, but not really hard to require special efforts. It should invoke a convenient interface to get the solutions in a form that is (correct and) well suited for further analysis in the familiar environment of the given CAS as the personal mathematical assistant.",
 paper = "Grab06.pdf",
 keywords = "axiomref"
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@misc{Grab91,
  author = "Grabmeier, Johannes and Huber, K. and Krieger, U.",
  title = "Das ComputeralgebraSystem AXIOM bei kryptologischen und
    verkehrstheoretischen Untersuchungen des Forschungsinstituts
    der Deutschen Bundespost TELEKOM",
  type = "technical report",
  number = "TR 75.91.20",
  location = "Heidelberg, Germany",
  year = "1991",
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keywords = "axiomref",
beebe = "Grabmeier:1991:CSA"
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— Grabmeier:1991:CSA —

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@TechReport{Grabmeier:1991:CSA,
  author = "J. Grabmeier and K. Huber and U. Krieger",
  title = "{Das Computeralgebra-System AXIOM bei kryptologischen
    und verkehrstheoretischen Untersuchungen des
    Forschungsinstituts der Deutschen Bundespost TELEKOM}",
  type = "Technischer Report",
  number = "TR 75.91.20",
  institution = "IBM Wissenschaftliches Zentrum",
  address = "Heidelberg, Germany",
  pages = "??",
  year = "1991",
  bibsource = "/usr/local/src/bib/bibliography/Theory/Comp.Alg.bib;
    http://www.math.utah.edu/pub/tex/bib/axiom.bib",
}

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@article{Grab91b,
  author = "Grabmeier, Johannes",
  title = "Axiom, ein Computeralgebrasystem mit abstrakten Datentypen",
  journal = "mathPAD",
  volume = "1",
  number = "3",
  pages = "13-15",
  year = "1991",
  paper = "Grab91b.pdf",
  keywords = "axiomref"
}

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\bibitem[Grabmeier 03]{GKW03} Grabmeier, Johannes; Kaltoven, Erich;

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Weispfenning, Volker (eds)

Computer algebra handbook: foundations, applications, systems.

Springer-Verlag, Berlin, Germany / Heidelberg, Germany / London, UK / etc.,

2003. ISBN 3-540-65466-6 637pp Includes CDR0M

link = "\url{http://www.springer.com/sgw/cda/frontpage/0,11855,1-102-22-1477871-0,00.html}",

keywords = "axiomref"

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@book{Gree01,
  author = "Green, Edward L.",
  title = "Symbolic Computation: Solving Equations in Algebra, Geometry, and
    Engineering",
  booktitle = "Proc. AMS-IMS-SIAM Joint Summer Research Conference on Symbolic
    Computation",
  volume = "232",
  publisher = "American Mathematical Society",
  year = "2001",
  abstract =
    "This volume contains papers related to the research conference,
    ‘Symbolic Computation: Solving Equations in Algebra, Analysis, and
    Engineering,’ held at Mount Holyoke College (MA). It provides a broad
    range of active research areas in symbolic computation as it applies
    to the solution of polynomial systems. The conference brought together
    pure and applied mathematicians, computer scientists, and engineers,
    who use symbolic computation to solve systems of equations or who
    develop the theoretical background and tools needed for this
    purpose. Within this general framework, the conference focused on
    several themes: systems of polynomials, systems of differential
    equations, noncommutative systems, and applications.",
  keywords = "axiomref"
}
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@InProceedings{Grie71,
  author = "Griesmer, James H. and Jenks, Richard D.",
  title = "SCRATCHPAD/1 -- an interactive facility for symbolic mathematics",
  booktitle = "Proc. second ACM Symposium on Symbolic and Algebraic
    Manipulation",
  series = "SYMSAC 71",
  year = "1971",
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pages = "42--58",
doi = "http://dx.doi.org/10.1145806266",
link = "\url{http://delivery.acm.org/10.1145/810000/806266/p42-griesmer.pdf}",
abstract = "
  The SCRATCHPAD/1 system is designed to provide an interactive symbolic
  computational facility for the mathematician user. The system features
  a user language designed to capture the style and succinctness of
  mathematical notation, together with a facility for conveniently
  introducing new notations into the language. A comprehensive system
  library incorporates symbolic capabilities provided by such systems as
  SIN, MATHLAB, and REDUCE.",
paper = "Grie71.pdf",
keywords = "axiomref",
beebe = "Griesmer:1971:SIF"
}

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— Griesmer:1971:SIF —

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@InProceedings{Griesmer:1971:SIF,
  author = "J. H. Griesmer and R. D. Jenks",
  title = "{SCRATCHPAD/1} --- an interactive facility for
    symbolic mathematics",
  crossref = "Petrick:1971:PSS",
  pages = "42--58",
  year = "1971",
  DOI = "http://dx.doi.org/10.1145806266",
  bibdate = "Thu Jul 26 08:45:53 2001",
  bibsource = "/usr/local/src/bib/bibliography/Theory/obscure.bib;
    http://www.math.utah.edu/pub/tex/bib/axiom.bib",
  link = "\url{http://delivery.acm.org/10.1145/810000/806266/p42-griesmer.pdf}",
}

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— axiom.bib —

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@techreport{Grie72a,
  author = "Griesmer, James H. and Jenks, Richard D.",
  title = "Experience with an online symbolic math system SCRATCHPAD",
  institution = "IBM",
  year = "1972",
  isbn = "0-903796-02-3",
  keywords = "axiomref",
  beebe = "Griesmer:1972:EOSb"
}

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— Griesmer:1972:EOSb —

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@InProceedings{Griesmer:1972:EOSb,
  author = "J. Griesmer and R. Jenks",
  title = "Experience with an online symbolic math. system
           {SCRATCHPAD}",
  crossref = "Online:1972:OCP",
  pages = "??--??",
  year = "1972",
  bibsource = "/usr/local/src/bib/bibliography/Distributed/QLD.bib;
              http://www.math.utah.edu/pub/tex/bib/axiom.bib",
  bydate = "Le",
  byrev = "Le",
  date = "00/00/00",
  descriptors = "Formula manipulation",
  enum = "1209",
  language = "English",
  location = "PKI-OG: Li-Ord.Le",
  references = "0",
  revision = "21/04/91",
}
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— axiom.bib —

```
@article{Grie72,
  author = "Griesmer, James H. and Jenks, Richard D.",
  title = "SCRATCHPAD: A capsule view",
  journal = "ACM SIGPLAN Notices",
  volume = "7",
  number = "10",
  pages = "93-102",
  year = "1972",
  comment = "Proc. Symp. Two-dimensional man-machine communications",
  doi = "http://dx.doi.org/10.1145807019",
  abstract =
    "SCRATCHPAD is an interactive system for algebraic manipulation
    available under the CP/CMS time-sharing system at Yorktown Heights. It
    features an extensible declarative language for the interactive
    formulation of symbolic computations. The system is a large and
    complex body of LISP programs incorporating significant portions of
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other symbolic systems. Here we present a capsule view of SCRATCHPAD,
its language and its capabilities. This is followed by an example
which illustrates its use in an application involving the solution of
an integral equation.",
keywords = "axiomref",
beebe = "Griesmer:1972:SCV"
}

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— Griesmer:1972:SCV —

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@Article{Griesmer:1972:SCV,
  author =      "James H. Griesmer and Richard D. Jenks",
  title =      "{SCRATCHPAD}: {A} capsule view",
  journal =     j-SIGPLAN,
  volume =     "7",
  number =     "10",
  pages =      "93--102",
  year =       "1972",
  CODEN =      "SINODQ",
  DOI =        "http://dx.doi.org/10.1145807019",
  ISSN =       "0362-1340 (print), 1523-2867 (print), 1558-1160
               (electronic)",
  ISSN-L =     "0362-1340",
  bibdate =    "Thu Jul 26 10:33:16 2001",
  bibsource =  "http://www.math.utah.edu/pub/tex/bib/axiom.bib",
  note =       "Proceedings of the symposium on Two-dimensional
               man-machine communication, Mark B. Wells and James B.
               Morris (eds.)",
  acknowledgement = ack-nhfb,
  bookpages =  "iii + 160",
  fjournal =    "ACM SIGPLAN Notices",
  link =       "\url{http://portal.acm.org/browse_dl.cfm?idx=J706}",
}

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— axiom.bib —

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@article{Grie74,
  author = "Griesmer, James H. and Jenks, Richard D.",
  title = "A solution to problem \#4: the lie transform",
  journal = "SIGSAM Bulletin",
  volume = "8",
  number = "4",

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pages = "12-13",
year = "1974",
abstract =
  "The following SCRATCHPAD conversation for carrying out the Lie
  Transform computation represents a slight modification of one written
  by Dr. David Barton, when he was a summer visitor during 1972 at the
  Watson Research Center.",
keywords = "axiomref"
}

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— axiom.bib —

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@techreport{Grie75,
  author = "Griesmer, James H. and Jenks, Richard D. and Yun, David Y.Y",
  title = "SCRATCHPAD User's Manual",
  institution = "IBM",
  year = "1975",
  type = "Research Report",
  number = "RA70",
  keywords = "axiomref"
}

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— axiom.bib —

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@article{Grie75a,
  author = "Griesmer, James H. and Jenks, Richard D. and Yun, David Y.Y.",
  title = "A SCRATCHPAD solution to problem \#7",
  journal = "SIGSAM",
  volume = "9",
  number = "3",
  pages = "13-17",
  year = "1975"
}

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@article{Grie75b,

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author = "Griesmer, James H. and Jenks, Richard D. and Yun, David Y.Y.",
title = "A FORMAT statement in SCRATCHPAD",
journal = "SIGSAM",
volume = "9",
number = "3",
pages = "24-25",
year = "1975",
abstract =
  "Algebraic manipulation covers branches of software, particularly list
  processing, mathematics, notably logic and number theory, and
  applications largely in physics. The lectures will deal with all of these
  to a varying extent.",
keywords = "axiomref"
}

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— ignore —

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\bibitem[Griesmer 76]{GJY76} Griesmer, J.H.; Jenks, R.D.; Yun, D.Y.Y
  title = "A Set of SCRATCHPAD Examples",
  April 1976 (private copy)
  keywords = "axiomref"

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— axiom.bib —

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@article{Grie79,
  author = "Griesmer, James H.",
  title = "The state of symbolic computation",
  journal = "SIGSAM Bulletin",
  volume = "13",
  number = "3",
  pages = "25-28",
  year = "1979",
  keywords = "axiomref"
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@article{Grun94,

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author = "Gruntz, Dominik and Monagan, Michael B.",
title = "Introduction to Gauss",
journal = "SIGSAM Bulletin",
volume = "28",
number = "3",
pages = "3-19",
year = "1994",
link = "\url{http://ftp.cecm.sfu.ca/personal/monaganm/papers/Gauss.pdf}",
abstract =
  "The Gauss package offers Maple users a new approach to programming
  based on the idea of parameterized types (domains) which is central to
  the AXIOM system. This approach to programming is now regarded by many
  as the right way to go in computer algebra systems design. In this
  article, we describe how Gauss is designed and show examples of usage.
  We end with some comments about how Gauss is being used in Maple.",
paper = "Grun94.pdf",
keywords = "axiomref",
beebe = "Gruntz:1994:IG"
}

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— Gruntz:1994:IG —

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@Article{Gruntz:1994:IG,
  author =      "D. Gruntz and M. Monagan",
  title =      "Introduction to {Gauss}",
  journal =     j-SIGSAM,
  volume =     "28",
  number =     "2",
  pages =      "3--19",
  month =      aug,
  year =       "1994",
  CODEN =      "SIGSBZ",
  ISSN =       "0163-5824 (print), 1557-9492 (electronic)",
  ISSN-L =     "0163-5824",
  bibdate =    "Tue Dec 12 09:33:35 MST 1995",
  bibsource =  "http://www.math.utah.edu/pub/tex/bib/axiom.bib",
  abstract =   "The Gauss package offers Maple users a new approach to
  programming based on the idea of parameterized types
  (domains) which is central to the AXIOM system. This
  approach to programming is now regarded by many as the
  right way to go in computer algebra systems design. We
  describe how Gauss is designed and show examples of
  usage. We end with some comments about how Gauss is
  being used in Maple.",
  acknowledgement = ack-nhfb,
  affiliation = "Inst. for Sci. Comput., Eidgenossische Tech.

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        Hochschule, Zurich, Switzerland",
classification = "C6110 (Systems analysis and programming); C6130
                (Data handling techniques); C7310 (Mathematics
                computing)",
fjournal =      "SIGSAM Bulletin",
keywords =      "AXIOM system; Computer algebra systems design; Gauss
                package; Maple users; Parameterized types;
                Programming",
language =      "English",
pubcountry =    "USA",
thesaurus =      "Programming environments; Software packages; Symbol
                manipulation; Systems analysis; Type theory",
}

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— axiom.bib —

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@phdthesis{Grun96,
  author = "Gruntz, Dominik",
  title = "On Computing Limits in a Symbolic Manipulation System",
  school = "Swiss Federal Institute of Technology Zurich",
  year = "1996",
  link = "\url{http://www.cybertester.com/data/gruntz.pdf}",
  abstract = "
    This thesis presents an algorithm for computing (one-sided) limits
    within a symbolic manipulation system. Computing limits is an
    important facility, as limits are used both by other functions such as
    the definite integrator and to get directly some qualitative
    information about a given function.

    The algorithm we present is very compact, easy to understand and easy
    to implement. It overcomes the cancellation problem other algorithms
    suffer from. These goals were achieved using a uniform method, namely
    by expanding the whole function into a series in terms of its most
    rapidly varying subexpression instead of a recursive bottom up
    expansion of the function. In the latter approach exact error terms
    have to be kept with each approximation in order to resolve the
    cancellation problem, and this may lead to an intermediate expression
    swell. Our algorithm avoids this problem and is thus suited to be
    implemented in a symbolic manipulation system.",
  paper = "Grun96.pdf",
  keywords = "axiomref"
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@misc{Gute16,
  author = "Gutenberg Self-Publishing Press",
  title = "OpenAxiom",
  link = "\url{http://self.gutenberg.org/articles/openaxiom}",
  year = "2016",
  keywords = "axiomref"
}
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— axiom.bib —

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@article{Harr98,
  author = "Harrison, J. and Thery, L.",
  title = "A Skeptic's approach to combining HOL and Maple",
  journal = "J. Autom. Reasoning",
  volume = "21",
  number = "3",
  pages = "279-294",
  year = "1998",
  link = "\url{http://www.cl.cam.ac.uk/~jrh13/papers/cas.ps.gz}",
  abstract =
    "We contrast theorem provers and computer algebra systems, pointing
    out the advantages and disadvantages of each, and suggest a simple way
    to achieve a synthesis of some of the best features of both. Our
    method is based on the systematic separation of search for a solution
    and checking the solution, using a physical connection between
    systems. We describe the separation of proof search and checking in
    some detail, relating it to proof planning and to the complexity class
    NP, and discuss different ways of exploiting a physical link between
    systems. Finally, the method is illustrated by some concrete examples
    of computer algebra results proved formally in the HOL theorem prover
    with the aid of Maple.",
  paper = "Harr98.pdf",
  keywords = "axiomref"
}
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— axiom.bib —

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@misc{Harr12,
  author = "Harriss, Edmund and Daly, Timothy",
  title = "Have we ever lost mathematics?",
  link = "\url{https://maxwelldemon.com/2012/05/09/have-we-ever-lost-mathematics/}",
  year = "2012",
  keywords = "axiomref"
}
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— ignore —

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\bibitem[Hassner 87]{HBW87} Hassner, Martin; Burge, William H.;
Watt, Stephen M.
  title = "Construction of Algebraic Error Control Codes (ECC) on the Elliptic Riemann Surface",
in [Wit87], pp5-8
  keywords = "axiomref"
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— axiom.bib —

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@inproceedings{Kead93a,
  author = "Keady, G. and Richardson, M.G.",
  title = "An application of IRENA to systems of nonlinear equations arising
          in equilibrium flows in networks",
  booktitle = "Proc. ISSAC 1993",
  series = "ISSAC '93",
  year = "1993",
  abstract =
    "IRENA - an interface from REDUCE to NAG - runs under the REDUCE
    Computer Algebra (CA) system and provides an interactive front end to
    the NAG Fortran Library.

    Here IRENA is tested on a problem closer to an engineering problem
    than previously published examples. We also illustrate the use of the
    {\tt codeonly} switch, which is relevant to larger scale problems. We
    describe progress on an issue raised in the 'Future Developments'
    section in our {\sl SIGSAM Bulletin} article [2]: the progress improves
    the practical effectiveness of IRENA.",
  paper = "Kead93a.pdf",
  keywords = "axiomref"
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@inproceedings{Hawk95,
  author = "Hawkes, Evatt and Keady, Grant",
  title = "Two more links to NAG numerics involving CA systems",
  booktitle = "IMACS Applied Computer Algebra Conference",
  location = "University of New Mexico",
  year = "1995",
  algebra =
    "\newline\refto{domain ASP1 Asp1}
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    \newline\refto{domain ASP9 Asp9}",
  abstract =
    "The 'more' in the title is because this paper is a sequel to papers
    by Keving Broughan, [BKRRD,BK]. For some years GK has had interests in
    (i) interactive front-ends to numeric computation, such as the
    NAG/IMSL library computation, and (ii) Fortran code generation for
    Argument SubPrograms (ASPs), such as those needed by some NAG/IMSL
    routines. Demonstrations of three links to the NAG library are
    described in [BKRRD]. A description of a link to NAG from Macsyma
    which was mentioned, but not in a sufficiently advanced state to
    demonstrate in early 1991, is given in [BK]. The situation at the end
    of 1991 was that there were links to NAG involving each of Macsyma,
```

REDUCE and Mathematica. The links are called Naglink, IRENA and InterCall, respectively. The principal authors of IRENA are Mike Dewar and Mike Richardson. InterCall is not specific to the NAG library; indeed InterCall is used with calls to IMSL and to elsewhere at the conference venue, the University of New Mexico.

The two further links to NAG library treated in this paper are AXIOM2.0 and genmex/ESC, genmex allows calls to NAG from Matlab. genmex can be regarded as similar to InterCall: genmex uses Matlab's mex files in a similar way to InterCall's use of Mathematica's MathLink. Again genmex is not specific to the NAG library. Mike Dewar is an author both of IRENA and the AXIOM2.0 link to the NAG library: see [D] for discussion of the differences between the IRENA project and the AXIOM-NAG link project.",

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paper = "Hawk95.pdf",
keywords = "axiomref"
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@inproceedings{Hear80,
  author = "Hearn, Anthony C.",
  title = "Symbolic Computation and its Application to High Energy Physics",
  booktitle = "Proc. 1980 CERN School of Computing",
  pages = "390-406",
  year = "1980",
  link = "\url{http://www.iaea.org/inis/collection/NCLCollectionStore/_Public/12/631/12631585.pdf}",
  abstract =
    "It is clear that we are in the middle of an electronic revolution
    whose effect will be as profound as the industrial revolution. The
    continuing advances in computing technology will provide us with
    devices which will make present day computers appear primitive. In
    this environment, the algebraic and other non-numerical capabilities
    of such devices will become increasingly important. These lectures
    will review the present state of the field of algebraic computation
    and its potential for problem solving in high energy physics and
    related areas. We shall begin with a brief description of the
    available systems and examine the data objects which they consider.
    As an example of the facilities which these systems can offer, we
    shall then consider the problem of analytic integration, since this
    is so fundamental to many of the calculational techniques used by high
    energy physicists. Finally, we shall study the implications which the
    current developments in hardware technology hold for scientific
    problem solving.",
  paper = "Hear80.pdf"
}
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— axiom.bib —

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@article{Hear95,
  author = "Hearn, Anthony C. and Eberhard, Schrufer",
  title = "A computer algebra system based on order-sorted algebra",
  journal = "J. Symbolic Computing",
  volume = "19",
  number = "1-3",
  pages = "65-77",
  year = "1995",
  abstract =
    "This paper presents the prototype design of an algebraic computation
    system that manipulates algebraic quantities as generic objects using
    order-sorted algebra as the underlying model. The resulting programs
    have a form that is closely related to the algorithmic description of
    a problem, but with the security of full type checking in a compact,
    natural style.",
  keywords = "axiomref"
}
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— axiom.bib —

```
@book{Heck93,
  author = "Heck, Andre",
  title = "Introduction to Maple",
  year = "1993",
  publisher = "Springer-Verlag",
  abstract =
    "This is an introductory book on one of the most powerful computer
    algebra systems, viz, Maple: The primary emphasis in this book is on
    learning those things that can be done with Maple and how it can be
    used to solve mathematical problems. In this book usage of Maple as a
    programming language is not discussed at a higher level than that of
    defining simple procedures and using simple language constructs.
    However, the Maple data structures are discussed in detail."

    This book is divided into eighteen chapters spanning a variety of
    topics. Starting with an introduction to symbolic computation and
    other similar computer algebra systems, this book covers several
    topics like polynomials and rational functions, series,
    differentiation and integration, differential equations, linear
```

algebra, 2-D and 3-D graphics, etc. The applications covered include kinematics of the Stanford manipulator, a 3-component model for cadmium transfer through the human body, molecular-orbital Hckel theory, prolate spheroidal coordinates and Moore-Penrose inverses.

At the end of each chapter, a good number of excercises is given. A list of relevant references is also given at the end of the book.

This book is very useful to all users of Maple package.",

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keywords = "axiomref"
}
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```
\bibitem[Heck 01]{Hec01} Heck, A.
  title = "Variables in computer algebra, mathematics and science",
  The International Journal of Computer Algebra in Mathematics Education
  Vol. 8 No. 3 pp195-210 (2001)
  keywords = "axiomref"
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@phdthesis{Hemm03,
  author = "Hemmecke, Ralf",
  title = "Involutive Bases for Polynomial Ideals",
  school = "Johannes Kepler University, RISC",
  year = "2003",
  abstract =
    "This thesis contributes to the theory of polynomial involutive
    bases. Firstly, we present the two existing theories of involutive
    divisions, compare them, and come up with a generalised approach of
    {\sl suitable partial divisions}. The thesis is built on this
    generalized approach. Secondly, we treat the question of choosing a
    ‘‘good’’ suitable partial division in each iteration of the involutive
    basis algorithm. We devise an efficient and flexible algorithm for
    this purpose, the {\sl Sliced Division} algorithm. During the
    involutive basis algorithm, the Sliced Division algorithm contributes
    to an early detection of the involutive basis property and a
    minimisation of the number of critical elements. Thirdly, we give new
    criteria to avoid unnecessary reductions in an involutive basis
    algorithm. We show that the termination property of an involutive
    basis algorithm which applies our criteria is independent of the
    prolongation selection strategy used during its run. Finally, we
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    present an implementation of the algorithm and results of this thesis
    in our software package CALIX."
}

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@misc{RISC06,
  author = "Hemmecke, Ralf and Rubey, Martin",
  title = "AXIOM Workshop 2006",
  link = "\url{http://axiom-wiki.newsynthesis.org/WorkShopRISC2006}",
  year = "2006",
  location = "Hagenberg, Austria",
  abstract =
    "Axiom is a computer algebra system with a long tradition. It recently
    became free software.

    The workshop aims at a cooperation of Axiom developers with developers
    of packages written for other Computer Algebra Systems or developers
    of stand-alone packages. Furthermore, the workshop wants to make the
    potential of Axiom and Aldor more widely known in order to attract new
    users and new developers.",
  keywords = "axiomref"
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@misc{RISC07,
  author = "Hemmecke, Ralf and Rubey, Martin",
  title = "AXIOM Workshop 2007",
  link = "\url{http://axiom-wiki.newsynthesis.org/WorkShopRISC2007}",
  year = "2007",
  location = "Hagenberg, Austria",
  abstract =
    "The workshop aims at a cooperation of Axiom developers with deelopers
    of packages written for other Computer Algebra Systems, and
    mathematicians that would like to use a computer algebra system to
    perform experiments.

    One goal of the workshop is to learn about the mathematical theory,
    the design of packages written for other CAS and to make those
    functionalities available in Axiom." ,
  keywords = "axiomref"
}

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}

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@misc{Hera16,
  author = "Heras, Jonathan and Martin-Mateos, Franciso Jesus and
    Pascual, Vico",
  title = "A Hierarchy of Mathematical Structures in ACL2",
  link = "\url{http://staff.computing.dundee.ac.uk/jheras/papers/ahomsia.pdf}",
  abstract =
    "In this paper, we present a methodology which allows one to deal with
    {\sl mathematical structures} in the ACL2 theorem prover. Namely, we
    cope with the representation of mathematical structures, the
    certification that an object fulfills the axioms characterizing an
    algebraic structure and the generation of generic theories about
    concrete structures. As a by-product, an {\sl ACL2 algebraic
    hierarchy} has been obtained. Our framework has been tested with the
    definition of {\sl homology groups}, an example coming from
    Homological Algebra which involves several notions related to
    Universal Algebra. The method presented here, when compared to a
    from-scratch approach, is preferred when working with complex
    mathematical structures; for instance, the ones coming from Algebraic
    Topology. The final aim of this work is the verification of Computer
    Algebra systems, a field where our hierarchy fits better than the ones
    developed in other systems.",
  paper = "Hera16.pdf",
  keywords = "axiomref"
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@article{Hera15,
  author = "Heras, Jonathan and Martin-Mateos, Franciso Jesus and
    Pascual, Vico",
  title = "Modelling algebraic structures and morphisms in ACL2",
  journal = "Appl. Algebra Eng. Commun. Comput.",
  volume = "26",
  number = "3",
  pages = "277-303",
  year = "2015",
  abstract =
    "In this paper, we present how algebraic structures and morphisms can
```

be modelled in the ACL2 theorem prover. Namely, we illustrate a methodology for implementing a set of tools that facilitates the formalisations related to algebraic structures -- as a result, an algebraic hierarchy ranging from setoids to vector spaces has been developed. The resultant tools can be used to simplify the development of generic theories about algebraic structures. In particular, the benefits of using the tools presented in this paper, compared to a from-scratch approach, are especially relevant when working with complex mathematical structures; for example, the structures employed in Algebraic Topology. This work shows that ACL2 can be a suitable tool for formalising algebraic concepts coming, for instance, from computer algebra systems.",

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keywords = "axiomref"
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@misc{Here96,
  author = "Hereman, Willy",
  title = "The Incredible World of Symbolic Mathematics
    A Review of Computer Algebra Systems",
  year = "1996",
  link = "\url{https://inside.mines.edu/~whereman/papers/Hereman-PhysicsWorld-9-March1996.pdf}",
  paper = "Here96.pdf",
  keywords = "axiomref"
}

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@article{Here97,
  author = "Hereman, Willy",
  title = "Review of Symbolic Software for Lie Symmetry Analysis",
  journal = "Math. Comput. Modelling",
  volume = "25",
  number = "8/9",
  pages = "115-132",
  year = "1997",
  abstract =
    "Sophus Lie (1842-1899) pioneered the study of continuous
    transformation groups that leave systems of differential equations
    invariant. Lie's work [1-3] brought diverse and ad hoc integration
    methods for solving special classes of differential equations under a

```

common conceptual umbrella. Indeed, Lie's infinitesimal transformation method provides a widely applicable technique to find closed form solutions of ordinary differential equations (ODEs). Standard solution methods for first-order or linear ODEs can be characterized in terms of symmetries. Through the group classification of ODEs, Lie succeeded in identifying all ODEs that can either be reduced to lower-order ones or be completely integrated via group theoretic techniques.

Applied to partial differential equations (PDEs), Lie's method [2] leads to group-invariant solutions and conservation laws. Exploiting the symmetries of PDEs, new solutions can be derived from known ones, and PDEs can be classified into equivalence classes. Furthermore, group-invariant solutions obtained via Lie's approach may provide insight into the physical models themselves, and explicit solutions can serve as benchmarks in the design, accuracy testing, and comparison of numerical algorithms.

Nowadays, the concept of symmetry plays a key role in the study and development of mathematics and physics. Indeed, the theory of Lie groups and Lie algebras is applied to diverse fields of mathematics including differential geometry, algebraic topology, bifurcation theory, to name a few. Lie's original ideas greatly influenced the study of physically important systems of differential equations in classical and quantum mechanics, fluid dynamics, elasticity, and many other applied areas [4-81].

The application of Lie group methods to concrete physical systems involves tedious computations. Even the calculation of the continuous symmetry group of a modest system of differential equations is prone to errors, if done with pencil and paper. Computer algebra systems (CAS) such as Mathematica, MACSYMA, Maple, REDUCE, AXIOM and MuPAD are extremely useful for such computations. Symbolic packages [9-11], written in the language of these CAS, can find the determining equations of the Lie symmetry group. The most sophisticated packages then reduce these into an equivalent but more suitable system, subsequently solve that system in closed form, and go on to calculate the infinitesimal generators that span the Lie algebra of symmetries.

In Section 2, we discuss methods and algorithms used in the computation of Lie symmetries. We address the computation of determining systems, their reduction to standard form, solution techniques, and the computation of the size of the symmetry group. In Section 3, we look beyond Lie-point symmetries, addressing contact and generalized symmetries, as well as nonclassical or conditional symmetries.

Section 4 is devoted to a review of modern Lie symmetry programs, classified according to the underlying CAS. The review focuses on Lie symmetry software for classical Lie-point symmetries,

contact (or dynamical), generalized (or Lie-Backlund) symmetries, nonclassical (or conditional) symmetries. Most of these packages were written in the last decade. Researchers interested in details about pioneering work should consult [9,10,12]. In Section 5, two examples illustrate results that can be obtained with Lie symmetry software. In Section 6 we draw some conclusions.

Lack of space forces us to give only a few key references for the Lie symmetry packages. A comprehensive survey of the literature devoted to theoretical as well as computational aspects of Lie symmetries, with over 300 references, can be found elsewhere [11].",

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paper = "Here97.pdf",
keywords = "axiomref"
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@article{Hive04,
  author = "Hivert, Florent and Thiery, Nicolas M.",
  title = "MuPAD-Combinat, an open-source package for research in
    algebraic combinatorics",
  journal = "Seminaire Lotharingien de Combinatoire",
  volume = "51",
  number = "B51z",
  year = "2004",
  link = "\url{http://www.emis.de/journals/SLC/wpapers/s51thiery.pdf}",
  abstract =
    "In this article we give an overview of the MuPAD-Combinat open-source
    algebraic combinatorics package for the computer algebra system MuPAD
    2.0.0 and higher. This includes our motivations for developing yet
    another combinatorial software, a tutorial introduction with lots of
    examples, as well as notes on the general design. The material
    presented here is also available as a part of the MuPAD-Combinat
    handbook; further details and references on the algorithms used can be
    found there. The package and the handbook are available from the web
    page, together with download and installation instructions, mailing
    lists, etc.",
  paper = "Hive04.pdf",
  keywords = "axiomref"
}
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@article{Hoan00,
  author = "Hoang, Ngoc Minh and Petitot, Michel and Van er Hoeven, Joris",
  title = "Shuffle algebra and polylogarithms",
  journal = "Discrete Math.",
  volume = "225",
  number = "1-3",
  pages = "217-230",
  year = "2000",
  abstract =
    "Generalized polylogarithms are defined as iterated integrals with
    respect to the two differential forms  $\omega_0 = \frac{dz}{z}$  and
     $\omega_1 = \frac{dz}{(1-z)}$ . We give an algorithm which computes the
    monodromy of these special functions. This algorithm, implemented in
    AXIOM, is based on the computation of the associator  $\Phi_{KZ}$  of
    Drinfeld, in factorized form. The monodromy formulae involve special
    constants, called multiple zeta values. We prove that the algebra of
    polylogarithms is isomorphic to a shuffle algebra.",
  paper = "Hoan00.pdf",
  keywords = "axiomref"
}

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— axiom.bib —

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@article{Hoar08,
  author = "Hoarau, Emma and David, Claire",
  title = "Lie group computation of finite difference schemes",
  year = "2008",
  journal = "math.NA",
  link = "\url{http://arxiv.org/pdf/math/0611895.pdf}",
  abstract =
    "A Mathematica based program has been elaborated in order to determine
    the symmetry group of a finite difference equation. The package
    provides functions which enable use to solve the determining equations
    of the related Lie group.",
  paper = "Hoar08.pdf",
  keywords = "axiomref"
}

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— axiom.bib —

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@phdthesis{Hodo11,
  author = "Hodorog, Madalina",

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title = "Symbolic-Numeric Algorithms for Plane Algebraic Curves",
year = "2011",
school = "RISC Research Institute for Symbolic Computation",
abstract =
  "In computer algebra, the problem of computing topological invariants
  (i.e. delta-invariant, genus) of a plan complex algebraic curve is
  well-understood if the coefficients of the defining polynomial of the
  curve are exact data (i.e. integer numbers or rational numbers). The
  challenge is to handle this problem if the coefficients are inexact
  (i.e. numerical values).

  In this thesis, we approach the algebraic problem of computing
  invariants of a plane complex algebraic curve defined by a polynomial
  with both exact and inexact data. For the inexact data, we associate a
  positive real number called {\sl tolerance} or {\sl noise}, which
  measures the error level in the coefficients. We deal with an {\sl
  ill-posed} problem in the sense that, tiny changes in the input data
  lead to dramatic modifications in the output solution.

  For handling the ill-posedness of the problem we present a {\sl
  regularization} method, which estimates the invariants of a plane
  complex algebraic curve. Our regularization method consists of a set
  of {\sl symbolic-numeric algorithms} that extract structural
  information on the input curve, and of a {\sl parameter choice rule},
  i.e. a function in the noise level. We first design the following
  symbolic-numeric algorithms for computing the invariants of a plane
  complex algebraic curve:
  \begin{itemize}
  \item we compute the link of each singularity of the curve by numerical
  equation solving
  \item we compute the Alexander polynomial of each link by using
  algorithms from computational geometry (i.e. an adapted version of
  the Bentley-Ottmann algorithm) and combinatorial objects from knot
  theory.
  \item we derive a formula for the delta-invariant and for the genus
  \end{itemize}

  We then prove that the symbolic-numeric algorithms together with the
  parameter choice rule compute approximate solutions, which satisfy the
  {\sl convergence for noisy data property}. Moreover, we perform
  several numerical experiments, which support the validity for the
  convergence statement.

  We implement the designed symbolic-numeric algorithms in a new
  software package called {\sl Genom3ck}, developed using the {\sl Axel}
  free algebraic modeler and the {\sl Mathmagix} free computer algebra
  system. For our purpose, both of these systems provide modern
  graphical capabilities, and algebraic and geometric tools for
  manipulating algebraic curves and surfaces defined by polynomials with
  both exact and inexact data. Together with its main functionality to

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compute the genus, the package {\sl Genom3ck} computes also other type
of information on a plane complex algebraic curve, such as the
singularities of the curve in the projective plane and the topological
type of each singularity.",
paper = "Hodo11.pdf"
}

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— axiom.bib —

```

@misc{Hoep95,
  author = "Hoepfner, Sabine",
  title = "Linear differential equations of second order in fields of
    positive characteristic",
  comment = "Essen: Univ. Essen, FB Math 71 S.",
  year = "1995",
  abstract =
    "Let be a differential field of characteristic $p>2$ of the type
    $\mathbb{F}_p(x), \frac{d}{dx}$. The equation studied is
    $y^{\prime\prime}+ay^{\prime}+by=0$ with $a,b \in K$. The goal is to
    produce a Liouvillian extension $L \supset K$ which contains two
    independent solutions. This is done by solving the associated Riccati
    equation $u^{\prime}+u^2+au+b=0$. Unlike the characteristic zero
    situation, the Riccati equation has one or two solutions in an
    algebraic extension of degree $\le 2$ of $K$. Full solutions are
    given for the various cases that do occur. The paper ends with a
    program in AXIOM which computes the solutions of the Riccati equation
    and the Liouvillian extensions.",
  keywords = "axiomref"
}

```

— axiom.bib —

```

@misc{Hoev13,
  author = "Hoeven, Joris van der and Lecerf, Gregoire",
  title = "Interfacing Mathemagix with C++",
  link = "\url{http://www.texmacs.org/joris/mmxcpp/mmxcpp.pdf}",
  abstract =
    "In this paper, we give a detailed description of the interface
    between the Mathemagix language and C++. In particular, we describe
    the mechanism which allows us to import a C++ template library
    (which only permits static instantiation) as a fully generic
    Mathemagix template library.",

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    paper = "Hoev13.pdf",
    keywords = "axiomref"
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— axiom.bib —

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@misc{Hoev15,
  author = "Hoeven, Joris van der and Lecerf, Gregoire",
  title = "Interfacing Mathemagix with C++",
  link = "\url{http://www.texmacs.org/joris/mmxcpp/mmxcpp.html}",
  abstract =
    "In this paper, we give a detailed description of the interface
    between the Mathemagix language and C++. In particular, we describe
    the mechanism which allows us to import a C++ template library
    (which only permits static instantiation) as a fully generic
    Mathemagix template library.",
  keywords = "axiomref"
}

```

— axiom.bib —

```

@book{Hold11,
  author = "Hohold, Tom and van Lint, Jacobus H. and Pellikaan, Ruud",
  title = "Algebraic Geometry Codes",
  year = "2011",
  pages = "871-961",
  booktitle = "Handbook of Coding Theory",
  publisher = "V.S Pless, W.C. Huffman and R.A. Brualdi (eds)",
  volume = "I",
  link = "\url{http://www.win.tue.nl/~ruudp/paper/31.pdf}",
  algebra = "\newline\refto{category PRSPCAT ProjectiveSpaceCategory}",
  isbn = "97804444814722",
  paper = "Hold11.pdf"
}

```

— ignore —

```

\bibitem[Huguet 89]{HP89} Huguet, L.; Poli, A. (eds).

```

Applied Algebra, Algebraic Algorithms and Error-Correcting Codes.
 5th International Conference AAEEC-5 Proceedings.
 Springer-Verlag, Berlin, Germany / Heidelberg, Germany / London, UK / etc.,
 1989. ISBN 3-540-51082-6. LCCN QA268.A35 1987
 keywords = "axiomref"

— axiom.bib —

```
@book{Hous92,
  author = "Houstis, E.N. and Gaffney, P.W.",
  title = "Programming environments for high-level scientific problem solving",
  year = "1992",
  publisher = "Elsevier",
  isbn = "978-0444891761",
  abstract =
    "Programming environments, as the name suggests, are intended to
    provide a unified, extensive range of capabilities for a person
    wishing to solve a problem using a computer. In this particular
    proceedings volume, the problem considered is a high-level scientific
    computation. In other words, a scientific problem whose solution
    usually requires sophisticated computing techniques and a large
    allocation of computing resources.",
  keywords = "axiomref"
}
```

— axiom.bib —

```
@article{Hous00,
  author = "Houstis, Elias N. and Rice, John R.",
  title = "Future problem solving environments for computational science",
  journal = "Math. Comput. Simul.",
  volume = "54",
  number = "4-5",
  pages = "243-257",
  year = "2000",
  abstract =
    "We review the current state of the Problem Solving Environment (PSE)
    field and make projections for the future. First, we describe the
    computing context, the definition of a PSE and the goals of a PSE. The
    state-of-the-art is summarized along with the principal components and
    paradigms for building PSEs. The discussion of the future is given in
    three parts: future trends, scenarios for 2010/2025, and research
```

```

    issues to be addressed.",
    keywords = "axiomref"
}

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J

— ignore —

```

\bibitem{Jacob 93}{JOS93} Jacob, G.; Oussous, N. E.; Steinberg, S. (eds)
Proceedings SC 93
International IMACS Symposium on Symbolic Computation. New Trends and
Developments. LIFL Univ. Lille, Lille France, 1993
    keywords = "axiomref"

```

— axiom.bib —

```

@article{Jacq97,
  author = "Jacquemard, Alain and Khechichine-Mourtada, F.Z. and Mourtada, A.",
  title = "Formal algorithms applied to the study of the cyclicity of a
          generic algebraic polycycle with four hyperbolic crests",
  journal = "Nonlinearity",
  volume = "10",
  number = "1",
  pages = "19-53",
  year = "1997",
  comment = "french",
  abstract =
    "Drawing on the work of Mourtada, we show that a family of vector
    fields with a generic algebraic polycycle of four hyperbolic apices
    possesses a maximum capacity of four limit cycles. This cyclicity is
    attained in an opening connecting the parameters which the edge
    contains, in particular a generic line of singularities of dovetail
    type. We also give an asymptotic estimation of the volume of this
    opening, as well as an explicit example of a family of polynomial
    vector fields replicating the above-described conditions and
    possessing five limit cycles. The methods employed are very diverse:
    geometrical arguments (Thoms theory of catastrophes and the theory of
    algebraic singularities), developments from Puiseux, the number of
    major roots by Descartes law and calculated exactly by Sturm series,
    and other specific methods for formal calculus, such as for example

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the cylindrical algebraic decomposition and the resolution of algebraic systems via the construction of Grbner bases. The calculations have been executed formally, that is to say without making the least appeal to numerical approximation, in using the formal calculus system AXIOM.",
 keywords = "axiomref"
 }

— axiom.bib —

```
@article{Jacq02,
  author = "Jacquemard, Alain and Teixeira, M.A.",
  title = "Effective algebraic geometry and normal forms of reversible
    mappings",
  journal = "Rev. Mat. Complut.",
  volume = "15",
  number = "1",
  pages = "31-55",
  year = "2002",
  abstract =
    "The authors consider a problem coming from the theory of discrete
    dynamical systems (iterations of diffeomorphisms in
     $\mathbb{R}^3$ ). Namely, how to characterize the behaviour of the
    trajectories near the fixed points and the stability of this behaviour
    under perturbations. The authors use the computer in the context of
    reversible mappings. When such a mapping has some degree of degeneracy
    they apply Grbner bases techniques and the theory of (formal) normal
    forms in the space of the coefficients of the jets of the
    mapping. They show that a language with typed objects like AXIOM is
    very convenient to solve such problems.",
  keywords = "axiomref"
}
```

— axiom.bib —

```
@article{Jaes93,
  author = "Jaeschke, Gerhard",
  title = "On Strong Pseudoprimes to Several Bases",
  journal = "Mathematics of Computation",
  volume = "61",
  number = "204",
  year = "1993",
```

```

pages = "915-926",
algebra = "\newline\refto{package PRIMES IntegerPrimesPackage}",
abstract =
  "With  $\psi_k$  denoting the smallest strong pseudoprime to all of the
  first  $k$  primes taken as bases we determine the exact values for
 $\psi_5$ ,  $\psi_6$ ,  $\psi_7$ ,  $\psi_8$ , and give upper bounds for
 $\psi_9$ ,  $\psi_{10}$ ,  $\psi_{11}$ . We discuss the methods and
  underlying facts for obtaining these results",
paper = "Jaes93.pdf",
keywords = "axiomref"
}

```

— axiom.bib —

```

@article{Jage96,
  author = "Jager, Bram De and van Asch, Bram",
  title = "Symbolic Solutions for a Class of Partial Differential Equations",
  journal = "J. Symbolic Computation",
  volume = "22",
  year = "1996",
  pages = "459-468",
  link = "\url{http://www.mate.tue.nl/mate/pdfs/1610.pdf}",
  abstract =
    "An algorithm to generate solutions for members of a class of
    completely integrable partial differential equations has been derived
    from a constructive proof of Frobenius' Theorem. The algorithm is
    implemented as a procedure in the computer algebra system
    Maple. Because the implementation uses the facilities of Maple for
    solving sets of ordinary differential equations and for sets of
    nonlinear equations, and those facilities are limited, the problems
    that actually can be solved are restricted in size and
    complexity. Several examples, some derived from industrial practice,
    are presented to illustrate the use of the algorithm and to
    demonstrate the advantages and shortcomings of the implementation.",
  paper = "Jage96.pdf",
  keywords = "axiomref"
}

```

— ignore —

```

\bibitem[Janssen 88]{Jan88} Jan{\ss}en, R. (ed)
Trends in Computer Algebra, International Symposium

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Bad Neuenahr, May 19-21, 1987, Proceedings, volume 296 of Lecture Notes in
Computer Science.
Springer-Verlag, Berlin, Germany / Heidelberg, Germany / London, UK / etc.,
1988 ISBN 3-540-18928-9, 0-387-18928-9 LCCN QA155.7.E4T74 1988
keywords = "axiomref"

— ignore —

```
\bibitem[Jenks 69]{Jen69} Jenks, R. D.
  title = "META/LISP: An interactive translator writing system",
  Research Report International Business Machines, Inc., Thomas J.
  Watson Research Center, Yorktown Heights, NY, USA, 1969 RC2968 July 1970
  keywords = "axiomref"
```

— axiom.bib —

```
@techReport{Jenk71,
  author = "Jenks, Richard D.",
  title = "META/PLUS: The syntax extension facility for SCRATCHPAD",
  type = "Research Report",
  number = "RC 3259",
  institution = "IBM Research",
  year = "1971",
  keywords = "axiomref",
  beebe = "Jenks:1971:MPS"
}
```

— Jenks:1971:MPS —

```
@TechReport{Jenks:1971:MPS,
  author = "R. D. Jenks",
  title = "{META\slash PLUS}: The Syntax Extension Facility for
  {SCRATCHPAD}",
  type = "Research Report",
  number = "RC 3259",
  institution = "International Business Machines Inc., Thomas J. Watson
  Research Center",
  address = "Yorktown Heights, NY, USA",
```

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pages =      "??",
month =      feb,
year =       "1971",
bibdate =    "Sat Dec 30 08:53:02 1995",
bibsource =  "/usr/local/src/bib/bibliography/Ai/lisp.bib;
              http://www.math.utah.edu/pub/tex/bib/axiom.bib",
}

```

— axiom.bib —

```

@misc{Jenk72,
  author = "Jenks, Richard D.",
  title = "SCRATCHPAD",
  volume = "??",
  number = "24",
  pages = "16-17",
  month = "October",
  year = "1972",
  abstract =
    "The following SCRATCHPAD solution of Problem \#2 was run on a 1280K
    virtual machine under CP/CMS time sharing system on a System/360
    model 67. The conversation below is a modification of a program
    originally written by Yngve Sundblad, August 1972. The program uses
    symmetrised formulae, saves certain intermediate results, but does not
    eliminate numerical factors in denominators",
  keywords = "axiomref"
}

```

— axiom.bib —

```

@article{Jenk74,
  author = "Jenks, Richard D.",
  title = "The SCRATCHPAD language",
  journal = "ACM SIGPLAN Notices",
  comment = "reprinted in SIGSAM Bulletin, Vol 8, No. 2, pp 20-30 May 1974",
  volume = "9",
  number = "4",
  pages = "101-111",
  year = "1974",
  doi = "http://dx.doi.org/10.1145807051",
  abstract =
    "SCRATCHPAD is an interactive system for symbolic mathematical

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computation. Its user language, originally intended as a
special-purpose non-procedural language, was designed to capture the
style and succinctness of common mathematical notations, and to serve
as a useful, effective tool for on-line problem solving. This paper
describes extensions to the language which enable it to serve also as
a high-level programming language, both for the formal description of
mathematical algorithms and their efficient implementation.",
keywords = "axiomref",
beebe = "Jenks:1974:SL"
}

```

— Jenks:1974:SL —

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@Article{Jenks:1974:SL,
  author = "R. D. Jenks",
  title = "The {SCRATCHPAD} language",
  journal = j-SIGPLAN,
  volume = "9",
  number = "4",
  pages = "101--111",
  month = apr,
  year = "1974",
  CODEN = "SINODQ",
  DOI = "http://dx.doi.org/10.1145807051",
  ISSN = "0362-1340 (print), 1523-2867 (print), 1558-1160
    (electronic)",
  ISSN-L = "0362-1340",
  bibdate = "Sat Apr 25 11:46:37 MDT 1998",
  bibsource = "http://www.math.utah.edu/pub/tex/bib/axiom.bib",
  acknowledgement = ack-nhfb,
  classification = "C6140D (High level languages); C7310 (Mathematics
    computing)",
  conflocation = "Santa Monica, CA, USA; 28-29 March 1974",
  conftitle = "ACM SIGPLAN Symposium on Very High Level Languages",
  corpsource = "IBM Thomas J. Watson Res. Center, Yorktown Heights,
    NY, USA",
  fjournal = "ACM SIGPLAN Notices",
  link = "\url{http://portal.acm.org/browse_dl.cfm?idx=J706}",
  keywords = "formal description; formal programming language; high
    level programming language; interactive system;
    mathematical algorithms; natural sciences applications
    of computers; online problem solving; problem oriented
    languages; SCRATCHPAD language; symbolic mathematical
    computation; user language",
  sponsororg = "ACM",
  treatment = "A Application; P Practical",

```


}

— axiom.bib —

```
@article{Jenk76a,
  author = "Jenks, Richard D.",
  title = "Problem \#11: generation of Runge-Kutta equations",
  journal = "SIGSAM Bulletin",
  volume = "10",
  number = "1",
  year = "1976",
  abstract =
    "Generate a set of equations for an explicit k-th order, m stage,
    Runge-Kutta method for integrating an autonomous system of ordinary
    differential equations, k and m as large as possible. The number of
    conditions and variables for various k and m are given in Table
    1. Tabulate the costs C(m,k)."
```

}

— axiom.bib —

```
@inproceedings{Jenk76,
  author = "Jenks, Richard D.",
  title = "A pattern compiler",
  booktitle = "Proc. 1976 ACM Symposium on Symbolic and Algebraic Computation",
  series = "SYMSAC '76",
  year = "1976",
  publisher = "ACM Press",
  doi = "http://dx.doi.org/10.1145806324",
  abstract =
    "A pattern compiler for the SCRATCHPAD system provides an efficient
    implementation of sets of user-defined pattern-replacement rules for
    symbolic mathematical computation such as tables of integrals or
    summation identities. Rules are compiled together, with common search
    paths merged and factored out and with the resulting code optimized
    for efficient recognition over all patterns. Matching principally
    involves structural comparison of expression trees and evaluation of
    predicates. Pattern recognizers are ‘fully compiled’; if values of
    match variables can be determined by solving equations at compile time.
    Recognition times for several pattern matchers are compared.",
  keywords = "axiomref",
  beebe = "Jenks:1976:PC"
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}

— Jenks:1976:PC —

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@InProceedings{Jenks:1976:PC,
  author =      "Richard D. Jenks",
  editor =      "Richard D. Jenks",
  booktitle =   "Symsac '76: proceedings of the 1976 ACM Symposium on
                  Symbolic and Algebraic Computation, August 10--12,
                  1976, Yorktown Heights, New York",
  title =       "A pattern compiler",
  publisher =   pub-ACM,
  address =     pub-ACM:adr,
  pages =       "60--65",
  year =        "1976",
  DOI =         "http://dx.doi.org/10.1145806324",
  ISBN =        "????",
  ISBN-13 =     "????",
  LCCN =         "QA155.7.E4 .A15 1976; QA9.58 .A11 1976",
  bibdate =     "Thu Jul 26 08:56:43 2001",
  bibsource =   "http://www.math.utah.edu/pub/tex/bib/axiom.bib",
  acknowledgement = ack-nhfb,
  bookpages =   "384",
  keywords =    "Scratchpad",
}
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— axiom.bib —

```
@article{Jenk77,
  author = "Jenks, Richard D.",
  title = "On the design of a mode-based symbolic system",
  journal = "SIGGAM Bulletin",
  volume = "11",
  number = "1",
  pages = "16-19",
  year = "1977",
  abstract =
    "This paper is a preliminary report on the design and implementation
    of a mode-based symbolic programming system and compiler which allows
    programming with rewrite rules and LET and IS pattern-match constructs.
    An important feature of this design is the provision for mode-valued
    variables which allow algebraic domains to be run-time parameters."
```

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    keywords = "axiomref"
}

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— axiom.bib —

```

@inproceedings{Jenk79,
  author = "Jenks, Richard D.",
  title = "MODLISP: An Introduction",
  booktitle = "Proc. ISSAC 1979",
  series = "EUROSAM 79",
  pages = "466-480",
  year = "1979",
  publisher = "Springer-Verlag",
  isbn = "3-540-09519-5",
  comment = "IBM Research Report RC 8073 Jan 1980",
  keywords = "axiomref"
}

```

— axiom.bib —

```

@article{Jenk79a,
  author = "Jenks, Richard D.",
  title = "SCRATCHPAD/360: reflections on a language design",
  journal = "SIGSAM",
  volume = "13",
  number = "1",
  pages = "16-26",
  year = "1979",
  comment = "IBM Research RC 7405",
  abstract =
    "The key concepts of the SCRATCHPAD language are described, assessed,
    and illustrated by an example. The language was originally intended as
    an interactive problem solving language for symbolic mathematics.
    Nevertheless, as this paper intends to show, it can be used as a
    programming language as well.",
  keywords = "axiomref"
}

```

— axiom.bib —

```

@InProceedings{Jenk81,
  author = "Jenks, Richard D. and Trager, Barry M.",
  title = "A Language for Computational Algebra",
  year = "1981",
  booktitle = "Proc. Symp. on Symbolic and Algebraic Manipulation",
  series = "SYMSAC 1981",
  location = "Snowbird, Utah",
  comment = "IBM Research Report 8930",
  abstract =
    "This paper reports ongoing research at the IBM Research Center on the
    development of a language with extensible parameterized types and
    generic operators for computational algebra. The language provides an
    abstract data type mechanism for defining algorithms which work in as
    general a setting as possible. The language is based on the notions of
    domains and categories. Domains represent algebraic
    structures. Categories designate collections of domains having common
    operations with stated mathematical properties. Domains and categories
    are computed objects which may be dynamically assigned to variables,
    passed as arguments, and returned by functions. Although the language
    has been carefully tailored for the application of algebraic
    computation, it actually provides a very general abstract data type
    mechanism. Our notion of a category to group domains with common
    properties appears novel among programming languages (cf. image
    functor of RUSSELL) and leads to a very powerful notion of abstract
    algorithms missing from other work on data types known to the authors.",
  keywords = "axiomref"
}

```

— ignore —

```

\bibitem[Jenks 81a]{JT81a} Jenks, R.D.; Trager, B.M.
  title = "A Language for Computational Algebra",
  SIGPLAN Notices, New York: Association for Computing Machinery, Nov 1981
  keywords = "axiomref"

```

— axiom.bib —

```

@inproceedings{Jenk84a,
  author = "Jenks, Richard D.",
  title = "The New SCRATCHPAD Language and System for Computer Algebra",
  booktitle = "Proc. 1984 MACSYMA Users Conference",
  year = "1984",

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pages = "409-??",
keywords = "axiomref",
beebe = "Jenks:1984:NSL"
}

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— Jenks:1984:NSL —

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@InProceedings{Jenks:1984:NSL,
  author =      "Richard D. Jenks",
  title =      "The New {SCRATCHPAD} Language and System for Computer
                Algebra",
  crossref =    "Golden:1984:PMU",
  pages =      "409--??",
  year =       "1984",
  bibsource =   "/usr/local/src/bib/bibliography/Theory/Comp.Alg.1.bib;
                http://www.math.utah.edu/pub/tex/bib/axiom.bib",
}

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— axiom.bib —

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@inproceedings{Jenk84b,
  author = "Jenks, Richard D.",
  title = "A primer: 11 keys to New Scratchpad",
  booktitle = "Proc. EUROSAM ISSAC 1984",
  year = "1984",
  publisher = "Springer-Verlag",
  pages = "123-147",
  isbn = "0-387-13350-X",
  abstract =
    "This paper is an abbreviated primer for the language of new
    SCRATCHPAD, a new implementation of SCRATCHPAD which has been under
    design and development by the Computer Algebra Group at the IBM
    Research Center during the past 6 years. The basic design goals of the
    new SCRATCHPAD language and interface to the user are to provide:
    \begin{itemize}
    \item a ‘‘typeless’’ interactive language suitable for on-line solution
    of mathematical problems by novice users with little or no programming
    required, and
    \item a programming language suitable for the formal description of
    algorithms and algebraic structures which can be compiled into run-time
    efficient object code.
    \end{itemize}

```

The new SCRATCHPAD language is introduced by 11 keys with each successive key introducing an additional capability of the language. The language is thus described as a ‘‘concentric’’ language with each of the 11 levels corresponding to a language subset. These levels are more than just a pedagogic device, since they correspond to levels at which the system can be effectively used. Level 1 is sufficient for naive interactive use; levels 2-8 progressively introduce interactive users to capabilities of the language; levels 9-11 are for system programmers and advanced users. Levels 2, 4, 6, and 7 give users the full power of LISP with a high-level language; level 8 introduces ‘‘type declarations;’’ level 9 allows polymorphic functions to be defined and compiled; levels 10-11 give users an Ada-like facility for defining types and packages (those of new SCRATCHPAD are dynamically constructable, however). One language is used for both interactive and system programming language use, although several freedoms such as abbreviation and optional type-declarations allowed at top-level are not permitted in system code. The interactive language (levels 1-8) is a blend of original SCRATCHPAD [GRJY75], some proposed extensions [JENK74], work by Loos [LOOS74], SETL [DEWA79], SMP [COWO81], and new ideas; the system programming language (levels 1-11) superficially resembles Ada but is more similar to CLU [LISK74] in its semantic design.

This presentation of the language in this paper omits many details to be covered in the SCRATCHPAD System Programming Manual [SCRA84] and an expanded version of this paper will serve as a primer for SCRATCHPAD users [JESU84].”,

```
keywords = "axiomref",
beebe = "Jenks:1984:PKN"
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}
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— Jenks:1984:PKN —

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@InProceedings{Jenks:1984:PKN,
  author = "Richard D. Jenks",
  title = "A primer: 11 keys to {New Scratchpad}",
  crossref = "Fitch:1984:E",
  pages = "123--147",
  year = "1984",
  bibsource = "/usr/local/src/bib/bibliography/Theory/Comp.Alg.1.bib;
http://www.math.utah.edu/pub/tex/bib/axiom.bib",
}
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— axiom.bib —

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@misc{Jenk84c,
  author = "Jenks, Richard D. and Sundaresan, Christine J.",
  title = "The 11 Keys to SCRATCHPAD: A Primer",
  year = "1984",
  keywords = "axiomref"
}
```

— axiom.bib —

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@misc{Acad16,
  author = "Academic Search",
  title = "A Primer: 11 Keys to New Scratchpad",
  link = "\url{http://libra.msra.cn/publication/645035/a-primer-11-keys-to-new-scratchpad}",
  year = "2016",
  keywords = "axiomref"
}
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— axiom.bib —

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@misc{Jenk86a,
  author = "Jenks, Richard D.",
  title = "Basic Algebraic Facilities of the Scratchpad II Computer
          Algebra System",
  institution = "IBM Research",
  year = "1986",
  keywords = "axiomref"
}
```

— axiom.bib —

```
@misc{Jenk86b,
  author = "Jenks, Richard D.",
  title = "Scratchpad II Examples from INPUT files",
  institution = "IBM Research",
}
```

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year = "1986",
keywords = "axiomref"
}

```

— axiom.bib —

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@techreport{Jenk86,
  author = "Jenks, Richard D. and Sutor, Robert S. and Watt, Stephen M.",
  title = "Scratchpad II: An Abstract Datatype System for Mathematical
    Computation",
  institution = "IBM Research",
  year = "1986",
  type = "Research Report",
  number = "RC 12327 (\#55257)",
  link = "\url{http://www.csd.uwo.ca/~watt/pub/reprints/1987-ima-spadadt.pdf}",
  abstract = "
    Scratchpad II is an abstract datatype language and system that is
    under development in the Computer Algebra Group, Mathematical Sciences
    Department, at the IBM Thomas J. Watson Research Center. Some features
    of APL that made computation particularly elegant have been borrowed.
    Many different kinds of computational objects and data structures are
    provided. Facilities for computation include symbolic integration,
    differentiation, factorization, solution of equations and linear
    algebra. Code economy and modularity is achieved by having
    polymorphic packages of functions that may create datatypes. The use
    of categories makes these facilities as general as possible.",
  paper = "Jenk86.pdf",
  keywords = "axiomref",
  beebe = "Jenks:1986:SIA"
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— Jenks:1986:SIA —

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@TechReport{Jenks:1986:SIA,
  author = "Richard D. Jenks and Robert S. Sutor and Stephen M.
    Watt",
  title = "Scratchpad {II}: an abstract datatype system for
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  number = "RC 12327 (\#55257)",
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address =      "Yorktown Heights, NY, USA",
pages =       "23",
year =        "1986",
bibdate =     "Thu Oct 31 17:23:28 2002",
bibsource =   "http://www.math.utah.edu/pub/tex/bib/axiom.bib",
acknowledgement = ack-nhfb,
keywords =    "Abstract data types (Computer science); Operating
               systems (Computers)",
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@inproceedings{Jenk87,
  author = "Jenks, Richard D. and Sutor, Robert S. and Watt, Stephen M.",
  title = "Scratchpad II: an Abstract Datatype System for
          Mathematical Computation'",
  booktitle = "Proceedings Trends in Computer Algebra",
  series = "Lecture Notes in Computer Science 296",
  pages = "157-182",
  publisher = "Springer-Verlag",
  isbn = "0-387-18928-9",
  year = "1987",
  comment = "IBM Research Report RC 12327 (\#55257) See Jenks86.pdf",
  abstract =
    "Scratchpad II is an abstract datatype language and system that is
    under development in the Computer Algebra Group, Mathematical Sciences
    Department, at the IBM Thomas J. Watson Research Center. Many
    different kinds of computational objects and data structures are
    provided. Facilities for computation include symbolic integration,
    differentiation, factorization, solution of equations and linear
    algebra. Code economy and modularity is achieved by having polymorphic
    packages of functions that may create datatypes. The use of categories
    makes these facilities as general as possible.",
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@inproceedings{Jenk88,
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booktitle = "Mathematical Aspects of Scientific Software",
year = "1988",
pages = "157-182",
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isbn = "0-387-18928-9",
abstract =
  "Scratchpad II is an abstract datatype language and system that is
  under development in the Computer Algebra Group, Mathematical Sciences
  Department, at the IBM Thomas J. Watson Research Center. Many
  different kinds of computational objects and data structures are
  provided. Facilities for computation include symbolic integration,
  differentiation, factorization, solution of equations and linear
  algebra. Code economy and modularity is achieved by having polymorphic
  packages of functions that may create datatypes. The use of categories
  makes these facilities as general as possible.",
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— Jenks:1988:SIA —

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@InProceedings{Jenks:1988:SIA,
  author =      "R. D. Jenks and R. S. Sutor and S. M. Watt",
  title =      "{Scratchpad II}: An Abstract Datatype System for
  Mathematical Computation",
  crossref =   "Janssen:1988:TCA",
  pages =      "12--37",
  year =       "1988",
  bibsource =  "/usr/local/src/bib/bibliography/Theory/Comp.Alg.bib;
  http://www.math.utah.edu/pub/tex/bib/axiom.bib",
}

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— ignore —

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\bibitem[Jenks 88a]{Jen88a} Jenks, R. D.
  title = "A Guide to Programming in BOOT",
  Computer Algebra Group, Mathematical Sciences Department, IBM Research
  Draft September 5, 1988
  keywords = "axiomref"

```

— ignore —

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\bibitem[Jenks 88b]{Jen88b} Jenks, Richard
  title = "The Scratchpad II Computer Algebra System Interactive Environment Users Guide",
  Spring 1988
  keywords = "axiomref"
```

— axiom.bib —

```
@article{Jenk88d,
  author = "Jenks, Richard D.",
  title = "Scratchpad II: A computer algebra language and system",
  journal = "The Journal of the Acoustical Society of America",
  year = "1988",
  volume = "83",
  number = "S1",
  pages = "S106",
  abstract =
    "The Scratchpad II system represents a new generation of systems for
    doing symbolic mathematics, based on modern algebra and abstract data
    types. A large number of facilities are provided, for example:
    symbolic integration, 'infinite' power series, differential operators,
    Cartesian tensors, and solution of nonlinear systems. Scratchpad II
    has been designed from the outset to be extendible. The system
    introduces a new data abstraction notion, the 'category,' to express
    intricate interrelationships between data types. The result design
    permits the compilation of algorithms described in their most natural
    mathematical setting. The use of categories guarantees user defined
    types and packages are compatible with each other and with built in
    facilities. This system provides a single highlevel language with an
    interpreter and compiler. The language can be used by the naive user
    for convenient interactive mathematics calculations and by the
    advanced user for the efficient implementation of
    algorithms. Scratchpad II is built on Lisp/VM and runs on IBM/370
    class mainframes. An implementation of the system on the RT/PC is
    expected soon.",
  keywords = "axiomref"
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— axiom.bib —

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@book{Jenk92,
  author = "Jenks, Richard D. and Sutor, Robert S.",
  title = "AXIOM: The Scientific Computation System",
  publisher = "Springer-Verlag, Berlin, Germany",
  year = "1992",
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@Book{Jenks:1992:ASC,
  author = "Richard D. Jenks and Robert S. Sutor",
  title = "{AXIOM}: The Scientific Computation System",
  publisher = pub-SV,
  address = pub-SV:adr,
  pages = "xxiv + 742",
  year = "1992",
  ISBN = "0-387-97855-0 (New York), 3-540-97855-0 (Berlin)",
  ISBN-13 = "978-0-387-97855-0 (New York), 978-3-540-97855-8 (Berlin)",
  LCCN = "QA76.95.J46 1992",
  bibdate = "Fri Dec 29 18:16:15 1995",
  bibsource = "/usr/local/src/bib/bibliography/Theory/Comp.Alg.bib;
  http://www.math.utah.edu/pub/tex/bib/axiom.bib",
}
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— axiom.bib —

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@InProceedings{Jenk94,
  author = "Jenks, Richard D. and Trager, Barry M.",
  title = "How to make AXIOM into a scratchpad",
  booktitle = "Proceedings of the ACM-SIGSAM 1989 International
    Symposium on Symbolic and Algebraic Computation, ISSAC '94",
  series = "ISSAC 94",
  year = "1994",
  pages = "32-40",
  isbn = "0-89791-638-7",
  publisher = "ACM Press",
  address = "New York, NY, USA",
  abstract =
```

"Scratchpad [GrJe71] was a computer algebra system developed in the early 1970s. Like M&M (Maple [CGG91ab] and Mathematical [W01S92]) and other systems today, Scratchpad had one principal representation for mathematical formulae based on "expression trees". Its user interface design was based on a pattern-matching paradigm with infinite rewrite-rule semantics, providing what we believe to be the most natural paradigm for interactive symbolic problem solving. Like M&M, however, user programs were interpreted, often resulting in poor performance relative to similar facilities coded in standard programming languages such as FORTRAN and C.

Scratchpad development stopped in 1976 giving way to a new system design ([JenR79], [JeTr81]) that evolved into AXIOM [JeSu92]. AXIOM has a strongly-typed programming language for building a library of parameterized types and algorithms, and a type-inferencing interpreter that accesses the library and can build any of an infinite number of types for interactive use.

We suggest that the addition of an expression tree type to AXIOM can allow users to operate with the same freedom and convenience of untyped systems without giving up the expressive power and run-time efficiency provided by the type system. We also present a design that supports a multiplicity of programming styles, from the Scratchpad pattern-matching paradigm to functional programming to more conventional procedural programming. The resulting design seems to us to combine the best features of Scratchpad with current AXIOM and to offer a most attractive, flexible, and user-friendly environment for interactive problem solving.

Section 2 is a discussion of design issues contrasting AXIOM with other symbolic systems. Sections 3 and 4 is an assessment of AXIOMs current design for building libraries and interactive use. Section 5 describes a new interface design for AXIOM, its resulting paradigms, and its underlying semantic model. Section 6 compares this work with others."

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paper = "Jenk94.pdf",
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@InProceedings{Jenks:1994:HMA,
author = "Richard D. Jenks and Barry M. Trager",
title = "How to make {AXIOM} into a {Scratchpad}",
crossref = "ACM:1994:IP1",
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pages =      "32--40",
year =       "1994",
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              computation)",
bibdate =    "Tue Sep 17 06:29:18 MDT 1996",
bibsource =  "http://www.math.utah.edu/pub/tex/bib/axiom.bib",
ZMnumber =   "0945.68543",
abstract =   "Scratchpad (Griesmer and Jenks, 1971) was a computer
              algebra system that had one principal representation
              for mathematical formulae based on expression trees.
              Its user interface design was based on a
              pattern-matching paradigm with infinite rewrite rule
              semantics, providing what we believe to be the most
              natural paradigm for interactive symbolic problem
              solving. Like M and M, however, user programs were
              interpreted, often resulting in poor performance
              relative to similar facilities coded in standard
              programming languages such as FORTRAN and C. Scratchpad
              development stopped in 1976 giving way to a new system
              design that evolved into AXIOM. AXIOM has a
              strongly-typed programming language for building a
              library of parameterized types and algorithms, and a
              type-inferencing interpreter that accesses the library
              and can build any of an infinite number of types for
              interactive use. We suggest that the addition of an
              expression tree type to AXIOM can allow users to
              operate with the same freedom and convenience of
              untyped systems without giving up the expressive power
              and run-time efficiency provided by the type system. We
              also present a design that supports a multiplicity of
              programming styles, from the Scratchpad
              pattern-matching paradigm to functional programming to
              more conventional procedural programming.",
acknowledgement = ack-nhfb,
affiliation =  "IBM Thomas J. Watson Res. Center, Yorktown Heights,
              NY, USA",
classification = "C6180 (User interfaces); C7310 (Mathematics
              computing)",
keywords =    "AXIOM; C; Computer algebra system; Expression trees;
              FORTRAN; Functional programming; Infinite rewrite rule
              semantics; Library; Mathematical formulae;
              Pattern-matching; Procedural programming; Run-time
              efficiency; Scratchpad; Strongly-typed programming
              language; Symbolic problem solving; Type-inferencing
              interpreter; Untyped systems; User interface design;
              User programs",
language =    "English",
thesaurus =   "Mathematics computing; Pattern matching; Program
              interpreters; Programming; Symbol manipulation; User
              interfaces",

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@phdthesis{Joha14,
  author = "Johansson, Fredrik",
  title = "Fast and Rigorous Computation of Special Functions to High
          Precision",
  school = "Johannes Kepler University, Linz, Austria RISC",
  year = "2014",
  abstract =
    "The problem of efficiently evaluating special functions to high
    precision has been considered by numerous authors. Important tools
    used for this purpose include algorithms for evaluation of linearly
    recurrent sequences, and algorithms for power series arithmetic.

    In this work, we give new baby-step, giant-step algorithms for
    evaluation of linearly recurrent sequences involving an expensive
    parameter (such as a high-precision real number) and for computing
    compositional inverses of power series. Our algorithms do not have the
    best asymptotic complexity, but they are faster than previous
    algorithms in practice over a large input range.

    Using a combination of techniques, we also obtain efficient new
    algorithms for numerically evaluating the gamma function  $\Gamma(z)$ 
    and the Hurwitz zeta function  $\zeta(s,a)$ , or Taylor series
    expansions of those functions, with rigorous error bounds. Our methods
    achieve softly optimal complexity when computing a large number of
    derivatives to proportionally high precision.

    Finally, we show that isolated values of the integer partition
    function  $p(n)$  can be computed rigorously with softly optimal
    complexity by means of the Hardy-Ramanujan-Rademacher formula and
    careful numerical evaluation.

    We provide open source implementations which run significantly faster
    than previous published software. The implementations are used for
    record computations of the partition function, including the
    tabulation of several billion Ramanujan-type congruences, and of
    Taylor series associated with the Reimann zeta function.",
  paper = "Joha14.pdf"
}
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— axiom.bib —

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@article{Joha02,
  author = "Johansson, Leif and Lambe, Larry and Skoldberg, Emil",
  title = "On Constructing Resolutions over the Polynomial Algebra",
  journal = "Homology, Homotopy and Applications",
  volume = "4",
  number = "2",
  year = "2002",
  pages = "315-336",
  link = "\url{http://projecteuclid.org/download/pdf/_1/euclid.hha/1139852468}",
  abstract =
    "Let  $k$  be a field, and  $A$  be a polynomial algebra over  $k$ .
    Let  $I \subseteq A$  be an ideal. We present a novel method for
    computing resolutions of  $A/I$  over  $A$ . The method is a synthesis
    of Groebner basis techniques and homological perturbation theory.
    The examples in this paper were computed using computer algebra.",
  paper = "Joha02.pdf",
  keywords = "axiomref"
}
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@article{John94,
  author = "Johnson, M.E. and Rogers, C. and Schief, W.K. and Seiler, W.M.",
  title = "On moving pseudospherical surfaces: a generalised Weingarten
    system and its formal analysis",
  journal = "Lie Groups Appl.",
  volume = "1",
  pages = "124-136",
  year = "1994",
  abstract =
    "The connection between the motion of certain curves in  $\mathbb{R}^3$ 
    and  $(1+1)$ -dimensional soliton equations is by now well-established. On the
    other hand, the sine-Gordon and other integrable equations may be
    readily derived via the classical geometry of stationary
    pseudospherical surfaces. Here, the motion of pseudospherical surfaces
    is considered in a natural orthonormal triad formulation. In one case,
    in a motion in which the Gaussian curvature of  $S$  remains constant in
    time, an integrable nonlinear evolution equation is derived which has
    its origin in the description of wave propagation in an anharmonic
    crystal. In a second case, wherein the Gaussian curvature is allowed
    to vary in time, a classical generalised Weingarten system is derived
    in connection with the purely normal propagation of a pseudospherical
    surface. This is linked to triply orthogonal coordinate systems of
```



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Bianchi type. The generalised Weingarten system incorporates an
integrable  $2+1$ -dimensional sine-Gordon equation. The arbitrariness of the
solutions of the generalised Weingarten system is determined via a
completion procedure.",
keywords = "axiomref"
}

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— ignore —

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\bibitem[Rainer 14]{Rain14} Joswig, Rainer
title = "2014: 30+ Years Common Lisp the Language",
link = "\url{http://lisp.de/30ycltl}",
keywords = "axiomref"

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— axiom.bib —

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@misc{Joyn08,
author = "Joyner, David and Stein, William",
title = "Open Source Mathematical Software: A White Paper",
year = "2008",
link = "\url{http://citeseerx.ist.psu.edu/viewdoc/download?doi=10.1.1.124.7499}",
abstract =
  "Open source software has had a profound effect on computing during
  the last decade, especially on web servers (Apache), web browsers
  (Firefox), operating systems (Linux and OS X), and programming
  languages (GC C, Java, Python, Perl, etc.). The purpose of this paper
  is to put forward the case that open source development methodologies
  might also have a positive effect on mathematical software,
  especially if the National Science Foundation (NSF) increases their
  support of open source mathematical software development. We argue
  that careful funding of open source mathematical software may lead to
  a lower total cost of ownership in the research and education
  community, and to more efficient and trustworthy mathematical software.",
paper = "Joyn08.pdf",
keywords = "axiomref"
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@misc{Joyn16,
  author = "Joyner, David",
  title = "Links to some open source mathematical programs",
  link = "\url{http://www.opensourcemath.org/opensource/_math.html}",
  keywords = "axiomref"
}
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@article{Joyn08a,
  author = "Joyner, David",
  title = "Open source computer algebra systems: Axiom",
  journal = "ACM Commun. Comput. Algebra",
  volume = "42",
  number = "1-2",
  pages = "39-47",
  year = "2008",
  abstract =
    "This survey will look at Axiom, a free and very powerful computer
    algebra system available. It is a general purpose CAS useful for
    symbolic computation, research, and the development of new
    mathematical algorithms. Axiom is similar in some ways to Maxima,
    covered in the survey, but different in many ways as well. Axiom,
    Maxima, and SAGE, are the largest of the general-purpose open-source
    CASs.",
  keywords = "axiomref",
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— Joyner:2008:OSC —

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@Article{Joyner:2008:OSC,
  author = "David Joyner",
  title = "Open source computer algebra systems: {Axiom}",
  journal = j-ACM-COMM-COMP-ALGEBRA,
  volume = "42",
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ISSN =      "1932-2232 (print), 1932-2240 (electronic)",
ISSN-L =    "1932-2232",
bibdate =   "Tue Aug 12 17:30:40 MDT 2008",
bibsource = "http://portal.acm.org/;
             http://www.math.utah.edu/pub/tex/bib/axiom.bib",
abstract =  "This survey will look at Axiom, a free and very
             powerful computer algebra system available. It is a
             general purpose CAS useful for symbolic computation,
             research, and the development of new mathematical
             algorithms. Axiom is similar in some ways to Maxima,
             covered in the survey [J1], but different in many ways
             as well. Axiom, Maxima, and SAGE [S], are the largest
             of the general-purpose open-source CAS's. If you want
             to 'take a test drive,' Axiom can be tested without
             installation via the web interface [AS] or the SAGE
             online interface [S].",
acknowledgement = ack-nhfb,
fjournal =   "ACM Communications in Computer Algebra",
}

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\bibitem[Joswig 03]{JT03} Joswig, Michael; Takayama, Nobuki
  title = "Algebra, geometry, and software systems",
  Springer-Verlag ISBN 3-540-00256-1 p291
  keywords = "axiomref"

```

— ignore —

```

\bibitem[Joyner 06]{J006} Joyner, David
  title = "OSCAS - Maxima",
  SIGSAM Communications in Computer Algebra, 157 2006
  link = "\url{http://sage.math.washington.edu/home/wdj/sigsam/oscas-cca1.pdf}",
  keywords = "axiomref"

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\bibitem[Joyner 14]{J014} Joyner, David

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title = "Links to some open source mathematical programs",
link = "\url{http://www.opensourcemath.org/opensource_math.html}",
keywords = "axiomref"

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— axiom.bib —

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@inproceedings{Kajl92,
  author = "Kajler, Norbert",
  title = "CAS/PI: a Portable and Extensible Interface for Computer
    Algebra Systems",
  year = "1992",
  booktitle = "Proc. ISSAC 1992",
  series = "ISSAC 1992",
  pages = "376-386",
  isbn = "0-89791-489-9 (soft cover) 0-89791-490-2 (hard cover)",
  abstract =
    "CAS/$\pi$ is a Computer Algebra System graphic user interface
    designed to be highly portable and extensible. It has been developed
    by composition of pre-existing software tools such as Maple, Sisyphe,
    or Ulysse systems, ZicVis 3-D plotting library, etc, using control
    integration technology and a set of high level graphic toolkits to
    build the formula editor and the dialog manager. The main aim of
    CAS/$\pi$ is to allow a wide range of runtime recon gurations and
    extensions. For instance, it is possible to add new tools to a running
    system, to modify connections between working tools, to extend the set
    of graphic symbols managed by the formula editor, to design new high
    level editing commands based on the syntax or semantics of
    mathematical formulas, to customize and extend the menu-button based
    user interface, etc. More generally, CAS/$\pi$ can be seen equally as
    a powerful system-independent graphic user interface enabling
    inter-systems communications, a toolkit to allow fast development of
    custom-made scientific software environments, or a very convenient
    framework for experimenting with computer algebra systems protocols
    and man-machine interfaces.",
  paper = "Kajl92.pdf",
  keywords = "axiomref"
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@article{Kajl94,
  author = "Kajler, Norbert and Soiffer, Neil",
  title = "Some human interaction issues in computer algebra",
  journal = "SIGSAM Bulletin",
  volume = "28",
  number = "1",
  pages = "18-28",
  year = "1994",
  abstract =
    "This paper addresses some of the current issues concerning the
    improvement of user interfaces for computer algebra systems. Some
    state of the art commercial software as well as research prototypes
    are presented, followed by a description of present research
    directions.",
  keywords = "axiomref"
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@misc{Kani16,
  author = "Kanigel, Robert",
  title = "OldQuotes",
  link = "\url{http://www.oldquotes.com}",
  year = "2016",
  abstract =
    "Sometimes in studying Ramanujan's work, George Andrews said at
    another time, 'I have wondered how much Ramanujan could have done if
    he had had MACSYMA or SCRATCHPAD or some other symbolic algebra package'"
}

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— axiom.bib —

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@book{Kapl05,
  author = "Kaplan, Michael",
  title = "Computer Algebra",
  publisher = "Springer, Berlin, Germany",
  year = "2005",
  isbn = "3-540-21379-1",
  comment = "German Language",
  keywords = "axiomref"
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@misc{Kapu81,
  author = "Kapur, D. and Musser, D.R. and Stepanov, A.A.",
  title = "Operators and Algebraic Structures",
  link = "\url{http://www.stepanovpapers.com/p59-kapur.pdf}",
  year = "1981",
  abstract =
    "Operators in functional languages such as APL and FFP are a useful
    programming concept. However, this concept cannot be fully
    exploited in these languages because of certain constraints. It is
    proposed that an operator should be associated with a structure having
    the algebraic properties on which the operator's behavior depends.
    This is illustrated by introducing a language that provides mechanisms
    for defining structures and operators on them. Using this language,
    it is possible to describe algorithms abstractly, thus emphasizing
    the algebraic properties on which the algorithms depend. The role
    that formal representation of mathematical knowledge can play in the
    development of programs is illustrated through an example. An
    approach for associating complexity measures with a structure and
    operators is also suggested. This approach is useful in analyzing
    the complexity of algorithms in an abstract setting.",
  paper = "Kapu81.pdf"
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@inproceedings{Kaue08,
  author = "Kauers, Manuel",
  title = "Integration of Algebraic Functions: A Simple Heuristic for
    Finding the Logarithmic Part",
  booktitle = "Proc ISSAC 2008",
  series = "ISSAC '08",
  year = "2008",
  pages = "133-140",
  isbn = "978-1-59593-904",
  link = "\url{http://www.risc.jku.at/publications/download/risc_3427/Ka01.pdf}",
  abstract =
    "A new method is proposed for finding the logarithmic part of an
    integral over an algebraic function. The method uses Groebner bases
    and is easy to implement. It does not have the feature of finding a
    closed form of an integral whenever there is one. But it very often
    does, as we will show by a comparison with the built-in integrators of
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    some computer algebra systems.",
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    keywords = "axiomref"
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@inproceedings{Kead93,
  author = "Keady, G. and Nolan, G.",
  title = "Production of Argument SubPrograms in the AXIOM -- NAG link:
          examples involving nonlinear systems",
  booktitle = "Proc. Workshop on Symbolic and Numeric Computation",
  location = "Helsinki",
  year = "1993",
  pages = "13-32",
  comment = "NAG Technical Report TR1/94",
  link = "\url{school.maths.uwa.edu.au/%7Ekeady/KeadyPapers/93Helsinki.ps}",
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\newline\ref{domain ASP9 Asp9}",
abstract =
  "Dewar's paper [6] earlier in this Proceedings 'sketches out the
  design of the AXIOM-NAG link' and gives a general account of new tools
  for generating Fortran. This paper is a sequel to [6]. Here we present
  'examples' of some of the items discussed in [6]. We have attempted to
  achieve some coherence by selecting our 'examples' from just the one
  application area - solving nonlinear systems.",
paper = "Kead93.pdf",
keywords = "axiomref",
beebe = "Keady:1994:PAS"
}

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— Keady:1994:PAS —

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@TechReport{Keady:1994:PAS,
  author = "G. Keady and G. Nolan",
  title = "Production of {Argument SubPrograms} in the {AXIOM}
  --- {NAG} link: examples involving nonlinear systems",
  number = "TR1/94 ATR/7 (NP2680)",
  institution = inst-NAG,
  address = inst-NAG:adr,
  pages = "??",
  year = "1994",
  bibdate = "Thu Jan 04 18:40:00 1996",
  bibsource = "http://www.math.utah.edu/pub/tex/bib/axiom.bib",
  link = "\url{http://www.nag.co.uk/doc/TechRep/axiomtr.html}",
  acknowledgement = ack-nhfb,
}

```

— ignore —

```

\bibitem[Kelsey 99]{Kel99} Kelsey, Tom
  title = "Formal Methods and Computer Algebra: A Larch Specification of AXIOM Categories and Functors",
  Ph.D. Thesis, University of St Andrews, 1999
  keywords = "axiomref"

```

— ignore —


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\bibitem[Kelsey 00a]{Kel00a} Kelsey, Tom
  title = "Formal specification of computer algebra",
  University of St Andrews, 6th April 2000
  link = "\url{http://www.cs.st-andrews.cs.uk/~tom/pub/fscbs.ps}",
  abstract = "
    We investigate the use of formal methods languages and tools in the
    design and development of computer algebra systems (henceforth CAS).
    We demonstrate that errors in CAS design can be identified and
    corrected by the use of (i) abstract specifications of types and
    procedures, (ii) automated proofs of properties of the specifications,
    and (iii) interface specifications which assist the verification of
    pre- and post conditions of implemented code.",
  paper = "Kel00a.pdf",
  keywords = "axiomref"

```

— ignore —

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\bibitem[Kelsey 00b]{Kel00b} Kelsey, Tom
  title = "Formal specification of computer algebra",
  (slides) University of St Andrews, Sept 21, 2000
  link = "\url{http://www.cs.st-andrews.cs.uk/~tom/pub/fscbstalk.ps}",
  keywords = "axiomref"

```

— ignore —

```

\bibitem[Kendall 99a]{Ken99a} Kendall, W.S.
  title = "Itovsn3 in AXIOM: modules, algebras and stochastic differentials",
  link = "\url{http://www2.warwick.ac.uk/fac/sci/statistics/staff/academic-research/kendall/pers}",
  keywords = "axiomref"

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— axiom.bib —

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@article{Kend01,
  author = "Kendall, Wilfrid S.",
  title = "Symbolic It~o calculus in AXIOM: an ongoing story",
  journal = "Statistics and Computing",
  volume = "11",

```

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pages = "25-35",
year = "2001",
link = "\url{http://www2.warwick.ac.uk/fac/sci/statistics/staff/academic-research/kendall/personal/ppt/32/}",
abstract =
  "Symbolic It\^o calculus refers both to the implementation of the
  It\^o calculus algebra package and to its application. This article
  reports on progress in the implementation of It\^o calculus in the
  powerful and innovative computer algebra package AXIOM, in the context
  of a decade of previous implementations and applications. It is shown
  how the elegant algebraic structure underlying the expressive and
  effective formalism of It\^o calculus can be implemented directly in
  AXIOM using the package's programmable facilities for "strong
  typing" of computational objects. An application is given of the use
  of the implementation to provide calculations for a new proof, based
  on stochastic differentials, of the Mardia-Dryden distribution from
  statistical shape theory.",
paper = "Kend01.pdf",
keywords = "axiomref",
beebe = "Kendall:2001:SIC"
}

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— Kendall:2001:SIC —

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@Article{Kendall:2001:SIC,
  author = "Wilfrid S. Kendall",
  title = "Symbolic {It\^o} calculus in {AXIOM}: an ongoing
  story",
  journal = "j-STAT-COMP",
  volume = "11",
  number = "1",
  pages = "25--35",
  year = "2001",
  CODEN = "STACE3",
  DOI = "http://dx.doi.org/10.1023/A:1026553731272",
  ISSN = "0960-3174",
  ISSN-L = "0960-3174",
  MRclass = "Database Expansion Item",
  MRnumber = "MR1837142",
  bibsource = "http://www.math.utah.edu/pub/tex/bib/axiom.bib",
  fjjournal = "Statistics and Computing",
  keywords = "computer algebra; coupling of random processes; Dryden
  density Mathematics; financial mathematics; Itovsn3;
  It\^o calculus; It\^o formula; Macsyma; Maple;
  Mardia; REDUCE; semimartingale; statistics of shape;
  stochastic calculus; stochastic integral; symbolic
  It\^o calculus; XIOM",
}

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}

 — axiom.bib —

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@article{Kend07,
  author = "Kendall, Wilfrid S.",
  title = "Coupling all the Levy Stochastic Areas of Multidimensional
    Brownian Motion",
  journal = "The Annals of Probability",
  volume = "35",
  number = "3",
  pages = "935-953",
  year = "2007",
  comment = "Author used Axiom for computation but says missed citation",
  link = "\url{http://arxiv.org/pdf/math/0512336v2.pdf}",
  abstract =
    "It is shown how to construct a successful co-adapted coupling of two
    copies of an  $n$ -dimensional Brownian motion  $(B_1, \dots, B_n)$  while
    simultaneously coupling all corresponding copies of the  $L^1$ -valued
    stochastic areas  $\int_0^t B_{id} B_j - \int_0^t B_j dB_i$ . It is conjectured
    that successful co-adapted couplings still exist when the  $L^1$ -valued
    stochastic areas are replaced by a finite set of multiply iterated
    path- and time-integrals, subject to algebraic compatibility of the
    initial conditions.",
  paper = "Kend07.pdf",
  keywords = "axiomref"
}
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 — axiom.bib —

```
@article{Kerb98,
  author = "Kerber, Manfred and Kohlhase, Michael and Volker, Sorge",
  title = "Integrating computer algebra into proof planning",
  journal = "J. Autom. Reasoning",
  volume = "21",
  number = "3",
  year = "1998",
  pages = "327-355",
  link = "\url{http://citeseerx.ist.psu.edu/viewdoc/download?doi=10.1.1.40.3914}",
  abstract =
    "Mechanized reasoning systems and computer algebra systems have
    different objectives. Their integration is highly desirable, since
```

formal proofs often involve both of the two different tasks proving and calculating. Even more important, proof and computation are often interwoven and not easily separable.

In this article, we advocate an integration of computer algebra into mechanized reasoning systems at the proof plan level. This approach allows us to view the computer algebra algorithms as methods, that is, declarative representations of the problem-solving knowledge specific to a certain mathematical domain. Automation can be achieved in many cases by searching for a hierarchic proof plan at the method level by using suitable domain-specific control knowledge about the mathematical algorithms. In other words, the uniform framework of proof planning allows us to solve a large class of problems that are not automatically solvable by separate systems.

Our approach also gives an answer to the correctness problems inherent in such an integration. We advocate an approach where the computer algebra system produces high-level protocol information that can be processed by an interface to derive proof plans. Such a proof plan in turn can be expanded to proofs at different levels of abstraction, so the approach is well suited for producing a high-level verbalized explication as well as for a low-level, machine-checkable, calculus-level proof. We present an implementation of our ideas and exemplify them using an automatically solved example.",

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paper = "Kerb98.pdf",
keywords = "axiomref"
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}
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— axiom.bib —

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@article{Kocb96,
  author = "Kocbach, Ladislav and Liska, Richard",
  title = "Generation and Verification of Algorithms for Symbolic\_Numeric
    Processing",
  journal = "J. Symbolic Computation",
  volume = "11",
  pages = "1-16",
  year = "1996",
  abstract =
    "Some large scale physical computations require algorithms performing
    symbolic computations with a particular class of algebraic formulas in
    a numerical code. Developing and implementing such algorithms in a
    numerical programming language is a tedious and error prone task. The
    algorithms can be developed in a computer algebra system and their
    correctness can be checked by comparison with built-in facilities of
    the system so that the system is used as an advanced debugging
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tool. After that a numerical code for the algorithms is automatically
generated from the same source code. The proposed methodology is
explained in detail on a simple example. Real applications to
calculation of matrix elements of Coulomb interaction and two-centre
exchange integrals needed in atomic collision codes, are
described. The method makes the developing and debugging of such
algorithms easier and faster.",
paper = "Kocb96.pdf",
keywords = "axiomref"
}

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— axiom.bib —

```

@article{Koep96,
  author = "Koepf, Wolfram",
  title = "Closed form Laurent-Puiseux series of algebraic functions",
  journal = "Appl. Algebra Eng. Commun. Comput.",
  volume = "7",
  number = "1",
  pages = "21-26",
  year = "1996",
  abstract =
    "Let  $y$  be an algebraic function defined by a polynomial equation
 $P(x,y)=0$  whose coefficients are polynomials in  $x$  over a field  $K$ 
which may be one of the fields  $\mathbb{Q}$ ,  $\mathbb{R}$ , or
 $\mathbb{C}$ . D. V. and G. V. Chudnovsky [J. Complexity 2, 271-294
(1986; Zbl 0629.68038); ibid. 3, 1-25 (1987; Zbl 0656.34003)] describe
a pair of algorithms to calculate the coefficients in the
Laurent-Puiseux developments of the branches of  $y$ : The first
algorithm returns a linear differential equation
 $[q_n(x)y^{(n)} + y_{n-1}(x)y^{(n-1)} + \dots + q_1(x)y' + q_0(x)y = 0]$ 
which is satisfied by all branches of  $y$  and whose coefficients are
polynomials in  $x$  over  $K$ , the other uses this differential equation
to get a linear recurrence relation for the Puiseux coefficients. The
author used this algorithms (the second in a simpler version) to
calculate the recurrence relation; if this relation contains only two
terms, an algorithm found by the author returns an explicit formula
for the Puiseux coefficients [J. Symb. Comp. 13, 581-603 (1992; Zbl
0758.30026)]. In this paper, the author gives examples to illustrate
his algorithms and to show that for many algebraic functions defined
by polynomials of low degree a closed form of their Puiseux
coefficients may be calculated. He points out that on the other side
the complexity of the resulting recurrence equation may be extremely
high even for an algebraic function defined by a sparse polynomial of
low degree.",
  keywords = "axiomref"
}

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}

— axiom.bib —

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@InProceedings{Koep99,
  author = "Koepf, Wolfram",
  title = "Orthogonal polynomials and computer algebra",
  booktitle = "Recent developments in complex analysis and computer algebra",
  series = "ISSAC 97",
  year = "1999",
  publisher = "Kluwer Academic Publishers",
  location = "Newark, DE",
  pages = "205-234",
  abstract =
    "Orthogonal polynomials have a long history, and are still important
    objects of consideration in mathematical research as well as in
    applications in Mathematical Physics, Chemistry, and
    Engineering. Quite a lot is known about them. Particularly well-known
    are differential equations, recurrence equations, Rodrigues formulas,
    generating functions and hypergeometric representations for the
    classical systems of Jacobi, Laguerre and Hermite which can be found
    in mathematical dictionaries. Less well-known are the corresponding
    representations for the classical discrete systems of Hahn,
    Krawtchouk, Meixner and Charlier, as well as addition theorems,
    connection relations between different systems and other identities
    for these and other systems of orthogonal polynomials. The ongoing
    research in this still very active subject of mathematics expands the
    knowledge database about orthogonal polynomials continuously. In the
    last few decades the classical families have been extended to a rather
    large collection of polynomial systems, the so-called Askey-Wilson
    scheme, and they have been generalized in other ways as well.
```

Recently new algorithmic approaches have been discovered to compute differential, recurrence and similar equations from series or integral representations. These methods turn out to be quite useful to prove or detect identities for orthogonal polynomial systems. Further algorithms to detect connection coefficients or to identify polynomial systems from given recurrence equations have been developed. Although some algorithmic methods had been known already in the last century, their use was rather limited due to the immense amount of calculations. Only the existence and distribution of computer algebra systems makes their use simple and useful for everybody.

In this plenary lecture an overview is given of how algorithmic methods implemented in computer algebra systems can be used to prove identities about and to detect new knowledge for orthogonal

```

polynomials and other hypergeometric type special functions.
Implementations for this type of algorithms exist in Maple,
Mathematica and REDUCE, and maybe also in other computer algebra
systems. Online demonstrations will be given using Maple V.5.",
paper = "Koep99.pdf",
keywords = "axiomref"
}

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— axiom.bib —

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@article{Koep99a,
  author = "Koepf, Wolfram",
  title = "Software for the algorithmic work with orthogonal polynomials
          and special functions",
  journal = "Electron. Trans. Numer. Anal.",
  volume = "9",
  year = "1999",
  link = "\url{http://arxiv.org/pdf/math/9809125v1.pdf}",
  abstract =
    "An overview of the MAPLE routines that can be used for hypergeometric
    and basic hypergeometric series with some discussion of how and why
    they work.",
  paper = "Koep99a.pdf",
  keywords = "axiomref"
}

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@misc{Koep14,
  author = "Koepf, Wolfram",
  title = "Methods of Computer Algebra for Orthogonal Polynomials",
  year = "2014",
  location = "Rutgers, NJ, USA",
  link = "\url{http://www.mathematik.uni-kassel.de/~koepf/Vortrag/2014-Zeilberger-Vortrag.pdf}",
  video1 = "https://vimeo.com/85573338",
  video2 = "https://vimeo.com/85573712",
  website = "http://www.caop.org",
  paper = "Koep14.pdf"
}

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— axiom.bib —

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@article{Kos191,
  author = "Koseleff, P.-V.",
  title = "Word games in free Lie algebras: several bases and formulas",
  journal = "Theoretical Computer Science",
  volume = "79",
  number = "1",
  pages = "241-256",
  year = "1991",
  abstract =
    "The author compares the efficiency of many methods which allow
    calculations in Lie algebras. Many construction methods exist for the
    base of free Lie algebras developed from finite sets. They use two
    algorithms for calculation of several Campbell-Hausdorf formulas.
    Diverse implementations are realised in LISP on Scratchpad II",
  keywords = "axiomref",
  beebe = "Koseleff:1991:WGF"
}
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— Koseleff:1991:WGF —

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@Article{Koseleff:1991:WGF,
  author = "P.-V. Koseleff",
  title = "Word games in free {Lie} algebras: several bases and
    formulas",
  journal = "j-THEOR-COMP-SCI",
  volume = "79",
  number = "1",
  pages = "241--256",
  month = "feb",
  year = "1991",
  CODEN = "TCSCDI",
  ISSN = "0304-3975 (print), 1879-2294 (electronic)",
  ISSN-L = "0304-3975",
  bibdate = "Tue Sep 17 06:44:07 MDT 1996",
  bibsource = "http://www.math.utah.edu/pub/tex/bib/axiom.bib",
  abstract = "The author compares the efficiency of many methods
    which allow calculations in Lie algebras. Many
    construction methods exist for the base of free Lie
    algebras developed from finite sets. They use two
    algorithms for calculation of several
    Campbell--Hausdorf formulas. Diverse implementations
    are realised in LISP on Scratchpad II.",
  acknowledgement = "ack-nhfb",
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affiliation = "IBM, Paris, France",
classification = "C6130 (Data handling techniques); C7310
                (Mathematics)",
fjournal = "Theoretical Computer Science",
link = "\url{http://www.sciencedirect.com/science/journal/03043975}",
keywords = "Bases; Campbell--Hausdorf formulas; Finite sets; Free
            Lie algebras; LISP; Scratchpad II",
language = "English",
pubcountry = "Netherlands",
thesaurus = "Mathematics computing; Symbol manipulation",
}

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— axiom.bib —

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@article{Kred08,
  author = "Kredel, Heinz",
  title = "On a Java computer algebra system,
           its performance and applications",
  journal = "Sci. Comput. Program.",
  volume = "70",
  number = "2-3",
  pages = "185-207",
  year = "2008",
  link = "\url{http://ac.els-cdn.com/S0167642307001736/1-s2.0-S0167642307001736-main.pdf}",
  abstract =
    "This paper considers Java as an implementation language for a
    starting part of a computer algebra library. It describes a design of
    basic arithmetic and multivariate polynomial interfaces and classes
    which are then employed in advanced parallel and distributed Groebner
    base algorithms and applications. The library is type-safe due to its
    design with Java's generic type parameters and thread-safe using
    Java's concurrent programming facilities. We report on the performance
    of the polynomial arithmetic and on applications built upon the core
    library.",
  paper = "Kred08.pdf",
  keywords = "axiomref"
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@InProceedings{Kred07,
  author = "Kredel, Heinz",

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title = "Evaluation of a Java computer algebra system",
booktitle = "Computer Mathematics, 8th Asian symposium",
series = "ASCM 2007",
year = "2007",
isbn = "978-3-540-87826-1",
location = "Singapore",
pages = "121-138",
link = "\url{http://krum.rz.uni-mannheim.de/kredel/oocas-casc2010-slides.pdf}",
abstract =
    "This paper evaluates the suitability of Java as an implementation
    language for the foundations of a computer algebra library. The design
    of basic arithmetic and multivariate polynomial interfaces and classes
    have been presented. The library is type-safe due to its design with
    Java's generic type parameters and thread-safe using Java's concurrent
    programming facilities. We evaluate some key points of our library and
    differences to other computer algebra systems.",
paper = "Kred07.pdf",
keywords = "axiomref"
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@InProceedings{Kred10,
  author = "Kredel, Heinz and Jolly, Raphael",
  title = "Generic, type-safe and object oriented computer algebra software",
  booktitle = "Proc. 12th International Workshop",
  series = "CASC 2010",
  year = "2010",
  isbn = "978-3-642-15273-3",
  location = "Berlin",
  pages = "162-177",
  link = "\url{http://krum.rz.uni-mannheim.de/kredel/oocas-casc2010-slides.pdf}",
  abstract =
    "Advances in computer science, in particular object oriented
    programming, and software engineering have had little practical impact
    on computer algebra systems in the last 30 years. The software design
    of existing systems is still dominated by ad-hoc memory management,
    weakly typed algorithm libraries and proprietary domain specific
    interactive expression interpreters. We discuss a modular approach to
    computer algebra software: usage of state-of-the-art memory management
    and run-time systems (e.g. JVM) usage of strongly typed, generic,
    object oriented programming languages (e.g. Java) and usage of general
    purpose, dynamic interactive expression interpreters (e.g. Python). To
    illustrate the workability of this approach, we have implemented and
    studied computer algebra systems in Java and Scala. In this paper we
    report on the current state of this work by presenting new examples.",
}

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    paper = "Kred10.pdf",
    keywords = "axiomref"
}

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— axiom.bib —

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@book{Kred11,
  author = "Kredel, Heinz",
  title = "Unique factorization domains in the Java computer algebra system",
  year = "2011",
  publisher = "Springer",
  booktitle = "Automated deduction in geometry (ADG 2008)",
  pages = "86-115",
  isbn = "978-3-642-21045-7",
  abstract =
    "This paper describes the implementation of recursive algorithms in
    unique factorization domains, namely multivariate polynomial greatest
    common divisors (gcd) and factorization into irreducible parts in the
    Java computer algebra library (JAS). The implementation of gcds,
    resultants and factorization is part of the essential building blocks
    for any computation in algebraic geometry, in particular in automated
    deduction in geometry. There are various implementations of these
    algorithms in procedural programming languages. Our aim is an
    implementation in a modern object oriented language with generic data
    types, as it is provided by Java programming language. We exemplify
    that the type design and implementation of JAS is suitable for the
    implementation of several greatest common divisor algorithms and
    factorization of multivariate polynomials. Due to the design we can
    employ this package in very general settings not commonly seen in other
    computer algebra systems. As for example, in the coefficient
    arithmetic for advanced Groebner basis computations like in polynomial
    rings over rational function fields or (finite, commutative) regular
    rings. The new package provides factory methods for the selection of
    one of the several implementations for non experts. Further we
    introduce a parallel proxy for gcd implementations which runs
    different implementations concurrently.",
  keywords = "axiomref"
}

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— axiom.bib —

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@InProceedings{Kred11a,

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author = "Kredel, Heinz and Jolly, Raphael",
title = "Algebraic structures as typed objects",
booktitle = "Proc. 13th International Workshop",
series = "CASC 2011",
year = "2011",
isbn = "978-3-642-23567-2",
location = "Berlin",
pages = "294-308",
link = "\url{http://krum.rz.uni-mannheim.de/kredel/to-cas-casc2011-slides.pdf}",
abstract =
  "Following the research direction of strongly typed, generic, object
  oriented computer algebra software, we examine the modeling of
  algebraic structures as typed objects in this paper. We discuss the
  design and implementation of algebraic and transcendental extension
  fields together with the modeling of real algebraic and complex
  algebraic extension fields. We will show that the modeling of the
  relation between algebraic and real algebraic extension fields using
  the delegation design concept has advantages over the modeling as
  sub-types using sub-class implementation. We further present a summary
  of design problems, which we have encountered so far with our
  implementation in Java and present possible solutions in Scala.",
paper = "Kred11a.pdf",
keywords = "axiomref"
}

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— axiom.bib —

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@book{Kreu14,
  author = "Kreuzer, Edwin",
  title = "Computerized Symbolic Manipulation in Mechanics",
  publisher = "Springer",
  year = "2014",
  abstract =
    "The aim of this book is to present important software tools, basic
    concepts, methods, and highly sophisticated applications of
    computerized symbolic manipulation to mechanics problems. An overview
    about general-purpose symbolic software is followed by general
    guidelines how to develop and implement high-quality computer algebra
    code. The theoretical background including modeling techniques for
    mechanical systems is provided which allows for the computer aided
    generation of the symbolic equation of motion for multibody
    systems. It is shown how the governing equations for different types
    of problems in structural mechanics can be automatically derived and
    how to implement finite element techniques via computer algebra
    software. Perturbation methods as a very powerful approach for
    nonlinear problems are discussed in detail and are demonstrated for a

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number of applications. The applications covered in this book
represent some of the most advanced topics in the rapidly growing
field of research on symbolic computation."
}

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— axiom.bib —

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@misc{Krip94,
  author = "Kripfganz, Jochen and Perlt, Holger",
  title = "Working with Mathematica. An Introduction with examples",
  comment = "Arbeiten mit Mathematica. Eine Einfuhrung mit Beispielen",
  book = "Hander",
  year = "1994",
  keywords = "axiomref"
}

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— axiom.bib —

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@article{Kuma00,
  author = "Kumar, P. and Pellegrino, S.",
  title = "Computation of kinematic paths and bifurcation points",
  journal = "Int. J. Solids Struct.",
  volume = "37",
  number = "46-47",
  pages = "7003-7027",
  year = "2000",
  abstract =
    "This article deals with the kinematic simulation of movable
    structures that go through special configurations of kinematic
    bifurcation, as they move. A series of algorithms are developed for
    structures that can be modelled using pin-jointed bars and that admit
    a single-parameter motion. These algorithms are able to detect and
    locate any bifurcation points that exist along the path of the
    structure and, at each bifurcation point, can determine all possible
    motions of the structure. The theory behind the algorithms is
    explained, and the analysis of a simple example is discussed in
    detail. Then, a simplified version of the particular problem that had
    motivated this work, the simulation of the folding and deployment of a
    thin membrane structure forming a solar sail, is analysed. For the
    particular cases that are considered, it is found that the entire
    process is inextensional, but a detailed study of the simulation
    results shows that in more general cases, it is likely that stretching

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    or wrinkling will occur.",
    keywords = "axiomref"
}

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— axiom.bib —

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@inproceedings{Kusc89,
  author = "Kusche, K. and Kutzler, B. and Mayr, H.",
  title = "Implementation of a geometry theorem proving package
    in SCRATCHPAD II",
  booktitle = "Proc. of Eurocal '87",
  series = "Lecture Notes in Computer Science 378",
  pages = "246-257",
  isbn = "3-540-51517-8",
  year = "1987",
  abstract =
    "The problem of automatically proving geometric theorems has gained a
    lot of attention in the last two years. Following the general approach
    of translating a given geometric theorem into an algebraic one,
    various powerful provers based on characteristic sets and Groebner
    bases have been implemented by groups at Academia Sinica Beijing
    (China), U. Texas at Austin (USA), General Electric Schenectady (USA),
    and Research Institute for Symbolic Computation Linz (Austria). So far,
    fair comparisons of the various provers were not possible, because the
    underlying hardware and the underlying algebra systems differed
    greatly. This paper reports on the first uniform implementation of all
    of these provers in the computer algebra system and language
    SCRATCHPAD II. We summarize the recent achievements in the area of
    automated geometry theorem proving, shortly review the SCRATCHPAD II
    system, describe the implementation of the geometry theorem proving
    package, and finally give a computing time statistics of 24 examples.",
  keywords = "axiomref",
  beebe = "Kusche:1989:IGT"
}

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— Kusche:1989:IGT —

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@InProceedings{Kusche:1989:IGT,
  author = "K. Kusche and B. Kutzler and H. Mayr",
  title = "Implementation of a geometry theorem proving package
    in {SCRATCHPAD} {II}",
  crossref = "Davenport:1989:EEC",

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pages = "246--257",
month = "",
year = "1989",
bibdate = "Tue Sep 17 06:46:18 MDT 1996",
bibsource = "http://www.math.utah.edu/pub/tex/bib/axiom.bib",
abstract = "The problem of automatically proving geometric
theorems has gained a lot of attention in the last two
years. Following the general approach of translating a
given geometric theorem into an algebraic one, various
powerful provers based on characteristic sets and
Gr{\o}bner bases have been implemented by groups at
Academia Sinica Beijing (China), U. Texas at Austin
(USA), General Electric Schenectady (USA), and Research
Institute for Symbolic Computation Linz (Austria). So
far, fair comparisons of the various provers were not
possible, because the underlying hardware and the
underlying algebra systems differed greatly. This paper
reports on the first uniform implementation of all
these provers in the computer algebra system and
language SCRATCHPAD II. The authors summarize the
recent achievements in the area of automated geometry
theorem proving, shortly review the SCRATCHPAD II
system, describe the implementation of the geometry
theorem proving package, and finally give computing
time statistics of 24 examples.",
acknowledgement = ack-nhfb,
affiliation = "Res. Inst. for Symbolic Comput., RISC-LINZ, Johannes
Kepler Univ., Linz, Austria",
classification = "C1230 (Artificial intelligence); C7310
(Mathematics)",
keywords = "Geometry theorem proving package; SCRATCHPAD II;
Characteristic sets; Gr{\o}bner bases; Computer
algebra system; Computing time statistics",
language = "English",
thesaurus = "Algebra; Computational geometry; Mathematics
computing; Symbol manipulation; Theorem proving",
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— ignore —

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\bibitem[Lahey 08]{Lah08} Lahey, Tim
title = "Sage Integration Testing",
link = "\url{http://github.com/tjl/sage_int_testing}",

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year = "2008",
keywords = "axiomref"
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@article{Lamb03a,
  author = "Lamban, Laureano and Pascual, Vico and Rubio, Julio",
  title = "An object-oriented interpretation of the EAT system",
  journal = "Appl. Algebra Eng. Commun. Comput.",
  volume = "14",
  number = "3",
  year = "2003",
  pages = "187-215",
  abstract =
    "In a previous paper we characterized, in the category theory setting,
    a class of implementations of abstract data types, which has been
    suggested by the way of programming in the EAT system. (EAT, Effective
    Algebraic Topology, is one of Sergeraerts systems for effective
    homology and homotopy computation.) This characterization was
    established using classical tools, in an unrelated way to the current
    mainstream topics in the field of algebraic specifications. Looking
    for a connection with these topics, we have found, rather
    unexpectedly, that our approach is related to some object-oriented
    formalisms, namely hidden specifications and the coalgebraic view. In
    this paper, we explore these relations making explicit the implicit
    object-oriented features of the EAT system and generalizing the data
    structure analysis we had previously done.",
  keywords = "axiomref"
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@article{Lamb89,
  author = "Lambe, Larry A.",
  title = "Scratchpad II as a tool for mathematical research",
  journal = "Notices of the AMS",
  year = "1989",
  pages = "143-147",
  keywords = "axiomref"
}

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— axiom.bib —

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@article{Lamb91,
  author = "Lambe, Larry A.",
  title = "Resolutions via homological perturbation",
  journal = "Journal of Symbolic Computation",
  volume = "12",
  number = "1",
  pages = "71-87",
  year = "1991",
  abstract =
    "The purpose of this paper is to review an algorithm for computing
    ‘‘small’’ resolutions in homological algebra, to provide examples of
    its use as promised in [L1], [LS], and to illustrate the use of
    computer algebra in an area not usually associated with that
    subject. Comparison of the complexes produced by the method discussed
    here with those produced by other methods shows that the algorithm
    generalizes several other approaches, [GL], [GLS1], [GLS2], [BL], [BL2].

    This is an expository note which is intended to help make homological
    perturbation theory more accesible and to encourage wider use of
    Computer Algebra in mathematical research.

    The class of objects presented here -- Finitely generated torsion-free
    nilpotent groups (of arbitrary nilpotency class) -- are given because
    of their simplicity. The examples point to the general phenomena that
    are to be expected when trying to derive complexes smaller than
    ‘‘standard complexes’’ in other homological contexts. The complexes
    produced are generally {\sl much} smaller than the bar construction, but
    larger than a {\sl minimal resolution}."
  paper = "Lamb91.pdf",
  keywords = "axiomref",
  beebe = "Lambe:1991:RHP"
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— Lambe:1991:RHP —

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@Article{Lambe:1991:RHP,
  author = "L. A. Lambe",
  title = "Resolutions via homological perturbation",
  journal = "j-J-SYMBOLIC-COMP",
  volume = "12",
  number = "1",
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pages =      "71--87",
month =      jul,
year =       "1991",
CODEN =      "JSYCEH",
ISSN =       "0747-7171 (print), 1095-855X (electronic)",
ISSN-L =     "0747-7171",
bibdate =    "Tue Sep 17 06:44:07 MDT 1996",
bibsource =  "http://www.math.utah.edu/pub/tex/bib/axiom.bib",
abstract =   "There is a trade-off between the size of the
              resolutions which arise from the perturbation method
              and the complexity of the new differential. In order to
              keep the modules relatively small, there is a
              considerable increase in the algebraic complexity of
              the resulting differentials. In order to study such
              complexes systematically, examples are needed. To
              facilitate such study, the Scratchpad system was used
              to set up and perform the necessary calculations.
              Because of the way Scratchpad is organized, this could
              be done in a way that minimizes programming effort and
              provides the natural mathematical environment for such
              calculations. The author discusses some of the general
              theory behind homological perturbation theory, gives an
              idea of what is needed to make calculations within that
              theory in Scratchpad, and calculates a resolution of
              the integers over the integral group ring of the 4*4
              upper triangular matrices with ones along the
              diagonal.",
acknowledgement = ack-nhfb,
affiliation =  "Illinois Univ., Chicago, IL, USA",
classification = "C4240 (Programming and algorithm theory); C7310
               (Mathematics)",
fjournal =    "Journal of Symbolic Computation",
link =        "\url{http://www.sciencedirect.com/science/journal/07477171}",
keywords =    "Algebraic complexity; Complexity; Homological
              perturbation; Integers; Mathematical environment;
              Resolutions; Scratchpad system",
language =    "English",
pubcountry =  "UK",
thesaurus =   "Computational complexity; Perturbation theory; Symbol
              manipulation",
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@article{Lamb92,
  author = "Lambe, Larry",

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title = "Next Generation Computer Algebra Systems AXIOM and the
        Scratchpad Concept: Applications to Research in Algebra",
publisher = "21st Nordic Congress of Mathematicians",
journal = "unknown",
year = "1992",
link = "\url{http://axiom-wiki.newsynthesis.org/public/refs/axiom-21cong.pdf}",
algebra = "\newline\ref{category FMCAT FreeModuleCat}",
abstract =
    "One way in which mathematicians deal with infinite amounts of data is
    symbolic representation. A simple example is the quadratic equation
    \[x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}\]
    a formula which uses symbolic representation to describe the solutions
    to an infinite class of equations. Most computer algebra systems can
    deal with polynomials with symbolic coefficients, but what if symbolic
    exponents are called for (e.g.  $1+t^i$ )? What if symbolic limits on
    summations are also called for, for example
    \[1+t+\ldots+t^i=\sum_j t^j\]
```

The ‘‘Scratchpad Concept’’ is a theoretical ideal which allows the implementation of objects at this level of abstraction and beyond in a mathematically consistent way. The Axiom computer algebra system is an implementation of a major part of the Scratchpad Concept. Axiom (formerly called Scratchpad) is a language with extensible parameterized types and generic operators which is based on the notions of domains and categories. By examining some aspects of the Axiom system, the Scratchpad Concept will be illustrated. It will be shown how some complex problems in homological algebra were solved through the use of this system.",

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paper = "Lambe92.pdf",
keywords = "axiomref",
bebe = "Lambe:1994:NGC"
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— Lambe:1994:NGC —

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@InProceedings{Lambe:1994:NGC,
  author = "Larry Lambe",
  editor = "Mats Gyllenberg and Lars Erik Persson",
  booktitle = "{Analysis, algebra, and computers in mathematical
               research: proceedings of the Twenty-first Nordic
               Congress of Mathematicians, Lule{\aa} University of
               Technology, Sweden, 1992}",
  title = "Next generation computer algebra systems {AXIOM} and
           the {Scratchpad} concept: Applications to research in
           algebra",
  volume = "156",

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publisher = pub-DEKKER,
address = pub-DEKKER:adr,
pages = "201--222",
year = "1994",
ISBN = "0-8247-9217-3",
ISBN-13 = "978-0-8247-9217-6",
LCCN = "QA299.6 .N67 1992",
MRclass = "18-04 (Machine computation, programs (category
theory)) 68W30 (Symbolic computation and algebraic
computation) 20-04 (Machine computation, programs
(group theory)) 18G15 (Ext and Tor, generalizations)
18G35 (Chain complexes (homological algebra)) 55U15
(Chain complexes) 20J05 (Homological methods in group
theory) 16E40 (Homology and cohomology theories for
assoc. rings)",
bibsource = "http://www.math.utah.edu/pub/tex/bib/axiom.bib",
series = "Lecture Notes in Pure and Applied Mathematics",
link = "\url{http://www.loc.gov/catdir/enhancements/fy0647/94002464-d.html}",
ZMnumber = "0832.18001",
abstract = "One way in which mathematicians deal with infinite
amounts of data is symbolic representation. A simple
example is the quadratic equation  $x = \frac{-b \pm \sqrt{b^2 - 4a} c}{2a}$ , a formula which uses symbolic
representation to describe the solutions to an infinite
class of equations. Most computer algebra systems can
deal with polynomials with symbolic coefficients, but
what if symbolic exponents are called for (e.g.,  $1 + t^i$ ) ? What if symbolic limits on summations are also
called for (e.g.,  $1 + t + \cdots + t^i = \sum_j t^j$ )
? The "Scratchpad concept" is a theoretical ideal
which allows the implementation of objects at this
level of abstraction and beyond in a mathematically
consistent way. The AXIOM computer algebra system is an
implementation of a major part of the Scratchpad
concept. AXIOM (formerly called Scratchpad) is a
language with extensible parameterized types and
generic operators which is based on the notion of
domains and categories [R. D. Jenks and R.
S. Sutor}, Axiom. The scientific computation system,
Springer, Berlin etc. (1992; Zbl 0758.68010)]. By
examining some aspects of the AXIOM system, the
Scratchpad concept will be illustrated. It will be
shown how some complex problems in homological algebra
[cf. the author, Contemp. Math. 134, 183-218 (1992; Zbl
0798.16028), J. Pure Appl. Algebra 84, No. 3, 311-329
(1993; Zbl 0766.55015)] were solved through the use of
this system.",
bookpages = "ix + 408",
keywords = "AXIOM; bar construction; computer algebra; domains;
Ext; generic operators; parameterized types;

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        perturbation lemma; Scratchpad; symbolic exponents;
        symbolic limits; Tor",
    language = "English",
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\bibitem[Lambe 93]{Lam93} Lambe, Larry
    title = "On Using Axiom to Generate Code",
    (preprint) 1993
    keywords = "axiomref"

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— axiom.bib —

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@article{Lamb93a,
    author = "Lambe, Larry and Luczak, Richard",
    title = "Object-Oriented Mathematical Programming and
        Symbolic/Numeric Interface",
    journal = "3rd Int. Conf. on Expert Systems in Numerical Computing",
    year = "1993",
    link = "\url{http://axiom-wiki.newsynthesis.org/public/refs/axiom-fem.pdf}",
    abstract =
        "The Axiom language is based on the notions of ‘‘categories’’,
        ‘‘domains’’, and ‘‘packages’’. These concepts are used to build an
        interface between symbolic and numeric calculations. In particular, an
        interface to the NAG Fortran Library and Axiom’s algebra and graphics
        facilities is presented. Some examples of numerical calculations in a
        symbolic computational environment are also included using the finite
        element method. While the examples are elementary, we believe that
        they point to very powerful methods for combining numeric and symbolic
        computational techniques.",
    paper = "Lamb93a.pdf",
    keywords = "axiomref"
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@article{Lamb93b,

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author = "Lambe, Larry A. and Radford, David E.",
title = "Algebraic Aspects of the Quantum Yang-Baxter Equation",
journal = "Journal of Algebra",
volume = "154",
pages = "228-288",
year = "1993",
link = "\url{pages.bangor.ac.uk/~mas019/papers/lamrad.pdf}",
abstract =
  "In this paper we examine a variety of algebraic contexts in which the
  quantum Yang-Baxter equation arises, and derive methods for generating
  new solutions from given ones. The solutions we describe are encoded
  in objects which have a module and a comodule structure over a
  bialgebra. Our work here is based in part on the ideas of [DR1,DR2].",
paper = "Lamb93b.pdf",
keywords = "axiomref"
}

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— axiom.bib —

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@book{Lamb97,
  author = "Lambe, Larry A. and Radford, David E.",
  title = "Introduction to the quantum Yang-Baxter equation and quantum
    groups: an algebraic approach",
  booktitle = "Mathematics and its Applications",
  publisher = "Kluwer Academic Publishers",
  year = "1997",
  abstract =
    "The quantum Yang-Baxter equation (QYBE) has roots in statistical
    mechanics and the inverse scattering method and leads to a natural
    construction of a bialgebra. It turns out to have important
    connections with knot theory and invariants of 3-manifolds. There are
    now available many reference books to quantum groups and these various
    applications. The book under review develops the algebraic
    underpinning and theory of the QYBE, including the constant form and
    the one and two parameter forms.

```

We give a brief description of the chapters. Chapter 1 (together with an Appendix) gives the algebraic preliminaries involving coalgebras, bialgebras, Hopf algebras, modules and comodules. Chapter 2 introduces the various forms of the QYBE, and the basic algebraic structures associated to them, including Faddeev-Reshetikhin-Takhtadzhian (FRT) construction. Chapter 3 explores various categorical settings for the constant form of the QYBE, the most basic being the category of left QYB modules over a bialgebra and the notion of algebras, coalgebras, etc. in this category. Chapter 4 develops universal mapping properties of the FRT construction and its reduced version, and the authors

investigate when the reduced FRT construction leads to a pointed bialgebra or a pointed Hopf algebra. Chapter 5 develops the quantum groups associated to $SL(2)$, i.e., the quantum universal enveloping algebra, and the quantum function algebra. Chapter 6 introduces quasitriangular Hopf algebras, and discusses how the finite-dimensional ones give rise to solutions of the QYBE through their representation theory. The most important example is the Drinfeld double of a finite-dimensional Hopf algebra. The authors note (through an exercise!) that every finite-dimensional Hopf algebra is the reduced FRT construction of some solution to the QYBE. Chapter 7 introduces coquasitriangular bialgebras, the most important being the FRT and the reduced FRT constructions. There are some generalizations here to the one-parameter form of the QYBE. Chapter 8 uses all the previously developed techniques to find solutions of the QYBE in certain cases, including the one-parameter form. Some of these were discovered by computer algebra methods. The final chapter 9 gives a brief discussion of certain categorical constructions and the QYBE is certain fairly abstract categories, motivated by the fact that the FRT construction is a coend.

This book fills an important niche in the literature involving the QYBE by highlighting the algebraic aspects and applications. Although this is basically a reference book, it includes so many important parts of the study of Hopf algebras that it could be used as a textbook for a certain type of course on Hopf algebras and quantum groups, and certainly as supplementary reading material for such a course. There are frequent exercises which would be useful for such purposes. Besides being a basic source book, the authors include some new results and some novel approaches to earlier results. All this makes this book a most welcome addition to the quantum group literature."

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keywords = "axiomref"
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@article{Lamb02,
  author = "Lambe, Larry A. and Seiler, Werner M.",
  title = "Differential equations, Spencer cohomology, and computing
    resolutions",
  journal = "Georgian Mathematical Journal",
  volume = "9",
  number = "4",
  pages = "723-774",
  year = "2002",
  link = "\url{https://www.emis.de/journals/GMJ/vol9/v9n4-11.pdf}"}
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abstract =
  "Spencer cohomology is one of the classical tools in the study of
  overdetermined systems of partial differential equations. The paper
  proposes a new point of view of the Spencer cohomology based on a dual
  approach via comodules that allows to relate the Spencer cohomology to
  standard constructions in homological algebra, and discuss concrete
  methods for its construction based on homological perturbation
  theory. It also gives a detailed introduction to the subject, and
  proposes a new algorithm of constructing the Spencer resolution by
  using symbolic computation systems.",
paper = "Lamb02.pdf",
keywords = "axiomref"
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@article{Lamb03,
  author = "Lambe, Larry A. and Luczak, Richard and Nehrbass, John W.",
  title = "A New Finite Difference Method for the Helmholtz Equation
    Using Symbolic Computation",
  journal = "Int. J. Comp. Eng. Sci.",
  volume = "4",
  year = "2003",
  link = "\url{http://pages.bangor.ac.uk/~mas019/papers/lln.pdf}",
  abstract =
    "A new finite difference method for the Helmholtz equation is
    presented. The method involves replacing the standard ‘‘weights’’ in the
    central difference quotients (Secs. 2.1, 2.2, and 2.3) by weights that
    are optimal in a sense that will be explained in the sections just
    mentioned. The calculation of the optimal weights involves some
    complicated and error-prone manipulations of integral formulas that is
    best done using computer-aided symbolic computation (SC). In addition,
    we discuss the important problem of interpolation involving meshes
    that have been refined in certain subregions. Analytic formulae are
    derived using SC for these interpolation schemes. Our results are
    discussed in Sec. 5. Some hints about the computer methods we used to
    accomplish these results are given in the Appendix. More information
    is available and access to that information is referenced.

    While we do not want to make SC the focus of this work, we also do not
    want to underestimate its value. Armed with robust and efficient SC
    libraries, a researcher can {\sl comfortably} and {\sl conveniently}
    experiment with ideas that he or she might not examine otherwise.",
  paper = "Lamb03.pdf",
  keywords = "axiomref"
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@InProceedings{Laza93,
  author = "Lazard, Daniel",
  title = "On the representation of rigid-body motions and its application
          to generalized platform manipulators",
  booktitle = "Proc. Workshop Computational Kinematics",
  year = "1993",
  location = "Dagstuhl Castle, Germany",
  publisher = "Kluwer Academic Publishers",
  pages = "175-181",
  abstract =
    "Different ways for representing rigid body motions (direct isometries)
     by a computer are presented. It turns out that the choice between them
     may have a dramatic effect on the difficulty of a computation or of a
     proof. As an application, a computational proof is given of the fact
     that the direct kinematic problem for the generalized Stewart platform
     has at most 40 complex solutions.",
  keywords = "axiomref"
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\bibitem[Lebedev 08]{Leb08} Lebedev, Yuri
  title = "OpenMath Library for Computing on Riemann Surfaces",
  PhD thesis, Nov 2008 Florida State University
  link = "\url{http://www.math.fsu.edu/~ylebedev/research/HyperbolicGeometry.html}",
  keywords = "axiomref"
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@inproceedings{LeBl91,
  author = "LeBlanc, S.E.",
  title = "The use of MathCAD and Theorist in the ChE classroom",
  booktitle = "Proc. ASEE Annual Meeting",
  year = "1991",
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pages = "287-299",
abstract =
  "MathCAD and Theorist are two powerful mathematical packages available
  for instruction in the ChE classroom. MathCAD is advertised as an
  'electronic scratchpad' and it certainly lives up to its billing. It
  is an extremely user-friendly collection of numerical routines that
  eliminates the drudgery of solving many of the types of problems
  encountered by undergraduate ChE's (and engineers in general). MathCAD
  is available for both the Macintosh and IBM PC compatibles. The PC
  version is available as a full-functioned student version for around
  US\$40 (less than many textbooks). Theorist is a symbolic mathematical
  package for the Macintosh. Many interesting and instructive things can
  be done with it in the ChE curriculum. One of its many attractive
  features includes the ability to generate high quality three
  dimensional plots that can be very instructive in examining the
  behavior of an engineering system. The author discusses the
  application and use of these packages in chemical engineering and give
  example problems and their solutions for a number of courses including
  stoichiometry, unit operations, thermodynamics and design.",
keywords = "axiomref",
beebe = "LeBlanc:1991:UMT"
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  author =      "S. E. LeBlanc",
  title =      "The use of {MathCAD} and {Theorist} in the {ChE}
               classroom",
  crossref =   "Anonymous:1991:PAC",
  pages =      "287--299 (vol. 1)",
  year =       "1991",
  bibdate =    "Tue Sep 17 06:37:45 MDT 1996",
  bibsource =  "http://www.math.utah.edu/pub/tex/bib/axiom.bib",
  abstract =   "MathCAD and Theorist are two powerful mathematical
               packages available for instruction in the ChE
               classroom. MathCAD is advertised as an 'electronic
               scratchpad' and it certainly lives up to its billing.
               It is an extremely user-friendly collection of
               numerical routines that eliminates the drudgery of
               solving many of the types of problems encountered by
               undergraduate ChE's (and engineers in general). MathCAD
               is available for both the Macintosh and IBM PC
               compatibles. The PC version is available as a
               full-functioned student version for around US\$40 (less
               than many textbooks). Theorist is a symbolic

```

```

mathematical package for the Macintosh. Many
interesting and instructive things can be done with it
in the ChE curriculum. One of its many attractive
features includes the ability to generate high quality
three dimensional plots that can be very instructive in
examining the behavior of an engineering system. The
author discusses the application and use of these
packages in chemical engineering and give example
problems and their solutions for a number of courses
including stoichiometry, unit operations,
thermodynamics and design.",
acknowledgement = ack-nhfb,
affiliation = "Toledo Univ., OH, USA",
classification = "C7450 (Chemical engineering); C7810C (Computer-aided
instruction)",
keywords = "Chemical engineering; MathCAD; Mathematical packages;
Numerical routines; Stoichiometry; Symbolic
mathematical package; Theorist; Thermodynamics; Unit
operations",
language = "English",
thesaurus = "Chemical engineering computing; Computer aided
instruction; Spreadsheet programs; Symbol
manipulation",
}

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@article{Lece02,
  author = "Lecerf, Gregoire",
  title = "Quadratic Newton iteration for systems with multiplicity",
  journal = "Found. Comput. Math.",
  volume = "2",
  number = "3",
  pages = "247-293",
  year = "2002",
  abstract =
    "The author proposes an efficient iterator with quadratic convergence
    that generalizes Newton iterator for multiple roots. It is based on an
     $m$ -adic topology where the ideal  $m$  can be chosen generic
    enough. Compared to the Newton iterator the proposed iterator
    introduces a small overhead that grows with the square of the
    multiplicity of the root.",
  paper = "Lece02.pdf",
  keywords = "axiomref"
}

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@InProceedings{Lece10,
  author = "Lecerf, Gregoire",
  title = "Mathemagix: toward large scale programming for symbolic and
          certified numeric computations",
  booktitle = "Mathematical software",
  series = "ICMS 2010",
  year = "2010",
  isbn = "978-3-642-15581-9",
  location = "Berlin",
  pages = "329-332",
  abstract =
    "Coordinated by Joris van der Hoeven from the 90's, the Mathemagix
    project aims at the design of a scientific programming language for
    symbolic and certified numeric algorithms. This language can be
    compiled and interpreted, and it features a strong type system with
    classes and categories. Several C++ libraries are also being
    developed, mainly with Bernard Mourrain and Philippe Trebuchet, for
    the elementary operations with polynomials, power series and matrices,
    with a special care towards efficiency and numeric stability.

    In my talk I will give an overview of the language, of the design and
    the contents of the C++ libraries, and I will illustrate possibilities
    offered for certified numeric computations with balls and intervals.",
  keywords = "axiomref"
}
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@misc{Lece96,
  author = "Lecerf, Gregoire",
  title = "Dynamic Evaluation and Real Closure Implementation in Axiom",
  year = "1996",
  link = "\url{http://lecerf.perso.math.cnrs.fr/software/drc/drc.ps}",
  paper = "Lece96.pdf",
  keywords = "axiomref"
}
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\bibitem[Lecerf 96a]{Le96a} Lecerf, Gr'egoire
  title = "The Dynamic Real Closure implemented in Axiom",
  link = "\url{http://lecerf.perso.math.cnrs.fr/software/drc/drc.ps}",
  keywords = "axiomref"
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@article{Leeu94,
  author = "van Leeuwen, Andre M.A.",
  title = "LiE, A software package for Lie group computations",
  journal = "Euromath Bulletin",
  volume = "1",
  number = "2",
  year = "1994",
  abstract =
    "A description is given of LiE, a specialized computer algebra package
    for computations concerning Lie groups and algebras, and their
    representations. The functionality of the package is demonstrated by
    sample computations, and the structure of the program and the
    algorithms implemented in it are discussed.",
  paper = "Leeu94.pdf",
  keywords = "axiomref"
}
```

— axiom.bib —

```
@article{Leti97,
  author = "Letichevskij, A. Alexander and Marinchenko, V. G.",
  title = "Objects in algebraic programming system",
  journal = "Cybern. Syst. Anal.",
  volume = "33",
  number = "2",
  pages = "283-299",
  year = "1997",
  comment = "translated from Russian",
  abstract =
    "The algebraic programming system (APS) developed at the
    V. M. Glushkov Institute of Cybernetics of the Academy of Sciences of
```

the Ukrainian SSR integrates the basic programming paradigms, including procedural, functional, algebraic, and logic programming.

Algebraic programming in APS relies on special data structures, the so-called graph terms, which permit using diverse data and knowledge representations in relevant application domains. In the language APLAN, graph terms are described by expressions or systems of expressions of a many-sorted algebra of data. They may represent both objects of the application domain and reasoning about these objects. The option of setting an arbitrary interpretation of the operations in the algebra of data makes it possible to use APS as a basis for various extensions.

Symbolic computation systems such as Scratchpad/AXIOM have acquired special importance. They provide various possibilities of manipulating typed mathematical objects, including objects of complex hierarchical structure. This is a natural requirement when working with algebraic objects. In particular, the properties of many algebraic structures (such as groups, rings, fields, etc.) are naturally hierarchical-modular.

The Institute of Cybernetics and the Kherson Teachers College have developed an instruction-oriented computer algebra system AIST. The AIST kernel is a hierarchical structure of mathematical concepts described in the APS language. However, construction of new applications on the basis of this hierarchical structure has proved difficult. The system kernel can be made more flexible by providing tools for flexible description of hierarchical structures of mathematical concepts.

In this article, we describe an extension of the language APLAN, which provides tools for the object-oriented style of programming. This is one of the possible ways of introducing types in APS. The object-oriented technology also can be used to develop a hierarchical system of mathematical objects.",

```
keywords = "axiomref"
}
```

— ignore —

```
\bibitem[Levelt 95]{Lev95} Levelt, A. H. M. (ed)
ISSAC '95: Proceedings of the 1995 International
Symposium on Symbolic and Algebraic Computation: July 10-12, 1995, Montreal,
Canada ISSAC-PROCEEDINGS-1995. ACM Press, New York, NY 10036, USA, 1995
ISBN 0-89791-699-9 LCCN QA76.95 I59 1995 ACM order number 505950
keywords = "axiomref"
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— axiom.bib —

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@article{Lew99,
  author = "Lewis, Robert H. and Wester, Michael",
  title = "Comparison of polynomial-orienged computer algebra systems",
  journal = "SIGSAM Bulletin",
  volume = "33",
  number = "4",
  pages = "5-13",
  year = "1999",
  link = "\url{https://home.bway.net/lewis/cacomp.ps}",
  abstract =
    "Exact symbolic computation with polynomials and matrices over
    polynomial rings has wide applicability to many fields [Hereman96,
    Lewis99]. By ‘‘exact symbolic’’, we mean computation with polynomials
    whose coefficients are integers (of any size), rational numbers, or
    from finite fields, as opposed to coefficients that are ‘‘floats’’ of a
    certain precision. Such computation is part of most computer algebra
    (CA) systems. Over the last dozen years, several large CA systems have
    become widely available, such as Axiom, Derive, Macsyma, Maple,
    Mathematica and Reduce. They tend to have great breadth, be produced
    by profit-making companies, and be relatively expensive, at least for
    a full blown non-student version. However, most if not all of these
    systems have difficulty computing with the polynomials and matrices
    that arise in actual research. Real problems tend to produce large
    polynomials and large matrices that the general CA systems cannot
    handle [Lewis99].
```

In the last few years, several smaller CA systems focused on polynomials have been produced at universities by individual researchers or small teams. They run on Macs, PCs and workstations. They are freeware or shareware. Several claim to be much more efficient than the large systems at exact polynomial computations. The list of these systems includes CoCoA, Fermat, MuPAD, Pari-Gp and Singular [CoCoA, Fermat, MuPAD, Pari-Gp, Singular].

In this paper, we compare these small systems to each other and to two of the large systems (Magma and Maple) on a set of problems involving exact symbolic computation with polynomials and matrices. The problems here involve:

```
\begin{itemize}
\item the ground rings  $\mathbb{Z}$ ,  $\mathbb{Q}$ ,  $\mathbb{Z}/p\mathbb{Z}$ 
and other finite fields
\item basic arithmetic of polynomials over the ground ring
\item basic arithmetic of rational functions over the ground ring
```

```

\item polynomial evaluation (substitution)
\item matrix normal form
\item determinants and characteristic polynomial
\item GCDs of multivariate polynomial
\item resultants
\end{itemize}",
paper = "Lew99.pdf",
keywords = "axiomref"
}

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— axiom.bib —

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@inproceedings{Laio95,
  author = "Liao, Hsin-Chao and Fateman, Richard J.",
  title = "Evaluation of the heuristic polynomial GCD",
  booktitle = "Proc. ISSAC 1995",
  year = "1995",
  pages = "240-247",
  link = "\url{http://www.cs.berkeley.edu/~fateman/papers/phil8.ps}",
  abstract =
    "The Heuristic Polynomial GCD procedure (GCDHEU) is used by the Maple
    computer algebra system, but not other. Because Maple has an
    especially efficient kernel that provides fast integer arithmetic, but
    a relatively slower interpreter for non-kernel code, the GCDHEU
    routine is especially effective in that it moves much of the
    computation into ‘bignum’ arithmetic and hence executes primarily in
    the kernel.

```

We speculated that in other computer algebra systems an implementation GCDHEU would not be advantageous. In particular, if {\sl all} the system code is compiled to run at ‘full speed’ in a (presumably more bulky) kernel that is entirely written in C or compiled Lisp, then there would seem to be no point in recasting the polynomial gcd problem into a bignum GCD problem. Manipulating polynomials that are vectors of coefficients would seem to be equivalent computationally to manipulating vectors of big digits.

Yet our evidence suggests that one can take advantage of the GCDHEU in a Lisp system as well. Given a good implementation of bignums, for most small problems and many large ones, a substantial speedup can be obtained by the appropriate choice of GCD algorithm, including often enough, the GCDHEU approach. Another major winner seem to be the subresultant polynomial remainder sequence algorithm. Because more sophisticated sparse algorithms are relatively slow on small problems and only occasionally emerge as superior (on larger problems) it seems the choice of a fast GCD algorithm is tricky.",


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    paper = "phil8.pdf"
}

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— axiom.bib —

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@phdthesis{Lixx05,
  author = "Li, Xin",
  title = "Efficient Management of Symbolic Computation with Polynomials",
  school = "University of Western Ontario",
  year = "2005",
  link = "\url{http://www.csd.uwo.ca/~moreno//Publications/XinLi-MSThesis-2005.pdf.gz}",
  abstract =
    "Symbolic polynomial computation is widely used to solve many applied
    or abstract mathematical problems. Some of them, such as solving
    systems of polynomial equations, have exponential complexity. Their
    implementation is, therefore, a challenging task.

    By using adapted data structures, asymptotically fast algorithms and
    effective code optimization techniques, we show how to reduce the
    practical and theoretical complexity of these computations. Our effort
    is divided into three categories: integrtrng the best known techniques
    into our implementation, investigating new directions, and measuring
    the interactions between numerous techniques.

    We chose AXIOM and ALDOR as our implementation and experimentation
    environment, since they are both strongly typed and highly efficient
    Computer Algebra Systems (CAS). Our implementation results show that
    our methods have great potential to improve the efficiency of exact
    polynomial computations with the selected CASs. The performance of our
    implementation is comparable to that of (often outperforming) the best
    available packages for polynomial computations.",
  paper = "Lixx05.pdf",
  keywords = "axiomref"
}

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— axiom.bib —

```

@InProceedings{Lixx06,
  author = "Li, Xin and Moreno Maza, Marc",
  title = "Efficient implementation of polynomial arithmetic in a
    multiple-level programming environment",

```

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booktitle = "Mathematical Software",
series = "ICMS 2006",
year = "2006",
isbn = "978-3-540-38084-9",
location = "Spain",
pages = "12-23",
link = "\url{http://www.math.kobe-u.ac.jp/icms2006/icms2006-video/slides/046.pdf}",
abstract =
    "The purpose of this study is to investigate implementation techniques
    for polynomial arithmetic in a multiple-level programming
    environment. Indeed, certain polynomial data types and algorithms can
    further take advantage of the features of lower level languages, such
    as their specialized data structures or direct access to machine
    arithmetic. Whereas, other polynomial operations, like Groebner basis
    over an arbitrary field, are suitable for generic programming in a
    high-level language.

    We are interested in the integration of polynomial data type
    implementations realized at different language levels, such as Lisp, C
    and Assembly. In particular, we consider situations for which code
    from different levels can be combined together within the same
    application in order to achieve high-performance. We have developed
    implementation techniques in the multiple-level programming
    environment provided by the computer algebra system AXIOM. For a given
    algorithm realizing a polynomial operation, available at the user
    level, we combine the strengths of each language level and the
    features of a specific machine architecture. Our experimentations show
    that this allows us to improve performances of this operation in a
    significant manner.",
paper = "Lixx06.pdf",
keywords = "axiomref",
beebe = "Li:2006:EIP"
}

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— Li:2006:EIP —

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@Article{Li:2006:EIP,
  author = "Xin Li and Moreno Maza",
  title = "Efficient Implementation of Polynomial Arithmetic in a
    Multiple-Level Programming Environment",
  journal = "j-LECT-NOTES-COMP-SCI",
  volume = "4151",
  pages = "12--23",
  year = "2006",
  CODEN = "LNCSD9",
  ISBN = "3-540-38084-1",
}

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ISBN-13 =      "978-3-540-38084-9",
ISSN =         "0302-9743 (print), 1611-3349 (electronic)",
ISSN-L =       "0302-9743",
bibdate =      "Mon Apr 19 08:40:16 2010",
bibsource =    "http://www.math.utah.edu/pub/tex/bib/axiom.bib",
note =         "Proceedings of the International Congress of
                Mathematical Software (ICMS 2006).",
link =         "\url{http://www.csd.uwo.ca/~moreno//Publications/Li-MorenoMaza-ICMS-06.pdf}",
acknowledgement = ack-nhfb,
fjournal =     "Lecture Notes in Computer Science",
keywords =     "Axiom symbolic-algebra system",
}

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— axiom.bib —

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@inproceedings{Lixx07,
  author = "Li, Xin and Maza, Marc Moreno and Schost, Eric",
  title = "On the Virtues of Generic Programming for Symbolic Computation",
  booktitle = "Computational Science -- ICCS 2007",
  series = "Lecture Notes in Computer Science",
  isbn = "3-540-72585-7",
  pages = "251-258",
  year = "2007",
  abstract =
    "The purpose of this study is to measure the impact of C level code
    polynomial arithmetic on the performances of Axiom high-level algorithms,
    such as polynomial factorization. More precisely, given a high-level
    Axiom package P parameterized by a univariate polynomial domain U, we
    have compared the performances of P when applied to different U's,
    including an Axiom wrapper for our C level code.

    Our experiments show that when P relies on U for its univariate
    polynomial computations, our specialized C level code can provide a
    significant speed-up. For instance, the improved implementation of
    square-free factorization in Axiom is 7 times faster than the one in
    Maple and very close to the one in Magma. On the contrary, when P does
    not rely much on the operations of U and implements its private univariate
    polynomial operation, the P cannot benefit from our highly optimized C
    level code. Consequently, code which is poorly generic reduces the
    speed-up opportunities when applied to highly efficient and specialized.",
  keywords = "axiomref",
  paper = "Lixx07.pdf",
  beebe = "Li:2007:VGP"
}

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— Li:2007:VGP —

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@InProceedings{Li:2007:VGP,
  author = "Xin Li and Marc Moreno Maza and {\'}eric Schost",
  title = "On the Virtues of Generic Programming for Symbolic
    Computation",
  crossref = "Shi:2007:CSib",
  pages = "251--258",
  year = "2007",
  DOI = "http://dx.doi.org/10.1007/978-3-540-72586-2_35",
  bibdate = "Tue Aug 12 10:36:21 2014",
  bibsource = "http://www.math.utah.edu/pub/tex/bib/axiom.bib;
    http://www.math.utah.edu/pub/tex/bib/magma.bib;
    http://www.math.utah.edu/pub/tex/bib/maple-extract.bib",
  acknowledgement = ack-nhfb,
  keywords = "Axiom; Magma; Maple",
}
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— axiom.bib —

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@phdthesis{Lixx09a,
  author = "Li, Xin",
  title = "Toward High-Performance Polynomial System Solvers Based On
    Triangular Decompositions",
  school = "University of Western Ontario",
  year = "2009",
  link = "\url{http://www.csd.uwo.ca/~moreno/Publications/XinLiPhDThesis-2008.pdf}",
  abstract =
    "This thesis is devoted to the design and implementation of polynomial
    system solvers based on symbolic computation. Solving systems of
    non-linear, algebraic or differential equations, is a fundamental
    problem in mathematical sciences. It has been studied for centuries
    and still stimulates many research developments, in particular on the
    front of high-performance computing.

    Triangular decompositions are a highly promising technique with the
    potential to produce high-performance polynomial system solvers. This
    thesis makes several contributions to this effort.

    We propose asymptotically fast algorithms for the core operations on
    which triangular decompositions rely. Complexity results and comparative
    implementation show that these new algorithms provide substantial
    performance improvements."
```

We present a fundamental software library for polynomial arithmetic in order to support the implementation of high-performance solvers based on triangular decompositions. We investigate strategies for the integration of this library in high-level programming environments where triangular decompositions are usually implemented.

We obtain a high performance library combining highly optimized C routines and solving procedures written in the Maple computer algebra system. The experimental result shows that our approaches are very effective, since our code often outperforms pre-existing solvers in a significant manner.",

```
paper = "Lixx09a.pdf",
keywords = "axiomref"
}
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— axiom.bib —

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@InProceedings{Lixx09,
  author = "Li, Xin and Moreno Maza, Marc and Schost, Eric",
  title = "Fast arithmetic for triangular sets: from theory to practice",
  booktitle = "Proc 32nd ISSAC",
  series = "ISSAC 2007",
  year = "2007",
  isbn = "978-1-59593-743-8",
  location = "Canada",
  pages = "269-276",
  link = "\url{http://www.csd.uwo.ca/~moreno/Publications/LiMorenoSchost-ISSAC-2007.pdf}",
  abstract =
    "We study arithmetic operations for triangular families of
    polynomials, concentrating on multiplication in dimension zero. By a
    suitable extension of fast univariate Euclidean division, we obtain
    theoretical and practical improvements over a direct recursive
    approach; for a family of special cases, we reach quasi-linear
    complexity. The main outcome we have in mind is the acceleration of
    higher-level algorithms, by interfacing our low-level implementation
    with languages such as AXIOM or Maple. We show the potential for huge
    speed-ups, by comparing two AXIOM implementations of vanHoeij and
    Monagan's modular GCD algorithm.",
  paper = "Lixx09.pdf",
  keywords = "axiomref"
}
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— ignore —

```
\bibitem[Li 10]{YL10} Li, Yue; Dos Reis, Gabriel
  title = "A Quantitative Study of Reductions in Algebraic Libraries",
  PASC0 2010
  link = "\url{http://www.axiomatics.org/~gdr/concurrency/quant-pasco10.pdf}",
  keywords = "axiomref"
```

— axiom.bib —

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@inproceedings{Lixx11,
  author = "Li, Yue and Dos Reis, Gabriel",
  title = "An Automatic Parallelization Framework for Algebraic
    Computation Systems",
  booktitle = "Proc. ISSAC 2011",
  pages = "233-240",
  isbn = "978-1-4503-0675-1",
  year = "2011",
  link = "\url{http://www.axiomatics.org/~gdr/concurrency/oa-conc-issac11.pdf}",
  abstract =
    "This paper proposes a non-intrusive automatic parallelization
    framework for typeful and property-aware computer algebra systems.
    Automatic parallelization remains a promising computer program
    transformation for exploiting ubiquitous concurrency facilities
    available in modern computers. The framework uses semantics-based
    static analysis to extract reductions in library components based on
    algebraic properties. An early implementation shows up to 5 times
    speed-up for library functions and homotopy-based polynomial system
    solver. The general framework is applicable to algebraic computation
    systems and programming languages with advanced type systems that
    support user-defined axioms or annotation systems.",
  paper = "Lixx11.pdf",
  keywords = "axiomref"
}
```

— ignore —

```
\bibitem[Ligatsikas 96]{Liga96} Ligatsikas, Zenon; Rioboo, Renaud;
  Roy, Marie Francoise
  title = "Generic computation of the real closure of an ordered field",
```

Math. and Computers in Simulation 42 pp 541-549 (1996)

```
abstract = "
  This paper describes a generalization of the real closure computation
  of an ordered field (Rioboo, 1991) enabling to use different techniques
  to code a single real algebraic number.",
paper = "Liga96.pdf",
keywords = "axiomref"
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— ignore —

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\bibitem[Linton 93]{Lin93} Linton, Steve
  title = "Vector Enumeration Programs, version 3.04",
  link = "\url{http://www.cs.st-andrews.ac.uk/~sal/nme/nme_toc.html#SEC1}",
  keywords = "axiomref"
```

— axiom.bib —

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@misc{Lint98,
  author = "Linton, Stephen",
  title = "The GAP 4 Type System Organising Algebraic Algorithms",
  link = "\url{http://www.gap-system.org/Doc/Talks/kobe.ps}",
  paper = "Lint98.pdf",
  keywords = "axiomref"
}
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— axiom.bib —

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@misc{Linu15,
  author = "Linux Links",
  title = "Axiom",
  link = "\url{http://www.linuxlinks.com/article/20080810064748970/Axiom.html}",
  year = "2015",
  keywords = "axiomref"
}
```

— ignore —

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\bibitem[Liska 97]{LD97} Liska, Richard; Drska, Ladislav; Limpouch, Jiri;
Sinor, Milan; Wester, Michael; Winkler, Franz
  title = "Computer Algebra - algorithms, systems and applications",
June 2, 1997
  link = "\url{http://kfe.fjfi.cvut.cz/~liska/ca/all.html}",
  keywords = "axiomref"
```

— axiom.bib —

```
@InProceedings{Lomo02,
  author = "Lomonaco, Samuel J. and Kauffman, Louis H.",
  title = "Quantum hidden subgroup algorithms: a mathematical perspective",
  booktitle = "Quantum computation and information",
  series = "AMS special session",
  year = "2002",
  isbn = "0-8218-2140-7",
  location = "Washington",
  pages = "139-202",
  link = "\url{https://arxiv.org/pdf/quant-ph/0201095v3.pdf}",
  abstract =
    "The ultimate objective of this paper is to create a stepping stone to
    the development of new quantum algorithms. The strategy chosen is to
    begin by focusing on the class of abelian quantum hidden subgroup
    algorithms, i.e., the class of abelian algorithms of the Shor/Simon
    genre. Our strategy is to make this class of algorithms as
    mathematically transparent as possible. By the phrase ‘‘mathematically
    transparent’’ we mean to expose, to bring to the surface, and to make
    explicit the concealed mathematical structures that are inherently and
    fundamentally a part of such algorithms. In so doing, we create
    symbolic abelian quantum hidden subgroup algorithms that are analogous
    to the those symbolic algorithms found within such software packages
    as Axiom, Cayley, Maple, Mathematica, and Magma."
```

As a spin-off of this effort, we create three different generalizations of Shors quantum factoring algorithm to free abelian groups of finite rank. We refer to these algorithms as wandering (or vintage $\mathbb{Z}_Q$) Shor algorithms. They are essentially quantum algorithms on free abelian groups of finite rank n which, with each iteration, first select a random cyclic direct summand $\mathbb{Z}$ of the group and then apply one iteration of the standard Shor algorithm to produce a random character of the approximating finite group $\widetilde{A} = \mathbb{Z}_Q$, called the group probe. These characters are then in turn used to find either the order P of a


```

maximal cyclic subgroup  $\mathbb{Z}_P$  of the hidden quotient group
 $H_{\varphi}$ , or the entire hidden quotient group  $H_{\varphi}$ . An
integral part of these wandering quantum algorithms is the selection
of a very special random transversal  $t_{\mu}:\widetilde{A}\rightarrow A$ ,
which we refer to as a Shor transversal. The algorithmic time
complexity of the first of these wandering Shor algorithms is found to
be  $O(n^2(\lg Q)^3 (\lg Q)^{n+1})$ .",
paper = "Lomo02.pdf",
keywords = "axiomref"
}

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— axiom.bib —

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@inproceedings{Luck86,
  author = "Lucks, Michael",
  title = "A fast implementation of polynomial factorization",
  booktitle = "Proc. 1986 Symposium on Symbolic and Algebraic Computation",
  series = "SYMSAC '86",
  year = "1986",
  location = "Waterloo, Ontario",
  pages = "228-232",
  publisher = "ACM Press",
  isbn = "0-89791-199-7",
  doi = "http://dx.doi.org/10.1145/32439.32485",
  abstract =
    "A new package for factoring polynomials with integer coefficients is
    described which yields significant improvements over previous
    implementations in both time and space requirements. For multivariate
    problems, the package features an inexpensive method for early
    detection and correction of spurious factors. This essentially solves
    the multivariate extraneous factor problem and eliminates the need to
    factor more than one univariate image, except in rare cases. Also
    included is an improved technique for coefficient prediction which is
    successful more frequently than prior versions at short-circuiting the
    expensive multivariate Hensel lifting stage. In addition some new
    approaches are discussed for the univariate case as well as for the
    problem of finding good integer substitution values. The package has
    been implemented both in Scratchpad II and in an experimental version
    of muMATH.",
  keywords = "axiomref",
  beebe = "Lucks:1986:FIP"
}

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— Lucks:1986:FIP —

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@InProceedings{Lucks:1986:FIP,
  author =      "Michael Lucks",
  editor =      "Bruce W. Char",
  booktitle =   "Proceedings of the 1986 Symposium on Symbolic and
                  Algebraic Computation: Symsac '86, July 21--23, 1986,
                  Waterloo, Ontario",
  title =       "A fast implementation of polynomial factorization",
  publisher =   pub-ACM,
  address =     pub-ACM:adr,
  pages =       "228--232",
  year =        "1986",
  DOI =         "http://dx.doi.org/10.1145.32485",
  ISBN =        "0-89791-199-7",
  ISBN-13 =     "978-0-89791-199-3",
  LCCN =         "QA155.7.E4 A281 1986",
  bibdate =     "Thu Jul 26 09:06:12 2001",
  bibsource =   "http://www.math.utah.edu/pub/tex/bib/axiom.bib",
  note =        "ACM order number 505860.",
  acknowledgement = ack-nhfb,
  keywords =     "Scratchpad",
}

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— axiom.bib —

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@mastersthesis{Luek77,
  author = "Lueken, E.",
  title = "Ueberlegungen zur Implementierung eines Formelmanipulationssystems",
  school = {Technischen Universit{"a"}t Carolo-Wilhelmina zu Braunschweig},
  address = "Braunschweig, Germany",
  year = "1977",
  keywords = "axiomref",
  beebe = "Lueken:1977:UIF"
}

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— Lueken:1977:UIF —

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@MastersThesis{Lueken:1977:UIF,
  author =      "E. Lueken",
  title =       "{Ueberlegungen zur Implementierung eines

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Formelmanipulationssystemes}",
school = "Technischen Universit{\{a\}t Carolo-Wilhelmina zu
          Braunschweig",
address = "Braunschweig, Germany",
pages = "??",
year = "1977",
bibsourc = "/usr/local/src/bib/bibliography/Misc/TUBScsd.bib;
            http://www.math.utah.edu/pub/tex/bib/axiom.bib",
descriptor = "Alpak, Altran, Formac, Funktion, G.g.t., Kanonische
              Darstellung von Polynomen, Macsyma, Mathlab, Polynom,
              Rationale Funktion, Reduce, Sac-1, Scratchpad",
}

```

— axiom.bib —

```

@article{Lync91,
  author = "Lynch, R. and Mavromatis, H. A.",
  title = "New quantum mechanical perturbation technique using an
           'electronic scratchpad' on an inexpensive computer",
  journal = "American Journal of Pyhsics",
  volume = "59",
  number = "3",
  pages = "270-273",
  year = "1991",
  abstract =
    "The authors have developed a new method for doing numerical quantum
    mechanical perturbation theory. It has the flavor of
    RayleighSchrödinger perturbation theory (division of the Hamiltonian
    into an unperturbed Hamiltonian and a perturbing term, use of the
    basis formed by the eigenfunctions of the unperturbed Hamiltonian)
    while turning out to be a variational technique. Furthermore, it is
    easily implemented by means of the widely used electronic
    scratchpad, MathCAD 2.0, using an inexpensive computer. As an
    example of the method, the problem of a harmonic oscillator with a
    quartic perturbing term is examined.",
  keywords = "axiomref",
  beebe = "Lynch:1991:NQM"
}

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— Lynch:1991:NQM —

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@Article{Lynch:1991:NQM,

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author =      "R. Lynch and H. A. Mavromatis",
title =      "New quantum mechanical perturbation technique using an
              'electronic scratchpad' on an inexpensive computer",
journal =     j-AMER-J-PHYSICS,
volume =     "59",
number =     "3",
pages =      "270--273",
month =      mar,
year =       "1991",
CODEN =      "AJPIAS",
ISSN =       "0002-9505 (print), 1943-2909 (electronic)",
ISSN-L =     "0002-9505",
bibdate =    "Tue Sep 17 06:44:07 MDT 1996",
bibsource =  "http://www.math.utah.edu/pub/tex/bib/axiom.bib",
abstract =    "The authors have developed a new method for doing
              numerical quantum mechanical perturbation theory. It
              has the flavor of Rayleigh--Schrödinger
              perturbation theory (division of the Hamiltonian into
              an unperturbed Hamiltonian and a perturbing term, use
              of the basis formed by the eigenfunctions of the
              unperturbed Hamiltonian) while turning out to be a
              variational technique. Furthermore, it is easily
              implemented by means of the widely used 'electronic
              scratchpad,' MathCAD 2.0, using an inexpensive
              computer. As an example of the method, the problem of a
              harmonic oscillator with a quartic perturbing term is
              examined.",
acknowledgement = ack-nhfb,
affiliation =  "Dept. of Phys., King Fahd Univ. of Pet. and Miner.,
              Dhahran, Saudi Arabia",
classification = "A0150H (Instructional computer use); A0210 (Algebra,
              set theory, and graph theory); A0230 (Function theory,
              analysis); A0365D (Functional analytical methods);
              A0365F (Algebraic methods); A0365G (Solutions of wave
              equations: bound state); C7810C (Computer-aided
              instruction)",
fjournal =    "American Journal of Physics",
keywords =    "Electronic scratchpad; Eigenvalues; Eigenfunctions;
              Quantum mechanical perturbation technique;
              Rayleigh--Schrödinger perturbation theory;
              Hamiltonian; Variational technique; MathCAD 2.0;
              Harmonic oscillator",
language =    "English",
pubcountry =  "USA",
thesaurus =   "Computer aided instruction; Eigenvalues and
              eigenfunctions; Harmonic oscillators; Perturbation
              theory; Quantum theory; Variational techniques",
}

```

M

— ignore —

```
\bibitem[Mahboubi 05]{Mah05} Mahboubi, Assia
  title = "Programming and certifying the CAD algorithm inside the coq system",
  Mathematics, Algorithms, Proofs, volume 05021 of Dagstuhl
  Seminar Proceedings, Schloss Dagstuhl (2005)
  keywords = "axiomref"
```

— axiom.bib —

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@article{Math89,
  author = "Mathews, J.",
  title = "Symbolic computational algebra applied to Picard iteration",
  journal = "Mathematics and computer education",
  volume = "23",
  number = "2",
  pages = "117-122",
  year = "1989",
  link = "\url{http://mathfaculty.fullerton.edu/mathews/articles/1989PicardIteration.pdf}",
  abstract =
    "The term ‘‘Picard iteration’’ occurs two places in undergraduate
    mathematics. In numerical analysis it is used when discussing fixed
    point iteration for finding a numerical approximation to the equation
     $y = g(x)$ . In differential equations, Picard iteration is a
    constructive procedure for establishing the existence of a solution to
    a differential equation  $y' = f(x,y)$ ."
```

The first type of Picard iteration uses computations to generate a sequence of numbers which converges to a solution. We will not present this application, but mention that it involves the traditional role of the computer as a ‘‘number cruncher.’’

The second application of Picard iteration illustrates how to use a computer to generate a sequence of functions which converges to a solution. The purpose of this article is to show the step by step process in translating mathematical theory into the symbolic manipulation setting. Systems such as MACSYM, ALTRAN, REDUCE, SMP, MAPLE, SCRATCHPAD and muMATH are being introduced in undergraduate mathematics courses to assist in keeping track of equations during complicated manipulations.",

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paper = "Math89.pdf",
keywords = "axiomref",
beebe = "Mathews:1989:SCA"
}

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— Mathews:1989:SCA —

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@Article{Mathews:1989:SCA,
  author = "J. Mathews",
  title = "Symbolic computational algebra applied to {Picard}
           iteration",
  journal = j-MATH-COMP-EDU,
  volume = "23",
  number = "2",
  pages = "117--122",
  month = "Spring",
  year = "1989",
  CODEN = "MCEDDA",
  ISSN = "0730-8639",
  bibdate = "Tue Sep 17 06:48:10 MDT 1996",
  bibsource = "http://www.math.utah.edu/pub/tex/bib/axiom.bib",
  abstract = "Picard iteration occurs in differential equations as a
              constructive procedure for establishing the existence
              of a solution to a differential equation. This
              application of Picard iteration illustrates how to use
              a computer to generate a sequence of functions which
              converges to a solution. The article shows the step by
              step process in translating mathematical theory into
              the symbolic manipulation setting. Systems such as
              MACSYMA, ALTRAN, REDUCE, SMP, MAPLE, SCRATCHPAD, and
              muMATH are being introduced in undergraduate
              mathematics courses to assist in keeping track of
              equations during complicated manipulations. The product
              muMATH is illustrated because of its availability. It
              runs on all 16-bit computers which are IBM compatible.
              The way has been opened to see how computers can be
              used as a symbol cruncher.",
  acknowledgement = ack-nhfb,
  affiliation = "California State Univ., Fullerton, CA, USA",
  classification = "C4130 (Interpolation and function approximation);
                   C4170 (Differential equations); C6130 (Data handling
                   techniques); C7310 (Mathematics)",
  fjournal = "Mathematics and computer education",
  keywords = "Differential equations; IBM compatible; Mathematical
              theory; Mathematics computing; MuMATH; Picard
              iteration; Symbol cruncher; Symbolic manipulation;

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        Undergraduate mathematics",
    language = "English",
    pubcountry = "USA",
    thesaurus = "Differential equations; Iterative methods; Mathematics
        computing; Microcomputer applications; Symbol
        manipulation",
}

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— axiom.bib —

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@misc{Maxi16,
    author = "Maxima",
    title = "Other Free Computer Algebra Systems",
    link = "\url{http://maxima.sourceforge.net/compalg.html}",
    year = "2016",
    abstract =
        "Axiom is a general purpose Computer Algebra system. It is useful for
        doing mathematics by computer and for research and development of
        mathematical algorithms. It defines a strongly typed, mathematically
        correct type hierarchy. It has a programming language and a built-in
        compiler.

        There is also an interesting Rosetta Stone which offers translations
        of many basic operations for several computer algebra systems,
        including Maxima.",
    keywords = "axiomref"
}

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— axiom.bib —

```

@techreport{Maza00,
    author = "Maza, Marc Moreno",
    title = "On Triangular Decompositions of Algebraic Varieties",
    institution = "Numerical Algorithms Group",
    year = "2000",
    month = "June",
    type = "technical report",
    number = "TR 4/99",
    link = "\url{http://www.csd.uwo.ca/~moreno/Publications}",
    algebra =
        "\newline\refto{category TSETCAT TriangularSetCategory}
        \newline\refto{category RSETCAT RegularTriangularSetCategory}

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\newline\refto{category NTSCAT NormalizedTriangularSetCategory}
\newline\refto{category SFRTCAT SquareFreeRegularTriangularSetCategory}
\newline\refto{package RSDCMPK RegularSetDecompositionPackage}",
abstract =
  "Different kinds of triangular decompositions of algebraic varieties
  are presented. The main result is an efficient method for obtaining
  them. Our strategy is based on a lifting theorem for polynomial
  computations module regular chains.",
paper = "Maza00.pdf",
keywords = "axiomref"
}

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— axiom.bib —

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@article{Mcge10,
  author = "McGettrick, Michael",
  title = "One dimensional quantum walks with memory",
  journal = "Quantum Inf. Comput.",
  volume = "10",
  number = "5-6",
  pages = "509-524",
  year = "2010",
  link = "\url{https://arxiv.org/pdf/0911.1653v2.pdf}",
  abstract =
    "We investigate the quantum versions of a one-dimensional random walk,
    whose corresponding Markov chain is of order 2. This corresponds to
    the walk having a memory of one previous step. We derive the
    amplitudes and probabilities for these walks, and point out how they
    differ from bot classical random walks, and quantum walks without
    memory.",
  paper = "Mcge10.pdf",
  keywords = "axiomref"
}

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— ignore —

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\bibitem[McJones 11]{McJ11} McJones, Paul
  title = "Software Presentation Group -- Common Lisp family",
  link = "\url{http://www.softwarepreservation.org/projects/LISP/common_lisp_family}",
  keywords = "axiomref"

```

— axiom.bib —

```
@article{Mela90,
  author = "Melachrinoudis, E.; Rumpf, D. L.",
  title = "Teaching advantages of transparent computer software -- MathCAD",
  journal = "CoED",
  volume = "10",
  number = "1",
  pages = "71-76",
  year = "1990",
  abstract =
    "The case is presented for using mathematical scratchpad software,
    such as MathCAD, in undergraduate and graduate engineering
    courses. The pedagogical benefits, especially relative to the usual
    black box engineering software, are described. Several examples of
    student written projects are presented. The projects solve problems in
    operations research, control theory and statistical regression
    analysis.",
  keywords = "axiomref",
  beebe = "Melachrinoudis:1990:TAT"
}
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— Melachrinoudis:1990:TAT —

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@Article{Melachrinoudis:1990:TAT,
  author = "E. Melachrinoudis and D. L. Rumpf",
  title = "Teaching advantages of transparent computer software
    --- {MathCAD}",
  journal = "j-COED",
  volume = "10",
  number = "1",
  pages = "71--76",
  month = "jan # "-" # mar",
  year = "1990",
  CODEN = "CWLJDP",
  ISSN = "0736-8607",
  bibdate = "Tue Sep 17 06:46:18 MDT 1996",
  bibsource = "http://www.math.utah.edu/pub/tex/bib/axiom.bib",
  abstract = "The case is presented for using mathematical
    scratchpad software, such as MathCAD, in undergraduate
    and graduate engineering courses. The pedagogical
    benefits, especially relative to the usual black box
    engineering software, are described. Several examples
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```

        of student written projects are presented. The projects
        solve problems in operations research, control theory
        and statistical regression analysis.",
    acknowledgement = ack-nhfb,
    affiliation = "Dept. of Ind. Eng., Northeastern Univ., Boston, MA,
        USA",
    classification = "C7110 (Education); C7310 (Mathematics); C7400
        (Engineering); C7810C (Computer-aided instruction)",
    fjournal = "CoED",
    keywords = "Black box engineering software; Control theory;
        Graduate engineering courses; MathCAD; Mathematical
        scratchpad software; Operations research; Pedagogical
        benefits; Statistical regression analysis; Student
        written projects; Transparent computer software;
        Undergraduate",
    language = "English",
    pubcountry = "USA",
    thesaurus = "CAD; Educational computing; Engineering computing;
        Mathematics computing; Teaching",
}

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— axiom.bib —

```

@misc{Mele88,
    author = "Melenk, H. and Moller, H. M. and Neun, W.",
    title = "On Groebner Bases Computation on a Supercomputer Using REDUCE",
    link = "\url{https://opus4.kobv.de/opus4-zib/files/10/SC-88-02.pdf}",
    abstract =
        "Groebner bases are the main tool for solving systems of algebraic
        equations and some other problems in connection with polynomial ideals
        using Computer Algebra Systems. The procedure for the computation of
        Groebner bases in REDUCE 3.3 has been modified in order to solve more
        complicated algebraic systems of equations by some general
        improvements and by some tools based on the specific resources of the
        CRAY X-MP. We present this modification and illustrate it by examples.",
    paper = "Mele88.pdf",
    keywords = "axiomref"
}

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— axiom.bib —

```

@article{Mich01,

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```

author = "Michler, Gerhard O.",
title = "The character values of multiplicity-free irreducible constituents
        of a transitive permutation representation",
journal = "Kyushu J. Math.",
volume = "55",
number = "1",
pages = "75-106",
year = "2001",
link = "\url{https://www.jstage.jst.go.jp/article/kyushujm/55/1/55\_1\_75/\_pdf}",
abstract =
    "Let  $G$  be a finite group and  $M$  a subgroup of  $G$ . Let the
    permutation character of  $G$  on the set of right cosets of  $M$  be
    denoted by  $(1_M)^G$ . Any ordinary irreducible constituent of
 $(1_M)^G$  with multiplicity one is called a multiplicity-free
    constituent of  $(1_M)^G$ . In the paper under review the author gives
    an efficient algorithm to compute the values of any multiplicity-free
    constituent of  $(1_M)^G$ . The character value is in terms of the
    double coset decomposition of  $M$  and related concepts concerning
    intersection matrices. The author uses this algorithm to determine the
    values of the multiplicity-free constituents of  $(1_C)^{J_1}$ , where
 $J_1$  is the smallest Janko group of order 175560 and  $C$  is the
    centralizer of an involution in  $J_1$ . Using these, he then is able to
    compute the character table of  $J_1$  which is already known.",
paper = "Mich01.pdf",
keywords = "axiomref"
}

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— axiom.bib —

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@misc{Mill95,
author = "Miller, Bruce R.",
title = "An expression formatter for MACSYMA",
year = "1995",
abstract =
    "A package for formatting algebraic expressions in MACSYA is described.
    It provides facilities for user-directed hierarchical structuring of
    expressions, as well as for directing simplifications to selected
    subexpressions. It emphasizes a semantic rather than syntactic description
    of the desired form. The package also provides utilities for obtaining
    efficiently the coefficients of polynomials, trigonometric sums and
    power series. Similar capabilities would be useful in other computer
    algebra systems.",
paper = "Mill95.pdf",
keywords = "axiomref"
}

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— axiom.bib —

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@misc{Mino07,
  author = "Minoiu, N. and Netto, M and Mammar, S",
  title = "Assistance control based on a composite Lyapunov function for
    lane departure avoidance",
  booktitle = "Proc. 15 Med. Conf. on Control \& Automation",
  year = "2007",
  abstract =
    "This paper presents a vehicle steering assistance designed to avoid
    lane departure during driver inattention periods. Activated for a
    driver loss of concentration during a lane keeping maneuver the
    steering assistance drives the vehicle back to the center of the
    lane. In order to ensure a vehicle trajectory as close as possible to
    the centerline, the control law has been developed based on invariant
    sets theory and on composite Lyapunov functions. The computation has
    been performed using LMI methods, which allow in addition imposing a
    maximum bound for the control steering angle.",
  keywords = "axiomref"
}
```

— ignore —

```
\bibitem[Miola 90]{Mio90} Miola, A. (ed)
  title = "Design and Implementation of Symbolic Computation Systems",
  International Symposium DISCO '90, Capri, Italy, April 10-12, 1990, Proceedings
  volume 429 of Lecture Notes in Computer Science,
  Springer-Verlag, Berlin, Germany / Heidelberg, Germany / London, UK / etc.,
  1990 ISBN 0-387-52531-9 (New York), 3-540-52531-9 (Berlin) LCCN QA76.9.S88I576
  1990
  keywords = "axiomref"
```

— ignore —

```
\bibitem[Miola 93]{Mio93} Miola, A. (ed)
  title = "Design and Implementation of Symbolic Computation Systems",
  International Symposium DISCO '93 Gmunden, Austria, September 15-17, 1993:
  Proceedings.
  Springer-Verlag, Berlin, Germany / Heidelberg, Germany / London, UK / etc.,
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1993 ISBN 3-540-57235-X LCCN QA76.9.S88I576 1993
 keywords = "axiomref"

— axiom.bib —

```
@book{Mish93,
  author = "Mishra, Bhubaneswar",
  title = "Algorithmic Algebra",
  publisher = "Springer-Verlag",
  series = "Texts and Monographs in Computer Sciences",
  year = "1993",
  abstract =
    "This book is based on a graduate course in computer science taught in
    1987. The following topics are covered: computational ideal theory,
    solving systems of polynomial equations, elimination theory, real
    algebra, as well as an introduction chapter and two chapters with the
    needed algebraic background. The book is self-contained and the proofs
    are given with many details.

    It is clear that this book is only an introduction to the topic and
    does not cover the many improvements that appeared in the last 7 years
    about for example the computation of Groebner basis, polynomial
    solving, multivariate resultants and algorithms in real
    algebra. Choices had to be made to keep the content of a reasonable
    size and the complexity issues are not considered.

    The choice of topics is excellent, there are many exercises and
    examples. It is a very useful book.",
  keywords = "axiomref"
}
```

— axiom.bib —

```
@InProceedings{Miss05,
  author = "Missura, Stephan A.",
  title = {Theories = Signatures + Propositions Used as Types},
  booktitle = "Integrating Symbolic Mathematics and Artificial Intelligence",
  volume = "958",
  pages = "144-155",
  year = "2005",
  abstract =
    "Languages that distinguish between types and structures use explicit
```

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    components for the carrier type(s) in structures. Examples are the
    function language Standard ML and most algebraic specification
    systems. Hence, they have to use general sum types or signatures to
    give types to structures and be able to build, for instance, the
    algebraic hierarchy.",
    paper = "Miss05",
    keywords = "axiomref"
}

```

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```

\bibitem[Missura 94]{Miss94} Missura, Stephan A.; Weber, Andreas
    title = "Using Commutativity Properties for Controlling Coercions",
    link = "\url{http://cg.cs.uni-bonn.de/personal-pages/weber/publications/pdf/WeberA/MissuraWeber94a.pdf}",
    abstract = "
        This paper investigates some soundness conditions which have to be
        fulfilled in systems with coercions and generic operators. A result of
        Reynolds on unrestricted generic operators is extended to generic
        operators which obey certain constraints. We get natural conditions
        for such operators, which are expressed within the theoretic framework
        of category theory. However, in the context of computer algebra, there
        arise examples of coercions and generic operators which do not fulfil
        these conditions. We describe a framework -- relaxing the above
        conditions -- that allows distinguishing between cases of ambiguities
        which can be resolved in a quite natural sense and those which
        cannot. An algorithm is presented that detects such unresolvable
        ambiguities in expressions.",
    paper = "Miss94.pdf",
    keywords = "axiomref"

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— ignore —

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\bibitem[Monagan 87]{Mon87} Monagan, Michael B.
    title = "Support for Data Structures in Scratchpad II",
    in [Wit87], pp17-18
    keywords = "axiomref"

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— axiom.bib —

```

@inproceedings{Mona93,
  author = "Monagan, Michael B.",
  title = "Gauss: a parameterized domain of computation system with
          support for signature functions",
  booktitle = "Design and Implementation of Symbolic Computation Systems",
  series = "Lecture Notes in Computer Science 722",
  pages = "81-94",
  isbn = "3-540-57235-X",
  year = "1993",
  link = "\url{http://www.cecm.sfu.ca/~monaganm/papers/DISC093.pdf}",
  abstract =
    "The fastest known algorithms in classical algebra make use of signature
    functions. That is, reducing computation with formulae to computing with
    the integers module  $p$ , by substituting random numbers for variables,
    and mapping constants module  $p$ . This idea is exploited in specific
    algorithms in computer algebra systems, e.g. algorithms for polynomial
    greatest common divisors. It is also used as a heuristic to speed up
    other calculations. But none exploit it in a systematic manner. The
    goal of this work was twofold. First, to design an AXIOM like system
    in which these signature functions can be constructed automatically,
    hence better exploited, and secondly, to exploit them in new ways. In
    this paper we report on the design of such a system, Gauss.",
  paper = "Mona93.pdf",
  keywords = "axiomref",
  beebe = "Monagan:1993:GPD"
}

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— Monagan:1993:GPD —

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@InProceedings{Monagan:1993:GPD,
  author = "M. B. Monagan",
  title = "{Gauss}: a parameterized domain of computation system
          with support for signature functions",
  crossref = "Miola:1993:DIS",
  pages = "81--94",
  month = "",
  year = "1993",
  bibdate = "Fri Dec 29 12:46:02 MST 1995",
  bibsource = "http://www.math.utah.edu/pub/tex/bib/axiom.bib",
  abstract = "The fastest known algorithms in classical algebra make
             use of signature functions. That is, reducing
             computation with formulae to computing with the
             integers modulo  $p$ , by substituting random numbers for
             variables, and mapping constants modulo  $p$ . This idea is
             exploited in specific algorithms in computer algebra
             systems, e.g. algorithms for polynomial greatest common

```

```

divisors. It is also used as a heuristic to speed up
other calculations. But none exploit it in a systematic
manner. The author designs an AXIOM like system in
which these signature functions can be constructed
automatically, hence better exploited. He exploits them
in new ways. He reports on the design of such a system,
Gauss.",
acknowledgement = ack-nhfb,
affiliation = "Inst. fur Wissenschaftliches Rechnen, ETH, Zurich,
              Switzerland",
classification = "C4130 (Interpolation and function approximation);
                  C7310 (Mathematics)",
keywords = "AXIOM like system; Classical algebra; Computation
            system; Computer algebra systems; Gauss; Heuristic;
            Integers modulo; Mapping constants modulo;
            Parameterized domain; Polynomial greatest common
            divisors; Random numbers; Signature functions",
language = "English",
thesaurus = "Polynomials; Symbol manipulation",
}

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— axiom.bib —

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@misc{Mona94,
  author = "Monagan, Michael B. and Gonnet, Gaston H.",
  title = "Signature Functions for Algebraic Numbers",
  link = "\url{http://lib.org/by/_djvu\_Papers/Computer\_algebra/Algebraic\%20numbers}",
  abstract =
    "In 1980 Schwartz gave a fast {\sl probabilistic} method which tests
    if a matrix of polynomials of  $\mathbb{Z}$  is singular or not. The
    method is based on the idea of {\sl signature functions} which are
    mappings of mathematical expressions into finite rings. In Schwartz's
    paper, they were polynomials over  $\mathbb{Z}$  into  $\text{GF}(p)$ . Because
    computation in  $\text{GF}(p)$  is very fast compared with computing with
    polynomials, Schwartz's method yields an enormous speedup both in
    theory and in practice. Therefore it is desirable to extend the class
    of expressions for which we can find effective signature functions. In
    the mid 80's Gonnet extended the class of expressions, for which
    signature functions can be found, to include a restricted class of
    elementary functions and integer roots. In this paper we present and
    compare methods for constructing signature functions for expressions
    containing {\sl algebraic numbers}. Some experimental results are
    given.",
  paper = "Mona94.djvu",
  keywords = "axiomref"
}

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— axiom.bib —

```
@InProceedings{Mona07,
  author = "Monagan, Michael and Pearce, Roman",
  title = "Polynomial division using dynamic arrays, heaps, and packed
          exponent vectors",
  booktitle = "Computer algebra in scientific computing",
  series = "CASC 2007",
  year = "2007",
  isbn = "978-3-540-75186-1",
  location = "Bonn",
  pages = "295-315",
  link = "\url{http://www.cecm.sfu.ca/~rpearcea/sdmp/sdmp\_paper.pdf}",
  abstract =
    "A common way of implementing multivariate polynomial multiplication
    and division is to represent polynomials as linked lists of terms
    sorted in a term ordering and to use repeated merging. This results in
    poor performance on large sparse polynomials.

    In this paper we use an auxiliary heap of pointers to reduce the
    number of monomial comparisons in the worst case while keeping the
    overall storage linear. We give two variations. In the first, the size
    of the heap is bounded by the number of terms in the quotient(s). In
    the second, which is new, the size is bounded by the number of terms
    in the divisor(s).

    We use dynamic arrays of terms rather than linked lists to reduce
    storage allocations and indirect memory references. We pack monomials
    in the array to reduce storage and to speed up monomial
    comparisons. We give a new packing for the graded reverse
    lexicographical ordering.

    We have implemented the heap algorithms in C with an interface to
    Maple. For comparison we have also implemented Yans ‘‘geobuckets’’ data
    structure. Our timings demonstrate that heaps of pointers are
    comparable in speed with geobuckets but use significantly less
    storage.",
  paper = "Mona07.pdf",
  keywords = "axiomref"
}
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— axiom.bib —

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@misc{Mont07,
  author = "Montes, Antonio",
  title = "On the canonical discussion of polynomial systems with parameters",
  year = "2007",
  link = "\url{http://arxiv.org/pdf/math/0601674.pdf}",
  abstract =
    "Given a parametric polynomial ideal  $I$ , the algorithm DISPG,
    introduced by the author in 2002, builds up a binary tree describing a
    dichotomic discussion of the different reduced Groebner bases
    depending on the values of the parameters, whose set of terminal
    vertices form a Comprehensive Groebner System (CGS). It is relevant
    to obtain CGSs having further properties in order to make them more
    useful for the applications. In this paper the interest is focused on
    obtaining a canonical CGS. We define the objective, show the
    difficulties and formulate a natural conjecture. If the conjecture is
    true then such a canonical CGS will exist and can be computed. We also
    give an algorithm to transform our original CGS in this direction and
    show its utility in applications.",
  paper = "Mont07.pdf",
  keywords = "axiomref"
}
```

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```
\bibitem[Mora 89]{Mor89} Mora, T. (ed)
Applied Algebra, Algebraic Algorithms and Error-Correcting
Codes, 6th International Conference, AAECC-6, Rome, Italy, July 4-8, 1998,
Proceedings, volume 357 of Lecture Notes in Computer Science
Springer-Verlag, Berlin, Germany / Heidelberg, Germany / London, UK / etc.,
1989 ISBN 3-540-51083-4, LCCN QA268.A35 1988 Conference held jointly with
ISSAC '88
  keywords = "axiomref"
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\bibitem[Moses 71]{Mos71} Moses, Joel
  title = "Algebraic Simplification: A Guide for the Perplexed",
  CACM August 1971 Vol 14 No. 8 pp527-537
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keywords = "axiomref"

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```
\bibitem[Moses 08]{Mos08} Moses, Joel
  title = "Macsyma: A Personal History",
  Invited Presentation in Milestones in Computer Algebra, May 2008, Tobago
  link = "\url{http://esd.mit.edu/Faculty_Pages/moses/Macsyma.pdf}",
  abstract = "
    The Macsyma system arose out of research on mathematical software in
    the AI group at MIT in the 1960's. Algorithm development in symbolic
    integration and simplification arose out of the interest of people,
    such as the author, who were also mathematics students. The later
    development of algorithms for the GCD of sparse polynomials, for
    example, arose out of the needs of our user community. During various
    times in the 1970's the computer on which Macsyma ran was one of the
    most popular notes on the ARPANET. We discuss the attempts in the late
    70's and the 80's to develop Macsyma systems that ran on popular
    computer architectures. Finally, we discuss the impact of the
    fundamental ideas in Macsyma on current research on large scale
    engineering systems.",
  paper = "Mos08.pdf",
  keywords = "axiomref"
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— axiom.bib —

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@misc{Mulders95,
  author = "Mulders, Thom",
  title = "Primitives: Orepoly and Lodo",
  year = "1995",
  link = "\url{ftp://ftp.inf.ethz.ch/org/cathode/workshops/jan95/abstracts/mulders.ps}",
  algebra =
    "\newline\refto{category OREPCAT UnivariateSkewPolynomialCategory}
    \newline\refto{category LODOCAT LinearOrdinaryDifferentialOperatorCategory}
    \newline\refto{domain AUTOMOR Automorphism}
    \newline\refto{domain ORESUP SparseUnivariateSkewPolynomial}
    \newline\refto{domain OREUP UnivariateSkewPolynomial}
    \newline\refto{domain LODO LinearOrdinaryDifferentialOperator}
    \newline\refto{domain LODO1 LinearOrdinaryDifferentialOperator1}
    \newline\refto{domain LODO2 LinearOrdinaryDifferentialOperator2}
    \newline\refto{package APPLYORE ApplyUnivariateSkewPolynomial}
    \newline\refto{package OREPCTO UnivariateSkewPolynomialCategoryOps}
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\newline\refto{package LODOF LinearOrdinaryDifferentialOperatorFactorizer}
\newline\refto{package LODOOPS LinearOrdinaryDifferentialOperatorsOps}",
paper = "Mulld95.pdf",
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N

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@InProceedings{Nay106,
  author = "Naylor, William and Padget, Julian",
  title = "From Untyped to Polymorphically Typed Objects in Mathematical
    Web Services",
  booktitle = "Lecture Notes in Computer Science",
  volume = "4108",
  pages = "222-236",
  year = "2006",
  abstract =
    "OpenMath is a widely recognized approach to the semantic markup of
    mathematics that is often used for communication between OpenMath
    compliant systems. The Aldor language has a sophisticated
    category-based type system that was specifically developed for the
    purpose of modelling mathematical structures, while the system itself
    supports the creation of small-footprint applications suitable for
    deployment as web services. In this paper we present our first results
    of how one may perform translations from generic OpenMath objects into
    values in specific Aldor domains, describing how the Aldor interfae
    domain ExpresstionTree is used to achieve this. We outline our Aldor
    implementation of an OpenMath translator, and describe an efficient
    extention of this to the Parser category. In addition, the Aldor
    service creation and invocation mechanism are explained. Thus we are
    in a position to develop and deploy mathematical web services whose
    descriptions may be directly derived from Aldor's rich type language.",
  paper = "NPxx.pdf",
  keywords = "axiomref"
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\bibitem[Naylor 95]{N95} Naylor, Bill
  title = "Symbolic Interface for an advanced hyperbolic PDE solver",
  link = "\url{http://www.sci.csd.uwo.ca/~bill/Papers/symbInterface2.ps}",
  abstract = "
    An Axiom front end is described, which is used to generate
    mathematical objects needed by one of the latest NAG routines, to be
    included in the Mark 17 version of the NAG Numerical library. This
    routine uses powerful techniques to find the solution to Hyperbolic
    Partial Differential Equations in conservation form and in one spatial
    dimension. These mathematical objects are non-trivial, requiring much
    mathematical knowledge on the part of the user, which is otherwise
    irrelevant to the physical problem which is to be solved. We discuss
    the individual mathematical objects, considering the mathematical
    theory which is relevant, and some of the problems which have been
    encountered and solved during the FORTRAN generation necessary to
    realise the object. Finally we display some of our results.",
  paper = "N95.pdf",
  keywords = "axiomref"

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— ignore —

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\bibitem[Naylor 00b]{ND00} Naylor, W.A.; Davenport, J.H.
  title = "A Monte-Carlo Extension to a Category-Based Type System",
  link = "\url{http://www.sci.csd.uwo.ca/~bill/Papers/monteCarCat3.ps}",
  abstract = "
    The normal claim for mathematics is that all calculations are 100\%
    accurate and therefore one calculation can rely completely on the
    results of sub-calculations, however there exist {\sl Monte-Carlo}
    algorithms which are often much faster than the equivalent
    deterministic ones where the results will have a prescribed
    probability (presumably small) of being incorrect. However there has
    been little discussion of how such algorithms can be used as building
    blocks in Computer Algebra. In this paper we describe how the
    computational category theory which is the basis of the type structure
    used in the Axiom computer algebra system may be extended to cover
    probabilistic algorithms, which use Monte-Carlo techniques. We follow
    this with a specific example which uses Straight Line Program
    representation.",
  paper = "ND00.pdf",
  keywords = "axiomref"

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@techreport{Ngxx80,
  author = "Ng, Edward W.",
  title = "Symbolic-Numeric Interface: A Review",
  type = "technical report",
  year = "1980",
  number = "NASA-CR-162690 HC A02/MF A01",
  institution = "NASA Jet Propulsion Lab",
  link = "\url{http://ntrs.nasa.gov/archive/nasa/casi.ntrs.nasa.gov/19800008508.pdf}",
  abstract =
    "This is a survey of recent activities that either used or encouraged the
    potential use of a combination of symbolic and numerical calculations.
    Symbolic calculations here primarily refer to the computer processing of
    procedures from classical algebra, analysis and calculus. Numerical
    calculations refer to both numerical mathematics research and scientific
    computation. This survey is intended to point out a large number of problem
    areas where a co-operation of symbolic and numeric methods is likely to
    bear many fruits. These areas include such classical operations as
    differentiation and integration, such diverse activities as function
    approximations and qualitative analysis, and such contemporary topics as
    finite element calculations and computational complexity. It is contended
    that other less obvious topics such as the fast Fourier transform, linear
    algebra, nonlinear analysis and error analysis would also benefit from a
    synergistic approach advocated here.",
  paper = "Ngxx80.pdf",
  keywords = "axiomref"
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@article{Norm75,
  author = "Norman, Arthur C.",
  title = "Computing with Formal Power Series",
  journal = "ACM Transactions on Mathematical Software",
  volume = "1",
  number = "4",
  pages = "346-356",
  year = "1975",
  doi = "10.1145/355656.355660",
  keywords = "axiomref",
  beebe = "Norman:1975:CFP"
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— Norman:1975:CFP —

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@Article{Norman:1975:CFP,
  author =      "A. C. Norman",
  title =       "Computing with Formal Power Series",
  journal =     j-TOMS,
  volume =      "1",
  number =      "4",
  pages =       "346--356",
  month =       dec,
  year =        "1975",
  CODEN =       "ACMSCU",
  ISSN =        "0098-3500 (print), 1557-7295 (electronic)",
  ISSN-L =      "0098-3500",
  bibdate =     "Sat Aug 27 00:22:26 1994",
  bibsource =   "http://www.math.utah.edu/pub/tex/bib/axiom.bib",
  acknowledgement = ack-nhfb,
  fjournal =    "ACM Transactions on Mathematical Software",
  link =        "\url{http://portal.acm.org/toc.cfm?idx=J782}",
  keywords =    "Scratchpad",
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— axiom.bib —

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@misc{Normxx,
  author = "Norman, Arthur C.",
  title = "Notes 13: How to Compute a Groebner Basis",
  link = "\url{http://people.math.umass.edu/~norman/462/_11/notes/m462notes13.pdf}",
  algebra =
    "\newline\refto{package AFALGGRO AffineAlgebraicSetComputeWithGroebnerBasis}
    \newline\refto{package GBEUCLID EuclideanGroebnerBasisPackage}
    \newline\refto{package GBF GroebnerFactorizationPackage}
    \newline\refto{package GBINTERN GroebnerInternalPackage}
    \newline\refto{package GB GroebnerPackage}
    \newline\refto{package GROESOL GroebnerSolve}
    \newline\refto{package INTERGB InterfaceGroebnerPackage}
    \newline\refto{package LGROBP LinGroebnerPackage}
    \newline\refto{package PGROEB PolyGroebner}",
  paper = "Normxx.pdf"
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@article{Norm75a,

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author = "Norman, Arthur C.",
title = "The SCRATCHPAD Power Series Package",
journal = "SIGSAM",
volume = "9",
number = "1",
pages = "12-20",
year = "1975",
comment = "IBM T.J. Watson Research RC4998",
keywords = "axiomref"
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@misc{Norm94,
author = "Norman, Arthur C.",
title = "Algebraic Manipulation",
paper = "Norm94.pdf",
keywords = "axiomref"
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— axiom.bib —

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@InProceedings{Norm96,
author = "Norman, Arthur C.",
title = "Memory tracing of algebraic calculations",
booktitle = "Proc. 1996 ISSAC",
series = "ISSAC 1996",
year = "1996",
publisher = "ACM Press",
location = "New York, NY",
pages = "113-119",
link = "\url{http://opus.bath.ac.uk/16452/1/NormanFitch96a.ps}",
abstract =
  "We present a software tool which allows us to visualize details of the
  use of memory during the execution of an algebra system. We apply this
  to gain a better understanding of the behaviour of REDUCE, and hence
  to make proposals for ways in which the execution can be improved. The
  same tool will soon be used in the performance engineering of a
  version of axiom.",
paper = "Norm96.pdf",
keywords = "axiomref"
}

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@phdthesis{Nguy09,
  author = "Nguyen, Minh Van",
  title = "Exploring Cryptography Using the Sage Computer Algebra System",
  school = "Victoria University",
  year = "2009",
  abstract =
    "Cryptography has become indispensable in areas such as e-commerce,
    the legal safe-guarding of medical records, and secure electronic
    communication. Hence, it is incumbent upon software engineers to
    understand the concepts and techniques underlying the cryptosystems
    that they implement. An educator needs to consider which topics to
    cover in a course on cryptography as well as how to present the
    concepts and techniques to be covered in the course. This thesis
    contributes to the field of cryptography pedagogy by discussing and
    implementing small-scale cryptosystems whose encryption and
    decryption processes can be stepped through by hand. Our
    implementation has been accepted and integrated into the code base of
    the computer algebra system Sage. As Sage is free and open source,
    students and educators of cryptology need not worry about paying
    license fees in order to use Sage, but can instead concentrate on
    exploring cryptography using Sages built-in support for cryptography.",
  paper = "Nguy09.pdf",
  keywords = "axiomref"
}
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O

— axiom.bib —

```
@InProceedings{Oance05,
  author = "Oancea, Cosmin E. and Watt, Stephen M.",
  title = "Domains and expressions: an interface between two approaches to
    computer algebra",
  booktitle = "Proc. 2005 ISSAC",
  series = "ISSAC 2005",
  year = "2005",
  isbn = "1-59593-095-7",
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location = "Beijing, China",
pages = "261-268",
link = "\url{http://www.csd.uwo.ca/~watt/pub/reprints/2005-issac-alma.pdf}",
abstract =
  "This paper describes a method to use compiled, strongly typed Aldor
  domains in the interpreted, expression-oriented Maple
  environment. This represents a non-traditional approach to structuring
  computer algebra software: using an efficient, compiled language,
  designed for writing large complex mathematical libraries, together
  with a top-level system based on user-interface priorities and ease of
  scripting.

  We examine what is required to use Aldor libraries to extend Maple in
  an effective and natural way. Since the computational models of Maple
  and Aldor differ significantly, new run-time code must implement a
  non-trivial semantic correspondence. Our solution allows Aldor
  functions to run tightly coupled to the Maple environment, able to
  directly and efficiently manipulate Maple data objects. We call the
  overall system Alma.",
paper = "Oanc05.pdf",
keywords = "axiomref"
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@misc{ORMS,
  author = "Unknown",
  title = "Oberwolfach References on Mathematical Software",
  link = "\url{http://orms.mfo.de/project?id=234}",
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@article{OKee96,
  author = "O'Keefe, Christine M. and Storme, Leo",
  title = {Arcs in  $PG(n,q)$  fixed by  $\{A_5\}$  and  $\{A_6\}$ },
  journal = "J. Geom.",
  volume = "55",
  number = "1-2",
  pages = "123-138",
  year = "1996",
}

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abstract =
  "A  $k$ -arc in a projective space is a set of points, no three of which
  are collinear. The author determines the  $k$  arcs in  $PG(n,q)$  which
  are fixed by primitive groups isomorphic to  $A_5$  or  $A_6$ . The best
  known examples are  $q+1$  arcs: in general these are normal rational
  curves and are conics in  $PG(2,q)$  and twisted cubics in  $PG(3,q)$ .
  The other cases turn out to be 6-arcs or 10-arcs.",
keywords = "axiomref"
}

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— axiom.bib —

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@inproceedings{Ollivier89,
  author = "Ollivier, F.",
  title = "Inversibility of rational mappings and structural
  identifiability in automatics",
  booktitle = "Proc. SIGSAM 1989",
  series = "SIGSAM '89",
  pages = "43-54",
  isbn = "0-89791-325-6",
  year = "1989",
  abstract =
    "We investigate different methods for testing whether a rational
    mapping  $f$  from  $k^n$  to  $k^m$  admits a rational inverse, or whether
    a polynomial mapping admits a polynomial one. We give a new solution,
    which seems much more efficient in practice than previously known ones
    using 'tag' variables and standard basis, and a majoration for the
    degree of the standard basis calculations which is valid for both
    methods in the case of a polynomial map which is birational. We
    further show that a better bound can be given for our method, under
    some assumptions on the form of  $f$ . Our method can also extend to
    check whether a given polynomial belong to the subfield generated by a
    finite set of fractions.

    We then illustrate our algorithm, with a application to structural
    identifiability. The implantation has been done in the IBM computer
    algebra system Scratchpad II.",
  paper = "Ollivier89.pdf",
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— Ollivier:1989:IRM —

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@InProceedings{Ollivier:1989:IRM,
  author = "F. Ollivier",
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    identifiability in automatics",
  crossref = "ACM:1989:PAI",
  pages = "43--54",
  month = "",
  year = "1989",
  bibdate = "Tue Sep 17 06:46:18 MDT 1996",
  bibsource = "http://www.math.utah.edu/pub/tex/bib/axiom.bib",
  abstract = "The author investigates different methods for testing
    whether a rational mapping  $f$  from  $k/\sup n/$  to  $k/\sup m/$ 
    admits a rational inverse, or whether a polynomial
    mapping admits a polynomial one. He gives a new
    solution, which seems much more efficient in practice
    than previously known ones using 'tag' variables and
    standard basis, and a majoration for the degree of the
    standard basis calculations which is valid for both
    methods in the case of a polynomial map which is
    birational. He shows that a better bound can be given
    for the method, under some assumption on the form of  $f$ .
    The method can also extend to check whether a given
    polynomial belongs to the subfield generated by a
    finite set of fractions. The author illustrates the
    algorithm with an application to structural
    identifiability. The implementation has been done in
    the IBM computer algebra system Scratchpad II.",
  acknowledgement = ack-nhfb,
  affiliation = "Lab. d'Inf. de l'X, Ecole Polytech., Palaiseau,
    France",
  classification = "C1110 (Algebra); C1120 (Analysis); C7310
    (Mathematics)",
  keywords = "Computer algebra system; Fractions; IBM;
    Inversibility; Polynomial inverse; Polynomial mapping;
    Rational inverse; Rational mappings; Scratchpad II;
    Structural identifiability",
  language = "English",
  thesaurus = "Inverse problems; Mathematics computing; Polynomials;
    Set theory; Symbol manipulation",
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\bibitem[Online 72]{Onl72}.

Online 72: conference proceedings ... international conference on online
interactive computing, Brunel University, Uxbridge, England, 4-7 September

1972 ISBN 0-903796-02-3 LCCN QA76.55.054 1972 Two volumes.
 keywords = "axiomref"

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\bibitem[OpenMath]{OpenMa}.
  title = "OpenMath Technical Overview",
  link = "\url{http://www.openmath.org/overview/technical.html}",
  keywords = "axiomref"
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— axiom.bib —

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@InProceedings{Page07,
  author = "Page, William S.",
  title = "Axiom - Open Source Computer Algebra System",
  booktitle = "Poster ISSAC 2007 Proceedings",
  series = "ISSAC 2007",
  year = "2007",
  volume = "41",
  pages = "114",
  abstract =
    "Axiom has been in development since 1971. Originally called
    Scratchpad II, it was developed by IBM under the direction of Richard
    Jenks[1]. The project evolved over a period of 20 years as a research
    platform for developing new ideas in computational mathematics.
    ScratchPad also attracted the interest and contributions of a large
    number of mathematicians and computer scientists outside of IBM. In
    the 1990s, the Scratchpad project was renamed to Axiom, and sold to
    the Numerical Algorithms Group (NAG) in England who marketed it as a
    commercial system. NAG withdrew Axiom from the market in October 2001
    and agreed to release Axiom as free software, under an open source
    license."
```

Tim Daly (a former ScratchPad developer at IBM) setup a public open source Axiom project[2] in October 2002 with a primary goal to improve the documentation of Axiom through the extensive use of literate programming[3]. The first free open source version of Axiom was released in 2003. Since that time the project has attracted a small

```

but very active group of developers and a growing number of users.

This exhibit includes a laptop computer running a recent version
of Axiom, Internet access (if available) to the Axiom Wiki website[4],
and CDs containing Axiom software for free distribution[5].",
keywords = "axiomref",
beebe = "Page:2007:AOS"
}

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— Page:2007:AOS —

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@Article{Page:2007:AOS,
  author =      "William S. Page",
  title =      "{Axiom}: open source computer algebra system",
  journal =     j-ACM-COMM-COMP-ALGEBRA,
  volume =     "41",
  number =     "3",
  pages =      "114--114",
  month =      sep,
  year =       "2007",
  CODEN =      "????",
  DOI =        "http://doi.acm.org/10.1145/1358190.1358206",
  ISSN =       "1932-2232 (print), 1932-2240 (electronic)",
  ISSN-L =     "1932-2232",
  bibdate =    "Wed Jun 18 09:23:01 MDT 2008",
  bibsource =  "http://portal.acm.org/;
http://www.math.utah.edu/pub/tex/bib/axiom.bib",
  abstract =   "Axiom has been in development since 1971. Originally
called Scratchpad II, it was developed by IBM under the
direction of Richard Jenks[1]. The project evolved over
a period of 20 years as a research platform for
developing new ideas in computational mathematics.
ScratchPad also attracted the interest and
contributions of a large number of mathematicians and
computer scientists outside of IBM. In the 1990s, the
Scratchpad project was renamed to Axiom, and sold to
the Numerical Algorithms Group (NAG) in England who
marketed it as a commercial system. NAG withdrew Axiom
from the market in October 2001 and agreed to release
Axiom as free software, under an open source
license.\par

```

Tim Daly (a former ScratchPad developer at IBM) setup a public open source Axiom project[2] in October 2002 with a primary goal to improve the documentation of Axiom through the extensive use of literate programming[3].

The first free open source version of Axiom was released in 2003. Since that time the project has attracted a small but very active group of developers and a growing number of users.\par

This exhibit includes a laptop computer running a recent version of Axiom, Internet access (if available) to the Axiom Wiki website[4], and CDs containing Axiom software for free distribution[5].",

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acknowledgement = ack-nhfb,
fjournal =      "ACM Communications in Computer Algebra",
issue =        "161",
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@misc{Page08,
  author = "Page, Bill",
  title = "Algebraist Network",
  link = "\url{http://lambda-the-ultimate.org/node/2737}",
  year = "2008",
  keywords = "axiomref"
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@misc{Paulxx,
  author = "Paule, Peter and Kartashova, Lena and Kauers, Manuel and
           Schneider, Carsten and Winkler, Franz",
  title = "Hot Topics in Symbolic Computation",
  publisher = "Springer",
  link = "\url{http://www.risc.jku.at/publications/download/risc_3845/chapter1.pdf}",
  paper = "Paulxx.pdf"
}
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@article{Peti99,
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author = "Petitjean, S.",
title = "Algebraic Geometry and Computer Vision: Polynomial Systems, Real
        and Complex Roots",
journal = "J. of Mathematical Imaging and Vision",
volume = "10",
number = "1",
year = "1999",
link = "\url{http://www.loria.fr/~petitjea/papers/jmiv99.pdf}",
abstract =
    "We review the different techniques known for doing exact computations
    on polynomial systems. Some are based on the use of Groebner bases and
    linear algebra, others on the more classical resultants and its modern
    counterparts. Many theoretical examples of the use of these techniques
    are given. Furthermore, a full set of examples of applications in the
    domain of artificial vision, where many constraints boil down to
    polynomial systems, are presented. Emphasis is also put on very recent
    methods for determining the number of (isolated) real and complex
    roots of such systems.",
paper = "Peti99.pdf",
keywords = "axiomref"
}

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— ignore —

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\bibitem[Petitot 90]{Pet90} Petitot, Michel
    title = "Types r\'ecursifs en scratchpad, application aux polyn\^omes non commutatifs",
    LIFL, 1990
    keywords = "axiomref"

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— axiom.bib —

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@inproceedings{Peti93,
    author = "Petitot, Michel",
    title = "Experience with Axiom",
    booktitle = "Proc. Int. IMACS Symposium on Symbolic Computation",
    pages = "240",
    year = "1993",
    keywords = "axiomref",
    beebe = "Petitot:1993:EA"
}

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@InProceedings{Petitot:1993:EA,
  author = "M. Petitot",
  title = "Experience with {Axiom}",
  crossref = "Jacob:1993:PSI",
  pages = "240",
  month = "",
  year = "1993",
  bibdate = "Tue Sep 17 06:35:39 MDT 1996",
  bibsource = "http://www.math.utah.edu/pub/tex/bib/axiom.bib",
  abstract = "The computer algebra system Axiom (formerly
    Scratchpad) allows a strict typing of the manipulated
    data. Using examples from noncommutative algebra
    (polynomials in noncommutative variables, Lie
    polynomials, Poincar{\e}--Birkoff--Witt basis) the
    authors show the interest and the limits of some
    essential notions in Axiom: the genericity, the
    inheritance, the distinction domain/category and the
    type inference.",
  acknowledgement = ack-nhfb,
  affiliation = "LIFL, Lille I Univ., Villeneuve d'Ascq, France",
  classification = "C1230 (Artificial intelligence); C4130
    (Interpolation and function approximation); C7310
    (Mathematics)",
  keywords = "Category; Computer algebra system Axiom; Distinction
    domain; Genericity; Inheritance; Lie polynomials;
    Manipulated data; Noncommutative algebra;
    Poincar{\e}--Birkoff--Witt basis; Polynomials;
    Scratchpad; Type inference",
  language = "English",
  thesaurus = "Inference mechanisms; Lie algebras; Mathematics
    computing; Polynomials",
}
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— ignore —

```
\bibitem[Petric 71]{Pet71} Petric, S. R. (ed)
  Proceedings of the second symposium on Symbolic and
  Algebraic Manipulation, March 23-25, 1971, Los Angeles, California, ACM Press,
  New York, NY 10036, USA, 1971. LCCN QA76.5.S94 1971
  keywords = "axiomref"
```

— ignore —

```
\bibitem[Pinch 93]{Pin93} Pinch, R.G.E.
  title = "Some Primality Testing Algorithms",
  Devlin, Keith (ed.)
  Computers and Mathematics November 1993, Vol 40, Number 9 pp1203-1210
  keywords = "axiomref"
```

— ignore —

```
\bibitem[Poll (b)]{Polxx} Poll, Erik
  title = "The type system of Axiom",
  paper = "Polxx.pdf",
  keywords = "axiomref"
```

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```
@InProceedings{Prit06,
  author = "Pritchard, F. Leon and Sit, William Y.",
  title = "On initial value problems for ordinary differential-algebraic
    equations",
  booktitle = "Radon Series on Computational and Applied Mathematics",
  year = "2006",
  pages = "283-340",
  isbn = "978-3-11-019323-7",
  abstract =
    "This paper addresses polynomial implicit ODEs in an autonomous
    context. These ODEs are defined by a system of the form
    \[f_i(z_1,\cdots,z_n,\dot{z}_1,\cdots,\dot{z}_n)=0,\quad i=1,\cdots,m\]
    where  $f_i$  is (for  $i=1,\ldots,m$ ) a polynomial in  $2n$  variables
     $(X,P)=(X_1,\ldots,X_n,P_1,\ldots,P_n)$ . This covers in particular
    quasilinear systems, often encountered in applications and defined
    by polynomials  $f_i$  in which the total degree in the variables  $P$ 
    is at most one. The approach is close to the geometrical framework
    of Rabier and Rheinboldt, profiting from the polynomial form of
    the system.
```

The authors develop an algorithm for the symbolic computation of the set of consistent initial values via ideal-theoretic results;

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    this is based on a stationary algebraic process of ‘prolongation’,
    together with the notions of the completion of a given ideal and the
    algebraic index of the system, defined as the number of steps taken
    by the process to stabilize. Over- and under-determined systems are
    also accommodated in their framework.",
    keywords = "axiomref"
}

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@inproceedings{Purt86,
  author = "Purtilo, J.",
  title = "Applications of a software interconnection system in
          mathematical problem solving environments",
  booktitle = "Proc.1986 Symposium on Symbolic and Algebraic Computation",
  series = "SYMSAC '86",
  pages = "16-23",
  year = "1986",
  publisher = "ACM Press",
  isbn = "0-89791-199-7",
  doi = "http://dx.doi.org/10.1145/32439.32443",
  keywords = "axiomref",
  beebe = "Purtilo:1986:ASI"
}

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— Purtilo:1986:ASI —

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@InProceedings{Purtilo:1986:ASI,
  author = "J. Purtilo",
  editor = "Bruce W. Char",
  booktitle = "Proceedings of the 1986 Symposium on Symbolic and
              Algebraic Computation: Symsac '86, July 21--23, 1986,
              Waterloo, Ontario",
  title = "Applications of a software interconnection system in
          mathematical problem solving environments",
  publisher = pub-ACM,
  address = pub-ACM:adr,
  pages = "16--23",
  year = "1986",
  DOI = "http://dx.doi.org/10.1145.32443",
  ISBN = "0-89791-199-7",
  ISBN-13 = "978-0-89791-199-3",
}

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LCCN =      "QA155.7.E4 A281 1986",
bibdate =   "Thu Jul 26 09:26:18 2001",
bibsource = "http://www.math.utah.edu/pub/tex/bib/axiom.bib",
note =      "ACM order number 505860.",
acknowledgement = ack-nhfb,
keywords =  "Scratchpad",
}

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— axiom.bib —

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@book{Rals78,
  author = "Ralston, Anthony and Rabinowitz, Philip",
  title = "A First Course in Numerical Analysis",
  year = "1978",
  publisher = "McGraw-Hill",
  isbn = "0-07-051158-6",
}

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— axiom.bib —

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@article{Riga99,
  author = "Rigal, Alain",
  title = "High-order compact schemes: Application to bidimensional unsteady
    diffusion-convection problems.",
  journal = "C. R. Acad. Sci.",
  volume = "328",
  number = "6",
  pages = "535-538",
  year = "1999",
  abstract =
    "For unsteady 2D diffusion-convection problems, we present two classes
    of compact difference schemes of order 2 in time and 4 in space. These
    finite difference schemes are essentially derived from 1D schemes,
    extensively analyzed in our previous paper [J. Comput. Phys. 114,
    No. 1, 59-76 (1994; Zbl 0807.65056)]. We propose two approaches:
    construction of 2D schemes as product of 1D schemes and global
    formulation of 2D schemes. Part II by M. Fourni [C. R. Acad. Sci.,
    Paris, Sr. I, Math. 328, No. 6, 539-542 (1999; reviewed below)]

```

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    focuses on the development and analysis of global schemes with the
    assistance of symbolic computation software (AXIOM).",
    keywords = "axiomref"
}

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— axiom.bib —

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@article{Riob09,
  author = "Rioboo, Renaud",
  title = "Invariants for the FoCaL language",
  journal = "Ann. Math. Artif. Intell.",
  volume = "56",
  number = "3-4",
  pages = "273-296",
  year = "2009",
  abstract =
    "We present a FoCaL formalization for quotient structures which are
    common in mathematics. We first present a framework for stating
    invariant properties of the data manipulated by running programs. A
    notion of equivalence relation is then encoded for the FoCaL library.
    It is implemented through projections functions, this enables us to
    provide canonical representations which are commonly used in Computer
    Algebra but seldom formally described. We further provide a FoCaL
    formalization for the code used inside the library for modular
    arithmetic through the certification of quotient groups and quotient
    rings which are involved in the model. We finally instantiate our
    framework to provide a trusted replacement of the existing FoCaL
    library.",
  keywords = "axiomref"
}

```

— ignore —

```

\bibitem[Rioboo 03a]{Riob03a} Rioboo, Renaud
  title = "Quelques aspects du calcul exact avec des nombres r\'eels",
  Ph.D. Thesis, Laboratoire d'Informatique Th\'eorique et Programmation
  paper = "Riob03a.ps",
  paper = "Riob03a.pdf",
  keywords = "axiomref"

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— ignore —

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\bibitem[Rioboo 03]{Riob03} Rioboo, Renaud
  title = "Towards Faster Real Algebraic Numbers",
  J. of Symbolic Computation 36 pp 513-533 (2003)
  abstract = "
    This paper presents a new encoding scheme for real algebraic number
    manipulations which enhances current Axiom's real closure. Algebraic
    manipulations are performed using different instantiations of
    sub-resultant-like algorithms instead of Euclidean-like algorithms.
    We use these algorithms to compute polynomial gcds and Bezout
    relations, to compute the roots and the signs of algebraic
    numbers. This allows us to work in the ring of real algebraic integers
    instead of the field of real algebraic numbers avoiding many
    denominators.",
  paper = "Riob03.pdf",
  keywords = "axiomref"
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— axiom.bib —

```
@article{Roan03,
  author = "Roanes-Lozano, Eugenio and Roanes-Macias, Eugenio and
    Villar-Mena, M.",
  title = "A bridge between dynamic geometry and computer algebra",
  journal = "Math. Comput. Modelling",
  volume = "37",
  number = "9-10",
  pages = "1005-1028",
  year = "2003",
  link = "\url{http://ac.els-cdn.com/S0895717703001158/1-s2.0-S0895717703001158-main.pdf}",
  abstract =
    "Both Computer Algebra Systems (CASs) and dynamic geometry systems
    (DGSs) have reached a high level of development. Some CASs (like Maple
    or Derive) include specific and powerful packages devoted to Euclidean
    geometry, but CASs have incorporated neither mouse drawing
    capabilities nor dynamic capabilities. Meanwhile, the well-known DGSs
    do not provide algebraic facilities."
```

Maples and Derives paramGeo packages and the DGS-CAS translator (all freely available from the authors) make it possible to draw a geometric configuration with the mouse (using The Geometers Sketchpad 3 or 4) and to obtain the coordinates, equations, etc., of the drawn configuration in Maples or Derives syntax. To obtain complicated formulae, coordinates of points or equations of loci, to perform automatic theorem proving and to perform automatic discovery directly

from sketches are examples of straightforward applications. Moreover, this strategy could be adapted to other CASs and DGSs.

This work clearly has a didactic application in geometric problems exploration. Nevertheless, its main interest is to provide a convenient time-saving way to introduce data when dealing with rule and compass geometry, which has a wider scope than only educational purposes.",
 paper = "Roan03.pdf",
 keywords = "axiomref"
 }

— axiom.bib —

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@article{Roan10,
  author = "Roanes-Lozano, Eugenio and val Labeke, Nicolas and
           Roanes-Macias, Eugenio",
  title = "Connecting the 3D DGS Calques3D with the CAS Maple",
  journal = "Math. Comput. Simul.",
  volume = "80",
  number = "6",
  pages = "1153-1176",
  year = "2010",
  link = "\url{http://nvl.calques3d.org/publications/2010.MatCom.Connecting.pdf}",
  abstract =
    "Many (2D) Dynamic Geometry Systems (DGSs) are able to export numeric
    coordinates and equations with numeric coefficients to Computer
    Algebra Systems (CASs). Moreover, different approaches and systems
    that linke (2D) DGSs with CASs, so that symbolic coordinates and
    equations with symbolic coefficients can be exported from the DGS to
    the CAS, already exist. Although the 2D DGS Calques3D can export
    numeric coordinates and equations with numeric coefficients to Maple
    and Mathematica, it cannot export symbolic coordinates and equations
    with symbolic coefficients. A connetion between the 3D DGS Calques3D
    and the CAS Maple, that can handle symbolic coordinates and equations
    with symbolic coefficients, is presented here. Its main interest is to
    provide a convenient time-saving way to explore problems and directly
    obtain both algebraic and numeric data when dealing with a 3D
    extension of ‘ruler and compass geometry’. This link has not only
    educational purposes but mathematical ones, like mechanical theorem
    proving in geometry, geometry discovery (hypothesis completion),
    geometric loci finding -- As far as we know, there is no comparable
    ‘symbolic’ link in the 3D case, except the prototype 3D-LD
    (restricted to determining algebraic surfaces as geometric loci).",
  paper = "Roan10.pdf",
  keywords = "axiomref"
```

}

— axiom.bib —

```
@article{Roes99,
  author = "Roesner, K. G.",
  title = "Supersonic flow around accelerated and decelerated bodies,
    analysed by analytical methods",
  journal = "Z. Angew. Math. Mech.",
  volume = "79",
  number = "3",
  pages = "815-816",
  year = "1999",
  abstract =
    "By an extensive use of the computer algebra system AXIOM, a power
    series expansion with respect to the radial variable  $r$  is used to
    describe the accelerated or decelerated supersonic flow field around
    the tip of slender conical bodies. The set of coupled nonlinear
    differential equations for the coefficient functions, depending on
     $\theta$  and  $t$ , is derived in closed form, and the first and second
    approximation of the coefficient functions are determined numerically.",
  keywords = "axiomref"
}
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— axiom.bib —

```
\article{Rojas13,
  author = "Rojas-Bruna, Carlos",
  title = "Trace forms and ideals on commutative algebras satisfying an
    identity of degree four",
  journal = "Rocky Mt. J. Math.",
  year = "2013",
  volume = "43",
  number = "4",
  pages = "1325-1336",
  abstract =
    "This paper deals with the variety of commutative algebras satisfying
    the identity
    
$$\begin{aligned} & [((xy)z)t - ((xy)t)z + ((yt)x)z - ((yt)z)x + ((yz)t)x - ((yz)x)t = 0] \\ & \text{These algebras appeared in the classification of the degree four} \\ & \text{identities in Carini et al. We prove the existence of a trace} \\ & \text{form. Moreover, if we assume the existence of degenerate trace form,} \end{aligned}$$

```



```

the  $A$  satisfies the identity  $((yx)x)x=y((xx)x)$ , a generalization
of right-alternativity. Finally we prove that  $Ass[A]$  and  $N(A)$  are
ideals in these algebras.",
keywords = "axiomref"
}

```

— axiom.bib —

```

@misc{Robi93,
  author = "Robidoux, Nicolas",
  title = "Does Axiom Solve Systems of O.D.E's Like Mathematica?",
  year = "1993",
  link = "\url{http://axiom-wiki.newsynthesis.org/public/refs/Robidoux.pdf}",
  abstract = "
    If I were demonstrating Axiom and were asked this question, my reply
    would be 'No, but I am not sure that this is a bad thing'. And I
    would illustrate this with the following example.

    Consider the following system of O.D.E.'s
    \[
    \begin{array}{rc1}
    \frac{dx_1}{dt} & = & \left(1+\frac{\cos t}{2+\sin t}\right)x_1\backslash\backslash \\
    \frac{dx_2}{dt} & = & x_1 - x_2
    \end{array}
    \]
    This is a very simple system:  $x_1$  is actually uncoupled from  $x_2$ ",
  paper = "Robi93.pdf",
  keywords = "axiomref"
}

```

— axiom.bib —

```

@article{Roes95,
  author = "Roesner, K. G.",
  title = "Verified solutions for parameters of an exact solution for
    non-Newtonian liquids using computer algebra",
  journal = "Zeitschrift fur Angewandte Mathematik und Physik",
  volume = "75",
  number = "suppl. 2",
  pages = "S435-S438",
  year = "1995",
  abstract =

```

"An exact solution of the time independent velocity field for the Taylor--Couette flow of a polymer solution is derived solving the resulting first order ordinary differential equation of fifth degree analytically. Intensive use is made of computer algebra systems AXIOM and MACSYMA to find the exact solution. The coaxial cylinders in the Taylor--Couette flow problem are assumed to rotate at different angular velocities. The geometrical and kinematic parameters can be chosen arbitrarily. The model equation for the material law of the viscoelastic liquid is based on the thermodynamic model for dilute solutions due to Lhuillier and Ouibrahim (1980) which is an analogy to the earlier paper of Frankel and Acrivos (1970). In the present investigation the influence of the parameters of the viscoelastic model on the velocity profile in the cylindrical gap is studied and the range of validity of the constitutive equation is investigated.",
 keywords = "axiomref",
 beebe = "Roesner:1995:VSP"
 }

— Roesner:1995:VSP —

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@Article{Roesner:1995:VSP,
  author =      "K. G. Roesner",
  title =       "Verified solutions for parameters of an exact solution
                for {non-Newtonian} liquids using computer algebra",
  journal =     "j-ZEIT-ANGE-MATH-PHYS",
  volume =     "75",
  number =     "suppl. 2",
  pages =      "S435--S438",
  month =      "",
  year =       "1995",
  ISSN =       "0044-2267",
  bibdate =    "Fri Dec 29 12:46:02 MST 1995",
  bibsource =  "http://www.math.utah.edu/pub/tex/bib/axiom.bib",
  abstract =    "An exact solution of the time independent velocity
                field for the Taylor--Couette flow of a polymer
                solution is derived solving the resulting first order
                ordinary differential equation of fifth degree
                analytically. Intensive use is made of computer algebra
                systems AXIOM and MACSYMA to find the exact solution.
                The coaxial cylinders in the Taylor--Couette flow
                problem are assumed to rotate at different angular
                velocities. The geometrical and kinematic parameters
                can be chosen arbitrarily. The model equation for the
                material law of the viscoelastic liquid is based on the
                thermodynamic model for dilute solutions due to
                Lhuillier and Ouibrahim (1980) which is an analogy to
```

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the earlier paper of Frankel and Acrivos (1970). In the
present investigation the influence of the parameters
of the viscoelastic model on the velocity profile in
the cylindrical gap is studied and the range of
validity of the constitutive equation is
investigated.",
acknowledgement = ack-nhfb,
affiliation = "Inst. fur Mech., Tech. Hochschule Darmstadt, Germany",
classification = "A0210 (Algebra, set theory, and graph theory); A0230
(Function theory, analysis); A0270 (Computational
techniques); A4710 (General fluid dynamics theory,
simulation and other computational methods); A4715
(Laminar flows); A4730 (Rotational flow, vortices,
buoyancy and other flows involving body forces); A4750
(Non-Newtonian dynamics)",
keywords = "AXIOM; Coaxial cylinders; Computer algebra;
Constitutive equation; Cylindrical gap; Dilute
solutions; Exact solution; First order ordinary
differential equation; Geometrical parameters;
Kinematic parameters; MACSYMA; Material law;
NonNewtonian liquids; Polymer solution; Taylor--Couette
flow; Thermodynamic model; Time independent velocity
field; Velocity profile; Viscoelastic liquid;
Viscoelastic model",
language = "English",
thesaurus = "Algebra; Couette flow; Differential equations; Flow
simulation; Non-Newtonian fluids; Physics computing;
Polymer solutions; Rotational flow; Thermodynamics",
}

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— axiom.bib —

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@article{Saf00,
  author = "Safouhi, Hassan",
  title = {The  $\$HD\$$  and  $\$H\overline{\{D\}}\$$  methods for accelerating the
    convergence of three-center nuclear attraction and four-center
    two-electron Coulomb integrals over  $\$B\$$  functions and their
    convergence properties.},
  journal = "J. Comput. Phys.",
  volume = "165",
  number = "2",
  pages = "473-495",
  year = "2000",
}

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abstract =
  "Three-center nuclear attraction and four-center two-electron
  Coulomb integrals over Slater-type orbitals are required for $ab$
  initio and density functional theory molecular structure
  calculations. They occur in many millions of terms, even for small
  molecules and require rapid and accurate evaluation. The $B$
  functions are used as a basis set of atomic orbitals. These
  functions are well adapted to the Fourier transform method that
  allowed analytical expressions for the integrals of interest to be
  developed. Rapid and accurate evaluation of these analytical
  expressions is now made possible by applying the $HD$ and
  $\overline{D}$ methods for accelerating the convergence of the
  semi-infinite oscillatory integrals. The convergence properties of
  the new methods are analyzed. The numerical results section shows
  the high predetermined accuracy and the substantial gain in the
  calculation times obtained using the new methods.",
keywords = "axiomref"
}

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@article{Safo01,
  author = "Safouhi, Hassan",
  title = "Numerical evaluation of three-center two-electron Coulomb and
  hybrid integrals over $B$ functions using the $HD$ and
  $\overline{D}$ methods and convergence properties",
  journal = "J. Math. Chem.",
  volume = "29",
  number = "3",
  pages = "213-232",
  year = "2001",
  abstract =
    "The $B$ function is a product of a spherical $K$ Bessel function and
    a spherical harmonic. The integrals to be evaluated contain
    complicated expressions of finite hypergeometric functions and
    spherical $J$ Bessel functions. Sequence transformation techniques are
    used for the numerical evaluation of the integrals. Numerical examples
    illustrate the accuracy and the efficiency of the method.",
  keywords = "axiomref"
}

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— axiom.bib —

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@misc{SALSA,
  author = "SALSA",
  title = "Solvers for Algebraic Systems and Applications",
  link = "\url{http://www.ens-lyon.fr/LIP/Arenaire/SYMB/teams/salsa/proposal-salsa.pdf}",
  algebra =
    "\newline\ref{category TSETCAT TriangularSetCategory}
    \newline\ref{category RSETCAT RegularTriangularSetCategory}
    \newline\ref{category NTSCAT NormalizedTriangularSetCategory}
    \newline\ref{category SFRTCAT SquareFreeRegularTriangularSetCategory}
    \newline\ref{package RSDCMPK RegularSetDecompositionPackage}",
  paper = "SALSA.pdf"
}

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— axiom.bib —

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@techreport{Salv89,
  author = "Salvy, Bruno",
  title = "Examples of automatic asymptotic expansions",
  institution = "Inst. Nat. Recherche Inf. Autom.",
  type = "technical report",
  number = "114",
  year = "1989",
  comment = "SIGSAM Bulletin Vol 25 No 2 1991 pp4-17",
  abstract =
    "We describe the current state of a Maple library, gdev, designed to
    perform asymptotic expansions for a large class of expressions. Many
    examples are provided, along with a short sketch of the underlying
    principles. At the time when this report is written, a striking
    feature of these examples is that none of them can be computed
    directly with any of today's most widespread symbolic computation
    systems (Macsyma, Mathematica, Maple or Scratchpad II).",
  paper = "Salv89.pdf",
  keywords = "axiomref",
  beebe = "Salvy:1989:EAA"
}

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— Salvy:1989:EAA —

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@TechReport{Salvy:1989:EAA,
  author = "B. Salvy",
  title = "Examples of automatic asymptotic expansions",
  number = "114",
}

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institution = "Inst. Nat. Recherche Inf. Autom.",
address =    "Le Chesnay, France",
pages =     "18",
month =     "dec",
year =      "1989",
bibdate =   "Tue Sep 17 06:46:18 MDT 1996",
bibsource = "http://www.math.utah.edu/pub/tex/bib/axiom.bib",
abstract =   "Describes the current state of a Maple library, gdev,
              designed to perform asymptotic expansions for a large
              class of expressions. Many examples are provided, along
              with a short sketch of the underlying principles. A
              striking feature of these examples is that none of them
              can be computed directly with any of the most
              widespread symbolic computation systems (Macsyma,
              Mathematica, Maple or Scratchpad II).",
acknowledgement = ack-nhfb,
classification = "C1120 (Analysis); C6130 (Data handling techniques);
                 C7310 (Mathematics)",
keywords =     "Asymptotic expansions; Gdev; Maple library; Symbolic
                 computation systems",
language =     "English",
pubcountry =   "France",
thesaurus =    "Mathematical analysis; Mathematics computing;
                 Subroutines; Symbol manipulation",
}

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— axiom.bib —

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@article{Salv91,
  author = "Salvy, Bruno",
  title = "Examples of automatic asymptotic expansions",
  journal = "SIGSAM Bulletin",
  volume = "25",
  number = "2",
  pages = "4-17",
  year = "1991",
  abstract =
    "We describe the current state of a Maple library, gdev, designed to
    perform asymptotic expansions for a large class of expressions. Many
    examples are provided, along with a short sketch of the underlying
    principles. At the time when this report is written, a striking
    feature of these examples is that none of them can be computed
    directly with any of today's most widespread symbolic computation
    systems (Macsyma, Mathematica, Maple or Scratchpad II).",
  keywords = "axiomref",
  beebe = "Salvy:1991:EAA"
}

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}

— Salvy:1991:EAA —

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@Article{Salvy:1991:EAA,
  author = "B. Salvy",
  title = "Examples of automatic asymptotic expansions",
  journal = j-SIGSAM,
  volume = "25",
  number = "2",
  pages = "4--17",
  month = apr,
  year = "1991",
  CODEN = "SIGSBZ",
  ISSN = "0163-5824 (print), 1557-9492 (electronic)",
  ISSN-L = "0163-5824",
  bibdate = "Tue Sep 17 06:44:07 MDT 1996",
  bibsource = "http://www.math.utah.edu/pub/tex/bib/axiom.bib",
  abstract = "Describes the current state of a Maple library, gdev,
    designed to perform asymptotic expansions for a large
    class of expressions. Many examples are provided, along
    with a short sketch of the underlying principles. At
    the time when this report is written, a striking
    feature of these examples is that none of them can be
    computed directly with any of today's most widespread
    symbolic computation systems (Macsyma, Mathematica,
    Maple or Scratchpad II).",
  acknowledgement = ack-nhfb,
  affiliation = "LIX, Ecole Polytech., Palaiseau, France",
  classification = "C6130 (Data handling techniques); C7310
    (Mathematics)",
  fjournal = "SIGSAM Bulletin",
  keywords = "Automatic asymptotic expansions; Expressions; Gdev;
    Maple library; Symbolic computation systems",
  language = "English",
  pubcountry = "USA",
  thesaurus = "Symbol manipulation",
}
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— axiom.bib —

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@article{Sant95,
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author = "Santas, Philip S.",
title = "A type system for computer algebra",
journal = "J. Symbolic Computation",
volume = "19",
number = "1-3",
pages = "79-109",
year = "1995",
abstract =
  "This paper presents a type system for support of subtypes,
  parameterized types with sharing and categories in a computer algebra
  environment. By modeling representation of instances in terms of
  existential types, we obtain a simplified model, and build a basis for
  defining subtyping among algebraic domains. The inheritance at
  category level has been formalized; this allows the automatic
  inference of type classes. By means of type classes and existential
  types we construct subtype relations without involving coercions. A
  type sharing mechanism works in parallel and allows the consistent
  extension and combination of domains. The expressiveness of the system
  is further increased by viewing domain types as special case of
  package types, forming weak and strong sums respectively. The
  introduced system, although awkward at first sight, is simpler than
  other proposed systems for computer algebra without including some of
  their problems. The system can be further extended in other to support
  more constructs and increase its flexibility.",
paper = "Sant95.pdf",
keywords = "axiomref"
}

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— axiom.bib —

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@inproceedings{Sant96,
  author = "Santas, Philip S.",
  title = "Conditional Categories and Domains",
  booktitle = "Proc. DISCO 1996",
  year = "1996",
  pages = "112-125",
  isbn = "3-540-61697-7",
  keywords = "axiomref"
}

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— axiom.bib —


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@book{Sant05,
  author = "Santas, Philip S.",
  title = "Conditional Categories and Domains",
  booktitle = "Design and Implementation of Symbolic Computation Systems",
  year = "2005",
  series = "Lecture Notes in Computer Science",
  volume = "1128",
  publisher = "Springer",
  abstract =
    "We extend the Type system defined in [Sant95] with Axiom-like
    Conditional Categories with the additional property of Static Typing
    and Checking. Categories and Domains may contain conditionals in their
    bodies, which are elaborated by our compiler by techniques used in
    standard typing. We define an appropriate calculus and discuss its
    properties. Examples inspired by the Axiom library illustrate the
    power of our approach and its application in constructing algebraic
    concepts. The full calculus has been implemented and tested with our
    LA compiler which generated executable files.",
  keywords = "axiomref"
}

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— axiom.bib —

```

@article{Saun80,
  author = "Saunders, B. David",
  title = "A Survey of Available Systems",
  journal = "SIGSAM Bull.",
  issue_date = "November 1980",
  volume = "14",
  number = "4",
  month = "November",
  year = "1980",
  issn = "0163-5824",
  pages = "12--28",
  numpages = "17",
  link = "\url{http://doi.acm.org/10.1145/1089235.1089237}",
  doi = "10.1145/1089235.1089237",
  acmid = "1089237",
  publisher = "ACM",
  address = "New York, NY, USA",
  paper = "Saun80.pdf",
  keywords = "axiomref,survey"
}

```

— ignore —

```
\bibitem[Schu 92]{Sch92} Sch\"u, J.
  title = "Implementing des Cartan-Kuranishi-Theorems in AXIOM",
  Master's diploma thesis (in german), Institut f\"ur Algorithmen und
  Kognitive Systeme, Universit\"at Karlsruhe 1992
  keywords = "axiomref"
```

— axiom.bib —

```
@book{Schw95,
  author = "Schwardmann, Ulrich",
  title = "Computer algebra systems",
  comment = "Computeralgebra-Systeme, German",
  publisher = "Addison-Wesley",
  year = "1995",
  keywords = "axiomref"
}
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— axiom.bib —

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@inproceedings{Schw88,
  author = "Schwarz, Fritz",
  title = "Programming with abstract data types: the symmetry package
          SPDE in Scratchpad'",
  booktitle = "Trends in Computer Algebra",
  series = "Lecture Notes in Computer Science 296",
  year = "1988",
  pages = "167-176",
  isbn = "3-540-18928-9",
  abstract =
    "The main problem which occurs in developing Computer Algebra packages
    for special areas in mathematics is the complexity. The unique concept
    which is advocated to cope with that problem is the introduction of
    suitable abstract data types. The corresponding decomposition into
    modules makes it much easier to develop, maintain and change the
    program. After introducing the relevant concepts from software
    engineering they are elaborated by means of the symmetry analysis of
    differential equations and the Scratchpad package SPDE which
    abbreviates Symmetries of Partial Differential Equations.",
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keywords = "axiomref",
beebe = "Schwarz:1988:PAD"
}

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— Schwarz:1988:PAD —

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@InCollection{Schwarz:1988:PAD,
  author =      "F. Schwarz",
  title =       "Programming with abstract data types: the symmetry
                package {SPDE} in {Scratchpad}",
  crossref =     "Janssen:1988:TCA",
  pages =       "167--176",
  year =        "1988",
  bibsource =    "/usr/local/src/bib/bibliography/Theory/cathode.bib;
                http://www.math.utah.edu/pub/tex/bib/axiom.bib",
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@inproceedings{Schw89,
  author = "Schwarz, Fritz",
  title = "A factorization algorithm for linear ordinary
          differential equations",
  booktitle = "Proc. SYMSAC 1989",
  series = "SYMSAC '89",
  isbn = "0-89791-325-6",
  year = "1989",
  pages = "17-25",
  abstract =
    "The reducibility and factorization of linear homogeneous differential
    equations are of great theoretical and practical importance in
    mathematics. Although it has been known for a long time that
    factorization is in principle a decision procedure, its use in an
    automatic differential equation solver requires a more detailed
    analysis of the various steps involved. Especially important are
    certain auxiliary equations, the so-called associated equations. An
    upper bound for the degree of its coefficients is derived. Another
    important ingredient is the computation of optimal estimates for the
    size of polynomial and rational solutions of certain differential
    equations with rotational coefficients. Applying these results, the
    design of the factorization algorithm LODEF and its implementation in
    the Scratchpad II Computer Algebra System is described.",
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paper = "Schw89.pdf",
keywords = "axiomref",
beebe = "Schwarz:1989:FAL"
}

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— Schwarz:1989:FAL —

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@InProceedings{Schwarz:1989:FAL,
  author =      "F. Schwarz",
  title =      "A factorization algorithm for linear ordinary
                differential equations",
  crossref =    "ACM:1989:PAI",
  pages =      "17--25",
  month =      "",
  year =       "1989",
  bibdate =    "Tue Sep 17 06:46:18 MDT 1996",
  bibsource =   "http://www.math.utah.edu/pub/tex/bib/axiom.bib",
  abstract =    "The reducibility and factorization of linear
                homogeneous differential equations are of great
                theoretical and practical importance in mathematics.
                Although it has been known for a long time that
                factorization is in principle a decision procedure, its
                use in an automatic differential equation solver
                requires a more detailed analysis of the various steps
                involved. Especially important are certain auxiliary
                equations, the so-called associated equations. An upper
                bound for the degree of its coefficients is derived.
                Another important ingredient is the computation of
                optimal estimates for the size of polynomial and
                rational solutions of certain differential equations
                with rotational coefficients. Applying these results,
                the design of the factorization algorithm LODEF and its
                implementation in the Scratchpad II Computer Algebra
                System is described.",
  acknowledgement = ack-nhfb,
  affiliation = "GMD, Inst. F1, St. Augustin, West Germany",
  classification = "C1120 (Analysis); C4170 (Differential equations);
                  C7310 (Mathematics)",
  keywords =    "Associated equations; Automatic differential equation
                solver; Factorization algorithm; Linear ordinary
                differential equations; LODEF; Optimal estimates;
                Polynomial solutions; Rational solutions; Rotational
                coefficients; Scratchpad II Computer Algebra System;
                Upper bound",
  language =    "English",
  thesaurus =   "Linear differential equations; Mathematics computing;

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    Polynomials; Symbol manipulation",
}

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— axiom.bib —

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@article{Schw91,
  author = "Schwarz, Fritz",
  title = "Monomial orderings and Groebner bases",
  journal = "SIGSAM Bulletin",
  volume = "25",
  number = "1",
  year = "1991",
  pages = "10-23",
  abstract =
    "Let there be given a set of monomials in n variables and some order
    relations between them. The following {\sl fundamental problem of
    monomial ordering} is considered. Is it possible to decide whether
    these ordering relations are consistent and if so to extend them to an
    {\sl admissible} ordering for all monomials? The answer is given in
    terms of the algorithm {\sl MACOT} which constructs a matrix of so
    called {\sl cotes} which establishes the desired ordering
    relations. The main area of application of this algorithm, i.e. the
    construction of Groebner bases for different orderings and of
    universal Groebner bases is treated in the last section.",
  keywords = "axiomref",
  beebe = "Schwarz:1991:MOG"
}

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— Schwarz:1991:MOG —

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@Article{Schwarz:1991:MOG,
  author = "F. Schwarz",
  title = "Monomial orderings and {Gr{"o"}bner} bases",
  journal = "j-SIGSAM",
  volume = "25",
  number = "1",
  pages = "10--23",
  month = "jan",
  year = "1991",
  CODEN = "SIGSBZ",
  ISSN = "0163-5824 (print), 1557-9492 (electronic)",
  ISSN-L = "0163-5824",
}

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bibdate =      "Tue Sep 17 06:44:07 MDT 1996",
bibsource =    "http://www.math.utah.edu/pub/tex/bib/axiom.bib",
abstract =     "Let there be given a set of monomials in n variables
                and some order relations between them. The following
                fundamental problem of monomial ordering is considered.
                Is it possible to decide whether these ordering
                relations are consistent and if so to extend them to an
                admissible ordering for all monomials? The answer is
                given in terms of the algorithm MACOT which constructs
                a matrix of so-called cotes which establishes the
                desired ordering relations. The main area of
                application of this algorithm, i.e. the construction of
                Gr{"o"}bner bases for different orderings and of
                universal Gr{"o"}bner bases, is presented. An
                implementation in Scratchpad is also briefly
                described.",
acknowledgement = ack-nhfb,
affiliation =   "GMD Inst., St. Augustin, Germany",
classification = "C1110 (Algebra); C4140 (Linear algebra); C7310
                (Mathematics)",
fjournal =     "SIGSAM Bulletin",
keywords =     "Computer algebra; Thomas theorem; Multivariate
                polynomial; Gr{"o"}bner bases; Monomial ordering;
                Ordering relations; Admissible ordering; MACOT; Matrix;
                Cotes; Scratchpad",
language =     "English",
pubcountry =   "USA",
thesaurus =    "Algebra; Matrix algebra; Polynomials; Symbol
                manipulation",
}

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@InProceedings{Schw02,
  author = "Schwarz, Fritz",
  title = "ALLTYPES: An algebraic language and type system",
  booktitle = "1st Int. Congress on Mathematical Software",
  year = "2002",
  isbn = "981-238-048-5",
  location = "Beijing China",
  pages = "486-500",
  link = "\url{http://www.scai.fraunhofer.de/content/dam/scai/de/documents/Mitarbeiterinnen-und-Mitarbeiter.}",
  abstract =
    "The software system ALLTYPES provides an environment that is
    particularly designed for developing software in differential
    algebra. Its most important features may be described as follows: A

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set of about thirty parametrized algebraic types is defined. Data
objects represented by these types may be manipulated by more than one
hundred polymorphic functions. Reusability of code is achieved by
genericity and multiple inheritance. The user may extend the system by
defining new types and polymorphic functions. A language comprising
seven basic language constructs is defined for implementing
mathematical algorithms. The easy manipulation of types is
particularly supported due to a special portion of the language
dedicated to manipulating typed objects, i.e. for performing
user-defined or automatic type coercions. Type inquiries are also
included in the language.",
paper = "Schw02.pdf",
keywords = "axiomref"
}

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— axiom.bib —

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@InProceedings{Schw94,
  author = "Schwarz, Fritz",
  title = "Computer algebra software for scientific applications",
  booktitle = "Computerized symbolic manipulation in mechanics",
  year = "1994",
  publisher = "Springer-Verlag",
  pages = "67-117",
  series = "CISM Courses Lecture 343",
  abstract =
    "The central subject of this article are two basic questions: How to
    make the process of developing computer algebra software on a large
    scale ( $10^4$  to  $10^5$ ) lines of code or more) more efficient and
    how to improve the quality of the result. Taking procedures from well
    established engineering sciences as a guide, two fundamental
    principles turned out to be of overwhelming importance: Modularization
    and limitation of growth through reuse. Important means for achieving
    these goals turned out to be concept of an abstract data type and the
    principles of object-oriented design. It is advocated to install an
    additional abstraction level between the mathematics and the machine
    in order to render it possible to develop (computer algebra) system
    independent mathematical software. Basic constituents of this level
    are a type system and a high-level language.",
  keywords = "axiomref"
}

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— axiom.bib —

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@phdthesis{Seil94,
  author = "Seiler, Werner Markus",
  title = "Analysis and Application of the Formal Theory of
    Partial Differential Equations",
  school = "Universitat Karlsruhe",
  year = "1994",
  link =
    "\url{http://www.mathematik.uni-kassel.de/~seiler/Papers/Diss/diss.ps.gz}",
  abstract = "
    An introduction to the formal theory of partial differential equations
    is given emphasizing the properties of involutive symbols and
    equations. An algorithm to complete any differential equation to an
    involutive one is presented. For an involutive equation possible
    values for the number of arbitrary functions in its general solution
    are determined. The existence and uniqueness of solutions for analytic
    equations is proven. Applications of these results include an
    analysis of symmetry and reduction methods and a study of gauge
    systems. It is show that the Dirac algorithm for systems with
    constraints is closely related to the completion of the equation of
    motion to an involutive equation. Specific examples treated comprise
    the Yang-Mills Equations, Einstein Equations, complete and Jacobian
    systems, and some special models in two and three dimensions. To
    facilitate the involved tedious computations an environment for
    geometric approaches to differential equations has been developed in
    the computer algebra system Axiom. The appendices contain among others
    brief introductions into Carten-K{"a"}hler Theory and Janet-Riquier
    Theory.",
  paper = "Seil94.pdf",
  keywords = "axiomref"
}
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— axiom.bib —

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@inproceedings{Seil94a,
  author = "Seiler, Werner Markus",
  title = "Completion to involution in AXIOM",
  booktitle = "Proc. Rhine Workshop on Computer Algebra",
  year = "1994",
  pages = "103-104",
  abstract =
    "We have implemented an algorithm to complete a given system of
    partial differential equations to an involutive one in the computer
    algebra system AXIOM. An earlier version of this program has been
    described in Schu, Seiler, and Calmet (1992). The new version is much
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more efficient due to better simplification and many new features. It
also provides procedures for the analysis of the arbitrariness of the
general solution. The goal of the implementation was not to transform
simply an algorithm into a program but to start with the construction
of an environment for symbolic computations within the geometric
theory of differential equations. The modular structure allows an easy
extension e.g. by a package for the symmetry analysis.",
keywords = "axiomref",
beebe = "Seiler:1994:CIA"
}

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— Seiler:1994:CIA —

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@InProceedings{Seiler:1994:CIA,
  author = "W. M. Seiler",
  title = "Completion to involution in {AXIOM}",
  crossref = "Calmet:1994:RWC",
  pages = "103--104",
  month = "",
  year = "1994",
  bibdate = "Tue Sep 17 06:32:41 MDT 1996",
  bibsource = "http://www.math.utah.edu/pub/tex/bib/axiom.bib",
  abstract = "We have implemented an algorithm to complete a given
system of partial differential equations to an
involutive one in the computer algebra system AXIOM. An
earlier version of this program has been described in
Schu, Seiler, and Calmet (1992). The new version is
much more efficient due to better simplification and
many new features. It also provides procedures for the
analysis of the arbitrariness of the general solution.
The goal of the implementation was not to transform
simply an algorithm into a program but to start with
the construction of an environment for symbolic
computations within the geometric theory of
differential equations. The modular structure allows an
easy extension e.g. by a package for the symmetry
analysis.",
  acknowledgement = ack-nhfb,
  affiliation = "Inst. fur Algorithmen und Kognitive Syst., Karlsruhe
Univ., Germany",
  classification = "C4170 (Differential equations); C7310 (Mathematics
computing)",
  keywords = "AXIOM; Computer algebra system; Involution; Partial
differential equations; Symbolic computations; Symmetry
analysis",
  language = "English",

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thesaurus = "Mathematics computing; Partial differential equations;
             Symbol manipulation",
}
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— axiom.bib —

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@article{Seil94b,
  author = "Seiler, Werner Markus",
  title = "Pseudo differential operators and integrable systems in AXIOM",
  journal = "Computer Physics Communications",
  volume = "79",
  number = "2",
  pages = "329-340",
  year = "1994",
  abstract =
    "An implementation of the algebra of pseudo differential operators in
    the computer algebra system Axiom is described. In several exmaples
    the application of the package to typical computations in the theory
    of integrable systems is demonstrated.",
  paper = "Seil94b.pdf",
  keywords = "axiomref",
  beebe = "Seiler:1994:PDO"
}
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— Seiler:1994:PDO —

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@Article{Seiler:1994:PDO,
  author = "Werner M. Seiler",
  title = "Pseudo differential operators and integrable systems
           in {AXIOM}",
  journal = "j-COMP-PHYS-COMM",
  volume = "79",
  number = "2",
  pages = "329--340",
  month = "apr",
  year = "1994",
  CODEN = "CPHCBZ",
  DOI = "http://dx.doi.org/10.1016/0010-4655(94)90076-0",
  ISSN = "0010-4655 (print), 1879-2944 (electronic)",
  ISSN-L = "0010-4655",
  bibdate = "Mon Feb 13 21:29:43 MST 2012",
  bibsource = "http://www.math.utah.edu/pub/tex/bib/axiom.bib;
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link = http://www.math.utah.edu/pub/tex/bib/compphyscomm1990.bib",
abstract = "\url{http://www.sciencedirect.com/science/article/pii/0010465594900760}",
        "An implementation of the algebra of pseudo
        differential operators in the computer algebra system
        AXIOM is described. In several examples the application
        of the package to typical computations in the theory of
        integrable systems is demonstrated.",
acknowledgement = ack-nhfb,
affiliation = "Inst. fur Algorithmen und Kognitive Syst., Karlsruhe
        Univ., Germany",
classification = "C4170 (Differential equations); C7310
        (Mathematics)",
fjournal = "Computer Physics Communications",
link = "\url{http://www.sciencedirect.com/science/journal/00104655}",
keywords = "AXIOM; Computer algebra; PDO; Pseudo differential
        operators",
language = "English",
pubcountry = "Netherlands",
thesaurus = "Mathematics computing; Partial differential equations;
        Symbol manipulation",
}

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— axiom.bib —

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@misc{Seil95,
  author = "Seiler, Werner Markus",
  title = "Applying AXIOM to partial differential equations",
  institution = {Universit\at Karlsruhe, Fakult\at f\ur Informatik},
  year = "1995",
  type = "Internal Report",
  number = "95-17",
  link = "\url{http://axiom-wiki.newsynthesis.org/public/refs/Axiom-pdf.pdf}",
  abstract =
    "We present an Axiom environment called JET for geometric computations
    with partial differential equations within the framework of the jet
    bundle formalism. This comprises especially the completion of a given
    differential equation to an involutive one according to the
    Cartan-Kuranishi Theorem and the setting up of the determining system
    for the generators of classical and non-classical Lie
    symmetries. Details of the implementations are described and
    applications are given. An appendix contains tables of all exported
    functions.",
  paper = "Seil95.pdf",
  keywords = "axiomref"
}

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— axiom.bib —

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@misc{Seil95a,
  author = "Seiler, Werner Markus and Calmet, J.",
  title = "JET -- An Axiom Environment for Geometric Computations with
    Differential Equations",
  link = "\url{http://axiom-wiki.newsynthesis.org/public/refs/axiom-jet95.pdf}",
  abstract =
    "JET is an environment within the computer algebra system Axiom to
    perform such computations. The current implementation emphasises the
    two key concepts involution and symmetry. It provides some packages
    for the completion of a given system of differential equations to an
    equivalent involutive one based on the Cartan-Kuranishi theorem and
    for setting up the determining equations for classical and
    non-classical point symmetries.",
  paper = "Seil95a.pdf",
  keywords = "axiomref"
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@article{Seil97,
  author = "Seiler, Werner M.",
  title = "Computer Algebra and Differential Equations: An Overview",
  journal = "mathPAD7",
  volume = "7",
  pages = "34-49",
  year = "1997",
  link = "\url{http://www.mathematik.uni-kassel.de/~seiler/Papers/Postscript/CADERep.ps.gz}",
  abstract =
    "We present an informal overview of a number of approaches to
    differential equations which are popular in computer algebra. This
    includes symmetry and completion theory, local analysis, differential
    ideal and Galois theory, dynamical systems and numerical analysis. A
    large bibliography is provided.",
  paper = "Seil97.pdf",
  keywords = "axiomref"
}
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— axiom.bib —

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@misc{Seil99,
  author = "Seiler, Werner Markus",
  title = "DETools: A Library for Differential Equations",
  year = "1999",
  abstract =
    "This article tries to give at least a brief introduction. The MuPAD
    library is extended on two levels. The first one consists of a new
    library detools containing a number of routines for treating
    differential equations. This includes support for the graphical
    presentation of the output of the numerical routines in MuPAD, some
    methods for analysing or generating differential equations and also
    routines for solving some classes of partial differential
    equations. The use of this new library will be described in this
    article. The second level is somewhat more advanced and requires a
    certain familiarity with the object-oriented domains.",
  paper = "Seil99.pdf",
  keywords = "axiomref"
}
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— axiom.bib —

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@misc{Seil01,
  author = "Seiler, Werner Markus",
  title = "Involution - The Formal Theory of Differential Equations and
    its Applications in Computer Algebra and Numerical Analysis",
  year = "2001",
  link =
    "\url{http://www.mathematik.uni-kassel.de/~seiler/Papers/Habil/habil.ps.gz}",
  paper = "Seil01.pdf",
  keywords = "axiomref"
}
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— axiom.bib —

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@techreport{Sene87,
  author = "Senechaud, P. and Siebert, F. and Villard, Gilles",
  title = "Scratchpad II: Pr{\'e}sentation d'un nouveau langage de
    calcul formel",
  type = "Technical Report",
  number = "640-M",
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institution = "TIM 3 (IMAG)",
address = "Grenoble, France",
year = "1987",
keywords = "axiomref",
beebe = "Senechaud:1987:SIP"
}

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— Senechaud:1987:SIP —

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@TechReport{Senechaud:1987:SIP,
author = "P. Senechaud and F. Siebert and G. Villard",
title = "Scratchpad {II}: Pr{\'e}sentation d'un nouveau langage
de calcul formel",
number = "640-M",
institution = "TIM 3 (IMAG)",
address = "Grenoble, France",
pages = "??",
month = feb,
year = "1987",
bibsourc = "/usr/local/src/bib/bibliography/Theory/Comp.Alg.1.bib;
http://www.math.utah.edu/pub/tex/bib/axiom.bib",
}

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— axiom.bib —

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@techreport{Sene87a,
author = "Senechaud, Pascale and Siebert, F.",
title = "Etude dl l'algorithme de Kovacic et son implantation sur
Scratchpad II",
type = "Technical Report",
number = "639",
institution = "Institut IMAG, Informatique et Mathematiques Appliquees
de Grenoble",
address = "Grenoble, France",
year = "1987",
keywords = "axiomref"
}

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— axiom.bib —

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@Article{Shan88,
  author = "Shannon, D. and Sweedler, M.",
  title = "Using Gr{\o}bner bases to determine algebra membership,
          split surjective algebra homomorphisms determine birational
          equivalence",
  journal = "Journal of Symbolic Computation",
  volume = "6",
  number = "2-3",
  pages = "267-273",
  year = "1988",
  abstract =
    "This paper presents a simple algorithm, based on Groebner bases, to
    test if a given polynomial  $g$  of  $k(X_1, \dots, X_n)$  lies in
     $k(f_1, \dots, f_m)$  where  $k$  is a field,  $X_1, \dots, X_n$  are
    indeterminates over  $k$  and  $f_1, \dots, f_m$  in  $k(X_1, \dots, X_n)$ .
    If so, the algorithm produces a polynomial  $P$  of  $m$  variables
    where  $g = P(f_1, \dots, f_m)$ . Say  $\Omega(B)$  to  $k(X_1, \dots, X_n)$  is a
    homomorphism where  $\Omega(b_i) = f_i$ , for algebra generators ( $b_i$ )
    contained in / implied by  $B$ . If  $\Omega$  is onto, the algorithm
    gives a homomorphism  $\lambda: k(X_1, \dots, X_n) \rightarrow B$ , where the
    composite  $\Omega \circ \lambda$  is the identity map. In particular, the
    algorithm computes the inverse of algebra automorphisms of the
    polynomial ring. A variation of the test if
     $k(f_1, \dots, f_m) = k(X_1, \dots, X_n)$  tells if
     $k(f_1, \dots, f_m) = k(X_1, \dots, X_n)$ . Existing computer algebra
    systems, such as IBM's SCRATCHPAD II, have Groebner basis packages
    which allow the user to specify a term ordering sufficient to carry
    out the algorithm.",
  keywords = "axiomref",
  beebe = "Shannon:1988:UGB"
}

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— Shannon:1988:UGB —

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@Article{Shannon:1988:UGB,
  author = "D. Shannon and M. Sweedler",
  title = "Using {Gr{\o}bner} bases to determine algebra
          membership, split surjective algebra homomorphisms
          determine birational equivalence",
  journal = "J-SYMBOLIC-COMP",
  volume = "6",
  number = "2-3",
  pages = "267--273",
  month = "oct # "-" # dec",
  year = "1988",
  CODEN = "JSYCEH",

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ISSN = "0747-7171 (print), 1095-855X (electronic)",
ISSN-L = "0747-7171",
bibdate = "Tue Sep 17 06:48:10 MDT 1996",
bibsource = "http://www.math.utah.edu/pub/tex/bib/axiom.bib",
abstract = "This paper presents a simple algorithm, based on
Gr{\o}bner bases, to test if a given polynomial g of
k(X/sub 1/,\ldots{}, X/sub n/) lies in k(f/sub
1/, \ldots{}, f/sub m/) where k is a field, X/sub
i/, \ldots{}, X/sub n/ are indeterminates over k and
f/sub 1/, \ldots{}, f/sub m/ in k(X/sub 1/, \ldots{},
X/sub n/). If so, the algorithm produces a polynomial P
of m variables where g=P(f/sub 1/, \ldots{}, f/sub m/).
Say omega:B to k(X/sub 1/, \ldots{}, X/sub n/) is a
homomorphism where omega (b/sub i/)=f/sub i/, for
algebra generators (b/sub i/) contained in/implied by
B. If omega is onto, the algorithm gives a homomorphism
lambda:k(X/sub 1/, \ldots{}, X/sub n/) to B, where the
composite omega lambda is the identity map. In
particular, the algorithm computes the inverse of
algebra automorphisms of the polynomial ring. A
variation of the test if k(f/sub 1/, \ldots{}, f/sub
m/)=k(X/sub 1/, \ldots{}, X/sub n/), tells if k(f/sub
1/, \ldots{}, f/sub m/)=k(X/sub 1/, \ldots{}, X/sub n/).
Existing computer algebra systems, such as IBM'S
SCRATCHPAD II, have Gr{\o}bner basis packages which
allow the user to specify a term ordering sufficient to
carry out the algorithm.",
acknowledgement = ack-nhfb,
affiliation = "Dept. of Math., Transylvania Univ., Lexington, KY,
USA",
classification = "C4130 (Interpolation and function approximation);
C6130 (Data handling techniques); C7310 (Mathematics)",
fjournal = "Journal of Symbolic Computation",
link = "\url{http://www.sciencedirect.com/science/journal/07477171}",
keywords = "IBM; Gr{\o}bner bases; Algebra membership; Split
surjective algebra homomorphisms; Birational
equivalence; Polynomial; Homomorphism; Algebra
generators; Identity map; Algebra automorphisms;
Computer algebra systems; SCRATCHPAD II",
language = "English",
pubcountry = "UK",
thesaurus = "Polynomials; Symbol manipulation",
}

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@misc{SIGS16,
  author = "SIGSAM, ACM",
  title = "Axiom",
  link = "\url{http://www.sigsam.org/software/axiom.html}",
  year = "2016",
  contact = "Infodir\_SIGSAM@acm.org",
  abstract =
    "Axiom is a free, open source, general-purpose computer algebra
    system. It features a strongly typed language. The system has an
    interactive interpreter and a compiler. It includes over 1100
    supported categories, domains, and packages covering large areas of
    Mathematics."
}

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— axiom.bib —

```

@article{Sing93,
  author = "Singer, Michael F. and Ulmer, Felix",
  title = "Galois groups of second and third order linear differential
    equations",
  journal = "J. Symb. Comput.",
  volume = "16",
  number = "1",
  pages = "9-36",
  year = "1993",
  abstract =
    "The authors discuss the first problem of Galois theory of differential
    equations. Let  $F$  be an ordinary (for simplicity) differential field
    and  $L(y)=0$  be an ordinary linear differential equation over  $F$ . How
    can one calculate the Galois group of  $L$  over  $F$ ? The authors
    suppose a new approach to the problem. They reduce it to the problem
    of finding solutions of linear differential equations in  $F$  and to
    the factorization problem of such equations over  $F$ . These allow them
    to give simple necessary and sufficient conditions for a second order
    linear differential equation to have Liouvillian solutions and for a
    third order linear differential equation to have Liouvillian solutions
    or to be solvable in terms of second order equations.",
  paper = "Sing93.pdf",
  keywords = "axiomref"
}

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```
@article{Sing93a,
  author = "Singer, Michael F. and Ulmer, Felix",
  title = "Liouvillian and algebraic solutions of second and third order
          linear differential equations",
  journal = "J. Symb. Comput.",
  volume = "16",
  number = "1",
  pages = "37-73",
  year = "1993",
  abstract =
    "Let  $F$  be an ordinary differential field of characteristic 0 and
     $L \in F\langle y \rangle$  be a linear homogeneous polynomial. How can one find the
    Liouvillian solutions of  $L(y)=0$ ? In the paper this problem is
    reduced to the problems of (1) factorization and (2) finding  $u$ 
    solutions such that  $\frac{u^{\prime}}{y} \in F$  of  $L$  and some
    polynomials associated with it (symmetric powers of  $L$ ).

    Now there are the algorithms for the solution of the last problems for
     $F=\mathbb{Q}(x)$  [see D. Yu. Grigorev, J. Symb. Comput. 10, 7-37
    (1990; Zbl 0728.68067) and M. F. Singer, Am. J. Math. 103, 661-682
    (1981; Zbl 0477.12026)].

    For polynomials  $L$  of the second and third order the authors provide
    full investigation of the most difficult case when the solution  $u$  of
     $L(y)$  is algebraic. They show that one can compute the minimal
    polynomial  $P(y) \in F[y]$  of  $u$ . We note that the authors
    essentially used the tools of representation theory, invariant theory
    and computer algebra.",
  paper = "Sing93a.pdf",
  keywords = "axiomref"
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@inproceedings{Sitx89,
  author = "Sit, William Y.",
  title = "On Goldman's algorithm for solving first-order multinomial
          autonomous systems",
  booktitle = "Proc. Algebraic Algorithms and Error-Correcting Codes, AAECC-6",
  series = "Lecture Notes in Computer Science 357",
  location = "Rome, Italy",
  year = "1988",
  isbn = "3-540-51083-4",
  pages = "386-395",
  abstract =
    "In this article, a brief exposition of a method for finding first
```

integrals for first order multinomial autonomous systems (FOMAS) of ordinary differential equations with constant coefficients will be given. The method is a simplified as well as a redesigned version based on a paper of Goldman (1987). We shall see how it can be applied to FOMAS with parametric coefficients. The algorithm is currently being implemented by the author, using the SCRATCHPAD II computer algebra language and system at the IBM T.J. Watson Research Center.

FOMAS occur and are of interest in many disciplines and their first integrals (or trajectories of motion) are generally difficult to find. Examples of FOMAS are too numerous to list, some well-known ones are the Riccati equation, the Lotka-Volterra equations for competing populations, Selkov's model for chemical reactions, the Lorenz system of the Rayleigh-Bernard problem, and Hamiltonian systems (where the Hamiltonian is a sum of monomial terms with constant coefficients).

Let $Y = (y_1, \dots, y_n)$ be n functions depending on the variable τ . A monomial in Y is a product of the form $y_1^{k_1} \cdots y_n^{k_n}$, where $k_1, \dots, k_n$ are constants. If $K = (k_1, \dots, k_n)$, we shall denote the monomial in Y by Y^K , and K is called the exponent vector for the monomial. By convention, exponent vectors are column vectors, but whenever convenient, we shall write exponent vectors as row vectors. We say that Y satisfies a first-order multinomial autonomous system (FOMAS) if for each i , $1 \leq i \leq n$, y_i satisfies a first order differential equation of the form:

$$[y_i'] = f_i(Y) \quad (1)$$

where f_i is a linear combination of monomials in Y with coefficients which may be either constants or parametric constants. For example, the Lotka-Volterra equations for three competing species considered by Schwarz and Steeb (1984), form a FOMAS:

$$[x_1'] = x_1(1 + ax_2 + bx_3)$$

$$[x_2'] = x_2(1 - ax_1 + bx_3)$$

$$[x_3'] = x_3(1 - bx_1 - cx_2)$$

When the exponent vectors occurring in f_i are all non-negative integers, as in the example above, a FOMAS reduces to a polynomial autonomous system (FOPAS).

A computer program was developed by Schwarz (1986) to compute the first integrals of FOPAS's which are themselves polynomials in $y_1, \dots, y_n$. Schwarz's algorithm literally takes a general polynomial of a fixed degree d in n variables and substitutes it into (1). This method does not work well on a FOMAS, because in a FOMAS, the exponent vectors need not have integral components. Also, it will not find integrals with exponent vectors that involve fractional or irrational numbers.

Goldman (1987) proved a theorem which gives necessary and sufficient conditions for the existence of a multinomial first integral for FOMAS. The proof also contained the outline of an algorithm for

finding such integrals. In Goldman's paper, he introduced the notion of an integral array, which is a certain matrix satisfying some 10 conditions. He gave a few hints and several examples but did not elaborate on how such an integral array can be found in general (except in the case $q=2$). Assuming such an array is found, he can compute the integral, in most cases, by solving systems of linear equations, or at worse in certain cases, by solving a system of algebraic equations. It was not clear when algebraic conditions are necessary.

In this brief exposition, Goldman's method will be expanded to a complete algorithm with a new simplified notation. The integral arrays are replaced by addition schemes (which is equivalent to integral arrays with some conditions removed). The generation of addition schemes is a combinatorial problem unrelated, in a sense, to FOMAS. When the first integral is a polynomial, the addition scheme is trivial to compute. We shall now begin by explaining some details of this theory.",

keywords = "axiomref",

beebe = "Sit:1989:GAS"

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— Sit:1989:GAS —

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@InProceedings{Sit:1989:GAS,
  author = "W. Y. Sit",
  title = "On {Goldman}'s algorithm for solving first-order
    multinomial autonomous systems",
  crossref = "Mora:1989:AAA",
  pages = "386--395",
  month = "",
  year = "1989",
  bibdate = "Tue Sep 17 06:46:18 MDT 1996",
  bibsource = "http://www.math.utah.edu/pub/tex/bib/axiom.bib",
  abstract = "A brief exposition of a method for finding first
    integrals for first order multinomial autonomous
    systems (FOMAS) of ordinary differential equations with
    constant coefficients is given. The method is a
    simplified as well as a redesigned version based on a
    paper of Goldman (1987). The author shows how it can be
    applied to FOMAS with parametric coefficients. The
    algorithm is currently being implemented using the
    SCRATCHPAD II computer algebra language and system at
    the IBM TJ Watson Research Center.",
  acknowledgement = ack-nhfb,
  affiliation = "Dept. of Math., City Coll. of New York, NY, USA",
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classification = "B0290P (Differential equations); B0290R (Integral
                  equations); C4170 (Differential equations); C4180
                  (Integral equations); C7310 (Mathematics)",
keywords =       "Computer algebra language; Constant coefficients;
                  First integrals; First order multinomial autonomous
                  systems; FOMAS; Goldman algorithm; IBM; Ordinary
                  differential equations; SCRATCHPAD II",
language =       "English",
thesaurus =       "Differential equations; Integral equations;
                  Mathematics computing",
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@article{Sitx92,
  author = "Sit, William Y.",
  title = "An algorithm for solving parametric linear systems",
  journal = "Journal of Symbolic Computations",
  volume = "13",
  number = "4",
  pages = "353-394",
  year = "1992",
  abstract =
    "We present a theoretical foundation for studying parametric systems of
    linear equations and prove an efficient algorithm for identifying all
    parametric values (including degenerate cases) for which the system is
    consistent. The algorithm gives a small set of regimes where for each
    regime, the solutions of the specialized systems may be given
    uniformly. For homogeneous linear systems, or for systems where the
    right hand side is arbitrary, this small set is irredundant. We discuss
    in detail practical issues concerning implementations, with particular
    emphasis on simplification of results. Examples are given based on a
    close implementation of the algorithm in SCRATCHPAD II. We also give a
    complexity analysis of the Gaussian elimination method and compare
    that with our algorithm.",
  paper = "Sitx92.pdf",
  keywords = "axiomref",
  beebe = "Sit:1992:ASP"
}

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— Sit:1992:ASP —

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@Article{Sit:1992:ASP,

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author =      "W. Y. Sit",
title =      "An algorithm for solving parametric linear systems",
journal =    j-J-SYMBOLIC-COMP,
volume =     "13",
number =     "4",
pages =      "353--394",
month =      apr,
year =       "1992",
CODEN =      "JSYCEH",
ISSN =       "0747-7171 (print), 1095-855X (electronic)",
ISSN-L =     "0747-7171",
bibdate =    "Tue Sep 17 06:41:20 MDT 1996",
bibsource =  "http://www.math.utah.edu/pub/tex/bib/axiom.bib",
abstract =   "The author presents a theoretical foundation for
studying parametric systems of linear equations and
proves an efficient algorithm for identifying all
parametric values (including degenerate cases) for
which the system is consistent. The algorithm gives a
small set of regimes where for each regime, the
solutions of the specialized systems may be given
uniformly. For homogeneous linear systems, or for
systems where the right hand side is arbitrary, this
small set is irredundant. He discusses in detail
practical issues concerning implementations, with
particular emphasis on simplification of results.
Examples are given based on a close implementation of
the algorithm in SCRATCHPAD II. He also gives a
complexity analysis of the Gaussian elimination method
and compares that with the algorithm.",
acknowledgement = ack-nhfb,
affiliation = "Dept. of Math., City Coll. of New York, NY, USA",
classification = "B0290H (Linear algebra); C4140 (Linear algebra);
C4240 (Programming and algorithm theory); C7310
(Mathematics)",
fjournal =   "Journal of Symbolic Computation",
link =       "\url{http://www.sciencedirect.com/science/journal/07477171}",
keywords =   "Complexity analysis; Efficient algorithm; Gaussian
elimination method; Linear equations; Parametric linear
systems; SCRATCHPAD II",
language =   "English",
pubcountry = "UK",
thesaurus =  "Computational complexity; Matrix algebra; Symbol
manipulation",
}

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\bibitem[Sit 06]{Sit06} Sit, Emil
  title = "Tools for Repeatable Research",
  link = "\url{http://www.emilsit.net/blog/archives/tools-for-repeatable-research}",
  keywords = "axiomref"
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@inproceedings{Smed92,
  author = "Smedley, Trevor J.",
  title = "Using pictorial and object oriented programming for computer
    algebra",
  booktitle = "Applied computing --- technological challenges of the
    1990's: proceedings of the 1992 ACM\slash SIGAPP
    Symposium on Applied Computing, Kansas City Convention
    Center, March 1--3, 1992",
  publisher = "ACM Press",
  pages = "1243-1247",
  year = "1992",
  isbn = "0-89791-502-X",
  keywords = "axiomref",
  beebe = "Smedley:1992:UPO"
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— Smedley:1992:UPO —

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@InProceedings{Smedley:1992:UPO,
  author = "Trevor J. Smedley",
  editor = "Hal Berghel and others",
  booktitle = "Applied computing --- technological challenges of the
    1990's: proceedings of the 1992 ACM\slash SIGAPP
    Symposium on Applied Computing, Kansas City Convention
    Center, March 1--3, 1992",
  title = "Using pictorial and object oriented programming for
    computer algebra",
  publisher = pub-ACM,
  address = pub-ACM:adr,
  pages = "1243--1247",
  year = "1992",
  DOI = "http://dx.doi.org/10.1145.130154",
  ISBN = "0-89791-502-X",
  ISBN-13 = "978-0-89791-502-1",
  LCCN = "QA76.76.A65 S95 1992",
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bibdate =      "Thu Jul 26 09:02:03 2001",
bibsource =    "http://www.math.utah.edu/pub/tex/bib/axiom.bib",
acknowledgement = ack-nhfb,
bookpages =    "xiii + 1257 (2 volumes)",
keywords =     "Scratchpad",
}

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@article{Smit93,
  author = "Smith, Geoff C.",
  title = "Group theory results with machine generated proofs",
  journal = "An. Univ. Timis., Ser. Mat.-Inform.",
  volume = "31",
  number = "2",
  pages = "273-280",
  year = "1993",
  abstract =
    "There are a variety of theorems in group theory which admit of proofs
    by machine. This talk illustrates these techniques in action. Examples
    are given of this phenomenon, drawn from the theory of group
    presentations, and from the theory of  $p$ -groups. The systems involved
    include AXIOM, CAYLEY and QUOTPIC",
  keywords = "axiomref"
}

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— axiom.bib —

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@InProceedings{Smit07,
  author = "Smith, Jacob and Dos Reis, Gabriel and Jarvi, Jaakko",
  title = "Algorithmic differentiation in Axiom",
  booktitle = "ACM SIGSAM Proceedings",
  series = "ISSAC 2007",
  year = "2007",
  pages = "347-354",
  isbn = "978-1-59593-743-8",
  keywords = "axiomref",
  abstract = "
    This paper describes the design and implementation of an algorithmic
    differentiation framework in the Axiom computer algebra system. Our
    implementation works by transformations on Spad programs at the level
    of the typed abstract syntax tree -- Spad is the language for extending

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```

Axiom with libraries. The framework illustrates an algebraic theory
of algorithmic differentiation, here only for Spad programs, but
we suggest that the theory is general. In particular, if it is
possible to define a compositional semantics for programs, we define
the exact requirements for when a program can be algorithmically
differentiated. This leads to a general algorithmic differentiation
system, and is not confined to functions which compute with basic
data types, such as floating point numbers.",
paper = "Smit07.pdf",
keywords = "axiomref",
beebe = "Smith:2007:ADA"
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— Smith:2007:ADA —

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@InProceedings{Smith:2007:ADA,
  author =      "Jacob Smith and Gabriel {Dos Reis} and Jaakko
                J{\\"a}rvi",
  title =      "Algorithmic differentiation in {Axiom}",
  crossref =   "Brown:2007:PIS",
  pages =      "347--354",
  year =       "2007",
  DOI =        "http://doi.acm.org/10.1145/1277548.1277595",
  bibdate =    "Fri Jun 20 08:46:50 MDT 2008",
  bibsource =  "http://portal.acm.org/;
                http://www.math.utah.edu/pub/tex/bib/axiom.bib",
  abstract =   "This paper describes the design and implementation of
                an algorithmic differentiation framework in the Axiom
                computer algebra system. Our implementation works by
                transformations on Spad programs at the level of the
                typed abstract syntax tree -- Spad is the language for
                extending Axiom with libraries. The framework
                illustrates an algebraic theory of algorithmic
                differentiation, here only for Spad programs, but we
                suggest that the theory is general. In particular, if
                it is possible to define a compositional semantics for
                programs, we define the exact requirements for when a
                program can be algorithmically differentiated. This
                leads to a general algorithmic differentiation system,
                and is not confined to functions which compute with
                basic data types, such as floating point numbers.",
  acknowledgement = ack-nhfb,
  keywords =    "algorithmic differentiation; axiom; program
                transformation; symbolic-numeric computation",
}

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@article{Spit11,
  author = "Spitters, Bas and van der Weegen, Eelis",
  title = "Type classes for mathematics in type theory",
  journal = "Math. Struct. Comput. Sci.",
  volume = "21",
  number = "4",
  pages = "795-825",
  year = "2011",
  link = "\url{https://arxiv.org/pdf/1102.1323.pdf}",
  abstract =
    "The introduction of first-class type classes in the Coq system calls
    for a re-examination of the basic interfaces used for mathematical
    formalisation in type theory. We present a new set of type classes for
    mathematics and take full advantage of their unique features to make
    practical a particularly flexible approach that was formerly thought
    to be infeasible. Thus, we address traditional proof engineering
    challenges as well as new ones resulting from our ambition to build
    upon this development a library of constructive analysis in which any
    abstraction penalties inhibiting efficient computation are reduced to
    a minimum.

    The basis of our development consists of type classes representing a
    standard algebraic hierarchy, as well as portions of category theory
    and universal algebra. On this foundation, we build a set of
    mathematically sound abstract interfaces for different kinds of
    numbers, succinctly expressed using categorical language and universal
    algebra constructions.

    Strategic use of type classes lets us support these high-level
    theory-friendly definitions, while still enabling efficient
    implementations unhindered by gratuitous indirection, conversion or
    projection.

    Algebra thrives on the interplay between syntax and semantics. The
    Prolog-like abilities of type class instance resolution allow us to
    conveniently define a quote function, thus facilitating the use of
    reflective techniques.",
  paper = "Spit11.pdf",
  keywords = "axiomref"
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@InProceedings{Schu92,
  author = "Schu, J. and Seiler, Werner Markus and Calmet, Jacques",
  title = "Algorithmic Methods For Lie Pseudogroups'",
  booktitle = "Proc. Modern Group Analysis: Advanced Analytical and
    Computational Methods in Mathematical Physics",
  pages = "337-344",
  location = "Acireale (Italy)",
  year = "1992",
  publisher = "Kluwer",
  link = "\url{http://www.iks.kti.edu/fileadmin/User/calmet/papers/Acireale-93.ps.gz}",
  abstract =
    "The authors recall the concepts of involutive non-linear systems of
    partial differential equations, the classical Cartan-Riquier-Janet
    criterion for involutiveness, and the Cartan-Kuranishi prolongation
    theorem. Then the algorithm of prolongation into involutive case is
    clearly outlined, the arising computational problems are discussed,
    experience with implementation in the computer algebra system AXIOM
    (1992) is described, and comparison with Maple and REDUCE algorithms
    is made.",
  paper = "Schu92.pdf",
  keywords = "axiomref"
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@book{Stau95,
  author = "Stauffer, Deitrich",
  title = "Annual Review of Computational Physics I",
  publisher = "unknown",
  section = "5.2.2",
  year = "1995",
  abstract =
    "Based on our experiences with IRENA, we decided to use generic
    inter-process communication tools for the link to AXIOM. This has the
    added advantage that we can operate across a network. The main
    technique we use is the {\sl Remote Procedure Call} (RPC) [Sun
    Microsystems Inc., 1988] which allows us to interact with a server on
    another machine (or on the local machine). RPC takes care of
    differences in data representation (e.g. the byte-order of floating
    point numbers) on different architectures.
```

AXIOM is a multi-process package. Normally when a user starts up the system they start up the various components which then interact via standard socket operations. If they are using the line, they start up

a new process: the NAG Manager (NAGMAN for short). Additionally, there will be a NAG daemon (NAGD) running on any machine on which the user may wish to execute NAG routines (which could include the local host). NAGMAN communicates with the running AXIOM system via a socket down which is transmitted the details of and data for the particular routine to be called. NAGMAN calls a NAGD on another machine via RPC and eventually returns the results to AXIOM.

NAGD consists of the server program, and a set of stub codes designed to call individual NAG routines. It is, in effect, a remotely-callable version of the NAG library. There is no reason why AXIOM should be the only system to use it, and indeed there are plans to incorporate the ability to call NAGD into other systems.

An ASP is treated just like any other piece of data by the AXIOM-NAG link. The source code is passed to NAGD and compiled. (There are various optimisations to prevent the same code being compiled multiple times, but the details needn't concern us here.) This compiled code is linked with the NAG Library to make the executable. Thus if a user calls the same NAG routine with different ASPs the routine will be relinked each time.

It would be nice if this were not necessary. The authors of the link considered two other possibilities:

```
\begin{itemize}
\item Have AXIOM simulate the ASPs, so that the NAG Library would call
back to AXIOM when it wanted to call an ASP. This was rejected as
being far too slow across a network.
\item Give NAGD the ability to interpret AXIOM or Fortran code. Thus
the NAG routine would call a function which would evaluate a
representation of an ASP to get the required values. This may happen
in the future if data interchange mechanisms between systems are
standardised, but was rejected for the time being since such a system
would have to be tailored to match each Fortran compiler that NAGD used.
\end{itemize}
```

By transmitting source code for ASPs we allow the remote Fortran compiler to take care of low-level portability problems.

Both IRENA and the AXIOM-NAG Link make using the NAG Library easier in a functional kind of way. They present a more natural, interactive interface and, what is more, they cut out some of the tedium of writing code by generating values for some parameters (array dimensions etc.) and computing derivatives and jacobians. Unfortunately the user still has to choose which algorithm to use, and then the control parameters to give a result of the required accuracy.

This is not a trivial task. The NAG Library has some thirty quadrature routines, and maybe forty top-level ODE-solving routines. The criteria

for selecting a routine can include features of the problem such as whether a function is continuous, or they can include requirements from the user such as whether speed or accuracy is important. The user should also think carefully about whether the problem is being presented in the best way: perhaps an integral can be solved more easily if the range of integration is split into several pieces for example.

Many users of numerical software would like to be presented with a "black box" for solving a given class of problems. In practice many users will always try solving a problem with the "simplest" method first, in the hope that they can escape having to think seriously about their problem! This is not always a bad idea anyway, since the feedback from the routine can help determine a better technique (e.g. by identifying the regions where problems occur).

We contend that computer algebra has an important role to play in this area. The ability of algebra systems to analyse the symbolic features of a problem make them ideal agents for knowledge-based (or expert) systems to use in selecting routines. Their ability (via IRENA or AXIOM-NAG) to drive the numerical codes is a bonus.",

}

— ignore —

```
\bibitem[Steele]{Steele} Steele, Guy L.; Gabriel, Richard P.
  title = "The Evolution of Lisp",
  link = "\url{http://www.dreamsongs.com/Files/HOPL2-Uncut.pdf}",
  keywords = "axiomref"
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\bibitem[Sage 14]{Sage14} Stein, William
  title = "Sage",
  link = "\url{http://www.sagemath.org/doc/reference/interfaces/sage/interfaces/axiom.html}",
  keywords = "axiomref"
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@misc{Steil6,
  author = "Stein, William",
  title = "Axiom as an OSCAS",
  link = "\url{http://wiki.sagemath.org/Axiom\_as\_an\_OSCAS}",
  keywords = "axiomref"
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@article{Stor95,
  author = "Storme, L. and van Maldeghem, H.",
  title = "Cyclic caps in PG(3,q)",
  journal = "Geom. Dedicata",
  volume = "56",
  number = "3",
  pages = "271-284",
  year = "1995",
  link = "\url{https://cage.ugent.be/~hvm/artikels/44.pdf}",
  abstract =
    "A  $k$ -cap  $KK$  on  $PG(n,q)$  is a set of  $k$  points, no three of which
    are collinear.  $KK$  is complete if it cannot be extended to a  $k+1$ -cap. If  $KK$  is invariant under a cyclic subgroup (which acts regularly
    on  $KK$ ) of  $PGL(n+1,q)$ , the  $KK$  is cyclic.

    This article investigates cyclic complete  $k$ -caps in
     $PG(3,q)$ . Namely, the different types of complete  $k$ -caps  $KK$  in
     $PG(3,q)$  stabilized by a cyclic projective group  $GG$  of order  $k$ ,
    acting regularly on the points of  $KK$ , are determined. We show that in
     $PG(3,q)$ ,  $q$  even, the elliptic quadric is the only cyclic complete
     $k$ -cap. For  $q$  odd, it is shown that besides the elliptic quadric,
    there also exist cyclic  $k$ -caps containing  $k/2$  points of two
    disjoint elliptic quadrics or two disjoint hyperbolic quadrics and
    that there exist cyclic  $k$ -caps stabilized by a transitive cyclic
    group  $GG$  fixed precisely one point and one plane of
     $PG(3,q)$ . Concrete examples of such caps, found using AXIOM and
    CAYLEY, are presented.",
  paper = "Stor95.pdf",
  keywords = "axiomref"
}
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— axiom.bib —

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@book{Stro99,
  author = "Stroeker, Roelof J. and Kaashoek, Johan F.",
  title = "Discovering mathematics with Maple. An interactive exploration for
          mathematicians, engineers and econometricians",
  year = "1999",
  publisher = "Birkhauser",
  abstract =
    "During the past decade, the mathematical computer software packages
    such as Mathematica, Maple, MATLAB (Axiom, Derive, Macsyma, MuPad are
    some further examples of such software) [see Macsyma 2.3. Lite the
    student edition (1998; Zbl 0911.68089); B. W. Char, K. O. Geddes,
    G. H. Gonnet, B. L. Leong, M. B. Monagan, and S. M. Watt, Maple V
    Library reference manual (1991; Zbl 0763.68046); J. L. Zachary,
    Introduction to scientific programming. Computational problem solving
    using Mathematica and C (1997; Zbl 0891.68053); The student edition of
    MATLAB. Student user guide. The problem-solving tool for engineers,
    mathematicians, and scientists (1992; Zbl 0782.65001); H. Benker,
    Ingenieurmathematik mit Computeralgebra-Systemen. AXIOM, DERIVE,
    MACSYMA, MAPLE, MATHCAD, MATHEMATICA, MATLAB und MuPAD in der
    Anwendung (1998; Zbl 0909.68109); W. Koepf, Hohere Analysis mit DERIVE
    (1994; Zbl 0819.26003)] have greatly facilitated mathematical
    experiments and have thus become popular tools for the modern
    mathematician. It is a pity that most of these packages are quite
    expensive, and that the frequently upgraded versions are not free for
    the owners of the earlier versions (fortunately, there are inexpensive
    student versions of some of these packages). There is a constant
    demand of instructional textbooks by users of these packages. This
    demand is reflected in the growing number of such textbooks. Many of
    these books provide software support (diskette, CD-ROM, access by
    ftp). Such a textbook should meet, in my opinion, the following
    criteria: (1) The size should be small, not bulky like the complete
    technical descriptions of the software. (2) There should be a lot of
    examples of the use of the software covering a wide range of
    mathematical topics. Electronic versions of these examples should be
    made available for free to the users of the textbook
    (e.g. diskette/CD-ROM, access by ftp). (3) There should be a good
    supply of exercises covering the basic mathematical applications. (4)
    The book should be visually pleasing, easy to read, have good indexes
    and provide pointers to other books and electronic sources of
    information. The book under review provides, in addition to the actual
    text, an interactive exploratorium of its topics, based on the
    mechanism of Maple worksheets. These worksheets can be ‘‘opened’’ by
    the Maple program and they form a mixture of usual text, hypertext,
    and Maple commands and have a nice style appearance. They also can be
    ‘‘exported’’ in a file and included in a file for further treatment.
    The book meets all the aforementioned criteria (1)-(4) with elegance.
    There are many exercises which cover all the usual mathematical topics
    from linear algebra to differential equations and statistics. A
    valuable feature is an appendix with hints and answers for all
    exercises. One of the highlights of the book is the examination of
```

Riemann's non-differentiable function

$$\sum_{k=1}^{\infty} k^{-2} \sin(\pi kx)$$
 which is differentiable only at the rational points p/q with p and q odd and relatively prime, where its derivative is $-1/2$.

The book is intended for students of mathematics, engineering sciences, and econometry. This book is an ideal guide for this purpose and it could probably be used along, without the bulky technical documentation of the Maple language. Note that Maple has a comprehensive on-line help program, which contains large parts of the original documentation.",

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@inproceedings{Suto85,
  author = "Sutor, Robert S.",
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  booktitle = "Research Contributions from the Euro. Conf. on Comp. Alg.",
  series = "Lecture Notes in Computer Science Volume 204",
  volume = "2",
  pages = "32-33",
  year = "1985",
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@InProceedings{Sutor:1985:SIC,
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@article{Suto87,
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    "The Scratchpad II system is an abstract datatype programming
    language, a compiler for the language, a library of packages of
    polymorphic functions and parametrized abstract datatypes, and an
    interpreter that provides sophisticated type inference and coercion
    facilities . Although originally designed for the implementation of
    symbolic mathematical algorithms, Scratchpad 11 is a general purpose
    programming language . This paper discusses aspects of the
    implementation of the interpreter and how it attempts to provide a
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        library of packages of polymorphic functions and
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        language. This paper discusses aspects of the
        implementation of the interpreter and how it attempts
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@misc{Suto87b,
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@techreport{Suto87c,
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  mathematical algorithms, Scratchpad II is a general purpose
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\bibitem[Sutor 88]{Su88} Sutor, Robert S.
  title = "A guide to programming in the scratchpad 2 interpreter",
  IBM Manual, March 1988

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— axiom.bib —

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author = "R. D. Jenks R. S. Sutor and S. M. Watt",
title = "\"Scratchpad II: An abstract Datatype system for
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pages = "12-37",
year = "1988"
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— Sutor:1988:SIA —

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@InProceedings{Sutor:1988:SIA,
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@misc{Swmath,
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    "Axiom is a general purpose Computer Algebra System (CAS). It is
    useful for research and developement of mathematical algorithms.
    It defines a strongly typed, mathematically correct type hierarchy.
    It has a programming language and a built-in compiler.",
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@misc{Sympy,
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  link = "\url{https://github.com/sympy/sympy/wiki/SymPy-vs.-Axiom}",
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@InProceedings{Thom01,
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  title = "Logic and dependent types in the Aldor Computer Algebra System",
  booktitle = "Symbolic computation and automated reasoning",
  series = "CALCULEMUS 2000",
  year = "2001",
  location = "St. Andrews, Scotland",
  pages = "205-233",
  link = "\url{http://axiom-wiki.newsynthesis.org/public/refs/aldor-calc2000.pdf}",
  abstract =
    "We show how the Aldor type system can represent propositions of
    first-order logic, by means of the 'propositions as types'
    correspondence. The representation relies on type casts (using
    pretend) but can be viewed as a prototype implementation of a modified
    type system with {\sl type evaluation} reported elsewhere. The logic
    is used to provide an axiomatisation of a number of familiar Aldor
    categories as well as a type of vectors.",
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@misc{Thomxx,
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  This paper introduces the \verb|Aldor--| language, which is a
  functional programming language with dependent types and a powerful,
  type-based, overloading mechanism. The language is built on a subset
  of Aldor, the 'library compiler' language for the Axiom computer
  algebra system. \verb|Aldor--| is designed with the intention of
  incorporating logical reasoning into computer algebra computations.

  The paper contains a formal account of the semantics and type system
  of \verb|Aldor--|; a general discussion of overloading and how the
  overloading in \verb|Aldor--| fits into the general scheme; examples
  of logic within \verb|Aldor--| and notes on the implementation of the
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@misc{Tour98,
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  year = "1998",
  link = "\url{http://axiom-wiki.newsynthesis.org/public/refs/Aldor-T1998_04.pdf}",
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@misc{Tour95,
  author = "Tournier, Evelyne",
  title = "Summary of organisation, history and possible future",
  year = "1995",
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@article{Tuom96,
  author = "Tuomela, Jukka",
  title = "On the construction of arbitrary order schemes for the many
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  journal = "BIT",
  volume = "36",
  number = "1",
  pages = "158-165",
  year = "1996",
  abstract =
    "The paper is devoted to a problem which was of an interest in the
    beginning of the theory of difference methods. The elementary
    constructed explicit high-order approximations for the wave equation
    (on the simplest cubic grid in space) assume that the solution is very
    smooth and that no boundary conditions are given. Stability is also
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@misc{Unkn16,
  author = "Unknown",
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  paper = "Unkn16.pdf"
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@article{Hoev12,
  author = "van der Hoeven, Joris",
  title = "Overview of the Mathemagix type system",
  journal = "ASCM 2102",
  year = "2012",
  link = "\url{http://www.texmacs.org/joris/mmxtyping/mmxtyping.pdf}",
  abstract =
    "The goal of the {\tt MATHEMAGIX} project is to develop a new and free
    software for computer algebra and computer analysis, base on a
    strongly typed and compiled language. In this paper, we focus on the
    underlying type system of this language, which allows for heavy
    overloading, including parameterized overloading with parameters in so
    called ‘‘categories’’. The exposition is informal and aims at giving
    the reader an overview of the main concepts, ideas and differences
    with existing languages. In a forthcoming paper, we intend to describe
    the formal semantics of the type system in more detail.",
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\bibitem{van der Hoeven 14}{JvdH14} van der Hoeven, Joris
  title = "Computer algebra systems and TeXmacs",
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@article{Hoei94,
  author = "{van Hoeij}, M.",
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  number = "4",
  year = "1994",
  pages = "353-363",
  issn = "0747-7171",
  abstract = "
    Algorithms for computing integral bases of an algebraic function field
    are implemented in some computer algebra systems. They are used e.g.
  "

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for the integration of algebraic functions. The method used by Maple 5.2 and AXIOM is given by Trager in [Trag84]. He adapted an algorithm of Ford and Zassenhaus [Ford, 1978], that computes the ring of integers in an algebraic number field, to the case of a function field.

It turns out that using algebraic geometry one can write a faster algorithm. The method we will give is based on Puiseux expansions. One can see this as a variant on the Coates' algorithm as it is described in [Davenport, 1981]. Some difficulties in computing with Puiseux expansions can be avoided using a sharp bound for the number of terms required which will be given in Section 3. In Section 5 we derive which denominator is needed in the integral basis. Using this result 'intermediate expression swell' can be avoided.

The Puiseux expansions generally introduce algebraic extensions. These extensions will not appear in the resulting integral basis.",
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@Article{vanHoeij:1994:ACI,
  author = "M. van Hoeij",
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  number = "4",
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  month = oct,
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  ISSN = "0747-7171 (print), 1095-855X (electronic)",
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  bibsource = "http://www.math.utah.edu/pub/tex/bib/axiom.bib",
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             algebraic function field are implemented in some
             computer algebra systems. They are used e.g. for the
             integration of algebraic functions. The method used by
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             adapted an algorithm of Ford and Zassenhaus (1978),
             that computes the ring of integers in an algebraic
             number field, to the case of a function field. It turns
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        out that using algebraic geometry one can write a
        faster algorithm. The method we give is based on
        Puiseux expansions. One can see this as a variant on
        the Coates' algorithm as it is described by Davenport
        (1981). Some difficulties in computing with Puiseux
        expansions can be avoided using a sharp bound for the
        number of terms required which are given. We derive
        which denominator is needed in the integral basis.
        Using this result 'intermediate expression swell' can
        be avoided. The Puiseux expansions generally introduce
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@misc{Hoei08,
  author = "{van Hoeij}, Mark and Novocin, Andrew",
  title = "A Reduction Algorithm for Algebraic Function Fields",
  year = "2008",
  month = "April",
  link = "\url{http://andy.novocin.com/pro/algext.pdf}",
  abstract = "
    Computer algebra systems often produce large expressions involving
    complicated algebraic numbers. In this paper we study variations of
    the {\tt polred} algorithm that can often be used to find better
    representations for algebraic numbers. The main new algorithm
    presented here is an algorithm that treats the same problem for the
    function field case.",
  paper = "Hoei08.pdf"
}

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\bibitem[Vasconcelos 99]{Vas99} Vasconcelos, Wolmer
  title = "Computational Methods in Commutative Algebra and Algebraic Geometry",
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  ISBN 3-540-21311-2
  keywords = "axiomref"
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@article{Wang90,
  author = "Wang, Dongming",
  title = "A Class of Cubic Differential Systems with 6-tuple Focus",
  journal = "J. Differential Equations",
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  volume = "87",
  pages = "305-315",
  year = "1990",
  abstract =
    "This paper presents a class of cubic differential systems with the
    origin as a 6-tuple focus from which 6 limit cycles may be
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  year = "1991",
  comment = "See Wang89 for the implementation code",
  abstract =
    "In this paper we describe a mechanical procedure for computing the
    Liapunov functions and Liapunov constants for a class of differential
    systems. These functions and constants are used for establishing the
    stability criteria, the conditions for the existence of a center and
    for the investigation of limit cycles. Some problems for handling the
    computer constants, which are usually large polynomials in terms of
    the coefficients of the differential system, and an approach towards
    their solution by using computer algebraic methods are proposed. This
    approach has been successfully applied to check some known results
    mechanically. The author has implemented a system DEMS on an HP1000
    and in Scratchpad II on an IBM4341 for computing and manipulating the
    Liapunov functions and Liapunov constants. As examples, two particular
    cubic systems are discussed in detail. The explicit algebraic
    relations between the computed Liapunov constants and the conditions
    given by Saharnikov are established, which leads to a rediscovery of
    the incompleteness of his conditions. A class of cubic systems with
    6-tuple focus is presented to demonstrate the feasibility of the
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ISSN-L =      "0747-7171",
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               and for the investigation of limit cycles. Some
               problems for handling the computer constants, which are
               usually large polynomials in terms of the coefficients
               of the differential system, and an approach towards
               their solution by using computer algebraic methods are
               proposed. This approach has been successfully applied
               to check some known results mechanically. The author
               has implemented a system DEMS on an HP1000 and in
               Scratchpad II on an IBM4341 for computing and
               manipulating the Liapunov functions and Liapunov
               constants. As examples, two particular cubic systems
               are discussed in detail. The explicit algebraic
               relations between the computed Liapunov constants and
               the conditions given by Saharnikov are established,
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               Scratchpad II; Stability criteria",

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@misc{Wang95,
  author = "Wang, Dongming",
  title = "Characteristic Sets and Zero Structure of Polynomial Sets",
  institution = "Johannes Kepler University",
  comment = "Lecture Notes",
  link = "\url{http://www-polsys.lip6.fr/~wang/papers/CharSet.ps.gz}",
  abstract =
    "This paper provides a tutorial on the theory and method of
    characteristic sets and some relevant topics. The basic algorithms as
    well as their generalization for computing the characteristic set and
    characteristic series of a set of multivariate polynomials are
    presented. The characteristic set, which is of certain triangular form,
    reflects in general the major part of zeros, and the characteristic
    series, which is a sequence of polynomial sets of triangular form,
    furnishes a complete zero decomposition of the given polynomial
    set. Using this decomposition, a complete solution to the algebraic
    decision problem and a method for decomposing any algebraic variety
    into irreducible components are described. Some applications of the
    method are indicated.",
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@book{Wang01,
  author = "Wang, Dongming",
  title = "Elimination Methods",
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    "The development of polynomial-elimination techniques from classical

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theory to modern algorithms has undergone a tortuous and rugged path. This can be observed L. van der Waerden's elimination of the "elimination theory" chapter from his classic *Modern Algebra* in later editions, A. Weil's hope to eliminate "from algebraic geometry the last traces of elimination theory," and S. Abhyankar's suggestion to "eliminate the eliminators of elimination theory." The renaissance and recognition of polynomial elimination owe much to the advent and advance of modern computing technology, based on which effective algorithms are implemented and applied to diverse problems in science and engineering. In the last decade, both theorists and practitioners have more and more realized the significance and power of elimination methods and their underlying theories. Active and extensive research has contributed a great deal of new developments on algorithms and software tools to the subject, that have been widely acknowledged. Their applications have taken place from pure and applied mathematics to geometric modeling and robotics, and to artificial neural networks. This book provides a systematic and uniform treatment of elimination algorithms that compute various zero decompositions for systems of multivariate polynomials. The central concepts are triangular sets and systems of different kinds, in terms of which the decompositions are represented. The prerequisites for the concepts and algorithms are results from basic algebra and some knowledge of algorithmic mathematics.",

keywords = "axiomref"

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@InProceedings{Wang02,
  author = "Wang, Dongming",
  title = "Epsilon: A Library of Software Tools for Polynomial Elimination",
  booktitle = "Proc. 1st Int. Congress of Mathematical Software",
  series = "ICMS 2002",
  year = "2002",
  location = "Beijing China",
  pages = "379-389",
  link = "\url{https://hal.inria.fr/inria-00107607/file/A02-R-314.pdf}",
  abstract =
    "This article presents a Maple library of functions for decomposing
    systems of multivariate polynomials into triangular systems of
    various kinds (regular, simple, or irreducible), with an application
    package for manipulating and proving geometric theorems.",
  paper = "Wang02.pdf",
  keywords = "axiomref"
}
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— ignore —

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\bibitem[Wang 92]{Wan92} Wang, Paul S. (ed)
International System Symposium on Symbolic and
Algebraic Computation 92 ACM Press, New York, NY 10036, USA, 1992
ISBN 0-89791-489-9 (soft cover), 0-89791-490-2 (hard cover),
LCCN QA76.95.I59 1992
keywords = "axiomref"
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— ignore —

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\bibitem[Watanabe 90]{WN90} Watanabe, Shunro; Nagata, Morio; (ed)
ISSAC '90 Proceedings of the
International Symposium on Symbolic and Algebraic Computation ACM Press,
New York, NY, 10036, USA. 1990 ISBN 0-89791-401-5 LCCN QA76.95.I57 1990
keywords = "axiomref"
```

— ignore —

```
\bibitem[Watt 85]{Wat85} Watt, Stephen
title = "Bounded Parallelism in Computer Algebra",
PhD Thesis, University of Waterloo
link = "\url{http://www.csd.uwo.ca/~watt/pub/reprints/1985-smw-phd.pdf}",
keywords = "axiomref"
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— axiom.bib —

```
@misc{Watt86,
author = "Watt, Stephen M. and Della Dora, J. and Wityak, Sandy (ed)",
title = "Algebra Snapshot: Linear Ordinary Differential Operators",
source = "Scratchpad II Newsletter: Vol 1 Num 2 (Jan 1986)",
link = "\url{http://www.csd.uwo.ca/~watt/pub/reprints/1986-snews-lodo.pdf}",
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paper = "Watt86.pdf",
keywords = "axiomref"
}

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— ignore —

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\bibitem[Watt 87]{Wat87} Watt, Stephen
  title = "Domains and Subdomains in Scratchpad II",
in [Wit87], pp3-5
  keywords = "axiomref"

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— ignore —

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\bibitem[Watt 87a]{WB87} Watt, Stephen M.; Burge, William H.
  title = "Mapping as First Class Objects",
in [Wit87], pp13-17
  keywords = "axiomref"

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— axiom.bib —

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@misc{Watt87,
  author = "Watt, Stephen M. and Jenks, Richard D.",
  title = "Abstract Datatypes, Multiple Views and Multiple Inheritance in
    Scratchpad II",
  year = "1987",
  link = "\url{https://cs.uwaterloo.ca/~smatt/pub/reprints/1987-itl-spadviews.pdf}",
  abstract =
    "Scratchpad II is an abstract datatype language developed at Yorktown
    Heights for the implementation of a new computer algebra system. It
    provides packages of polymorphic functions and parameterized, abstract
    datatypes with operator overloading and multiple inheritance. To
    express the intricate inter-relationships between the datatypes
    necessary for the description of mathematical objects, a number of
    techniques based on the notion of {\sl category} have been
    used. Categories are used to enforce relationships between type
    parameters and to provide the mechanism for multiple inheritance. They
    also allow the language to be statically type checked and the

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generation of efficient code. This paper describes the role of
categories in Scratchpad II.",
paper = "Watt87.pdf",
keywords = "axiomref"
}

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— axiom.bib —

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@inproceedings{Watt89,
  author = "Watt, Stephen M.",
  title = "A fixed point method for power series computation",
  booktitle = "Proc. ISSAC '88",
  series = "Lecture Notes in Computer Science 358",
  location = "Rome, Italy",
  pages = "206-217",
  isbn = "3-540-51084-2",
  year = "1988",
  abstract =
    "This paper presents a novel technique for manipulating structures
    which represents infinite power series.

```

When power series are implemented using lazy evaluation, many operations can be written as simple recursive procedures. For example, the programs to generate the series for the elementary transcendental functions are almost transliterations of the defining integral equations. However, a naive lazy algorithm provides an implementation which may be orders of magnitude slower than a method which manipulates the coefficients explicitly.

The technique described here allows a power series to be defined in a very natural but computationally inefficient way and transforms it to an equivalent, efficient form. This is achieved by using a fixed point operator on the delayed part to remove redundant calculations.

This paper describes this fixed point method and the class of problems to which it is applicable. It has been used in Scratchpad II to improve the performance of a number of operations on infinite series, including division, reversion, special functions and the solution of linear and non-linear ordinary differential equations.

A few examples are given of the method and of the speed up obtained. To illustrate, the computation of the first n terms of $\exp(x)$ for a dense, infinite series x is reduced from $O(n^4)$ to $O(n^2)$ coefficient operations, the same as required by the standard on-line algorithms.",

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keywords = "axiomref",

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beebe = "Watt:1989:FPM"
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— Watt:1989:FPM —

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@InProceedings{Watt:1989:FPM,
  author =      "S. M. Watt",
  title =      "A fixed point method for power series computation",
  crossref =    "Gianni:1989:SAC",
  pages =      "206--217",
  month =      "",
  year =       "1989",
  bibdate =    "Tue Sep 17 06:46:18 MDT 1996",
  bibsource =  "http://www.math.utah.edu/pub/tex/bib/axiom.bib",
  abstract =   "Presents a novel technique for manipulating structures
               which represent infinite power series. The technique
               described allows a power series to be defined in a very
               natural but computationally inefficient way and
               transforms it to an equivalent, efficient form. This is
               achieved by using a fixed point operator on the delayed
               part to remove redundant calculations. The paper
               describes this fixed point method and the class of
               problems to which it is applicable. It has been used in
               Scratchpad II to improve the performance of a number of
               operations on infinite series, including division,
               reversion, special functions and the solution of linear
               and non-linear ordinary differential equations. A few
               examples are given of the method and of the speed up
               obtained. To illustrate, the computation of the first n
               terms of exp(u) for a dense, infinite series u is
               reduced from O(n/sup 4/) to O(n/sup 2/) coefficient
               operations, the same as required by the standard
               on-line algorithms.",
  acknowledgement = ack-nhfb,
  affiliation = "IBM Thomas J. Watson Res. Center, Yorktown Heights,
               NY, USA",
  classification = "C4240 (Programming and algorithm theory); C7310
               (Mathematics)",
  keywords =    "Delayed part; Fixed point method; Fixed point
               operator; Infinite power series; Power series
               computation; Redundant calculations; Scratchpad II",
  language =    "English",
  thesaurus =    "Computational complexity; Mathematics computing",
}

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— axiom.bib —

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@inproceedings{Watt90,
  author = "Watt, Stephen M. and Jenks, Richard D. and Sutor, Robert S. and
    Trager, Barry M.",
  title = "The Scratchpad II type system: Domains and Subdomains",
  booktitle = "Computing Tools for Scientific Problem Solving",
  year = "1990",
  publisher = "Academic Press",
  link = "\url{https://cs.uwaterloo.ca/~smwatt/pub/reprints/1990-miola-spadtypes.pdf}",
  abstract =
    "Scratchpad II is a language developed at Yorktown Heights for the
    implementation of a new computer algebra system. The need to model the
    intricate relationships among the datatypes representing mathematical
    objects has provided a number of challenges in the design of a type
    system for the programming language.

    In languages in which a datatype constructor may take multiple
    parameters, ensuring compatibility between them is extremely
    important. Scratchpad II addresses this issue by basing its
    implementation of abstract datatypes on {\sl categories}. Categories
    provide a convenient and useful method for specifying requirements on
    operations from datatypes. These requirements can be very complex when
    modelling mathematics.

    We show how categories provide multiple inheritance and how
    inheritance of specification is separated from inheritance of
    implementation. We also present implications of the type system on
    compilation of efficient code and flexibility of a weakly typed
    interactive user interface.

    Finally, the mechanisms of Scratchpad II are compared with those of
    traditional abstract datatype and object-oriented programming
    languages.",
  paper = "Watt90.pdf",
  keywords = "axiomref"
}
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\bibitem[Watt 91]{Wat91} Watt, Stephen M. (ed)
  Proceedings of the 1991 International Symposium on
  Symbolic and Algebraic Computation, ISSAC'91, July 15-17, 1991, Bonn, Germany,
  ACM Press, New York, NY 10036, USA, 1991 ISBN 0-89791-437-6
  LCCN QA76.95.I59 1991
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keywords = "axiomref"

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\bibitem[Watt 94a]{Wat94a} Watt, Stephen M.; Dooley, S.S.; Morrison, S.C.;
Steinback, J.M.; Sutor, R.S.
title = "A\# User's Guide",
Version 1.0.0 0(\$\epsilon{}^1\$) June 8, 1994
keywords = "axiomref"

— axiom.bib —

@techreport{Watt94,
author = "Watt, Stephen M. and Broadbery, Peter A. and Dooley, Samuel S.
and Iglio, Pietro",
title = "A First Report on the A\# Compiler (including benchmarks)",
institution = "IBM Research",
year = "1994",
type = "technical report",
number = "RC19529 (85075)",
link = "\url{http://www.algor.org/docs/reports/i94acomp/i94acomp.pdf}",
abstract =
"The $A^{\#}$ compiler allows users of computer algebra to develop
programs in a context where multiple programming languages are
employed. The compiler translates programs written in the $A^{\#}$
programming language to a low-level intermediate language, Foam,
from which it can generate stand-alone programs, native object
libraries to be linked with other applications, or code to be read
into closed environments. In addition, Foam code may be directly
executed using an interpreter provided with the $A^{\#}$ compiler.

The $A^{\#}$ programming language provides support for object-oriented
and functional programming styles. It is ‘higher-order’ in the sense
that both types and functions are first class, and may be manipulated
in the same ways as any other values. The primary considerations in
the formulation of the language have been generality, composability,
and efficiency. The language has been designed to admit a number of
important optimizations, allowing compilation to machine code which is
in many instances of efficiency comparable to that produced by a C or
Fortran compiler.

The original motivation for $A^{\#}$ comes from the field of computer

```

    algebra: to provide an improved extension language for the Axiom
      computer algebra system.",
    paper = "Watt94.pdf",
    keywords = "axiomref"
}

```

— ignore —

```

\bibitem[Watt 94c]{Wat94c} Watt, Stephen M.
  title = "A\# Language Reference Version 0.35",
  IBM Research Division Technical Report RC19530 May 1994
  keywords = "axiomref"

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— ignore —

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\bibitem[Watt 95]{Wat95} Watt, S.M.; Broadbery, P.A.; Dooley, S.S.; Iglio, P.
  Steinbach, J.M.; Morrison, S.C.; Sutor, R.S.
  title = "AXIOM Library Compiler Users Guide",
  The Numerical Algorithms Group (NAG) Ltd, 1994
  keywords = "axiomref"

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— axiom.bib —

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@misc{Watt95a,
  author = "Watt, Stephen M.",
  title = "The A\# Programming Language and Compiler",
  year = "1995",
  link = "\url{ftp://ftp.inf.ethz.ch/org/cathode/workshops/march93/Watt.tex}",
  paper = "Watt95a.pdf",
  keywords = "axiomref"
}

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— ignore —

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\bibitem[Watt 01]{Wat01} Watt, Stephen M.; Broadbery, Peter A.; Iglio, Pietro;
Morrison, Scott C.; Steinbach, Jonathan M.
  title = "FOAM: A First Order Abstract Machine Version 0.35",
  IBM T. J. Watson Research Center (2001)
  paper = "Wat01.pdf"
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— axiom.bib —

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@misc{Watt00,
  author = "Watt, Stephen M.",
  title = "Aldor: The language and recent directions",
  year = "2000",
  institution = "University of Western Ontario",
  link = "\url{http://www.aldor.org/docs/reports/sa2000/aldortalk-sa2000.pdf}",
  paper = "Watt00.pdf",
  keywords = "axiomref"
}
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— axiom.bib —

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@misc{Watt00a,
  author = "Watt, Stephen M.",
  title = "Aldor: An Introduction to the Language",
  year = "2000",
  institution = "University of Western Ontario",
  link = "\url{http://www.aldor.org/docs/reports/ukqcd-2000/intro1-ukqcd00.pdf}",
  paper = "Watt00a.pdf",
  keywords = "axiomref"
}
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— axiom.bib —

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@misc{Watt00b,
  author = "Watt, Stephen M.",
  title = "Aldor: Interfaces",
  year = "2000",
  institution = "University of Western Ontario",
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link = "\url{http://www.aldor.org/docs/reports/ukqcd-2000/intro2-ukqcd00.pdf}",
paper = "Watt00b.pdf",
keywords = "axiomref"
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— axiom.bib —

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@InProceedings{Watt07,
  author = "Watt, Stephen",
  title = "What Happened to Languages for Symbolic Mathematical Computation?",
  booktitle = "Proc. Prog. Lang. for Mechanized Mathematics",
  series = "PLMMS 07",
  year = "2007",
  location = "RISC-Linz, Austria",
  pages = "81-90",
  link = "\url{http://www.csd.uwo.ca/~watt/pub/reprints/2007-plmms-what\_happened.pdf}",
  abstract =
    "While the state of the art is relatively sophisticated in programming
    language support for computer algebra, there has been less development
    in programming language support for symbolic computation over the past
    two decades. We summarize certain advances in programming languages
    for computer algebra and propose a set of directions and challenges
    for programming languages for symbolic computation.",
  paper = "Watt07.pdf",
  keywords = "axiomref"
}

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@phdthesis{Webe93b,
  author = "Weber, Andreas",
  title = "Type Systems for Computer Algebra",
  school = "University of Tübingen",
  year = "1993",
  abstract =
    "An important feature of modern computer algebra systems is the support
    of a rich type system with the possibility of type inference. Basic
    features of such a type system are polymorphism and coercion between
    types. Recently the use of order-sorted rewrite systems was proposed
    as a general framework. We will give a quite simple example of a
    family of types arising in computer algebra whose coercion relations
    cannot be captured by a finite set of first-order rewrite rules.",

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paper = "Webe93b.pdf",
keywords = "axiomref"
}

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\bibitem[Weber 92b]{Webe92b} Weber, Andreas
  title = "Structuring the Type System of a Computer Algebra System",
  link = "\url{http://cg.cs.uni-bonn.de/personal-pages/weber/publications/pdf/WeberA/Weber92a.pdf}",
  abstract = "
    Most existing computer algebra systems are pure symbol manipulating
    systems without language support for the occurring types. This is
    mainly due to the fact taht the occurring types are much more
    complicated than in traditional programming languages. In the last
    decade the study of type systems has become an active area of
    research. We will give a proposal for a type system showing that
    several problems for a type system of a symbolic computation system
    can be solved by using results of this research. We will also provide
    a variety of examples which will show some of the problems that remain
    and that will require further research.",
  paper = "Webe92b.pdf",
  keywords = "axiomref"

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@inproceedings{Webe05,
  author = "Weber, Andreas",
  title = "A Type-Coercion Problem in Computer Algebra",
  booktitle = "Artificial Intelligence and Symbolic Mathematical Computing",
  series = "Lecture Notes in Computer Science 737",
  year = "2005",
  publisher = "Springer",
  pages = "188-194",
  abstract =
    "An important feature of modern computer algebra systems is the
    support of a rich type system with the possibility of type inference.

    Basic features of such a system are polymorphism and coercion between
    types. Recently the use of order-sorted rewrite systems was proposed
    as a general framework.

    We will give a quite simple example of a family of types arising in

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    computer algebra whose coercion relations cannot be captured by a
    finite set of first-order rewrite rules.",
    paper = "Webe05.pdf"
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\bibitem[Weber 93b]{Webe93b} Weber, Andreas
  title = "Type Systems for Computer Algebra",
  link = "\url{http://cg.cs.uni-bonn.de/personal-pages/weber/publications/pdf/WeberA/Weber93b.pdf}",
  abstract = "
    We study type systems for computer algebra systems, which frequently
    correspond to the ‘‘pragmatically developed’’ typing constructs used
    in AXIOM. A central concept is that of {\sl type classes} which
    correspond to AXIOM categories. We will show that types can be
    syntactically described as terms of a regular order-sorted signature
    if no type parameters are allowed. Using results obtained for the
    functional programming language Haskell we will show that the problem
    of {\sl type inference} is decidable. This result still holds if
    higher-order functions are present and {\sl parametric polymorphism}
    is used. These additional typing constructs are useful for further
    extensions of existing computer algebra systems: These typing concepts
    can be used to implement category theoretic constructs and there are
    many well known constructive interactions between category theory and
    algebra.",
    paper = "Webe93b.pdf",
    keywords = "axiomref"

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— axiom.bib —

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@InProceedings{Webe94,
  author = "Weber, Andreas",
  title = "Algorithms for Type Inference with Coercions",
  booktitle = "Proc ISSAC 94",
  series = "ISSAC 94",
  pages = "324-329",
  year = "1994",
  abstract =
    "This paper presents algorithms that perform a type inference for a
    type system occurring in the context of computer algebra. The type
    system permits various classes of coercions between types and the
    algorithms are complete for the precisely defined system, which can be

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seen as a formal description of an important subset of the type system supported by the computer algebra program Axiom.

```

Previously only algorithms for much more restricted cases of coercions
have been described or the frameworks used have been so general that
the corresponding type inference problems were known to be
undecidable.",
paper = "Webe94.pdf",
keywords = "axiomref"
}

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@article{Webe95,
  author = "Weber, Andreas",
  title = "On coherence in computer algebra",
  journal = "J. Symb. Comput.",
  volume = "19",
  number = "1-3",
  pages = "25-38",
  year = "1995",
  link = "\url{http://cg.cs.uni-bonn.de/personal-pages/weber/publications/pdf/WeberA/Weber94e.pdf}",
  abstract = "
    Modern computer algebra systems (e.g. AXIOM) support a rich type
    system including parameterized data types and the possibility of
    implicit coercions between types. In such a type system it will be
    frequently the case that there are different ways of building
    coercions between types. An important requirement is that all
    coercions between two types coincide, a property which is called {\sl
    coherence}. We will prove a coherence theorem for a formal type system
    having several possibilities of coercions covering many important
    examples. Moreover, we will give some informal reasoning why the
    formally defined restrictions can be satisfied by an actual system.",
  paper = "Webe95.pdf",
  keywords = "axiomref",
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@InProceedings{Weber:1993:CCA,
  author = "A. Weber",

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title = "On coherence in computer algebra",
crossref = "Miola:1993:DIS",
pages = "95--106",
month = "",
year = "1993",
bibdate = "Fri Dec 29 12:46:02 MST 1995",
bibsource = "http://www.math.utah.edu/pub/tex/bib/axiom.bib",
abstract = "Modern computer algebra systems (e.g. AXIOM) support a
rich type system including parameterized data types and
the possibility of implicit coercions between types. In
such a type system it will be frequently the case that
there are different ways of building coercions between
types. An important requirement is that all coercions
between two types coincide, a property which is called
coherence. The author proves a coherence theorem for a
formal type system having several possibilities of
coercions covering many important examples. Moreover,
he gives some informal reasoning why the formally
defined restrictions can be satisfied by an actual
system.",
acknowledgement = ack-nhfb,
affiliation = "Wilhelm-Schickard-Inst. fur Inf., Tübingen Univ.,
Germany",
classification = "C4210 (Formal logic); C4240 (Programming and
algorithm theory); C7310 (Mathematics)",
keywords = "AXIOM; Coherence; Coherence theorem; Computer algebra;
Formal type system; Informal reasoning; Parameterized
data types; Type system",
language = "English",
thesaurus = "Symbol manipulation; Theorem proving; Type theory",
}

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— ignore —

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\bibitem[Weber 96]{Webe96} Weber, Andreas
title = "Computing Radical Expressions for Roots of Unity",
link = "\url{http://cg.cs.uni-bonn.de/personal-pages/weber/publications/pdf/WeberA/Weber96a.pdf}",
abstract = "
We present an improvement of an algorithm given by Gauss to compute a
radical expression for a  $p$ -th root of unity. The time complexity of
the algorithm is  $O(p^3 m^6 \log p)$ , where  $m$  is the largest prime
factor of  $p-1$ .",
paper = "Webe96.pdf",
keywords = "axiomref"

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\bibitem[Weber 99]{Webe99} Weber, Andreas
  title = "Solving Cyclotomic Polynomials by Radical Expressions",
  link = "\url{http://cg.cs.uni-bonn.de/personal-pages/weber/publications/pdf/WeberA/WeberKeckeisen99a.pdf}",
  abstract = "
    We describe a Maple package that allows the solution of cyclotomic
    polynomials by radical expressions. We provide a function that is an
    extension of the Maple {\code{sl solve}} command. The major algorithmic
    ingredient of the package is an improvement of a method due to Gauss
    which gives radical expressions for roots of unity. We will give a
    summary for computations up to degree 100, which could be done within
    a few hours of cpu time on a standard workstation.",
  paper = "Webe99.pdf",
  keywords = "axiomref"
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\bibitem[Wei-Jiang 12]{WJ12} Wei-Jiang
  title = "Top free algebra System",
  link = "\url{http://wei-jiang.com/it/software/top-free-algebra-system-by-mathematica-by-maple}",
  keywords = "axiomref"
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@misc{Westxx,
  author = "Wester, Michael J.",
  title = "Computer Algebra Synonyms",
  link = "\url{http://math.unm.edu/~wester/cas/synonyms.pdf}",
  abstract =
    "The following is a collection of synonyms for various operations in
    the seven general purpose computer algebra systems {\code{Axiom}}, {\code{Derive}},
    {\code{Macsyma}}, {\code{Maple}}, {\code{Mathematica}}, {\code{MuPAD}},
    and {\code{Reduce}}. This collection does not attempt to be
    comprehensive, but hopefully it will be useful in giving an indication
    of how to translate between the syntaxes used by the different systems
    in many common situations. Note that for a blank entry means that
    there is no exact translation of a particular operation for the
    indicated system, but it may still be possible to work around this"
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    lack with a related functionality.",
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@book{West99,
  author = "Wester, Michael J.",
  title = "Computer Algebra Systems. A practical guide",
  year = "1999",
  publisher = "Wiley",
  isbn = "0-471-98353-5",
  abstract =
    "In this book some of the most popular general purpose computer
    algebra systems (CAS), such as Mathematica, Maple, Derive, Axiom,
    MuPAD, and Macsyma, are examined. The strengths and weaknesses of
    these programs are compared and contrasted, and tutorial information
    for using these systems in various ways is given. The different
    packages are quantitatively compared using standard test suites,
    giving the possibility to asses the most appropriate for a particular
    user or application. The origins of these systems are revealed and
    many of their behaviors analyzed. This furnishes a feel for where the
    current computer algebra system state of the art stays and what can be
    expected for existing and future systems. The book is organized in
    several chapters written by different authors. Chapters 1,2, and 3 are
    organized as reviews, comparisons, and critiques of CAS
    capabilities. Then more technical issues are discussed considering
    different approaches taken by different CAS: simplifying square roots
    of square roots by denesting (chapter 4), complex number calculation
    (chapter 5), efficiently computing Chebyshev polynomials (chapter 6),
    solving single equations and systems of polynomial equations (chapters
    7, 8), computing limits (chapter 9), multiple integration (chapter
    10), solving ordinary differential equation (chapter 11), integration
    of nonlinear evolution equations (chapter 12), code generation
    (chapter 13), evaluation of expressions and programs in the embedded
    computer algebra programming language (chapter 14), and computer
    algebra in education (chapter 15). Chapter 16 covers the origin of CA,
    and, finally chapter 17 gives a list of most CAS available today.",
  keywords = "axiomref"
}

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\bibitem[Wexelblat 87]{Wex87} Wexelblat, Richard L. (ed)
Proceedings of the SIGPLAN '87 Symposium on
Interpreter and Interpretive Techniques, St. Paul, Minnesota, June 24-26, 1987
ACM Press, New York, NY 10036, USA, 1987 ISBN 0-89791-235-7
LCCN QA76.7.S54 v22:7 SIGPLAN Notices, vol 22, no 7 (July 1987)
keywords = "axiomref"
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@misc{Wikixx,
author = "Unknown",
title = "List of open-source software for mathematics",
link = "\url{https://en.wikipedia.org/wiki/List\_of\_open-source\_software\_for\_mathematics}",
keywords = "axiomref"
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@InProceedings{Will95,
author = "Williamson, Clifton J.",
title = "On the algebraic construction of tri-diagonal matrices with given
characteristic polynomial",
booktitle = "4th Conf. of Canadian Number Theory Association",
year = "1995",
location = "Halifax, Nova Scotia, Canada",
pages = "417-431",
abstract =
"Let  $K$  be a field of characteristic zero and assume
 $[f(x)=x^{n-s_1}x^{n-1}+\cdots+s_n \text{ in } L[x]]$ 
where  $L=K(s_1,\cdots,s_n)$ . Call, for the purposes of this review, an
 $n \times n$  matrix 1-tridiagonal if its entries above and below the
main diagonal are all 1, the entries on it are  $d_1,\cdots,d_n$ , and
all other entries are 0. Under which conditions can
 $d_1,d_2,\cdots,d_n$  be chosen in a radical extension of  $L$  such that
the resulting 1-tridiagonal matrix has  $f$  as its characteristic
polynomial? The authors answer: for  $n=3$  always. The  $d_i,s_i$ 
are related by a system of algebraic equations of the form
 $\overline{f}_i(d_1,d_2,d_3)=s_i$ . If this is satisfied, then by
a resultant- or Groebner basis computation with respect to the
lexicographic order,  $d_3$  (say) must be a root of a certain
```



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irreducible polynomial  $\varphi$  of degree 6. Joining this
with a Galois group computation for  $\varphi$  over  $L$ , it is shown
that a solvable and necessarily transitive  $\text{Gal}(\varphi)$  will be a
necessary and sufficient condition for expressibility of  $d_3$  (clear)
and, thanks to the form of the Groebner basis, also of  $d_1, d_2$  in
radicals. Upon a discriminant computation of  $\varphi$  and use of
results of G. Butler and J. McKay [Commun. Algebra 11,
863-911 (1983; Zbl 0518.20003)] and J. McKay and L. Soicher [J. Number
Theory 20, 273-281 (1985; Zbl 0579.12006)], this allows the author to
deduce the mentioned result. For  $n=4$ , though the author cannot get as
complete information about Galois groups etc., he can deduce e.g. that
if  $s_1, \dots, s_4$  are algebraically independent over  $\mathbb{Q}$ ,
then  $d_1, \dots, d_4$  are not expressible in radicals over
 $\mathbb{Q}(s_1, \dots, s_4)$ . He mentions, however, a solvable subcase.",
paper = "Will95.pdf",
keywords = "axiomref"
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@misc{Wink89,
  author = "Winkler, Franz",
  title = "Equational Theorem Proving and Rewrite Rule Systems",
  year = "1989",
  publisher = "Springer-Verlag",
  link = "\url{http://www.risc.jku.at/publications/download/risc_3527/paper_47.pdf}",
  abstract =
    "Equational theorem proving is interesting both from a mathematical
    and a computational point of view. Many mathematical structures like
    monoids, groups, etc. can be described by equational axioms. So the
    theory of free monoids, free groups, etc. is the equational theory
    defined by these axioms. A decision procedure for the equational
    theory is a solution for the word problem over the associated
    algebraic structure. From a computational point of view, abstract data
    types are basically described by equations. Thus, proving properties
    of an abstract data type amounts to proving theorems in the associated
    equational theory.

```

One approach to equational theorem proving consists in associating a direction with the equational axioms, thus transforming them into rewrite rules. Now in order to prove an equation $a=b$, the rewrite rules are applied to both sides, finally yielding reduced versions $a^{\{ \prime \}}$ and $b^{\{ \prime \}}$ of the left and right hand sides, respectively. If $a^{\{ \prime \}}$ and $b^{\{ \prime \}}$ agree syntactically, then the equation holds in the equational theory. However, in general this argument cannot be reversed; $a^{\{ \prime \}}$ and $b^{\{ \prime \}}$ might be different even if $a=b$ is a

theorem. The reason for this problem is that the rewrite system might not have the Church-Rosser property. So the goal is to take the original rewrite system and transform it into an equivalent one which has the desired Church-Rosser property.

We show how rewrite systems can be used for proving theorems in equational and inductive theories, and how an equational specification of a problem can be turned into a rewrite program.",

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@article{Wink93,
  author = "Winkler, Franz",
  title = "Algebraic Computation in Geometry",
  year = "1993",
  journal = "IMACS Symposium SC-1993",
  link = "\url{http://www.risc.jku.at/publications/download/risc_3777/paper_35.pdf}",
  abstract =
    "Computation with algebraic curves and surfaces are very well suited
    for being treated with computer algebra. Many aspects of computer
    algebra need to be combined for successfully solving problems in this
    area, e.g. computations with algebraic coefficients, solution of
    algebraic equations and elimination theory, and derivation of power
    series approximations of branches. We will describe the application of
    computer algebra to problems arising in algebraic geometry. The
    program system CASA, which has been developed by the author and a
    group of students will be introduced.",
  paper = "Wink93.pdf"
}
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— axiom.bib —

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@book{Wink96,
  author = "Winkler, Franz",
  title = "Polynomial Algorithms in Computer Algebra",
  year = "1996",
  publisher = "Springer-Verlag",
  isbn = "3-211-82759-5",
  abstract =
    "This book surveys algorithms and results of Computer Algebra that are
```

concerned with polynomials. It introduces algorithms from the bottom up, starting from very basic problems in computation over the integers, and finally leading to, e.g. advanced topics in factorization, solution of polynomial equations and constructive algebraic geometry. It is not based on a particular computer algebra program system.

After two introductory chapters, the book contains six chapters with the following respective topics: computation by homomorphic images, gcd computation, factorization and decomposition of polynomials, linear systems and Hankel systems, Grbner bases. The last three chapters are concerned with applications of polynomial algorithms to higher level problems in computer algebra. In particular, a decision algorithm in the elementary theory of real closed fields, a description of Gossers algorithm for solving summation problems, and an algorithm for deciding whether an algebraic curve can be parametrized by rational functions (and if so for computing such a parametrization) is given. Along the way, the complexity of many of the algorithms is investigated. Each chapter ends with rich bibliographical notes.

The book was originally developed from course material. It can easily be used as a textbook on the topic. Most subsections contain exercises. Solutions of some of the exercises are provided."

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@misc{Wity85,
  author = "Wityak, Sandra",
  title = "The Scratchpad II Newsletter",
  volume = "1",
  number = "1",
  year = "1985",
  month = "September",
  institution = "IBM Research",
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@misc{Wity86,
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author = "Wityak, Sandra",
title = "The Scratchpad II Newsletter",
volume = "1",
number = "2",
year = "1986",
month = "January",
institution = "IBM Research",
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@misc{Wity86a,
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title = "The Scratchpad II Newsletter",
volume = "1",
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month = "May",
institution = "IBM Research",
keywords = "axiomref"
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— ignore —

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\bibitem[WWW1]{WWW1}.
Software Preservation Group
link = "\url{http://www.softwarerepresentation.org/projects/LISP/common_lisp_family}",
keywords = "axiomref"

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@misc{WikiG,
author = "Wikipedia Authors",
title = "Gamma Function",
year = "2016",
link = "\url{https://en.wikipedia.org/wiki/Gamma\_function}",

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    algebra = "\newline\ref{package DFSFUN DoubleFloatSpecialFunctions}"
}

```

Y

— ignore —

```

\bibitem[Yap 00]{Yap00} Yap, Chee Keng
  title = "Fundamental Problems of Algorithmic Algebra",
  Oxford University Press (2000) ISBN0-19-512516-9
  keywords = "axiomref",

```

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\bibitem[Yapp 07]{Yapp07} Yapp, Clifford; Hebisch, Waldek; Kaminski, Kai
  title = "Literate Programming Tools Implemented in ANSI Common Lisp",
  link = "\url{http://brlcad.org/~starseeker/cl-web-v0.8.lisp.pamphlet}",
  keywords = "axiomref"

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— axiom.bib —

```

@inproceedings{Yunx76,
  author = "Yun, David Y.Y",
  title = "Algebraic Algorithms using p-adic Constructions",
  booktitle = "Proc. 1976 Symp. on Symbolic and Algebraic Computation",
  series = "SYMSAC '76",
  publisher = "ACM",
  year = "1976",
  pages = "248-259",
  link = "\url{http://lib.org/by/_djvu\_Papers/Computer\_algebra/Algebraic\%20numbers}",
  paper = "Yunx76.djvu",
  keywords = "axiomref"
}

```

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```
\bibitem[Yun 83]{Yun83} Yun, David Y.Y.
  title = "Computer Algebra and Complex Analysis",
  Computational Aspects of Complex Analysis pp379-393
  D. Reidel Publishing Company H. Werner et. al. (eds.)
  keywords = "axiomref"
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Z

— axiom.bib —

```
@mastersthesis{Zeng92,
  author = "Zenger, Christoph",
  title = "Gr{\o}bnerbasen f{\u}r Differentialformen und ihre
    Implementierung in AXIOM'",
  type = "Diplomarbeit",
  school = "Universit{\a}t Karlsruhe",
  address = "Karlsruhe, Germany",
  year = "1992",
  keywords = "axiomref",
  beebe = "Zenger:1992:GFD"
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@MastersThesis{Zenger:1992:GFD,
  author = "Ch. Zenger",
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    Implementierung in AXIOM}",
  type = "Diplomarbeit",
  school = "Universit{\a}t Karlsruhe",
  address = "Karlsruhe, Germany",
  pages = "??",
  year = "1992",
  bibsource = "/usr/local/src/bib/bibliography/Theory/Comp.Alg.bib;
    http://www.math.utah.edu/pub/tex/bib/axiom.bib",
}
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```
@article{Zeng97,
  author = "Zenger, Christoph",
  title = "Indexed types",
  journal = "Theor. Comput. Sci.",
  volume = "187",
  numbers = "1-2",
  pages = "147-165",
  year = "1997",
  abstract =
    "A new extension of the Hindley/Milner type system is proposed. The
    type system has algebraic types, that have not only type parameters
    but also value parameters (indices). This allows for example to
    parameterize matrices and vectors by their size and to check size
    compatibility statically. This is especially of interest in computer
    algebra.",
  paper = "Zeng97.pdf",
  keywords = "axiomref"
}
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@misc{Zimm95,
  author = "Zimmermann, Paul",
  title = "Wester's test suite in MuPAD 1.2.2",
  year = "1995",
  abstract =
    "A few months ago, Michael Wester made a review of the mathematical
    capabilities of different computer algebra systems, namely Axiom,
    Derive, Macsyma, Maple, Mathematica and Reduce. This review, which is
    available by anonymous ftp from math.unm.edu, file pub/cas/Paper.ps,
    consists of 131 tests in different domains of mathematics (arithmetic,
    algebraic equations, differential equations, integration, operator
    computation, series expansions, limits).

    We describe in this paper the problems that can be solved with MuPAD
    1.2.2, and how to solve them. The problems marked as [New] are solved
    using new functionalities of the version 1.2.2 with respect to 1.2.1",
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  keywords = "axiomref"
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  abstract =
    "In December 1994, Michael Wester made a review of the mathematical
    capabilities of different computer algebra systems, namely Axiom,
    Derive, Macsyma, Maple, Mathematica and Reduce. This review, which is
    available by anonymous ftp from math.unm.edu, file pub/cas/Paper.ps,
    consists of 131 tests in different domains of mathematics (arithmetic,
    algebraic equations, differential equations, integration, operator
    computation, series expansions, limits).

    We describe in this paper the problems that can be solved with MuPAD
    1.3, and how to solve them. The problems marked as [New] are solved
    using new functionalities of the version 1.3 with respect to 1.2.2",
  paper = "Zimm96.pdf",
  keywords = "axiomref"
}
```

— ignore —

```
\bibitem[Zip92]{Zip92} Zippel, Richard
  title = "Algebraic Computation",
  (unpublished) Cornell University Ithaca, NY Sept 1992
  keywords = "axiomref"
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— axiom.bib —

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@misc{Zipp89,
  author = "Zippel, Richard",
  title = "The Weyl Computer Algebra Substrate",
  year = "1989",
  paper = "Zipp89.pdf",
  link = "\url{https://ecommons.cornell.edu/bitstream/handle/1813/6917/90-1077.pdf}",
  paper = "Zipp89.pdf",
  keywords = "axiomref"
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}

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```
@misc{Zipp93,
  author = "Zippel, Richard",
  title = "The Weyl Computer Algebra Substrate",
  booktitle = "Proc. of DISCO 1993",
  series = "DISCO 93",
  pages = "303=307",
  year = "1993",
  paper = "Zipp93.pdf",
  keywords = "axiomref"
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— ignore —

```
\bibitem[Zwi92]{Zwi92} Zwillinger, Daniel
  title = "Handbook of Integration",
  Jones and Bartlett, 1992, ISBN 0-86720-293-9
  keywords = "axiomref"
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1.40 Axiom Citations of External Sources

A

— axiom.bib —

```
@article{Abla98,
  author = "Ablamowicz, Rafal",
  title = "Spinor Representations of Clifford Algebras: A Symbolic Approach",
  journal = "Computer Physics Communications",
  volume = "115",
  number = "2-3",
  month = "December",
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    year = "1998",
    pages = "510-535"
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@article{Abra06,
  author = "Abramov, Sergey A.",
  title = "In Memory of Manuel Bronstein",
  journal = "Programming and Computer Software",
  volume = "32",
  number = "1",
  pages = "56-58",
  publisher = "Pleiades Publishing Inc",
  year = "2006",
  paper = "Abra06.pdf"
}

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— ignore —

```

\bibitem[Abramowitz 64]{AS64} Abramowitz, Milton; Stegun, Irene A.
  title = "Handbook of Mathematical Functions",
  (1964) Dover Publications, NY ISBN 0-486-61272-4

```

— ignore —

```

\bibitem[Abramowitz 68]{AS68} Abramowitz M; Stegun I A
  title = "Handbook of Mathematical Functions",
  Dover Publications. (1968)

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@book{Altm05,

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author = "Altmann, Simon L.",
title = "Rotations, Quaternions, and Double Groups",
publisher = "Dover Publications, Inc.",
year = "2005",
isbn = "0-486-44518-6"
}

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\bibitem[Ames 77]{Ames77} Ames W F
title = "Nonlinear Partial Differential Equations in Engineering",
Academic Press (2nd Edition). (1977)

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— ignore —

```

\bibitem[Amos 86]{Amos86} Amos D E
title = "Algorithm 644: A Portable Package for Bessel Functions of a Complex Argument and Nonn
ACM Trans. Math. Softw. 12 265--273. (1986)

```

— ignore —

```

\bibitem[Anderson 00]{And00} Anderson, Edward
title = "Discontinuous Plane Rotations and the Symmetric Eigenvalue Problem",
LAPACK Working Note 150, University of Tennessee, UT-CS-00-454,
December 4, 2000.

```

— ignore —

```

\bibitem[Anthony 82]{ACH82} Anthony G T; Cox M G; Hayes J G
title = "DASL - Data Approximation Subroutine Library",
National Physical Laboratory. (1982)

```

— axiom.bib —

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@article{Aubr99,
  author = "Aubry, Phillippe and Lazard, Daniel and {Moreno Maza}, Marc",
  title = "On the Theories of Triangular Sets",
  year = "1999",
  pages = "105-124",
  journal = "Journal of Symbolic Computation",
  volume = "28",
  link = "\url{http://www.csd.uwo.ca/~moreno/Publications/Aubry-Lazard-MorenoMaza-1999-JSC.pdf}",
  papers = "Aubr99.pdf",
  algebra =
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  abstract =
    "Different notions of triangular sets are presented. The relationship
     between these notions are studied. The main result is that four
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     equivalent."
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@misc{Aubr96,
  author = "Aubry, Phillippe and Maza, Marc Moreno",
  title = "Triangular Sets for Solving Polynomial Systems:
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  year = "1996",
  link = "\url{http://www.lip6.fr/lip6/reports/1997/lip6.1997.009.ps.gz}",
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    "\newline\ref{category TSETCAT TriangularSetCategory}
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  abstract =
    "Four methods for solving polynomial systems by means of triangular
     sets are presented and implemented in a unified way. These methods are
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    those of Wu, Lazard, Kalkbrener, and Wang. They are compared on
    various examples with emphasis on efficiency, conciseness and
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@article{Aubr99a,
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  journal = "J. Symb. Comput.",
  volume = "28",
  number = "1-2",
  pages = "125-154",
  year = "1999",
  link = "\url{http://www.csd.uwo.ca/~moreno/Publications/Aubry-MorenoMaza-1999-JSC.pdf}",
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    \newline\ref{category NTSCAT NormalizedTriangularSetCategory}
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    "Four methods for solving polynomial systems by means of triangular
    sets are presented and implemented in a unified way. These methods are
    those of Wu, Lazard, Kalkbrener, and Wang. They are compared on
    various examples with emphasis on efficiency, conciseness and
    legibility of the outputs.",
  paper = "Aubr99a.pdf",
  keywords = "axiomref"
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@phdthesis{Aubr99b,
  author = "Aubry, Philippe",
  title = "Ensembles triangulaires de polynomes et resolution de systemes
    algebriques. Implantation en Axiom",
  school = "l'Universite de Paris VI",

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year = "1999",
month = "January",
comment = "French",
paper = "Aubr99b.pdf"
}

```

B

— ignore —

```

\bibitem[Bailey 66]{Bai66} Bailey P B
  title = "Sturm-Liouville Eigenvalues via a Phase Function",
  SIAM J. Appl. Math . 14 242--249. (1966)

```

— ignore —

```

\bibitem[Baker 96]{BGM96} Baker, George A.; Graves-Morris, Peter
  title = "Pade Approximants",
  Cambridge University Press, March 1996 ISBN 9870521450072

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```

\bibitem[Baker 10]{Ba10} Baker, Martin
  title = "3D World Simulation",
  link = "\url{http://www.euclideanspace.com}",

```

— axiom.bib —

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@misc{Bake14,
  author = "Baker, Martin",
  title = "Axiom Architecture",
  year = "2014",

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link = "\url{http://www.euclideanspace.com/prog/scratchpad/internals/ccode}",
keywords = "axiomref"
}

```

— ignore —

```

\bibitem[Banks 68]{BK68} Banks D O; Kurowski I
  title = "Computation of Eigenvalues of Singular Sturm-Liouville Systems",
  Math. Computing. 22 304--310. (1968)

```

— ignore —

```

\bibitem[Bard 74]{Bard74} Bard Y
  title = "Nonlinear Parameter Estimation",
  Academic Press. 1974

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— ignore —

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\bibitem[Barrodale 73]{BR73} Barrodale I; Roberts F D K
  title = "An Improved Algorithm for Discrete  $\ell_1$  Linear Approximation",
  SIAM J. Numer. Anal. 10 839--848. (1973)

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\bibitem[Barrodale 74]{BR74} Barrodale I; Roberts F D K
  title = "Solution of an Overdetermined System of Equations in the  $\ell_1$ -norm.",
  Comm. ACM. 17, 6 319--320. (1974)

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— axiom.bib —

```

@article{Bart72,
  author = "Barton, D.R. and Fitch, John P.",
  title = "A Review of Algebraic Manipulative Programs and their Application",
  journal = "The Computer Journal",
  volume = "15",
  number = "4",
  pages = "362-381",
  year = "1972",
  link = "\url{http://comjnl.oxfordjournals.org/content/15/4/362.full.pdf+html}",
  abstract =
    "This paper describes the applications area of computer programs that
    carry out formal algebraic manipulation. The first part of the paper
    is tutorial and several typical problems are introduced which can be
    solved using algebraic manipulative systems. Sample programs for the
    solution of these problems using several algebra systems are then
    presented. Next, two more difficult examples are used to introduce the
    reader to the true capabilities of an algebra program and these are
    proposed as a means of comparison between rival algebra systems. A
    brief review of the technical problems of algebraic manipulation is
    given in the final section.",
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@misc{Batu03,
  author = "Batut, C. and Belabas, K. and Bernardi, D. and Cohen, H. and
    Olivier, M.",
  title = "User's Guide to PARI/GP",
  link = "\url{http://math.mit.edu/~brubaker/PARI/PARIusers.pdf}",
  paper = "Batu03.pdf",
  keywords = "axiomref"
}

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\bibitem[Beauzamy 92]{Bea92} Beauzamy, Bernard
  title = "Products of polynomials and a priori estimates for coefficients in polynomial decompositions: a s
  J. Symbolic Computation (1992) 13, 463-472
  paper = "Bea92.pdf"

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\bibitem[Beauzamy 93]{Bea93} Beauzamy, Bernard; Trevisan, Vilmar;
Wang, Paul S.
  title = "Polynomial Factorization: Sharp Bounds, Efficient Algorithms",
J. Symbolic Computation (1993) 15, 393-413
  paper = "Bea93.pdf"
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@bookBeez15, author = "Judson, Tom and Beezer, Rob", title = "Abstract Algebra: Theory
and Applications", year = "2015", publisher = "Tom Judson", link = "http://abstract.ups.edu/download/aata-20150812-sage-6.8.pdf", paper = "Beez15.pdf"
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```
@article{Bert95,
  author = "Bertrand, Laurent",
  title = "Computing a hyperelliptic integral using arithmetic in the
          jacobian of the curve",
  journal = "Applicable Algebra in Engineering, Communication and Computing",
  volume = "6",
  pages = "275-298",
  year = "1995",
  abstract = "
    In this paper, we describe an efficient algorithm for computing an
    elementary antiderivative of an algebraic function defined on a
    hyperelliptic curve. Our algorithm combines B.M. Trager's integration
    algorithm and a technique for computing in the Jacobian of a
    hyperelliptic curve introduced by D.G. Cantor. Our method has been
    implemented and successfully compared to Trager's general algorithm."
}
```

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\bibitem[Berzins 87]{BBG87} Berzins M; Brankin R W; Gladwell I.
  title = "Design of the Stiff Integrators in the NAG Library",
Technical Report. TR14/87 NAG. (1987)
```

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\bibitem[Berzins 90]{Ber90} Berzins M
  title = "Developments in the NAG Library Software for Parabolic Equations",
  Scientific Software Systems. (ed J C Mason and M G Cox)
  Chapman and Hall. 59--72. (1990)
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\bibitem[Birkhoff 62]{BR62} Birkhoff, G; Rota, G C
  title = "Ordinary Differential Equations",
  Ginn \& Co., Boston and New York. (1962)
```

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  title = "Bounds for the Height of a Factor of a Polynomial in Terms of Bombieri's Norms: I. The Largest F
  J. Symbolic Computation (1993) 16, 115-130
  paper = "Boyd93a.pdf"
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\bibitem[Boyd 93b]{Boyd93b} Boyd, David W.
  title = "Bounds for the Height of a Factor of a Polynomial in Terms of Bombieri's Norms: II. The Smallest
  J. Symbolic Computation (1993) 16, 131-145
  paper = "Boyd93b.pdf"
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@inproceedings{Brad86,
  author = "Bradford, Russell J. and Hearn, Anthony C. and Padget, Julian and
  Schrufer, Eberhard",
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title = "Enlarging the REDUCE domain of computation",
booktitle = "Proc SYMSAC 1986",
series = "SYMSAC '86",
publisher = "ACM",
year = "1986",
pages = "100-106",
isbn = "0-89791-199-7",
abstract =
    "We describe the methods available in the current REDUCE system for
    introducing new mathematical domains, and illustrate these by discussing
    several new domains that significantly increase the power of the overall
    system."
}

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@article{Bram02a,
  author = "Braman, Karen and Byers, Ralph and Mathias, Roy",
  title = "The Multi-Shift QR Algorithm Part I: Maintaining Well Focused
    Shifts, and Level 3 Performance",
  journal = "SIAM Journal of Matrix Analysis",
  year = "2002",
  volume = "23",
  pages = "929-947",
  abstract =
    "This paper presents a small-bulge multishift variation of the
    multishift QR algorithm that avoids the phenomenon of shift blurring,
    which retards convergence and limits the number of simultaneous
    shifts. It replaces the large diagonal bulge in the multishift QR
    sweep with a chain of many small bulges. The small-bulge multishift QR
    sweep admits nearly any number of simultaneous shifts -- even hundreds
    -- without adverse effects on the convergence rate. With enough
    simultaneous shifts, the small-bulge multishift QR algorithm takes
    advantage of the level 3 BLAS, which is a special advantage for
    computers with advanced architectures."
}

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@article{Bram02b,
  author = "Braman, Karen and Byers, Ralph and Mathias, Roy",

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title = "The Multi-Shift QR Algorithm Part II: Aggressive Early Deflation",
journal = "SIAM Journal of Matrix Analysis",
year = "2002",
volume = "23",
pages = "948-973",
abstract =
  "Aggressive early deflation is a QR algorithm strategy that takes
  advantage of matrix perturbations outside of the subdiagonal entries
  of the Hessenberg QR iterate. It identifies and deflates converged
  eigenvalues long before the classic small-subdiagonal strategy
  would. The new deflation strategy enhances the performance of
  conventional large-bulge multishift QR algorithms, but it is
  particularly effective in combination with the small-bulge multishift
  QR algorithm. The small-bulge multishift QR sweep with aggressive
  early deflation maintains a high rate of execution of floating point
  operations while significantly reducing the number of operations
  required."
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  title = "Multiple-Precision Zero-Finding Methods and the Complexity of Elementary Function Evaluation, And",
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  title = "The Fast Fourier Transform",

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  title = "Note on Irreducibility Testing",
  Mathematics of Computation, vol. 35, num. 35, Oct. 1980, 1379-1381
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@misc{Bron95,
  author = "Bronstein, Manuel",
  title = "On radical solutions of linear ordinary differential equations",
  year = "1995",
  link = "\url{ftp://ftp.inf.ethz.ch/org/cathode/workshops/jan95/abstracts/bronstein.ps}",
  algebra =
    "\newline\refto{category OREPCAT UnivariateSkewPolynomialCategory}
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    \newline\refto{domain OREUP UnivariateSkewPolynomial}
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    \newline\refto{domain LODO2 LinearOrdinaryDifferentialOperator2}
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    \newline\refto{package LODOF LinearOrdinaryDifferentialOperatorFactorizer}
    \newline\refto{package LODOOPS LinearOrdinaryDifferentialOperatorsOps}",
  paper = "Bron95.pdf"
}
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\bibitem[Bronstein 98a]{Bro98a} Bronstein, M.; Grabmeier, J.; Weispfenning, V. (eds)
  title = "Symbolic Rewriting Techniques",
  Progress in Computer Science and Applied Logic 15, Birkhauser-Verlag, Basel
  ISBN 3-7643-5901-3 (1998)
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@article{Bron90,
  author = "Bronstein, Manuel",
  title = "The Transcendental Risch Differential Equation",
  journal = "J. Symbolic Computation",
  volume = "9",
  pages = "49-60",
  year = "1990",
  comment = "IBM Research Report RC13460 IBM Corp. Yorktown Heights, NY",
  algebra =
    "\newline\refto{package RDETR TranscendentalRischDE}
    \newline\refto{package RDETRS TranscendentalRischDESystem}",
  abstract =
    "We present a new rational algorithm for solving Risch differential
    equations in towers of transcendental elementary extensions. In
    contrast to a recent algorithm by Davenport we do not require a
    progressive reduction of the denominators involved, but use weak
    normality to obtain a formula for the denominator of a possible
    solution. Implementation timings show this approach to be faster than
    a Hermite-like reduction.",
  paper = "Bron90.pdf",
  keywords = "axiomref"
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@techreport{Bron98,
  author = "Bronstein, Manuel",
  title = "The lazy hermite reduction",
  type = "Rapport de Recherche",
  number = "RR-3562",
```

```

year = "1998",
institution = "French Institute for Research in Computer Science",
abstract = "
    The Hermite reduction is a symbolic integration technique that reduces
    algebraic functions to integrands having only simple affine
    poles. While it is very effective in the case of simple radical
    extensions, its use in more general algebraic extensions requires the
    precomputation of an integral basis, which makes the reduction
    impractical for either multiple algebraic extensions or complicated
    ground fields. In this paper, we show that the Hermite reduction can
    be performed without {\sl a priori} computation of either a primitive
    element or integral basis, computing the smallest order necessary for
    a particular integrand along the way.",
paper = "Bron98.pdf"
}

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@misc{Bro98b,
  author = "Bronstein, Manuel",
  title = "Symbolic Integration Tutorial",
  series = "ISSAC'98",
  year = "1998",
  address = "INRIA Sophia Antipolis",
  link = "\url{http://www-sop.inria.fr/cafe/Manuel.Bronstein/publications/issac98.pdf}",
  paper = "Bro98b.pdf"
}

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@article{Burg74,
  author = "William H. Burge",
  title = "Stream Processing Functions",
  year = "1974",
  month = "January",
  journal = "IBM Journal of Research and Development",
  volume = "19",
  issue = "1",
  pages = "12-25",
  papers = "Burg74.pdf",
  abstract = "
    One principle of structured programming is that a program should be

```

separated into meaningful independent subprograms, which are then combined so that the relation of the parts to the whole can be clearly established. This paper describes several alternative ways to compose programs. The main method used is to permit the programmer to denote by an expression the sequence of values taken on by a variable. The sequence is represented by a function called a stream, which is a functional analog of a coroutine. The conventional while and for loops of structured programming may be composed by a technique of stream processing (analogous to list processing), which results in more structured programs than the originals. This technique makes it possible to structure a program in a natural way into its logically separate parts, which can then be considered independently."

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@article{Cant87,
  author = "Cantor, D.",
  title = "Computing in the Jacobian of a HyperellipticCurve",
  journal = "Mathematics of Computation",
  volume = "48",
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\bibitem[Carlson 77a]{Car77a} Carlson B C
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  title = "Exercices de Mathématiques Quatrieme Année. De Bure Frères",
  Paris 1829 (reprinted Oeuvres, II Série, Tome IX,
  Gauthier-Villars, Paris, 1891).
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@misc{Chew95,
  author = "Chew, Paul and Constable, Robert L. and Pingali, Keshav and
           Vavasis, Steve and Zippel, Richard",
  title = "Collaborative Mathematics Environment",
  link = "\url{http://www.cs.cornell.edu/rz/MathBus95/TechSummary.html}",
  keywords = "axiomref"
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\bibitem[Ch\`eze 07]{Chez07} Ch\`eze, Guillaume; Lecerf, Gr\`egoire
  title = "Lifting and recombination techniques for absolute factorization",
  Journal of Complexity, Vol 23 Issue 3 June 2007 pp 380-420
  link = "\url{http://www.sciencedirect.com/science/article/pii/S0885064X07000465}",
  abstract = "
    In the vein of recent algorithmic advances in polynomial factorization
    based on lifting and recombination techniques, we present new faster
    algorithms for computing the absolute factorization of a bivariate
    polynomial. The running time of our probabilistic algorithm is less
    than quadratic in the dense size of the polynomial to be factored.",
  paper = "Chez07.pdf"
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title = "On Quaternions and Octonions",
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  The State of the Art in Numerical Analysis. (ed D A H Jacobs)
  Academic Press. 627--668. (1977)
  keywords = "survey",
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  title = "On the Estimation of Sparse Jacobian Matrices",
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\bibitem[Davenport]{Dav} Davenport, James  
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  link = "\url{http://www.nag.co.uk/doc/TechRep/axiomtr.html}",
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\bibitem[DeBoor 72]{DeB72} De Boor C  
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  title = "Numerical Methods for Unconstrained Optimixation and Nonlinear Equations",  
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  title = "An Algorithm for Smoothing, Differentiating and Integration of Experimental Data Using Spline Functions",  
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Hanson R J
  title = "A Proposal for an Extended set of Fortran Basic Linear Algebra Subprograms",
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Hammarling, Sven; Hanson, Richard J.
  title = "An Extended Set of FORTRAN Basic Linear Algebra Subroutines",
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\bibitem[Dongarra 88a]{REF-DON88a} Dongarra, Jack J.; Du Croz, Jeremy;
Hammarling, Sven; Hanson, Richard J.
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    title = "ALGORITHM 656: An Extended Set of Basic Linear Algebra Subprograms: Model Implementation and Test",
    journal = "ACM Transactions on Mathematical Software", Vol 14, No 1, March 1988,
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\bibitem[Dongarra 90]{REF-DON90} Dongarra, Jack J.; Du Croz, Jeremy;
Hammarling, Sven; Duff, Iain S.
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    title = "A Set of Level 3 Basic Linear Algebra Subprograms",
    journal = "ACM Transactions on Mathematical Software", Vol 16, No 1, March 1990,
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\bibitem[Dongarra 90a]{REF-DON90a} Dongarra, Jack J.; Du Croz, Jeremy;
Hammarling, Sven; Duff, Iain S.
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    title = "ALGORITHM 679: A Set of Level 3 Basic Linear Algebra Subprograms: Model Implementation and Test",
    journal = "ACM Transactions on Mathematical Software", Vol 16, No 1, March 1990,
    pp 18-28
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@misc{blas01,
  author = "Dongarra, Jack and et al.",
  title = "Basic Linear Algebra Subprograms Technical (BLAST) Forum Standard",
  year = "2001",
  link = "\url{http://www.netlib.org/blas/blast-forum/blas-report.pdf}",
  paper = "blas01.pdf"
}
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\bibitem[Ducos 00]{Duc00} Ducos, Lionel
  title = "Optimizations of the subresultant algorithm",
  Journal of Pure and Applied Algebra V145 No 2 Jan 2000 pp149-163
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  title = "MA28 -- a set of Fortran subroutines for sparse unsymmetric linear equations",
  A.E.R.E. Report R.8730. HMSO. (1977)
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@article{Duva96a,
  author = {Duval, Dominique and Gonz\'alez-Vega, L.},
  title = "Dynamic Evaluation and Real Closure",
  journal = "Mathematics and Computers in Simulation",
  volume = "42",
  pages = "551-560",
  year = "1996",
  abstract = "
    The aim of this paper is to present how the dynamic evaluation method
    can be used to deal with the real closure of an ordered field. Two
    kinds of questions, or tests, may be asked in an ordered field:
    equality tests  $(a=b?)$  and sign tests  $(a > b?)$ . Equality tests are
    handled through splittings, exactly as in the algebraic closure of a
    field. Sign tests are handled through a structure called ‘‘Tarski data
    type’’.",
  paper = "Duva96a.pdf"
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@article{Duva96,
  author = "Duval, D. and Reynaud, Jean-Claude",
  title = "Sketches and Computations over Fields",
  journal = "Mathematics and Computers in Simulation",
  volume = "42",
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pages = "363-373",
year = "1996",
abstract =
  "The goal of this short paper is to describe one possible use of
  sketches in computer algebra. We show that sketches are a powerful
  tool for the description of mathematical structures and for the
  description of computations.",
paper = "Duva96.pdf"
}

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@article{Duva94a,
  author = "Duval, D. and Reynaud, J.C.",
  title = "Sketches and Computation (Part I):
          Basic Definitions and Static Evaluation",
  journal = "Mathematical Structures in Computer Science",
  volume = "4",
  pages = "185-238",
  publisher = "Cambridge University Press",
  year = "1994",
  link = "\url{http://journals.cambridge.org/abstract_S0960129500000438}",
  abstract =
    "We define a categorical framework, based on the notion of {\sl
    sketch}, for specification and evaluation in the sense of algebraic
    specifications and algebraic programming. This framework goes far
    beyond our initial motivations, which was to specify computation with
    algebraic numbers. We begin by redefining sketches in order to deal
    explicitly with programs. Expressions and terms are carefully defined
    and studied, then {\sl quasi-projective sketches} are introduced. We
    describe {\sl static evaluation} in these sketches: we propose a
    rigorous basis for evaluation in the corresponding structures. These
    structures admit an initial model, but are not necessarily
    equational. In Part II (Duval and Reynaud 1994), we study a more
    general process, called {\sl dynamic evaluation}, for structures that
    may have no initial model.",
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}

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@article{Duva94b,

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author = "Duval, Dominique and Reynaud, Jean-Claude",
title = "Sketches and Computation (Part II):
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journal = "Mathematical Structures in Computer Science",
volume = "4",
pages = "239-271",
publisher = "Cambridge University Press",
year = "1994",
link = "\url{http://journals.cambridge.org/abstract_S096012950000044X}",
abstract =
    "In the first part of this paper (Duval and Reynaud 1994), we defined a
    categorical framework, based on the notion of {\sl sketch}, for
    specification and evaluation in the senses of algebraic specification
    and algebraic programming. {\sl Static evaluation} in {\sl
    quasi-projective sketches} was defined in Part I; in this paper, {\sl
    dynamic evaluation} is introduced. It deals with more general
    structures, which may have no initial model. Until now, this process
    has not been used in algebraic specification systems, but computer
    algebra systems are beginning to use it as a basic tool. Finally, we
    give some applications of dynamic evaluation to computation in field
    extensions.",
paper = "Duva94b.pdf"
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@article{Duva94c,
  author = "Duval, Dominique",
  title = "Algebraic Numbers: An Example of Dynamic Evaluation",
  journal = "J. Symbolic Computation",
  volume = "18",
  pages = "429-445",
  year = "1994",
  link = "\url{http://www.sciencedirect.com/science/article/pii/S0747717106000551}",
  abstract = "
    Dynamic evaluation is presented through examples: computations
    involving algebraic numbers, automatic case discussion according to
    the characteristic of a field. Implementation questions are addressed
    too. Finally, branches are presented as ‘‘dual’’ to binary functions,
    according to the approach of sketch theory.",
  paper = "Duva94c.pdf",
  keywords = "axiomref"
}

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@misc{Fate08,
  author = "Fateman, Richard J.",
  title = "Revisiting numeric/symbolic indefinite integration of rational
    functions, and extensions",
  link = "\url{http://www.eecs.berkeley.edu/~fateman/papers/integ.pdf}",
  abstract = "
    We know we can solve this problem: Given any rational function
     $f(x)=p(x)/q(x)$ , where  $p$  and  $q$  are univariate polynomials over
    the rationals, compute its  $\int$  integral, using if
    necessary, algebraic numbers. But in many circumstances an approximate
    result is more likely to be of use. Furthermore, it is plausible that
    it would be more useful to solve the problem to allow definite
    integration, or introduce additional parameters so that we can solve
    multiple definite integrations. How can a computer algebra system
    best answer the more useful questions? Finally, what if the integrand
    is not a ratio of polynomials, but something more challenging?",
  paper = "Fate08.pdf"
}

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@misc{Fitc74,
  author = "Fitch, J.P.",
  title = "CAMAL Users Manual",
  institution = "University of Cambridge Computer Laboratory",
  year = "1974"
}

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@misc{Flet01,
  author = "Fletcher, John P.",
  title = "Symbolic processing of Clifford Numbers in C++",
  year = "2001",
  journal = "Paper 25, AGACSE 2001."
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@misc{Flet09,
  author = "Fletcher, John P.",
  title = "Clifford Numbers and their inverses calculated using the matrix
    representation",
  publisher = "Chemical Engineering and Applied Chemistry, School of
    Engineering and Applied Science, Aston University, Aston Triangle,
    Birmingham B4 7 ET, U. K.",
  link = "\url{http://www.ceac.aston.ac.uk/research/staff/jpf/papers/paper24/index.php}"
}
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  title = "Practical Methods of Optimization",
  Vol 2. Constrained Optimization. Wiley. (1981)
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@article{Floy63,
  author = "Floyd, R. W.",
  title = "Semantic Analysis and Operator Precedence",
  journal = "JACM",
  volume = "10",
  number = "3",
  pages = "316-333",
  year = "1963"
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\bibitem[Forsythe 57]{For57} Forsythe G E,
  title = "Generation and use of orthogonal polynomials for data fitting with a digital computer
  J. Soc. Indust. Appl. Math. 5 74--88. (1957)
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\bibitem[Fortenbacher 90]{REF-For90} Fortenbacher, A.
  "Efficient type inference and coercion in computer algebra"
  Design and Implementation of Symbolic Computation Systems (DISCO 90)
  A. Miola, (ed) vol 429 of Lecture Notes in Computer Science
  Springer-Verlag, pp56-60
  abstract = "
    Computer algebra systems of the new generation, like Scratchpad, are
    characterized by a very rich type concept, which models the
    relationship between mathematical domains of computation. To use these
    systems interactively, however, the user should be freed of type
    information. A type inference mechanism determines the appropriate
    function to call. All known models which allow to define a semantics
    for type inference cannot express the rich "mathematical" type
    structure, so presently type inference is done heuristically. The
    following paper defines a semantics for a subproblem thereof, namely
    coercion, which is based on rewrite rules. From this definition, and
    efficient coercion algorithm for Scratchpad is constructed using graph
    techniques."
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  title = "Chebyshev Polynomials in Numerical Analysis",
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  title = "Smooth Interpolation of Large Sets of Scattered Data",
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\bibitem[Fritsch 82]{Fri82} Fritsch F N
  title = "PCHIP Final Specifications",
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  title = "A Method for Constructing Local Monotone Piecewise Cubic Interpolants",
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  title = "Introduction to Numerical Analysis",
  Addison-Wesley. 181--187. (1965)
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@misc{Fult08,
  author = "Fulton, William",
  title = "Algebraic Curves: An Introduction to Algebraic Geometry",
  link = "\url{http://www.math.lsa.umich.edu/~wfulton/CurveBook.pdf}",
  year = "2008",
  algebra =
    "\newline\ref{package GPAFF GeneralPackageForAlgebraicFunctionField}
\newline\ref{package PAFFFF PackageForAlgebraicFunctionFieldOverFiniteField}
\newline\ref{package PAFF PackageForAlgebraicFunctionField}",
  paper = "Fult08.pdf"
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\bibitem[Garcia 95]{Ga95} Garcia, A.; Stichtenoth, H.
  title = "A tower of Artin-Schreier extensions of function fields attaining the Drinfeld-Vladut bound",
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\bibitem[Gathen 90a]{Gat90a} {von zur Gathen}, Joachim; Giesbrecht, Mark
  "Constructing Normal Bases in Finite Fields"
  J. Symbolic Computation pp 547-570 (1990)
  abstract = "
    An efficient probabilistic algorithm to find a normal basis in a
    finite field is presented. It can, in fact, find an element of
    arbitrary prescribed additive order. It is based on a density estimate
    for normal elements. A similar estimate yields a probabilistic
    polynomial-time reduction from finding primitive normal elements to
    finding primitive elements.",
  paper = "Gat90a.pdf"
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\bibitem[Gathen 90]{Gat90} von zur Gathen, Joachim
  title = "Functional Decomposition Polynomials: the Tame Case",
  Journal of Symbolic Computation (1990) 9, 281-299
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@book{Gath99,
  author = {{von zur Gathen}, Joachim and Gerhard, J\"urgen},
  title = "Modern Computer Algebra",
  publisher = "Cambridge University Press",
  year = "1999",
  isbn = "0-521-64176-4",
  algebra = "\newline\refto{package PGCD PolynomialGcdPackage}"
}
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  title = "A Computational Procedure for Incomplete Gamma Functions",  
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  title = "An Error Analysis of Goertzel's (Watt's) Method for Computing Fourier Coefficients",  
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  title = "Algorithm AS 75. Basic Procedures for Large Sparse or Weighted Linear Least-squares P  
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title = "An Algorithm for the Integration of Unequally Spaced Data",
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title = "Minimization subject to bounds on the variables",
Report NAC 72. National Physical Laboratory. (1976)

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  title = "Algorithms for the Solution of the Nonlinear Least-squares Problem",  
  SIAM J. Numer. Anal. 15 977--992. (1978)
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  title = "Conjugate-gradient Methods for Large-scale Nonlinear Optimization",  
  Technical Report SOL 79-15. Department of Operations Research,  
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  title = "Practical Optimization",  
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  title = "User's Guide for SOL/QPSOL Version 3.2",
  Report SOL 84-5. Department of Operations Research, Stanford University. 1984
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  title = "Procedures for Optimization Problems with a Mixture of Bounds and General Linear Constraints",
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Saunders M A.; Wright M H.
  title = "User's Guide for LSSOL (Version 1.0)",
  Report SOL 86-1. Department of Operations Research, Stanford University. 1986
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  title = "Some Theoretical Properties of an Augmented Lagrangian Merit Function",
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  title = "Computational Techniques for Ordinary Differential Equations",  
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  title = "Vectorisation of one dimensional quadrature codes",  
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  title = "The NAG Library Boundary Value Codes",  
  Numerical Analysis Report. 134 Manchester University. (1987)
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  title = "The consistency of the continuum hypothesis",  
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  title = "Integrals of multinomial systems of ordinary differential equations",  
  J. of Pure and Applied Algebra, 45, 225-240 (1987)  
  link = "\url{http://www.sciencedirect.com/science/article/pii/0022404987900727/pdf?md5=5a0c706...}  
  paper = "Gold87.pdf"
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@article{Goll90,
  author = "H. Gollan and Grabmeier, Johannes",
  title = "Algorithms in Representation Theory and their Realization
          in the Computer Algebra System Scratchpad",
  journal = "Bayreuther Mathematische Schriften",
  volume = "33",
  year = "1990",
  pages = "1-23",
  keywords = "axiomref"
}
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@misc{Grab91a,
  author = "Grabmeier, Johannes",
  title = "Groups, finite fields and algebras, constructions and calculations",
  location = "IBM Europe Institute",
  year = "1991"
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\bibitem[Grabmeier]{Grab} Grabmeier, J.
  title = "On Plesken's root finding algorithm",
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\bibitem[Grabmeier 87]{GK87} Grabmeier, J.; Kerber, A.;
  title = "The Evaluation of Irreducible Polynomial Representations of the General Linear Groups and of the
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\bibitem[Grabmeier 92]{REF-GS92} Grabmeier, J.; Scheerhorn, A.
  title = "Finite fields in Axiom",
  AXIOM Technical Report TR7/92 (ATR/5)(NP2522),
  Numerical Algorithms Group, Inc., Downer's
  Grove, IL, USA and Oxford, UK, 1992
  link = "\url{http://www.nag.co.uk/doc/TechRep/axiomtr.html}",
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  title = "Elements of the Differential and Integral Calculus",
  link = "\url{http://djm.cc/library/Elements_Differential_Integral_Calculus_Granville_edited_2.}
  paper = "Gran1911.pdf"
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\bibitem[Gruntz 93]{Gru93} Gruntz, Dominik
  title = "Limit computation in computer algebra",
  link = "\url{http://algo.inria.fr/seminars/sem92-93/gruntz.pdf}",
  abstract = "
    The automatic computation of limits can be reduced to two main
    sub-problems. The first one is asymptotic comparison where one must
    decide automatically which one of two functions in a specified class
    dominates the other one asymptotically. The second one is asymptotic
    cancellation and is usually exemplified by
    \[e^x[\exp(1/x+e^{-x})-\exp(1/x)],\quad x \rightarrow \infty\]

    In this example, if the sum is expanded in powers of  $1/x$ , the
    expansion always yields  $O(x^{-k})$ , and this is not enough to
    conclude.

    In 1990, J.Shackell found an algorithm that solved both these problems
    for the case of  $\exp\text{-}\log$  functions, i.e. functions build by recursive
    application of exponential, logarithm, algebraic extension and field
    operations to one variable and the rational numbers. D. Gruntz and
    G. Gonnet propose a slightly different algorithm for  $\exp\text{-}\log$ 
    functions. Extensions to larger classes of functions are also
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discussed.",
 paper = "Gru93.pdf"

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@article{Hach95,
  author = "Hach\`e, G. and Le Brigand, D.",
  title = "Effective construction of algebraic geometry codes",
  journal = "IEEE Transaction on Information Theory",
  volume = "41",
  number = "6",
  month = "November",
  year = "1995",
  pages = "1615--1628",
  link = "\url{https://hal.inria.fr/inria-00074404/file/RR-2267.pdf}",
  algebra =
    "\newline\ref{package GPAFF GeneralPackageForAlgebraicFunctionField}
\newline\ref{package PAFFFF PackageForAlgebraicFunctionFieldOverFiniteField}
\newline\ref{package PAFF PackageForAlgebraicFunctionField}",
  abstract =
    "We intend to show that algebraic geometry codes (AG-codes, introduced
    by Goppa in 1977 [5]) can be constructed easily using blowing-up
    theory. This work is based on a paper by Le Brigand and Risler. Given
    a plane curve, we desingularize the curve by means of blowing-up, and
    then using the desingularisation trees and the monoidal
    transformations associated to the blowing-up morphisms, we compute the
    adjoint divisor of the curve. Finally we show how to use the algorithm
    of Brill-Noether to compute a basis of the vector space associated to
    a divisor of the curve. Two examples of constructions of AG-codes are
    given at the end.",
  paper = "Hach95.pdf",
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@article{Hach95a,
  author = "Hach\`e, G.",
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title = "Computation in algebraic function fields for effective
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journal = "Lecture Notes in Computer Science",
volume = "948",
year = "1995",
pages = "262--278"
}

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@phdthesis{Hach96,
  author = "Hach\`e, G.",
  title = "Construction effective des codes g\`eom\`etriques",
  school = "l'Universit\`e Pierre et Marie Curie (Paris 6)",
  year = "1996",
  month = "Septembre"
}

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\bibitem[Hall 76]{HW76} Hall G.; Watt J M. (eds),
  title = "Modern Numerical Methods for Ordinary Differential Equations",
  Clarendon Press. (1976)

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\bibitem[Hamdy 04]{Ham04} Hamdy, S.
  title = "LiDIA A library for computational number theory",
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  link = "\url{http://www.cdc.informatik.tu-darmstadt.de/TI/LiDIA}",

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\bibitem[Hammersley 67]{HH67} Hammersley J M; Handscomb D C.
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Methuen. (1967)
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@misc{Hath1896,
  author = "Hathway, Arthur S.",
  title = "A Primer Of Quaternions",
  year = "1896"
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@book{Haya05,
  author = "Hayashi, K. and Kangkook, J. and Lascu, O. and Pienaar, H. and
    Schreitmueeller, S. and Tarquinio, T. and Thompson, J.",
  title = "AIX 5L Practical Performance Tools and Tuning Guide",
  publisher = "IBM",
  year = "2005",
  link = "\url{http://www.redbooks.ibm.com/redbooks/pdfs/sg246478.pdf}",
  paper = "Haya05.pdf"
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\bibitem[Hayes 70]{Hay70} Hayes J G.
  title = "Curve Fitting by Polynomials in One Variable",
Numerical Approximation to Functions and Data.
(ed J G Hayes) Athlone Press, London. (1970)
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\bibitem[Hayes 74]{Hay74} Hayes J G.
  title = "Numerical Methods for Curve and Surface Fitting",
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  title = "The Least-squares Fitting of Cubic Spline Surfaces to General Data Sets",
  J. Inst. Math. Appl. 14 89--103. (1974)
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@misc{Hear87,
  author = "Hearn, Anthony",
  title = "REDUCE User's Manual",
  version = "3.3",
  institution = "Rand Corporation",
  year = "1987"
}
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@misc{Hebi10,
  author = "Hebisch, Waldemar and Rubey, Martin",
  title = "Extended Rate, more GFUN",
  year = "2010",
  link = "\url{https://arxiv.org/abs/math/0702086v2}",
  algebra =
    "\newline\ref{to}{package GUESS Guess}
    \newline\ref{to}{package GUESSAN GuessAlgebraicNumber}
    \newline\ref{to}{package GUESSF GuessFinite}
    \newline\ref{to}{package GUESSF1 GuessFiniteFunctions}
    \newline\ref{to}{package GUESSINT GuessInteger}
    \newline\ref{to}{package GUESSP GuessPolynomial}
    \newline\ref{to}{package GUESSUP GuessUnivariatePolynomial}",
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    "We present a software package that guesses formulae for sequences of,
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    for example, rational numbers or rational functions, given the first
    few terms. We implement an algorithm due to Bernhard Beckermann and
    George Labahn, together with some enhancements to render our package
    efficient. Thus we extend and complement Christian Krattenthaler's
    program Rate, the parts concerned with guessing of Bruno Salvy and
    Paul Zimmermann's GFUN, the univariate case of Manuel Kauers' Guess.m
    and Manuel Kauers' and Christoph Koutschan's qGeneratingFunctions.m.",
    paper = "Heb110.pdf"
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\bibitem[Henrici 56]{Hen56} Henrici, Peter
    title = "Automatic Computations with Power Series",
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    January 1956, 10-15

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@book{Hous81,
    author = "Householder, Alston S.",

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title = "Principles of Numerical Analysis",
publisher = "Dover Publications, Mineola, NY",
year = "1981",
isbn = "0-486-45312-X"
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\bibitem[Huang 96]{HI96} Huang, M.D.; Ierardi, D.
  title = "Efficient algorithms for Riemann-Roch problem and for addition in the jacobian of a c
  Proceedings 32nd Annual Symposium on Foundations of Computer Sciences.
  IEEE Comput. Soc. Press, pp. 678--687.

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@misc{Hubb,
  author = "Hubbard, John H. and Lundell, Benjamin",
  title = "A First Look at Differential Algebra",
  link = "\url{http://www.math.cornell.edu/~hubbard/diffalg1.pdf}",
  algebra = "\newline\refto{category DVARCAT DifferentialVariableCategory}",
  abstract =
    "The object of the paper is to prove that the differential equation
    \[u'(t)=t-[u(t)]^2\]
    has no solutions which can be written using elementary functions, or
    anti-derivatives of elementary functions, or exponentials of such
    anti-derivatives, or anti-derivative of those, etc. We should note
    that Equation 1 can be solved using power series, integrals which
    depend on a parameter, or Bessel functions of order  $1/3$ . However,
    as we will see, none of these methods of solution are ‘‘algebraic’’ in
    nature.

```

We aim to give a precise definition of ‘‘algebraic’’ by developing the theory of `\sl differential algebra`, which is largely the work of Ritt. Other contributors are Liouville, Picard, Vessiot, Kolchin, Rosenlicht, ... The part of differential Galois theory which leads to a proof of Abel’s celebrated result that a general polynomial equation of degree five or higher cannot be solved by radicals. In effort to derive these two areas in parallel, we will also explain why the polynomial equation

$$x^5-4x^2-2=0$$

has no solutions which can be written as radicals of solutions to lower degree polynomial equations.",

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    paper = "Hubb.pdf"
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@InProceedings{Hube03,
  author = "Hubert, Evelyne",
  title = "Notes on Triangular Sets and Triangulation-Decomposition I:
    Polynomial Systems",
  booktitle = "Symbolic and Numerical Scientific Computing",
  series = "Lecture Notes in Computer Science 2630",
  year = "2003",
  pages = "1-39",
  link = "\url{http://www-sop.inria.fr/members/Evelyne.Hubert/publications/sncsp.pdf}",
  abstract =
    "This is the first in a series of two tutorial articles devoted to
    triangulation- decomposition algorithms. The value of these notes
    resides in the uniform presen- tation of triangulation-decomposition
    of polynomial and differential radical ideals with detailed proofs of
    all the presented results. We emphasize the study of the mathematical
    objects manipulated by the algorithms and show their properties in
    independently of those. We also detail a selection of algorithms, one
    for each task. We address here polynomial systems and some of the
    material we develop here will be used in the second part, devoted to
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@InProceedings{Hube03a,
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  booktitle = "Symbolic and Numerical Scientific Computing",
  series = "Lecture Notes in Computer Science 2630",
  year = "2003",
  pages = "40-87",
  link = "\url{http://www-sop.inria.fr/members/Evelyne.Hubert/publications/sncsd.pdf}",
  abstract =
    "This is the second in a series of two tutorial articles devoted to

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triangulation-decomposition algorithms. The value of these notes
resides in the uniform presentation of triangulation-decomposition of
polynomial and differential radical ideals with detailed proofs of all
the presented results. We emphasize the study of the mathematical
objects manipulated by the algorithms and show their properties
independently of those. We also detail a selection of algorithms, one
for each task. The present article deals with differential systems. It
uses results presented in the first article on polynomial systems but
can be read independently.",
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@misc{IBMx91,
  author = "Computer Algebra Group",
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  year = "1991"
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\bibitem[IBM]{IBM}.
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IBM Corporation, Publishing Systems Information Development,
Dept. G68, P.O. Box 1900, Boulder, Colorado, USA 80301-9191.

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@article{Ioak94,
  author = "Ioakimidis, N.I.",
  title = "Symbolic computations for the solution of inverse/design

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        problems with Maple",
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    pages = "63-68",
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@book{Iyan60,
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    isbn = "978-0262090261",
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@article{Jaco51,
    author = "Jacobson, Nathan",
    title = "General Representation Theory of Jordan Algebras",
    journal = "Trans. of the American Mathematical Society",
    volume = "70",
    number = "3",
    year = "1951",
    pages = "509-530",
    link = "\url{http://www.math.uci.edu/~brusso/jacobson1951.pdf}",
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@article{Jaco68,
  author = "Jacobson, Nathan",
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  journal = "Bull. Amer. Math. Soc",
  volume = "79",
  number = "3",
  year = "1973",
  pages = "509-514",
  link = "\url{http://projecteuclid.org/euclid.bams/1183534656}",
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  author = "Jenks, Richard D.",
  title = "A History of the SCRATCHPAD Project (1977-1986)",
  institution = "IBM Research",
  year = "1986",
  month = "May",
  type = "Scratchpad II Newsletter",
  volume = "1",
  number = "3",
}
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@miscJoyn08a, author = "Joyner, W. D.", title = "Open Source Mathematical Software: A
White Paper", link = "http://wdjoyner.com/writing/research/oscas-nsf-white-paper12.
tex", year = "2008", paper = "Joyn08a.tex", keywords = "axiomref"
@bookJuds15, author = "Judson, Tom", title = "Abstract Algebra: Theory and Applica-
tions", year = "2015", publisher = "Tom Judson", link = "http://abstract.ups.edu/
download/aata-20150812.pdf", paper = "Juds15.pdf"
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@phdthesis{Kalk91,
  author = "Kalkbrener, M.",
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  abstract =
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    radicals, unmixed and primary decompositions of ideals can be lifted
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  author = "Karpinski, Stefan",
  title = "Re: Symbolic Math: try a translation of Axiom to Julia?",

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link = "\url{https://groups.google.com/forum/#!msg/julia-dev/NTfS9fJuIcE/MINQuQuGfoUJ}",
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publisher = "Center for the Study of Language and Information, Stanford CA",
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  author = "Lamport, Leslie",
  title = "LaTeX: A Document Preparation System",
  publisher = "Addison-Wesley Publishing Company, Reading, Massachusetts",
  year = "1986",
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  author = "Lawson, C.L. and Hanson R.J. and Kincaid, D.R. and Krogh, F.T.",
  title = "Algorithm 539: Basic linear algebra subprograms for FORTRAN usage",
  journal = "ACM Transactions on Mathematical Software",
  volume = "5",
  number = "3",
  month = "September",
  year = "1979",
  pages = "308-323"
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  author = "Lawson, C. L. and Hanson, R. J. and Kincaid, D.R. and
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  title = "Basic Linear Algebra Subprograms for Fortran Usage",
  journal = "Transactions on Mathematical Software",
  volume = "5",
  number = "3",
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  month = "September",
  pages = "308-325",
  abstract =
    "A package of 38 low level subprograms for many of the basic
    operations of numerical linear algebra is presented. The package is
    intended to be used with Fortran. The operations in the package
    include dot product, elementary vector operations, Givens
    transformations, vector copy and swap, vector norm, vector scaling,
    and the determination of the index of the vector component of largest
    magnitude.",
  paper = "Laws79.pdf"
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@article{Laza91,
  author = "Lazard, Daniel",
  title = "A new method for solving algebraic systems of positive dimension",
  journal = "Discrete. Applied. Mathematics",
  volume = "33",
  year = "1991",
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  abstract =
    "A new algorithm is presented for solving algebraic systems of
    equations, which is designed from the structure which is wanted for
    the result. This algorithm is not yet implemented; thus technical
    details and proofs are omitted, for emphasising on the relation
    between the algorithm design and a good representation of the
    result. The algorithm is based on a new theorem of decomposition for
    algebraic varieties.",
  paper = "Laza91.pdf"
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@article{Laza92,
  author = "Lazard, Daniel",
  title = "Solving Zero-dimensional Algebraic Systems",
  journal = "J. of Symbolic Computation",
  volume = "13",
  pages = "117-131",
  year = "1992",
  abstract =
    "It is shown that a good output for a solver of algebraic systems of
    dimension zero consists of a family of ‘‘triangular sets of
    polynomials’’. Such an output is simple, readable, and consists
    of all information which may be wanted.

    Different algorithms are described for handling triangular systems
    and obtaining them from Groebner bases. These algorithms are
    practicable, and most of them are polynomial in the number of
    solutions",
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    paper = "Laza92.pdf"
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@article{Laza90,
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  title = "Integration of rational functions: Rational computation of the
    logarithmic part",
  journal = "Journal of Symbolic Computation",
  volume = "9",
  number = "2",
  year = "1990",
  month = "February",
  pages = "113-115",
  abstract = "
    A new formula is given for the logarithmic part of the integral of a
    rational function, one that strongly improves previous algorithms and
    does not need any computation in an algebraic extension of the field
    of constants, nor any factorisation since only polynomial arithmetic
    and GCD computations are used. This formula was independently found
    and implemented in SCRATCHPAD by B.M. Trager.",
  paper = "Laza90.pdf",
  keywords = "axiomref"
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@article{LeBr88,
  author = "Le Brigand, D.; Risler, J.J.",
  title = "Algorithme de Brill-Noether et codes de Goppa",
  journal = "Bull. Soc. Math. France",
  volume = "116",
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  pages = "231--253"
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@book{Lege11,
  author = "Legendre, George L. and Grazini, Stefano",
  title = "Pasta by Design",
  publisher = "Thames and Hudson",
  isbn = "978-0-500-51580-8",
  year = "2011"
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@misc{Leap03,
  author = "Leopardi, Paul",
  title = "A quick introduction to Clifford Algebras",
  publisher = "School of Mathematics, University of New South Wales",
  year = "2003",
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@misc{Lehn10,
  author = "Lehner, Franz",
  title = "FriCAS Tutorium",
  year = "2010",
  link = "\url{https://www.math.tugraz.at/mathc/compmath2/Demo/fricas-tutorium-0.6.pdf}",
  paper = "Lehn10.pdf"
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  title = "Algorithms for sparse matrix eigenvalue problems",
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  title = "Structured Programming: Theory and Practice",
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@techreport{Lisk79,
  author = "Liskov, Barbara and Atkinson, Russ and Bloom, Toby and
Moss, Eliot and Schaffert, Craig and Scheifler, Bob and
Snyder, Alan",
  title = "CLU Reference Manual",
  institution = "Massachusetts Institute of Technology",
  year = "1979",
  paper = "Lisk79.pdf"
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@misc{Loet09,
  author = "Loetzsch, Martin and Bleys, Joris and Wellens, Pieter",
  title = "Understanding the Dynamics of Complex Lisp Programs",
  year = "2009",
  link = "\url{http://www.martin-loetzsch.de/papers/loetzsch09understanding.pdf}",
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@misc{Loet00,
  author = "Loetzsch, M.",
  title = "GTFL - A graphical terminal for Lisp",
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\bibitem[LTU10]{LTU10}.
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@book{Luke69a,
  author = "Luke, Yudell L.",
  title = "The Special Functions and their Approximations",
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  publisher = "Academic Press",
  year = "1969",
  booktitle = "Mathematics in Science and Engineering Volume 53-I",
  algebra = "\newline\refto{package DFSFUN DoubleFloatSpecialFunctions}"
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@book{Luke69b,
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booktitle = "Mathematics in Science and Engineering Volume 53-I",
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author = "Luke, Yudell L.",
title = "Algorithms for the Computation of Mathematical Functions",
publisher = "Academic Press",
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@misc{Mars07,
  author = "Marshak, U.",
  title = "HT-AJAX - AJAX framework for Hunchentoot",
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@misc{Maxi16a,
  author = "Maxima",
  title = "Symbolic Integration: The Algorithms",
  link = "\url{http://maxima.sourceforge.net/docs/tutorial/en/gaertner-tutorial-revision/Pages/SI001.htm}"
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@article{Maza95,
  author = "Maza, Marc Moreno and Rioboo, Renaud",
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    "Some methods for polynomial system solving require efficient
    techniques for computing univariate polynomial gcd over algebraic
    extensions of a field. Currently used techniques compute {\sl generic}
    univariate polynomial gcd before {\sl specializing} the result using
    algebraic relations in the ring of coefficients. This strategy
    generates very big intermediate data and fails for many problems. We
    present here a new approach which takes permanently into account those
    algebraic relations. It is based on a property of subresultant
    remainder sequences and leads to a great increase of the speed of
    computations and thus the size of accessible systems.",
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@phdthesis{Maza97,
  author = "Maza, Marc Moreno",
  title = "Calculs de pgcd au-dessus des tours d'extensions simples et
    resolution des systemes d'equations algebriques",
  school = "Universite P.etM. Curie",
  year = "1997",
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link = "\url{http://www.csd.uwo.ca/~moreno//Publications/MorenoMaza-Thesis-1997.ps.gz}",
abstract =
  "This thesis is dedicated to polynomial system solving by means of
  triangular sets. A first prt presents two algorithms to compute
  polynomial gcds over tower of simple extensions. The first one was
  designed by Renaud Rioboo and applies to algebraic towers. The second
  one is a generalization of the previous one to the most general case
  of seperable towers. These algorithms lead to an efficient
  implementation of two methods suggested by Daniel Lazard to solve
  polynomial systems by means of triangular sets. These programs solved
  problems that were previously unreachable. The second method was only
  sketched by its author. So, a second part of this thesis presents the
  necessary developements to describe a right implementation. Moreover,
  a theorecal and unified presentation, together with an experimental
  comparison with similar methods due to Wu Wen-Tsun, Dongming Wang and
  Michael Kalkbrener were realized by Philippe Aubry and are reported in
  a third part of this document.",
paper = "Maza97.pdf",
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@techreport{Maza98,
  author = "Maza, M. Moreno",
  title = "A new algorithm for computing triangular decomposition of
    algebraic varieties",
  institution = "Numerical Algorithms Group (NAG)",
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    "\newline\ref{category TSETCAT TriangularSetCategory}
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@article{Mill68,
  author = "Millen, J. K.",
  title = "CHARYBDIS: A LISP program to display mathematical expressions
    on typewriter-like devices",
  year = "1968",
  journal = "Interactive Systems for Experimental and Applied Mathematics",
  publisher = "M. Klerer and J. Reinfelds, eds., Academic Press, New York",
  pages = "79--90",
  abstract = "
    CHARYBDIS (from CHARacter-composed sYmbolic DISplay) is a LISP program
    to display mathematical expressions on typewriter-like devices such as
    line printers, teletypes, and scopes which display lines of characters,
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    title = "Evaluation of Permanents",  
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@article{Mul97,
  author = "Mulders, Thom",
  title = "A Note on Subresultants and the Lazard/Rioboo/Trager Formula in
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  journal = "Journal of Symbolic Computation",
  year = "1997",
  volume = "24",
  number = "1",
  month = "July",
  pages = "45-50",
  abstract = "
    An ambiguity in a formula of Lazard, Rioboo and Trager, connecting
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  title = "MINOS 5.0 User's Guide",
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  title = "On the Efficiency of a Polynomial Irreducibility Test",
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@article{Naud98,
  author = "Naudin, Patrice and Quitte, Claude",
  title = "Univariate polynomial factorization over finite fields",
  journal = "Theor. Comput. Sci.",
  volume = "191",
  number = "1-2",
  pages = "1-36",
  year = "1998",
  abstract =
    "This paper is a tutorial introduction to univariate polynomial
    factorization over finite fields. The authors recall the classical
    methods that induced most factorization algorithms (Berlekamps and
    the Cantor-Zassenhaus ones) and some refinements which can be applied
    to these methods. Explicit algorithms are presented in a form suitable
    for almost immediate implementation. They give a detailed description
    of an efficient implementation of the Cantor-Zassenhaus algorithm used
    in the release 2 of the Axiom computer algebra system.",
  paper = "Naud98.pdf"
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@misc{OCAM14,
  author = "unknown",
  title = "The OCAML website",
  link = "\url{http://ocaml.org}"
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\bibitem{Ollagnier 94}{Olla94} Ollagnier, Jean Moulin
  title = "Algorithms and methods in differential algebra",
  link = "\url{http://www.lix.polytechnique.fr/~moulin/papiers/atelier.pdf}",
  paper = "Olla94.pdf"
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  title = "NIST Handbook of Mathematical Functions",
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  journal = "Bulletin de la Classe Physico-Math'\{e\}matiques de
    l'Acad'\{e\}mie Imp'\{e\}riale des Sciences de St. P'\{e\}tersbourg,",
  volume = "IV",
  pages = "145-167,286-300",
  year = "1845"
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@misc{Pagal7,
  author = "Pagani, Kurt",
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  year = "2017",
  link = "\url{https://groups.google.com/forum/#topic/fricas-devel/kteSoeR_Iyg}"
}
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@misc{Gode16,
  author = "Godek, Panicz Maciej",
  title = "A Pamphlet Against R",
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  year = "2016",
  link = "\url{http://github.com/panicz/pamphlet/raw/master/pamphlet.pdf}",
}
```

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abstract =
  "R is a programming language that has been gaining popularity in the
  domains such as machine learning and artificial intelligence over the
  recent years. It has been praised for its simplicity and elasticity,
  and its creators were even assigned a rockstar status.

  In my pamphlet, I claim that this ‘‘blazing new technology’’ is
  actually a step backwards in the development of the domains that it
  tries to tackle, and the better means of expression, or better tools
  for doing the job, were available at least since the 70s, and recently
  they were only getting better.

  This pamphlet explores and expresses various {\sl computational
  intelligence methods} using {\sl Guile}, a very pleasant
  implementation of the {\sl Scheme} programming language.",
paper = "Gode16.pdf"
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  title = "The Symmetric Eigenvalue Problem",
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  title = "Functional Documents for Computer Systems",
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@book{Paul81,
  author = "Paul, Richard",
  title = "Robot Manipulators",
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  publisher = "MIT Press",
  isbn = "0-262-16082-X"
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@book{Pear56,
  author = "Pearcey, T.",
  title = "Table of the Fresnel Integral",
  publisher = "Cambridge University Press",
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  title = "NPL Algorithms Library",
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@misc{Pfei12,
  author = "Pfeil, Greg",
  title = "Common Lisp Type Hierarchy",
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  link = "\url{http://sellout.github.io/2012/03/03/common-lisp-type-hierarchy}"
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@phdthesis{Phil02,
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author = "Philippe, M. Trebuchet",
title = "Toward a fast and numerically stable algebraic equation solving",
comment = "Vers une resolution stable et rapide des equations algebriques",
school = "l'Universite de Paris 6",
year = "2002",
month = "December",
abstract =
  "Polynomial systems can be found in many industrial applications. They
  are also in the heart of effective algebraic geometry. A fundamental
  tool for studying them is the Groebner bases. The knowledge of this
  particular base of the ideal generated by the polynomials composing the
  system allows us to compute in  $A=K[x_1,\ldots,x_n]/I$ , the quotient
  algebra, and this is necessary when we try to solve. Nevertheless,
  Groebner bases computations rely heavily on the introduction of
  monomial ordering. This introduces a certain rigidity in the
  computation and thus numerical instability. We propose a new algorithm
  that tries to remedy that problem. It generalises Groebner bases
  computation and is much less numerically instable. To do this, we
  decrease the requirement of monomial ordering, and use a new normal
  form criterion. We then give an algorithm and prove its termination
  and correctness when the input polynomial system is
  0-dimensional. After, we compare it with the previously known methods
  and show how it can be seen as a generalisation of them. Next, we
  detail how we implemented it in C++ using the Synaps library. We also
  describe the sparse matrix elimination algorithm we used in our
  program. Finally, we present some of the experiments we have done with
  our program in domains like computer vision, algorithmic geometry,
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 author = "Pratt, Vaughan R.",
 title = "Top down operator precedence",
 booktitle = "Proc. 1st annual ACM SIGACT-SIGPLAN Symposium on Principles
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series = "POPL'73",
pages = "41-51",
year = "1973",
link = "\url{http://hall.org.ua/halls/wizzard/pdf/Vaughan.Pratt.TDOP.pdf}",
paper = "Prat73.pdf",
keywords = "axiomref"
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@book{Pres07,
  author = "Press, William H. and Teukolsky, Saul A. and
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  title = "Noweb--A Simple, Extensible Tool for Literate Programming",
  link = "\url{http://www.eecs.harvard.edu/~nr/noweb}",
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@phdthesis{Rave91,
  author = "Ravenscroft, Robert Andrews Jr.",
  title = "Generating Function Algorithms for Symbolic Computation",
  school = "Brown University",
  year = "1991",
  file = "Rave91.pdf",
  abstract =
    "Various algebraic models have been used to implement symbolic
    procedures for solving summations and recurrences. Surprisingly, one
    model that has received little attention in the development of
    symbolic algorithms is generating functions. Generating function
    methods for discrete mathematics provide an ad hoc collection of
    techniques for solving summations and recurrences. In this work we
    consider their systematic application to symbolic computation.
```

To demonstrate the feasibility and capabilities of generating functions as an algebraic model of computation, we develop and implement algorithms for manipulating generating functions and the

sequences that they encode. In particular, we concentrate on algorithms for solving summations and recurrences. Techniques are developed to handle special functions and hybrid terms, which are terms involving two or more factors.

Given various classes of summations and recurrences we develop a theory to characterize the form of their solution. This allows us to demonstrate the ability of our algorithms to solve problems of these classes. This knowledge also assists us in developing algorithms that know where to look for a solution rather than doing an ad hoc search for a solution."

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@article{Redf27,
  author = "Redfield, J. Howard",
  title = "The Theory of Group-Reduced Distributions",
  journal = "American J. Math.",
  volume = "49",
  number = "3",
  year = "1927",
  pages = "433-455"
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  title = "Free Lie Algebras",
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  title = "An Inequality for Products of Polynomials",
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  title = "Crafting a Repository of Knowledge Based on Transformation",
  link = "\url{http://www.apmaths.uwo.ca/~djeffrey/Offprints/IntegrationRules.pdf}",
  abstract = "
    We describe the development of a repository of mathematical knowledge
    based on transformation rules. The specific mathematical problem is
    indefinite integration. It is important that the repository be not
    confused with a look-up table. The database of transformation rules is
    at present encoded in Mathematica, but this is only one convenient
    form of the repository, and it could be readily translated into other
    formats. The principles upon which the set of rules is compiled is
    described. One important principle is minimality. The benefits of the
    approach are illustrated with examples, and with the results of
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comparisons with other approaches.",
 paper = "Rixx.pdf"

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 title = "The identity problem for elementary functions and constants",
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  In Mathematics and Computers in Simulation Volume 42, Issue 4-6,
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@article{Ritt50,
  author = "Ritt, Joseph Fels",
  title = "Differential Algebra",
  journal = "AMS Colloquium Publications",
  year = "1950",
  volume = "33",
  isbn = "978-0-8218-4638-4",
  algebra = "\newline\refto{category DVARCAT DifferentialVariableCategory}",
  paper = "Ritt50.pdf"
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\bibitem[Rote 01]{Rote01} Rote, G\ "unter
  title = "Division-free algorithms for the determinant and the Pfaffian",
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@InProceedings{Rube06,
  author = "Rubey, Martin",
  title = "Extented rate, more GFUN",
  booktitle = "4th Colloquium on mathematics and computer science",
  series = "DMTCS",
  year = "2006",
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  pages = "431-434",
  link = "\url{http://mathinfo06.iecn.u-nancy.fr/papers/dmAG431-434.pdf}",
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"\newline\refto{package GUESS Guess}
\newline\refto{package GUESSAN GuessAlgebraicNumber}
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\newline\refto{package GUESSINT GuessInteger}
\newline\refto{package GUESSP GuessPolynomial}
\newline\refto{package GUESSUP GuessUnivariatePolynomial}",
abstract =
  "We present a software package that guesses formulas from sequences
  of, for example, rational numbers or rational functions, given the
  first few terms. Thereby we extend and complement C. Krattenthaler's
  program RATE [RATE: a Mathematics guessing machine] and the relevant
  parts of B. Salvy and P. Zimmermann's GFUN.",
paper = "Rube06.pdf"
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  title = "Simultaneous iteration method for symmetric matrices",
  Num. Math. 16 205--223. (1970)

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@book{Saun79,
  author = "MacLane, Saunders and Birkhoff, Garrett",
  title = "Algebra, Second Edition",
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@misc{Saloi6,
  author = "Salomone, Matthew",
  title = "Exploring Abstract Algebra II",
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@article{Scha61,
  author = "Schafer, R.D.",
  title = "An Introduction to Nonassociative Algebras",
  year = "1961",
  algebra = "\newline\refto{category NARNG NonAssociativeRng}",
  link = "\url{http://www.gutenberg.org/ebooks/25156}",
  journal = "Advanced Subject Matter Institute",
  abstract =
    "These are notes for my lectures in July, 1961, at the Advanced
    Subject Matter Institute in Algebra which was held at Oklahoma State
    University in the summer of 1961.

    Students at the Institute were provided with reprints of my paper,
    {\sl Structure and representation of nonassociate algebras} (Bulletin
    of the American Mathematical Society, vol. 61 (1955), pp469-484),
    together with copies of a selective bibliography of more recent papers
    on non-associative algebras. These notes supplement the 1955 Bulletin
```



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        article, bringing the statements there up to date and providing
        detailed proofs of a selected group of theorems. The proofs illustrate
        a number of important techniques used in the study of nonassociative
        algebras.",
    paper = "Scha61.pdf"
}

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@article{Scha86,
    author = "Schaffert, C. and Cooper, T.",
    title = "An Introduction to Trellis/Owl",
    journal = "SIGPLAN Notices",
    volume = "21",
    number = "11",
    publisher = "ACM",
    year = "1986",
    pages = "9-16"
}

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@book{Scha66,
    author = "Schafer, R.D.",
    title = "An Introduction to Nonassociative Algebras",
    year = "1966",
    publisher = "Academic Press, New York",
    algebra =
        "\newline\ref{category NARNG NonAssociativeRng}"
        "\newline\ref{category NASRING NonAssociativeRing}"
        "\newline\ref{domain ALGSC AlgebraGivenByStructuralConstants}"
}

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@book{Scha10,
    author = "Schafer, R.D.",

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title = "An Introduction to Nonassociative Algebras",
year = "2010",
publisher = "Benediction Classics",
algebra = "\newline\refto{category NARNG NonAssociativeRng}",
isbn = "978-1849025904",
abstract =
  "Concise study presents in a short space some of the important ideas
  and results in the theory of non-associative algebras, with particular
  emphasis on alternative and (commutative) Jordan algebras. Written as
  an introduction for graduate students and other mathematicians meeting
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@book{Segg93,
  author = "{von Seggern}, David Henry",
  title = "CRC Standard Curves and Surfaces",
  publisher = "CRC Press",
  year = "1993",
  isbn = "0-8493-0196-3"
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  title = "JET -- An Axiom Environment for Geometric Computations with Differential Equations",
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  abstract = "
    Bracket coefficients for polynomials are introduced. These are like
    specific precision floating point numbers together with error
    terms. Working in terms of bracket coefficients, an algorithm that
    computes a Gr{\\"obner} basis with floating point coefficients is
    presented, and a new criterion for determining whether a bracket
    coefficient is zero is proposed. Given a finite set  $S$  of polynomials
    with real coefficients, let  $G_\mu$  be the result of the algorithm for
     $S$  and a precision  $\mu$ , and  $G$  be a true Gr{\\"obner} basis of
     $S$ . Then, as  $\mu$  approaches infinity,  $G_\mu$  converges to  $G$ 
```

coefficientwise. Moreover, there is a precision M such that if $\mu \geq M$, then the sets of monomials with non-zero coefficients of G_μ and G are exactly the same. The practical usefulness of the algorithm is suggested by experimental results.",
 paper = "Shir96.pdf"

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@misc{Simo97,
  author = "Simon, Barry",
  title = "The PC Is Now Axiomatic",
  publisher = "PC Mag",
  year = "1997",
  month = "March",
  day = "25",
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    differential equations. These include methods for finding power series
    solutions, elementary and liouvillian solutions, first integrals, Lie
    theoretic methods, transform methods, asymptotic methods. A brief
    discussion of difference equations is also included.",
  paper = "Sing89.pdf",
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  year = "1990",
  link = "\url{http://daly.axiom-developer.org/clm.pdf}",
  publisher = "Digital Equipment Corporation",
  isbn = "1-555558-041-6",
  algebra = "\newline\refto{package DFSFUN DoubleFloatSpecialFunctions}",
  paper = "Stee90.pdf"
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  author = "Stichtenoth, H.",
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  institution = "IBM Research",
  year = "1986",
  month = "January",
  type = "Scratchpad II Newsletter",
  volume = "1",
  number = "2",
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  publisher = "C.J. Clay and Sons, Cambridge University Press Warehouse,
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author = "Thurston, William P.",

title = "On Proof and Progress in Mathematics",

journal = "Bulletin AMS",

volume = "30",

number = "2",

month = "April",

year = "1994",

link = "\url{http://www.ams.org/journals/bull/1994-30-02/S0273-0979-1994-00502-6/S0273-0979-1994-00502-6.}]",

paper = "Thur94.pdf"

}

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  title = "Chebyshev-series",
  Modern Computing Methods
  Chapter 8. NPL Notes on Applied Science (2nd Edition). 16 HMSO. 1961
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@InProceedings{Mart99,
  author = "Martin, Ursula",
  title = "Computers, reasoning and mathematical practice",
  booktitle = "Computational Logic",
  publisher = "Springer",
  year = "1999",
  location = "Marktoberdorf, Germany",
  pages = "301-346",
  link = "\url{http://citeseerx.ist.psu.edu/viewdoc/download?doi=10.1.1.50.2061}",
  abstract =
    "We identify three main objectives for computer aided reasoning
    enhancing the techniques available for mathematical experimentation,
    developing community standards for experiment and modelling and
    developing methods which will make computation more acceptable as part
    of a proof. We discuss three areas of research which address these:
    the use of theorem proving techniques to enhance or extend
    mathematical software systems, support for formal methods techniques
    to increase the reliability of such systems, and the use of computer
    aided formal reasoning in support of mathematical practice. This last
    includes activities such as formalizing systems for computational
    mathematics or visualization so that they can still be used informally
    but generate a formal development, and developing techniques to
    provide assistance in the initial stages of developing a new theory.
```

By mathematics here we mean the activities of working research mathematicians, producing new results in pure or applied mathematics, although we touch briefly on some questions concerning the applications of computational mathematics and simulation in research science and engineering. We have left out several other related areas entirely: logical questions of decidability, soundness or completeness, theoretical computer science issues of semantics, computability or complexity, foundational issues such as constructivity, computer aided proofs about software and hardware, and the use of computers in mathematical education at all levels and in heuristic discovery. In particular foundational questions about computation have transformed mathematical logic, computation has made constructive proof feasible, and effective notions of practice for proofs about hardware and software are by no means well understood. However these matters fall outside the scope of this paper.",

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    year = "2007",

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link = "\url{http://mathforum.org/library/drmath/view/71229.html}",
algebra =
  "\newline\refto{package GPAFF GeneralPackageForAlgebraicFunctionField}
\newline\refto{package PAFFFF PackageForAlgebraicFunctionFieldOverFiniteField}
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  algebra =
    "\newline\refto{package GPAFF GeneralPackageForAlgebraicFunctionField}
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  title = "Givens Rotations",
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  abstract = "
    let  $L$  be a linear differential operator with polynomial
    coefficients. We show that there is an isomorphism of differential
    operators  $D_\alpha$  and an integral transform  $H_\alpha$ 
    (called the Hermite transform) on functions for which  $D_\alpha(L)f(x)=0$ 
    implies  $L(H_\alpha(f))(x)=0$ . We
    present an algorithm that computes the Hermite transform of a rational
    function and use it to find  $n+1$  linearly independent solutions of
     $Ly=0$  when  $(D_\alpha(L)f)(x)=0$  has a rational
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  paper = "Zak02.pdf"
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@misc{Zdan14,
  author = "Zdancewic, Steve and Martin, Milo M.K.",
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link = "\url{http://www.cis.upenn.edu/~stevez/vellvm}"
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author = "Zhi, Lihong",
title = "Optimal Algorithm for Algebraic Factoring",
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link = "\url{http://www.mmrc.iss.ac.cn/~lzhi/Publications/zopfacs.pdf}",
abstract = "
  This paper presents an optimized method for factoring multivariate
  polynomials over algebraic extension fields which defined by an
  irreducible ascending set. The basic idea is to convert multivariate
  polynomials to univariate polynomials and algebraic extensions fields
  to algebraic number fields by suitable integer substitutions, then
  factorize the univariate polynomials over the algebraic number fields.
  Finally, construct multivariate factors of the original polynomial by
  Hensel lemma and TRUEFACTOR test. Some examples with timing are
  included.",
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@misc{OCon08,
author = "O'Connor, Christine",
title = "Christine Jeanne O'Connor Obituary",
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link = "\url{http://www.cargainfuneralhomes.com/home/index.cfm/obituaries/view/fh_id/10350/id/183842}"
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@misc{Hale13,
author = "Hales, Thomas C.",
title = "Mathematics in the Age of the Turing Machine",
year = "2013",
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    This article gives a survey of mathematical proofs that rely on
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@misc{Martxx,
  author = "Martin, W.A. and Fateman, R.J.",
  title = "The Macsyma System",
  link = "\url{http://groups.csail.mit.edu/mac/classes/symbolic/spring13/readings/simplification}",
  abstract = "
    MACSYMA is a system for symbolic manipulation of algebraic expressions
    which is being developed at Project MAC, M.I.T. This paper discusses
    its philosophy, goals, and current achievements.

    Drawing on the past work of Maring, Moses, and Engelman, it extends
    the capabilities of automated algebraic manipulation systems in
    several areas, including

    a) limit calculations
    b) symbolic integration
    c) solution of equations
    d) canonical simplification
    e) user-level pattern matching
    f) user-specified expression manipulation
    g) programming and bookkeeping assistance

    MACSYMA makes extensive use of the power of its rational function
    subsystem. The facilities derived from this are discussed in
    considerable detail.

    An appendix briefly notes some highlights of the overall system.",
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@misc{Andr00,

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author = "Andrews, George E. and Knopfmacher, Arnold and Paule, Peter
        and Zimmermann, Burkhard",
title = "Engel Expansions of q-Series by Computer Algebra",
year = "2000",
link = "\url{http://citeseerx.ist.psu.edu/viewdoc/summary?doi=10.1.1.32.207}",
abstract = "
    The  $q$ -Engle Expansion is an algorithm that leads to unique series
    expansions of  $q$ -series. Various examples related to classical
    partition theorems, including the Rogers-Ramanujan identities together
    with the elegant generalization found by Garrett, Ismail and Stanton,
    have been described recently. The object of this paper is to present
    the computer algebra package Engel, written in Mathematics, that has
    already played a significant role in this work. The package now is made
    freely available via the web and should help to intensify research in
    this new branch of  $q$ -series theory. Among various illustrative
    examples we present a new infinite Rogers-Ramanujan type family that
    has been discovered by using the package.",
paper = "Andr00.pdf"
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@misc{Abra99,
  author = "Abramov, Sergei A. and van Hoeij, Mark",
  title = "Integration of Solutions of Linear Functional Equations",
  year = "1999",
  link = "\url{http://www.math.fsu.edu/~hoeij/papers/itsf99/ab_final.pdf}",
  abstract = "
    We introduce the notion of the adjoint Ore ring and give a definition
    of adjoint polynomial, operator and equation. We apply this for
    integrating solutions of Ore equations.",
  paper = "Abra99.pdf"
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@misc{Bail97,
  author = "Bailey, David and Borwein, Peter and Plouffe, Simon",
  title = "On the Rapid Computation of Various Polylogarithmic Constants",
  year = "1997",

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link = "\url{http://www.ams.org/journals/mcom/1997-66-218/S0025-5718-97-00856-9/S0025-5718-97-
abstract = "
    We give algorithms for the computation of the  $d$ -th digit of certain
    transcendental numbers in various bases. These algorithms can be
    easily implemented (multiple precision arithmetic is not needed),
    require virtually no memory, and feature run times that scale nearly
    linearly with the order of the digit desired. They make it feasible to
    compute, for example, the billionth binary digit of  $\log(2)$  or  $\pi$  on
    a modest work station in a few hours run time.

    We demonstrate this technique by computing the ten billionth
    hexadecimal digit of  $\pi$ , the billionth hexadecimal digits of
 $\pi^2$ ,  $\log(2)$ , and  $\log^2(2)$ , and the ten billionth decimal digit
    of  $\log(9/10)$ .

    These calculations rest on the observation that very special types of
    identities exist for certain numbers like  $\pi$ ,  $\pi^2$ ,  $\log(2)$  and
 $\log^2$ . These are essentially polylogarithmic ladders in an integer
    base. A number of these identities that we derive in this work appear
    to be new, for example the critical identity for  $\pi$ :

    
$$\pi = \sum_{i=0}^{\infty} \frac{1}{16^i} \left( \frac{4}{8i+1} - \frac{2}{8i+4} - \frac{1}{8i+5} - \frac{1}{8i+6} \right)$$

paper = "Bail97.pdf"
}

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@misc{Thie15,
  author = "Thiery, Nicolas M.",
  title = "Open Digital Research Environment Toolkit for the Advancement of Mathematics",
  year = "2015",
  link = "\url{http://opendreamkit.org}",
  abstract =
    "OpenDreamKit will deliver a flexible toolkit enabling research groups
    to set up Virtual Research Environments, customised to meet the varied
    needs of research projects in pure mathematics and applications, and
    supporting the full research life-cycle from exploration, through
    proof and publication, to archival and sharing of data and code.

    OpenDreamKit will be built out of a sustainable ecosystem of
    community-developed open software, databases, and services,
    including popular tools such as LINBOX, MPIR, SAGE (sagemath.org),
    GAP, PARI/GP, LMFDB, and SINGULAR. We will extend the JUPYTER Notebook
    environment to provide a flexible user interface. By improving and

```

unifying existing building blocks, OpenDreamKit will maximise both sustainability and impact, with beneficiaries extending to scientific computing, physics, chemistry, biology and more, and including researchers, teachers, and industrial practitioners.

We will define a novel component-based VRE architecture and adapt existing mathematical software, databases, and user interface components to work well within it on varied platforms. Interfaces to standard HPC and grid services will be built in. Our architecture will be informed by recent research into the sociology of mathematical collaboration, so as to properly support actual research practice. The ease of set up, adaptability and global impact will be demonstrated in a variety of demonstrator VREs.

We will ourselves study the social challenges associated with large-scale open source code development and publications based on executable documents, to ensure sustainability.

OpenDreamKit will be conducted by a Europe-wide steered by demand collaboration, including leading mathematicians, computational researchers, and software developers with a long track record of delivering innovative open source software solutions for their respective communities. All produced code and tools will be open source.",

paper = "Thie15.pdf"

}

Chapter 2

Beebe Bibliography

— beebe.bib —

```
%%% -*-BibTeX-*-
%%% =====
%%% BibTeX-file{
%%%   author      = "Nelson H. F. Beebe",
%%%   version     = "1.18",
%%%   date        = "15 August 2014",
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%%%   filename    = "axiom.bib",
%%%   address     = "University of Utah
%%%                 Department of Mathematics, 110 LCB
%%%                 155 S 1400 E RM 233
%%%                 Salt Lake City, UT 84112-0090
%%%                 USA",
%%%   telephone  = "+1 801 581 5254",
%%%   FAX         = "+1 801 581 4148",
%%%   URL         = "http://www.math.utah.edu/~beebe",
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%%%                 beebe at computer.org (Internet)",
%%%   codetable   = "ISO/ASCII",
%%%   keywords    = "bibliography; AXIOM;
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%%%                 publications about the AXIOM (formerly
%%%                 known as Scratchpad) symbolic algebra
%%%                 system. It also covers publications about
%%%                 Scratchpad and a few about MathCAD.
```


NAG maintains an AXIOM code and documentation repository at http://www.nag.co.uk/symbolic/AX/Upload_Readme.html. All of the NAG technical reports listed in this bibliography can be found at that address.

```

%%%
%%%      In 2006, AXIOM became free software, with a
%%%      Web site at
%%%
%%%      http://www.axiom-developer.org
%%%
%%%      This bibliography has been collected from the
%%%      author's personal bibliography files, from
%%%      the very large computer science bibliography
%%%      collection on ftp.ira.uka.de in
%%%      /pub/bibliography to which many people of
%%%      have contributed, and from several
%%%      Internet-accessible library catalogs, notably
%%%      those of the University of California,
%%%      Library of Congress, OCLC, plus the IEEE
%%%      INSPEC (1989--1995) database.
%%%
%%%      This bibliography is sorted by year, and
%%%      within each year, by author and title key,
%%%      with 'bibsrt -byyear'. Cross-referenced
%%%      proceedings entries appear at the end,
%%%      because of a restriction in the current
%%%      BibTeX.
%%%
%%%      The checksum field above contains a CRC-16
%%%      checksum as the first value, followed by the
%%%      equivalent of the standard UNIX wc (word
%%%      count) utility output of lines, words, and
%%%      characters. This is produced by Robert
%%%      Solovay's checksum utility.",
%%% }
%%% =====

%%% Acknowledgement abbreviations:

@String{ack-nhfb = "Nelson H. F. Beebe,
                  University of Utah,
                  Department of Mathematics, 110 LCB,
                  155 S 1400 E RM 233,
                  Salt Lake City, UT 84112-0090, USA,
                  Tel: +1 801 581 5254,
                  FAX: +1 801 581 4148,
                  e-mail: \path|beebe@math.utah.edu|,
                          \path|beebe@acm.org|,
                          \path|beebe@computer.org| (Internet),
                  URL: \path|http://www.math.utah.edu/~beebe/|"}

%%% =====
%%% Institution abbreviations:

@String{inst-NAG      = "Numerical Algorithms Group, Inc."}

```

```

@String{inst-NAG:adr          = "Downer's Grove, IL, USA and Oxford,
                                UK"}

%%% =====
%%% Journal abbreviations:

@String{j-ACM-COMM-COMP-ALGEBRA = "ACM Communications in Computer Algebra"}

@String{j-AMER-J-PHYSICS        = "American Journal of Physics"}

@String{j-COED                  = "CoED"}

@String{j-COMP-PHYS-COMM        = "Computer Physics Communications"}

@String{j-ELECTRONIK            = "Elektronik"}

@String{j-IFIP-TRANS-A          = "IFIP Transactions. A. Computer Science and
                                Technology"}

@String{j-J-SYMBOLIC-COMP       = "Journal of Symbolic Computation"}

@String{j-LECT-NOTES-COMP-SCI   = "Lecture Notes in Computer Science"}

@String{j-MATH-COMP-EDU         = "Mathematics and computer education"}

@String{j-SIGPLAN               = "ACM SIG{\-}PLAN Notices"}

@String{j-SIGSAM                = "SIGSAM Bulletin (ACM Special Interest
                                Group on Symbolic and Algebraic
                                Manipulation)"}

@String{j-STAT-COMP             = "Statistics and Computing"}

@String{j-THEOR-COMP-SCI        = "Theoretical Computer Science"}

@String{j-TOMS                  = "ACM Transactions on Mathematical Software"}

@String{j-ZEIT-ANGE-MATH-PHYS   = "Zeitschrift fur Angewandte Mathematik
                                und Physik"}

%%% =====
%%% Publisher abbreviations:

@String{pub-ACM                 = "ACM Press"}

@String{pub-ACM:adr             = "New York, NY 10036, USA"}

@String{pub-AP                  = "Academic Press"}

```

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@String{pub-AP:adr          = "New York, NY, USA"}

@String{pub-DEKKER         = "Marcel Dekker"}

@String{pub-DEKKER:adr     = "New York, NY, USA"}

@String{pub-SV             = "Spring{\-}er-Ver{\-}lag"}

@String{pub-SV:adr        = "Berlin, Germany~/ Heidelberg,
                          Germany~/ London, UK~/ etc."}

@String{pub-VIEWEG         = "Friedrich Vieweg und Sohn"}

@String{pub-VIEWEG:adr     = "Braunschweig, Germany"}

%%% =====
%%% Series abbreviations:

@String{ser-LNCS           = "Lecture Notes in Computer Science"}

%%% =====
%%% Bibliography entries:

\getchunk{Griesmer:1971:SIF}
\getchunk{Jenks:1971:MPS}
\getchunk{Griesmer:1972:E0Sb}
\getchunk{Griesmer:1972:SCV}
\getchunk{Jenks:1974:SL}
\getchunk{Norman:1975:CFP}
\getchunk{Jenks:1976:PC}
\getchunk{Lueken:1977:UIF}
\getchunk{Andrews:1984:RS}
\getchunk{Davenport:1984:S}
\getchunk{Jenks:1984:NSL}
\getchunk{Jenks:1984:PKN}
\getchunk{Sutor:1985:SIC}
\getchunk{Gebauer:1986:BAS}
\getchunk{Jenks:1986:SIA}
\getchunk{Lucks:1986:FIP}
\getchunk{Purtilo:1986:ASI}
\getchunk{Burge:1987:ISS}
\getchunk{Senechaud:1987:SIP}
\getchunk{Sutor:1987:TICa}
\getchunk{Sutor:1987:TICb}
\getchunk{Andrews:1988:ASP}
\getchunk{Davenport:1988:CA}
\getchunk{Gebauer:1988:IBA}
\getchunk{Jenks:1988:SIA}
\getchunk{Schwarz:1988:PAD}
\getchunk{Shannon:1988:UGB}

```

\getchunk{Sutor:1988:SIA}
\getchunk{Boehm:1989:TIP}
\getchunk{Bronstein:1989:SRE}
\getchunk{Burge:1989:ISS}
\getchunk{Dicrescenzo:1989:AEA}
\getchunk{Gianni:1989:ASS}
\getchunk{Kusche:1989:IGT}
\getchunk{Mathews:1989:SCA}
\getchunk{Ollivier:1989:IRM}
\getchunk{Salvy:1989:EAA}
\getchunk{Schwarz:1989:FAL}
\getchunk{Sit:1989:GAS}
\getchunk{Wang:1989:PCL}
\getchunk{Watt:1989:FPM}
\getchunk{Davenport:1990:SVA}
\getchunk{Fateman:1990:ATD}
\getchunk{Fortenbacher:1990:ETI}
\getchunk{Fouche:1990:ILK}
\getchunk{Melachrinoudis:1990:TAT}
\getchunk{Augot:1991:MDS}
\getchunk{Bronstein:1991:RDE}
\getchunk{Burge:1991:SRI}
\getchunk{Davenport:1991:SVA}
\getchunk{Goodwin:1991:UMT}
\getchunk{Grabmeier:1991:CSA}
\getchunk{Koseleff:1991:WGF}
\getchunk{Lambe:1991:RHP}
\getchunk{LeBlanc:1991:UMT}
\getchunk{Lynch:1991:NQM}
\getchunk{Salvy:1991:EAA}
\getchunk{Schwarz:1991:MOG}
\getchunk{Wang:1991:MMC}
\getchunk{Anonymous:1992:PEH}
\getchunk{Camion:1992:PCG}
\getchunk{Dalmas:1992:PFL}
\getchunk{Davenport:1992:AS}
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\getchunk{Duval:1992:EPS}
\getchunk{Gil:1992:CJC}
\getchunk{Grabmeier:1992:FFA}
\getchunk{Jenks:1992:ASC}
\getchunk{Lambe:1994:NGC}
\getchunk{Rioboo:1992:RAC}
\getchunk{Sit:1992:ASP}
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\getchunk{Bronstein:1993:FPF}
\getchunk{Davenport:1993:PTR}

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\getchunk{Goodloe:1993:ADT}
\getchunk{Monagan:1993:GPD}
\getchunk{Petitot:1993:EA}
\getchunk{Weber:1993:CCA}
\getchunk{Beneke:1994:DFM}
\getchunk{Brown:1994:CSC}
\getchunk{Gruntz:1994:IG}
\getchunk{Jenks:1994:HMA}
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\getchunk{Seiler:1994:PDO}
\getchunk{vanHoeij:1994:ACI}
\getchunk{Anonymous:1995:GAM}
\getchunk{Arnault:1995:CCN}
\getchunk{Boulanger:1995:OOM}
\getchunk{Broadbery:1995:IDE}
\getchunk{Duval:1995:DEA}
\getchunk{Roesner:1995:VSP}
\getchunk{Benker:1998:ICS}
\getchunk{Doye:1999:ACA}
\getchunk{Kendall:2001:SIC}
\getchunk{Daly:2002:AOS}
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\getchunk{Grabmeier:2003:CAH}
\getchunk{Jenks:2003:AVS}
\getchunk{Caviness:2004:MRD}
\getchunk{Daly:2005:AVAb}
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\getchunk{Daly:2005:AV Aj}
\getchunk{Daly:2006:AVA}
\getchunk{Li:2006:EIP}
\getchunk{Li:2007:VGP}
\getchunk{Portes:2007:AVA}
\getchunk{Page:2007:AOS}
\getchunk{Smith:2007:ADA}
\getchunk{Joyner:2008:OSC}
\getchunk{Petrick:1971:PSS}
\getchunk{Online:1972:OCP}
\getchunk{Golden:1984:PMU}
\getchunk{Fitch:1984:E}
\getchunk{Buchberger:1985:EEC}
\getchunk{Wexelblat:1987:IIT}
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\getchunk{ACM:1989:PAI}

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\getchunk{Huguet:1989:AAA}
\getchunk{Davenport:1989:EEC}
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\getchunk{Mora:1989:AAA}
\getchunk{Watanabe:1990:IPI}
\getchunk{Miola:1990:DIS}
\getchunk{Cohen:1991:EIS}
\getchunk{Watt:1991:PIS}
\getchunk{Anonymous:1991:PAC}
\getchunk{Wang:1992:ISS}
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\getchunk{Calmet:1994:RWC}
\getchunk{Levelt:1995:IPI}
\getchunk{Dooley:1999:IJS}
\getchunk{Brown:2007:PIS}
\getchunk{Shi:2007:CSib}
\getchunk{Proceedings}

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— Anonymous:1992:PEH —

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@Article{Anonymous:1992:PEH,
  author = "Anonymous",
  title = "Programming Environments for High-Level Scientific
    Problem Solving. {IFIP} {TC2}\slash {WG} 2.5 Working
    Conference",
  journal = j-IFIP-TRANS-A,
  volume = "A-2",
  pages = "??--??",
  year = "1992",
  CODEN = "ITATEC",
  ISSN = "0926-5473",
  bibdate = "Tue Sep 17 06:41:20 MDT 1996",
  bibsource = "http://www.math.utah.edu/pub/tex/bib/axiom.bib",
  acknowledgement = ack-nhfb,
  confdate = "23--27 Sept. 1991",
  conflocation = "Karlsruhe, Germany",
  confsponsor = "IFIP",
  fjournal = "IFIP Transactions. A. Computer Science and
    Technology",
  pubcountry = "Netherlands",
}

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— Anonymous:1995:GAM —

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@Article{Anonymous:1995:GAM,
  author =      "Anonymous",
  title =      "{GAMM} 94 Annual Meeting",
  journal =     j-ZEIT-ANGE-MATH-PHYS,
  volume =     "75",
  number =     "suppl. 2",
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  CODEN =      "ZAMMAX",
  ISSN =       "0044-2267",
  bibdate =    "Fri Dec 29 12:46:02 MST 1995",
  bibsource =  "http://www.math.utah.edu/pub/tex/bib/axiom.bib",
  acknowledgement = ack-nhfb,
  confdate =   "4--8 April 1994",
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  pubcountry = "Germany",
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%% Cross-referenced entries must come last:
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@Proceedings{Petrick:1971:PSS,
  editor =      "S. R. Petrick",
  booktitle =   "{Proceedings of the second symposium on Symbolic and
                  Algebraic Manipulation, March 23--25, 1971, Los
                  Angeles, California}",
  title =       "{Proceedings of the second symposium on Symbolic and
                  Algebraic Manipulation, March 23--25, 1971, Los
                  Angeles, California}",
  publisher =   pub-ACM,
  address =     pub-ACM:adr,
  pages =       "x + 464",
  year =        "1971",
  LCCN =        "QA76.5 .S94 1971",
  bibdate =     "Sat Dec 30 08:56:27 1995",
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@Proceedings{Online:1972:OCP,
  key = "Online'72",
  booktitle = "{Online 72: conference proceedings \ldots{}
international conference on online interactive
computing, Brunel University, Uxbridge, England, 4--7
September 1972}",
  title = "{Online 72: conference proceedings \ldots{}
international conference on online interactive
computing, Brunel University, Uxbridge, England, 4--7
September 1972}",
  publisher = "Online Computer Systems Ltd",
  address = "Uxbridge, England",
  pages = "various",
  month = sep,
  year = "1972",
  ISBN = "0-903796-02-3",
  ISBN-13 = "978-0-903796-02-6",
  LCCN = "QA76.55 .054 1972",
  bibdate = "Fri Dec 29 18:31:29 1995",
  bibsource = "http://www.math.utah.edu/pub/tex/bib/axiom.bib",
  note = "Two volumes.",
  acknowledgement = ack-nhfb,
}

```

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@Proceedings{Golden:1984:PMU,
  editor = "V. Ellen Golden and M. A. Hussain",
  booktitle = "{Proceedings of the 1984 MACSYMA Users' Conference:
Schenectady, New York, July 23--25, 1984}",
  title = "{Proceedings of the 1984 MACSYMA Users' Conference:
Schenectady, New York, July 23--25, 1984}",
  publisher = "General Electric",
  address = "Schenectady, NY, USA",
  pages = "xv + 567",
  year = "1984",
  bibdate = "Sat Dec 30 09:01:01 1995",
  bibsource = "http://www.math.utah.edu/pub/tex/bib/axiom.bib",
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}

```

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@Proceedings{Fitch:1984:E,
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  booktitle = "{EUROSAM '84: International Symposium on Symbolic and
Algebraic Computation, Cambridge, England, July 9--11,
1984}",
  title = "{EUROSAM '84: International Symposium on Symbolic and
Algebraic Computation, Cambridge, England, July 9--11,
1984}",
  volume = "174",
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}

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address =      pub-SV:adr,
pages =       "xi + 396",
year =        "1984",
ISBN =        "0-387-13350-X",
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LCCN =        "QA155.7.E4 I57 1984",
bibdate =     "Fri Dec 29 18:17:16 1995",
bibsource =   "http://www.math.utah.edu/pub/tex/bib/axiom.bib",
series =      "Lecture Notes In Computer Science",
acknowledgement = ack-nhfb,
}

@Proceedings{Buchberger:1985:EEC,
  editor =      "Bruno Buchberger and Bob F. Caviness",
  booktitle =   "{EUROCAL '85: European Conference on Computer Algebra,
Linz, Austria, April 1--3, 1985: proceedings}",
  title =      "{EUROCAL '85: European Conference on Computer Algebra,
Linz, Austria, April 1--3, 1985: proceedings}",
  volume =     "204",
  publisher =   pub-SV,
  address =     pub-SV:adr,
  pages =      "various",
  year =       "1985",
  ISBN =       "0-387-15983-5 (vol. 1), 0-387-15984-3 (vol. 2)",
  ISBN-13 =    "978-0-387-15983-6 (vol. 1), 978-0-387-15984-3 (vol.
2)",
  LCCN =       "QA155.7.E4 E86 1985",
  bibdate =    "Fri Dec 29 18:07:46 1995",
  bibsource =  "http://www.math.utah.edu/pub/tex/bib/axiom.bib",
  note =       "Two volumes.",
  series =     "Lecture Notes in Computer Science",
  acknowledgement = ack-nhfb,
}

@Proceedings{Wexelblat:1987:IIT,
  editor =      "Richard L. Wexelblat",
  booktitle =   "{Proceedings of the SIGPLAN '87 Symposium on
Interpreters and Interpretive Techniques, St. Paul,
Minnesota, June 24--26, 1987}",
  title =      "{Proceedings of the SIGPLAN '87 Symposium on
Interpreters and Interpretive Techniques, St. Paul,
Minnesota, June 24--26, 1987}",
  publisher =   pub-ACM,
  address =     pub-ACM:adr,
  pages =      "vii + 291",
  year =       "1987",
  ISBN =       "0-89791-235-7",
  ISBN-13 =    "978-0-89791-235-8",
  LCCN =       "QA76.7 .S54 v.22:7",
  bibdate =    "Thu Jan 04 18:40:07 1996",

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bibsource = "http://www.math.utah.edu/pub/tex/bib/axiom.bib",
note = "SIGPLAN Notices, vol. 22, no. 7 (July 1987).",
price = "US\$23.00",
acknowledgement = ack-nhfb,
keywords = "design; interpreters (computer programs) ---
            congresses; languages",
subject = "D.0 Software, GENERAL",
}

@Proceedings{Janssen:1988:TCA,
  editor = "R. Jan{\ss}en",
  booktitle = "{Trends in Computer Algebra, International Symposium
Bad Neuenahr, May 19--21, 1987, Proceedings}",
  title = "{Trends in Computer Algebra, International Symposium
Bad Neuenahr, May 19--21, 1987, Proceedings}",
  volume = "296",
  publisher = pub-SV,
  address = pub-SV:adr,
  pages = "??",
  year = "1988",
  ISBN = "3-540-18928-9, 0-387-18928-9",
  ISBN-13 = "978-3-540-18928-2, 978-0-387-18928-4",
  LCCN = "QA155.7.E4T74 1988",
  bibsource = "/usr/local/src/bib/bibliography/Theory/Comp.Alg.bib;
http://www.math.utah.edu/pub/tex/bib/axiom.bib",
  series = "Lecture Notes in Computer Science",
  notes = "{\footnotesize Die Beitr{\a}ge in diesem Band geben
einen guten {\u}berblick {\u}ber den aktuellen Stand
der Forschung in verschiedenen f{\u}r die \CA\
wichtigen Teilgebieten, so z.B. Faktorisierung von
Polynomen, konstruktive Galois Theory,
Termersetzungssysteme und das System Scratchpad II.
\hfill F. Schwarz}",
}

@Proceedings{ACM:1989:PAI,
  editor = "{ACM}",
  booktitle = "{Proceedings of the ACM-SIGSAM 1989 International
Symposium on Symbolic and Algebraic Computation, ISSAC
'89}",
  title = "{Proceedings of the ACM-SIGSAM 1989 International
Symposium on Symbolic and Algebraic Computation, ISSAC
'89}",
  publisher = pub-ACM,
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bibdate =      "Tue Sep 17 06:46:18 MDT 1996",
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}

@Proceedings{Huguet:1989:AAA,
  editor =      "L. Huguet and A. Poli",
  booktitle =   "{Applied Algebra, Algebraic Algorithms and
Error-Correcting Codes. 5th International Conference,
AAECC-5 Proceedings}",
  title =       "{Applied Algebra, Algebraic Algorithms and
Error-Correcting Codes. 5th International Conference,
AAECC-5 Proceedings}",
  publisher =   pub-SV,
  address =     pub-SV:adr,
  pages =       "417",
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  ISBN =        "3-540-51082-6",
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  LCCN =        "QA268.A35 1987",
  bibdate =     "Tue Sep 17 06:46:18 MDT 1996",
  bibsource =   "http://www.math.utah.edu/pub/tex/bib/axiom.bib",
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  confdate =    "15--19 June 1987",
  conflocation = "Menorca, Spain",
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@Proceedings{Davenport:1989:EEC,
  editor =      "J. H. Davenport",
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Proceedings}",
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Proceedings}",
  publisher =   pub-SV,
  address =     pub-SV:adr,
  pages =       "viii + 499",
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  bibdate =     "Tue Sep 17 06:46:18 MDT 1996",
  bibsource =   "http://www.math.utah.edu/pub/tex/bib/axiom.bib",
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@Proceedings{Gianni:1989:SAC,
  editor = "P. (Patrizia) Gianni",
  booktitle = "{Symbolic and Algebraic Computation. International
    Symposium ISSAC '88, Rome, Italy, July 4--8, 1988.
    Proceedings}",
  title = "{Symbolic and Algebraic Computation. International
    Symposium ISSAC '88, Rome, Italy, July 4--8, 1988.
    Proceedings}",
  volume = "358",
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  address = pub-SV:adr,
  pages = "xi + 543",
  year = "1989",
  ISBN = "3-540-51084-2",
  ISBN-13 = "978-3-540-51084-0",
  LCCN = "QA76.95 .I57 1988",
  bibdate = "Tue Sep 17 06:46:18 MDT 1996",
  bibsource = "http://www.math.utah.edu/pub/tex/bib/axiom.bib",
  note = "Conference held jointly with AAEECC-6.",
  series = "Lecture Notes In Computer Science",
  acknowledgement = ack-nhfb,
  confdate = "4--8 July 1988",
  conflocation = "Rome, Italy",
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}

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@Proceedings{Mora:1989:AAA,
  editor = "T. Mora",
  booktitle = "{Applied Algebra, Algebraic Algorithms and
    Error-Correcting Codes. 6th International Conference,
    AAEECC-6, Rome, Italy, July 4--8, 1988. Proceedings}",
  title = "{Applied Algebra, Algebraic Algorithms and
    Error-Correcting Codes. 6th International Conference,
    AAEECC-6, Rome, Italy, July 4--8, 1988. Proceedings}",
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@Proceedings{Watanabe:1990:IPI,
  editor =      "Shunro Watanabe and Morio Nagata",
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                  on Symbolic and Algebraic Computation}",
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                  on Symbolic and Algebraic Computation}",
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  confsponsor = "Inf. Processing Soc. Japan; Japan Soc. Software Sci.
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  pubcountry =  "USA",
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@Proceedings{Miola:1990:DIS,
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confdate = "6--8 July 1993",
conflocation = "Kiev, Ukraine",
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  editor =      "G. Jacob and N. E. Oussous and S. Steinberg",
  booktitle =   "{Proceedings SC 93. International IMACS Symposium on
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  title =      "{Proceedings SC 93. International IMACS Symposium on
                  Symbolic Computation. New Trends and Developments}",
  publisher =   "LIFL Univ. Lille",
  address =     "Lille, France",
  pages =       "vii + 239",
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                  Systems International Symposium. DISCO '93 Gmunden,
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                  Computation}",
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                  Canada}",
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                  30, 2007, Proceedings, Part II}",
  title =       "{Computational Science --- ICCS 2007: 7th
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